

$$\Rightarrow \frac{(\theta_A - \theta_x)}{6} = \frac{(\theta_A - \theta_B)}{20} \Rightarrow (100 - \theta) = \frac{6}{20} \times (100 - 0)$$

$$\Rightarrow \theta_x = 70^\circ\text{C}$$

4. (d) In conducting rod given heat transmits so burning temperature does not reach soon. In wooden rod heat doesn't conduct.

5. (b) $Q = mL = KA \frac{(\theta_1 - \theta_2)}{l} t \Rightarrow m = \frac{1}{L} \times KA \frac{(\theta_1 - \theta_2)}{l} \times t$

$$= \frac{1}{80} \times 0.2 \times 4 \times \frac{(100 - 0)}{\sqrt{4}} \times 10 \times 60 \quad (\because l^2 = 4 \Rightarrow l = \sqrt{4})$$

$$= \frac{0.2 \times 4 \times 100 \times 600}{80 \times 2} = 300 \text{ gm}$$

6. (d) $\lambda_m T = 2892 \times 10^{-6} \Rightarrow T = \frac{2892 \times 10^{-6}}{14.46 \times 10^{-6}} = 200 \text{ K}$

7. (a) $\lambda_m \propto \frac{1}{T} \Rightarrow \lambda_{m1} T_1 = \lambda_{m2} T_2$

$$\Rightarrow T_2 = \frac{\lambda_{m1} T_1}{\lambda_{m2}} = \frac{1.4 \times 10^{-6} \times 1000}{2.8 \times 10^{-6}} = 2000 \text{ K}$$

8. (a) We know $\lambda_{\max} T = b$

$$\Rightarrow T = \frac{b}{\lambda_{\max}} = \frac{2898 \times 10^{-6}}{289.8 \times 10^{-9}} = 10^4 \text{ K}$$

According to Stefan's Law

$$E = \sigma T^4 = (5.67 \times 10^{-8})(10^4)^4 = 5.67 \times 10^8 \text{ W/m}^2$$

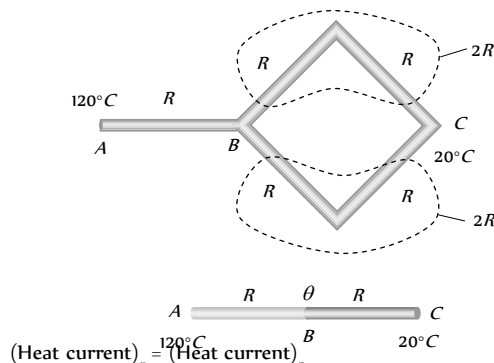
9. (a) The rate of heat loss is proportional to the difference in temperature. The difference of temperature between the tea in cup A and the surrounding is reduced, so it loses less heat. the tea in cup B loses more heat because of large temperature difference. Hence the tea in cup A will be hotter.

10. (c) Rate of cooling $R_C = \frac{A\sigma(T^4 - T_0^4)}{mc} = \frac{A\sigma(T^4 - T_0^4)}{V\rho C}$

$$\Rightarrow R_C \propto \frac{A}{V} \propto \frac{1}{r} \propto \frac{1}{(\text{Diameter})} \quad (\because m = \rho V)$$

Since diameter of A is half that of B so it's rate of cooling will be doubled that of B

11. (c) If thermal resistance of each rod is considered R then, the given combination can be redrawn as follows



$$\frac{(120 - 20)}{R} = \frac{(120 - \theta)}{R} \Rightarrow \theta = 70^\circ\text{C}$$

12. (b) No, In convection the hot liquid at the bottom becomes lighter and hence it rises up. In this way the base of the convection is the difference in weight and upthrust. In the state of weightlessness this difference does not occur, so convection is not possible.

13. (b)

14. (d) For the two sheets $H_1 = H_2$ (H = Rate of heat flow)

$$\Rightarrow \frac{(100 - \theta)}{R} = \frac{(\theta - 20)}{3R} \Rightarrow \theta = 80^\circ\text{C}$$

15. (a) Rate of heat loss per unit area due to radiation i.e. emissive power $e = \sigma(T^4 - T_0^4)$

$$= 0.6 \times \frac{17}{3} \times 10^{-8} \times [(400)^4 - (300)^4]$$

$$= 3.4 \times 10^{-8} \times (175 \times 10^8) = 3.4 \times 175 = 595 \text{ J/m}^2 \times \text{sec}$$

16. (a) Black bulb absorbs more heat in comparison with painted bulb. So air in black bulb expands more. Hence the level of alcohol in limb X falls while that in limb Y rises.

17. (a) Heat current $H = \frac{\Delta\theta}{R} \Rightarrow \frac{H_p}{H_s} = \frac{R_s}{R_p}$

$$\text{In first case : } R_s = R_1 + R_2 = \frac{l}{(3K)A} + \frac{l}{KA} = \frac{4}{3} \frac{l}{KA}$$

$$\text{In second case : } R_p = \frac{R_1 R_2}{R_1 + R_2} = \frac{\frac{l}{(3K)A} \times \frac{l}{KA}}{\left(\frac{l}{(3K)A} + \frac{l}{KA}\right)} = \frac{l}{4KA}$$

$$\therefore \frac{H_p}{H_s} = \frac{\frac{4l}{3KA}}{\frac{l}{4KA}} = \frac{16}{3}$$

17. (b) $\frac{Q}{t} = \frac{KA(\theta_1 - \theta_2)}{l} = \frac{0.2 \times 10 \times 20}{2 \times 10^{-3}} = 2 \times 10^4 \text{ J/sec}$

18. (c) All wavelengths are emitted.

19. (d)

20. (d) $\frac{Q_1}{Q_2} = \frac{T_1^4}{T_2^4} = \left(\frac{273 + 27}{273 + 127}\right)^4 = \left(\frac{300}{400}\right)^4 = \frac{81}{256}$

21. (b) In first step

$$\frac{62 - 61}{T} = K \left[\frac{62 - 61}{2} - 30^\circ \right] \Rightarrow \frac{1}{T} = K[81.5] \quad \dots(i)$$

In second step, suppose process takes T' min then

$$\frac{46 - 45.5}{T'} = K \left[\frac{46 - 45.5}{2} - 30 \right] \frac{0.5}{T'} = K[15.75] \quad \dots(ii)$$

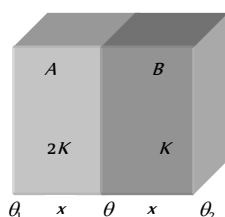
On diving equation (i) and (ii) $\frac{2T'}{T} = 2 \Rightarrow T' = T$

22. (b) Suppose conductivity of layer B is K , then it is $2K$ for layer A .
Also conductivity of

combination layers A and B is K ,

$$= \frac{2 \times 2K \times K}{(2K + K)} = \frac{4}{3} K$$

$$\text{Hence } \left(\frac{Q}{t} \right)_{\text{Combination}} = \left(\frac{Q}{t} \right)_A$$

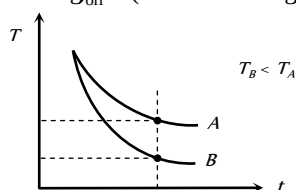


$$\Rightarrow \frac{4}{3} \frac{KA \times 60}{2x} = \frac{2K.A \times (\Delta\theta)_A}{x} \Rightarrow (\Delta\theta)_A = 20K$$

23. (b) As we know, Rate of cooling $\propto \frac{1}{\text{specific heat}(c)}$

$$\because c_{\text{oil}} < c_{\text{Water}}$$

$$\Rightarrow (\text{Rate of cooling})_{\text{oil}} > (\text{Rate of cooling})_{\text{Water}}$$



It is clear that, at a particular time after start cooling, temperature of oil will be less than that of water.

So graph B represents the cooling curve of oil and A represents the cooling curve of water



16

Simple Harmonic Motion

Periodic and Oscillatory (Vibratory) Motion



(1) A motion, which repeats itself over and over again after a regular interval of time is called a periodic motion

Revolution of earth around the sun (period one year), Rotation of earth about its polar axis (period one day), Motion of hour's hand of a clock (period 12-hour) etc are common examples of periodic motion.

(2) Oscillatory or vibratory motion is that motion in which a body moves to and fro or back and forth repeatedly about a fixed point in a definite interval of time. In such a motion, the body is confined within well-defined limits on either side of mean position. Oscillatory motion is also called as harmonic motion.

(i) Common examples are

(a) The motion of the pendulum of a wall clock

(b) The motion of a load attached to a spring, when it is pulled and then released.

(c) The motion of liquid contained in U-tube when it is compressed once in one limb and left to itself.

(d) A loaded piece of wood floating over the surface of a liquid when pressed down and then released executes oscillatory motion.

(ii) Harmonic oscillation is that oscillation which can be expressed in terms of single harmonic function (i.e. sine or cosine function). *Example* : $y = a \sin \omega t$ or $y = a \cos \omega t$

(iii) Non-harmonic oscillation is that oscillation which can not be expressed in terms of single harmonic function. It is a combination of two or more than two harmonic oscillations. *Example* : $y = a \sin \omega t + b \sin 2\omega t$.

Simple Harmonic Motion

(1) Simple harmonic motion is a special type of periodic motion, in which a particle moves to and fro repeatedly about a mean position.

(2) In linear S.H.M. a restoring force which is always directed towards the mean position and whose magnitude at any instant is directly proportional to the displacement of the particle from the mean position at that instant i.e. Restoring force $\propto$ Displacement of the particle from mean position.

$$F \propto -x \Rightarrow F = -kx$$

Where k is known as force constant. Its S.I. unit is Newton/meter and dimension is $[MT^{-2}]$.

(3) In stead of straight line motion, if particle or centre of mass of body is oscillating on a small arc of circular path, then for angular S.H.M.

Restoring torque (τ) $\propto$ - Angular displacement (θ)

Some Important Definitions

(1) **Time period (T)** : It is the least interval of time after which the periodic motion of a body repeats itself.

S.I. unit of time period is second.

(2) **Frequency (n)** : It is defined as the number of oscillations executed by body per second. S.I unit of frequency is hertz (Hz).

(3) **Angular Frequency (ω)** : Angular frequency of a body executing periodic motion is equal to product of frequency of the body with factor 2π . Angular frequency $\omega = 2\pi n$

Its unit is rad/sec .

(4) **Phase (ϕ)** : Phase of a vibrating particle at any instant is a physical quantity, which completely express the position and direction of motion, of the particle at that instant with respect to its mean position.

In oscillatory motion the phase of a vibrating particle is the argument of *sine* or *cosine* function involved to represent the generalised equation of motion of the vibrating particle.

$$y = a \sin \theta = a \sin(\omega t + \phi_0)$$

here, $\theta = \omega t + \phi_0$ = phase of vibrating particle.

ϕ = Initial phase or epoch. It is the phase of a vibrating particle at $t = 0$.

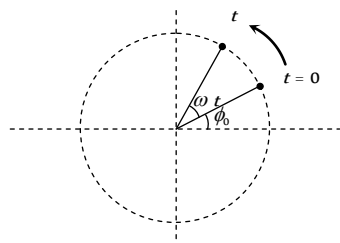


Fig. 16.2

(1) **Same phase** : Two vibrating particle are said to be in same phase, if the phase difference between them is an even multiple of π or path difference is an even multiple of $(\lambda / 2)$ or time interval is an even multiple of $(T / 2)$ because 1 time period is equivalent to $2\pi \text{ rad}$ or 1 wave length (λ) .

(2) **Opposite phase** : When the two vibrating particles cross their respective mean positions at the same time moving in opposite directions, then the phase difference between the two vibrating particles is 180° .

Opposite phase means the phase difference between the particle is an odd multiple of π (say $\pi, 3\pi, 5\pi, 7\pi, \dots$) or the path difference is an odd multiple of λ (say $\frac{\lambda}{2}, \frac{3\lambda}{2}, \dots$) or the time interval is an odd multiple of $(T / 2)$.

(3) **Phase difference** : If two particles performs S.H.M and their equation are

$$y_1 = a \sin(\omega t + \phi_1) \quad \text{and} \quad y_2 = a \sin(\omega t + \phi_2)$$

then phase difference $\Delta\phi = (\omega t + \phi_2) - (\omega t + \phi_1) = \phi_2 - \phi_1$

Displacement in S.H.M.

(1) The displacement of a particle executing S.H.M. at an instant is defined as the distance of particle from the mean position at that instant.

(2) Simple harmonic motion is also defined as the projection of uniform circular motion on any diameter of circle of reference.

(3) If the projection is taken on y -axis. then from the figure

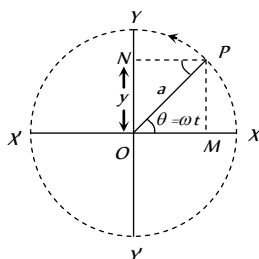


Fig. 16.3

$$y = a \sin \omega t = a \sin \frac{2\pi}{T} t = a \sin 2\pi n t = a \sin(\omega t \pm \phi)$$

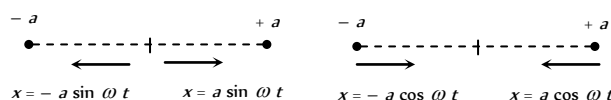
(i) $y = a \sin \omega t$ when the time is noted from the instant when the vibrating particle is at mean position.

(ii) $y = a \cos \omega t$ when the time is noted from the instant when the vibrating particle is at extreme position.

(iii) $y = a \sin(\omega t \pm \phi)$ when the vibrating particle is ϕ phase leading or lagging from the mean position.

(4) If the projection of P is taken on X -axis then equations of S.H.M. can be given as

$$x = a \cos(\omega t \pm \phi) = a \cos\left(\frac{2\pi}{T} t \pm \phi\right) = a \cos(2\pi n t \pm \phi)$$



(A)

(B)

Fig. 16.4

(5) Direction of displacement is always away from the equilibrium position, particle either is moving away from or is coming towards the equilibrium position.

Velocity in S.H.M.

(1) Velocity of the particle executing S.H.M. at any instant, is defined as the time rate of change of its displacement at that instant.

(2) In case of S.H.M. when motion is considered from the equilibrium position, displacement $y = a \sin \omega t$

$$\text{So } v = \frac{dy}{dt} = a\omega \cos \omega t = a\omega \sqrt{1 - \sin^2 \omega t} = \omega \sqrt{a^2 - y^2}$$

[As $\sin \omega t = y/a$]

(3) At mean position or equilibrium position ($y = 0$ and $\theta = \omega t = 0$), velocity of particle is maximum and it is $v_m = a\omega$.

(4) At extreme position ($y = \pm a$ and $\theta = \omega t = \pi/2$), velocity of oscillating particle is zero i.e. $v = 0$.

$$(5) \text{ From } v = \omega \sqrt{a^2 - y^2} \Rightarrow v^2 = \omega^2(a^2 - y^2) \Rightarrow \frac{v^2}{\omega^2} = a^2 - y^2$$

$$\Rightarrow \frac{v^2}{a^2 \omega^2} + \frac{y^2}{a^2} = 1$$

This is the equation of ellipse. Hence the graph between v and y is an ellipse.

For $\omega = 1$, graph between v and y is a circle.

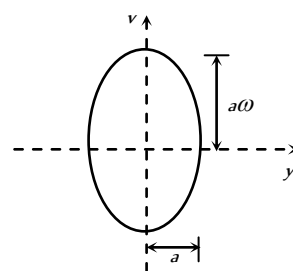


Fig. 16.5

(6) Direction of velocity is either towards or away from mean position depending on the position of particle.

Acceleration in S.H.M.

(1) The acceleration of the particle executing S.H.M. at any instant, is defined as the rate of change of its velocity at that instant. So acceleration

$$A = \frac{dv}{dt} = \frac{d}{dt}(a\omega \cos \omega t) = -\omega^2 a \sin \omega t = -\omega^2 y$$

$$[\text{As } y = a \sin \omega t]$$

(2) In S.H.M. as $|\text{Acceleration}| = \omega^2 y$ is not constant. So equations of transitory motion can not be applied.

(3) In S.H.M. acceleration is maximum at extreme position (at $y = \pm a$). Hence $|A_{\max}| = \omega^2 a$ when $|\sin \omega t| = \text{maximum} = 1$ i.e. at $t = \frac{T}{4}$ or

$$\omega t = \frac{\pi}{2}. \text{ From equation (ii) } |A_{\max}| = \omega^2 a \text{ when } y = a.$$

(i) In S.H.M. acceleration is minimum at mean position

From equation (i) $A_{\min} = 0$ when $\sin \omega t = 0$ i.e. at $t = 0$ or $t = \frac{T}{2}$ or $\omega t = \pi$. From equation (ii) $A_{\min} = 0$ when $y = 0$

(ii) Acceleration is always directed towards the mean position and so is always opposite to displacement

$$\text{i.e., } A \propto -y$$

Graph between acceleration (A) and displacement (y) is a straight line as shown

Slope of the line = $-\omega^2$

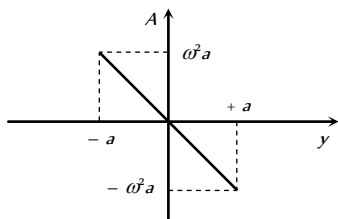


Fig. 16.6

Comparative Study of Displacement Velocity and Acceleration

(1) All the three quantities displacement, velocity and acceleration show harmonic variation with time having same period.

(2) The velocity amplitude is ω times the displacement amplitude

(3) The acceleration amplitude is ω^2 times the displacement amplitude

(4) In S.H.M. the velocity is ahead of displacement by a phase angle $\pi/2$

(5) In S.H.M. the acceleration is ahead of velocity by a phase angle $\pi/2$

(6) The acceleration is ahead of displacement by a phase angle of π

Table 16.1 : Various physical quantities in S.H.M. at different position :

Graph	Formula	At mean position	At extreme position
Displacement	$y = a \sin \omega t$	$y = 0$	$y = \pm a$
Velocity	$v = a\omega \cos \omega t$ $= a\omega \sin(\omega t + \frac{\pi}{2})$	$v_{\max} = a\omega$	$v_{\min} = 0$

	or $v = \omega \sqrt{a^2 - y^2}$		
Acceleration	$A = -a\omega^2 \sin \omega t$ $= a\omega^2 \sin(\omega t + \pi)$ or $ A = \omega^2 y$	$A_{\min} = 0$	$ A_{\max} = \omega^2 a$
Force	$F = -m\omega^2 a \sin \omega t$ or $F = m\omega^2 y$	$F_{\min} = 0$	$F_{\max} = m\omega^2 a$

Energy in S.H.M.

(i) **Potential energy** : This is an account of the displacement of the particle from its mean position.

(i) The restoring force $F = -ky$ against which work has to be done. Hence potential energy U is given by

$$U = \int dU = - \int dW = - \int_0^y F dx = \int_0^y ky dy = \frac{1}{2} ky^2 + U_0$$

where U_0 = Potential energy at equilibrium position.

$$\text{If } U_0 = 0 \text{ then } U = \frac{1}{2} m \omega^2 y^2 \quad [\text{As } \omega^2 = k/m]$$

$$(ii) \text{ Also } U = \frac{1}{2} m \omega^2 a^2 \sin^2 \omega t = \frac{1}{4} m \omega^2 a^2 (1 - \cos 2\omega t)$$

$$[\text{As } y = a \sin \omega t]$$

Hence potential energy varies periodically with double the frequency of S.H.M.

(iii) Potential energy maximum and equal to total energy at extreme positions

$$U_{\max} = \frac{1}{2} ka^2 = \frac{1}{2} m \omega^2 a^2 \text{ when } y = \pm a; \omega t = \pi/2; t = \frac{T}{4}$$

(iv) Potential energy is minimum at mean position

$$U_{\min} = 0 \quad \text{when } y = 0; \omega t = 0; t = 0$$

(2) **Kinetic energy** : This is because of the velocity of the particle

$$\text{Kinetic Energy } K = \frac{1}{2} mv^2 = \frac{1}{2} m \omega^2 (a^2 - y^2)$$

$$[\text{As } v = \omega \sqrt{a^2 - y^2}]$$

$$(i) \text{ Also } K = \frac{1}{2} m \omega^2 a^2 \cos^2 \omega t = \frac{1}{4} m \omega^2 a^2 (1 + \cos 2\omega t)$$

$$[\text{As } v = a\omega \cos \omega t]$$

Hence kinetic energy varies periodically with double the frequency of S.H.M.

(ii) Kinetic energy is maximum at mean position and equal to total energy at mean position.

$$K_{\max} = \frac{1}{2} m \omega^2 a^2 \text{ when } y = 0; t = 0; \omega t = 0$$

(iii) Kinetic energy is minimum at extreme position.

$$K_{\min} = 0 \text{ when } y = a; t = T/4, \omega t = \pi/2$$

(3) **Total mechanical energy** : Total mechanical energy always remains constant and it is equal to sum of potential energy and kinetic energy i.e. $E = U + K$

$$E = \frac{1}{2} m \omega^2 (a^2 - y^2) + \frac{1}{2} m \omega^2 y^2 = \frac{1}{2} m \omega^2 a^2$$

Total energy is not a position function.

(4) **Energy position graph**

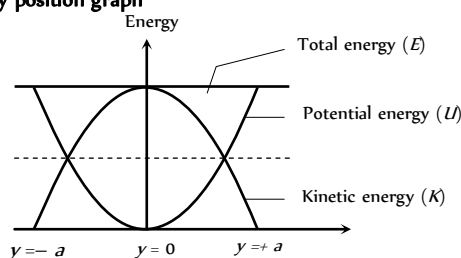


Fig. 16.7

(i) At $y = 0$; $U = 0$ and $K = E$

(ii) At $y = \pm a$; $U = E$ and $K = 0$

(iii) At $y = \pm \frac{a}{2}$; $U = \frac{E}{4}$ and $K = \frac{3E}{4}$

(iv) At $y = \pm \frac{a}{\sqrt{2}}$; $U = K = \frac{E}{2}$

Average Value of P.E. and K.E.

The average value of potential energy for complete cycle is given by

$$U_{\text{average}} = \frac{1}{T} \int_0^T U dt = \frac{1}{T} \int_0^T \frac{1}{2} m \omega^2 a^2 \sin^2(\omega t + \phi) dt = \frac{1}{4} m \omega^2 a^2$$

The average value of kinetic energy for complete cycle

$$K_{\text{average}} = \frac{1}{T} \int_0^T K dt = \frac{1}{T} \int_0^T \frac{1}{2} m \omega^2 a^2 \cos^2 \omega t dt = \frac{1}{4} m \omega^2 a^2$$

Thus average values of kinetic energy and potential energy of harmonic oscillator are equal and each equal to half of the total energy

$$K_{\text{average}} = U_{\text{average}} = \frac{1}{2} E = \frac{1}{4} m \omega^2 a^2$$

Differential Equation of S.H.M.

For S.H.M. (linear) Acceleration $\propto -$ (Displacement)

$$A \propto -y \text{ or } A = -\omega^2 y \text{ or } \frac{d^2 y}{dt^2} = -\omega^2 y$$

$$\text{or } m \frac{d^2 y}{dt^2} + ky = 0 \quad [\text{As } \omega = \sqrt{\frac{k}{m}}]$$

$$\text{For angular S.H.M. } \tau = -c\theta \text{ and } \frac{d^2 \theta}{dt^2} + \omega^2 \theta = 0$$

where $\omega^2 = \frac{c}{I}$ [As c = Restoring torque constant and I = Moment of inertia]

How to Find Frequency and Time Period of S.H.M.

Step 1 : When particle is in its equilibrium position, balance all forces acting on it and locate the equilibrium position mathematically.

Step 2 : From the equilibrium position, displace the particle slightly by a displacement y and find the expression of net restoring force on it.

Step 3 : Try to express the net restoring force acting on particle as a proportional function of its displacement from mean position. The final expression should be obtained in the form.

$$F = -ky$$

Here we put $-$ sign as direction of F is opposite to the displacement y . If a be the acceleration of particle at this displacement, we have

$$a = -\left(\frac{k}{m}\right)y$$

Step 4 : Comparing this equation with the basic differential equation

$$\text{of S.H.M. we get } \omega^2 = \frac{k}{m} \Rightarrow \omega = \sqrt{\frac{k}{m}} \text{ or } n = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

As ω is the angular frequency of the particle in S.H.M., its time period of oscillation can be given as $T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}$

(i) In different types of S.H.M. the quantities m and k will go on taking different forms and names. In general m is called inertia factor and k is called spring factor.

$$\text{Thus } T = 2\pi \sqrt{\frac{\text{Inertia factor}}{\text{Spring factor}}} \text{ or } n = \frac{1}{2\pi} \sqrt{\frac{\text{Spring factor}}{\text{Inertia factor}}}$$

(ii) In linear S.H.M. the spring factor stands for force per unit displacement and inertia factor for mass of the body executing S.H.M. and in Angular S.H.M. k stands for restoring torque per unit angular displacement and inertial factor for moment of inertia of the body executing S.H.M.

For linear S.H.M.

$$T = 2\pi \sqrt{\frac{m}{k}} = \sqrt{\frac{\text{Displacement}}{\text{Force/Displacement}}} = 2\pi \sqrt{\frac{\text{Displacement}}{\text{Acceleration}}}$$

Simple Pendulum

(1) An ideal simple pendulum consists of a heavy point mass body (bob) suspended by a weightless, inextensible and perfectly flexible string from a rigid support about which it is free to oscillate.

(2) But in reality neither point mass nor weightless string exist, so we can never construct a simple pendulum strictly according to the definition.

(3) Suppose simple pendulum of length l is displaced through a small angle θ from its mean (vertical) position. Consider mass of the bob is m and linear displacement from mean position is x

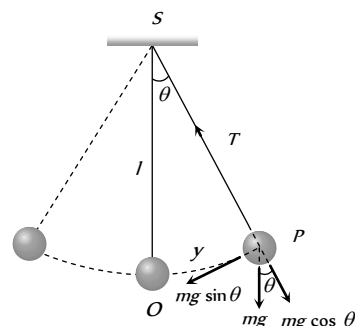


Fig. 16.8

Restoring force acting on the bob

$$F = -mg \sin \theta \quad \text{or} \quad F = -mg \theta = -mg \frac{x}{l}$$

$$(\text{When } \theta \text{ is small } \sin \theta \approx \theta = \frac{\text{Arc}}{\text{Length}} = \frac{OP}{l} = \frac{x}{l})$$

$$\therefore \frac{F}{x} = \frac{-mg}{l} = k \quad (\text{Spring factor})$$

$$\text{So } T = 2\pi \sqrt{\frac{\text{Inertia factor}}{\text{Spring factor}}} = 2\pi \sqrt{\frac{m}{mg/l}} = 2\pi \sqrt{\frac{l}{g}}$$

Factor Affecting Time Period of Simple Pendulum

(1) **Amplitude** : The period of simple pendulum is independent of amplitude as long as its motion is simple harmonic. But if θ is not small, $\sin \theta \neq \theta$ then motion will not remain simple harmonic but will become oscillatory. In this situation if θ is the amplitude of motion. Time period

$$T = 2\pi \sqrt{\frac{l}{g} \left[1 + \frac{1}{2^2} \sin^2 \left(\frac{\theta_0}{2} \right) + \dots \right]} \approx T_0 \left[1 + \frac{\theta_0^2}{16} \right]$$

(2) **Mass of the bob** : Time period of simple pendulum is also independent of mass of the bob. This is why

(i) If the solid bob is replaced by a hollow sphere of same radius but different mass, time period remains unchanged.

(ii) If a girl is swinging in a swing and another sits with her, the time period remains unchanged.

(3) **Length of the pendulum** : Time period $T \propto \sqrt{l}$ where l is the distance between point of suspension and center of mass of bob and is called effective length.

(i) When a sitting girl on a swinging swing stands up, her center of mass will go up and so l and hence T will decrease.

(ii) If a hole is made at the bottom of a hollow sphere full of water and water comes out slowly through the hole and time period is recorded till the sphere is empty, initially and finally the center of mass will be at the center of the sphere. However, as water drains off the sphere, the center of mass of the system will first move down and then will come up. Due to this l and hence T first increase, reaches a maximum and then decreases till it becomes equal to its initial value.

(iii) Different graphs

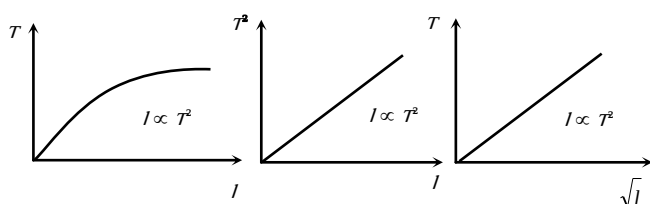


Fig. 16.9

(4) **Effect of g** : $T \propto \frac{1}{\sqrt{g}}$ i.e. as g increase T decreases.

(i) As we go high above the earth surface or we go deep inside the mines the value of g decrease, hence time period of pendulum (T) increases.

(ii) If a clock, based on simple pendulum is taken to hill (or on any other planet), g will decrease so T will increase and clock will become slower.

(iii) Different graphs

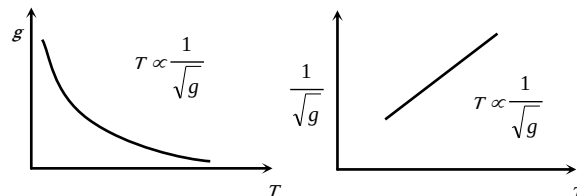


Fig. 16.10

(5) **Effect of temperature on time period** : If the bob of simple pendulum is suspended by a wire then effective length of pendulum will increase with the rise of temperature due to which the time period will increase.

$l = l_0 (1 + \alpha \Delta \theta)$ (If $\Delta \theta$ is the rise in temperature, l_0 = initial length of wire, l = final length of wire)

$$\frac{T}{T_0} = \sqrt{\frac{l}{l_0}} = (1 + \alpha \Delta \theta)^{1/2} \approx 1 + \frac{1}{2} \alpha \Delta \theta$$

$$\text{So } \frac{T}{T_0} - 1 = \frac{1}{2} \alpha \Delta \theta \quad \text{i.e. } \frac{\Delta T}{T} \approx \frac{1}{2} \alpha \Delta \theta$$

Oscillation of Pendulum in Different Situations

(1) **Oscillation in liquid** : If bob of a simple pendulum of density ρ is made to oscillate in some fluid of density σ (where $\sigma < \rho$) then time period of simple pendulum gets increased.

As thrust will oppose its weight hence $mg_{\text{eff.}} = mg - \text{Thrust}$

$$\text{or } g_{\text{eff.}} = g - \frac{V\sigma g}{V\rho} \quad \text{i.e. } g_{\text{eff.}} = g \left[1 - \frac{\sigma}{\rho} \right]$$

$$\Rightarrow \frac{g_{\text{eff.}}}{g} = \frac{\rho - \sigma}{\rho}$$

$$\Rightarrow \frac{T'}{T} = \sqrt{\frac{g}{g_{\text{eff.}}}} = \sqrt{\frac{\rho}{\rho - \sigma}} > 1$$

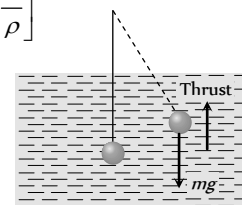


Fig. 16.11

(2) **Oscillation under the influence of electric field** : If a bob of mass m carries a positive charge q and pendulum is placed in a uniform electric field of strength E

(i) If electric field directed vertically upwards.

Effective acceleration

$$g_{\text{eff.}} = g - \frac{qE}{m}$$

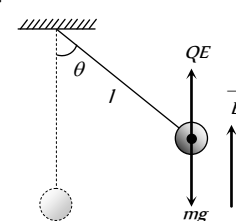


Fig. 16.12

$$\text{So } T = 2\pi \sqrt{\frac{l}{g - \frac{qE}{m}}}$$

(ii) If electric field is vertically downward then

$$g_{\text{eff.}} = g + \frac{qE}{m}$$

$$T = 2\pi \sqrt{\frac{l}{g + \frac{qE}{m}}}$$

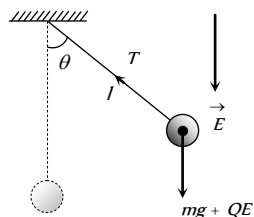


Fig. 16.13

(3) **Pendulum in a lift** : If the pendulum is suspended from the ceiling of the lift.

(i) If the lift is at rest or moving down ward /up ward with constant velocity.

$$T = 2\pi \sqrt{\frac{l}{g}}$$

$$\text{and } n = \frac{1}{2\pi} \sqrt{\frac{g}{l}}$$

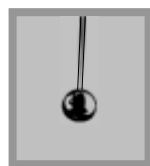


Fig. 16.14

(ii) If the lift is moving up ward with constant acceleration a

$$T = 2\pi \sqrt{\frac{l}{g + a}}$$

$$\text{and } n = \frac{1}{2\pi} \sqrt{\frac{g + a}{l}}$$

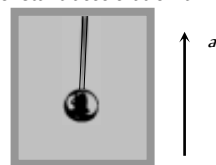


Fig. 16.15

Time period decreases and frequency increases

(iii) If the lift is moving down ward with constant acceleration a

$$T = 2\pi \sqrt{\frac{l}{g - a}}$$

$$\text{and } n = \frac{1}{2\pi} \sqrt{\frac{g - a}{l}}$$

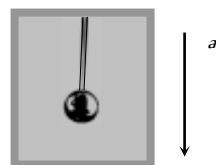


Fig. 16.16

Time period increase and frequency decreases

(iv) If the lift is moving down ward with acceleration $a = g$

$$T = 2\pi \sqrt{\frac{l}{g - g}} = \infty$$

$$\text{and } n = \frac{1}{2\pi} \sqrt{\frac{g - g}{l}} = 0$$

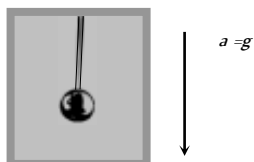


Fig. 16.17

It means there will be no oscillation in a pendulum.

Similar is the case in a satellite and at the centre of earth where effective acceleration becomes zero and pendulum will stop.

(4) **Pendulum in an accelerated vehicle** : The time period of simple pendulum whose point of suspension moving horizontally with acceleration a

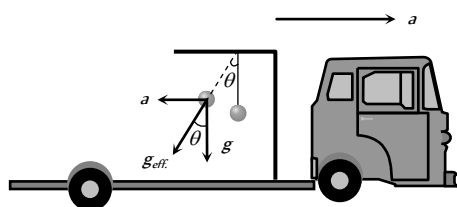


Fig. 16.18

In this case effective acceleration $g_{\text{eff.}} = \sqrt{g^2 + a^2}$

$$T = 2\pi \sqrt{\frac{l}{(g^2 + a^2)^{1/2}}} \quad \text{and} \quad \theta = \tan^{-1}(a/g)$$

If simple pendulum suspended in a car that is moving with constant speed v around a circle of radius r .

$$T = 2\pi \sqrt{\frac{l}{g^2 + \left(\frac{v^2}{r}\right)^2}}$$

Some Other Types of Pendulum

(1) **Infinite length pendulum** : If the length of the pendulum is comparable to the radius of earth then

$$T = 2\pi \sqrt{\frac{1}{g \left[\frac{1}{l} + \frac{1}{R} \right]}}$$

(i) If $l \ll R$, then $\frac{1}{l} \gg \frac{1}{R}$ so $T = 2\pi \sqrt{\frac{l}{g}}$

(ii) If $l \gg R (\rightarrow \infty)$ then $\frac{1}{l} < \frac{1}{R}$

$$\text{so } T = 2\pi \sqrt{\frac{R}{g}} = 2\pi \sqrt{\frac{6.4 \times 10^6}{10}} \cong 84.6 \text{ minutes}$$

and it is the maximum time period which an oscillating simple pendulum can have

(iii) If $l = R$ so $T = 2\pi \sqrt{\frac{R}{2g}} \cong 1 \text{ hour}$

(2) **Second's Pendulum** : It is that simple pendulum whose time period of vibrations is two seconds.

Putting $T = 2 \text{ sec}$ and $g = 9.8 \text{ m/sec}^2$ in $T = 2\pi \sqrt{\frac{l}{g}}$ we get

$$l = \frac{4 \times 9.8}{4\pi^2} = 0.993 \text{ m} = 99.3 \text{ cm}$$

Hence length of second's pendulum is 99.3 cm or nearly 1 meter on earth surface.

For the moon the length of the second's pendulum will be 1/6 meter

[As $g_{\text{moon}} = \frac{g_{\text{Earth}}}{6}$]

(3) **Compound pendulum** : Any rigid body suspended from a fixed support constitutes a physical pendulum. Consider the situation when the body is displaced through a small angle θ . Torque on the body about O is given by

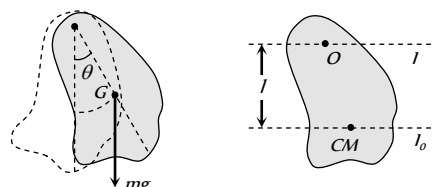


Fig. 16.19

$$\tau = mgl \sin \theta \quad \dots(i)$$

where l = distance between point of suspension and centre of mass of the body.

If I be the M.I. of the body about O . Then $\tau = I\alpha \quad \dots(ii)$

From (i) and (ii), we get $I \frac{d^2\theta}{dt^2} = -mgl \sin \theta$ as θ and $\frac{d^2\theta}{dt^2}$ are

oppositely directed $\Rightarrow \frac{d^2\theta}{dt^2} = -\frac{mgl}{I} \theta$ since θ is very small

Comparing with the equation $\frac{d^2\theta}{dt^2} = -\omega^2 \theta$, we get

$$\omega = \sqrt{\frac{mgl}{I}} \Rightarrow T = 2\pi \sqrt{\frac{I}{mgl}}$$

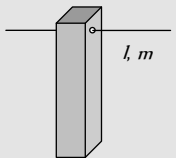
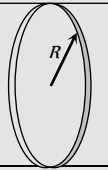
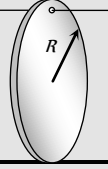
Also $I = I_{cm} + ml^2$ (Parallel axis theorem)

$= mk^2 + ml^2$ (where k = radius of gyration)

$$\therefore T = 2\pi \sqrt{\frac{mk^2 + ml^2}{mgl}} = 2\pi \sqrt{\frac{\frac{K^2}{l} + l}{g}} = 2\pi \sqrt{\frac{l_{eff}}{g}}$$

l_{eff} = Effective length of pendulum = Distance between point of suspension and centre of mass.

Table 16. 2: Some common physical pendulum

Body	Time period
Bar 	$T = 2\pi \sqrt{\frac{2l}{3g}}$
Ring 	$T = 2\pi \sqrt{\frac{2R}{g}}$
Disc 	$T = 2\pi \sqrt{\frac{3R}{2g}}$

Spring System

When a spring is stretched or compressed from its normal position ($x = 0$) by a small distance x , then a restoring force is produced in the spring because it obeys Hooke's law

$$i.e. F \propto -x \Rightarrow F = -kx$$

where k is called spring constant.

(i) It's S.I. unit *Newton/metre*, C.G.S unit *Dyne/cm* and dimension is $[MT^{-2}]$

(ii) Actually k is a measure of the stiffness/softness of the spring.

(iii) For massless spring constant restoring elastic force is same every where

(iv) When a spring compressed or stretched then work done is stored in the form of elastic potential energy in it.

(v) Spring constant depend upon radius and length of the wire used in spring.

(vi) The spring constant k is inversely proportional to the spring length.

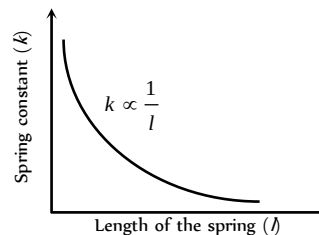


Fig. 16.20

$$k \propto \frac{1}{\text{Extension}} \propto \frac{1}{\text{Length of spring}}$$

That means if the length of spring is halved then its force constant becomes double.

(vii) When a spring of length l is cut in two pieces of length l_1 and l_2 such that $l_1 = nl_2$.

If the constant of a spring is k then spring constant of first part

$$k_1 = \frac{k(n+1)}{n}$$

Spring constant of second part $k_2 = (n+1)k$

and ratio of spring constant $\frac{k_1}{k_2} = \frac{1}{n}$

Spring Pendulum

A point mass suspended from a mass less spring or placed on a frictionless horizontal plane attached with spring (fig.) constitutes a linear harmonic spring pendulum

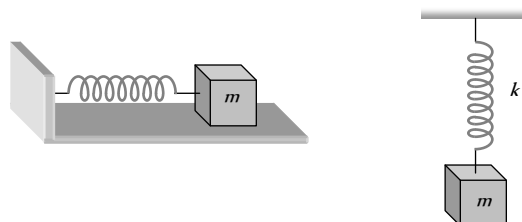


Fig. 16.21

$$\text{Time period } T = 2\pi \sqrt{\frac{\text{Inertia factor}}{\text{Spring factor}}} = 2\pi \sqrt{\frac{m}{k}}$$

$$\text{and Frequency } n = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

(i) Time period of a spring pendulum depends on the mass suspended $\Rightarrow T \propto \sqrt{m}$ or $n \propto \frac{1}{\sqrt{m}}$ i.e. greater the mass greater will be the inertia

and so lesser will be the frequency of oscillation and greater will be the time period.

(2) The time period depends on the force constant k of the spring i.e.

$$T \propto \frac{1}{\sqrt{k}} \text{ or } n \propto \sqrt{k}$$

(3) Time of a spring pendulum is independent of acceleration due to gravity. That is why a clock based on spring pendulum will keep proper time every where on a hill or moon or in a satellite and time period of a spring pendulum will not change inside a liquid if damping effects are neglected.

(4) **Massive spring** : If the spring has a mass M and mass m is suspended from it, effective mass is given by $m_{eff} = m + \frac{M}{3}$. Hence

$$T = 2\pi\sqrt{\frac{m_{eff}}{k}}$$

(5) **Reduced mass** : If two masses of mass m and m are connected by a spring and made to oscillate on horizontal surface, the reduced mass m is given by $\frac{1}{m_r} = \frac{1}{m_1} + \frac{1}{m_2}$ so that

$$T = 2\pi\sqrt{\frac{m_r}{k}}$$

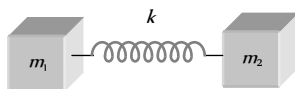


Fig. 16.22

(6) If a spring pendulum, oscillating in a vertical plane is made to oscillate on a horizontal surface, (or on inclined plane) time period will remain unchanged.

(7) Equilibrium position for a spring in a horizontal plain is the position of natural length of spring as weight is balanced by reaction. While in case of vertical motion equilibrium position will be $l + y_0$ with $ky_0 = mg$

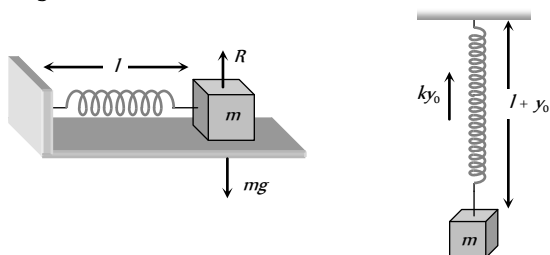


Fig. 16.23

If the stretch in a vertically loaded spring is y_0 then for equilibrium of mass m , $ky_0 = mg$ i.e. $\frac{m}{k} = \frac{y_0}{g}$

So that
$$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{y_0}{g}}$$

Time period does not depend on ' g ' because along with g , y_0 will also change in such a way that $\frac{y_0}{g} = \frac{m}{k}$ remains constant

Oscillation of Spring Combination

(i) **Series combination** : If two springs of spring constants K_1 and K_2 are joined in series as shown then

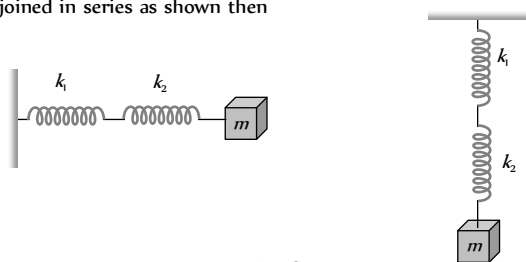


Fig. 16.24

(i) In series combination equal forces act on spring but extension in springs are different.

(ii) Spring constants of combination

$$\frac{1}{k_s} = \frac{1}{k_1} + \frac{1}{k_2} \Rightarrow k_s = \frac{k_1 k_2}{k_1 + k_2}$$

(iii) If n springs of different force constant $k_1, k_2, k_3, \dots$ respectively then

$$\frac{1}{k_s} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3} + \dots$$

If all spring have same spring constant then $k_s = \frac{k}{n}$

(iv) Time period of combination
$$T = 2\pi\sqrt{\frac{m}{k_s}} = 2\pi\sqrt{\frac{m(k_1 + k_2)}{k_1 k_2}}$$

(2) **Parallel combination** : If the springs are connected in parallel as shown

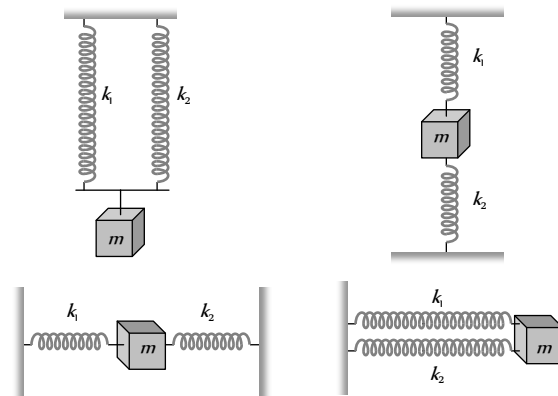


Fig. 16.25

(i) In parallel combination different forces act on different springs but extension in springs are same

(ii) Spring constants of combination $k_p = k_1 + k_2$

(iii) If n springs of different force constant are connected in parallel having force constant $k_1, k_2, k_3, \dots$ respectively then $k_p = k_1 + k_2 + k_3 + \dots$

If all spring have same spring constant then $k_p = nk$

(iv) Time period of combination
$$T_p = 2\pi\sqrt{\frac{m}{k_p}} = 2\pi\sqrt{\frac{m}{k_1 + k_2}}$$

Various Formulae of S.H.M.

(1) **S.H.M. of a liquid in U tube** : If a liquid of density ρ contained in a vertical U tube performs S.H.M. in its two limbs. Then time period

$$T = 2\pi\sqrt{\frac{L}{2g}} = 2\pi\sqrt{\frac{h}{g}}$$

where L = Total length of liquid column,

h = Height of undisturbed liquid in each limb ($L = 2h$)

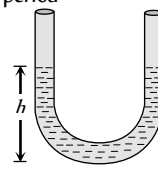


Fig. 16.26

(2) **S.H.M. of a floating cylinder** : If l is the length of cylinder dipping in liquid then

Time period
$$T = 2\pi\sqrt{\frac{l}{g}}$$

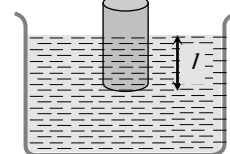


Fig. 16.27

(3) **S.H.M. of a small ball rolling down in hemi-spherical bowl**

$$T = 2\pi\sqrt{\frac{R-r}{g}}$$

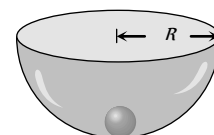


Fig. 16.28

R = Radius of the bowl

r = Radius of the ball

(4) S.H.M. of a piston in a cylinder

$$T = 2\pi\sqrt{\frac{Mh}{PA}}$$

M = mass of the piston

A = area of cross section

h = height of cylinder

P = pressure in a cylinder

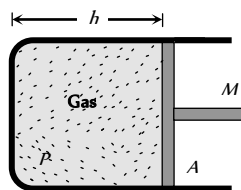


Fig. 16.29

(5) S.H.M. of a body in a tunnel dug along any chord of earth

$$T = 2\pi\sqrt{\frac{R}{g}} = 84.6 \text{ minutes}$$

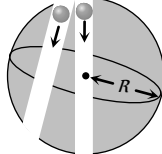


Fig. 16.30

(6) **Torsional pendulum** : In a torsional pendulum an object is suspended from a wire. If such a wire is twisted, due to elasticity it exerts a restoring torque $\tau = C\theta$.

In this case time period is given by

$$T = 2\pi\sqrt{\frac{I}{C}}$$

where I = Moment of inertia of disc

$$C = \text{Torsional constant of wire} = \frac{\pi\eta r^4}{2l}$$

η = Modulus of elasticity of wire and r = Radius of wire

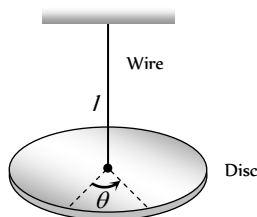


Fig. 16.31

(7) **Longitudinal oscillations of an elastic wire** : Wire/string pulled a distance Δl and left. It executes longitudinal oscillations. Restoring force

$$F = -AY\left(\frac{\Delta l}{l}\right)$$

Y = Young's modulus

A = Area of cross-section

$$\text{Hence } T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{ml}{AY}}$$

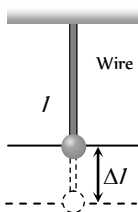


Fig. 16.32

Free, Damped, Forced and Maintained Oscillations



(1) Free oscillation

(i) The oscillation of a particle with fundamental frequency under the influence of restoring force are defined as free oscillations

(ii) The amplitude, frequency and energy of oscillation remains constant

(iii) Frequency of free oscillation is called natural frequency because it depends upon the nature and structure of the body.

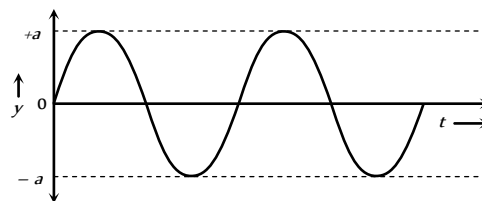


Fig. 16.33

(2) Damped oscillation

(i) The oscillation of a body whose amplitude goes on decreasing with time are defined as damped oscillation

(ii) In these oscillation the amplitude of oscillation decreases exponentially due to damping forces like frictional force, viscous force, hysteresis etc.

(iii) Due to decrease in amplitude the energy of the oscillator also goes on decreasing exponentially

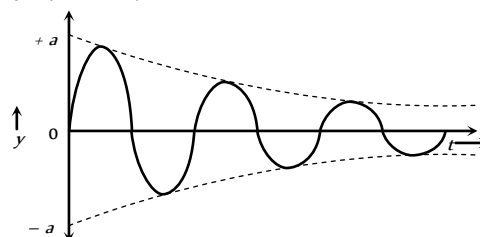


Fig. 16.34

(iv) The force produces a resistance to the oscillation is called damping force.

If the velocity of oscillator is v then

Damping force $F_d = -bv$, b = damping constant

(v) Resultant force on a damped oscillator is given by

$$F = F_R + F_d = -Kx - Kv \Rightarrow \frac{md^2x}{dt^2} + b\frac{dx}{dt} + Kx = 0$$

(vi) Displacement of damped oscillator is given by

$$x = x_m e^{-bt/2m} \sin(\omega't + \phi) \text{ where } \omega' = \text{angular frequency of}$$

the damped oscillator $= \sqrt{\omega_0^2 - (b/2m)^2}$

The amplitude decreases continuously with time according to

$$x = x_m e^{-(b/2m)t}$$

(vii) For a damped oscillator if the damping is small then the mechanical energy decreases exponentially with time as

$$E = \frac{1}{2} Kx_m^2 e^{-bt/m}$$

(3) Forced oscillation

(i) The oscillation in which a body oscillates under the influence of an external periodic force are known as forced oscillation

(ii) The amplitude of oscillator decrease due to damping forces but on account of the energy gained from the external source it remains constant.

(iii) **Resonance** : When the frequency of external force is equal to the natural frequency of the oscillator. Then this state is known as the state of resonance. And this frequency is known as resonant frequency.

(iv) While swinging in a swing if you apply a push periodically by pressing your feet against the ground, you find that not only the oscillations can now be maintained but the amplitude can also be increased. Under this condition the swing has forced or driven oscillation.

(v) In forced oscillation, frequency of damped oscillator is equal to the frequency of external force.

(vi) Suppose an external driving force is represented by

$$F(t) = F \cos \omega_d t$$

The motion of a particle under combined action of

(a) Restoring force $(-Kx)$

(b) Damping force $(-bv)$ and

(c) Driving force $F(t)$ is given by $ma = -Kx - bv + F_0 \cos \omega_d t$

$$\Rightarrow m^2 \frac{d^2 x}{dt^2} + Kx + b \frac{dx}{dt} = F_0 \cos \omega_d t$$

The solution of this equation gives $x = x_0 \sin(\omega_d t + \phi)$ with

$$\text{amplitude } x_0 = \frac{F_0/m}{\sqrt{(\omega^2 - \omega_0^2)^2 + (b\omega/m)^2}} \text{ and } \tan \theta = \frac{(\omega^2 - \omega_0^2)}{b\omega/m}$$

where $\omega_0 = \sqrt{\frac{K}{m}}$ = Natural frequency of oscillator.

(vii) **Amplitude resonance** : The amplitude of forced oscillator depends upon the frequency ω_d of external force.

When $\omega = \omega_d$, the amplitude is maximum but not infinite because of presence of damping force. The corresponds frequency is called resonant frequency (ω_0).

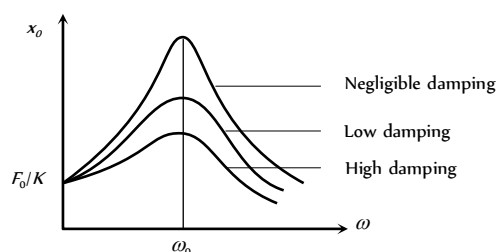


Fig. 16.35

(viii) **Energy resonance** : At $\omega = \omega_0$, oscillator absorbs maximum kinetic energy from the driving force system this state is called energy resonance.

At resonance the velocity of a driven oscillator is in phase with the driving term.

The sharpness of the resonance of a driven oscillator depends on the damping.

In the driven oscillator, the power input of the driving term is maximum at resonance.

(4) **Maintained oscillation** : The oscillation in which the loss of oscillator is compensated by the supplying energy from an external source are known as maintained oscillation.

Super Position of S.H.M's (Lissajous Figures)

If two S.H.M's act in perpendicular directions, then their resultant motion is in the form of a straight line or a circle or a parabola etc. depending on the frequency ratio of the two S.H.M. and initial phase difference. These figures are called Lissajous figures.

Let the equations of two mutually perpendicular S.H.M's of same frequency be

$$x = a_1 \sin \omega t \text{ and } y = a_2 \sin(\omega t + \phi)$$

then the general equation of Lissajou's figure can be obtained as

$$\frac{x^2}{a_1^2} + \frac{y^2}{a_2^2} - \frac{2xy}{a_1 a_2} \cos \phi = \sin^2 \phi$$

$$\text{For } \phi = 0^\circ : \frac{x^2}{a_1^2} + \frac{y^2}{a_2^2} - \frac{2xy}{a_1 a_2} = 0 \Rightarrow \left(\frac{x}{a_1} - \frac{y}{a_2} \right)^2 = 0$$

$$\Rightarrow \frac{x}{a_1} = \frac{y}{a_2} \Rightarrow y = \frac{a_2}{a_1} x$$

This is a straight line passes through origin

and its slope is $\frac{a_2}{a_1}$.

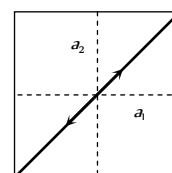


Fig. 16.36

Table 16.3 : Lissajou's figures in other conditions

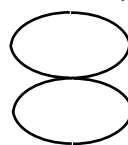
(with $\frac{\omega_1}{\omega_2} = 1$)

Phase diff. (ϕ)	Equation	Figure
$\frac{\pi}{4}$	$\frac{x^2}{a_1^2} + \frac{y^2}{a_2^2} - \frac{\sqrt{2}xy}{a_1 a_2} = \frac{1}{2}$	Oblique ellipse
$\frac{\pi}{2}$	$\frac{x^2}{a_1^2} + \frac{y^2}{a_2^2} = 1$	$a_1 = a_2$ (Circle) $a_1 \neq a_2$ (Ellipse)
$\frac{3\pi}{4}$	$\frac{x^2}{a_1^2} + \frac{y^2}{a_2^2} - \frac{\sqrt{2}xy}{a_1 a_2} = \frac{1}{2}$	Oblique ellipse
π	$\frac{x}{a_1} + \frac{y}{a_2} = 0$ $\Rightarrow y = -\frac{a_2}{a_1} x$	Straight line

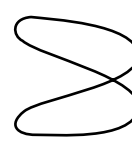
For the frequency ratio $\omega_1 : \omega_2 = 2 : 1$ the two perpendicular S.H.M's are

$$x = a_1 \sin(\omega t + \phi) \text{ and } y = a_2 \sin \omega t$$

Different Lissajou's figures as follows



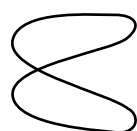
$\phi = 0, \pi, 2\pi$
Figure of eight



$\phi = \pi/4, 3\pi/4$
Double parabola



$\phi = \pi/2$
Parabola



$\phi = 5\pi/4, 7\pi/4$
Double parabola



$\phi = 3\pi/2$
Parabola

Fig. 16.37

Tips & Tricks

✍ Suppose a body of mass m vibrates separately with two different springs (of spring constants k and k) with time period T_1 and T_2 respectively. $T_1 = 2\pi\sqrt{\frac{m}{k_1}}$ and $T_2 = 2\pi\sqrt{\frac{m}{k_2}}$

If the same body vibrates with series combination of these two springs then for the system time period $T = \sqrt{T_1^2 + T_2^2}$

If the same body vibrates with parallel combination of these two springs then time period of the system $T = \frac{T_1 T_2}{\sqrt{T_1^2 + T_2^2}}$

✍ The pendulum clock runs slow due to increase in its time period whereas it becomes fast due to decrease in time period.

✍ If infinite spring with force constant $k, 2k, 4k, 8k, \dots$ respectively are connected in series. The effective force constant of the spring will be $k/2$.

✍ Percentage change in time period with l and g .

If g is constant and length varies by $n\%$. Then % change in time period $\frac{\Delta T}{T} \times 100 = \frac{n}{2} \times 100$

If l is constant and g varies by $n\%$. Then % change in time period $\frac{\Delta T}{T} \times 100 = -\frac{n}{2} \times 100$

(Valid only for small percentage change say 5%).

✍ Suppose a spring of force constant k oscillates with time period T . If it is divided into n equal parts then spring constant of each part will become nk and time period of oscillation of each part will become $\frac{T}{\sqrt{n}}$.

If these n parts connected in parallel then $k_{eff} = n^2 k$. So time period of the system becomes $T' = \frac{T}{n}$

✍ If a particle performs S.H.M. whose velocity is v_1 at a x_1 distance from mean position and velocity v_2 at distance x_2

$$\omega = \sqrt{\frac{v_1^2 - v_2^2}{x_2^2 - x_1^2}}; T = 2\pi\sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$$

$$a = \sqrt{\frac{v_1^2 x_2^2 - v_2^2 x_1^2}{v_1^2 - v_2^2}}; v_{max} = \sqrt{\frac{v_1^2 x_2^2 - v_2^2 x_1^2}{x_2^2 - x_1^2}}$$

✍ If $y_1 = a \sin \omega t$ and $y_2 = b \cos \omega t$ are two S.H.M. then by the superimposition of these two S.H.M. we get $\vec{y} = \vec{y}_1 + \vec{y}_2$

$\Rightarrow y = a \sin \omega t + b \cos \omega t \Rightarrow y = A \sin(\omega t + \phi)$ this is also the equation of S.H.M.; where $A = \sqrt{a^2 + b^2}$ and $\phi = \tan^{-1}(b/a)$

✍ In the absence of resistive force the work done by a simple pendulum in one complete oscillation is zero

✍ If θ is the angular amplitude of pendulum then

Height rises by the bob $h = l(1 - \cos \theta)$

Velocity at mean position

$$v = \sqrt{2gl(1 - \cos \theta)}$$

Work done in displacement

$$W = U = mgl(1 - \cos \theta)$$

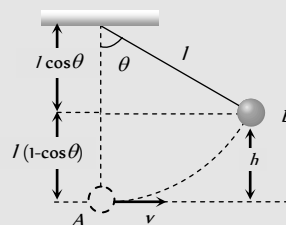
K.E. at mean position

$$KE_{mean} = mgl(1 - \cos \theta)$$

Tension in the string of pendulum

$$\text{At mean position: } T_A(\text{max}) = mg + \frac{mv^2}{l} = (3mg - 2mg \cos \theta)$$

$$\text{At extreme position: } T = mg \cos \theta$$



Ordinary Thinking

Objective Questions

Displacement of S.H.M. and Phase

- The phase of a particle executing simple harmonic motion is $\frac{\pi}{2}$ when it has [MP PET 1985]
 - Maximum velocity
 - Maximum acceleration
 - Maximum energy
 - Maximum displacement
- A particle starts S.H.M. from the mean position. Its amplitude is A and time period is T . At the time when its speed is half of the maximum speed, its displacement y is [Haryana CEE 1996; CBSE PMT 1996; MH CET 2002]
 - $\frac{A}{2}$
 - $\frac{A}{\sqrt{2}}$
 - $\frac{A\sqrt{3}}{2}$
 - $\frac{2A}{\sqrt{3}}$
- The amplitude and the periodic time of a S.H.M. are 5 cm and 6 sec respectively. At a distance of 2.5 cm away from the mean position, the phase will be
 - $5\pi/12$
 - $\pi/4$
 - $\pi/3$
 - $\pi/6$
- Two equations of two S.H.M. are $y = a \sin(\omega t - \alpha)$ and $y = b \cos(\omega t - \alpha)$. The phase difference between the two is



762 Simple Harmonic Motion

[MP PMT 1985]

- | | |
|----------------|--------------------|
| (a) 0° | (b) α° |
| (c) 90° | (d) 180° |



5. The amplitude and the time period in a S.H.M. is 0.5 cm and 0.4 sec respectively. If the initial phase is $\pi/2$ radian, then the equation of S.H.M. will be
 (a) $y = 0.5 \sin 5\pi t$ (b) $y = 0.5 \sin 4\pi t$
 (c) $y = 0.5 \sin 2.5\pi t$ (d) $y = 0.5 \cos 5\pi t$
6. The equation of S.H.M. is $y = a \sin(2\pi mt + \alpha)$, then its phase at time t is [DPMT 2001]
 (a) $2\pi mt$ (b) α
 (c) $2\pi mt + \alpha$ (d) 2π
7. A particle is oscillating according to the equation $X = 7 \cos 0.5\pi t$, where t is in second. The point moves from the position of equilibrium to maximum displacement in time
 (a) 4.0 sec (b) 2.0 sec
 (c) 1.0 sec (d) 0.5 sec
8. A simple harmonic oscillator has an amplitude a and time period T . The time required by it to travel from $x = a$ to $x = a/2$ is [CBSE PMT 1992; SCRA 1996; BHU 1997]
 (a) $T/6$ (b) $T/4$
 (c) $T/3$ (d) $T/2$
9. Which of the following expressions represent simple harmonic motion [Roorkee 1999]
 (a) $x = A \sin(\omega t + \delta)$ (b) $x = B \cos(\omega t + \phi)$
 (c) $x = A \tan(\omega t + \phi)$ (d) $x = A \sin \omega t \cos \omega t$
10. A $1.00 \times 10^{-20}\text{ kg}$ particle is vibrating with simple harmonic motion with a period of $1.00 \times 10^{-5}\text{ sec}$ and a maximum speed of $1.00 \times 10^3\text{ m/s}$. The maximum displacement of the particle is
 (a) 1.59 mm (b) 1.00 m
 (c) 10 m (d) None of these
11. The phase (at a time t) of a particle in simple harmonic motion tells [AMU (Engg.) 1999]
 (a) Only the position of the particle at time t
 (b) Only the direction of motion of the particle at time t
 (c) Both the position and direction of motion of the particle at time t
 (d) Neither the position of the particle nor its direction of motion at time t
12. A particle is moving with constant angular velocity along the circumference of a circle. Which of the following statements is true
 (a) The particle so moving executes S.H.M.
 (b) The projection of the particle on any one of the diameters executes S.H.M.
 (c) The projection of the particle on any of the diameters executes S.H.M.
 (d) None of the above
13. A particle is executing simple harmonic motion with a period of T seconds and amplitude a metre. The shortest time it takes to reach a point $\frac{a}{\sqrt{2}}\text{ m}$ from its mean position in seconds is [EAMCET (Med.) 2000]
 (a) T (b) $T/4$
 (c) $T/8$ (d) $T/16$
14. A simple harmonic motion is represented by $F(t) = 10 \sin(20t + 0.5)$. The amplitude of the S.H.M. is [DPMT 1998; CBSE PMT 2000; MH CET 2001]
 (a) $a = 30$ (b) $a = 20$
 (c) $a = 10$ (d) $a = 5$
15. Which of the following equation does not represent a simple harmonic motion [Kerala (Med.) 2002]
 (a) $y = a \sin \omega t$ (b) $y = a \cos \omega t$
 (c) $y = a \sin \omega t + b \cos \omega t$ (d) $y = a \tan \omega t$
16. A particle in S.H.M. is described by the displacement function $x(t) = a \cos(\omega t + \theta)$. If the initial ($t = 0$) position of the particle is 1 cm and its initial velocity is $\pi\text{ cm/s}$. The angular frequency of the particle is $\pi\text{ rad/s}$, then its amplitude is [CPMT 1989]
 (a) 1 cm (b) $\sqrt{2}\text{ cm}$
 (c) 2 cm (d) 2.5 cm
17. A particle executes a simple harmonic motion of time period T . Find the time taken by the particle to go directly from its mean position to half the amplitude [UPSEAT 2002]
 (a) $T/2$ (b) $T/4$
 (c) $T/8$ (d) $T/12$
18. A particle executing simple harmonic motion along y -axis has its motion described by the equation $y = A \sin(\omega t) + B$. The amplitude of the simple harmonic motion is [Orissa JEE 2003]
 (a) A [AMU (Med.) 1999] (b) B
 (c) $A + B$ (d) $\sqrt{A^2 + B^2}$
19. A particle executing S.H.M. of amplitude 4 cm and $T = 4\text{ sec}$. The time taken by it to move from positive extreme position to half the amplitude is [BHU 1995]
 (a) 1 sec (b) $1/3\text{ sec}$
 (c) $2/3\text{ sec}$ (d) $\sqrt{3}/2\text{ sec}$
20. Which one of the following is a simple harmonic motion [CBSE PMT 1994]
 (a) Wave moving through a string fixed at both ends
 (b) Earth spinning about its own axis [AMU (Engg.) 1999]
 (c) Ball bouncing between two rigid vertical walls
 (d) Particle moving in a circle with uniform speed
21. A particle is moving in a circle with uniform speed. Its motion is [CPMT 1978; CBSE PMT 1994]
 (a) Periodic and simple harmonic
 (b) Periodic but not simple harmonic
 (c) A periodic
 (d) None of the above
22. Two simple harmonic motions are represented by the equations $y_1 = 0.1 \sin\left(100\pi t + \frac{\pi}{3}\right)$ and $y_2 = 0.1 \cos \pi t$. The phase difference of the velocity of particle 1 with respect to the velocity of particle 2 is [AIEEE 2005]

- (a) $-\frac{\pi}{3}$ (b) $\frac{\pi}{6}$
(c) $-\frac{\pi}{6}$ (d) $\frac{\pi}{3}$
23. Two particles are executing S.H.M. The equation of their motion are $y_1 = 10 \sin\left(\omega t + \frac{\pi T}{4}\right)$, $y_2 = 25 \sin\left(\omega t + \frac{\sqrt{3}\pi T}{4}\right)$. What is the ratio of their amplitude [DCE 1996]
(a) 1 : 1 (b) 2 : 5
(c) 1 : 2 (d) None of these
24. The periodic time of a body executing simple harmonic motion is 3 sec. After how much interval from time $t = 0$, its displacement will be half of its amplitude [BHU 1998]
(a) $\frac{1}{8}$ sec (b) $\frac{1}{6}$ sec
(c) $\frac{1}{4}$ sec (d) $\frac{1}{3}$ sec
25. A system exhibiting S.H.M. must possess [KCET 1994]
(a) Inertia only
(b) Elasticity as well as inertia
(c) Elasticity, inertia and an external force
(d) Elasticity only
26. If $x = a \sin\left(\omega t + \frac{\pi}{6}\right)$ and $x' = a \cos \omega t$, then what is the phase difference between the two waves [RPET 1996]
(a) $\pi/3$ (b) $\pi/6$
(c) $\pi/2$ (d) π

Velocity of Simple Harmonic Motion

1. A simple pendulum performs simple harmonic motion about $X = 0$ with an amplitude A and time period T . The speed of the pendulum at $X = \frac{A}{2}$ will be [MP PMT 1987]
(a) $\frac{\pi A \sqrt{3}}{T}$ (b) $\frac{\pi A}{T}$
(c) $\frac{\pi A \sqrt{3}}{2T}$ (d) $\frac{3\pi^2 A}{T}$
2. A body is executing simple harmonic motion with an angular frequency 2 rad/s . The velocity of the body at 20 mm displacement, when the amplitude of motion is 60 mm, is [CPMT 1999]
(a) 40 mm/s (b) 60 mm/s
(c) 113 mm/s (d) 120 mm/s
3. A body of mass 5 gm is executing S.H.M. about a point with amplitude 10 cm. Its maximum velocity is 100 cm/sec. Its velocity will be 50 cm/sec at a distance [CPMT 1976]
(a) 5 (b) $5\sqrt{2}$
(c) $5\sqrt{3}$ (d) $10\sqrt{2}$
4. A simple harmonic oscillator has a period of 0.01 sec and an amplitude of 0.2 m. The magnitude of the velocity in m sec^{-1} at the centre of oscillation is [JIPMER 1997]
(a) 20π (b) 100
(c) 40π (d) 100π
5. A particle executes S.H.M. with a period of 6 second and amplitude of 3 cm. Its maximum speed in cm/sec is [AIIMS 1982]
(a) $\pi/2$ (b) π
(c) 2π (d) 3π
6. A particle is executing S.H.M. If its amplitude is 2 m and periodic time 2 seconds, then the maximum velocity of the particle will be
(a) $\pi \text{ m/s}$ (b) $\sqrt{2\pi} \text{ m/s}$
(c) $2\pi \text{ m/s}$ (d) $4\pi \text{ m/s}$
7. A S.H.M. has amplitude 'a' and time period T . The maximum velocity will be [MP PMT 1985; CPMT 1997; UPSEAT 1999]
(a) $\frac{4a}{T}$ (b) $\frac{2a}{T}$
(c) $2\pi\sqrt{\frac{a}{T}}$ (d) $\frac{2\pi a}{T}$
8. A body is executing S.H.M. When its displacement from the mean position is 4 cm and 5 cm, the corresponding velocity of the body is 10 cm/sec and 8 cm/sec. Then the time period of the body is [CPMT 1991; MP PMT 1996; UPSEAT 1997; AIIMS 1998]
(a) 2π sec (b) $\pi/2$ sec
(c) π sec (d) $3\pi/2$ sec
9. A particle has simple harmonic motion. The equation of its motion is $x = 5 \sin\left(4t - \frac{\pi}{6}\right)$, where x is its displacement. If the displacement of the particle is 3 units, then its velocity is [MP PMT 1994]
(a) $\frac{2\pi}{3}$ (b) $\frac{5\pi}{6}$
(c) 20 (d) 16
10. A particle is executing S.H.M. with an amplitude of 50 mm and time period of 2 sec, then its maximum velocity is [AIIMS 1998; MH CET 2000; DPMT 2000]
(a) 0.10 m/s (b) 0.15 m/s
(c) 0.8 m/s (d) 0.26 m/s
11. If the displacement of a particle executing SHM is given by $y = 0.30 \sin(220t + 0.64)$ in metre, then the frequency and maximum velocity of the particle is [AFMC 1998]
(a) 35 Hz, 66 m/s (b) 45 Hz, 66 m/s
(c) 58 Hz, 113 m/s (d) 35 Hz, 132 m/s



12. The maximum velocity and the maximum acceleration of a body moving in a simple harmonic oscillator are 2 m/s and 4 m/s^2 . Then angular velocity will be
[Pb. PMT 1998; MH CET 1999, 2003]
- (a) 3 rad/sec (b) 0.5 rad/sec
(c) 1 rad/sec (d) 2 rad/sec
13. If a particle under S.H.M. has time period 0.1 sec and amplitude $2 \times 10^{-3}\text{ m}$. It has maximum velocity
[RPET 2000]
- (a) $\frac{\pi}{25}\text{ m/s}$ (b) $\frac{\pi}{26}\text{ m/s}$
(c) $\frac{\pi}{30}\text{ m/s}$ (d) None of these
14. A particle executing simple harmonic motion has an amplitude of 6 cm . Its acceleration at a distance of 2 cm from the mean position is 8 cm/s^2 . The maximum speed of the particle is [EAMCET (Engg.) 2000]
- (a) 8 cm/s (b) 12 cm/s
(c) 16 cm/s (d) 24 cm/s
15. A particle executes simple harmonic motion with an amplitude of 4 cm . At the mean position the velocity of the particle is 10 cm/s . The distance of the particle from the mean position when its speed becomes 5 cm/s is
[EAMCET (Med.) 2000]
- (a) $\sqrt{3}\text{ cm}$ (b) $\sqrt{5}\text{ cm}$
(c) $2(\sqrt{3})\text{ cm}$ (d) $2(\sqrt{5})\text{ cm}$
16. Two particles P and Q start from origin and execute Simple Harmonic Motion along X -axis with same amplitude but with periods 3 seconds and 6 seconds respectively. The ratio of the velocities of P and Q when they meet is
[EAMCET 2001]
- (a) $1:2$ (b) $2:1$
(c) $2:3$ (d) $3:2$
17. A particle is performing simple harmonic motion with amplitude A and angular velocity ω . The ratio of maximum velocity to maximum acceleration is
[Kerala (Med.) 2002]
- (a) ω (b) $1/\omega$
(c) ω (d) $A\omega$
18. The angular velocities of three bodies in simple harmonic motion are $\omega_1, \omega_2, \omega_3$ with their respective amplitudes as A_1, A_2, A_3 . If all the three bodies have same mass and velocity, then
- (a) $A_1\omega_1 = A_2\omega_2 = A_3\omega_3$ (b) $A_1\omega_1^2 = A_2\omega_2^2 = A_3\omega_3^2$
(c) $A_1^2\omega_1 = A_2^2\omega_2 = A_3^2\omega_3$ (d) $A_1^2\omega_1^2 = A_2^2\omega_2^2 = A_3^2\omega_3^2$
19. The velocity of a particle performing simple harmonic motion, when it passes through its mean position is
[MH CET (Med.) 2002; BCECE 2004]
- (a) Infinity (b) Zero
(c) Minimum (d) Maximum
20. The velocity of a particle in simple harmonic motion at displacement y from mean position is
- (a) $\omega\sqrt{a^2 + y^2}$ (b) $\omega\sqrt{a^2 - y^2}$
(c) ωy (d) $\omega^2\sqrt{a^2 - y^2}$
21. A particle is executing the motion $x = A\cos(\omega t - \theta)$. The maximum velocity of the particle is
[BHU 2003; CPMT 2004]
- (a) $A\omega\cos\theta$ (b) $A\omega$
(c) $A\omega\sin\theta$ (d) None of these
22. A particle executing simple harmonic motion with amplitude of 0.1 m . At a certain instant when its displacement is 0.02 m , its acceleration is 0.5 m/s^2 . The maximum velocity of the particle is (in m/s)
- (a) 0.01 (b) 0.05
(c) 0.5 (d) 0.25
23. The amplitude of a particle executing SHM is 4 cm . At the mean position the speed of the particle is 16 cm/sec . The distance of the particle from the mean position at which the speed of the particle becomes $8\sqrt{3}\text{ cm/s}$, will be
[Pb. PET 2003]
- (a) $2\sqrt{3}\text{ cm}$ (b) $\sqrt{3}\text{ cm}$
(c) 1 cm (d) 2 cm
24. The maximum velocity of a simple harmonic motion represented by $y = 3\sin\left(100t + \frac{\pi}{6}\right)$ is given by
[BCECE 2005]
- (a) 300 (b) $\frac{3\pi}{6}$
(c) 100 (d) $\frac{\pi}{6}$
25. The displacement equation of a particle is $x = 3\sin 2t + 4\cos 2t$. The amplitude and maximum velocity will be respectively
- (a) $5, 10$ (b) $3, 2$
(c) $4, 2$ (d) $3, 4$
26. Velocity at mean position of a particle executing S.H.M. is v , they velocity of the particle at a distance equal to half of the amplitude
- (a) $4v$ (b) $2v$
(c) $\frac{\sqrt{3}}{2}v$ (d) $\frac{\sqrt{3}}{4}v$
27. The instantaneous displacement of a simple pendulum oscillator is given by $x = A\cos\left(\omega t + \frac{\pi}{4}\right)$. Its speed will be maximum at time
[BHU 2002]
- (a) $\frac{\pi}{4\omega}$ (b) $\frac{\pi}{2\omega}$
(c) $\frac{\pi}{\omega}$ (d) $\frac{2\pi}{\omega}$

Acceleration of Simple Harmonic Motion

1. Which of the following is a necessary and sufficient condition for S.H.M.
[NCERT 1974]

- (a) Constant period
(b) Constant acceleration
(c) Proportionality between acceleration and displacement from equilibrium position
(d) Proportionality between restoring force and displacement from equilibrium position
2. If a hole is bored along the diameter of the earth and a stone is dropped into hole [CPMT 1984]
(a) The stone reaches the centre of the earth and stops there
(b) The stone reaches the other side of the earth and stops there
(c) The stone executes simple harmonic motion about the centre of the earth
(d) The stone reaches the other side of the earth and escapes into space
3. The acceleration of a particle in S.H.M. is [MP PMT 1993]
(a) Always zero
(b) Always constant
(c) Maximum at the extreme position
(d) Maximum at the equilibrium position
4. The displacement of a particle moving in S.H.M. at any instant is given by $y = a \sin \omega t$. The acceleration after time $t = \frac{T}{4}$ is (where T is the time period) [MP PET 1984]
(a) $a\omega$ (b) $-a\omega$
(c) $a\omega^2$ (d) $-a\omega^2$
5. The amplitude of a particle executing S.H.M. with frequency of 60 Hz is 0.01 m. The maximum value of the acceleration of the particle is
[DPMT 1998; CBSE PMT 1999; AFMC 2001; Pb. PMT 2001; Pb. PET 2001, 02; CPMT 1993, 95, 04; RPMT 2005; MP PMT 2005]
(a) $144\pi^2 m/sec^2$ (b) $144m/sec^2$
(c) $\frac{144}{\pi^2} m/sec^2$ (d) $288\pi^2 m/sec^2$
6. A small body of mass 0.10 kg is executing S.H.M. of amplitude 1.0 m and period 0.20 sec. The maximum force acting on it is
(a) 98.596 N (b) 985.96 N
(c) 100.2 N (d) 76.23 N
7. A body executing simple harmonic motion has a maximum acceleration equal to 24 metres/sec^2 and maximum velocity equal to 16 metres/sec . The amplitude of the simple harmonic motion is
[MP PMT 1995; DPMT 2002; RPET 2003; Pb. PET 2004]
(a) $\frac{32}{3} \text{ metres}$ (b) $\frac{3}{32} \text{ metres}$
(c) $\frac{1024}{9} \text{ metres}$ (d) $\frac{64}{9} \text{ metres}$
8. For a particle executing simple harmonic motion, which of the following statements is not correct
[MP PMT 1997; AIIMS 1999; Kerala PMT 2005]
(a) The total energy of the particle always remains the same
(b) The restoring force is always directed towards a fixed point
(c) The restoring force is maximum at the extreme positions
(d) The acceleration of the particle is maximum at the equilibrium position
9. A particle of mass 10 grams is executing simple harmonic motion with an amplitude of 0.5 m and periodic time of $(\pi/5)$ seconds. The maximum value of the force acting on the particle is [MP PET 1999; MP PMT 1999]
(a) 25 N (b) 5 N
(c) 2.5 N (d) 0.5 N
10. The displacement of an oscillating particle varies with time (in seconds) according to the equation $y \text{ (cm)} = \sin \frac{\pi}{2} \left(\frac{t}{2} + \frac{1}{3} \right)$. The maximum acceleration of the particle is approximately
(a) 5.21 cm/s^2 (b) 3.62 cm/s^2
(c) 1.81 cm/s^2 (d) 0.62 cm/s^2
11. A particle moving along the x-axis executes simple harmonic motion, then the force acting on it is given by [CBSE PMT 1994]
(a) $-A Kx$ (b) $A \cos(Kx)$
(c) $A \exp(-Kx)$ (d) $A Kx$
Where A and K are positive constants
12. A body is vibrating in simple harmonic motion with an amplitude of 0.06 m and frequency of 15 Hz. The velocity and acceleration of body is [AFMC 1999]
(a) 5.65 m/s and $5.32 \times 10^2 \text{ m/s}^2$
(b) 6.82 m/s and $7.62 \times 10^2 \text{ m/s}^2$
(c) 8.91 m/s and $8.21 \times 10^2 \text{ m/s}^2$
(d) 9.82 m/s and $9.03 \times 10^2 \text{ m/s}^2$
13. A particle executes harmonic motion with an angular velocity and maximum acceleration of 3.5 rad/sec and 7.5 m/s respectively. The amplitude of oscillation is [AIIMS 1999; Pb. PET 1999]
(a) 0.28 m (b) 0.36 m
(c) 0.53 m (d) 0.61 m
14. A 0.10 kg block oscillates back and forth along a horizontal surface. Its displacement from the origin is given by: $x = (10 \text{ cm}) \cos[(10 \text{ rad/s})t + \pi/2 \text{ rad}]$. What is the maximum acceleration experienced by the block [AMU (Engg.) 2000]
(a) 10 m/s^2 (b) $10 \pi \text{ m/s}^2$
(c) $\frac{10\pi}{2} \text{ m/s}^2$ (d) $\frac{10\pi}{3} \text{ m/s}^2$
15. In S.H.M. maximum acceleration is at [RPET 2001; BVP 2003]
(a) Amplitude (b) Equilibrium
(c) Acceleration is constant (d) None of these



16. A particle is executing simple harmonic motion with an amplitude of 0.02 metre and frequency 50 Hz. The maximum acceleration of the particle is [MP PET 2001]
- (a) 100 m/s^2 (b) $100\pi^2 \text{ m/s}^2$
 (c) 100 m/s^2 (d) $200\pi^2 \text{ m/s}^2$
17. Acceleration of a particle, executing SHM, at its mean position is [MH CET (Med.) 2002]
- (a) Infinity (b) Varies
 (c) Maximum (d) Zero
18. Which one of the following statements is true for the speed v and the acceleration a of a particle executing simple harmonic motion
- (a) When v is maximum, a is maximum
 (b) Value of a is zero, whatever may be the value of v
 (c) When v is zero, a is zero
 (d) When v is maximum, a is zero
19. What is the maximum acceleration of the particle doing the SHM $y = 2 \sin\left[\frac{\pi}{2} + \phi\right]$ where 2 is in cm [DCE 2003]
- (a) $\frac{\pi}{2} \text{ cm/s}^2$ (b) $\frac{\pi^2}{2} \text{ cm/s}^2$
 (c) $\frac{\pi}{4} \text{ cm/s}^2$ (d) $\frac{\pi}{4} \text{ cm/s}^2$
20. A particle executes linear simple harmonic motion with an amplitude of 2 cm. When the particle is at 1 cm from the mean position the magnitude of its velocity is equal to that of its acceleration. Then its time period in seconds is [Kerala PET 2005]
- (a) $\frac{1}{2\pi\sqrt{3}}$ (b) $2\pi\sqrt{3}$
 (c) $\frac{2\pi}{\sqrt{3}}$ (d) $\frac{\sqrt{3}}{2\pi}$
21. In simple harmonic motion, the ratio of acceleration of the particle to its displacement at any time is a measure of [UPSEAT 2001]
- (a) Spring constant (b) Angular frequency
 (c) (Angular frequency) (d) Restoring force
3. A particle is vibrating in a simple harmonic motion with an amplitude of 4 cm. At what displacement from the equilibrium position, is its energy half potential and half kinetic [NCERT 1984; MNR 1995; RPMT 1995; DCE 2000; UPSEAT 2000]
- (a) 1 cm (b) $\sqrt{2}$ cm
 (c) 3 cm (d) $2\sqrt{2}$ cm
4. For a particle executing simple harmonic motion, the kinetic energy K is given by $K = K_0 \cos^2 \omega t$. The maximum value of potential energy is [CPMT 1981]
- (a) K_0 [CBSE PMT 2004] (b) Zero
 (c) $\frac{K_0}{2}$ (d) Not obtainable
5. The potential energy of a particle with displacement X is $U(X)$. The motion is simple harmonic, when (K is a positive constant)
- (a) $U = -\frac{KX^2}{2}$ (b) $U = KX^2$
 (c) $U = K$ (d) $U = KX$
6. The kinetic energy and potential energy of a particle executing simple harmonic motion will be equal, when displacement (amplitude = a) is [MP PMT 1987; CPMT 1990; DPMT 1996; MH CET 1997, 99; AFMC 1999; CPMT 2000]
- (a) $\frac{a}{2}$ (b) $a\sqrt{2}$
 (c) $\frac{a}{\sqrt{2}}$ (d) $\frac{a\sqrt{2}}{3}$
7. The total energy of the body executing S.H.M. is E . Then the kinetic energy when the displacement is half of the amplitude, is [RPMT 1994, 96; CBSE PMT 1995; JIPMER 2002]
- (a) $\frac{E}{2}$ (b) $\frac{E}{4}$
 (c) $\frac{3E}{4}$ (d) $\frac{\sqrt{3}}{4} E$
8. The potential energy of a particle executing S.H.M. is 2.5 J, when its displacement is half of amplitude. The total energy of the particle be
- (a) 18 J (b) 10 J
 (c) 12 J (d) 2.5 J
9. The angular velocity and the amplitude of a simple pendulum is ω and a respectively. At a displacement X from the mean position if its kinetic energy is T and potential energy is V , then the ratio of T to V is [CBSE PMT 1991]
- (a) $X^2\omega^2/(a^2 - X^2\omega^2)$ (b) $X^2/(a^2 - X^2)$
 (c) $(a^2 - X^2\omega^2)/X^2\omega^2$ (d) $(a^2 - X^2)/X^2$
10. When the potential energy of a particle executing simple harmonic motion is one-fourth of its maximum value during the oscillation, the displacement of the particle from the equilibrium position in terms of its amplitude a is [CBSE PMT 1993; EAMCET (Engg.) 1995;

Energy of Simple Harmonic Motion

1. The total energy of a particle executing S.H.M. is proportional to [CPMT 1974, 78; EAMCET 1994; RPET 1999; MP PMT 2001; Pb. PMT 2002; MH CET 2002]
- (a) Displacement from equilibrium position
 (b) Frequency of oscillation
 (c) Velocity in equilibrium position
 (d) Square of amplitude of motion
2. A particle executes simple harmonic motion along a straight line with an amplitude A . The potential energy is maximum when the displacement is [CPMT 1982]
- (a) $\pm A$ (b) Zero
 (c) $\pm \frac{A}{2}$ (d) $\pm \frac{A}{\sqrt{2}}$

MP PMT 1994, 2000; MP PET 1995, 96, 2002]

- (a) $a/4$ (b) $a/3$
(c) $a/2$ (d) $2a/3$

11. A particle of mass 10 gm is describing S.H.M. along a straight line with period of 2 sec and amplitude of 10 cm . Its kinetic energy when it is at 5 cm from its equilibrium position is

- (a) $37.5\pi^2\text{ ergs}$ (b) $3.75\pi^2\text{ ergs}$
(c) $375\pi^2\text{ ergs}$ (d) $0.375\pi^2\text{ ergs}$

12. When the displacement is half the amplitude, the ratio of potential energy to the total energy is

[CPMT 1999; JIPMER 2000; Kerala PET 2002]

- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$
(c) 1 (d) $\frac{1}{8}$

13. The P.E. of a particle executing SHM at a distance x from its equilibrium position is

[Roorkee 1992; CPMT 1997; RPMT 1999]

- (a) $\frac{1}{2}m\omega^2x^2$ (b) $\frac{1}{2}m\omega^2a^2$
(c) $\frac{1}{2}m\omega^2(a^2 - x^2)$ (d) Zero

14. A vertical mass-spring system executes simple harmonic oscillations with a period of 2 s . A quantity of this system which exhibits simple harmonic variation with a period of 1 s is

- (a) Velocity
(b) Potential energy
(c) Phase difference between acceleration and displacement
(d) Difference between kinetic energy and potential energy

15. For any S.H.M., amplitude is 6 cm . If instantaneous potential energy is half the total energy then distance of particle from its mean position is

[RPET 2000]

- (a) 3 cm (b) 4.2 cm
(c) 5.8 cm (d) 6 cm

16. A body of mass 1 kg is executing simple harmonic motion. Its displacement $y(\text{cm})$ at t seconds is given by $y = 6 \sin(100t + \pi/4)$. Its maximum kinetic energy is

[EAMCET (Engg.) 2000]

- (a) 6 J (b) 18 J
(c) 24 J (d) 36 J

17. A particle is executing simple harmonic motion with frequency f . The frequency at which its kinetic energy change into potential energy is

[MP PET 2000]

- (a) $f/2$ (b) f
(c) $2f$ (d) $4f$

18. There is a body having mass m and performing S.H.M. with amplitude a . There is a restoring force $F = -Kx$, where x is the displacement. The total energy of body depends upon

[CBSE PMT 2001]

- (a) K, x (b) K, a
(c) K, a, x (d) K, a, v

19. The total energy of a particle executing S.H.M. is 80 J . What is the potential energy when the particle is at a distance of $3/4$ of amplitude from the mean position

[Kerala (Engg.) 2001]

- (a) 60 J (b) 10 J
(c) 40 J (d) 45 J

20. In a simple harmonic oscillator, at the mean position

[AIEEE 2002]

- (a) Kinetic energy is minimum, potential energy is maximum
(b) Both kinetic and potential energies are maximum
(c) Kinetic energy is maximum, potential energy is minimum
(d) Both kinetic and potential energies are minimum

21. Displacement between maximum potential energy position and maximum kinetic energy position for a particle executing S.H.M. is

- (a) $-a$ (b) $+a$
(c) $\pm a$ (d) $\pm \frac{a}{4}$

22. When a mass M is attached to the spring of force constant k , then the spring stretches by l . If the mass oscillates with amplitude l , what will be maximum potential energy stored in the spring

- (a) $\frac{kl}{2}$ (b) $2kl$
(c) $\frac{1}{2}Mgl$ [SCRA 1998] (d) Mgl

23. The potential energy of a simple harmonic oscillator when the particle is half way to its end point is (where E is the total energy)

- (a) $\frac{1}{8}E$ (b) $\frac{1}{4}E$
(c) $\frac{1}{2}E$ (d) $\frac{2}{3}E$

24. A body executes simple harmonic motion. The potential energy (P.E.), the kinetic energy (K.E.) and total energy (T.E.) are measured as a function of displacement x . Which of the following statements is true

[AIEEE 2003]

- (a) P.E. is maximum when $x = 0$
(b) K.E. is maximum when $x = 0$
(c) T.E. is zero when $x = 0$
(d) K.E. is maximum when x is maximum

25. If $\langle E \rangle$ and $\langle U \rangle$ denote the average kinetic and the average potential energies respectively of mass describing a simple harmonic motion, over one period, then the correct relation is

- (a) $\langle E \rangle = \langle U \rangle$ (b) $\langle E \rangle = 2\langle U \rangle$
(c) $\langle E \rangle = -2\langle U \rangle$ (d) $\langle E \rangle = -\langle U \rangle$

26. The total energy of a particle, executing simple harmonic motion is

- (a) $\propto x$ (b) $\propto x^2$



- (c) Independent of x (d) $\propto x^{1/2}$
27. The kinetic energy of a particle executing S.H.M. is 16 J when it is at its mean position. If the mass of the particle is 0.32 kg , then what is the maximum velocity of the particle

[MH CET 2004]

- (a) 5 m/s (b) 15 m/s
(c) 10 m/s (d) 20 m/s
28. Consider the following statements. The total energy of a particle executing simple harmonic motion depends on its
(1) Amplitude (2) Period (3) Displacement
Of these statements [RPMT 2001; BCECE 2005]
- (a) (1) and (2) are correct
(b) (2) and (3) are correct
(c) (1) and (3) are correct
(d) (1), (2) and (3) are correct
29. A particle starts simple harmonic motion from the mean position. Its amplitude is a and total energy E . At one instant its kinetic energy is $3E/4$. Its displacement at that instant is

[Kerala PET 2005]

- (a) $a/\sqrt{2}$ (b) $a/2$
(c) $\frac{a}{\sqrt{3/2}}$ (d) $a/\sqrt{3}$
30. A particle executes simple harmonic motion with a frequency f . The frequency with which its kinetic energy oscillates is [IIT JEE 1973, 87; Manipal MEE 1995; MP PET 1997; DCE 1997; DCE 1999; UPSEAT 2000; RPET 2002; RPMT 2004; BHU 2005]
- (a) $f/2$ (b) f
(c) $2f$ (d) $4f$

31. The amplitude of a particle executing SHM is made three-fourth keeping its time period constant. Its total energy will be
- (a) $\frac{E}{2}$ (b) $\frac{3}{4}E$
(c) $\frac{9}{16}E$ (d) None of these

32. A particle of mass m is hanging vertically by an ideal spring of force constant K . If the mass is made to oscillate vertically, its total energy is [CPMT 1978; RPET 1999]

- (a) Maximum at extreme position
(b) Maximum at mean position
(c) Minimum at mean position
(d) Same at all position
33. A body is moving in a room with a velocity of 20 m/s perpendicular to the two walls separated by 5 meters . There is no friction and the collisions with the walls are elastic. The motion of the body is [MP PMT 1999]
- (a) Not periodic
(b) Periodic but not simple harmonic
(c) Periodic and simple harmonic
(d) Periodic with variable time period

34. A body is executing Simple Harmonic Motion. At a displacement x its potential energy is E_1 and at a displacement y its potential energy is E_2 . The potential energy E at displacement $(x+y)$ is [EAMCET 2001]

- (a) $\sqrt{E} = \sqrt{E_1} - \sqrt{E_2}$ (b) $\sqrt{E} = \sqrt{E_1} + \sqrt{E_2}$
(c) $E = E_1 + E_2$ (d) $E = E_1 - E_2$

Time Period and Frequency

1. A particle moves such that its acceleration a is given by $a = -bx$, where x is the displacement from equilibrium position and b is a constant. The period of oscillation is

[NCERT 1984; CPMT 1991; MP PMT 1994; MNR 1995; UPSEAT 2000]

- (a) $2\pi\sqrt{b}$ (b) $\frac{2\pi}{\sqrt{b}}$
(c) $\frac{2\pi}{b}$ (d) $2\sqrt{\frac{\pi}{b}}$

2. The equation of motion of a particle is $\frac{d^2y}{dt^2} + Ky = 0$, where K is positive constant. The time period of the motion is given by

- (a) $\frac{2\pi}{K}$ (b) $2\pi K$
(c) $\frac{2\pi}{\sqrt{K}}$ (d) $2\pi\sqrt{K}$

3. A tunnel has been dug through the centre of the earth and a ball is released in it. It will reach the other end of the tunnel after

- (a) 84.6 minutes
(b) 42.3 minutes
(c) 1 day
(d) Will not reach the other end [RPMT 2004]

4. The maximum speed of a particle executing S.H.M. is 1 m/s and its maximum acceleration is 1.57 m/sec^2 . The time period of the particle will be [DPMT 2002]

- (a) $\frac{1}{1.57} \text{ sec}$ (b) 1.57 sec
(c) 2 sec (d) 4 sec

5. The motion of a particle executing S.H.M. is given by $x = 0.01 \sin 100\pi(t + .05)$, where x is in metres and time is in seconds. The time period is [CPMT 1990]

- (a) 0.01 sec (b) 0.02 sec
(c) 0.1 sec (d) 0.2 sec

6. The kinetic energy of a particle executing S.H.M. is 16 J when it is in its mean position. If the amplitude of oscillations is 25 cm and the mass of the particle is 5.12 kg , the time period of its oscillation is

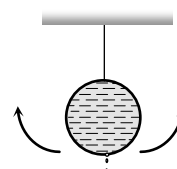
[Haryana CEE 1996; AFMC 1998]

- (a) $\frac{\pi}{5} \text{ sec}$ (b) $2\pi \text{ sec}$
(c) $20\pi \text{ sec}$ (d) $5\pi \text{ sec}$

7. The acceleration of a particle performing S.H.M. is 12 cm/sec^2 at a distance of 3 cm from the mean position. Its time period is [MP PET 1996; MP PMT 1997]
- (a) 0.5 sec (b) 1.0 sec
(c) 2.0 sec (d) 3.14 sec
8. To make the frequency double of an oscillator, we have to [CPMT 1999]
- (a) Double the mass
(b) Half the mass
(c) Quadruple the mass
(d) Reduce the mass to one-fourth
9. What is constant in S.H.M. [UPSEAT 1999]
- (a) Restoring force (b) Kinetic energy
(c) Potential energy (d) Periodic time
10. If a simple harmonic oscillator has got a displacement of 0.02 m and acceleration equal to 2.0 ms^{-2} at any time, the angular frequency of the oscillator is equal to [CBSE PMT 1992; RPMT 1996]
- (a) 10 rads^{-1} (b) 0.1 rads^{-1}
(c) 100 rads^{-1} (d) 1 rads^{-1}
11. The equation of a simple harmonic motion is $X = 0.34 \cos(3000t + 0.74)$ where X and t are in mm and sec . The frequency of motion is [Kerala (Engg.) 2002]
- (a) 3000 (b) $3000/2\pi$
(c) $0.74/2\pi$ (d) $3000/\pi$
12. Mark the wrong statement [MP PMT 2003]
- (a) All S.H.M.'s have fixed time period
(b) All motion having same time period are S.H.M.
(c) In S.H.M. total energy is proportional to square of amplitude
(d) Phase constant of S.H.M. depends upon initial conditions
13. A particle in SHM is described by the displacement equation $x(t) = A \cos(\omega t + \theta)$. If the initial ($t = 0$) position of the particle is 1 cm and its initial velocity is $\pi \text{ cm/s}$, what is its amplitude? The angular frequency of the particle is $\pi \text{ s}^{-1}$ [DPMT 2004]
- (a) 1 cm (b) $\sqrt{2} \text{ cm}$
(c) 2 cm (d) 2.5 cm
14. A particle executes SHM in a line 4 cm long. Its velocity when passing through the centre of line is 12 cm/s . The period will be
- (a) 2.047 s (b) 1.047 s
(c) 3.047 s (d) 0.047 s
15. The displacement x (in metre) of a particle in, simple harmonic motion is related to time t (in seconds) as
- $$x = 0.01 \cos\left(\pi t + \frac{\pi}{4}\right)$$
- The frequency of the motion will be [UPSEAT 2004]
- (a) 0.5 Hz (b) 1.0 Hz
(c) $\frac{\pi}{2} \text{ Hz}$ (d) $\pi \text{ Hz}$
16. A simple harmonic wave having an amplitude a and time period T is represented by the equation $y = 5 \sin \pi(t + 4)m$. Then the value of amplitude (a) in (m) and time period (T) in second are [Pb. PET 2004]
- (a) $a = 10, T = 2$ (b) $a = 5, T = 1$
(c) $a = 10, T = 1$ (d) $a = 5, T = 2$
17. A particle executing simple harmonic motion of amplitude 5 cm has maximum speed of 31.4 cm/s . The frequency of its oscillation is
- (a) 3 Hz (b) 2 Hz
(c) 4 Hz (d) 1 Hz
18. The displacement x (in metres) of a particle performing simple harmonic motion is related to time t (in seconds) as
- $$x = 0.05 \cos\left(4\pi t + \frac{\pi}{4}\right)$$
- The frequency of the motion will be
- (a) 0.5 Hz (b) 1.0 Hz
(c) 1.5 Hz (d) 2.0 Hz

Simple Pendulum

1. The period of a simple pendulum is doubled, when [CPMT 1974; MNR 1980; AFMC 1995; Pb. PET/PMT 2002]
- (a) Its length is doubled
(b) The mass of the bob is doubled
(c) Its length is made four times
(d) The mass of the bob and the length of the pendulum are doubled
2. The period of oscillation of a simple pendulum of constant length at earth surface is T . Its period inside a mine is [CPMT 1973; DPMT 2001]
- (a) Greater than T (b) Less than T
(c) Equal to T (d) Cannot be compared
3. A simple pendulum is made of a body which is a hollow sphere containing mercury suspended by means of a wire. If a little mercury is drained off, the period of pendulum will [NCERT 1972; BHU 1979]
- (a) Remains unchanged
(b) Increase
(c) Decrease
(d) Become erratic
4. A pendulum suspended from the ceiling of a train has a period T , when the train is at rest. When the train is accelerating with a uniform acceleration a , the period of oscillation will [NCERT 1980; CPMT 1997] [Pb. PET 2000]
- (a) Increase (b) Decrease
(c) Remain unaffected (d) Become infinite
5. The mass and diameter of a planet are twice those of earth. The period of oscillation of pendulum on this planet will be (If it is a second's pendulum on earth) [IIT 1973; DCE 2002]
- (a) $\frac{1}{\sqrt{2}} \text{ sec}$ (b) $2\sqrt{2} \text{ sec}$
(c) 2 sec (d) $\frac{1}{2} \text{ sec}$





6. A simple pendulum is set up in a trolley which moves to the right with an acceleration a on a horizontal plane. Then the thread of the pendulum in the mean position makes an angle θ with the vertical

- (a) $\tan^{-1} \frac{a}{g}$ in the forward direction
 (b) $\tan^{-1} \frac{a}{g}$ in the backward direction
 (c) $\tan^{-1} \frac{g}{a}$ in the backward direction
 (d) $\tan^{-1} \frac{g}{a}$ in the forward direction

7. Which of the following statements is not true? In the case of a simple pendulum for small amplitudes the period of oscillation is

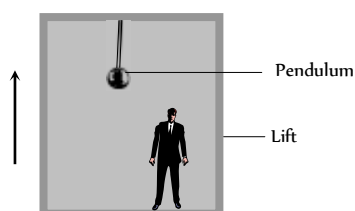
- (a) Directly proportional to square root of the length of the pendulum
 (b) Inversely proportional to the square root of the acceleration due to gravity
 (c) Dependent on the mass, size and material of the bob
 (d) Independent of the amplitude

8. The time period of a second's pendulum is 2 sec. The spherical bob which is empty from inside has a mass of 50 gm. This is now replaced by another solid bob of same radius but having different mass of 100 gm. The new time period will be

- (a) 4 sec (b) 1 sec
 (c) 2 sec (d) 8 sec

9. A man measures the period of a simple pendulum inside a stationary lift and finds it to be T sec. If the lift accelerates upwards with an acceleration $g/4$, then the period of the pendulum will be

- (a) T
 (b) $\frac{T}{4}$
 (c) $\frac{2T}{\sqrt{5}}$
 (d) $2T\sqrt{5}$



10. A simple pendulum is suspended from the roof of a trolley which moves in a horizontal direction with an acceleration a , then the time period is given by $T = 2\pi\sqrt{\frac{l}{g'}}$, where g' is equal to

- (a) g (b) $g - a$
 (c) $g + a$ (d) $\sqrt{g^2 + a^2}$

11. A second's pendulum is placed in a space laboratory orbiting around the earth at a height $3R$, where R is the radius of the earth. The time period of the pendulum is

- (a) Zero (b) $2\sqrt{3}$ sec
 (c) 4 sec (d) Infinite

[CPMT 1989; RPMT 1995]

12. The bob of a simple pendulum of mass m and total energy E will have maximum linear momentum equal to

[CPMT 1983]

[MP PMT 1986]

- (a) $\sqrt{\frac{2E}{m}}$ (b) $\sqrt{2mE}$
 (c) $2mE$ (d) mE^2

13. The length of the second pendulum on the surface of earth is 1 m. The length of seconds pendulum on the surface of moon, where g is 1/6th value of g on the surface of earth, is

[CPMT 1971]

- (a) $1/6$ m (b) 6 m
 (c) $1/36$ m (d) 36 m

14. If the length of second's pendulum is decreased by 2%, how many seconds it will lose per day

[CPMT 1992]

- (a) 3927 sec (b) 3727 sec
 (c) 3427 sec (d) 864 sec

15. The period of simple pendulum is measured as T in a stationary lift. If the lift moves upwards with an acceleration of $5g$, the period will be

[MNR 1979]

- (a) The same (b) Increased by 3/5
 (c) Decreased by 2/3 times (d) None of the above

16. The length of a simple pendulum is increased by 1%. Its time period will

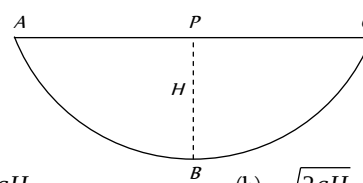
[MP PET 1994; RPET 2001]

- (a) Increase by 1% (b) Increase by 0.5%
 (c) Decrease by 0.5% (d) Increase by 2%

17. A simple pendulum with a bob of mass ' m ' oscillates from A to C and back to A such that PB is H . If the acceleration due to gravity is g , then the velocity of the bob as it passes through B is

[NCERT 1990; BHU 2001]

[CBSE PMT 1995; DPMT 1995; Pb. PMT 1996]



- (a) mgH (b) $\sqrt{2gH}$
 (c) $2gH$ (d) Zero

18. Identify correct statement among the following

[BHU 1997]

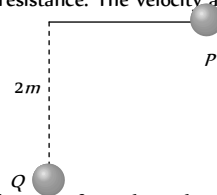
[Manipal MEE 1995]

- (a) The greater the mass of a pendulum bob, the shorter is its frequency of oscillation
 (b) A simple pendulum with a bob of mass M swings with an angular amplitude of 40° . When its angular amplitude is 20° , the tension in the string is less than $Mg \cos 20^\circ$.
 (c) As the length of a simple pendulum is increased, the maximum velocity of its bob during its oscillation will also decrease
 (d) The fractional change in the time period of a pendulum on changing the temperature is independent of the length of the pendulum

19. The bob of a pendulum of length l is pulled aside from its equilibrium position through an angle θ and then released. The bob will then pass through its equilibrium position with a speed v , where v equals [Haryana CEE 1996]
- (a) $\sqrt{2gl(1 - \sin\theta)}$ (b) $\sqrt{2gl(1 + \cos\theta)}$
(c) $\sqrt{2gl(1 - \cos\theta)}$ (d) $\sqrt{2gl(1 + \sin\theta)}$
20. A simple pendulum executing S.H.M. is falling freely along with the support. Then
- (a) Its periodic time decreases
(b) Its periodic time increases
(c) It does not oscillate at all
(d) None of these
21. A pendulum bob has a speed of 3 m/s at its lowest position. The pendulum is 0.5 m long. The speed of the bob, when the length makes an angle of 60° to the vertical, will be (If $g = 10 \text{ m/s}^2$)
- (a) 3 m/s (b) $\frac{1}{3} \text{ m/s}$
(c) $\frac{1}{2} \text{ m/s}$ (d) 2 m/s
22. The time period of a simple pendulum is 2 sec . If its length is increased 4 times, then its period becomes [CBSE PMT 1999; DPMT 1999]
- (a) 16 sec (b) 12 sec
(c) 8 sec (d) 4 sec
23. If the metal bob of a simple pendulum is replaced by a wooden bob, then its time period will [AIIMS 1998, 99]
- (a) Increase
(b) Decrease
(c) Remain the same
(d) First increase then decrease
24. In a simple pendulum, the period of oscillation T is related to length of the pendulum l as [EAMCET (Med.) 1995]
- (a) $\frac{l}{T} = \text{constant}$ (b) $\frac{l^2}{T} = \text{constant}$
(c) $\frac{l}{T^2} = \text{constant}$ (d) $\frac{l^2}{T^2} = \text{constant}$
25. A pendulum has time period T . If it is taken on to another planet having acceleration due to gravity half and mass 9 times that of the earth then its time period on the other planet will be
- (a) $\sqrt{T}$ (b) T
(c) $T^{1/3}$ (d) $\sqrt{2} T$
26. A simple pendulum is executing simple harmonic motion with a time period T . If the length of the pendulum is increased by 21%, the percentage increase in the time period of the pendulum of increased length is [BHU 1994, 96; Pb. PMT 1995; AFMC 2001; AIIMS 2001; AIEEE 2003]
- (a) 10% (b) 21%
(c) 30% (d) 50%
27. If the length of simple pendulum is increased by 300%, then the time period will be increased by [RPMT 1999]
- (a) 100% (b) 200%
(c) 300% (d) 400%
28. The length of a seconds pendulum is [RPET 2000]
- (a) 99.8 cm (b) 99 cm
(c) 100 cm (d) None of these
29. The time period of a simple pendulum in a lift descending with constant acceleration g is [DCE 1998; MP PMT 2001]
- (a) $T = 2\pi\sqrt{\frac{l}{g}}$ (b) $T = 2\pi\sqrt{\frac{l}{2g}}$
(c) Zero [MP PET 1996] (d) Infinite
30. A chimpanzee swinging on a swing in a sitting position, stands up suddenly, the time period will [KCET (Engg./Med.) 2000; AIEEE 2002; DPMT 2004]
- (a) Become infinite (b) Remain same
(c) Increase (d) Decrease
31. The acceleration due to gravity at a place is $\pi^2 \text{ m/sec}^2$. Then the time period of a simple pendulum of length one metre is
- (a) $\frac{2}{\pi} \text{ sec}$ (b) $2\pi \text{ sec}$
(c) 2 sec (d) $\pi \text{ sec}$
32. A plate oscillated with time period ' T '. Suddenly, another plate put on the first plate, then time period [AIEEE 2002]
- (a) Will decrease (b) Will increase
(c) Will be same (d) None of these
33. A simple pendulum of length l has a brass bob attached at its lower end. Its period is T . If a steel bob of same size, having density x times that of brass, replaces the brass bob and its length is changed so that period becomes $2T$, then new length is
- (a) $2l$ (b) $4l$
(c) $4lx$ (d) $\frac{4l}{x}$
34. In a seconds pendulum, mass of bob is 30 gm . If it is replaced by 90 gm mass. Then its time period will [CMEET Bihar 1995] [Orissa PMT 2001]
- (a) 1 sec (b) 2 sec
(c) 4 sec (d) 3 sec
35. The time period of a simple pendulum when it is made to oscillate on the surface of moon [J & K CET 2004]
- (a) Increases (b) Decreases
(c) Remains unchanged (d) Becomes infinite



36. A simple pendulum is attached to the roof of a lift. If time period of oscillation, when the lift is stationary is T . Then frequency of oscillation, when the lift falls freely, will be [DCE 2002]
- (a) Zero (b) T
(c) $1/T$ (d) None of these
37. A simple pendulum, suspended from the ceiling of a stationary van, has time period T . If the van starts moving with a uniform velocity the period of the pendulum will be [RPMT 2003]
- (a) Less than T (b) Equal to $2T$
(c) Greater than T (d) Unchanged
38. If the length of the simple pendulum is increased by 44%, then what is the change in time period of pendulum [MH CET 2004; UPSEAT 2005]
- (a) 22% (b) 20%
(c) 33% (d) 44%
39. To show that a simple pendulum executes simple harmonic motion, it is necessary to assume that [CPMT 2001]
- (a) Length of the pendulum is small
(b) Mass of the pendulum is small
(c) Amplitude of oscillation is small
(d) Acceleration due to gravity is small
40. The height of a swing changes during its motion from 0.1 m to 2.5 m. The minimum velocity of a boy who swings in this swing is
- (a) 5.4 m/s (b) 4.95 m/s
(c) 3.14 m/s (d) Zero
41. The amplitude of an oscillating simple pendulum is 10 cm and its period is 4 sec. Its speed after 1 sec after it passes its equilibrium position, is
- (a) Zero (b) 0.57 m/s
(c) 0.212 m/s (d) 0.32 m/s
42. A simple pendulum consisting of a ball of mass m tied to a thread of length l is made to swing on a circular arc of angle θ in a vertical plane. At the end of this arc, another ball of mass m is placed at rest. The momentum transferred to this ball at rest by the swinging ball is [NCERT 1977]
- (a) Zero (b) $m\theta\sqrt{\frac{g}{l}}$
(c) $\frac{m\theta}{l}\sqrt{\frac{l}{g}}$ (d) $\frac{m}{l}2\pi\sqrt{\frac{l}{g}}$
43. A simple pendulum hangs from the ceiling of a car. If the car accelerates with a uniform acceleration, the frequency of the simple pendulum will [Pb. PMT 2000]
- (a) Increase (b) Decrease
(c) Become infinite (d) Remain constant
44. The periodic time of a simple pendulum of length 1 m and amplitude 2 cm is 5 seconds. If the amplitude is made 4 cm, its periodic time in seconds will be [MP PMT 1985]
- (a) 2.5 (b) 5
(c) 10 (d) $5\sqrt{2}$
45. The ratio of frequencies of two pendulums are 2 : 3, then their length are in ratio [DCE 2005]
- (a) $\sqrt{2/3}$ (b) $\sqrt{3/2}$
(c) 4/9 (d) 9/4
46. Two pendulums begin to swing simultaneously. If the ratio of the frequency of oscillations of the two is 7 : 8, then the ratio of lengths of the two pendulums will be [J & K CET 2005]
- (a) 7 : 8 (b) 8 : 7
(c) 49 : 64 (d) 64 : 49
47. A simple pendulum hanging from the ceiling of a stationary lift has a time period T . When the lift moves downward with constant velocity, the time period is T , then [Orissa JEE 2005]
- (a) T_2 is infinity (b) $T_2 > T_1$
(c) $T_2 < T_1$ (d) $T_2 = T_1$
48. If the length of a pendulum is made 9 times and mass of the bob is made 4 times then the value of time period becomes [BHU 2005]
- (a) $3T$ (b) $3/2T$
(c) $4T$ (d) $2T$
49. A simple pendulum is taken from the equator to the pole. Its period [CPMT 1997]
- (a) Decreases
(b) Increases
(c) Remains the same
(d) Decreases and then increases
50. A pendulum of length $2m$ lift at P . When it reaches Q , it losses 10% of its total energy due to air resistance. The velocity at Q is
- (a) 6 m/sec
(b) 1 m/sec
(c) 2 m/sec
(d) 8 m/sec
51. There is a simple pendulum hanging from the ceiling of a lift. When the lift is stand still, the time period of the pendulum is T . If the resultant acceleration becomes $g/4$, then the new time period of the pendulum is [DCE 2004]
- (a) 0.8 T (b) 0.25 T
(c) 2 T (d) 4 T
52. The period of a simple pendulum measured inside a stationary lift is found to be T . If the lift starts accelerating upwards with acceleration of $g/3$, then the time period of the pendulum is [RPMT 2000; DPMT 2000]
- (a) $\frac{T}{\sqrt{3}}$ (b) $\frac{T}{3}$
(c) $\frac{\sqrt{3}}{2}T$ (d) $\sqrt{3}T$



53. Time period of a simple pendulum will be double, if we
[MH CET 2003]
- Decrease the length 2 times
 - Decrease the length 4 times
 - Increase the length 2 times
 - Increase the length 4 times
54. Length of a simple pendulum is l and its maximum angular displacement is θ , then its maximum K.E. is
[RPMT 1995; BHU 2003]

- $mgl \sin \theta$
- $mgl(1 + \sin \theta)$
- $mgl(1 + \cos \theta)$
- $mgl(1 - \cos \theta)$

55. The velocity of simple pendulum is maximum at
[RPMT 2004]

- Extremes
- Half displacement
- Mean position
- Every where

56. A simple pendulum is vibrating in an evacuated chamber, it will oscillate with
[Pb. PMT 2004]

- Increasing amplitude
- Constant amplitude
- Decreasing amplitude
- First (c) then (a)

57. The time period of a simple pendulum of length L as measured in an elevator descending with acceleration $\frac{g}{3}$ is
[CPMT 2000]

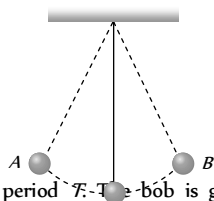
- $2\pi \sqrt{\frac{3L}{g}}$
- $\pi \sqrt{\left(\frac{3L}{g}\right)}$
- $2\pi \sqrt{\left(\frac{3L}{2g}\right)}$
- $2\pi \sqrt{\frac{2L}{3g}}$

58. If a body is released into a tunnel dug across the diameter of earth, it executes simple harmonic motion with time period

- $T = 2\pi \sqrt{\frac{R_e}{g}}$
- $T = 2\pi \sqrt{\frac{2R_e}{g}}$
- $T = 2\pi \sqrt{\frac{R_e}{2g}}$
- $T = 2$ seconds

59. What is the velocity of the bob of a simple pendulum at its mean position, if it is able to rise to vertical height of 10 cm ($g = 9.8 \text{ m/s}^2$)

- 2.2 m/s
- 1.8 m/s
- 1.4 m/s
- 0.6 m/s



60. A simple pendulum has time period T . The bob is given negative charge and surface below it is given positive charge. The new time period will be
[AFMC 2004]

- Less than T
- Greater than T
- Equal to T
- Infinite

61. What effect occurs on the frequency of a pendulum if it is taken from the earth surface to deep into a mine
[AFMC 2005]

- Increases
- Decreases
- First increases then decrease
- None of these

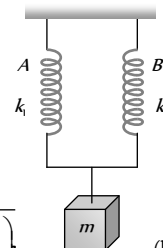
Spring Pendulum

1. Two bodies M and N of equal masses are suspended from two separate massless springs of force constants k and k respectively. If the two bodies oscillate vertically such that their maximum velocities are equal, the ratio of the amplitude M to that of N is

[IIT-JEE 1988; MP PET 1997, 2001; MP PMT 1997; BHU 1998; Pb. PMT 1998; MH CET 2000, 03; AIEEE 2003]

- $\frac{k_1}{k_2}$
- $\sqrt{\frac{k_1}{k_2}}$
- $\frac{k_2}{k_1}$
- $\sqrt{\frac{k_2}{k_1}}$

2. A mass m is suspended by means of two coiled spring which have the same length in unstretched condition as in figure. Their force constant are k and k respectively. When set into vertical vibrations, the period will be [MP PMT 2001]



- $2\pi \sqrt{\left(\frac{m}{k_1 k_2}\right)}$
- $2\pi \sqrt{m \left(\frac{k_1}{k_2}\right)}$
- $2\pi \sqrt{\left(\frac{m}{k_1 - k_2}\right)}$
- $2\pi \sqrt{\left(\frac{m}{k_1 + k_2}\right)}$

3. A spring has a certain mass suspended from it and its period for vertical oscillation is T . The spring is now cut into two equal halves and the same mass is suspended from one of the halves. The period of vertical oscillation is now
[BHU 2000]

[MP PET 1995]

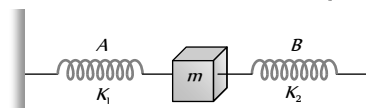
- $\frac{T}{2}$
- $\frac{T}{\sqrt{2}}$
- $\sqrt{2}T$
- $2T$

4. Two masses m_1 and m_2 are suspended together by a massless spring of constant k . When the masses are in equilibrium, m_1 is removed without disturbing the system. Then the angular frequency of oscillation of m_2 is

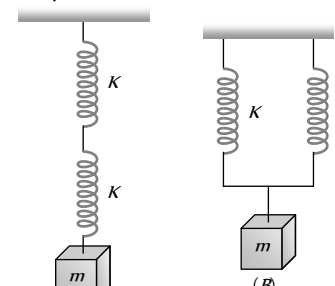
- (a) $\sqrt{\frac{k}{m_1}}$ (b) $\sqrt{\frac{k}{m_2}}$
 (c) $\sqrt{\frac{k}{m_1 + m_2}}$ (d) $\sqrt{\frac{k}{m_1 m_2}}$

5. In arrangement given in figure, if the block of mass m is displaced, the frequency is given by

[BHU 1994; Pb. PET 2001]

- 
- (a) $n = \frac{1}{2\pi} \sqrt{\left(\frac{k_1 - k_2}{m}\right)}$ (b) $n = \frac{1}{2\pi} \sqrt{\left(\frac{k_1 + k_2}{m}\right)}$
 (c) $n = \frac{1}{2\pi} \sqrt{\left(\frac{m}{k_1 + k_2}\right)}$ (d) $n = \frac{1}{2\pi} \sqrt{\left(\frac{m}{k_1 - k_2}\right)}$

6. Two identical spring of constant K are connected in series and parallel as shown in figure. A mass m is suspended from them. The ratio of their frequencies of vertical oscillations will be [MP PET 1993; BHU 1997]

- 
- (a) 2 : 1 (b) 1 : 1
 (c) 1 : 2 (d) 4 : 1

7. A mass m is suspended from the two coupled springs connected in series. The force constant for springs are K_1 and K_2 . The time period of the suspended mass will be

[CBSE PMT 1990; Pb. PET 2002]

- (a) $T = 2\pi \sqrt{\left(\frac{m}{K_1 + K_2}\right)}$ (b) $T = 2\pi \sqrt{\left(\frac{m}{K_1 K_2}\right)}$
 (c) $T = 2\pi \sqrt{\left(\frac{m(K_1 + K_2)}{K_1 K_2}\right)}$ (d) $T = 2\pi \sqrt{\left(\frac{m K_1 K_2}{K_1 + K_2}\right)}$

8. A spring is stretched by 0.20 m , when a mass of 0.50 kg is suspended. When a mass of 0.25 kg is suspended, then its period of oscillation will be ($g = 10\text{ m/s}^2$)

- (a) 0.328 sec (b) 0.628 sec
 (c) 0.137 sec (d) 1.00 sec

9. A mass M is suspended from a spring of negligible mass. The spring is pulled a little and then released so that the mass executes simple harmonic oscillations with a time period T . If the mass is increased by m then the time period becomes $\left(\frac{5}{4}T\right)$. The ratio of $\frac{m}{M}$ is [CPMT 1991]

- (a) $9/16$ (b) $25/16$
 (c) $4/5$ (d) $5/4$

10. A spring having a spring constant ' K ' is loaded with a mass ' m '. The spring is cut into two equal parts and one of these is loaded again with the same mass. The new spring constant is [NCERT 1990; KCET 1999;

Kerala PMT 2004; BCECE 2004]

- (a) $K/2$ (b) K
 (c) $2K$ (d) K^2

11. A weightless spring which has a force constant oscillates with frequency n when a mass m is suspended from it. The spring is cut into two equal halves and a mass $2m$ is suspended from it. The frequency of oscillation will now become

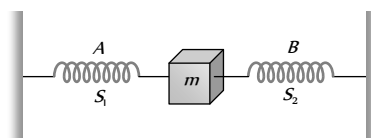
[CPMT 1988]

- (a) n (b) $2n$
 (c) $n/\sqrt{2}$ (d) $n(2)^{1/2}$

12. A mass M is suspended from a light spring. An additional mass m added displaces the spring further by a distance x . Now the combined mass will oscillate on the spring with period [CPMT 1989, 1998; UPSEAT 1998]

- (a) $T = 2\pi \sqrt{(mg/x(M+m))}$
 (b) $T = 2\pi \sqrt{((M+m)x/mg)}$
 (c) $T = (\pi/2) \sqrt{(mg/x(M+m))}$
 (d) $T = 2\pi \sqrt{((M+m)/mgx)}$

13. In the figure, S_1 and S_2 are identical springs. The oscillation frequency of the mass m is f . If one spring is removed, the frequency will become [CPMT 1971]



- (a) f (b) $f \times 2$
 (c) $f \times \sqrt{2}$ (d) $f/\sqrt{2}$

14. The vertical extension in a light spring by a weight of 1 kg suspended from the wire is 9.8 cm . The period of oscillation

[CPMT 1981; MP PMT 2003]

- (a) $20\pi\text{ sec}$ (b) $2\pi\text{ sec}$
 (c) $2\pi/10\text{ sec}$ (d) $200\pi\text{ sec}$

15. A particle of mass 200 gm executes S.H.M. The restoring force is provided by a spring of force constant 80 N/m . The time period of oscillations is [MP PET 1994]

- (a) 0.31 sec (b) 0.15 sec
 (c) 0.05 sec (d) 0.02 sec

16. The length of a spring is l and its force constant is k . When a weight W is suspended from it, its length increases by x . If the spring is cut into two equal parts and put in parallel and the same weight W is suspended from them, then the extension will be

- (a) $2x$ (b) x

(c) $\frac{x}{2}$ (d) $\frac{x}{4}$

17. A block is placed on a frictionless horizontal table. The mass of the block is m and springs are attached on either side with force constants K_1 and K_2 . If the block is displaced a little and left to oscillate, then the angular frequency of oscillation will be

(a) $\left(\frac{K_1 + K_2}{m}\right)^{1/2}$ (b) $\left[\frac{K_1 K_2}{m(K_1 + K_2)}\right]^{1/2}$
(c) $\left[\frac{K_1 K_2}{(K_1 - K_2)m}\right]^{1/2}$ (d) $\left[\frac{K_1^2 + K_2^2}{(K_1 + K_2)m}\right]^{1/2}$

18. A uniform spring of force constant k is cut into two pieces, the lengths of which are in the ratio 1 : 2. The ratio of the force constants of the shorter and the longer pieces is

[Manipal MEE 1995]

(a) 1 : 3 (b) 1 : 2
(c) 2 : 3 (d) 2 : 1

19. A mass $m = 100 \text{ gms}$ is attached at the end of a light spring which oscillates on a frictionless horizontal table with an amplitude equal to 0.16 metre and time period equal to 2 sec. Initially the mass is released from rest at $t = 0$ and displacement $x = -0.16 \text{ metre}$. The expression for the displacement of the mass at any time t is [MP PMT 1995]

(a) $x = 0.16 \cos(\pi t)$ (b) $x = -0.16 \cos(\pi t)$
(c) $x = 0.16 \sin(\pi t + \pi)$ (d) $x = -0.16 \sin(\pi t + \pi)$

20. A block of mass m , attached to a spring of spring constant k , oscillates on a smooth horizontal table. The other end of the spring is fixed to a wall. The block has a speed v when the spring is at its natural length. Before coming to an instantaneous rest, if the block moves a distance x from the mean position, then

(a) $x = \sqrt{m/k}$ (b) $x = \frac{1}{v} \sqrt{m/k}$
(c) $x = v \sqrt{m/k}$ (d) $x = \sqrt{mv/k}$

21. The force constants of two springs are K_1 and K_2 . Both are stretched till their elastic energies are equal. If the stretching forces are F_1 and F_2 , then $F_1 : F_2$ is

[MP PET 2002]

(a) $K_1 : K_2$ (b) $K_2 : K_1$
(c) $\sqrt{K_1} : \sqrt{K_2}$ (d) $K_1^2 : K_2^2$

22. A mass m is vertically suspended from a spring of negligible mass; the system oscillates with a frequency n . What will be the frequency of the system if a mass $4m$ is suspended from the same spring

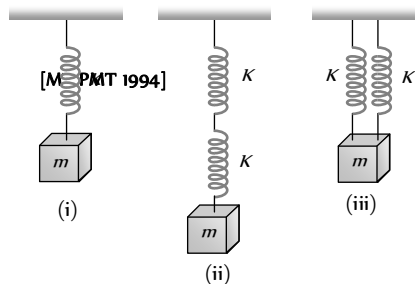
(a) $n/4$ (b) $4n$
(c) $n/2$ (d) $2n$

23. If the period of oscillation of mass m suspended from a spring is 2 sec, then the period of mass $4m$ will be

[AIIMS 1998]

(a) 1 sec (b) 2 sec
(c) 3 sec (d) 4 sec

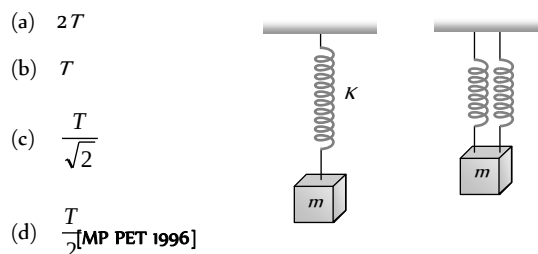
24. Five identical springs are used in the following three configurations. The time periods of vertical oscillations in configurations (i), (ii) and (iii) are in the ratio [AMU 1995]



(a) $1 : \sqrt{2} : \frac{1}{\sqrt{2}}$ (b) $2 : \sqrt{2} : \frac{1}{\sqrt{2}}$
(c) $\frac{1}{\sqrt{2}} : 2 : 1$ (d) $2 : \frac{1}{\sqrt{2}} : 1$

25. A mass m performs oscillations of period T when hanged by spring of force constant K . If spring is cut in two parts and arranged in parallel and same mass is oscillated by them, then the new time period will be

[CPMT 1995; RPET 1997; RPMT 2003]



(a) $2T$
(b) T
(c) $\frac{T}{\sqrt{2}}$
(d) $\frac{T}{2}$ [MP PET 1996]

26. If a watch with a wound spring is taken on to the moon, it

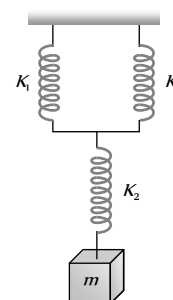
[AFMC 1993]

(a) Runs faster (b) Runs slower
(c) Does not work (d) Shows no change

27. What will be the force constant of the spring system shown in the figure

[RPET 1996; Kerala (Med./ Engg.) 2005]

(a) $\frac{K_1}{2} + K_2$
(b) $\left[\frac{1}{2K_1} + \frac{1}{K_2}\right]^{-1}$ [CBSE PMT 1998]
(c) $\frac{1}{2K_1} + \frac{1}{K_2}$
(d) $\left[\frac{2}{K_1} + \frac{1}{K_2}\right]^{-1}$

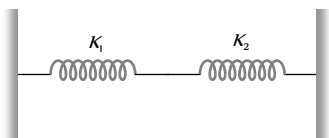


28. Two springs have spring constants K_A and K_B and $K_A > K_B$. The work required to stretch them by same extension will be

(a) More in spring A (b) More in spring B

- (c) Equal in both (d) Nothing can be said

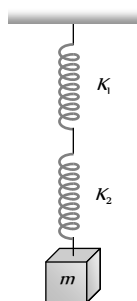
29. The effective spring constant of two spring system as shown in figure will be [RPMT 1999]



- (a) $K_1 + K_2$ (b) $K_1 K_2 / K_1 + K_2$
 (c) $K_1 - K_2$ (d) $K_1 K_2 / K_1 - K_2$
30. A mass m attached to a spring oscillates every 2 sec. If the mass is increased by 2 kg, then time-period increases by 1 sec. The initial mass is [CBSE PMT 2000; AIIMS 2000; MP PET 2000; DPMT 2001; Pb. PMT 2003]

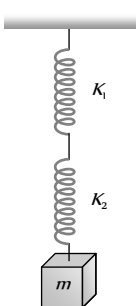
- (a) 1.6 kg (b) 3.9 kg
 (c) 9.6 kg (d) 12.6 kg
31. A mass M is suspended by two springs of force constants K_1 and K_2 respectively as shown in the diagram. The total elongation (stretch) of the two springs is [MP PMT 2000; RPET 2001]

- (a) $\frac{Mg}{K_1 + K_2}$
 (b) $\frac{Mg(K_1 + K_2)}{K_1 K_2}$
 (c) $\frac{Mg K_1 K_2}{K_1 + K_2}$
 (d) $\frac{K_1 + K_2}{K_1 K_2 Mg}$



32. The frequency of oscillation of the springs shown in the figure will be [AIIMS 2001; Pb. PET 2002]

- (a) $\frac{1}{2\pi} \sqrt{\frac{K}{m}}$
 (b) $\frac{1}{2\pi} \sqrt{\frac{(K_1 + K_2)m}{K_1 K_2}}$
 (c) $2\pi \sqrt{\frac{K}{m}}$
 (d) $\frac{1}{2\pi} \sqrt{\frac{K_1 K_2}{m(K_1 + K_2)}}$



33. The scale of a spring balance reading from 0 to 10 kg is 0.25 m long. A body suspended from the balance oscillates vertically with a period of $\pi/10$ second. The mass suspended is (neglect the mass of the spring) [Kerala (Engg.) 2001]

- (a) 10 kg (b) 0.98 kg
 (c) 5 kg (d) 20 kg
34. If a spring has time period T , and is cut into n equal parts, then the time period of each part will be [AIIEE 2002]

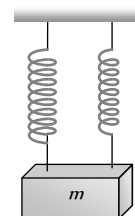
- (a) $T\sqrt{n}$ (b) $T/\sqrt{n}$
 (c) nT (d) T

35. One-fourth length of a spring of force constant K is cut away. The force constant of the remaining spring will be [MP PET 2002]

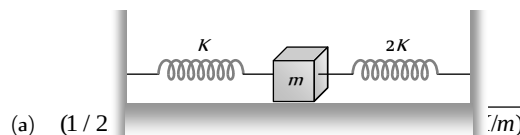
- (a) $\frac{3}{4}K$ (b) $\frac{4}{3}K$
 (c) K (d) $4K$

36. A mass m is suspended separately by two different springs of spring constant K and K gives the time-period t_1 and t_2 respectively. If same mass m is connected by both springs as shown in figure then time-period t is given by the relation [CBSE PMT 2002]

- (a) $t = t_1 + t_2$
 (b) $t = \frac{t_1 \cdot t_2}{t_1 + t_2}$
 (c) $t^2 = t_1^2 + t_2^2$
 (d) $t^{-2} = t_1^{-2} + t_2^{-2}$



37. Two springs of force constants K and $2K$ are connected to a mass as shown below. The frequency of oscillation of the mass is [RPMT 1996; DCE 2000]



- (a) $1/2$ (b) $(1/2\pi)\sqrt{3K/m}$ (c) $(1/2\pi)\sqrt{(m/K)}$ (d) $(1/2\pi)\sqrt{(m/K)}$

38. Two springs of constant k_1 and k_2 are joined in series. The effective spring constant of the combination is given by [CBSE PMT 2004]

- (a) $\sqrt{k_1 k_2}$ (b) $(k_1 + k_2)/2$
 (c) $k_1 + k_2$ (d) $k_1 k_2 / (k_1 + k_2)$

39. A particle at the end of a spring executes simple harmonic motion with a period t_1 , while the corresponding period for another spring is t_2 . If the period of oscillation with the two springs in series is T , then [AIIEE 2004]

- (a) $T = t_1 + t_2$ (b) $T^2 = t_1^2 + t_2^2$
 (c) $T^{-1} = t_1^{-1} + t_2^{-1}$ (d) $T^{-2} = t_1^{-2} + t_2^{-2}$

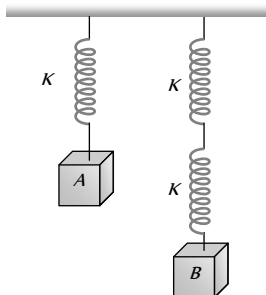
40. Infinite springs with force constant $k, 2k, 4k$ and $8k, \dots$ respectively are connected in series. The effective force constant of the spring will be [J & K CET 2004]

- (a) $2K$ (b) k
 (c) $k/2$ (d) 2048

41. To make the frequency double of a spring oscillator, we have to

- (a) Reduce the mass to one fourth
 (b) Quadruple the mass
 (c) Double of mass
 (d) Half of the mass

42. The springs shown are identical. When $A = 4\text{ kg}$, the elongation of spring is 1 cm . If $B = 6\text{ kg}$, the elongation produced by it is



- (a) 4 cm (b) 3 cm
(c) 2 cm (d) 1 cm
43. When a body of mass 1.0 kg is suspended from a certain light spring hanging vertically, its length increases by 5 cm . By suspending 2.0 kg block to the spring and if the block is pulled through 10 cm and released the maximum velocity in it in m/s is : (Acceleration due to gravity $= 10\text{ m/s}^2$)

[EAMCET 2003]

- (a) 0.5 (b) 1
(c) 2 (d) 4
44. Two springs with spring constants $K_1 = 1500\text{ N/m}$ and $K_2 = 3000\text{ N/m}$ are stretched by the same force. The ratio of potential energy stored in spring will be

[RPET 2001]

- (a) $2 : 1$ (b) $1 : 2$
(c) $4 : 1$ (d) $1 : 4$
45. If a spring extends by x on loading, then energy stored by the spring is (if T is the tension in the spring and K is the spring constant)

- (a) $\frac{T^2}{2x}$ (b) $\frac{T^2}{2K}$
(c) $\frac{2K}{T^2}$ (d) $\frac{2T^2}{K}$
46. A weightless spring of length 60 cm and force constant 200 N/m is kept straight and unstretched on a smooth horizontal table and its ends are rigidly fixed. A mass of 0.25 kg is attached at the middle of the spring and is slightly displaced along the length. The time period of the oscillation of the mass is

[MP PET 2003]

- (a) $\frac{\pi}{20}\text{ s}$ (b) $\frac{\pi}{10}\text{ s}$
(c) $\frac{\pi}{5}\text{ s}$ (d) $\frac{\pi}{\sqrt{200}}\text{ s}$
47. The time period of a mass suspended from a spring is T . If the spring is cut into four equal parts and the same mass is suspended from one of the parts, then the new time period will be

[MP PMT 2002; CBSE PMT 2003]

- (a) T (b) $\frac{T}{2}$
(c) $2T$ (d) $\frac{T}{4}$

48. A mass M is suspended from a spring of negligible mass. The spring is pulled a little and then released so that the mass executes S.H.M. of time period T . If the mass is increased by m , the time period becomes $5T/3$. Then the ratio of m/M is

- (a) $\frac{5}{3}$ (b) $\frac{3}{5}$
(c) $\frac{25}{9}$ (d) $\frac{16}{9}$

49. An object is attached to the bottom of a light vertical spring and set vibrating. The maximum speed of the object is 15 cm/sec and the period is 628 milli-seconds . The amplitude of the motion in centimeters is

[EAMCET 2003]

- (a) 3.0 (b) 2.0
(c) 1.5 (d) 1.0

50. When a mass m is attached to a spring, it normally extends by 0.2 m . The mass m is given a slight addition extension and released, then its time period will be

[MH CET 2001]

- (a) $\frac{1}{7}\text{ sec}$ (b) 1 sec
(c) $\frac{2\pi}{7}\text{ sec}$ (d) $\frac{2}{3\pi}\text{ sec}$

51. If a body of mass 0.98 kg is made to oscillate on a spring of force constant 4.84 N/m , the angular frequency of the body is

- (a) 1.22 rad/s (b) 2.22 rad/s
(c) 3.22 rad/s (d) 4.22 rad/s

52. A mass m is suspended from a spring of length l and force constant K . The frequency of vibration of the mass is f_1 . The spring is cut into two equal parts and the same mass is suspended from one of the parts. The frequency of vibration of mass is f_2 . Which of the following relations between the frequencies is correct

[NCERT 1983; CPMT 1986; MP PMT 1991; DCE 2002]

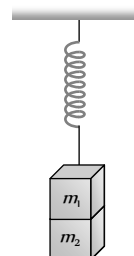
- (a) $f_1 = \sqrt{2}f_2$ (b) $f_1 = f_2$
(c) $f_1 = 2f_2$ (d) $f_2 = \sqrt{2}f_1$

53. A mass m oscillates with simple harmonic motion with frequency $f = \frac{\omega}{2\pi}$ and amplitude A on a spring with constant K , therefore

- (a) The total energy of the system is $\frac{1}{2}KA^2$
(b) The frequency is $\frac{1}{2\pi}\sqrt{\frac{K}{M}}$
(c) The maximum velocity occurs, when $x = 0$
(d) All the above are correct

54. Two masses m_1 and m_2 are suspended together by a massless spring of constant K . When the masses are in equilibrium, m_1 is removed without disturbing the system. The amplitude of oscillations is

- (a) $\frac{m_1g}{K}$
(b) $\frac{m_2g}{K}$



(c) $\frac{(m_1 + m_2)g}{K}$

(d) $\frac{(m_1 - m_2)g}{K}$

55. A spring executes SHM with mass of 10 kg attached to it. The force constant of spring is 10 N/m . If at any instant its velocity is 40 cm/sec , the displacement will be (where amplitude is 0.5 m)

- (a) 0.09 m (b) 0.3 m
(c) 0.03 m (d) 0.9 m

Superposition of S.H.M's and Resonance

1. The S.H.M. of a particle is given by the equation $y = 3 \sin \omega t + 4 \cos \omega t$. The amplitude is [MP PET 1993]

- (a) 7 (b) 1
(c) 5 (d) 12

2. If the displacement equation of a particle be represented by $y = A \sin PT + B \cos PT$, the particle executes

[MP PET 1986]

- (a) A uniform circular motion
(b) A uniform elliptical motion
(c) A S.H.M.
(d) A rectilinear motion

3. The motion of a particle varies with time according to the relation $y = a(\sin \omega t + \cos \omega t)$, then

- (a) The motion is oscillatory but not S.H.M.
(b) The motion is S.H.M. with amplitude a
(c) The motion is S.H.M. with amplitude $a\sqrt{2}$
(d) The motion is S.H.M. with amplitude $2a$

4. The resultant of two rectangular simple harmonic motions of the same frequency and unequal amplitudes but differing in phase by $\frac{\pi}{2}$ is [BHU 2003;

CPMT 2004; MP PMT 1989, 2005; BCECE 2005]

- (a) Simple harmonic (b) Circular
(c) Elliptical (d) Parabolic

5. The composition of two simple harmonic motions of equal periods at right angle to each other and with a phase difference of π results in the displacement of the particle along

- (a) Straight line (b) Circle
(c) Ellipse (d) Figure of eight

6. Two mutually perpendicular simple harmonic vibrations have same amplitude, frequency and phase. When they superimpose, the resultant form of vibration will be

[MP PMT 1992]

- (a) A circle (b) An ellipse
(c) A straight line (d) A parabola

7. The displacement of a particle varies according to the relation $x = 4(\cos \pi t + \sin \pi t)$. The amplitude of the particle is

- (a) 8 (b) -4
(c) 4 (d) $4\sqrt{2}$

8. A S.H.M. is represented by $x = 5\sqrt{2}(\sin 2\pi t + \cos 2\pi t)$. The amplitude of the S.H.M. is [MH CET 2004]

- (a) 10 cm (b) 20 cm
(c) $5\sqrt{2}\text{ cm}$ (d) 50 cm

9. Resonance is an example of

[CBSE PMT 1999; BHU 1999; 2005]

- (a) Tuning fork (b) Forced vibration
(c) Free vibration (d) Damped vibration

10. In case of a forced vibration, the resonance wave becomes very sharp when the [CBSE PMT 2003]

- (a) Restoring force is small
(b) Applied periodic force is small
(c) Quality factor is small
(d) Damping force is small

11. Amplitude of a wave is represented by

$$A = \frac{c}{a + b - c}$$

Then resonance will occur when [CPMT 1984]

- (a) $b = -c/2$ (b) $b = 0$ and $a = -c$
(c) $b = -a/2$ (d) None of these

12. A particle with restoring force proportional to displacement and resisting force proportional to velocity is subjected to a force $F \sin \omega t$. If the amplitude of the particle is maximum for $\omega = \omega_1$ and the energy of the particle is maximum for $\omega = \omega_2$, then (where ω natural frequency of oscillation of particle)

- (a) $\omega_1 = \omega_0$ and $\omega_2 \neq \omega_0$ (b) $\omega_1 = \omega_0$ and $\omega_2 = \omega_0$
(c) $\omega_1 \neq \omega_0$ and $\omega_2 = \omega_0$ (d) $\omega_1 \neq \omega_0$ and $\omega_2 \neq \omega_0$

13. A simple pendulum is set into vibrations. The bob of the pendulum comes to rest after some time due to

[AFMC 2003; JIPMER 1999]

- (a) Air friction
(b) Moment of inertia [CBSE PMT 1990]
(c) Weight of the bob
(d) Combination of all the above

14. A simple pendulum oscillates in air with time period T and amplitude A . As the time passes [CPMT 2005]

- (a) T and A both decrease
(b) T increases and A is constant
(c) T increases and A decreases
(d) T decreases and A is constant

Critical Thinking

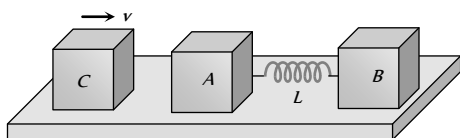
Objective Questions

- Two particles execute S.H.M. of same amplitude and frequency along the same straight line. They pass one another when going in opposite directions, and each time their displacement is half of their amplitude. The phase difference between them is
 - 30°
 - 60°
 - 90°
 - 120°
- The displacement of a particle varies with time as $x = 12 \sin \omega t - 16 \sin^3 \omega t$ (in cm). If its motion is S.H.M., then its maximum acceleration is
 - $12 \omega^2$
 - $36 \omega^2$
 - $144 \omega^2$
 - $\sqrt{192} \omega^2$
- A linear harmonic oscillator of force constant $2 \times 10^6 \text{ N/m}$ times and amplitude 0.01 m has a total mechanical energy of 160 joules . Its

[IIT JEE 1989; CPMT 1995; CBSE PMT 1996; KECT (Med.) 1999; AMU (Engg.) 2000; UPSEAT 2001]

 - Maximum potential energy is 100 J
 - Maximum K.E. is 100 J
 - Maximum P.E. is 160 J
 - Minimum P.E. is zero
- A particle of mass m is executing oscillations about the origin on the x -axis. Its potential energy is $U(x) = k[x]^3$, where k is a positive constant. If the amplitude of oscillation is a , then its time period T is
 - Proportional to $\frac{1}{\sqrt{a}}$
 - Independent of a
 - Proportional to $\sqrt{a}$
 - Proportional to $a^{3/2}$
- Two blocks A and B each of mass m are connected by a massless spring of natural length L and spring constant K . The blocks are initially resting on a smooth horizontal floor with the spring at its natural length as shown in figure. A third identical block C also of mass m moves on the floor with a speed v along the line joining A and B and collides with A . Then

[IIT-JEE 1993]



- The kinetic energy of the A - B system at maximum compression of the spring is zero
 - The kinetic energy of the A - B system at maximum compression of the spring is $mv^2/4$
 - The maximum compression of the spring is $v\sqrt{m/K}$
 - The maximum compression of the spring is $v\sqrt{m/2K}$
- A cylindrical piston of mass M slides smoothly inside a long cylinder closed at one end, enclosing a certain mass of gas. The cylinder is kept with its axis horizontal. If the piston is disturbed from its equilibrium position, it oscillates simple harmonically. The period of oscillation will be

[IIT-JEE 1981]

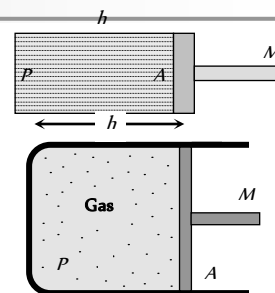
$$(a) T = 2\pi \sqrt{\left(\frac{Mh}{PA}\right)}$$

$$(b) T = 2\pi \sqrt{\left(\frac{MA}{Ph}\right)}$$

$$(c) T = 2\pi \sqrt{\left(\frac{M}{PAh}\right)}$$

[MP PMT 1999]

$$(d) T = 2\pi \sqrt{MPhA}$$



- A sphere of radius r is kept on a concave mirror of radius of curvature R . The arrangement is kept on a horizontal table (the surface of concave mirror is frictionless and sliding not rolling). If the sphere is displaced from its equilibrium position and left, then it executes S.H.M. The period of oscillation will be
 - $2\pi \sqrt{\left(\frac{(R-r)1.4}{g}\right)}$
 - $2\pi \sqrt{\left(\frac{R-r}{g}\right)}$
 - $2\pi \sqrt{\left(\frac{rR}{a}\right)}$
 - $2\pi \sqrt{\left(\frac{R}{gr}\right)}$

- The amplitude of vibration of a particle is given by $a_m = (a_0)/(\omega^2 - b\omega + c)$; where a_0, a, b and c are positive. The condition for a single resonant frequency is

[CPMT 1982]

- $b^2 = 4ac$
 - $b^2 > 4ac$
 - $b^2 = 5ac$
 - $b^2 = 7ac$
- [IIT-JEE 1998]

- A U tube of uniform bore of cross-sectional area A has been set up vertically with open ends facing up. Now $m \text{ gm}$ of a liquid of density d is poured into it. The column of liquid in this tube will oscillate with a period T such that
 - $T = 2\pi \sqrt{\frac{M}{g}}$
 - $T = 2\pi \sqrt{\frac{MA}{gd}}$
 - $T = 2\pi \sqrt{\frac{M}{gdA}}$
 - $T = 2\pi \sqrt{\frac{M}{2Adg}}$

- A particle is performing simple harmonic motion along x -axis with amplitude 4 cm and time period 1.2 sec . The minimum time taken by the particle to move from $x = -2 \text{ cm}$ to $x = +4 \text{ cm}$ and back again is given by

[AIIMS 1995]

- 0.6 sec
- 0.4 sec
- 0.3 sec
- 0.2 sec

- A large horizontal surface moves up and down in SHM with an amplitude of 1 cm . If a mass of 10 kg (which is placed on the surface) is to remain continually in contact with it, the maximum frequency of S.H.M. will be

[SCRA 1994; AIIMS 1995]

- 0.5 Hz
- 1.5 Hz
- 5 Hz
- 10 Hz

12. Due to some force F_1 a body oscillates with period $4/5$ sec and due to other force F_2 oscillates with period $3/5$ sec. If both forces act simultaneously, the new period will be

[RPET 1997]

- (a) 0.72 sec (b) 0.64 sec
(c) 0.48 sec (d) 0.36 sec
13. A horizontal platform with an object placed on it is executing S.H.M. in the vertical direction. The amplitude of oscillation is $3.92 \times 10^{-3} m$. What must be the least period of these oscillations, so that the object is not detached from the platform
- (a) 0.1256 sec (b) 0.1356 sec
(c) 0.1456 sec (d) 0.1556 sec
14. A particle executes simple harmonic motion (amplitude = A) between $x = -A$ and $x = +A$. The time taken for it to go from 0 to $A/2$ is T_1 and to go from $A/2$ to A is T_2 . Then

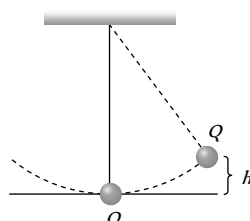
[IIT-JEE (Screening) 2001]

- (a) $T_1 < T_2$ (b) $T_1 > T_2$
(c) $T_1 = T_2$ (d) $T_1 = 2T_2$
15. A simple pendulum of length L and mass (bob) M is oscillating in a plane about a vertical line between angular limits $-\phi$ and $+\phi$. For an angular displacement θ ($|\theta| < \phi$), the tension in the string and the velocity of the bob are T and v respectively. The following relations hold good under the above conditions
- (a) $T \cos \theta = Mg$
(b) $T - Mg \cos \theta = \frac{Mv^2}{L}$
(c) The magnitude of the tangential acceleration of the bob $|a_T| = g \sin \theta$
(d) $T = Mg \cos \theta$
16. Two simple pendulums of length 5 m and 20 m respectively are given small linear displacement in one direction at the same time. They will again be in the phase when the pendulum of shorter length has completed oscillations.

[CBSE PMT 1998; JIPMER 2001, 02]

- (a) 5 (b) 1
(c) 2 (d) 3
17. The bob of a simple pendulum is displaced from its equilibrium position O to a position Q which is at height h above O and the bob is then released. Assuming the mass of the bob to be m and time period of oscillations to be 2.0 sec, the tension in the string when the bob passes through O is

- (a) $m(g + \pi\sqrt{2gh})$
(b) $m(g + \sqrt{\pi^2 gh})$
(c) $m\left(g + \sqrt{\frac{\pi^2}{2} gh}\right)$



$$(d) m\left(g + \sqrt{\frac{\pi^2}{2} gh}\right)$$

18. The metallic bob of a simple pendulum has the relative density ρ . The time period of this pendulum is T . If the metallic bob is immersed in water, then the new time period is given by

- (a) $T \frac{\rho-1}{\rho}$ (b) $T \frac{\rho}{\rho-1}$
(c) $T \sqrt{\frac{\rho-1}{\rho}}$ (d) $T \sqrt{\frac{\rho}{\rho-1}}$

[IIMS 1999]

19. A clock which keeps correct time at $20^\circ C$, is subjected to $40^\circ C$. If coefficient of linear expansion of the pendulum is $12 \times 10^{-6} / ^\circ C$. How much will it gain or lose in time

[BHU 1998]

- (a) 10.3 seconds / day (b) 20.6 seconds / day
(c) 5 seconds / day (d) 20 minutes / day
20. The period of oscillation of a simple pendulum of length L suspended from the roof of a vehicle which moves without friction down an inclined plane of inclination α , is given by

[IIT-JEE (Screening) 2000]

- (a) $2\pi \sqrt{\frac{L}{g \cos \alpha}}$ (b) $2\pi \sqrt{\frac{L}{g \sin \alpha}}$
(c) $2\pi \sqrt{\frac{L}{g}}$ (d) $2\pi \sqrt{\frac{L}{g \tan \alpha}}$

- [IIT 1986; UPSEAT 1998]
21. The bob of a simple pendulum executes simple harmonic motion in water with a period t , while the period of oscillation of the bob is t_0 in air. Neglecting frictional force of water and given that the density of the bob is $(4/3) \times 1000$ kg/m. What relationship between t and t_0 is true

[AIEEE 2004]

- (a) $t = t_0$ (b) $t = t_0 / 2$
(c) $t = 2t_0$ (d) $t = 4t_0$
22. A spring of force constant k is cut into two pieces such that one piece is double the length of the other. Then the long piece will have a force constant of

[IIT-JEE (Screening) 1999]

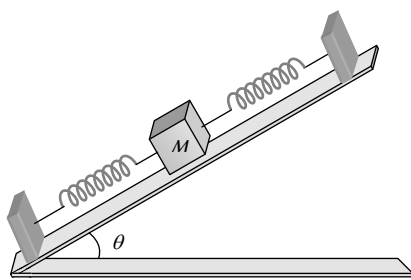
- (a) $(2/3)k$ (b) $(3/2)k$
(c) $3k$ (d) $6k$
23. One end of a long metallic wire of length L is tied to the ceiling. The other end is tied to massless spring of spring constant K . A mass m hangs freely from the free end of the spring. The area of cross-section and Young's modulus of the wire are A and Y respectively. If the mass is slightly pulled down and released, it will oscillate with a time period T equal to

[IIT 1993]

- [AMU 1995]
(a) $2\pi \sqrt{\frac{m}{K}}$ (b) $2\pi \sqrt{\frac{(YA + KL)m}{YAK}}$
(c) $2\pi \sqrt{\frac{mYA}{KL}}$ (d) $2\pi \sqrt{\frac{mL}{YA}}$

24. On a smooth inclined plane, a body of mass M is attached between two springs. The other ends of the springs are fixed to firm supports. If each spring has force constant K , the period of oscillation of the body (assuming the springs as massless) is

- (a) $2\pi\left(\frac{m}{2K}\right)^{1/2}$
 (b) $2\pi\left(\frac{2M}{K}\right)^{1/2}$
 (c) $2\pi\frac{Mg\sin\theta}{2K}$
 (d) $2\pi\left(\frac{2Mg}{K}\right)^{1/2}$



25. A particle of mass m is attached to a spring (of spring constant k) and has a natural angular frequency ω_0 . An external force $F(t)$ proportional to $\cos \omega t$ ($\omega \neq \omega_0$) is applied to the oscillator. The time displacement of the oscillator will be proportional to

- (a) $\frac{m}{\omega_0^2 - \omega^2}$ (b) $\frac{1}{m(\omega_0^2 - \omega^2)}$
 (c) $\frac{1}{m(\omega_1^2 + \omega^2)}$ (d) $\frac{m}{\omega_1^2 + \omega^2}$

26. A 15 g ball is shot from a spring gun whose spring has a force constant of 600 N/m. The spring is compressed by 5 cm. The greatest possible horizontal range of the ball for this compression is ($g = 10 \text{ m/s}^2$) [DPMT 2004]

- (a) 6.0 m (b) 10.0 m
 (c) 12.0 m (d) 8.0 m

27. An ideal spring with spring-constant K is hung from the ceiling and a block of mass M is attached to its lower end. The mass is released with the spring initially unstretched. Then the maximum extension in the spring is

[IIT-JEE (Screening) 2002]

- (a) $4 Mg/K$ (b) $2 Mg/K$
 (c) Mg/K (d) $Mg/2K$

28. The displacement y of a particle executing periodic motion is given by $y = 4 \cos^2(t/2) \sin(1000t)$. This expression may be considered to be a result of the superposition of independent harmonic motions [IIT 1992]

- (a) Two (b) Three
 (c) Four (d) Five

29. Three simple harmonic motions in the same direction having the same amplitude a and same period are superposed. If each differs in phase from the next by 45° , then

- (a) The resultant amplitude is $(1 + \sqrt{2})a$
 (b) The phase of the resultant motion relative to the first is 90°
 (c) The energy associated with the resulting motion is $(3 + 2\sqrt{2})$ times the energy associated with any single motion
 (d) The resulting motion is not simple harmonic

30. The function $\sin^2(\omega t)$ represents [AIEEE 2005]

- (a) A simple harmonic motion with a period $2\pi/\omega$
 (b) A simple harmonic motion with a period π/ω

- (c) A periodic but not simple harmonic motion with a period $2\pi/\omega$
 (d) A periodic but not simple harmonic, motion with a period π/ω

31. A simple pendulum has time period T . The point of suspension is now moved upward according to equation $y = kt^2$ where $k = 1 \text{ m/sec}^2$. If new time period is T' then ratio $\frac{T_1^2}{T_2^2}$ will be

- (a) 2/3 (b) 5/6
 (c) 6/5 (d) 3/2

32. A simple pendulum is hanging from a peg inserted in a vertical wall. Its bob is stretched in horizontal position from the wall and is left free to move. The bob hits on the wall the coefficient of restitution

[AIEEE 2004] After how many collisions the amplitude of vibration will become less than 60°

[UPSEAT 1999]

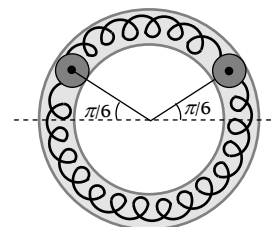
- (a) 6 (b) 3
 (c) 5 (d) 4

33. A brass cube of side a and density σ is floating in mercury of density ρ . If the cube is displaced a bit vertically, it executes S.H.M. Its time period will be

- (a) $2\pi\sqrt{\frac{\sigma a}{\rho g}}$ (b) $2\pi\sqrt{\frac{\rho a}{\sigma g}}$
 (c) $2\pi\sqrt{\frac{\rho g}{\sigma a}}$ (d) $2\pi\sqrt{\frac{\sigma g}{\rho a}}$

34. Two identical balls A and B each of mass 0.1 kg are attached to two identical massless springs. The spring mass system is constrained to move inside a rigid smooth pipe bent in the form of a circle as shown in the figure. The pipe is fixed in a horizontal plane. The centres of the balls can move in a circle of radius 0.06 m . Each spring has a natural length of $0.06\pi \text{ m}$ and force constant 0.1 N/m . Initially both the balls are displaced by an angle $\theta = \pi/6$ radian with respect to the diameter PQ of the circle and released from rest. The frequency of oscillation of the ball B is

- (a) $\pi \text{ Hz}$
 (b) $\frac{1}{\pi} \text{ Hz}$
 (c) $2\pi \text{ Hz}$ [IIT JEE 1999]
 (d) $\frac{1}{2\pi} \text{ Hz}$

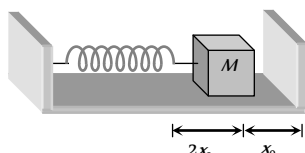


35. A disc of radius R and mass M is pivoted at the rim and is set for small oscillations. If simple pendulum has to have the same period as that of the disc, the length of the simple pendulum should be

- (a) $\frac{5}{4} R$ (b) $\frac{2}{3} R$
 (c) $\frac{3}{4} R$ (d) $\frac{3}{2} R$

36. One end of a spring of force constant k is fixed to a vertical wall and the other to a block of mass m resting on a smooth horizontal surface. There is another wall at a distance x_0 from the block. The

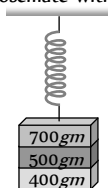
spring is then compressed by $2x_0$ and released. The time taken to strike the wall is



- (a) $\frac{1}{6}\pi\sqrt{\frac{k}{m}}$ (b) $\sqrt{\frac{k}{m}}$
(c) $\frac{2\pi}{3}\sqrt{\frac{m}{k}}$ (d) $\frac{\pi}{4}\sqrt{\frac{k}{m}}$

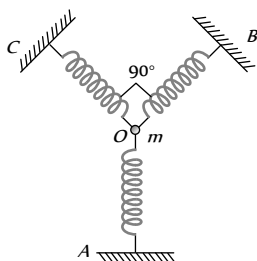
37. Three masses $700g$, $500g$, and $400g$ are suspended at the end of a spring as shown and are in equilibrium. When the $700g$ mass is removed, the system oscillates with a period of 3 seconds, when the $500g$ mass is also removed, it will oscillate with a period of

- (a) $1s$
(b) $2s$
(c) $3s$
(d) $\sqrt{\frac{12}{5}}s$



38. A particle of mass m is attached to three identical springs A , B and C each of force constant k as shown in figure. If the particle of mass m is pushed slightly against the spring A and released then the time period of oscillations is

- (a) $2\pi\sqrt{\frac{2m}{k}}$
(b) $2\pi\sqrt{\frac{m}{2k}}$
(c) $2\pi\sqrt{\frac{m}{k}}$
(d) $2\pi\sqrt{\frac{m}{3k}}$



39. A hollow sphere is filled with water through a small hole in it. It is then hung by a long thread and made to oscillate. As the water slowly flows out of the hole at the bottom, the period of oscillation will

[MP PMT 1994; KCET 1994;

RPET 1996; AFMC 2000;

CBSE PMT 2000; CPMT 2001; AIEEE 2005]

- (a) Continuously decrease
(b) Continuously increase
(c) First decrease and then increase to original value
(d) First increase and then decrease to original value

40. Two simple pendulums whose lengths are $100cm$ and $121cm$ are suspended side by side. Their bobs are pulled together and then released. After how many minimum oscillations of the longer pendulum, will the two be in phase again

- (a) 11 (b) 30
(c) 21 (d) 20

41. The amplitude of a damped oscillator becomes half in one minute. The amplitude after 3 minute will be $\frac{1}{X}$ times the original, where X is

[CPMT 1989; DPMT 2002]

- (a) 2×3 (b) 2^3
(c) 3^2 (d) 3×2^2

42. Which of the following function represents a simple harmonic oscillation

[AIIMS 2005]

- (a) $\sin\omega t - \cos\omega t$ (b) $\sin^2\omega t$
(c) $\sin\omega t + \sin 2\omega t$ (d) $\sin\omega t - \sin 2\omega t$

43. A uniform rod of length $2.0m$ is suspended through an end and is set into oscillation with small amplitude under gravity. The time period of oscillation is approximately

[AMU (Med.) 2000]

- (a) $1.60sec$ (b) $1.80sec$
(c) $2.0sec$ (d) $2.40sec$

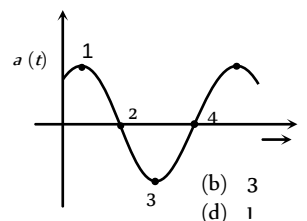
Graphical Questions

1. A particle is executing S.H.M. Then the graph of acceleration as a function of displacement is

- (a) A straight line (b) A circle
(c) An ellipse (d) A hyperbola

2. The acceleration a of a particle undergoing S.H.M. is shown in the figure. Which of the labelled points corresponds to the particle being at $-x_-$

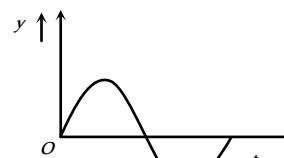
[AMU (Med.) 2000]



- (a) 4 (b) 3
(c) 2 (d) 1

3. The displacement time graph of a particle executing S.H.M. is as shown in the figure

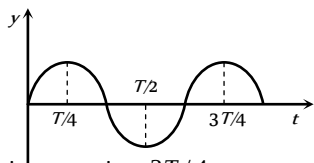
[KCET 2003]



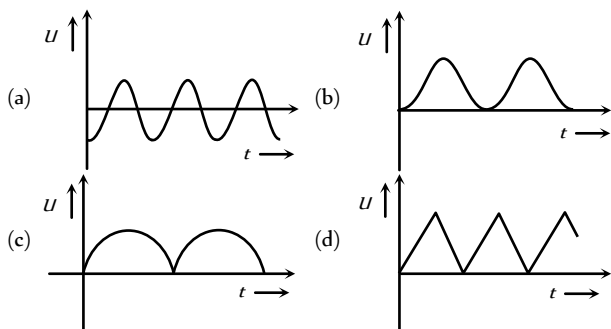
The corresponding force-time graph of the particle is

- (a) (b)
(c) (d) [DPMT 2005]

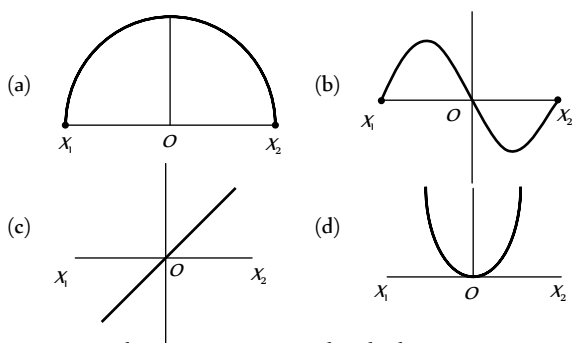
4. The graph shows the variation of displacement of a particle executing S.H.M. with time. We infer from this graph that



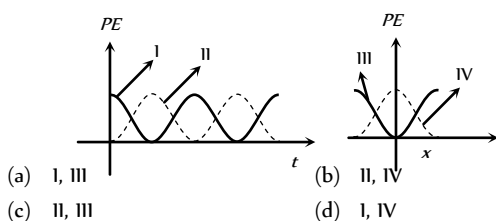
- (a) The force is zero at time $3T/4$
 (b) The velocity is maximum at time $T/2$
 (c) The acceleration is maximum at time T
 (d) The P.E. is equal to total energy at time $T/2$
5. As a body performs S.H.M., its potential energy U varies with time as indicated in [AMU (Med.) 2001]



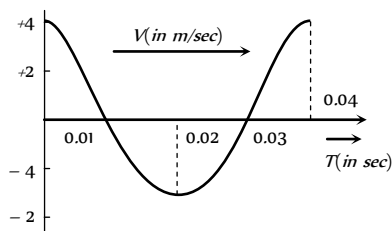
6. A particle of mass m oscillates with simple harmonic motion between points x_1 and x_2 , the equilibrium position being O . Its potential energy is plotted. It will be as given below in the graph



7. For a particle executing S.H.M. the displacement x is given by $x = A \cos \omega t$. Identify the graph which represents the variation of potential energy (P.E.) as a function of time t and displacement x

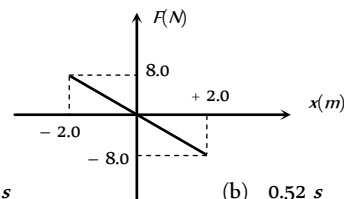


8. The velocity-time diagram of a harmonic oscillator is shown in the adjoining figure. The frequency of oscillation is [CPMT 1989]



- (a) 25 Hz (b) 50 Hz
 (c) 12.25 Hz (d) 33.3 Hz

9. A body of mass 0.01 kg executes simple harmonic motion (S.H.M.) about $x = 0$ under the influence of a force shown below : The period of the S.H.M. is [AMU (Med.) 2002]



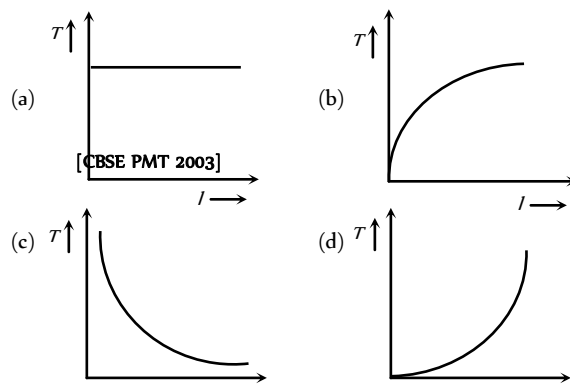
- (a) 1.05 s (b) 0.52 s
 (c) 0.25 s (d) 0.30 s

10. For a simple pendulum the graph between L and T will be.

[CPMT 1992]

- (a) Hyperbola (b) Parabola
 (c) A curved line (d) A straight line

11. In case of a simple pendulum, time period versus length is depicted by [DCE 1999, 2001]

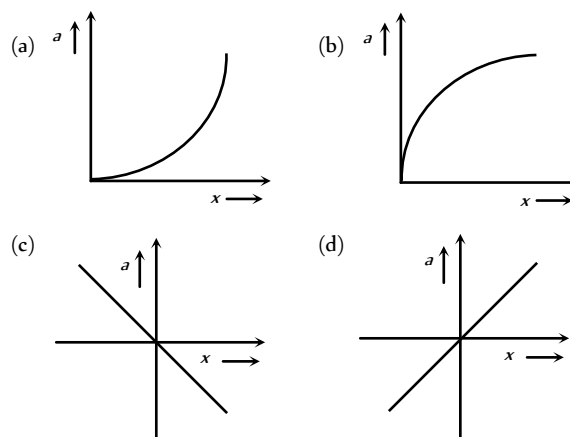


[CBSE PMT 2003]

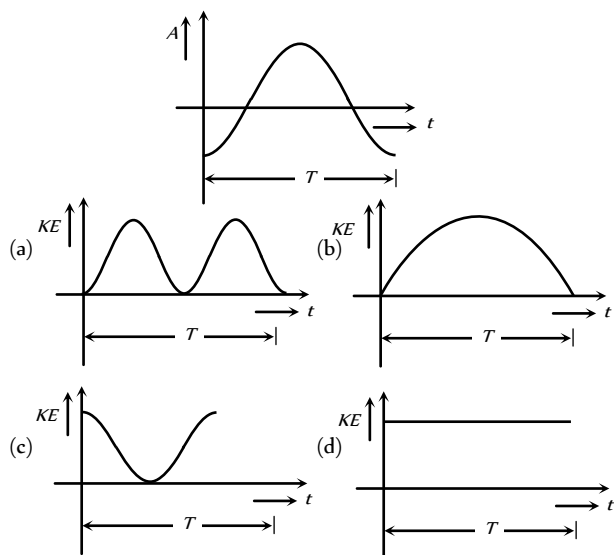
12. Graph between velocity and displacement of a particle, executing S.H.M. is [DPMT 2005]

- (a) A straight line (b) A parabola
 (c) A hyperbola (d) An ellipse

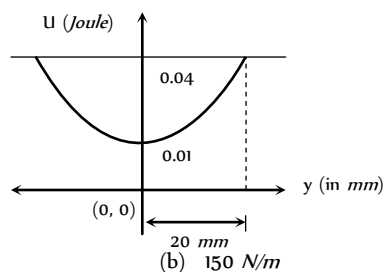
13. The variation of the acceleration a of the particle executing S.H.M. with displacement x is as shown in the figure [IIT-JEE (Screening) 2003]



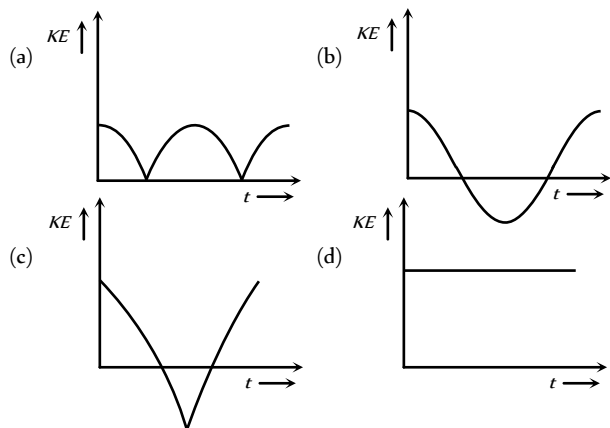
14. Acceleration A and time period T of a body in S.H.M. is given by a curve shown below. Then corresponding graph, between kinetic energy (K.E.) and time t is correctly represented by



15. The variation of potential energy of harmonic oscillator is as shown in figure. The spring constant is



- (a) $1 \times 10^5 \text{ N/m}$
(b) 150 N/m
(c) $0.667 \times 10^5 \text{ N/m}$
(d) $3 \times 10^5 \text{ N/m}$
16. A body performs S.H.M. Its kinetic energy K varies with time t as indicated by graph



Read the assertion and reason carefully to mark the correct option out of the options given below:

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
(c) If assertion is true but reason is false.
(d) If the assertion and reason both are false.
(e) If assertion is false but reason is true.

- Assertion : All oscillatory motions are necessarily periodic motion but all periodic motion are not oscillatory.
Reason : Simple pendulum is an example of oscillatory motion.
- Assertion : Simple harmonic motion is a uniform motion.
Reason : Simple harmonic motion is the projection of uniform circular motion.
- Assertion : Acceleration is proportional to the displacement. This condition is not sufficient for motion in simple harmonic.
Reason : In simple harmonic motion direction of displacement is also considered.
- Assertion : Sine and cosine functions are periodic functions.
Reason : Sinusoidal functions repeats it values after a definite interval of time.
- Assertion : The graph between velocity and displacement for a harmonic oscillator is a parabola.
Reason : Velocity does not change uniformly with displacement in harmonic motion.
- Assertion : When a simple pendulum is made to oscillate on the surface of moon, its time period increases.
Reason : Moon is much smaller as compared to earth.
- Assertion : Resonance is special case of forced vibration in which the natural frequency of vibration of the body is the same as the impressed frequency of external periodic force and the amplitude of forced vibration is maximum.
Reason : The amplitude of forced vibrations of a body increases with an increase in the frequency of the externally impressed periodic force.

[AIIMS 1994]

- Assertion : The graph of total energy of a particle in SHM w.r.t. position is a straight line with zero slope.
Reason : Total energy of particle in SHM remains constant throughout its motion.
- Assertion : The percentage change in time period is 1.5%, if the length of simple pendulum increases by 3%.
Reason : Time period is directly proportional to length of pendulum.
- Assertion : The frequency of a second pendulum in an elevator moving up with an acceleration half the acceleration due to gravity is 0.612 s.
Reason : The frequency of a second pendulum does not depend upon acceleration due to gravity.
- Assertion : Damped oscillation indicates loss of energy.
Reason : The energy loss in damped oscillation may be due to friction, air resistance etc.
- Assertion : In a S.H.M., kinetic and potential energies become equal when the displacement is $1/\sqrt{2}$ times the amplitude.

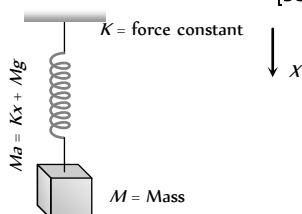


Assertion & Reason

For AIIMS Aspirants

- Reason : In SHM, kinetic energy is zero when potential energy is maximum.
13. Assertion : If the amplitude of a simple harmonic oscillator is doubled, its total energy becomes four times.
- Reason : The total energy is directly proportional to the square of amplitude of vibration of the harmonic oscillator.
14. Assertion : For an oscillating simple pendulum, the tension in the string is maximum at the mean position and minimum at the extreme position.
- Reason : The velocity of oscillating bob in simple harmonic motion is maximum at the mean position.
15. Assertion : The spring constant of a spring is k . When it is divided into n equal parts, then spring constant of one piece is k/n .
- Reason : The spring constant is independent of material used for the spring.
16. Assertion : The periodic time of a hard spring is less as compared to that of a soft spring.
- Reason : The periodic time depends upon the spring constant, and spring constant is large for hard spring.
17. Assertion : In extreme position of a particle executing S.H.M., both velocity and acceleration are zero.
- Reason : In S.H.M., acceleration always acts towards mean position.
18. Assertion : Soldiers are asked to break steps while crossing the bridge.
- Reason : The frequency of marching may be equal to the natural frequency of bridge and may lead to resonance which can break the bridge.
- [AIIMS 2001]
19. Assertion : The amplitude of oscillation can never be infinite.
- Reason : The energy of oscillator is continuously dissipated.
20. Assertion : In S.H.M., the motion is 'to and fro' and periodic.
- Reason : Velocity of the particle (v) = $\omega\sqrt{k^2 - x^2}$ (where x is the displacement and k is amplitude)
- [AIIMS 2002]
21. Assertion : The amplitude of an oscillating pendulum decreases gradually with time
- Reason : The frequency of the pendulum decreases with time [AIIMS 2003]
22. Assertion : In simple harmonic motion, the velocity is maximum when acceleration is minimum
- Reason : Displacement and velocity of S.H.M. differ in phase by $\pi/2$ [AIIMS 1999]
23. Assertion : Consider motion for a mass spring system under gravity, motion of M is not a simple harmonic motion unless Mg is negligibly small.
- Reason : For simple harmonic motion acceleration must be proportional to displacement and is directed towards the mean position

[SCRA 1994]



Answers

Displacement of S.H.M. and Phase

1	b,d	2	c	3	d	4	c	5	d
6	c	7	c	8	a	9	a,b,d	10	a
11	c	12	c	13	c	14	c	15	d
16	b	17	d	18	a	19	c	20	a
21	b	22	c	23	b	24	c	25	b
26	a								

Velocity of Simple Harmonic Motion

1	a	2	c	3	c	4	c	5	b
6	c	7	d	8	c	9	d	10	b
11	a	12	d	13	a	14	b	15	c
16	b	17	b	18	a	19	d	20	b
21	b	22	c	23	d	24	a	25	a
26	c	27	a						

Acceleration of Simple Harmonic Motion

1	d	2	c	3	c	4	d	5	a
6	a	7	a	8	d	9	d	10	d
11	a	12	a	13	d	14	a	15	a
16	d	17	d	18	d	19	b	20	c
21	c								

Energy of Simple Harmonic Motion

1	d	2	a	3	d	4	a	5	a
6	c	7	c	8	b	9	d	10	c
11	c	12	b	13	a	14	a	15	b
16	b	17	c	18	b	19	d	20	c
21	c	22	c	23	b	24	b	25	a
26	c	27	c	28	a	29	b	30	c
31	c	32	d	33	b	34	b		

Time Period and Frequency

1	b	2	c	3	b	4	d	5	b
6	a	7	d	8	d	9	d	10	a
11	b	12	b	13	b	14	b	15	a
16	d	17	d	18	d				

Simple Pendulum

1	c	2	a	3	b	4	b	5	b
6	b	7	c	8	c	9	c	10	d
11	d	12	b	13	a	14	d	15	d

16	b	17	b	18	c	19	c	20	c
21	d	22	d	23	c	24	c	25	d
26	a	27	a	28	b	29	d	30	d
31	c	32	c	33	b	34	b	35	a
36	a	37	d	38	b	39	c	40	d
41	a	42	a	43	a	44	b	45	d
46	d	47	b	48	a	49	a	50	a
51	c	52	c	53	c	54	d	55	c
56	b	57	c	58	a	59	c	60	a
61	b								

Spring Pendulum

1	d	2	d	3	b	4	b	5	b
6	c	7	c	8	b	9	a	10	c
11	a	12	b	13	d	14	c	15	a
16	d	17	a	18	d	19	b	20	c
21	c	22	c	23	d	24	a	25	d
26	d	27	b	28	a	29	a	30	a
31	b	32	d	33	b	34	b	35	b
36	d	37	c	38	d	39	b	40	c
41	a	42	b	43	b	44	a	45	b
46	a	47	b	48	d	49	c	50	c
51	b	52	d	53	d	54	a	55	b

Superposition of S.H.M's and Resonance

1	c	2	c	3	c	4	c	5	a
6	c	7	d	8	a	9	b	10	d
11	b	12	c	13	a	14	c		

Critical Thinking Questions

1	d	2	b	3	b,c	4	a	5	b,d
6	a	7	b	8	a	9	d	10	b
11	c	12	c	13	a	14	a	15	b,c
16	c	17	a	18	d	19	a	20	a
21	c	22	b	23	b	24	a	25	b
26	b	27	b	28	b	29	a,c	30	d
31	c	32	b	33	a	34	b	35	d
36	c	37	b	38	b	39	d	40	b
41	b	42	a	43	d				

Graphical Questions

1	a	2	d	3	d	4	d	5	b
6	d	7	a	8	a	9	d	10	b
11	b	12	d	13	c	14	a	15	b
16	a								

Assertion and Reason

1	b	2	e	3	a	4	a	5	e
6	b	7	c	8	a	9	c	10	c
11	b	12	b	13	a	14	b	15	e
16	a	17	e	18	a	19	a	20	b
21	c	22	b	23	e				

AS Answers and Solutions

Displacement of S.H.M. and Phase

1. (b,d) For S.H.M. displacement $y = a \sin \omega t$ and acceleration

$$A = -\omega^2 y \sin \omega t \quad \text{these are maximum at } \omega t = \frac{\pi}{2}.$$

2. (c) $v_{\max} = \omega A \Rightarrow v = \frac{\omega A}{2} = \omega \sqrt{A^2 - y^2}$

$$\Rightarrow A^2 - y^2 = \frac{A^2}{4} \Rightarrow y^2 = \frac{3A^2}{4} \Rightarrow y = \frac{\sqrt{3}A}{2}$$

3. (d) Equation of motion is $y = 5 \sin \frac{2\pi}{6} t$. For $y = 2.5 \text{ cm}$

$$2.5 = 5 \sin \frac{2\pi}{6} t \Rightarrow \frac{2\pi}{6} t = \frac{\pi}{6} \Rightarrow t = \frac{1}{2} \text{ sec}$$

$$\text{and phase} = \frac{2\pi}{6} = \frac{\pi}{6}.$$

4. (c) $y = a \sin(\omega t - \alpha) = a \cos\left(\omega t - \alpha - \frac{\pi}{2}\right)$

$$\text{Another equation is given } y = \cos(\omega t - \alpha)$$

$$\text{So, there exists a phase difference of } \frac{\pi}{2} = 90^\circ$$

5. (d) $y = a \sin(\omega t + \phi)$

$$= a \sin\left(\frac{2\pi}{T} t + \phi\right) \Rightarrow y = 0.5 \sin\left(\frac{2\pi}{0.4} t + \frac{\pi}{2}\right)$$

$$y = 0.5 \sin\left(5\pi t + \frac{\pi}{2}\right) = 0.5 \cos 5\pi t$$

6. (c) $y = a \sin(2\pi m t + \alpha)$. Its phase at time $t = 2\pi m t + \alpha$

7. (c) From given equation $\omega = \frac{2\pi}{T} = 0.5\pi \Rightarrow T = 4 \text{ sec}$

$$\text{Time taken from mean position to the maximum displacement} \\ = \frac{1}{4} T = 1 \text{ sec.}$$

8. (a) It is required to calculate the time from extreme position.
Hence, in this case equation for displacement of particle can be written as $x = a \sin\left(\omega t + \frac{\pi}{2}\right) = a \cos \omega t$

$$\Rightarrow \frac{a}{2} = a \cos \omega t \Rightarrow \omega t = \frac{\pi}{3} \Rightarrow \frac{2\pi}{T} \cdot t = \frac{\pi}{3} \Rightarrow t = \frac{T}{6}$$

9. (a,b,d) $x = a \sin \omega t \cos \omega t = \frac{a}{2} \sin 2\omega t$

10. (a) $v_{\max} = a\omega = a \times \frac{2\pi}{T} \Rightarrow a = \frac{v_{\max} \times T}{2\pi}$

$$a = \frac{1.00 \times 10^3 \times (1 \times 10^{-5})}{2\pi} = 1.59 \text{ mm}$$

11. (c)

12. (c)

13. (c) $y = a \sin \frac{2\pi}{T} t \Rightarrow \frac{a}{\sqrt{2}} = a \sin \frac{2\pi}{T} \cdot t$

$$\Rightarrow \sin \frac{2\pi}{T} t = \frac{1}{\sqrt{2}} = \sin \frac{\pi}{4} \Rightarrow \frac{2\pi}{T} t = \frac{\pi}{4} \Rightarrow t = \frac{T}{8}$$

14. (c)

15. (d) Standard equation of S.H.M. $\frac{d^2y}{dt^2} = -\omega^2 y$, is not satisfied by $y = a \tan \omega t$.

16. (b) $x = a \cos(\omega t + \theta)$ (i)

and $v = \frac{dx}{dt} = -a\omega \sin(\omega t + \theta)$ (ii)

Given at $t = 0$, $x = 1 \text{ cm}$ and $v = \pi$ and $\omega = \pi$

Putting these values in equation (i) and (ii) we will get

$$\sin \theta = \frac{-1}{a} \text{ and } \cos \theta = \frac{1}{a}$$

$$\Rightarrow \sin^2 \theta + \cos^2 \theta = \left(-\frac{1}{a}\right)^2 + \left(\frac{1}{a}\right)^2 \Rightarrow a = \sqrt{2} \text{ cm}$$

17. (d) $y = A \sin \omega t = \frac{A \sin 2\pi}{T} t \Rightarrow \frac{A}{2} = A \sin \frac{2\pi}{T} \Rightarrow t = \frac{T}{12}$

18. (a) The amplitude is a maximum displacement from the mean position.

19. (c) Equation of motion $y = a \cos \omega t$

$$\Rightarrow \frac{a}{2} = a \cos \omega t \Rightarrow \cos \omega t = \frac{1}{2} \Rightarrow \omega t = \frac{\pi}{3}$$

$$\Rightarrow \frac{2\pi}{T} = \frac{\pi}{3} \Rightarrow t = \frac{\frac{\pi}{3} \times T}{2\pi} = \frac{4}{3 \times 2} = \frac{2}{3} \text{ sec}$$

20. (a) Simple harmonic waves are set up in a string fixed at the, two ends.

21. (b)

22. (c) $v_1 = \frac{dy_1}{dt} = 0.1 \times 100\pi \cos\left(100\pi t + \frac{\pi}{3}\right)$

$$v_2 = \frac{dy_2}{dt} = -0.1\pi \sin \pi t = 0.1\pi \cos\left(\pi t + \frac{\pi}{2}\right)$$

Phase difference of velocity of first particle with respect to the velocity of 2nd particle at $t = 0$ is

$$\Delta \phi = \phi_1 - \phi_2 = \frac{\pi}{3} - \frac{\pi}{2} = -\frac{\pi}{6}$$

23. (b) $\frac{a_1}{a_2} = \frac{10}{25} = \frac{2}{5}$

24. (c) $y = a \sin \frac{2\pi}{T} t \Rightarrow \frac{a}{2} = a \sin \frac{2\pi}{3} \Rightarrow \frac{1}{2} = \sin \frac{2\pi}{3}$
 $\Rightarrow \sin \frac{2\pi}{3} = \sin \frac{\pi}{6} \Rightarrow \frac{2\pi}{3} = \frac{\pi}{6} \Rightarrow t = \frac{1}{4} \text{ sec}$

25. (b)

26. (a) $x = a \sin\left(\omega t + \frac{\pi}{6}\right)$ and $x' = a \cos \omega t = a \sin\left(\omega t + \frac{\pi}{2}\right)$
 $\therefore \Delta \phi = \left(\omega t + \frac{\pi}{2}\right) - \left(\omega t + \frac{\pi}{6}\right) = \frac{\pi}{3}$

Velocity of Simple Harmonic Motion

1. (a) Velocity of a particle executing S.H.M. is given by

$$v = \omega \sqrt{a^2 - x^2} = \frac{2\pi}{T} \sqrt{A^2 - \frac{A^2}{4}} = \frac{2\pi}{T} \sqrt{\frac{3A^2}{4}} = \frac{\pi A \sqrt{3}}{T}$$

2. (c) $v = \omega \sqrt{a^2 - y^2} = 2\sqrt{60^2 - 20^2} = 113 \text{ mm/s}$

3. (c) It is given $v_{\max} = 100 \text{ cm/sec}$, $a = 10 \text{ cm}$

$$\Rightarrow v_{\max} = a\omega \Rightarrow \omega = \frac{100}{10} = 10 \text{ rad/sec}$$

$$\text{Hence } v = \omega \sqrt{a^2 - y^2} \Rightarrow 50 = 10 \sqrt{(10)^2 - y^2}$$

$$\Rightarrow y = 5\sqrt{3} \text{ cm}$$

4. (c) At centre $v_{\max} = a\omega = a \cdot \frac{2\pi}{T} = \frac{0.2 \times 2\pi}{0.01} = 40\pi$

5. (b) $v_{\max} = a\omega = a \cdot \frac{2\pi}{T} = 3 \times \frac{2\pi}{6} = \pi \text{ cm/s}$

6. (c) $v_{\max} = \omega a = \frac{2\pi}{T} \times a \Rightarrow v_{\max} = \frac{2 \times \pi \times 2}{2} = 2\pi \text{ m/s}$

7. (d) $v_{\max} = a\omega = \frac{a \cdot 2\pi}{T} = \frac{2\pi a}{T}$

8. (c) $v = \omega \sqrt{a^2 - y^2} \Rightarrow 10 = \omega \sqrt{a^2 - (4)^2}$ and $8 = \omega \sqrt{a^2 - (5)^2}$

$$\text{On solving } \omega = 2 \Rightarrow \omega = \frac{2\pi}{T} = 2 \Rightarrow T = \pi \text{ sec}$$

9. (d) From the given equation, $a = 5$ and $\omega = 4$

$$\therefore v = \omega \sqrt{a^2 - y^2} = 4 \sqrt{(5)^2 - (3)^2} = 16$$

10. (b) $v_{\max} = a\omega = a \times \frac{2\pi}{T} = (50 \times 10^{-3}) \times \frac{2\pi}{2} = 0.15 \text{ m/s}$

11. (a) $n = \frac{\omega}{2\pi} = \frac{220}{2\pi} = 35 \text{ Hz}$
 $v_{\max} = \omega a = 220 \times 0.30 \text{ m/s} = 66 \text{ m/s}$

12. (d) $v_{\max} = a\omega$ and $A_{\max} = a\omega^2 \Rightarrow \omega = \frac{A_{\max}}{v_{\max}} = \frac{4}{2} = 2 \text{ rad/sec}$

13. (a) $v_{\max} = a\omega = \frac{a \times 2\pi}{T} = \frac{2 \times 10^{-3} \times 2\pi}{0.1} = \frac{\pi}{25} \text{ m/s}$

14. (b) $A = \omega^2 y \Rightarrow \omega = \sqrt{A/y} = \sqrt{\frac{8}{2}} = 2 \text{ rad/sec}$
 Now $v_{\max} = a\omega = 6 \times 2 = 12 \text{ cm/sec}$

15. (c) $v_{\max} = a\omega \Rightarrow \omega = \frac{v_{\max}}{a} = \frac{10}{4}$

$$\text{Now, } v = \omega \sqrt{a^2 - y^2} \Rightarrow v^2 = \omega^2 (a^2 - y^2) \Rightarrow y^2 = a^2 - \frac{v^2}{\omega^2}$$

$$\Rightarrow y = \sqrt{a^2 - \frac{v^2}{\omega^2}} = \sqrt{4^2 - \frac{5^2}{(10/4)^2}} = 2\sqrt{3} \text{ cm}$$

16. (b) The particles will meet at the mean position when P completes one oscillation and Q completes half an oscillation

$$\text{So } \frac{v_P}{v_Q} = \frac{a\omega_P}{a\omega_Q} = \frac{T_Q}{T_P} = \frac{6}{3} = \frac{2}{1}$$

17. (b) $\frac{v_{\max}}{A_{\max}} = \frac{a\omega}{a\omega^2} = \frac{1}{\omega}$

18. (a) Velocity is same. So by using $v = a\omega$

$$\Rightarrow A_1 \omega_1 = A_2 \omega_2 = A_3 \omega_3$$

19. (d) In S.H.M. at mean position velocity is maximum
So $v = a\omega$ (maximum)
20. (b)
21. (b)
22. (c) Acceleration $A = \omega^2 y \Rightarrow \omega = \sqrt{\frac{A}{y}} = \sqrt{\frac{0.5}{0.02}} = 5$
Maximum velocity $v_{\max} = a\omega = 0.1 \times 5 = 0.5$
23. (d) At mean position velocity is maximum
i.e., $v_{\max} = \omega a \Rightarrow \omega = \frac{v_{\max}}{a} = \frac{16}{4} = 4$
 $\therefore v = \omega \sqrt{a^2 - y^2} \Rightarrow 8\sqrt{3} = 4\sqrt{4^2 - y^2}$
 $\Rightarrow 192 = 16(16 - y^2) \Rightarrow 12 = 16 - y^2 \Rightarrow y = 2 \text{ cm.}$
24. (a) $v_{\max} = a\omega = 3 \times 100 = 300$
25. (a) $x = 3 \sin 2t + 4 \cos 2t$. From given equation
 $a_1 = 3, a_2 = 4$, and $\phi = \frac{\pi}{2}$
 $\therefore a = \sqrt{a_1^2 + a_2^2} = \sqrt{3^2 + 4^2} = 5 \Rightarrow v_{\max} = a\omega = 5 \times 2 = 10$
26. (c) Velocity in mean position $v = a\omega$, velocity at a distance of half amplitude.
 $v' = \omega \sqrt{a^2 - y^2} = \omega \sqrt{a^2 - \frac{a^2}{4}} = \sqrt{\frac{3}{2}} a\omega = \sqrt{\frac{3}{2}} v$
27. (a) $x = A \cos\left(\omega t + \frac{\pi}{4}\right)$ and $v = \frac{dx}{dt} = -A\omega \sin\left(\omega t + \frac{\pi}{4}\right)$
For maximum speed,
 $\sin\left(\omega t + \frac{\pi}{4}\right) = 1 \Rightarrow \omega t + \frac{\pi}{4} = \frac{\pi}{2}$ or $\omega t = \frac{\pi}{2} - \frac{\pi}{4} \Rightarrow t = \frac{\pi}{4\omega}$

Acceleration of Simple Harmonic Motion

1. (d) $F = -kx$
2. (c) The stone execute S.H.M. about centre of earth with time period $T = 2\pi \sqrt{\frac{R}{g}}$; where R = Radius of earth.
3. (c) Acceleration $= \omega^2 a$ at extreme position is maximum.
4. (d) $-a\omega^2$ when it is at one extreme point.
5. (a) Maximum acceleration $= a\omega^2 = a \times 4\pi^2 n^2$
 $= 0.01 \times 4 \times (\pi)^2 \times (60)^2 = 144\pi^2 \text{ m/sec}$
6. (a) Maximum acceleration
 $A_{\max} = a\omega^2 = \frac{a \times 4\pi^2}{T^2} = \frac{1 \times 4 \times (3.14)^2}{0.2 \times 0.2}$
 $F_{\max} = m \times A_{\max} = \frac{0.1 \times 4 \times (3.14)^2}{0.2 \times 0.2} = 98.596 \text{ N}$
7. (a) Maximum velocity $= a\omega = 16$
Maximum acceleration $= \omega^2 a = 24$

$$\Rightarrow a = \frac{(a\omega)^2}{\omega^2 a} = \frac{16 \times 16}{24} = \frac{32}{3} \text{ m}$$

8. (d) Acceleration $\propto$ - displacement, and direction of acceleration is always directed towards the equilibrium position.
9. (d) Maximum force $= m(a\omega^2) = ma\left(\frac{4\pi^2}{T^2}\right)$
 $= 0.5 \left(\frac{4\pi^2}{\pi^2 / 25}\right) \times 0.01 = 0.5 \text{ N}$
10. (d) $a_{\max} = \omega^2 a = \left(\frac{\pi}{4}\right)^2 a = 0.62 \text{ cm/sec}^2$ [$\because a=1$]
11. (a) For S.H.M. $F = -kx$.
 $\therefore$ Force = Mass $\times$ Acceleration $\propto -x$
 $\Rightarrow F = -Akx$; where A and k are positive constants.
12. (a) Velocity $v = a\omega = a \times 2\pi n$
 $= 0.06 \times 2\pi \times 15 = 5.65 \text{ m/s}$
Acceleration $A = \omega^2 a = 4\pi^2 n^2 a = 5.32 \times 10^2 \text{ m/s}^2$
13. (d) $A_{\max} = a\omega^2 \Rightarrow a = \frac{A_{\max}}{\omega^2} = \frac{7.5}{(3.5)^2} = 0.61 \text{ m}$
14. (a) $a = 10 \times 10^{-2} \text{ m}$ and $\omega = 10 \text{ rad/sec}$
 $A_{\max} = \omega^2 a = 10 \times 10^{-2} \times 10^2 = 10 \text{ m/sec}^2$
15. (a) $A_{\max} = \omega^2 a$
16. (d) $A_{\max} = 4\pi^2 n^2 a = 4\pi^2 \times (50)^2 \times 0.02 = 200\pi^2 \text{ m/s}^2$
17. (d) $A = -\omega^2 y$ at mean position $y = 0$
So acceleration is minimum (zero).
18. (d) In S.H.M. $v = \sqrt{a^2 - y^2}$ and $a = -\omega^2 y$ when $y = a$
 $\Rightarrow v_{\min} = 0$ and $a_{\max} = -\omega^2 a$
19. (b) Comparing given equation with standard equation,
 $y = a \sin(\omega t + \phi)$, we get, $a = 2 \text{ cm}$, $\omega = \frac{\pi}{2}$
 $\therefore A_{\max} = \omega^2 A = \left(\frac{\pi}{2}\right)^2 \times 2 = \frac{\pi^2}{2} \text{ cm/s}^2$.
20. (c) Velocity $v = \omega \sqrt{A^2 - x^2}$ and acceleration $= \omega^2 x$
Now given, $\omega^2 x = \omega \sqrt{A^2 - x^2} \Rightarrow \omega^2 \cdot 1 = \omega \sqrt{2^2 - 1^2}$
 $\Rightarrow \omega = \sqrt{3} \therefore T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{3}}$
21. (c) $a = -\omega^2 x \Rightarrow \left|\frac{a}{x}\right| = \omega^2$

Energy of Simple Harmonic Motion

1. (d) $E = \frac{1}{2} m \omega^2 a^2 \Rightarrow E \propto a^2$

2. (a) $P.E. = \frac{1}{2} m \omega^2 x^2$
It is clear P.E. will be maximum when x will be maximum i.e., at $x = \pm A$
3. (d) Let x be the point where $K.E. = P.E.$
Hence $\frac{1}{2} m \omega^2 (a^2 - x^2) = \frac{1}{2} m \omega^2 x^2$
 $\Rightarrow 2x^2 = a^2 \Rightarrow x = \frac{a}{\sqrt{2}} = \frac{4}{\sqrt{2}} = 2\sqrt{2} \text{ cm}$
4. (a) Since maximum value of $\cos^2 \omega t$ is 1.
 $\therefore K_{\max} = K_0 \cos^2 \omega t = K_0$
Also $K_{\max} = P.E._{\max} = K_0$
5. (a) $F = -kx \Rightarrow dW = Fdx = -kxdx$
So $\int_0^W dW = \int_0^x -kx dx \Rightarrow W = U = -\frac{1}{2} kx^2$
6. (c) Suppose at displacement y from mean position potential energy = kinetic energy
 $\Rightarrow \frac{1}{2} m(a^2 - y^2)\omega^2 = \frac{1}{2} m\omega^2 y^2$
 $\Rightarrow a^2 = 2y^2 \Rightarrow y = \frac{a}{\sqrt{2}}$
7. (c) Total energy in SHM $E = \frac{1}{2} m \omega^2 a^2$; (where a = amplitude)
Potential energy $U = \frac{1}{2} m \omega^2 (a^2 - y^2) = E - \frac{1}{2} m \omega^2 y^2$
When $y = \frac{a}{2} \Rightarrow U = E - \frac{1}{2} m \omega^2 \left(\frac{a^2}{4}\right) = E - \frac{E}{4} = \frac{3E}{4}$
8. (b) $\frac{\text{Potential energy } (U)}{\text{Total energy } (E)} = \frac{\frac{1}{2} m \omega^2 y^2}{\frac{1}{2} m \omega^2 a^2} = \frac{y^2}{a^2}$
So $\frac{2.5}{E} = \frac{\left(\frac{a}{2}\right)^2}{a^2} \Rightarrow E = 10J$
9. (d) Kinetic energy $T = \frac{1}{2} m \omega^2 (a^2 - x^2)$
and potential energy, $V = \frac{1}{2} m \omega^2 x^2 \therefore \frac{T}{V} = \frac{a^2 - x^2}{x^2}$
10. (c) $\frac{U}{U_{\max}} = \frac{\frac{1}{2} m \omega^2 y^2}{\frac{1}{2} m \omega^2 a^2} \Rightarrow \frac{1}{4} = \frac{y^2}{a^2} \Rightarrow y = \frac{a}{2}$
11. (c) Kinetic energy $K = \frac{1}{2} m \omega^2 (a^2 - y^2)$
 $= \frac{1}{2} \times 10 \times \left(\frac{2\pi}{2}\right)^2 [10^2 - 5^2] = 375 \pi^2 \text{ ergs}$
12. (b) $\frac{U}{E} = \frac{\frac{1}{2} m \omega^2 y^2}{\frac{1}{2} m \omega^2 a^2} = \frac{y^2}{a^2} = \frac{\left(\frac{a}{2}\right)^2}{a^2} = \frac{1}{4}$
13. (a)
14. (a) The time period of potential energy and kinetic energy is half that of SHM.
15. (b) If at any instant displacement is y then it is given that
 $U = \frac{1}{2} \times E \Rightarrow \frac{1}{2} m \omega^2 y^2 = \frac{1}{2} \times \left(\frac{1}{2} m \omega^2 a^2\right)$
 $\Rightarrow y = \frac{a}{\sqrt{2}} = \frac{6}{\sqrt{2}} = 4.2 \text{ cm}$
16. (b) So $a = 6 \text{ cm}$, $\omega = 100 \text{ rad/sec}$
 $K_{\max} = \frac{1}{2} m \omega^2 a^2 = \frac{1}{2} \times 1 \times (100)^2 \times (6 \times 10^{-2})^2 = 18 \text{ J}$
17. (c) In S.H.M., frequency of K.E. and P.E.
 $= 2 \times (\text{Frequency of oscillating particle})$
18. (b) Total energy $U = \frac{1}{2} Ka^2$
19. (d) $\frac{U}{E} = \frac{\frac{1}{2} m \omega^2 y^2}{\frac{1}{2} m \omega^2 a^2} = \frac{y^2}{a^2} \Rightarrow \frac{U}{80} = \frac{\left(\frac{3}{4}a\right)^2}{a^2} = \frac{9}{16} \Rightarrow U = 45 \text{ J}$
20. (c)
21. (c) Maximum potential energy position is $y = \pm a$
and maximum kinetic energy position is $y = 0$
22. (c) $Mg = Kl \Rightarrow U_{\max} = \frac{1}{2} Kl^2 = \frac{1}{2} mgl$
23. (b) $\frac{U}{E} = \frac{\frac{1}{2} m \omega^2 y^2}{\frac{1}{2} m \omega^2 a^2} = \frac{y^2}{a^2} = \frac{\left(\frac{a}{2}\right)^2}{a^2} = \frac{1}{4} \Rightarrow U = \frac{E}{4}$
24. (b) In S.H.M., at mean position i.e. at $x = 0$ kinetic energy will be maximum and $P.E.$ will be minimum. Total energy is always constant.
25. (a) In SHM for a complete cycle average value of kinetic energy and potential energy are equal i.e. $\langle E \rangle = \langle U \rangle$
 $= \frac{1}{4} m \omega^2 a^2$
26. (c) Total energy $= \frac{1}{2} m \omega^2 a^2 = \text{constant}$
27. (c) Kinetic energy at mean position,
 $K_{\max} = \frac{1}{2} m v_{\max}^2 \Rightarrow v_{\max} = \sqrt{\frac{2K_{\max}}{m}}$
 $= \sqrt{\frac{2 \times 16}{0.32}} = \sqrt{100} = 10 \text{ m/s.}$
28. (a) $E = \frac{1}{2} m a^2 \omega^2 = \frac{1}{2} m a^2 \left(\frac{4\pi^2}{T^2}\right) \Rightarrow E \propto \frac{a^2}{T^2}$
29. (b) $\frac{K}{E} = \frac{\frac{1}{2} m \omega^2 (a^2 - y^2)}{\frac{1}{2} m \omega^2 a^2} = \frac{a^2 - y^2}{a^2} = 1 - \frac{y^2}{a^2}$

$$\text{So, } \frac{\left(\frac{3E}{4}\right)}{E} = 1 - \frac{y^2}{a^2} \Rightarrow \frac{y^2}{a^2} = 1 - \frac{3}{4} = \frac{1}{4} \Rightarrow y = \frac{a}{2}.$$

30. (c) Kinetic energy $K = \frac{1}{2}mv^2 = \frac{1}{2}ma^2\omega^2 \cos^2 \omega t$
 $= \frac{1}{2}m\omega^2 a^2 (1 + \cos 2\omega t)$ hence kinetic energy varies periodically with double the frequency of S.H.M. i.e. 2ω .
31. (c) $E = \frac{1}{2}m\omega^2 a^2 \Rightarrow \frac{E'}{E} = \frac{a'^2}{a^2} \Rightarrow \frac{E'}{E} = \left(\frac{3a}{4a}\right)^2 \left(\because a' = \frac{3}{4}a\right)$
 $\Rightarrow E' = \frac{9}{16}E$
32. (d) In simple harmonic motion, energy changes from kinetic to potential and potential to kinetic but the sum of two always remains constant.
33. (b) Body collides elastically with walls of room. So, there will be no loss in its energy and it will remain colliding with walls of room, so its motion will be periodic.
 There is no change in energy of the body, hence there is no acceleration, so its motion is not SHM.
34. (b) $E_1 = \frac{1}{2}Kx^2 \Rightarrow x = \sqrt{\frac{2E_1}{K}}$, $E_2 = \frac{1}{2}Ky^2 \Rightarrow y = \sqrt{\frac{2E_2}{K}}$
 and $E = \frac{1}{2}K(x+y)^2 \Rightarrow x+y = \sqrt{\frac{2E}{K}}$
 $\Rightarrow \sqrt{\frac{2E_1}{K}} + \sqrt{\frac{2E_2}{K}} = \sqrt{\frac{2E}{K}} \Rightarrow \sqrt{E_1} + \sqrt{E_2} = \sqrt{E}$

Time Period and Frequency

1. (b) In the given case, $\frac{\text{Displacement}}{\text{Acceleration}} = \frac{1}{b}$
 $\therefore$ Time period $T = 2\pi\sqrt{\frac{\text{Displacement}}{\text{Acceleration}}} = \frac{2\pi}{\sqrt{b}}$
2. (c) On comparing with standard equation $\frac{d^2y}{dt^2} + \omega^2 y = 0$ we get $\omega^2 = K \Rightarrow \omega = \frac{2\pi}{T} = \sqrt{K} \Rightarrow T = \frac{2\pi}{\sqrt{K}}$.
3. (b) Ball execute S.H.M. inside the tunnel with time period
 $T = 2\pi\sqrt{\frac{R}{g}} = 84.63 \text{ min}$
 Hence time to reach the ball from one end to the other end of the tunnel $t = \frac{84.63}{2} = 42.3 \text{ min}$.
4. (d) Given max velocity $\omega a = 1$ and maximum acceleration $\omega^2 a = 1.57$
 $\therefore \frac{\omega^2 a}{\omega a} = 1.57 \Rightarrow \omega = 1.57 \Rightarrow \frac{2\pi}{T} = 1.57 \Rightarrow T = 4$
5. (b) $\omega = \frac{2\pi}{T} = 100\pi \Rightarrow T = 0.02 \text{ sec}$
6. (a) At mean position, the kinetic energy is maximum.

$$\text{Hence } \frac{1}{2}ma^2\omega^2 = 16$$

$$\text{On putting the values we get } \omega = 10 \Rightarrow T = \frac{2\pi}{\omega} = \frac{\pi}{5} \text{ sec}$$

7. (d) $T = 2\pi\sqrt{\frac{\text{Displacement}}{\text{Acceleration}}} = 2\pi\sqrt{\frac{3}{12}} = \pi = 3.14 \text{ sec}$
8. (d) $\omega = \sqrt{\frac{k}{m}} \Rightarrow \frac{\omega_2}{\omega_1} = \sqrt{\frac{m_1}{m_2}} \Rightarrow 2 = \sqrt{\frac{m_1}{m_2}} \Rightarrow m_2 = \frac{m_1}{4}$
9. (d)
10. (a) $\omega = \sqrt{\frac{\text{Acceleration}}{\text{Displacement}}} = \sqrt{\frac{2.0}{0.02}} = 10 \text{ rad s}^{-1}$
11. (b) From given equation $\omega = 3000$, $\Rightarrow n = \frac{\omega}{2\pi} = \frac{3000}{2\pi}$
12. (b)
13. (b) Given, $v = \pi \text{ cm/sec}$, $x = 1 \text{ cm}$ and $\omega = \pi \text{ s}^{-1}$
 using $v = \omega\sqrt{a^2 - x^2} \Rightarrow \pi = \pi\sqrt{a^2 - 1}$
 $\Rightarrow 1 = a^2 - 1 \Rightarrow a = \sqrt{2} \text{ cm}$.
14. (b) Length of the line = Distance between extreme positions of oscillation = 4 cm
 So, Amplitude $a = 2 \text{ cm}$.
 also $v_{\max} = 12 \text{ cm/s}$.
 $\therefore v_{\max} = \omega a = \frac{2\pi}{T} a$
 $\Rightarrow T = \frac{2\pi a}{v_{\max}} = \frac{2 \times 3.14 \times 2}{12} = 1.047 \text{ sec}$
15. (a) Comparing given equation with standard equation, $x = a \cos(\omega t + \phi)$ we get, $a = 0.01$ and $\omega = \pi$
 $\Rightarrow 2\pi n = \pi \Rightarrow n = 0.5 \text{ Hz}$
16. (d) $y = 5 \sin(\pi t + 4\pi)$, comparing it with standard equation
 $y = a \sin(\omega t + \phi) = a \sin\left(\frac{2\pi t}{T} + \phi\right)$
 $a = 5 \text{ m}$ and $\frac{2\pi t}{T} = \pi t \Rightarrow T = 2 \text{ sec}$.
17. (d) $v_{\max} = a\omega = a \times 2\pi n \Rightarrow n = \frac{v_{\max}}{2\pi a} = \frac{31.4}{2 \times 3.14 \times 5} = 1 \text{ Hz}$
18. (d) From the given equation $\omega = 2\pi n = 4\pi \Rightarrow n = 2 \text{ Hz}$

Simple Pendulum

1. (c) $T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow T \propto \sqrt{l}$
2. (a) Inside the mine g decreases
 hence from $T = 2\pi\sqrt{\frac{l}{g}}$; T increase

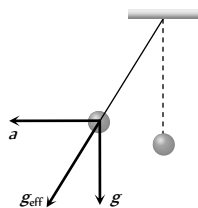
3. (b) When a little mercury is drained off, the position of c.g. of ball falls (w.r.t. fixed end) so that effective length of pendulum increases hence T increases.

4. (b) Initially time period was $T = 2\pi\sqrt{\frac{l}{g}}$.

When train accelerates, the effective value of g becomes

$\sqrt{g^2 + a^2}$ which is greater than g

Hence, new time period, becomes less than the initial time period.



5. (b) As we know $g = \frac{GM}{R^2}$

$$\Rightarrow \frac{g_{\text{earth}}}{g_{\text{planet}}} = \frac{M_e}{M_p} \times \frac{R_p^2}{R_e^2} \Rightarrow \frac{g_e}{g_p} = \frac{2}{1}$$

$$\text{Also } T \propto \frac{1}{\sqrt{g}} \Rightarrow \frac{T_e}{T_p} = \sqrt{\frac{g_p}{g_e}} \Rightarrow \frac{2}{T_p} = \sqrt{\frac{1}{2}}$$

$$\Rightarrow T_p = 2\sqrt{2} \text{ sec.}$$

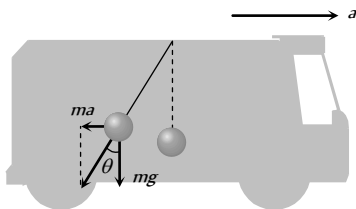
6. (b) In accelerated frame of reference, a fictitious force (pseudo force) ma acts on the bob of pendulum as shown in figure.

Hence,

$$\tan \theta = \frac{ma}{mg} = \frac{a}{g}$$

$\Rightarrow$

$$\theta = \tan^{-1}\left(\frac{a}{g}\right) \text{ in the backward direction.}$$



7. (c)

8. (c) $T = 2\pi\sqrt{\frac{l}{g}}$ (Independent of mass)

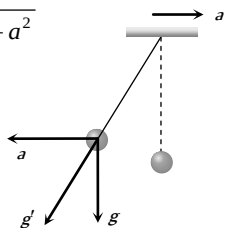
9. (c) In stationary lift $T = 2\pi\sqrt{\frac{l}{g}}$

$$\text{In upward moving lift } T' = 2\pi\sqrt{\frac{l}{g+a}}$$

(a = Acceleration of lift)

$$\Rightarrow \frac{T'}{T} = \sqrt{\frac{g}{g+a}} = \sqrt{\frac{g}{g+\frac{g}{4}}} = \sqrt{\frac{4}{5}} \Rightarrow T' = \frac{2T}{\sqrt{5}}$$

10. (d) $g' = \sqrt{g^2 + a^2}$



11. (d) In the given case effective acceleration $g' = 0 \Rightarrow T = \infty$

12. (b) $p_{\text{max}} = \sqrt{2mE_{\text{max}}}$

13. (a) $T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow \sqrt{\frac{l}{g}} = \text{constant}$

$$\Rightarrow l \propto g; \Rightarrow \frac{l_m}{1} = \frac{1}{6} \frac{g}{g} \Rightarrow l_m = \frac{1}{6} m$$

14. (d) $T \propto \sqrt{l} \Rightarrow \frac{\Delta T}{T} = \frac{1}{2} \frac{\Delta l}{l} = \frac{0.02}{2} = 0.01 \Rightarrow \Delta T = 0.01 T$

$$\text{Loss of time per day} = 0.01 \times 24 \times 60 \times 60 = 864 \text{ sec}$$

15. (d) $\frac{T'}{T} = \sqrt{\frac{g}{g'+a}} = \sqrt{\frac{g}{g+5g}} = \sqrt{\frac{1}{6}} \Rightarrow T' = \frac{T}{\sqrt{6}}$

16. (b) $T \propto \sqrt{l} \Rightarrow \frac{\Delta T}{T} = \frac{1}{2} \frac{\Delta l}{l} = \frac{1}{2} \times 1\% = 0.5\%$

17. (b) At B , the velocity is maximum using conservation of mechanical energy

$$\Delta PE = \Delta KE \Rightarrow mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gH}$$

18. (c)

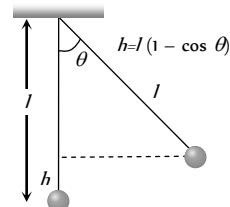
19. (c) If suppose bob rises up to a height h as shown then after releasing potential energy at extreme position becomes kinetic energy of mean position

$$\Rightarrow mgh = \frac{1}{2}mv_{\text{max}}^2 \Rightarrow v_{\text{max}} = \sqrt{2gh}$$

$$\text{Also, from figure } \cos \theta = \frac{l-h}{l}$$

$$\Rightarrow h = l(1 - \cos \theta)$$

$$\text{So, } v_{\text{max}} = \sqrt{2gl(1 - \cos \theta)}$$

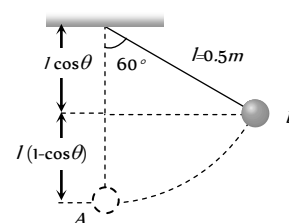


20. (c) $T = 2\pi\sqrt{\frac{l}{g}}$; for freely falling system effective $g = 0$

$$\text{so } T = \infty \text{ or } n = 0$$

It means that pendulum does not oscillate at all.

21. (d) Let bob velocity be v at point B where it makes an angle of 60° with the vertical, then using conservation of mechanical energy



$$KE_A + PE_A = KE_B + PE_B \quad v = 3 \text{ m/sec}$$

$$\Rightarrow \frac{1}{2}m \times 3^2 = \frac{1}{2}mv^2 + mgl(1 - \cos \theta)$$

$$\Rightarrow 9 = v^2 + 2 \times 10 \times 0.5 \times \frac{1}{2} \Rightarrow v = 2 \text{ m/s}$$

22. (d) $T \propto \sqrt{l} \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{l_1}{l_2}} \Rightarrow \frac{2}{T_2} = \sqrt{\frac{l}{4l}} \Rightarrow T_2 = 4 \text{ sec}$

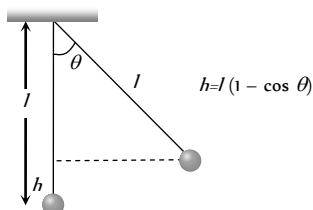
23. (c) Remains the same because time period of simple pendulum T is independent of mass of the bob

24. (c) $T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow \frac{l}{T^2} = \frac{g}{4\pi^2} = \text{constant}$

25. (d) $T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow T \propto \frac{1}{\sqrt{g}} \Rightarrow \frac{T_p}{T_e} = \sqrt{\frac{g_e}{g_p}} = \sqrt{\frac{2}{1}} \Rightarrow T' = \sqrt{2}T$
26. (a) If initial length $l_1 = 100$ then $l_2 = 121$
By using $T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{l_1}{l_2}}$
Hence, $\frac{T_1}{T_2} = \sqrt{\frac{100}{121}} \Rightarrow T_2 = 1.1 T_1$
% increase = $\frac{T_2 - T_1}{T_1} \times 100 = 10\%$
27. (a) $\frac{T_1}{T_2} = \sqrt{\frac{l_1}{l_2}} \Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{100}{400}}$ (If $l_1 = 100$ then $l_2 = 400$)
 $\Rightarrow T_2 = 2T_1$
Hence % increase = $\frac{T_2 - T_1}{T_1} \times 100 = 100\%$
28. (b) $T = 2\pi\sqrt{l/g} \Rightarrow l = \frac{gT^2}{4\pi^2} = \frac{9.8 \times 4}{4 \times \pi^2} = 99 \text{ cm}$
29. (d) This is the case of freely falling lift and in free fall of lift effective g for pendulum will be zero. So $T = 2\pi\sqrt{\frac{l}{0}} = \infty$
30. (d) After standing centre of mass of the oscillating body will shift upward therefore effective length will decrease and by $T \propto \sqrt{l}$, time period will decrease.
31. (c) $T = 2\pi\sqrt{l/g} = 2\pi\sqrt{\frac{1}{\pi^2}} = 2 \text{ sec}$
32. (c) Time period is independent of mass of pendulum.
33. (b) $T \propto \sqrt{l}$ Time period depends only on effective length. Density has no effect on time period. If length made 4 times then time period becomes 2 times.
34. (b) Time period is independent of mass of bob of pendulum.
35. (a) At the surface of moon, g decreases hence time period increases $\left(\text{as } T \propto \frac{1}{\sqrt{g}} \right)$
36. (a) When lift falls freely effective acceleration and frequency of oscillations be zero
 $g_{\text{eff}} = 0 \Rightarrow T' = \infty$, hence a frequency = 0.
37. (d) Effective value of ' g ' remains unchanged.
38. (b) $T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow \frac{T_2}{T_1} = \sqrt{\frac{l_2}{l_1}} = \sqrt{\frac{144}{100}} = \frac{12}{10}$
 $\Rightarrow T_2 = 1.2 T_1$
Hence % increase = $\frac{T_2 - T_1}{T_1} \times 100 = 20\%$
39. (c) If amplitude is large motion will not remain simple harmonic.
40. (d) Minimum velocity is zero at the extreme positions.
41. (a) At the time $t = \frac{T}{4} = \frac{4}{4} = 1 \text{ sec}$ after passing from mean position, the body reaches at it's extreme position. At extreme, position velocity of body becomes zero.
42. (a) No momentum will be transferred because, at extreme position the velocity of bob is zero.
43. (a) In this case frequency of oscillation is given by $n = \frac{1}{2\pi} \sqrt{\frac{g^2 + a^2}{l}}$ where a is the acceleration of car. If a increases then n also increases.
44. (b) As periodic time is independent of amplitude.
45. (d) Frequency $n \propto \frac{1}{\sqrt{l}} \Rightarrow \frac{n_1}{n_2} = \sqrt{\frac{l_2}{l_1}} \Rightarrow \frac{l_1}{l_2} = \frac{n_2^2}{n_1^2} = \frac{3^2}{2^2} = \frac{9}{4}$
46. (d) Suppose at $t = 0$, pendulums begins to swing simultaneously. Hence, they will again swing simultaneously if $n_1 T_1 = n_2 T_2$
 $\Rightarrow \frac{n_1}{n_2} = \frac{T_2}{T_1} = \sqrt{\frac{l_2}{l_1}} \Rightarrow \frac{l_1}{l_2} = \left(\frac{n_2}{n_1} \right)^2 = \left(\frac{8}{7} \right)^2 = \frac{64}{49}$
47. (b) $T \propto \frac{1}{\sqrt{g}}$ and g is same in both cases so time period remain same.
48. (a) $T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow T \propto \sqrt{l}$, hence if l made 9 times T becomes 3 times.
Also time period of simple pendulum does not depends on the mass of the bob.
49. (a) As we go from equator to pole the value of g increases. Therefore time period of simple pendulum $\left(T \propto \frac{1}{\sqrt{g}} \right)$ decreases. $\left(\because T \propto \frac{1}{\sqrt{g}} \right)$
50. (a) If v is velocity of pendulum at Q and 10% energy is lost while moving from P to Q
Hence, by applying conservation of between P and Q
 $\frac{1}{2}mv^2 = 0.9(mgh) \Rightarrow v^2 = 2 \times 0.9 \times 10 \times 2 \Rightarrow v = 6 \text{ m/sec}$
51. (c) $T \propto \frac{1}{\sqrt{g}} \Rightarrow \frac{T_2}{T_1} = \sqrt{\frac{g_1}{g_2}} = \sqrt{\left(\frac{g}{g/4} \right)} \Rightarrow T_2 = 2T_1 = 2T$
52. (c) For stationary lift $T_1 = 2\pi\sqrt{\frac{l}{g}}$
For ascending lift with acceleration a , $T_2 = 2\pi\sqrt{\frac{l}{g+a}}$
 $\Rightarrow \frac{T_1}{T_2} = \sqrt{\frac{g+a}{g}} \Rightarrow \frac{T}{T_2} = \sqrt{\frac{g+\frac{g}{3}}{g}} = \sqrt{\frac{4}{3}} \Rightarrow T_2 = \frac{\sqrt{3}}{2} T$
53. (c) $T \propto \sqrt{l}$

54. (d) Kinetic energy will be maximum at mean position.

From law of conservation of energy maximum kinetic energy at mean position = Potential energy at displaced position



$\Rightarrow$

$$K_{\max} = mgh = mgl(1 - \cos \theta)$$

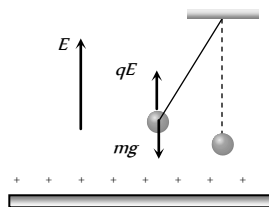
55. (c)
56. (b) As it is clear that in vacuum, the bob will not experience any frictional force. Hence, there shall be no dissipation therefore, it will oscillate with constant amplitude.
57. (c) The effective acceleration in a lift descending with acceleration $\frac{g}{3}$ is $g_{\text{eff}} = g - \frac{g}{3} = \frac{2g}{3}$

$$\therefore T = 2\pi \sqrt{\frac{L}{g_{\text{eff}}}} = 2\pi \sqrt{\frac{L}{2g/3}} = 2\pi \sqrt{\frac{3L}{2g}}$$

58. (a)
59. (c) According to the principle of conservation of energy, $\frac{1}{2}mv^2 = mgh$ or $v = \sqrt{2gh} = \sqrt{2 \times 9.8 \times 0.1} = 1.4 \text{ m/s}$.
60. (a) In this case time period of pendulum becomes

$$T' = 2\pi \sqrt{\frac{l}{g + \frac{qE}{m}}}$$

$$\Rightarrow T' < T$$



61. (b) In deep mine $g' = g \left(1 - \frac{d}{R}\right)$; i.e., g decreases so according to $n \propto \sqrt{g}$, frequency also decreases.

Spring Pendulum

1. (d) Maximum velocity $= a\omega = a\sqrt{\frac{k}{m}}$
- Given that $a_1\sqrt{\frac{K_1}{m}} = a_2\sqrt{\frac{K_2}{m}} \Rightarrow \frac{a_1}{a_2} = \sqrt{\frac{K_2}{K_1}}$
2. (d) Given spring system has parallel combination, so $k_{\text{eq}} = k_1 + k_2$ and time period $T = 2\pi \sqrt{\frac{m}{k_1 + k_2}}$
3. (b) $T = 2\pi \sqrt{\frac{m}{k}}$. Also spring constant $(k) \propto \frac{1}{\text{Length}(l)}$, when the spring is half in length, then k becomes twice.
- $$\therefore T' = 2\pi \sqrt{\frac{m}{2k}} \Rightarrow \frac{T'}{T} = \frac{1}{\sqrt{2}} \Rightarrow T' = \frac{T}{\sqrt{2}}$$

4. (b) $\omega = \sqrt{\frac{k}{m}}$

5. (b) With respect to the block the springs are connected in parallel combination.

$$\therefore \text{Combined stiffness } k = k_1 + k_2 \text{ and } n = \frac{1}{2\pi} \sqrt{\frac{k_1 + k_2}{m}}$$

6. (c) $n = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \Rightarrow \frac{n_s}{n_p} = \sqrt{\frac{k_s}{k_p}} \Rightarrow \frac{n_s}{n_p} = \sqrt{\frac{\left(\frac{k}{2}\right)}{2k}} = \frac{1}{2}$

7. (c) In series $k_{\text{eq}} = \frac{k_1 k_2}{k_1 + k_2}$ so time period $T = 2\pi \sqrt{\frac{m(k_1 + k_2)}{k_1 k_2}}$

8. (b) Force constant $k = \frac{F}{x} = \frac{0.5 \times 10}{0.2} = 25 \text{ N/m}$

$$\text{Now } T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{0.25}{25}} = 0.628 \text{ sec}$$

9. (a) $T = 2\pi \sqrt{\frac{m}{k}} \Rightarrow m \propto T^2 \Rightarrow \frac{m_2}{m_1} = \frac{T_2^2}{T_1^2}$

$$\Rightarrow \frac{M+m}{M} = \left(\frac{5}{4}\right)^2 \Rightarrow \frac{m}{M} = \frac{9}{16}$$

10. (c) Spring constant $(k) \propto \frac{1}{\text{Length of the spring}(l)}$
as length becomes half, k becomes twice is $2k$.

11. (a) $n = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \Rightarrow \frac{n}{n'} = \sqrt{\frac{k}{m} \times \frac{m'}{K'}} = \sqrt{\frac{k}{m} \times \frac{2m}{2K}} = 1 \Rightarrow n' = n$

12. (b) As mg produces extension x , hence $k = \frac{mg}{x}$

$$\therefore T = 2\pi \sqrt{\frac{(M+m)}{k}} = 2\pi \sqrt{\frac{(M+m)x}{mg}}$$

13. (d) For the given figure $f = \frac{1}{2\pi} \sqrt{\frac{k_{\text{eq}}}{m}} = \frac{1}{2\pi} \sqrt{\frac{2k}{m}}$ (i)

If one spring is removed, then $k = k$ and

$$f' = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \text{(ii)}$$

$$\text{From equation (i) and (ii), } \frac{f}{f'} = \sqrt{2} \Rightarrow f' = \frac{f}{\sqrt{2}}$$

14. (c) $\because mg = kx \Rightarrow \frac{m}{k} = \frac{x}{g} \Rightarrow T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{x}{g}}$

$$= 2\pi \sqrt{\frac{9.8 \times 10^{-2}}{9.8}} = \frac{2\pi}{10} \text{ sec}$$

15. (a) $T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{0.2}{80}} = 0.31 \text{ sec}$

16. (d) Spring is cut into two equal halves so spring constant of each part = $2k$

These parts are in parallel so $K_{eq} = 2K + 2K = 4K$

Extension force (i.e. W) is same hence by using $F = kx \Rightarrow$

$$4k \times x' = kx \Rightarrow x' = \frac{x}{4}$$

17. (a) In this case springs are in parallel, so $k_{eq} = k_1 + k_2$

$$\text{and } \omega = \sqrt{\frac{k_{eq}}{m}} = \sqrt{\frac{k_1 + k_2}{m}}$$

18. (d) Force constant $(k) \propto \frac{1}{\text{Length of the spring}(l)}$

$$\Rightarrow \frac{k_1}{k_2} = \frac{l_2}{l_1} = \frac{2}{1}$$

19. (b) Standard equation for given condition

$$x = a \cos \frac{2\pi}{T} t \Rightarrow x = -0.16 \cos(\pi t)$$

[As $a = -0.16$ meter, $T = 2$ sec]

20. (c) By using conservation of mechanical energy

$$\frac{1}{2} kx^2 = \frac{1}{2} mv^2 \Rightarrow x = v\sqrt{m/k}$$

21. (c) Given elastic energies are equal i.e., $\frac{1}{2} k_1 x_1^2 = \frac{1}{2} k_2 x_2^2$

$$\Rightarrow \frac{k_1}{k_2} = \left(\frac{x_2}{x_1}\right)^2 \text{ and using } F = kx$$

$$\Rightarrow \frac{F_1}{F_2} = \frac{k_1 x_1}{k_2 x_2} = \frac{k_1}{k_2} \times \sqrt{\frac{k_2}{k_1}} = \sqrt{\frac{k_1}{k_2}}$$

22. (c) $n = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \Rightarrow n \propto \frac{1}{\sqrt{m}} \Rightarrow \frac{n_1}{n_2} = \sqrt{\frac{m_2}{m_1}}$

$$\Rightarrow \frac{n}{n_2} = \sqrt{\frac{4m}{m}} \Rightarrow n_2 = \frac{n}{2}$$

23. (d) $T = 2\pi \sqrt{\frac{m}{k}} \Rightarrow \frac{T_2}{T_1} = \sqrt{\frac{m_2}{m_1}} = \sqrt{\frac{4m}{m}} = 2 \Rightarrow T_2 = 2 \times 2 = 4$ s

24. (a) $T \propto \frac{1}{\sqrt{k}} \Rightarrow T_1 : T_2 : T_3 = \frac{1}{\sqrt{k}} : \frac{1}{\sqrt{k/2}} : \frac{1}{\sqrt{2k}} = 1 : \sqrt{2} : \frac{1}{\sqrt{2}}$

25. (d) $T \propto \frac{1}{\sqrt{k}} \Rightarrow \frac{T_2}{T_1} = \sqrt{\frac{k_1}{k_2}} = \sqrt{\frac{k}{4k}} = \frac{1}{2} \Rightarrow T_2 = \frac{T_1}{2}$

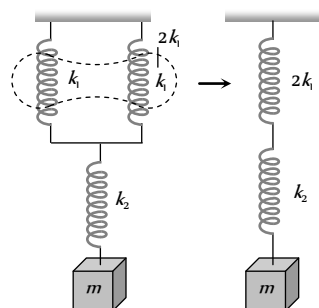
26. (d) The time period of oscillation of a spring does not depend on gravity.

27. (b) In series combination

$$\frac{1}{k_s} = \frac{1}{2k_1} + \frac{1}{k_2}$$

$\Rightarrow$

$$k_s = \left[\frac{1}{2k_1} + \frac{1}{k_2} \right]^{-1}$$



28. (a) Work done in stretching (W) $\propto$ Stiffness of spring (i.e. k)

$$\therefore k_A > k_B \Rightarrow W_A > W_B$$

29. (a) When external force is applied, one spring gets extended and another one gets contracted by the same distance hence force due to two springs act in same direction.

$$\text{i.e. } F = F_1 + F_2 \Rightarrow -kx = -k_1 x - k_2 x \Rightarrow k = k_1 + k_2$$

30. (a) $T = 2\pi \sqrt{\frac{m}{k}} \Rightarrow \frac{T_2}{T_1} = \sqrt{\frac{m_2}{m_1}} \Rightarrow \frac{3}{2} = \sqrt{\frac{m+2}{m}} \Rightarrow \frac{9}{4} = \frac{m+2}{m}$

$$\Rightarrow m = \frac{8}{5} \text{ kg} = 1.6 \text{ kg}$$

31. (b) For series combination $k_{eq} = \frac{k_1 k_2}{k_1 + k_2}$

$$F = k_{eq} x \Rightarrow mg = \left(\frac{k_1 k_2}{k_1 + k_2} \right) x \Rightarrow x = \frac{mg(k_1 + k_2)}{k_1 k_2}$$

32. (d) $n = \frac{1}{2\pi} \sqrt{\frac{k_{eq}}{m}} = \frac{1}{2\pi} \sqrt{\frac{k_1 k_2}{(k_1 + k_2)m}}$

33. (b) Using $F = kx \Rightarrow 10g = k \times 0.25 \Rightarrow k = \frac{10g}{0.25} = 98 \times 4$

$$\text{Now } T = 2\pi \sqrt{\frac{m}{k}} \Rightarrow m = \frac{T^2}{4\pi^2} k$$

$$\Rightarrow m = \frac{\pi^2}{100} \times \frac{1}{4\pi^2} \times 98 \times 4 = 0.98 \text{ kg}$$

34. (b) When spring is cut into n equal parts then spring constant of each part will be nk and so using $T \propto \frac{1}{\sqrt{k}}$, time period will be $T/\sqrt{n}$.

35. (b) By using $K \propto \frac{1}{l}$

Since one fourth length is cut away so remaining length is $\frac{3}{4}l$, hence k becomes $\frac{4}{3}$ times i.e., $k' = \frac{4}{3}k$.

36. (d) $t_1 = 2\pi \sqrt{\frac{m}{K_1}}$ and $t_2 = 2\pi \sqrt{\frac{m}{K_2}}$

Equivalent spring constant for shown combination is

$$K_1 + K_2. \text{ So time period } t \text{ is given by } t = 2\pi \sqrt{\frac{m}{K_1 + K_2}}$$

By solving these equations we get $t^{-2} = t_1^{-2} + t_2^{-2}$

37. (c) $n = \frac{1}{2\pi} \sqrt{\frac{K_{effective}}{m}} = \frac{1}{2\pi} \sqrt{\frac{(K+2K)}{m}} = \frac{1}{2\pi} \sqrt{\frac{3K}{m}}$

38. (d) In series combination

$$\frac{1}{k_s} = \frac{1}{k_1} + \frac{1}{k_2} = \frac{k_2 + k_1}{k_1 k_2} \Rightarrow k_s = \frac{k_1 k_2}{k_1 + k_2}$$

39. (b) $t_1 = 2\pi \sqrt{\frac{m}{k_1}}$ and $t_2 = 2\pi \sqrt{\frac{m}{k_2}}$

In series, effective spring constant, $k = \frac{k_1 k_2}{k_1 + k_2}$

$$\therefore \text{Time period, } T = 2\pi \sqrt{\frac{m(k_1 + k_2)}{k_1 k_2}} \quad \dots (ii)$$

$$\text{Now, } t_1^2 + t_2^2 = 4\pi^2 m \left(\frac{1}{k_1} + \frac{1}{k_2} \right) = \frac{4\pi^2 m (k_1 + k_2)}{k_1 k_2}$$

$$t_1^2 + t_2^2 = T^2. \quad [\text{Using equation (ii)}]$$

$$40. \quad (c) \quad \frac{1}{k_{eff}} = \frac{1}{k} + \frac{1}{2k} + \frac{1}{4k} + \frac{1}{8k} + \dots$$

$$= \frac{1}{k} \left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \right] = \frac{1}{k} \left(\frac{1}{1 - 1/2} \right) = \frac{2}{k}$$

(By using sum of infinite geometrical progression)

$$a + \frac{a}{r} + \frac{a}{r^2} + \dots \infty \text{ sum } (S) = \frac{a}{1-r}$$

$$\therefore k_{eff} = \frac{k}{2}.$$

$$41. \quad (a) \quad n \propto \sqrt{\frac{k}{m}}$$

$$42. \quad (b) \quad F = kx \Rightarrow mg = kx \Rightarrow m \propto kx$$

$$\text{Hence } \frac{m_1}{m_2} = \frac{k_1}{k_2} \times \frac{x_1}{x_2} \Rightarrow \frac{4}{6} = \frac{k}{k/2} \times \frac{1}{x_2}$$

$$\Rightarrow x_2 = 3 \text{ cm}.$$

$$43. \quad (b) \quad \text{Initially when } 1 \text{ kg mass is suspended then by using } F = kx$$

$$\Rightarrow mg = kx \Rightarrow k = \frac{mg}{x} = \frac{1 \times 10}{5 \times 10^{-2}} = 200 \frac{N}{m}$$

Further, the angular frequency of oscillation of 2 kg mass is

$$\omega = \sqrt{\frac{k}{M}} = \sqrt{\frac{200}{2}} = 10 \text{ rad/sec}$$

$$\text{Hence, } v_{\max} = a\omega = (10 \times 10^{-2}) \times 10 = 1 \text{ m/s}$$

$$44. \quad (a) \quad U = \frac{F^2}{2K} \Rightarrow U \propto \frac{1}{K} \Rightarrow \frac{U_1}{U_2} = \frac{K_2}{K_1} = 2$$

$$45. \quad (b) \quad U = \frac{1}{2} Kx^2 \text{ but } T = Kx$$

$$\text{So energy stored} = \frac{1}{2} \frac{(Kx)^2}{K} = \frac{1}{2} \frac{T^2}{K}$$

$$46. \quad (a) \quad \text{System is equivalent to parallel combination of springs} \\ \therefore K_{eq} = K_1 + K_2 = 400 \text{ and}$$

$$T = 2\pi \sqrt{\frac{m}{K_{eq}}} = 2\pi \sqrt{\frac{0.25}{400}} = \frac{\pi}{20}$$

$$47. \quad (b) \quad \text{By cutting spring in four equal parts force constant } (K) \text{ of each} \\ \text{parts becomes four times } \left(\because k \propto \frac{1}{l} \right) \text{ so by using}$$

$$T = 2\pi \sqrt{\frac{m}{K}}; \text{ time period will be half i.e. } T' = T/2$$

$$48. \quad (d) \quad T \propto \sqrt{m} \Rightarrow \frac{T_2}{T_1} = \sqrt{\frac{m_2}{m_1}} \Rightarrow \frac{5}{3} = \sqrt{\frac{M+m}{M}}$$

$$\Rightarrow \frac{25}{9} = \frac{M+m}{M} \Rightarrow \frac{m}{M} = \frac{16}{9}$$

$$49. \quad (c) \quad v_{\max} = a\omega = a \frac{2\pi}{T}$$

$$\Rightarrow a = \frac{v_{\max} T}{2\pi} = \frac{15 \times 628 \times 10^{-3}}{2 \times 3.14} = 1.5 \text{ cm}$$

$$50. \quad (c) \quad Kx = mg \Rightarrow \frac{m}{K} = \frac{x}{g}$$

$$\text{So } T = 2\pi \sqrt{\frac{m}{K}} = 2\pi \sqrt{\frac{x}{g}} = 2\pi \sqrt{\frac{0.2}{9.8}} = \frac{2\pi}{7} \text{ sec}$$

$$51. \quad (b) \quad \omega = \sqrt{k/m} = \sqrt{\frac{4.84}{0.98}} = 2.22 \text{ rad/sec}$$

$$52. \quad (d) \quad \text{When spring is cut into two equal parts then spring constant} \\ \text{of each part will be } 2K \text{ and so using } n \propto \sqrt{K}, \text{ new frequency} \\ \text{will be } \sqrt{2} \text{ times i.e. } f_2 = \sqrt{2} f_1.$$

$$53. \quad (d)$$

$$54. \quad (a) \quad \text{With mass } m_2 \text{ alone, the extension of the spring } l \text{ is given as} \\ m_2 g = kl \quad \dots (i)$$

With mass $(m_1 + m_2)$, the extension l' is given by

$$(m_1 + m_2)g = k(l + \Delta l) \quad \dots (ii)$$

The increase in extension is Δl which is the amplitude of vibration. Subtracting (i) from (ii), we get

$$m_1 g = k\Delta l \text{ or } \Delta l = \frac{m_1 g}{k}$$

$$55. \quad (b) \quad \text{Angular velocity } \omega = \sqrt{\left(\frac{k}{m}\right)} = \sqrt{\left(\frac{10}{10}\right)} = 1$$

$$\text{Now } v = \omega \sqrt{a^2 - y^2} \Rightarrow y^2 = a^2 - \frac{v^2}{\omega^2} = (0.5)^2 - \frac{(0.4)^2}{1^2}$$

$$\Rightarrow y^2 = 0.9 = y = 0.3 \text{ m}$$

Superposition of S.H.M.'s and Resonance

$$1. \quad (c) \quad \text{Resultant amplitude} = \sqrt{3^2 + 4^2} = 5$$

$$2. \quad (c) \quad y = A \sin PT + B \cos PT$$

$$\text{Let } A = r \cos \theta, \quad B = r \sin \theta$$

$$\Rightarrow y = r \sin(PT + \theta) \text{ which is the equation of SHM.}$$

$$3. \quad (c) \quad y = a(\cos \omega t + \sin \omega t) = a\sqrt{2} \left[\frac{1}{\sqrt{2}} \cos \omega t + \frac{1}{\sqrt{2}} \sin \omega t \right]$$

$$= a\sqrt{2} [\sin 45^\circ \cos \omega t + \cos 45^\circ \sin \omega t]$$

$$= a\sqrt{2} \sin(\omega t + 45^\circ) \Rightarrow \text{Amplitude} = a\sqrt{2}$$

$$4. \quad (c) \quad \text{If first equation is } y_1 = a_1 \sin \omega t \Rightarrow \sin \omega t = \frac{y_1}{a_1} \quad \dots (i)$$

$$\text{then second equation will be } y_2 = a_2 \sin \left(\omega t + \frac{\pi}{2} \right)$$

$$= a_2 \left[\sin \omega t \cos \frac{\pi}{2} + \cos \omega t \sin \frac{\pi}{2} \right] = a_2 \cos \omega t$$

$$\Rightarrow \cos \omega t = \frac{y_2}{a_2} \quad \dots (ii)$$

By squaring and adding equation (i) and (ii)

$$\sin^2 \omega t + \cos^2 \omega t = \frac{y_1^2}{a_1^2} + \frac{y_2^2}{a_2^2}$$

$$\Rightarrow \frac{y_1^2}{a_1^2} + \frac{y_2^2}{a_2^2} = 1; \text{ This is the equation of ellipse.}$$

5. (a) If $y_1 = a_1 \sin \omega t$ and $y_2 = a_2 \sin(\omega t + \pi)$

$$\Rightarrow \frac{y_1}{a_1} + \frac{y_2}{a_2} = 0 \Rightarrow y_2 = -\frac{a_2}{a_1} y_1$$

This is the equation of straight line.

6. (c) If $y_1 = a_1 \sin \omega t$ and $y_2 = a_2 \sin(\omega t + 0) = a_2 \sin \omega t$

$$\Rightarrow \frac{y_1^2}{a_1^2} + \frac{y_2^2}{a_2^2} - \frac{2y_1 y_2}{a_1 a_2} = 0 \Rightarrow y_2 = \frac{a_2}{a_1} y_1$$

This is the equation of straight line.

7. (d) For given relation

$$\text{Resultant amplitude} = \sqrt{4^2 + 4^2} = 4\sqrt{2}$$

8. (a) $x = 5\sqrt{2}(\sin 2\pi t + \cos 2\pi t)$

$$= 5\sqrt{2} \sin 2\pi t + 5\sqrt{2} \cos 2\pi t$$

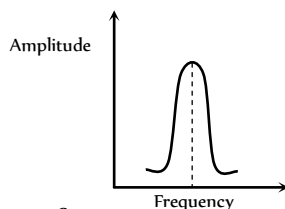
$$x = 5\sqrt{2} \sin 2\pi t + 5\sqrt{2} \sin \left(2\pi t + \frac{\pi}{2} \right)$$

Phase difference between constituent waves $\phi = \frac{\pi}{2}$

$$\therefore \text{Resultant amplitude } A = \sqrt{(5\sqrt{2})^2 + (5\sqrt{2})^2} = 10 \text{ cm.}$$

9. (b)

10. (d) Less damping force gives a taller and narrower resonance peak



11. (b) $A = \frac{c}{a+b-c}$; when $b = 0$, $a = c$ amplitude

$A \rightarrow \infty$. This corresponds to resonance.

12. (c) Energy of particle is maximum at resonant frequency i.e., $\omega_2 = \omega_0$. For amplitude resonance (amplitude maximum)

$$\text{frequency of driver force } \omega = \sqrt{\omega_0^2 - b^2 2m^2} \Rightarrow \omega_1 \neq \omega_0$$

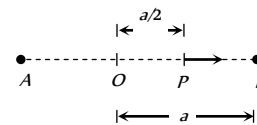
13. (a)

14. (c)

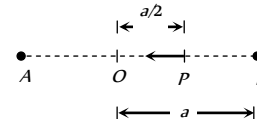
1. (d) $y = a \sin(\omega t + \phi_0)$. According to the question

$$y = \frac{a}{2} \Rightarrow \frac{a}{2} = a \sin(\omega t + \phi_0) \Rightarrow (\omega t + \phi_0) = \phi = \frac{\pi}{6} \text{ or } \frac{5\pi}{6}$$

Physical meaning of $\phi = \frac{\pi}{6}$: Particle is at point P and it is going towards B



Physical meaning of $\phi = \frac{5\pi}{6}$: Particle is at point P and it is going towards O



$$\text{So phase difference } \Delta\phi = \frac{5\pi}{6} - \frac{\pi}{6} = \frac{2\pi}{3} = 120^\circ$$

2. (b) $x = 12 \sin \omega t - 16 \sin^3 \omega t = 4[3 \sin \omega t - 4 \sin^3 \omega t]$

$$= 4[\sin 3\omega t] \text{ (By using } \sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta)$$

$$\therefore \text{maximum acceleration } A_{\max} = (3\omega)^2 \times 4 = 36\omega^2$$

3. (b,c) Harmonic oscillator has some initial elastic potential energy and amplitude of harmonic variation of energy is

$$\frac{1}{2} K a^2 = \frac{1}{2} \times 2 \times 10^6 \times (0.01)^2 = 100 \text{ J}$$

This is the maximum kinetic energy of the oscillator. Thus $K_{\max} = 100 \text{ J}$

This energy is added to initial elastic potential energy may give maximum mechanical energy to have value 160 J .

4. (a) $U = k|x|^3 \Rightarrow F = -\frac{dU}{dx} = -3k|x|^2 \dots(i)$

Also, for SHM $x = a \sin \omega t$ and $\frac{d^2x}{dt^2} + \omega^2 x = 0$

$$\Rightarrow \text{acceleration } a = \frac{d^2x}{dt^2} = -\omega^2 x \Rightarrow F = ma$$

$$= m \frac{d^2x}{dt^2} = -m\omega^2 x \dots(ii)$$

From equation (i) & (ii) we get $\omega = \sqrt{\frac{3kx}{m}}$

$$\Rightarrow T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{3kx}} = 2\pi \sqrt{\frac{m}{3k(a \sin \omega t)}} \Rightarrow T \propto \frac{1}{\sqrt{a}}$$

5. (b, d) Let the velocity acquired by A and B be V, then

$$mv = mV + mV \Rightarrow V = \frac{v}{2}$$

$$\text{Also } \frac{1}{2}mv^2 = \frac{1}{2}mV^2 + \frac{1}{2}mV^2 + \frac{1}{2}kx^2$$

Where x is the maximum compression of the spring. On

solving the above equations, we get $x = v \left(\frac{m}{2k} \right)^{1/2}$

At maximum compression, kinetic energy of the

$$A - B \text{ system} = \frac{1}{2}mV^2 + \frac{1}{2}mV^2 = mV^2 = \frac{mv^2}{4}$$

6. (a) Let the piston be displaced through distance x towards left, then volume decreases, pressure increases. If ΔP is increase in pressure and ΔV is decrease in volume, then considering the process to take place gradually (i.e. isothermal)

$$P_1 V_1 = P_2 V_2$$

$$\Rightarrow PV = (P + \Delta P)(V - \Delta V)$$

$$\Rightarrow PV = PV + \Delta PV - P\Delta V - \Delta P\Delta V$$

$$\Rightarrow \Delta P \cdot V - P \cdot \Delta V = 0 \quad (\text{neglecting } \Delta P \cdot \Delta V)$$

$$\Delta P(Ah) = P(Ax) \Rightarrow \Delta P = \frac{P \cdot x}{h}$$

This excess pressure is responsible for providing the restoring force (F) to the piston of mass M .

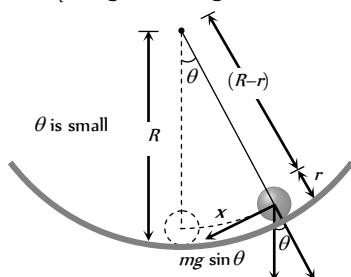
$$\text{Hence } F = \Delta P \cdot A = \frac{PAx}{h}$$

$$\text{Comparing it with } |F| = kx \Rightarrow k = M\omega^2 = \frac{PA}{h}$$

$$\Rightarrow \omega = \sqrt{\frac{PA}{Mh}} \Rightarrow T = 2\pi \sqrt{\frac{Mh}{PA}}$$

Short trick : by checking the options dimensionally. Option (a) is correct.

7. (b) Tangential acceleration, $a_t = -g \sin \theta = -g\theta$



$$a_t = -g \frac{x}{(R-r)}$$

Motion is S.H.M., with time period

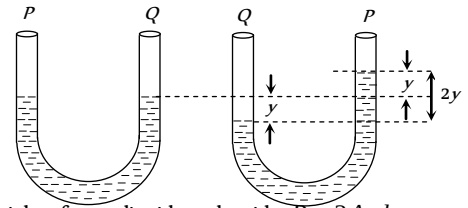
$$T = 2\pi \sqrt{\frac{\text{displacement}}{\text{acceleration}}} = 2\pi \sqrt{\frac{x}{\frac{gx}{(R-r)}}} = 2\pi \sqrt{\frac{R-r}{g}}$$

8. (a) For resonance amplitude must be maximum which is possible only when the denominator of expression is zero i.e.

$$a\omega^2 - b\omega + c = 0 \Rightarrow \omega = \frac{+b \pm \sqrt{b^2 - 4ac}}{2a}$$

For a single resonant frequency, $b = 4ac$.

9. (d) If the level of liquid is depressed by y cm on one side, then the level of liquid in column P is $2y$ cm higher than B as shown.



The weight of extra liquid on the side P = $2A y d g$.

This becomes the restoring force on mass M .

$$\therefore \text{Restoring acceleration} = \frac{-2A y d g}{M}$$

This relation satisfies the condition of SHM i.e. $a \propto -y$.

$$\text{Hence time period } T = 2\pi \sqrt{\frac{\text{Displacement}}{\text{Acceleration}}}$$

$$= 2\pi \sqrt{\frac{y}{\frac{2A y d g}{M}}} \Rightarrow T = 2\pi \sqrt{\frac{M}{2A d g}}$$

10. (b) Time taken by particle to move from $x=0$ (mean position) to $x=4$ (extreme position) = $\frac{T}{4} = \frac{1.2}{4} = 0.3$ s

Let t be the time taken by the particle to move from $x=0$ to $x=2$ cm

$$y = a \sin \omega t \Rightarrow 2 = 4 \sin \frac{2\pi}{T} t \Rightarrow \frac{1}{2} = \sin \frac{2\pi}{1.2} t$$

$$\Rightarrow \frac{\pi}{6} = \frac{2\pi}{1.2} t \Rightarrow t = 0.1 \text{ s. Hence time to move from } x=2$$

to $x=4$ will be equal to $0.3 - 0.1 = 0.2$ s

Hence total time to move from $x=2$ to $x=4$ and back again = $2 \times 0.2 = 0.4$ sec

11. (c) For body to remain in contact $a_{\max} = g$

$$\therefore \omega^2 A = g \Rightarrow 4\pi^2 n^2 A = g$$

$$\Rightarrow n^2 = \frac{g}{4\pi^2 A} = \frac{10}{4(3.14)^2 \cdot 0.01} = 25 \Rightarrow n = 5 \text{ Hz}$$

12. (c) Under the influence of one force $F_1 = m\omega_1^2 y$ and under the action of another force, $F_2 = m\omega_2^2 y$.

Under the action of both the forces $F = F_1 + F_2$

$$\Rightarrow m\omega^2 y = m\omega_1^2 y + m\omega_2^2 y$$

$$\Rightarrow \omega^2 = \omega_1^2 + \omega_2^2 \Rightarrow \left(\frac{2\pi}{T}\right)^2 = \left(\frac{2\pi}{T_1}\right)^2 + \left(\frac{2\pi}{T_2}\right)^2$$

$$\Rightarrow T = \frac{\sqrt{T_1^2 T_2^2}}{\sqrt{T_1^2 + T_2^2}} = \frac{\sqrt{\left(\frac{4}{5}\right)^2 \left(\frac{3}{5}\right)^2}}{\sqrt{\left(\frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2}} = 0.48 \text{ sec}$$

13. (a) By drawing free body diagram of object during the downward motion at extreme position, for equilibrium of mass

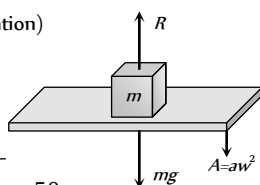
$$mg - R = mA \quad (A = \text{Acceleration})$$

For critical condition $R = 0$

$$\text{so } mg = mA \Rightarrow mg = ma\omega^2$$

$$\Rightarrow \omega = \sqrt{g/a} = \sqrt{\frac{9.8}{3.92 \times 10^{-3}}} = 50$$

$$\Rightarrow T = \frac{2\pi}{\omega} = \frac{2\pi}{50} = 0.1256 \text{ sec}$$



14. (a) Using $x = A \sin \omega t$

$$\text{For } x = A/2, \sin \omega T_1 = 1/2 \Rightarrow T_1 = \frac{\pi}{6\omega}$$

$$\text{For } x = A, \sin \omega(T_1 + T_2) = 1 \Rightarrow T_1 + T_2 = \frac{\pi}{2\omega}$$

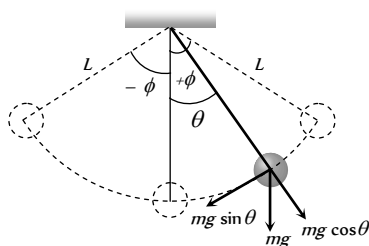
$$\Rightarrow T_2 = \frac{\pi}{2\omega} - T_1 = \frac{\pi}{2\omega} - \frac{\pi}{6\omega} = \frac{\pi}{3\omega} \text{ i.e. } T_1 < T_2$$

Alternate method : In S.H.M., velocity of particle also oscillates simple harmonically. Speed is more near the mean position and less near the extreme position. Therefore the time taken for the

particle to go from 0 to $\frac{A}{2}$ will be less than the time taken to

go from $\frac{A}{2}$ to A. Hence $T_1 < T_2$.

15. (b, c) From following figure it is clear that



$$T - Mg \cos \theta = \text{Centripetal force}$$

$$\Rightarrow T - Mg \cos \theta = \frac{Mv^2}{L}$$

Also tangential acceleration $|a_T| = g \sin \theta$.

16. (c) If t is the time taken by pendulums to come in same phase again first time after $t = 0$.

and N_S = Number of oscillations made by shorter length pendulum with time period T_S .

N_L = Number of oscillations made by longer length pendulum with time period T_L .

$$\text{Then } t = N_S T_S = N_L T_L$$

$$\Rightarrow N_S 2\pi \sqrt{\frac{5}{g}} = N_L \times 2\pi \sqrt{\frac{20}{g}} \quad (\because T = 2\pi \sqrt{\frac{l}{g}})$$

$$\Rightarrow N_S = 2N_L \text{ i.e. if } N_L = 1 \Rightarrow N_S = 2$$

17. (a) Tension in the string when bob passes through lowest point

$$T = mg + \frac{mv^2}{r} = mg + mv\omega \quad (\because v = r\omega)$$

$$\text{putting } v = \sqrt{2gh} \text{ and } \omega = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi$$

$$\text{we get } T = m(g + \pi \sqrt{2gh})$$

18. (d) When the bob is immersed in water its effective weight =

$$\left(mg - \frac{m}{\rho}g\right) = mg \left(\frac{\rho - 1}{\rho}\right) \therefore g_{\text{eff}} = g \left(\frac{\rho - 1}{\rho}\right)$$

$$\frac{T'}{T} = \sqrt{\frac{g}{g_{\text{eff}}}} \Rightarrow T' = T \sqrt{\frac{\rho}{\rho - 1}}$$

19. (a) Time period $T \propto \sqrt{l} \Rightarrow \frac{\Delta T}{T} = \frac{1}{2} \frac{\Delta l}{l} = \frac{1}{2} \alpha \Delta \theta$

Also according to thermal expansion $l' = (1 + \alpha \Delta \theta)l$

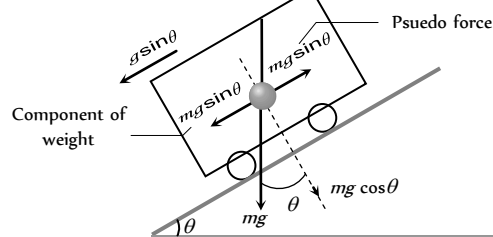
$$\frac{\Delta l}{l} = \alpha \Delta \theta. \text{ Hence } \frac{\Delta T}{T} = \frac{1}{2} \frac{\Delta l}{l} = \frac{1}{2} \alpha \Delta \theta$$

$$= \frac{1}{2} \times 12 \times 10^{-6} \times (40 - 20) = 12 \times 10^{-5}$$

$$\Rightarrow \Delta T = 12 \times 10^{-5} \times 86400 \text{ seconds / day}$$

$$\therefore \Delta T \approx 10.3 \text{ seconds / day}$$

20. (a) See the following force diagram.



Vehicle is moving down the incline so, its acceleration is $g \sin \theta$. Since vehicle is accelerating, a pseudo force $m(g \sin \theta)$ will act on bob of pendulum which cancel the $\sin \theta$ component of weight of the bob.

Hence net force on the bob is $F_{\text{net}} = mg \cos \theta$ or net acceleration of the bob is $g_{\text{eff}} = g \cos \theta$

$$\therefore \text{Time period } T = 2\pi \sqrt{\frac{l}{g_{\text{eff}}}} = 2\pi \sqrt{\frac{l}{g \cos \theta}}$$

21. (c) $\therefore t_0 = 2\pi \sqrt{\frac{l}{g}}$

Effective weight of bob inside water,

$$W' = mg - \text{thrust} = V\rho g - V\rho'g$$

$$\Rightarrow V\rho g_{\text{eff}} = V(\rho - \rho')g, \text{ where, } \rho = \text{Density of bob}$$

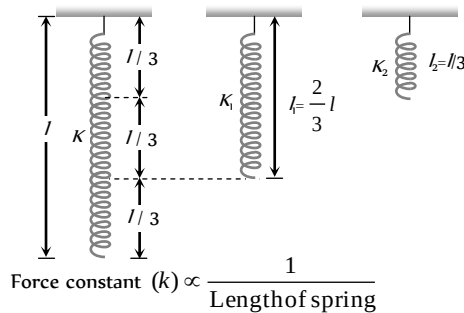
$$\Rightarrow g_{\text{eff}} = \left(1 - \frac{\rho'}{\rho}\right)g \quad \text{and } \rho' = \text{Density of water}$$

$$\therefore t = 2\pi\sqrt{\frac{l}{g_{\text{eff}}}} = 2\pi\sqrt{\frac{l}{(1 - \rho'/\rho)g}}$$

$$\therefore \frac{t}{t_0} = \sqrt{\frac{1}{1 - \rho'/\rho}} = \sqrt{\frac{1}{1 - \frac{3}{4}}} \quad \left(\because \rho' = 10^3 \text{ kg/m}^3\right) \quad \rho = \frac{4}{3} \times 10^3 \text{ kg/m}^3$$

$$\Rightarrow t = 2t_0.$$

22. (b)



$$\Rightarrow \frac{K}{K_1} = \frac{l_1}{l} = \frac{\frac{2}{3}l}{l} \Rightarrow K_1 = \frac{3}{2}K.$$

23. (b) The wire may be treated as a string for which force constant

$$k_1 = \frac{\text{Force}}{\text{Extension}} = \frac{YA}{L} \quad \left(\because Y = \frac{F}{A} \times \frac{L}{\Delta L}\right)$$

Spring constant of the spring $k_2 = K$

Hence spring constant of the combination (series)

$$k_{\text{eq}} = \frac{k_1 k_2}{k_1 + k_2} = \frac{(YA/L)K}{(YA/L) + K} = \frac{YAK}{YA + KL}$$

$$\therefore \text{Time period } T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{(YA + KL)m}{YAK}}^{1/2}$$

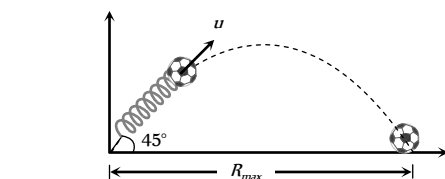
24. (a) Slope is irrelevant hence $T = 2\pi\sqrt{\frac{M}{2K}}^{1/2}$

25. (b) For forced oscillation,

$$x = x_0 \sin(\omega t + \phi) \text{ and } F = F_0 \cos \omega t$$

$$\text{where, } x_0 = \frac{F_0}{m(\omega_0^2 - \omega^2)} \propto \frac{1}{m(\omega_0^2 - \omega^2)}.$$

26. (b) For getting horizontal range, there must be some inclination of spring with ground to project ball.



$$\Rightarrow R_{\text{max}} = \frac{u^2}{g} \quad \dots(i)$$

But K.E. acquired by ball = P.E. of spring gun

$$\Rightarrow \frac{1}{2}mu^2 = \frac{1}{2}kx^2 \Rightarrow u^2 = \frac{kx^2}{m} \quad \dots(ii)$$

From equation (i) and (ii)

$$R_{\text{max}} = \frac{kx^2}{mg} = \frac{600 \times (5 \times 10^{-2})^2}{15 \times 10^{-3} \times 10} = 10 \text{ m}.$$

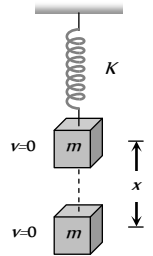
27. (b) Let x be the maximum extension of the spring. From energy conservation

Loss in gravitational potential energy

= Gain in potential energy of spring

$$Mgx = \frac{1}{2}Kx^2$$

$$\Rightarrow x = \frac{2Mg}{K}$$



$$28. (b) y = 4 \cos^2\left(\frac{t}{2}\right) \sin 1000 t$$

$$\Rightarrow y = 2(1 + \cos t) \sin 1000 t$$

$$\Rightarrow y = 2 \sin 1000 t + 2 \cos t \sin 1000 t$$

$$\Rightarrow y = 2 \sin 1000 t + \sin 999 t + \sin 1001 t$$

It is a sum of three S.H.M.

29. (a, c) Let simple harmonic motions be represented by

$$y_1 = a \sin\left(\omega t - \frac{\pi}{4}\right); y_2 = a \sin \omega t \text{ and}$$

$$y_3 = a \sin\left(\omega t + \frac{\pi}{4}\right). \text{ On superimposing, resultant SHM will}$$

$$\text{be } y = a \left[\sin\left(\omega t - \frac{\pi}{4}\right) + \sin \omega t + \sin\left(\omega t + \frac{\pi}{4}\right) \right]$$

$$= a \left[2 \sin \omega t \cos \frac{\pi}{4} + \sin \omega t \right]$$

$$= a [\sqrt{2} \sin \omega t + \sin \omega t] = a (1 + \sqrt{2}) \sin \omega t$$

Resultant amplitude $= (1 + \sqrt{2})a$ Energy is S.H.M. $\propto$ (Amplitude)

$$\therefore \frac{E_{\text{Resultant}}}{E_{\text{Single}}} = \left(\frac{A}{a}\right)^2 = (\sqrt{2} + 1)^2 = (3 + 2\sqrt{2})$$

$$\Rightarrow E_{\text{Resultant}} = (3 + 2\sqrt{2})E_{\text{Single}}$$

$$30. (d) y = \sin^2 \omega t = \frac{1 - \cos 2\omega t}{2} \Rightarrow \text{Period, } T = \frac{2\pi}{2\omega} = \frac{\pi}{\omega}$$

The given function is not satisfying the standard differential

equation of S.H.M. $\frac{d^2y}{dx^2} = -\omega^2 y$. Hence it represents periodic motion but not S.H.M.

$$31. (c) y = Kt^2 \Rightarrow \frac{d^2y}{dt^2} = a_y = 2K = 2 \times 1 = 2 \text{ m/s}^2 (\because K = 1 \text{ m/s}^2)$$

$$\text{Now, } T_1 = 2\pi\sqrt{\frac{l}{g}} \text{ and } T_2 = 2\pi\sqrt{\frac{l}{g+a_y}}$$

Dividing, $\frac{T_1}{T_2} = \sqrt{\frac{g+a_y}{g}} \Rightarrow \sqrt{\frac{6}{5}} \Rightarrow \frac{T_1^2}{T_2^2} = \frac{6}{5}$

32. (b) From the relation of restitution $\frac{h_n}{h_0} = e^{2n}$ and

$$h_n = h_0(1 - \cos 60^\circ) \Rightarrow \frac{h_n}{h_0} = 1 - \cos 60^\circ = \left(\frac{2}{\sqrt{5}}\right)^{2n}$$

$$\Rightarrow 1 - \frac{1}{2} = \left(\frac{4}{5}\right)^n \Rightarrow \frac{1}{2} = \left(\frac{4}{5}\right)^n$$

Taking log of both sides we get

$$\log 1 - \log 2 = n(\log 4 - \log 5)$$

$$0 - 0.3010 = n(0.6020 - 0.6990)$$

$$-0.3010 = -n \times 0.097 \Rightarrow n = \frac{0.3010}{0.097} = 3.1 \approx 3$$

33. (a) As a is the side of cube σ is its density.

Mass of cube is $a^3 \sigma$, its weight $= a^3 \sigma g$

Let h be the height of cube immersed in liquid of density ρ in equilibrium then,

$$F = a^2 h \rho g = Mg = a^3 \sigma g$$

If it is pushed down by y then the buoyant force

$$F' = a^2(h+y)\rho g$$

Restoring force is $\Delta F = F' - F = a^2(h+y)\rho g - a^3 \sigma g$

$$= a^2 y \rho g$$

$$\text{Restoring acceleration} = \frac{\Delta F}{M} = -\frac{a^2 y \rho g}{M} = -\frac{a^2 \rho g}{a^3 \sigma} y$$

Motion is S.H.M.

$$\Rightarrow T = 2\pi \sqrt{\frac{a^3 \sigma}{a^2 \rho g}} = 2\pi \sqrt{\frac{a \sigma}{\rho g}}$$

34. (b) As here two masses are connected by two springs, this problem is equivalent to the oscillation of a reduced mass m_r of a spring of effective spring constant.

$$T = 2\pi \sqrt{\frac{m_r}{K_{\text{eff}}}}$$

$$\text{Here } m_r = \frac{m_1 m_2}{m_1 + m_2} = \frac{m}{2} \Rightarrow K_{\text{eff}} = K_1 + K_2 = 2K$$

$$\therefore n = \frac{1}{2\pi} \sqrt{\frac{K_{\text{eff}}}{m_r}} = \frac{1}{2\pi} \sqrt{\frac{2K}{m}} \times 2 = \frac{1}{\pi} \sqrt{\frac{K}{m}} = \frac{1}{\pi} \sqrt{\frac{0.1}{0.1}} = \frac{1}{\pi} \text{ Hz}$$

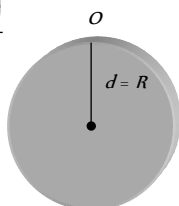
35. (d) Time period of a physical pendulum

$$T = 2\pi \sqrt{\frac{I_0}{mgd}} = 2\pi \sqrt{\frac{\left(\frac{1}{2}mR^2 + mR^2\right)}{mgR}}$$

$$= 2\pi \sqrt{\frac{3R}{2g}} \quad \dots(i)$$

$$T_{\text{simple pendulum}} = 2\pi \sqrt{\frac{l}{g}} \quad \dots(ii)$$

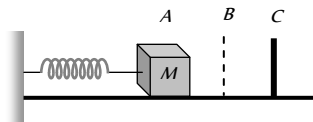
$$\text{Equating (i) and (ii), } l = \frac{3}{2} R.$$



36. (c) The total time from A to C

$$t_{AC} = t_{AB} + t_{BC}$$

$$= (T/4) + t_{BC}$$



where T = time period of oscillation of spring mass system

$$t_{BC} \text{ can be obtained from, } BC = AB \sin(2\pi/T)t_{BC}$$

$$\text{Putting } \frac{BC}{AB} = \frac{1}{2} \text{ we obtain } t_{BC} = \frac{T}{12}$$

$$\Rightarrow t_{AC} = \frac{T}{4} + \frac{T}{12} = \frac{2\pi}{3} \sqrt{\frac{m}{k}}$$

37. (b) When mass 700 gm is removed, the left out mass (500 + 400) gm oscillates with a period of 3 sec

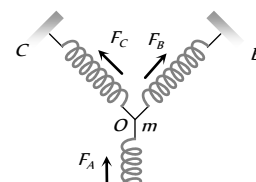
$$\therefore 3 = t = 2\pi \sqrt{\frac{(500+400)}{k}} \quad \dots(i)$$

When 500 gm mass is also removed, the left out mass is 400 gm.

$$\therefore t' = 2\pi \sqrt{\frac{400}{k}} \quad \dots(ii)$$

$$\Rightarrow \frac{3}{t'} = \sqrt{\frac{900}{400}} \Rightarrow t' = 2 \text{ sec}$$

38. (b) When the particle of mass m at O is pushed by y in the direction of A. The spring A will be compressed by y while spring B and C will be stretched by $y' = y \cos 45^\circ$. So that the total restoring force on the mass m along OA.



$$F_{\text{net}} = F_A + F_B \cos 45^\circ + F_C \cos 45^\circ$$

$$= ky + 2ky' \cos 45^\circ = ky + 2k(y \cos 45^\circ) \cos 45^\circ = 2ky$$

$$\text{Also } F_{\text{net}} = k'y \Rightarrow k'y = 2ky \Rightarrow k' = 2k$$

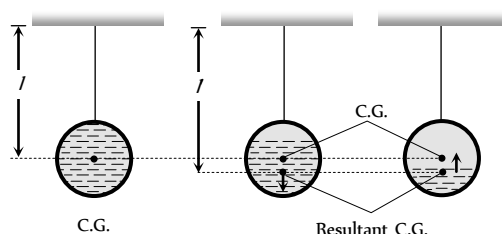
$$T = 2\pi \sqrt{\frac{m}{k'}} = 2\pi \sqrt{\frac{m}{2k}}$$

39. (d) The given system is like a simple pendulum, whose effective length (l) is equal to the distance between point of suspension and C.G. (Centre of Gravity) of the hanging body.

When water slowly flows out the sphere, the C.G. of the system is lowered, and hence l increases, which in turn increases time period (as $T \propto \sqrt{l}$).

After some time weight of water left in sphere become less than the weight of sphere itself, so the resultant C.G. gets clear the C.G. of sphere itself i.e. l decreases and hence T increases.

Finally when the sphere becomes empty, the resulting C.G. is the C.G. of sphere i.e. length becomes equal to the original length and hence the time period becomes equal to the same value as when it was full of water.



40. (b) Let T_1 and T_2 are the time period of the two pendulums

$$T_1 = 2\pi\sqrt{\frac{100}{g}} \text{ and } T_2 = 2\pi\sqrt{\frac{121}{g}}$$

$$(T_1 < T_2 \text{ because } l_1 < l_2).$$

Let at $t = 0$, they start swinging together. Since their time periods are different, the swinging will not be in unison always. Only when number of completed oscillation differs by an integer, the two pendulum will again begin to swing together.

Let longer length pendulum complete n oscillation and shorter length pendulum complete $(n+1)$ oscillation, for the unison swinging, then $(n+1)T_1 = nT_2$

$$(n+1) \times 2\pi\sqrt{\frac{100}{g}} = n \times 2\pi\sqrt{\frac{121}{g}} \Rightarrow n = 10$$

41. (b) Amplitude of damped oscillator

$$A = A_0 e^{-\lambda t}; \lambda = \text{constant}, t = \text{time}$$

$$\text{For } t = 1 \text{ min. } \frac{A_0}{2} = A_0 e^{-\lambda t} \Rightarrow e^{\lambda} = 2$$

$$\text{For } t = 3 \text{ min. } A = A_0 e^{-\lambda \times 3} = \frac{A_0}{(e^{\lambda})^3} = \frac{A_0}{2^3}$$

$$\Rightarrow X = 2^3$$

42. (a) The standard differential equation is satisfied by only the function $\sin \omega t - \cos \omega t$. Hence it represents S.H.M.

43. (d) This is the special case of physical pendulum and in this case

$$T = 2\pi\sqrt{\frac{2l}{3g}}$$

$$\Rightarrow T = 2 \times 3.14 \sqrt{\frac{2 \times 2}{3 \times 9.8}} = 2.31 \text{ sec} \approx 2.4 \text{ sec}$$

Graphical Questions

- (a) Because acceleration $\propto$ displacement.
- (d) Using acceleration $A = -\omega^2 x$
 $At - x_{\max}$ A will be maximum and positive.
- (d) Acceleration $= -\omega^2 y$. So $F = -m\omega^2 y$
 y is sinusoidal function.
So F will be also sinusoidal function with phase difference π
- (d) At time $\frac{T}{2}$; $v = 0 \therefore$ Total energy = Potential energy.
- (b) PE varies from zero to maximum. It is always positive sinusoidal function.

6. (d) Potential energy of particle performing SHM is given by:

$$PE = \frac{1}{2} m \omega^2 y^2 \text{ i.e. it varies parabolically such that at mean position it becomes zero and maximum at extreme position.}$$

7. (a) Potential energy is minimum (in this case zero) at mean position ($x = 0$) and maximum at extreme position ($x = \pm A$).

At time $t = 0$, $x = A$, hence potential should be maximum. Therefore graph I is correct. Further in graph III. Potential energy is minimum at $x = 0$, hence this is also correct.

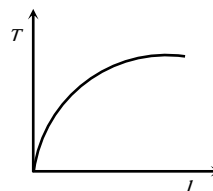
8. (a) $f = \frac{1}{T} = \frac{1}{0.04} = 25 \text{ Hz}$

9. (d) From graph, slope $K = \frac{F}{x} = \frac{8}{2} = 4$

$$T = 2\pi\sqrt{\frac{m}{K}} \Rightarrow T = 2\pi\sqrt{\frac{0.01}{4}} = 0.3 \text{ sec}$$

10. (b) $T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow T \propto \sqrt{l}$ (Equation of parabola)

11. (b) $T \propto \sqrt{l} \Rightarrow T^2 \propto l$



12. (d) In simple harmonic motion

$$y = a \sin \omega t \text{ and } v = a \omega \cos \omega t \text{ from this we have}$$

$$\frac{y^2}{a^2} + \frac{v^2}{a^2 \omega^2} = 1, \text{ which is a equation of ellipse.}$$

13. (c)

14. (a) In S.H.M. when acceleration is negative maximum or positive maximum, the velocity is zero so kinetic energy is also zero. Similarly for zero acceleration, velocity is maximum so kinetic energy is also maximum.

15. (b) Total potential energy = 0.04 J

$$\text{Resting potential energy} = 0.01 \text{ J}$$

$$\text{Maximum kinetic energy} = (0.04 - 0.01)$$

$$= 0.03 \text{ J} = \frac{1}{2} m \omega^2 a^2 = \frac{1}{2} k a^2$$

$$0.03 = \frac{1}{2} \times k \times \left(\frac{20}{1000}\right)^2$$

$$k = 0.06 \times 2500 \text{ N/m} = 150 \text{ N/m}.$$

16. (a) Kinetic energy varies with time but is never negative.

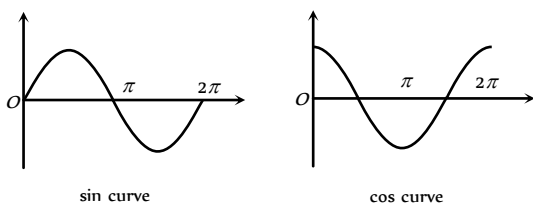
Assertion and Reason

1. (b) Both assertion and reason are correct but reason is not the correct explanation of assertion.

2. (e) In simple harmonic motion, $v = \omega\sqrt{a^2 - y^2}$ as y changes, velocity v will also change. So simple harmonic motion is not uniform motion. But simple harmonic motion may be defined as the projection of uniform circular motion along one of the diameter of the circle.

3. (a) In SHM, the acceleration is always in a direction opposite to that of the displacement *i.e.*, proportional to $(-y)$.

4. (a) A periodic function is one whose value repeats after a definite interval of time. $\sin\theta$ and $\cos\theta$ are periodic functions because they repeat itself after 2π interval of time.



It is also true that moon is smaller than the earth, but this statement is not explaining the assertion.

5. (e) In SHM, $v = \omega\sqrt{a^2 - y^2}$ or $v^2 = \omega^2 a^2 - \omega^2 y^2$.

Dividing both sides by $\omega^2 a^2$, $\frac{v^2}{\omega^2 a^2} + \frac{y^2}{a^2} = 1$. This is the equation of an ellipse. Hence the graph between v and y is an ellipse not a parabola.

6. (b) $T = 2\pi\sqrt{\frac{l}{g}}$. On moon, g is much smaller compared to g on earth. Therefore, T increases.

7. (c) Amplitude of oscillation for a forced, damped oscillator is $A = \frac{F_0/m}{\sqrt{(\omega^2 - \omega_0^2)^2 + (b\omega/m)^2}}$, where b is constant related to

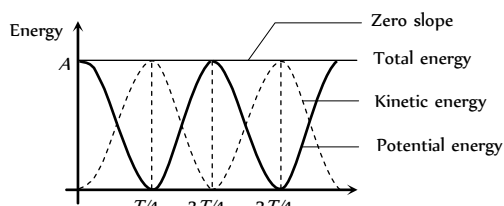
the strength of the resistive force, $\omega_0 = \sqrt{k/m}$ is natural frequency of undamped oscillator ($b = 0$).

When the frequency of driving force (ω) $\approx \omega_0$, then amplitude A is very larger.

For $\omega < \omega_0$ or $\omega > \omega_0$, the amplitude decrease.

8. (a) The total energy of S.H.M. = Kinetic energy of particle + potential energy of particle.

The variation of total energy of the particle in SHM with time is shown in a graph.



9. (c) Time period of simple pendulum of length l is,

$$T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow T \propto \sqrt{l} \Rightarrow \frac{\Delta T}{T} = \frac{1}{2} \frac{\Delta l}{l}$$

$$\therefore \frac{\Delta T}{T} = \frac{1}{2} \times 3 = 1.5\%$$

10. (c) Frequency of second pendulum $n = (1/2)\text{s}^{-1}$. When elevator is moving upwards with acceleration $g/2$, the effective acceleration due to gravity is

$$g = g + a = g + g/2 = 3g/2.$$

$$\text{As } n = \frac{1}{2\pi}\sqrt{\frac{g}{l}} \text{ so } n^2 \propto g.$$

$$\therefore \frac{n_1^2}{n_2^2} = \frac{g_1}{g} = \frac{3g/2}{g} = \frac{3}{2} \text{ or, } \frac{n_1}{n} = \sqrt{\frac{3}{2}} = 1.225$$

$$\text{or, } n_1 = 1.225n = 1.225 \times (1/2) = 0.612 \text{ s}^{-1}.$$

11. (b) Energy of damped oscillator at an any instant t is given by

$$E = E_0 e^{-bt/m} \text{ [where } E_0 = \frac{1}{2}kx^2 = \text{maximum energy]}$$

Due to damping forces the amplitude of oscillator will go on decreasing with time whose energy is expressed by above equation.

12. (b) In SHM, $K.E. = \frac{1}{2}m\omega^2(a^2 - y^2)$ and $P.E. = \frac{1}{2}m\omega^2 y^2$.

For $K.E. = P.E. \Rightarrow 2y^2 = a^2 \Rightarrow y = a/\sqrt{2}$. Since total energy remains constant through out the motion, which is $E = K.E. + P.E.$ So, when $P.E$ is maximum then $K.E$ is zero and viceversa.

13. (a) Total energy of the harmonic oscillator,

$$E = \frac{1}{2}m\omega^2 a^2 \text{ i.e., } E \propto a^2.$$

$$\text{Therefore } \frac{E'}{E} = \left(\frac{2a}{a}\right)^2 \text{ or, } E' = 4E.$$

14. (b) In simple pendulum, when bob is in deflection position, the

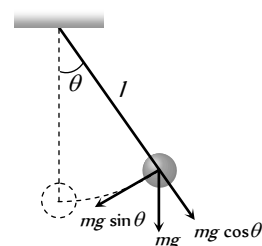
tension in the string is $T = mg \cos\theta + \frac{mv^2}{l}$. Since the

value of θ is different at different positions, hence tension in the string is not constant throughout the oscillation. At end points θ is maximum; the value of $\cos\theta$ is least, hence the value of tension in the string is least. At the mean position, the value of $\theta = 0^\circ$ and $\cos 0^\circ = 1$, so the value of tension is largest.

Also velocity is given by

$$v = \omega\sqrt{a^2 - y^2} \text{ which is}$$

maximum when $y = 0$, at mean position.



15. (e) Spring constant $\propto \frac{1}{\text{Length of spring}}$

$$\Rightarrow k' = \frac{k}{n}$$

Also, spring constant depends on material properties of the spring.

Hence assertion is false, but reason is true.

16. (a) The time period of a oscillating spring is given by,

$$T = 2\pi\sqrt{\frac{m}{k}} \Rightarrow T \propto \frac{1}{\sqrt{k}}. \text{ Since the spring constant is large}$$

for hard spring, therefore hard spring has a less periodic time as compared to soft spring.

17. (e) In simple harmonic motion the velocity is given by,

$$v = \omega\sqrt{a^2 - y^2} \text{ at extreme position, } y = a.$$

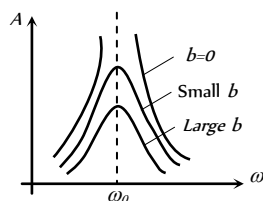
$\therefore v = 0$. But acceleration $A = -\omega^2 a$, which is maximum at extreme position.

18. (a) If the soldiers while crossing a suspended bridge march in steps, the frequency of marching steps of soldiers may match with the natural frequency of oscillations of the suspended bridge. In that situation resonance will take place, then the amplitude of oscillation of the suspended bridge will increase enormously, which may cause the collapsing of the bridge. To avoid situations the soldiers are advised to go out steps on suspended bridge.

19. (a) From equation, amplitude of oscillation

$$A = \frac{F_0 / m}{\sqrt{(\omega^2 - \omega_0^2)^2 + (b\omega / m)^2}}$$

In absence of damping force ($b = 0$), that the steady state amplitude approaches infinity as $\omega \rightarrow \omega_0$. That is, if there is no resistive force in the system and then it is possible to drive an oscillator with sinusoidal force at the resonance frequency, the amplitude of motion will build up without limit. This does not occur in practice because some damping is always present in real oscillation.



20. (b)

21. (c) The amplitude of an oscillating pendulum decreases with time because of friction due to air. Frequency of pendulum is

independent $\left(T = \frac{1}{2\pi} \sqrt{\frac{g}{l}} \right)$ of amplitude.

22. (b) $x = a \sin \omega t$ and $v = \frac{dx}{dt} = a\omega \cos \omega t$

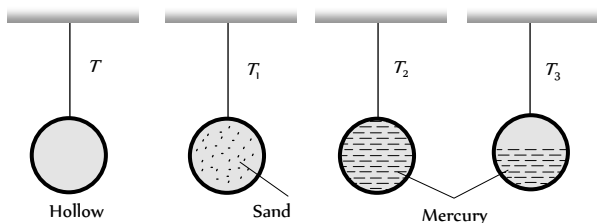
It is clear phase difference between ' x ' and ' v ' is $\pi/2$.

23. (e)

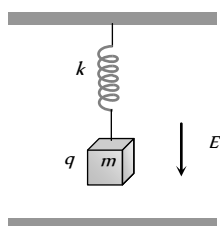
Simple Harmonic Motion

Self Evaluation Test -16

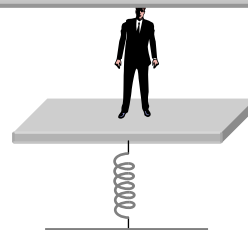
1. The period of a simple pendulum, whose bob is a hollow metallic sphere, is T . The period is T_1 when the bob is filled with sand, T_2 when it is filled with mercury and T_3 when it is half filled with mercury. Which of the following is true



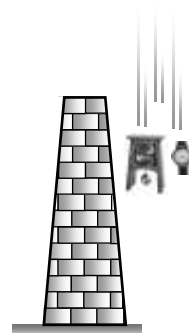
- (a) $T = T_1 = T_2 > T_3$ (b) $T_1 = T_2 = T_3 > T$
(c) $T > T_1 > T_2 = T_3$ (d) $T = T_1 = T_2 < T_3$
2. A pendulum clock that keeps correct time on the earth is taken to the moon it will run (it is given that $g_m = g_e/6$)
(a) At correct rate (b) 6 time faster
(c) $\sqrt{6}$ times faster (d) $\sqrt{6}$ times slowly
3. A pendulum has time period T in air. When it is made to oscillate in water, it acquired a time period $T' = \sqrt{2}T$. The density of the pendulum bob is equal to (density of water = 1)
(a) $\sqrt{2}$ (b) 2
(c) $2\sqrt{2}$ (d) None of these
4. An object of mass 0.2 kg executes simple harmonic along X -axis with frequency of $\frac{25}{\pi} \text{ Hz}$. At the position $x = 0.04 \text{ m}$, the object has kinetic energy of 0.5 J and potential energy of 0.4 J amplitude of oscillation in meter is equal to
(a) 0.05 (b) 0.06
(c) 0.01 (d) None of these
5. Time period of a block suspended from the upper plate of a parallel plate capacitor by a spring of stiffness k is T . When block is uncharged. If a charge q is given to the block then, the new time period of oscillation will be



- (a) T
(b) $> T$
(c) $< T$
(d) $\geq T$
6. A man weighing 60 kg stands on the horizontal platform of a spring balance. The platform starts executing simple harmonic motion of amplitude 0.1 m and frequency $\frac{2}{\pi} \text{ Hz}$. Which of the following statement is correct

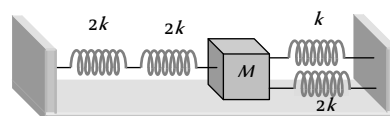


- (a) The spring balance reads the weight of man as 60 kg
(b) The spring balance reading fluctuates between 60 kg and 70 kg
(c) The spring balance reading fluctuates between 50 kg and 60 kg
(d) The spring balance reading fluctuates between 50 kg and 70 kg
7. A man having a wrist watch and a pendulum clock rises on a TV tower. The wrist watch and pendulum clock per chance fall from the top of the tower. Then
(a) Both will keep correct time during the fall.
(b) Both will keep incorrect time during the fall.
(c) Wrist watch will keep correct time and clock will become fast.
(d) Clock will stop but wrist watch will function normally.
8. A force of 6.4 N stretches a vertical spring by 0.1 m . The mass that must be suspended from the spring so that it oscillates with a period of $\left(\frac{\pi}{4}\right) \text{ sec}$ is



[Roorkee 1990]

- (a) $\left(\frac{\pi}{4}\right) \text{ kg}$ (b) 1 kg
(c) $\left(\frac{1}{\pi}\right) \text{ kg}$ (d) 10 kg
9. A spring with 10 coils has spring constant k . It is exactly cut into two halves, then each of these new springs will have a spring constant
(a) $k/2$ (b) $3k/2$
(c) $2k$ (d) $3k$
(e) $4k$
10. Four massless springs whose force constants are $2k$, $2k$, k and $2k$ respectively are attached to a mass M kept on a frictionless plane (as shown in figure). If the mass M is displaced in the horizontal direction, then the frequency of oscillation of the system is



(a) $\frac{1}{2\pi} \sqrt{\frac{k}{4M}}$

(b) $\frac{1}{2\pi} \sqrt{\frac{4k}{M}}$

(c) $\frac{1}{2\pi} \sqrt{\frac{k}{7M}}$

(d) $\frac{1}{2\pi} \sqrt{\frac{7k}{M}}$

11. Values of the acceleration A of a particle moving in simple harmonic motion as a function of its displacement x are given in the table below.

$A \text{ (mm s}^{-2}\text{)}$	16	8	0	-8	-16
$x \text{ (mm)}$	-4	-2	0	2	4

The period of the motion is

(a) $\frac{1}{\pi} \text{ s}$

(b) $\frac{2}{\pi} \text{ s}$

(c) $\frac{\pi}{2} \text{ s}$

(d) $\pi \text{ s}$

12. Two pendulums have time periods T and $\frac{5T}{4}$. They start S.H.M. at the same time from the mean position. What will be the phase difference between them after the bigger pendulum has complete one oscillation

(a) 45°

(b) 90°

(c) 60°

(d) 30°

13. The periodic time of a particle doing simple harmonic motion is 4 second. The time taken by it to go from its mean position to half the maximum displacement (amplitude) is

(a) 2 s

(b) 1 s

(c) $\frac{2}{3} \text{ s}$

(d) $\frac{1}{3} \text{ s}$

14. The displacement of a particle from its mean position (in metre) is given by $y = 0.2 \sin(10\pi t + 1.5\pi) \cos(10\pi t + 1.5\pi)$. The motion of particle is [CPMT 1998]

(a) Periodic but not S.H.M.

(b) Non-periodic

(c) Simple harmonic motion with period 0.1 s (d) Simple harmonic motion with period 0.2 s

15. The kinetic energy and the potential energy of a particle executing S.H.M. are equal. The ratio of its displacement and amplitude will be [RPMT 2003; CPMT 2001]

(a) $\frac{1}{\sqrt{2}}$

(b) $\frac{\sqrt{3}}{2}$

(c) $\frac{1}{2}$

(d) $\sqrt{2}$

16. Two simple pendulums of lengths 1.44 m and 1 m start swinging together. After how many vibrations will they again start swinging together [J & K CET 2005]

(a) 5 oscillations of smaller pendulum

(b) 6 oscillations of smaller pendulum

(c) 4 oscillations of bigger pendulum

(d) 6 oscillations of bigger pendulum

17. Equations $y_1 = A \sin \omega t$ and $y_2 = \frac{A}{2} \sin \omega t + \frac{A}{2} \cos \omega t$

represent S.H.M. The ratio of the amplitudes of the two motions is

(a) 1

(b) 2

(c) 0.5

(d) $\sqrt{2}$

18. A particle doing simple harmonic motion, amplitude = 4 cm , time period = 12 sec . The ratio between time taken by it in going from its mean position to 2 cm and from 2 cm to extreme position is

(a) 1

(b) $1/3$

(c) $1/4$

(d) $1/2$

19. On a planet a freely falling body takes 2 sec when it is dropped from a height of 8 m , the time period of simple pendulum of length 1 m on that planet is [Pb. PMT 2004]

(a) 3.14 sec

(b) 16.28 sec

(c) 1.57 sec

(d) None of these

20. If a simple pendulum is taken to place where g decreases by 2%, then the time period [Pb. PET 2002]

(a) Decreases by 1%

(b) Increases by 2%

(c) Increases by 2%

(d) Increases by 1%

21. Two simple pendulum first of bob mass M and length L second of bob mass M and length L . $M = M$ and $L = 2L$. If these vibrational energy of both is same. Then which is correct

(a) Amplitude of B greater than A (b) Amplitude of B smaller than A

(c) Amplitudes will be same

(d) None of these

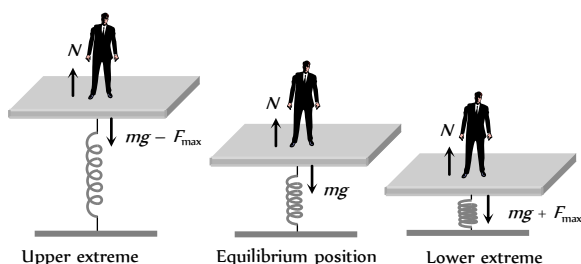
AS Answers and Solutions

(SET -16)

1. (d) Time period of pendulum doesn't depend upon mass but it depends upon length (distance between point of suspension and centre of mass).

In first three cases length are same so $T = T_1 = T_2$ but in last case centre of mass lowers which in turn increases the length. So in this case time period will be more than the other cases.

2. (d) $T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow \frac{T_e}{T_m} = \sqrt{\frac{g_m}{g_e}} = \sqrt{\frac{g_e/6}{g_e}} = \frac{1}{\sqrt{6}}$
 $\Rightarrow T_m = \sqrt{6}T_e$ i.e. clock becomes slower.
3. (b) The effective acceleration of a bob in water
 $= g' = g\left(1 - \frac{\sigma}{\rho}\right)$ where σ and ρ are the density of water and the bob respectively. Since the period of oscillation of the bob in air and water are given as $T = 2\pi\sqrt{\frac{l}{g}}$ and $T' = 2\pi\sqrt{\frac{l}{g'}}$
 $\therefore \frac{T}{T'} = \sqrt{\frac{g'}{g}} = \sqrt{\frac{g(1 - \sigma/\rho)}{g}} = \sqrt{1 - \frac{\sigma}{\rho}} = \sqrt{1 - \frac{1}{\rho}}$
 Putting $\frac{T}{T'} = \frac{1}{\sqrt{2}}$. We obtain, $\frac{1}{2} = 1 - \frac{1}{\rho} \Rightarrow \rho = 2$
4. (b) $E = \frac{1}{2}m\omega^2 A^2 \Rightarrow E = \frac{1}{2}m(2\pi f)^2 A^2 \Rightarrow A = \frac{1}{2\pi f}\sqrt{\frac{2E}{m}}$
 Putting $E = K + U$ we obtain,
 $A = \frac{1}{2\pi\left(\frac{25}{\pi}\right)}\sqrt{\frac{2 \times (0.5 + 0.4)}{0.2}} \Rightarrow A = 0.06 \text{ m}$
5. (a) The forces that act on the block are qE and mg . Since qE and mg are constant forces, the only variable elastic force changes by kx . Where x is the elongation in the spring $\Rightarrow$ unbalanced (restoring) force $= F = -kx$
 $\Rightarrow -m\omega^2 X = -KX \Rightarrow \omega = \sqrt{\frac{k}{M}} = T$.
6. (d) The maximum force acting on the body executing simple harmonic motion is
 $m\omega^2 a = m \times (2\pi f)^2 a = 60 \times \left(2\pi \times \frac{2}{\pi}\right)^2 \times 0.1 \text{ N}$
 $= 60 \times 16 \times 0.1 = 96 \text{ N} = \frac{96}{9.8} \approx 10 \text{ kgf}$ and this force is towards mean position.



The reaction of the force on the platform away from the mean position. It reduces the weight of man on upper extreme i.e. net weight $= (60 - 10) \text{ kgf}$.

This force adds to the weight at lower extreme position i.e. net weight becomes $= (60 + 10) \text{ kgf}$.

Therefore, the reading the weight recorded by spring balance fluctuates between 50 kgf and 70 kgf .

7. (d) Function of wrist watch depends upon spring action so it is not effected by gravity but pendulum clock has time period,
 $T = 2\pi\sqrt{\frac{l}{g}}$. During free fall effective acceleration becomes zero, so time period comes out to be infinity i.e. the clock stops.
8. (b) Force constant of a spring is given by $F = kx$
 $6.4 = k(0.1)$ or $k = 64 \text{ N/m}$
 $\therefore T = 2\pi\sqrt{\frac{m}{k}} \Rightarrow \frac{\pi}{4} = 2\pi\sqrt{\frac{m}{64}}; \frac{m}{64} = \left(\frac{1}{8}\right)^2; m = 1 \text{ kg}$
9. (b) $K \propto \frac{1}{l} \Rightarrow Kl = K' \times \frac{l}{2} \Rightarrow K' = 2K$
10. (b) The two springs on left side having spring constant of $2k$ each are in series, equivalent constant is $\frac{1}{\left(\frac{1}{2k} + \frac{1}{2k}\right)} = k$. The two springs on right hand side of mass M are in parallel. Their effective spring constant is $(k + 2k) = 3k$.
 Equivalent spring constants of value k and $3k$ are in parallel and their net value of spring constant of all the four springs is $k + 3k = 4k$
 $\therefore$ Frequency of mass is $n = \frac{1}{2\pi}\sqrt{\frac{4k}{M}}$
11. (d) $|A| = \omega x \Rightarrow \frac{|A|}{x} = \omega^2$
 From the given value $\frac{|A|}{x} = \omega^2 = 4 \Rightarrow \omega = 2$.
 Also $\omega = \frac{2\pi}{T} \Rightarrow 2 = \frac{2\pi}{T} \Rightarrow T = \pi \text{ sec}$
12. (b) $\frac{5T}{4} = T + \frac{T}{4}$

 By the time, the bigger pendulum makes one full oscillation, the smaller pendulum will make $\left(1 + \frac{1}{4}\right)$ oscillation. The bigger pendulum will be in the mean position and the smaller one will be in the positive extreme position. Thus, phase difference $= 90^\circ$
13. (d) $y = A \sin\left(\frac{2\pi}{T}\right) \cdot t$
 $\Rightarrow \frac{A}{2} = A \sin\left(\frac{2\pi}{4}\right) t \Rightarrow \frac{\pi t}{2} = \frac{\pi}{6} \Rightarrow t = \frac{1}{3} \text{ sec}$
14. (c) $y = 0.2 \sin(10\pi t + 1.5\pi) \cos(10\pi t + 1.5\pi)$

$$= 0.1 \sin 2(10\pi t + 1.5\pi) \quad [\because \sin 2A = 2 \sin A \cos A]$$

$$= 0.1 \sin(20\pi t + 3.0\pi)$$

$$\therefore \text{Time period, } T = \frac{2\pi}{\omega} = \frac{2\pi}{20\pi} = \frac{1}{10} = 0.1 \text{ sec}$$

15. (a) Given $K.E. = P.E. \Rightarrow \frac{1}{2}mv^2 = \frac{1}{2}kx^2$

$$\Rightarrow \frac{1}{2}m\omega^2(a^2 - x^2) = \frac{1}{2}m\omega^2x^2$$

$$\Rightarrow a^2 - x^2 = x^2 \Rightarrow x^2 = \frac{a^2}{2} \Rightarrow \frac{x}{a} = \frac{1}{\sqrt{2}}$$

16. (b) $n \propto \frac{1}{\sqrt{l}} \Rightarrow \frac{n_2}{n_1} = \sqrt{\frac{1.44}{1}} = \frac{1.2}{1} \Rightarrow n_2 = 1.2n_1$

For n be integer minimum value of n should be 5 and then $n = 6$ i.e., after 6 oscillations of smaller pendulum both will be in phase.

17. (d) $y_2 = \frac{A}{2} \sin \omega t + \frac{A}{2} \cos \omega t$

$$y_2 = \frac{A}{2} (\sin \omega t + \cos \omega t) = \frac{A}{2} \times \sqrt{2} [\sin(\omega t + 45^\circ)]$$

$$y_2 = \frac{A}{\sqrt{2}} \sin(\omega t + 45^\circ) \Rightarrow \frac{A_1}{A_2} = \frac{A}{A/\sqrt{2}} = \sqrt{2}$$

18. (d) $\omega = \frac{2\pi}{T} = \frac{2\pi}{12} = \frac{\pi}{6} \text{ rad/sec}$ (For $y = 2 \text{ cm}$) $2 = 4 \left(\sin \frac{\pi}{6} t_1 \right)$

By solving $t_1 = 1 \text{ sec}$ (For $y = 4 \text{ cm}$) $t_2 = 3 \text{ sec}$

So time taken by particle in going from 2 cm to extreme position is

$$t_2 - t_1 = 2 \text{ sec. Hence required ratio will be } \frac{1}{2}.$$

19. (a) On a planet, if a body dropped initial velocity ($u = 0$) from a height h and takes time t to reach the ground then

$$h = \frac{1}{2}g_p t^2 \Rightarrow g_p = \frac{2h}{t^2} = \frac{2 \times 8}{4} = 4 \text{ m/s}^2$$

Using $T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow T = 2\pi\sqrt{\frac{1}{4}} = \pi = 3.14 \text{ sec.}$

20. (d) $T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow T \propto \frac{1}{\sqrt{g}}$

$$\Rightarrow \frac{\Delta T}{T} \times 100 = -\frac{1}{2} \left(\frac{\Delta g}{g} \right) \times 100 = -\frac{1}{2}(-2\%) = 1\%.$$

21. (b) $n = \frac{1}{2\pi} \sqrt{\frac{g}{l}} \Rightarrow n \propto \frac{1}{\sqrt{l}} \Rightarrow \frac{n_1}{n_2} = \sqrt{\frac{l_2}{l_1}} = \sqrt{\frac{L_2}{2L_2}}$

$$\Rightarrow \frac{n_1}{n_2} = \frac{1}{\sqrt{2}} \Rightarrow n_2 = \sqrt{2} n_1 \Rightarrow n_2 > n_1$$

Energy $E = \frac{1}{2}m\omega^2 a^2 = 2\pi^2 m n^2 a^2$

$$\Rightarrow \frac{a_1^2}{a_2^2} = \frac{m_2 n_2^2}{m_1 n_1^2} \quad (\because E \text{ is same})$$

Given $n_2 > n_1$ and $m_1 = m_2 \Rightarrow a_1 > a_2$



Chapter 17 Waves and Sound

Waves



Ripple on a pond



Musical sound



Seismic waves
(Earth quake)



Tsunami

light waves from the sun warms the surface of our planet; the energy in seismic waves can crack our planet's crust.

Characteristics of Wave Motion

(1) When a wave motion passes through a medium, particles of the medium only vibrate simple harmonically about their mean position. They do leave their position and move with the disturbance.

(2) In wave motion, the phase of particles of medium keeps on changing.

(3) The velocity of the particle during their vibration is different at different position.

(4) The velocity of wave motion through a particular medium is constant. It depends only on the nature of medium not on the frequency, wavelength or intensity.

(5) Energy is propagated along with the wave motion without any net transport of the medium.

(6) For the propagation of wave, a medium should have following characteristics.

(i) Elasticity : So that particles can return to their mean position, after having been.

(ii) Inertia : So that particles can store energy and overshoot their mean position.

(iii) Minimum friction amongst the particles of the medium.

(iv) Uniform density of the medium.

Types of Waves

Waves can be classified in a number of ways based on the following characteristics

(i) **On the basis necessity of medium**

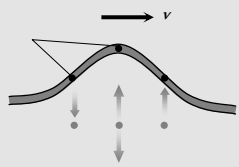
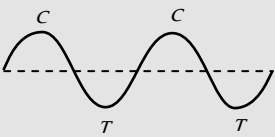
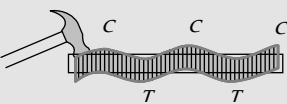
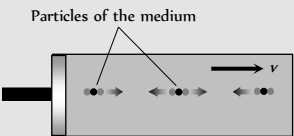
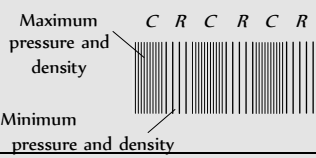
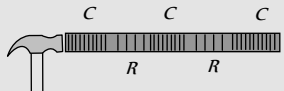
(i) **Mechanical waves** : Require medium for their propagation *e.g.* Waves on string and spring, waves on water surface, sound waves, seismic waves.

When a system is disturbed from equilibrium and the disturbance propagates from one region of the system to another. Wave can carry energy and momentum. The energy in

(ii) **Non-mechanical waves** : Do not require medium for their propagation are called *e.g.* Light, heat (Infrared), radio waves, γ -rays, X-rays *etc.*

(2) **On the basis of vibration of particle** : On the basis of vibration of particle of medium waves can be classified as transverse waves and longitudinal waves.

Table 17.1 : Transverse and longitudinal waves

Transverse waves	Longitudinal waves
Particles of the medium vibrates in a direction perpendicular to the direction of propagation of wave.  Transverse wave on a string It travels in the form of crests (C) and troughs (T).  Transverse waves can be transmitted through solids, they can be setup on the surface of liquids. But they can not be transmitted into liquids and gases.  Medium should possess the property of rigidity. Transverse waves can be polarised. Movement of string of a sitar or violin, movement of the membrane of a Tabla or Dholak, movement of kink on a rope, waves set-up on the surface of water.	Particles of a medium vibrate in the direction of wave motion.  Longitudinal wave in a fluid It travels in the form of compression (C) and rarefaction (R).  These waves can be transmitted through solids, liquids and gases because for these waves propagation, volume elasticity is necessary.  Medium should possess the property of elasticity. Longitudinal waves can not be polarised. Sound waves travel through air, Vibration of air column in organ pipes Vibration of air column above the surface of water in the tube of resonance apparatus

(3) On the basis of energy propagation

(i) **Progressive wave** : These waves advances in a medium with definite velocity. These waves propagate energy in the medium *e.g.* Sound wave and light waves.

(ii) **Stationary wave** : These waves remains stationary between two boundaries in medium. Energy is not propagated by these waves but it is confined in segments (or loops) *e.g.* Wave in a string, waves in organ pipes.

(4) On the basis of dimension

(i) **One dimensional wave** : Energy is transferred in a single direction only *e.g.* Wave propagating in a stretched string.

(ii) **Two dimensional wave** : Energy is transferred in a plane in two mutually perpendicular directions *e.g.* Wave propagating on the surface of water.

(iii) **Three dimensional wave** : Energy is transferred in space in all direction *e.g.* Light and sound waves propagating in space.

(5) Some other waves

(i) **Matter waves** : The waves associated with the moving particles are called matter waves.

(ii) **Audible or sound waves** : Range 20 Hz to 20 KHz. These are generated by vibrating bodies such as vocal cords, stretched strings or membrane.

(iii) **Infrasonic waves** : Frequency lie below 20 Hz and wavelengths are greater than 16.6 cm. *Example* : waves produced during earth quake, ocean waves *etc.*

(iv) **Ultrasonic waves** : Frequency greater than 20 KHz. Human ear cannot detect these waves, certain creatures such as mosquito, dog and bat show response to these. As velocity of sound in air is 332 m/sec so the wavelength $\lambda < 1.66$ cm.

These waves are used for navigation under water (SONAR).

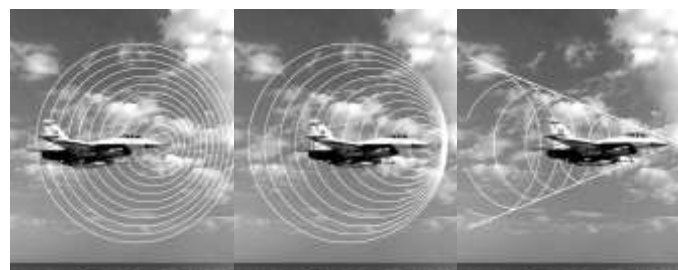
They are used in ultrasonography (in photography or scanning soft tissue of body).

Their used to repel mosquitoes or attract fishes

(v) **Shock waves** : When an object moves with a velocity greater than that of sound, it is termed as **Supersonic**. When such a supersonic body or plane travels in air, it produces energetic disturbance which moves in backward direction and diverges in the form of a cone. Such disturbance are called the shock waves.

The speed of supersonic is measured in Mach number. One mach number is the speed of sound.

$$\text{Mach Number} = \frac{\text{Velocity of source}}{\text{Velocity of sound}}$$



Important Terms Regarding wave motion

(i) **Amplitude (a)** : Maximum displacement of a vibrating particle of medium from it's mean position is called amplitude.

(2) **Wavelength (λ)** : It is equal to the distance travelled by the wave during the time in which any one particle of the medium completes one vibration about its mean position.

(i) Or distance travelled by the wave in one time period is known as wavelength.

(ii) Or is the distance between the two successive points with the same phase.

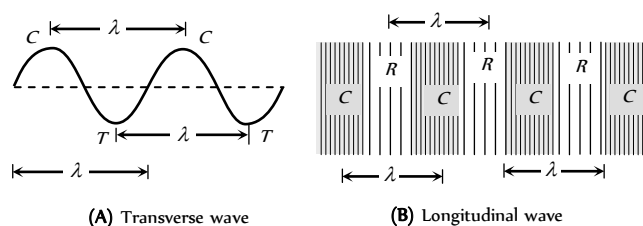


Fig. 17.1

(3) **Frequency (n)** : Frequency of vibration of a particle is defined as the number of vibrations completed by particle in one second.

It is the number of complete wavelengths traversed by the wave in one second.

Units of frequency are hertz (Hz) and per second.

(4) **Time period (T)** : Time period of vibration of particle is defined as the time taken by the particle to complete one vibration about its mean position.

It is the time taken by the wave to travel a distance equal to one wavelength

$$\text{Time period} = 1/\text{Frequency} \Rightarrow T = 1/n$$

(5) **Wave pulse** : It is a short wave produced in a medium when the disturbance created for a short time.

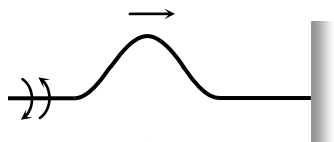
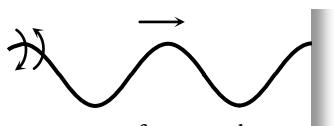


Fig. 17.2

(6) **Wave train** : A series of wave pulse is called wave train.



(7) **Wave front** : A wave front is a line or surface on which the disturbance has the same phase at all points. If the source is periodic, it produces a succession of wave front, all of the same shape. Ripples on a pond are the example of wave fronts.

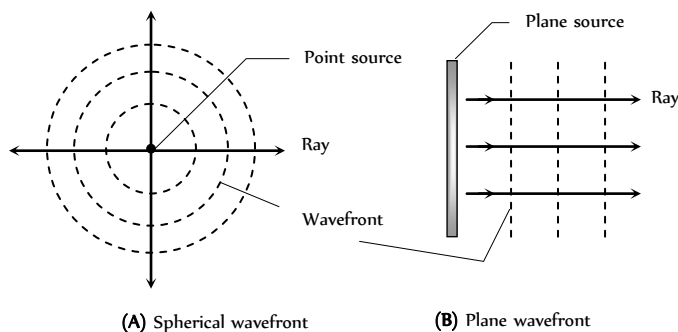


Fig. 17.4

(8) **Wave function** ; It is a mathematical description of the disturbance created by a wave. For a string, the wave function is a displacement for sound waves. It is a pressure or density fluctuation where as for light waves it is electric or magnetic field.

Now let us consider a one dimensional wave travelling along x -axis. During wave motion, a particle with equilibrium position x is displaced some distance y in the direction perpendicular to the x -axis. In this case y is a function of position (x) and time (t).

i.e. $y = f(x, t)$. This is called wave function .

Let the wave pulse is travelling with a speed v , after a time t , the pulse reaches a distance vt along the $+x$ -axis as shown. The wave function now can be represented as $y = f(x - vt)$

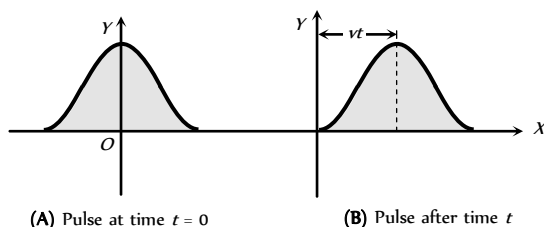


Fig. 17.5

If the wave pulse is travelling along $-x$ -axis then $y = f(x + vt)$

If order of a wave function to represent a wave, the three quantities x , v , t must appear in combinations $(x + vt)$ or $(x - vt)$

Thus $y = (x - vt)$, $\sqrt{(x - vt)}$, $Ae^{-B(x-vt)^2}$ etc. represents travelling waves while $y = (x^2 - v^2t^2)$, $(\sqrt{x} - \sqrt{vt})$, $A \sin (4x - 9t)$ etc. doesn't represents a wave.

(9) **Harmonic wave** : If a travelling wave is a sin or cos function of $(x \pm vt)$ the wave is said to be harmonic or plane progressive wave.

(10) **The wave equation** : All the travelling waves satisfy a differential equation which is called the wave equation. It is given by $\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}$;

$$\text{where } v = \frac{\omega}{k}$$

It is satisfied by any equation of the form $y = f(x \pm vt)$

(11) **Angular wave number or propagation constant (k)** : Number of wavelengths in the distance 2π is called the wave number or propagation constant i.e. $k = \frac{2\pi}{\lambda}$.

It is unit is rad/m .

(12) **Wave velocity (v)** : It is the distance travelled by the disturbance in one time period. It only depends on the properties of the medium and it independent of time and position.

$$v = n\lambda = \frac{\lambda}{T} = \frac{\omega}{2\pi} = \frac{\omega}{k}$$

(13) **Group velocity (v_g)** : The velocity with which the group of waves travels is known as group velocity

or the velocity with which a wave packet travels is known as group velocity $v_g = \frac{d\omega}{dk}$.

(14) **Phase (ϕ)** : The quantity which express at any instant, the displacement of the particle and it's direction of motion is called the phase of the particle.

If two particles of the medium, at any instant are at the same distance in the same direction from their equilibrium positions and are moving in the same direction then they are said to be in same phase e.g. In the following figure particles 1, 3 and 5 are in same phase and point 6, 7 are also in same phase.

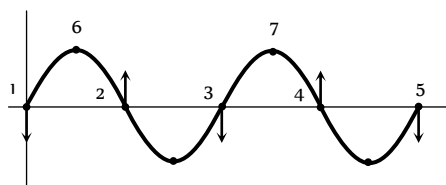


Fig. 17.6

(15) **Intensity of wave** : The wave intensity is defined as the average amount of energy flow in medium per unit time and per unit of its cross-sectional area. Its unit is W/m

$$\text{Hence intensity } (I) = \frac{\text{Energy}}{\text{Area} \times \text{Time}} = \frac{\text{Power}}{\text{Area}} = 2\pi n a \rho v$$

$$\Rightarrow I \propto a^2 \quad (\text{when } v, \rho = \text{constant})$$

where a = Amplitude, n = Frequency, v = Wave velocity,

ρ = Density of medium.

At a distance r from a point source of power P the intensity is given

$$\text{by } I = \frac{P}{4\pi r^2} \Rightarrow I \propto \frac{1}{r^2}$$

The human ear can hear sound of intensity up to $10^{-12} W/m$. This is called **threshold of intensity**. The upper limit of intensity of sound which can be tolerated by human ear is $1 W/m$. This is called **threshold of pain**.

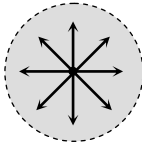


Fig. 17.7

(16) **Energy density** : The energy associated with unit volume of the medium is defined as energy density

$$\text{Energy density} = \frac{\text{Energy}}{\text{Volume}} = \frac{\text{Intensity}}{\text{Velocity}} = \frac{2\pi^2 n^2 a^2 \rho v}{v} = 2\pi^2 n^2 a^2 \rho$$

Velocity of Transverse Wave

The velocity of a transverse wave in a stretched string is given by $v = \sqrt{\frac{T}{m}}$; where T = Tension in the string; m = Linear density of string (mass per unit length).

(1) If A is the area of cross-section of the wire then $m = \rho A$

$$\Rightarrow v = \sqrt{\frac{T}{\rho A}} = \sqrt{\frac{S}{\rho}}; \text{ where } S = \text{Stress} = \frac{T}{A}$$

(2) If string is stretched by some weight then

$$T = Mg$$

$$\Rightarrow v = \sqrt{\frac{Mg}{m}}$$

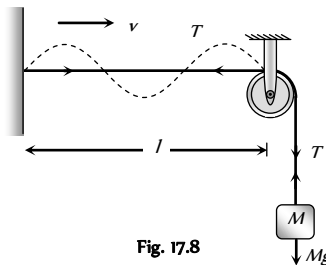


Fig. 17.8

(3) If suspended weight is immersed in a liquid of density σ and ρ = density of material of the suspended load then

$$T = Mg \left(1 - \frac{\sigma}{\rho} \right)$$

$$\Rightarrow v = \sqrt{\frac{Mg(1 - \sigma/\rho)}{m}}$$

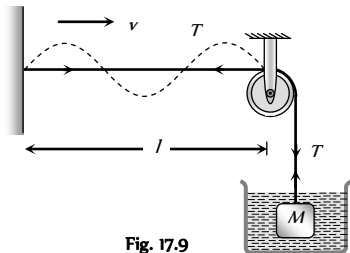


Fig. 17.9

(4) If two rigid supports of stretched string are maintained at temperature difference of $\Delta\theta$ then due to elasticity of string.

$$T = YA \alpha \Delta\theta$$

$$\Rightarrow v = \sqrt{\frac{YA \alpha \Delta\theta}{m}} = \sqrt{\frac{Y \alpha \Delta\theta}{d}}$$

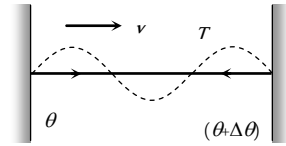


Fig. 17.10

where Y = Young's modulus of elasticity of string, A = Area of cross section of string, α = Temperature coefficient of thermal expansion, d =

$$\text{Density of wire} = \frac{m}{A}$$

$$(5) \text{ In a solid body : } v = \sqrt{\frac{\eta}{\rho}}$$

where η = Modulus of rigidity; ρ = Density of the material.

Velocity of Longitudinal Wave (Sound Wave)

(1) **Velocity of sound in any elastic medium** : It is given by

$$v = \sqrt{\frac{E}{\rho}} = \sqrt{\frac{\text{Elasticity of the medium}}{\text{Density of the medium}}}$$

(i) In solids $v = \sqrt{\frac{Y}{\rho}}$; where Y = Young's modulus of elasticity

(ii) In a liquid and gaseous medium $v = \sqrt{\frac{B}{\rho}}$; where B = Bulk modulus of elasticity of liquid or gaseous medium.

(iii) As solids are most elastic while gases least i.e. $E_S > E_L > E_G$. So the velocity of sound is maximum in solids and minimum in gases, hence $v_s > v_l > v_g$

$$5000 \text{ m/s} > 1500 \text{ m/s} > 330 \text{ m/s}$$

(iv) The velocity of sound in case of extended solid (crust of the earth)

$$v = \sqrt{\frac{B + \frac{4}{3}\eta}{\rho}}; B = \text{Bulk modulus}; \eta = \text{Modulus of rigidity}; \rho = \text{Density}$$

(2) **Newton's formula** : He assumed that when sound propagates through air temperature remains constant. i.e. the process is isothermal. For isothermal process

$$B = \text{Isothermal elasticity } (E_\theta) = \text{Pressure } (P) \Rightarrow v = \sqrt{\frac{B}{\rho}} = \sqrt{\frac{P}{\rho}}$$

For air at NTP : $P = 1.01 \times 10^5 \text{ N/m}^2$ and $\rho = 1.29 \text{ kg/m}^3$.

$$\Rightarrow v_{\text{air}} = \sqrt{\frac{1.01 \times 10^5}{1.29}} \approx 280 \text{ m/s}$$

However the experimental value of sound in air is 332 m/sec which is greater than that given by Newton's formula.

(3) **Laplace correction** : He modified Newton's formula assuming that propagation of sound in gaseous medium is adiabatic process. For adiabatic process

$$B = \text{Adiabatic elasticity } (E_\phi) = \gamma P$$

$$\Rightarrow v = \sqrt{\frac{B}{\rho}} = \sqrt{\frac{E_{\phi}}{\rho}} = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}}$$

$$\text{For air : } \gamma = 1.41 \Rightarrow v = \sqrt{1.41} \times 280 \approx 332 \text{ m/sec}$$

(4) **Relation between velocity of sound and root mean square velocity :**

If sound travel in a gaseous medium then $v = \sqrt{\frac{\gamma RT}{M}}$ and *r.m.s.* velocity

$$\text{of gas } v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

$$\text{So } \frac{v_{\text{rms}}}{v_{\text{sound}}} = \sqrt{\frac{3}{\gamma}} \quad \text{or } v_{\text{rms}} = [\gamma/3]^{1/2} v_{\text{sound}}$$

Factors Affecting Velocity of Sound in Gaseous Medium

(1) **Effect of pressure at constant temperature :** Velocity of sound is independent of the pressure of gas, because as pressure increases, density

also increases hence $\frac{P}{\rho}$ ratio remains constant. So from $v = \sqrt{\frac{\gamma P}{\rho}}$,

(2) **Effect of temperature :** With rise in temperature velocity of sound increases.

$$v = \sqrt{\frac{\gamma RT}{M}} \Rightarrow v \propto \sqrt{T} \Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}} = \sqrt{\frac{(273 + t_1^\circ\text{C})}{(273 + t_2^\circ\text{C})}}$$

When the temperature change is small then $v_t = v_0 + 0.61 t$

where v_t = Velocity of sound at $t^\circ\text{C}$

v_0 = Velocity of sound at $0^\circ\text{C} = 332 \text{ m/sec}$

t = Small temperature change

If $t = 1^\circ\text{C}$ then $v_t = (v_0 + 0.61) \text{ m/sec}$. Hence for 1°C rise, speed of sound in air increases by 0.61 m/sec .

$$(3) \text{ Effect of density : } v = \sqrt{\frac{\gamma P}{\rho}} \Rightarrow v \propto \frac{1}{\sqrt{\rho}}$$

(4) **Effect of humidity :** With increase in humidity, density of air decreases. So with rise in humidity velocity of sound increases.

Sound travels faster in humid air (rainy season) than in dry air (summer) at the same temperature because

$$\rho_{\text{moist air}} < \rho_{\text{dry air}} \Rightarrow v_{\text{moist air}} > v_{\text{dry air}}$$

(5) **Effect of wind velocity :** Because wind drifts the medium (air) along its direction of motion therefore the velocity of sound in a particular direction is the algebraic sum of the velocity of sound and the component of wind velocity in that direction. Resultant velocity of sound towards observer $v' = v + w \cos \theta$.

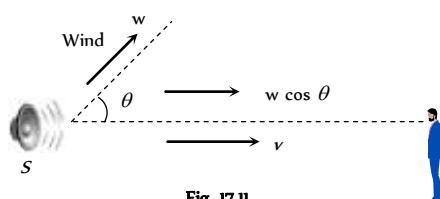


Fig. 17.11

(6) Sound of any frequency or wavelength travels through a given medium with the same velocity.

For a given medium velocity remains constant. All other factors like phase, loudness pitch, quality *etc.* have practically no effect on sound velocity.

Equation of a Plane Progressive Waves

(1) If during the propagation of a progressive wave, the particles of the medium perform SHM about their mean position, then the waves is known as a harmonic progressive wave.

(2) Suppose a plane simple harmonic wave travels from the origin along the positive direction of x -axis from left to right as shown in the figure.

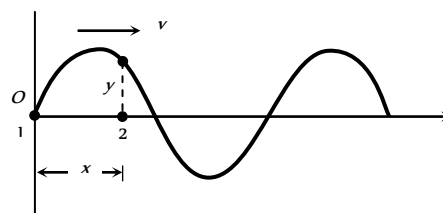


Fig. 17.12

The displacement y of a particle at O from its mean position at any time t is given by $y = a \sin \omega t$.

The wave reaches the particle 2 after time $t = \frac{x}{v}$. Hence displacement y of a particle 2 is given by

$$y = a \sin \omega \left(t - \frac{x}{v} \right) = a \sin (\omega t - kx) \quad \left(\because k = \frac{\omega}{v} \right)$$

The general equation of a plane progressive wave with initial phase is

$$y(x, t) = a \sin (\omega t \pm kx \pm \phi_0)$$

Labels for the equation above:
 Displacement: $y(x, t)$
 Amplitude: a
 Oscillating term: $\sin (\omega t \pm kx \pm \phi_0)$
 Phase: $\omega t \pm kx \pm \phi_0$
 Initial phase: ϕ_0
 Angular frequency: ω
 Position: x
 Propagation constant: k

(3) Various forms of progressive wave function.

$$(i) y = a \sin (\omega t - kx)$$

$$(ii) y = a \sin \left(\omega t - \frac{2\pi}{\lambda} x \right)$$

$$(iii) y = a \sin 2\pi \left[\frac{t}{T} - \frac{x}{\lambda} \right]$$

$$(iv) y = a \sin \frac{2\pi}{T} \left(t - x \frac{T}{\lambda} \right)$$

$$(v) y = a \sin \frac{2\pi}{\lambda} (v t - x)$$

$$(vi) y = a \sin \omega \left(t - \frac{x}{v} \right)$$

(4) **Particle velocity :** The rate of change of displacement y w.r.t. time t is known as particle velocity

$$\text{Hence from } y = a \sin (\omega t - kx)$$

Particle velocity $v_p = \frac{\partial y}{\partial t} = a\omega \cos(\omega t - kx)$

Maximum particle velocity $(v_p)_{\max} = a\omega$

Also $\frac{\partial y}{\partial t} = -\frac{\omega}{k} \times \frac{\partial y}{\partial x} \Rightarrow v_p = -v \times \text{Slope of wave at that point}$

(5) Important relations for numerical solving

(i) Angular frequency ω = co-efficient of t

(ii) Propagation constant k = co-efficient of x

Wave speed $v = \frac{\text{co-efficient of } t}{\text{co-efficient of } x} = \frac{\omega}{k}$

(iii) Wave length $\lambda = \frac{\text{co-efficient of } x}{2\pi}$

(iv) Time period $T = \frac{2\pi}{\text{co-efficient of } t}$

(v) Frequency $n = \frac{\text{co-efficient of } t}{2\pi}$

(vi) $(v_p)_{\max} = a\omega = a(2\pi n) = \frac{a2\pi}{T}$

(vii) If the sign between t and x terms is negative the wave is propagating along positive X -axis and if the sign is positive then the wave moves in negative X -axis direction.

(viii) Co-efficient of $\sin$ or $\cos$ functions i.e. Argument of $\sin$ or $\cos$ function is represented by phase i.e. $(\omega t - kx) = \text{Phase}$.

(ix) **Phase difference and path difference** : At any instant t , if ϕ_1 and ϕ_2 are the phases of two particles whose distances from the origin are x_1 and x_2 respectively then $\phi_1 = (\omega t - kx_1)$ and $\phi_2 = (\omega t - kx_2) \Rightarrow \phi_1 - \phi_2 = k(x_2 - x_1)$

$\Rightarrow$ Phase difference $(\Delta\phi) = \frac{2\pi}{\lambda} \cdot \text{Path difference } (\Delta x)$

(x) **Phase difference and time difference** : If the phases of a particle distance x from the origin is ϕ_1 at time t and ϕ_2 at time t_2 , then $\phi_1 = (\omega t_1 - kx)$ and $\phi_2 = (\omega t_2 - kx) \Rightarrow \phi_1 - \phi_2 = \omega(t_1 - t_2)$

$\Rightarrow$ Phase difference $(\Delta\phi) = \frac{2\pi}{T} \cdot \text{Time difference } (\Delta t)$

Pressure Waves

A longitudinal sound wave can be expressed either in terms of the longitudinal displacement of the particles of the medium or in terms of excess pressure produced due to compression or rarefaction. (at compression, the pressure is more than the normal pressure of the medium and at rarefaction the pressure is lesser than the normal). The first type is called the displacement wave and the second type the pressure wave.

If the displacement wave is represented by $y = a \sin(\omega t - kx)$ then

the corresponding pressure wave will be represented by $\Delta P = -B \frac{dy}{dx}$ (B

= Bulk modulus of elasticity of medium)

$\Rightarrow \Delta P = \Delta P_0 \cos(\omega t - kx)$

where ΔP_0 = pressure amplitude $= akB$

Pressure wave is $\left(\frac{\pi}{2}\right)$ out of phase with displacement wave. i.e.

pressure is maximum when displacement is minimum and vice-versa.

Reflection and Refraction of Waves

When waves are incident on a boundary between two media, a part of incident waves returns back into the initial medium (reflection) while the remaining is partly absorbed and partly transmitted into the second medium (refraction)

(1) **Rarer and denser medium** : A medium is said to be denser (relative to the other) if the speed of wave in this medium is less than the speed of the wave in other medium.

In comparison to air speed of sound is maximum in water, hence water is rarer medium for sound waves w.r.t. air. But it is not true for light (EM-waves). For light waves water is denser medium w.r.t. air.

(2) In reflection or refraction frequency remains same

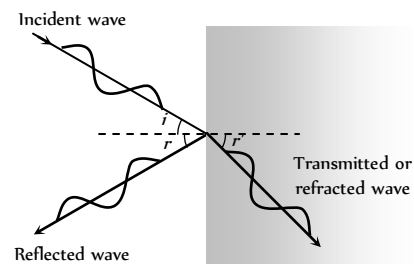


Fig. 17.13

(3) For reflection angle of incidence (i) = Angle of reflection (r)

(4) In case of refraction or transmission $\frac{\sin i}{\sin r} = \frac{v_i}{v_t}$

(5) **Boundary conditions** : Reflection of a wave pulse from some boundary depends on the nature of the boundary.

(i) **Rigid end** : When the incident wave reaches a fixed end, it exerts an upward pull on the end, according to Newton's law the fixed end exerts an equal and opposite downward force on the string. It results in an inverted pulse or phase change of π .

Crest (C) reflects as trough (T) and vice-versa, Time changes by $\frac{T}{2}$

and Path changes by $\frac{\lambda}{2}$

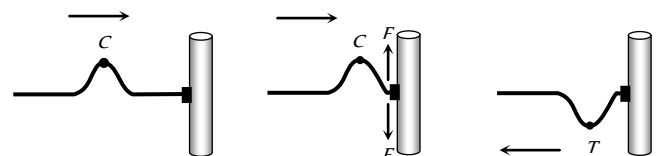


Fig. 17.14

(ii) **Free end** : When a wave or pulse is reflected from a free end, then there is no change of phase (as there is no reaction force).

Crest (C) reflects as crest (C) and trough (T) reflects as trough (T), Time changes by zero and Path changes by zero.

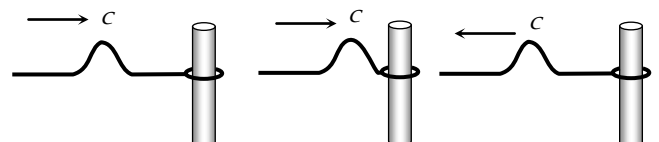


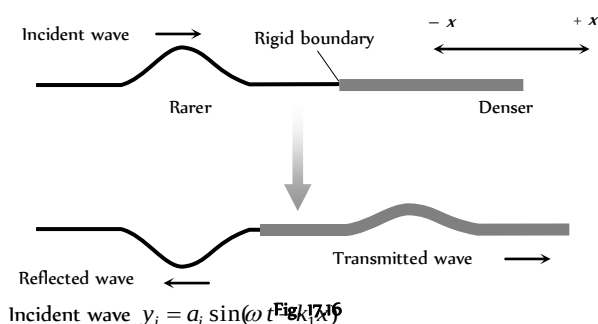
Fig. 17.15

(iii) **Exception** : Longitudinal pressure waves suffer no change in phase from rigid end *i.e.* compression pulse reflects as compression pulse. On the other hand if longitudinal pressure wave reflects from free end, it suffer a phase change of π *i.e.* compression reflects as rarefaction and vice-versa.

(iv) **Effect on different variables** : In case of reflection, because medium is same and hence, speed, frequency (ω) and wavelength λ (or k) do not changes. On the other hand in case of transmitted wave since medium changes and hence speed, wavelength (or k) changes but frequency (ω) remains the same.

(6) Wave in a combination of string

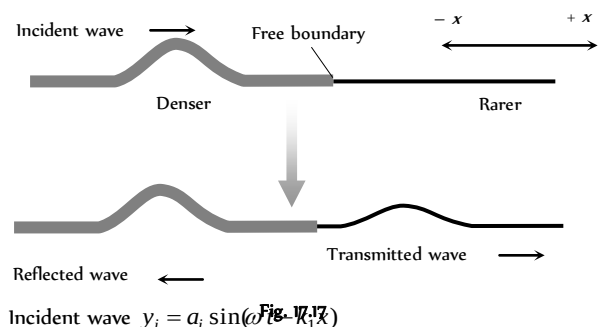
(i) Wave goes from rarer to denser medium



Reflected wave $y_r = a_r \sin[\omega t - k_1(-x) + \pi] = -a_r \sin(\omega t + k_1 x)$

Transmitted wave $y_t = a_t \sin(\omega t - k_2 x)$

(ii) Wave goes from denser to rarer medium



Reflected wave $y_r = a_r \sin[\omega t - k_1(-x) + 0] = a_r \sin(\omega t + k_1 x)$

Transmitted wave $y_t = a_t \sin(\omega t - k_2 x)$

(iii) **Ratio of amplitudes** : It is given as follows

$$\frac{a_r}{a_i} = \frac{k_1 - k_2}{k_1 + k_2} = \frac{v_2 - v_1}{v_2 + v_1} \quad \text{and} \quad \frac{a_t}{a_i} = \frac{2k_1}{k_1 + k_2} = \frac{2v_2}{v_1 + v_2}$$

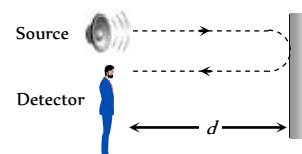
Echo



An echo is simply the repetition of speaker's own voice caused by reflection at a distance surface *e.g.* a cliff, a row of building or any other extended surface.

If there is a sound reflector at a distance d from source, then the time interval between original source and it's echo at the site of source will be

$$t = \frac{d}{v} + \frac{d}{v} = \frac{2d}{v}$$



As the persistence of hearing for human ear is 0.1 sec, therefore in order that an echo of short sound (*e.g.* clap or gun fire) will be heard if $t > 0.1$

$$0.1 \Rightarrow \frac{2d}{v} > 0.1 \Rightarrow d > \frac{v}{20}$$

If v = Speed of sound = 340 m/s then $d > 17$ m.

Principle of Superposition

(i) The displacement at any time due to any number of waves meeting simultaneously at a point in a medium is the vector sum of the individual displacements due each one of the waves at that point at the same time.

(2) If $\vec{y}_1, \vec{y}_2, \vec{y}_3, \dots$ are the displacements at a particular time at a particular position, due to individual waves, then the resultant displacement.

$$\vec{y} = \vec{y}_1 + \vec{y}_2 + \vec{y}_3 + \dots$$

(3) Important applications of superposition principle

(i) **Interference of waves** : Adding waves that differs in phase

(ii) **Formation of stationary waves** : Adding wave that differs in direction.

(iii) **Formation of beats** : Adding waves that differs in frequency.

(iv) **Formation of Lissajou's figure** : Adding two perpendicular simple harmonic motions. (See S.H.M. for more detail)

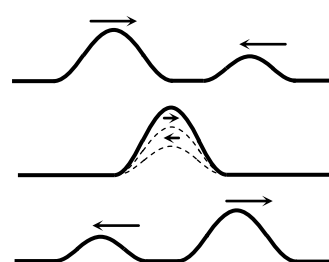


Fig. 17.19

Interference of Sound Waves

(1) When two waves of same frequency, same wavelength, same velocity (nearly equal amplitude) moves in the same direction, Their superimposition results in the interference.

(2) Due to interference the resultant intensity of sound at that point is different from the sum of intensities due to each wave separately.

(3) Interference is of two type (i) Constructive interference (ii) Destructive interference.

(4) In interference energy is neither created nor destroyed but is redistributed.

(5) For observable interference, the sources (producing interfering waves) must be coherent.

(6) Let at a given point two waves arrives with phase difference ϕ and the equation of these waves is given by

$$y_1 = a_1 \sin \omega t, \quad y_2 = a_2 \sin (\omega t + \phi) \quad \text{then by the principle of superposition } \vec{y} = \vec{y}_1 + \vec{y}_2$$

$$\Rightarrow y = a \sin \omega t + a \sin (\omega t + \phi) = A \sin (\omega t + \theta)$$

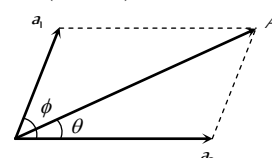


Fig. 17.20

where $A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \phi}$

$$\text{and } \tan \theta = \frac{a_2 \sin \phi}{a_1 + a_2 \cos \phi}$$

$$\text{since Intensity } (I) \propto (\text{Amplitude } A) \Rightarrow \frac{I_1}{I_2} = \left(\frac{a_1}{a_2}\right)^2$$

Therefore, the resultant intensity is given by

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

Table 17.2 : Constructive and destructive interference

Constructive interference	Destructive interference
When the waves meet at a point with same phase, constructive interference is obtained at that point (i.e. maximum sound).	When the wave meets at a point with opposite phase, destructive interference is obtained at that point (i.e. minimum sound).
Phase difference between the waves at the point of observation $\phi = 0^\circ$ or $2n\pi$	Phase difference $\phi = 180^\circ$ or $(2n-1)\pi$; $n = 1, 2, \dots$
Path difference between the waves at the point of observation $\Delta = n\lambda$ (i.e. even multiple of $\lambda/2$)	Path difference $\Delta = (2n-1)\frac{\lambda}{2}$ (i.e. odd multiple of $\lambda/2$)
Resultant amplitude at the point of observation will be maximum $A_{\max} = a_1 + a_2$ If $a_1 = a_2 = a_0 \Rightarrow A_{\max} = 2a_0$	Resultant amplitude at the point of observation will be minimum $A_{\min} = a_1 - a_2$ If $a_1 = a_2 \Rightarrow A_{\min} = 0$
Resultant intensity at the point of observation will be maximum $I_{\max} = I_1 + I_2 + 2\sqrt{I_1 I_2}$ $= (\sqrt{I_1} + \sqrt{I_2})^2$ If $I_1 = I_2 = I_0 \Rightarrow I_{\max} = 4I_0$	Resultant intensity at the point of observation will be minimum $I_{\min} = I_1 + I_2 - 2\sqrt{I_1 I_2}$ $= (\sqrt{I_1} - \sqrt{I_2})^2$ If $I_1 = I_2 = I_0 \Rightarrow I_{\min} = 0$

$$(7) \frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}}\right)^2 = \left(\frac{\frac{\sqrt{I_1}}{\sqrt{I_2}} + 1}{\frac{\sqrt{I_1}}{\sqrt{I_2}} - 1}\right)^2 = \left(\frac{a_1 + a_2}{a_1 - a_2}\right)^2 = \left(\frac{\frac{a_1}{a_2} + 1}{\frac{a_1}{a_2} - 1}\right)^2$$

Quink's Tube

This is an apparatus used to demonstrate the phenomenon of interference and also used to measure velocity of sound in air. This is made up of two U-tube A and B as shown in figure. Here the tube B can slide in and out from the tube A . There are two openings P and Q in the tube A . At opening P , a tuning fork or a sound source of known frequency n is placed and at the other opening a detector is placed to detect the resultant sound of interference occurred due to superposition of two sound waves coming from the tubes A and B .

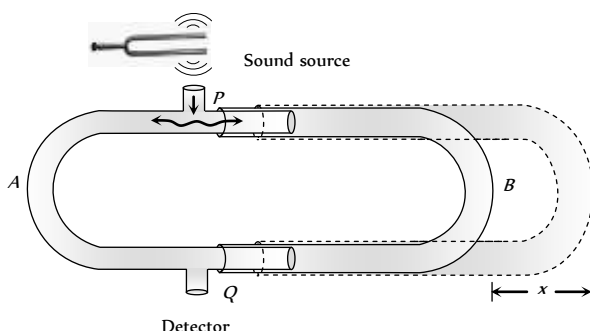


Fig. 17.21

Initially tube B is adjusted so that detector detects a maximum. At this instant if length of paths covered by the two waves from P and Q from the side of A and side of B are l and l' respectively then for constructive interference we must have

$$l_2 - l_1 = N\lambda \quad \dots (i)$$

If now tube B is further pulled out by a distance x so that next maximum is obtained and the length of path from the side of B is l'_2 then we have

$$l'_2 = l_2 + 2x \quad \dots (ii)$$

where x is the displacement of the tube. For next constructive interference of sound at point Q , we have

$$l'_2 = l_1 = (N+1)\lambda \quad \dots (iii)$$

From equation (i), (ii) and (iii), we get

$$l'_2 - l_2 = 2 \times x = \lambda \Rightarrow x = \frac{\lambda}{2}$$

Thus by experiment we get the wavelength of sound as for two successive points of constructive interference, the path difference must be λ . As the tube B is pulled out by x , this introduces a path difference $2x$ in the path of sound wave through tube B . If the frequency of the source is known, n , the velocity of sound in the air filled in tube can be given as $v = n_0 \cdot \lambda = 2n_0 x$

Standing Waves or Stationary Waves

When two sets of progressive wave trains of same type (both longitudinal or both transverse) having the same amplitude and same time period/frequency/wavelength travelling with same speed along the same straight line in opposite directions superimpose, a new set of waves are formed. These are called stationary waves or standing waves.



In practice, a stationary wave is formed when a wave train is reflected at a boundary. The incident and reflected waves then interfere to produce a stationary wave.

(i) Suppose that the two superimposing waves are incident wave $y_1 = a \sin(\omega t - kx)$ and reflected wave $y_2 = a \sin(\omega t + kx)$

(As y is the displacement due to a reflected wave from a free boundary)

Then by principle of superposition

$$y = y_1 + y_2 = a[\sin(\omega t - kx) + \sin(\omega t + kx)]$$

$$(\text{By using } \sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2})$$

$$\Rightarrow y = 2a \cos kx \sin \omega t$$

(If reflection takes place from rigid end, then equation of stationary wave will be $y = 2a \sin kx \cos \omega t$)

(2) As this equation satisfies the wave equation

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2} \text{ . It represents a wave.}$$

(3) As it is not of the form $f(ax \pm bt)$, the wave is not progressive.

(4) Amplitude of the wave $A_{SW} = 2a \cos kx$.

Table 17.3 : Amplitude in two different cases

Reflection at open end	Reflection at closed end
$A_{SW} = 2a \cos kx$	$A_{SW} = 2a \sin kx$
Amplitude is maximum when $\cos kx = \pm 1$	Amplitude is maximum when $\sin kx = \pm 1$
$\Rightarrow kx = 0, \pi, 2\pi, \dots, n\pi$	$\Rightarrow kx = \frac{\pi}{2}, \frac{3\pi}{2}, \dots, \frac{(2n-1)\pi}{2}$
$\Rightarrow x = 0, \frac{\lambda}{2}, \lambda, \dots, \frac{n\lambda}{2}$	$\Rightarrow x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \dots$
where $k = \frac{2\pi}{\lambda}$ and $n = 0, 1, 2, 3, \dots$	where $k = \frac{2\pi}{\lambda}$ and $n = 1, 2, 3, \dots$
Amplitude is minimum when $\cos kx = 0$	Amplitude is minimum when $\sin kx = 0$
$\Rightarrow kx = \frac{\pi}{2}, \frac{3\pi}{2}, \dots, \frac{(2n-1)\pi}{2}$	$\Rightarrow kx = \frac{\pi}{2}, \frac{3\pi}{2}, \dots, \frac{(2n-1)\pi}{2}$
$\Rightarrow x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \dots$	$\Rightarrow x = 0, \frac{\lambda}{2}, \lambda, \dots, \frac{n\lambda}{2}$

(5) **Nodes (N)** : The points where amplitude is minimum are called nodes.

(i) Distance between two successive nodes is $\frac{\lambda}{2}$.

(ii) Nodes are at permanent rest.

(iii) At nodes air pressure and density both are high.

(6) **Antinodes (A)** : The points of maximum amplitudes are called antinodes.

(i) The distance between two successive antinodes is $\frac{\lambda}{2}$

(ii) At nodes air pressure and density both are low.

(iii) The distance between a node (N) and adjoining antinode (A) is

$$\frac{\lambda}{4}.$$

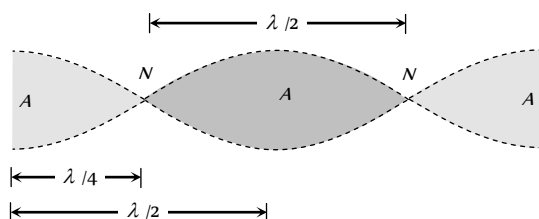


Fig. 17.22

Characteristics of Standing Waves

(1) Standing waves can be transverse or longitudinal.

(2) The disturbance confined to a particular region between the starting point and reflecting point of the wave.

(3) There is no forward motion of the disturbance from one particle to the adjoining particle and so on, beyond this particular region.

(4) The total energy associated with a stationary wave is twice the energy of each of incident and reflected wave. As in stationary waves nodes are permanently at rest. So no energy can be transmitted across then i.e. energy of one region (segment) is confined in that region. However this energy oscillates between elastic potential energy and kinetic energy of particles of the medium.

(5) The medium splits up into a number of segments. Each segment is vibrating up and down as a whole.

(6) All the particles in one particular segment vibrate in the same phase. Particles in two consecutive segments differ in phase by 180° .

(7) All the particles except those at nodes, execute simple harmonic motion about their mean position with the same time period.

(8) The amplitude of vibration of particles varies from zero at nodes to maximum at antinodes ($2a$).

(9) All points (except nodes) pass their mean position twice in one time period.

(10) Velocity of particles while crossing mean position varies from maximum ($\omega A_{SW} = \omega \cdot 2a$) at antinodes to zero at nodes.

(11) In standing waves, if amplitude of component waves are not equal. Resultant amplitude at nodes will be minimum (but not zero). Therefore, some energy will pass across nodes and waves will be partially standing.

(12) Application of stationary waves

(i) Vibration in stretched string

(ii) Vibration in organ pipes (closed and open)

(iii) Kundt's tube

Table 17.4 : Progressive v/s stationary wave

Progressive wave	Stationary wave
These waves transfers energy	These wave does not transfers energy
All particles have the same amplitude	Between a node and an antinode all particles have different amplitudes
Over one wavelength span all particles have difference phase	Between a node and an antinodes all particles have same phase.
No point is at rest	Nodes are always at rest
All particles do not cross the mean position simultaneously.	All particles cross the mean position simultaneously.

Terms Related to the Application of Stationary Wave

(1) **Note** : Any musical sound produced by the simple harmonic oscillations of the source is called note.

(2) **Tone** : Every musical sound consists of a number of components of different frequencies every component is known as a Tone.

(3) **Fundamental note and fundamental frequency** : The note of lowest frequency produced by an instrument is called fundamental note. The frequency of this note is called fundamental frequency.

(4) **Harmonics** : The frequency which are the integral multiple of the fundamental frequency are known as harmonics e.g. if n be the fundamental frequency, then the frequencies $n, 2n, 3n, \dots$ are termed as first, second, third harmonics.

(5) **Overtone** : The harmonics other than the first (fundamental note) which are actually produced by the instrument are called overtones. e.g. the tone with frequency immediately higher than the fundamental is defined as first overtone.

(6) **Octave** : The tone whose frequency is doubled the fundamental frequency is defined as Octave.

(i) If $n = 2n$ it means n is an octave higher than n or n is an octave lower than n .

(ii) If $n_2 = 2^3 n_1$, it means n is 3-octave higher or n is 3-octave lower.

(iii) Similarly if $n_2 = 2^n n_1$ it means n is n -octave higher n is n octave lower.

(7) **Unison** : If the interval is one *i.e.* two frequencies are equal then vibrating bodies are said to be in unison.

(8) **Resonance** : The phenomenon of making a body vibrate with its natural frequency under the influence of another vibrating body with the same frequency is called resonance.

Standing Waves on a String

(i) Consider a string of length l , stretched under tension T between two fixed points.

(2) If the string is plucked and then released, a transverse harmonic wave propagate along its length and is reflected at the end.

(3) The incident and reflected waves will superimpose to produce transverse stationary waves in a string.

(4) Nodes (N) are formed at rigid end and antinodes (A) are formed in between them.

(5) Number of antinodes = Number of nodes - 1

(6) Velocity of wave (incident or reflected wave) is given by $v = \sqrt{\frac{T}{m}}$; m = Mass per unit length of the wire

(7) Frequency of vibration (n) = Frequency of wave

$$= \frac{v}{\lambda} = \frac{1}{\lambda} \sqrt{\frac{T}{m}}$$

(8) For obtaining p loops (p -segments) in string, it has to be plucked at a distance $\frac{l}{2p}$ from one fixed end.

(9) **Fundamental mode of vibration**

(i) Number of loops $p = 1$

(ii) Plucking at $\frac{l}{2}$ (from one fixed end)

(iii) $l = \frac{\lambda_1}{2} \Rightarrow \lambda_1 = 2l$

(iv) Fundamental frequency or first harmonic

$$n_1 = \frac{1}{\lambda_1} \sqrt{\frac{T}{m}} = \frac{1}{2l} \sqrt{\frac{T}{m}}$$

(10) **Second mode of vibration** (First over tone or second harmonic)

(i) Number of loops $p = 2$

(ii) Plucking at $\frac{l}{2 \times 2} = \frac{l}{4}$

Fig. 17.24

(from one fixed end)

(iii) $l = \lambda_2$

(iv) Second harmonic or first over tone

$$n_2 = \frac{1}{\lambda_2} \sqrt{\frac{T}{m}} = \frac{1}{l} \sqrt{\frac{T}{m}} = 2n_1$$

(11) **Third normal mode of vibration** (Second over tone or third harmonic)

(i) Number of loops $p = 3$

(ii) Plucking at $\frac{l}{2 \times 3} = \frac{l}{6}$

(from one fixed end)

(iii) $l = \frac{3\lambda_3}{2} \Rightarrow \lambda_3 = \frac{2l}{3}$

(iv) Third harmonic or second over tone

$$n_3 = \frac{1}{\lambda_3} \sqrt{\frac{T}{m}} = \frac{3}{2l} \sqrt{\frac{T}{m}} = 3n_1$$

(12) **More about string vibration**

(i) In general, if the string is plucked at length $\frac{l}{2p}$, then it vibrates in p segments (loops) and we have the p harmonic is give

$$f_p = \frac{p}{2l} \sqrt{\frac{T}{m}}$$

(ii) All even and odd harmonics are present. Ratio of harmonic = 1 : 2 : 3

(iii) Ratio of over tones = 2 : 3 : 4

(iv) General formula for wavelength $\lambda = \frac{2l}{N}$; where $N = 1, 2, 3, \dots$ correspond to 1st, 2nd, 3rd modes of vibration of the string.

(v) General formula for frequency $n = N \times \frac{v}{2l}$

(vi) **Position of nodes** : $x = 0, \frac{l}{N}, \frac{2l}{N}, \frac{3l}{N}, \dots, l$

(vii) **Position of antinodes** : $x = \frac{l}{2N}, \frac{3l}{2N}, \frac{5l}{2N}, \dots, \frac{(2N-1)l}{2N}$

Melde's Experiment

(1) It is an experimental representation of transverse stationary wave.

(2) In Melde's experiment, one end of a flexible piece of string is tied to the end of a tuning fork. The other end passes over a smooth pulley carries a suitable load.

(3) If p is the number of loop's formed in stretched string and T is the tension in the string then Melde's law is $p\sqrt{T} = \text{constant}$

$$\Rightarrow \frac{p_1}{p_2} = \sqrt{\frac{T_2}{T_1}} \quad (\text{For comparing two cases})$$

Table 17.5 ; Two arrangements of connecting a string to turning fork

Transversely	Example
Prongs of tuning fork vibrates at right angles to the thread.	Prongs vibrated along the length of the thread.
Frequency of vibration of turning fork : frequency of vibration of the thread.	Frequency of turning fork = 2 × (Frequency of vibration of thread)
If number of loops in string is p then $l = \frac{p\lambda}{2} \Rightarrow \lambda = \frac{2l}{p}$	If number of loop so in string is p then $l = \frac{p\lambda}{2} \Rightarrow \lambda = \frac{2l}{p}$
$\Rightarrow$ Frequency of string $= \frac{v}{\lambda} = \frac{p}{2l} \sqrt{\frac{T}{m}} \left(\because v = \sqrt{\frac{T}{m}} \right)$	$\Rightarrow$ Frequency of string $= \frac{v}{\lambda} = \frac{p}{2l} \sqrt{\frac{T}{m}}$
$\Rightarrow$ Frequency of tuning fork $= \frac{p}{2l} \sqrt{\frac{T}{m}}$	$\Rightarrow$ Frequency of turning fork $I = \frac{p}{l} \sqrt{\frac{T}{m}}$
$\Rightarrow$ If $l, m, n \rightarrow$ constant then $p\sqrt{T} = \text{constant}$	$\Rightarrow$ If $l, m, n \rightarrow$ constant then $p\sqrt{T} = \text{constant}$

Sonometer

(1) It is an apparatus, used to produce resonance (matching frequency) of tuning fork (or any source of sound) with stretched vibrating string.

(2) It consists of a hollow rectangular box of light wood. The experimental fitted on the box as shown.

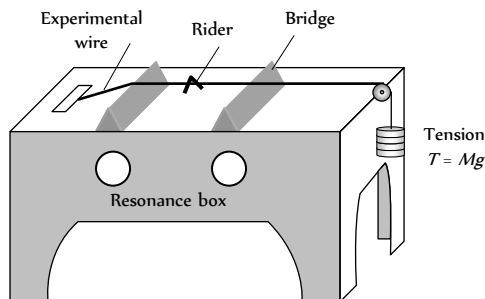


Fig. 17.26

(3) The box serves the purpose of increasing the loudness of the sound produced by the vibrating wire.

(4) If the length of the wire between the two bridges is l , then the frequency of vibration is $n = \frac{1}{2l} \sqrt{\frac{T}{m}} = \sqrt{\frac{T}{\pi r^2 d}}$

(r = Radius of the wire, d = Density of material of wire) m = mass per unit length of the wire)

(5) **Resonance** : When a vibrating tuning fork is placed on the box, and if the length between the bridges is properly adjusted then if $(n)_{\text{Fork}} = (n)_{\text{String}} \rightarrow$ rider is thrown off the wire.

(6) **Laws of string**

(i) **Law of length** : If T and m are constant then $n \propto \frac{1}{l}$

$$\Rightarrow n l = \text{constant} \Rightarrow n_1 l_1 = n_2 l_2$$

(ii) **Law of mass** : If T and l are constant then $n \propto \frac{1}{\sqrt{m}}$

$$\Rightarrow n \sqrt{m} = \text{constant} \Rightarrow \frac{n_1}{n_2} = \sqrt{\frac{m_2}{m_1}}$$

(iii) **Law of density** : If T, l and r are constant then $n \propto \frac{1}{\sqrt{d}}$

$$\Rightarrow n \sqrt{d} = \text{constant} \Rightarrow \frac{n_1}{n_2} = \sqrt{\frac{d_2}{d_1}}$$

(iv) **Law of tension** : If l and m are constant then $n \propto \sqrt{T}$

$$\Rightarrow \frac{n}{\sqrt{T}} = \text{constant} \Rightarrow \frac{n_1}{n_2} = \sqrt{\frac{T_2}{T_1}}$$

Vibration of Composite Strings

Suppose two strings of different material and lengths are joined end to end and tied between clamps as shown. Now after plucking, stationary waves are established only at those frequencies which matches with any one harmonic of both the independent string S and S' .

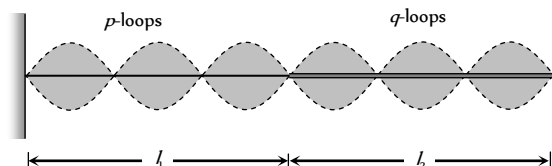


Fig. 17.27

As the frequency of the wave in both strings must be same so

$$\frac{p}{2l_1} = \sqrt{\frac{T}{m_1}} = \frac{q}{2l_2} \sqrt{\frac{T}{m_2}} \Rightarrow \frac{p}{q} = \frac{l_1}{l_2} \sqrt{\frac{m_1}{m_2}} = \frac{l_1}{l_2} \sqrt{\frac{\rho_1}{\rho_2}}$$

Standing Wave in a Organ Pipe

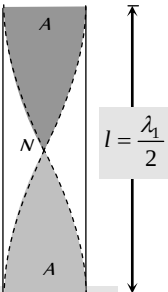
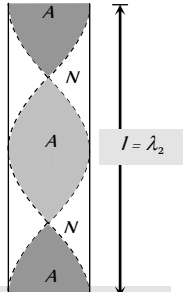
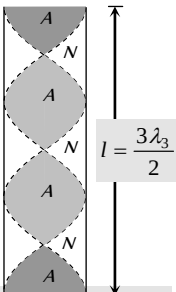
Organ pipes are the musical instrument which are used for producing musical sound by blowing air into the pipe. Longitudinal stationary waves are formed on account of superimposition of incident and reflected longitudinal waves.

$$\text{Equation of standing wave } y = 2a \cos \frac{2\pi vt}{\lambda} \sin \frac{2\pi x}{\lambda}$$

$$\text{Frequency of vibration } n = \frac{v}{\lambda}$$

Table 17.6 : Different mode of vibration in organ pipe

Closed organ pipe		
Fundamental mode	Third harmonic First over tone	Fifth harmonic Second over tone

$n_1 = \frac{v}{4l}$	$n_2 = \frac{v}{\lambda_2} = \frac{3v}{4l} = 3n_1$	$n_3 = \frac{5v}{4l} = 5n_1$
Open organ pipe		
Fundamental mode	Second harmonic	Third harmonic
		
$n_1 = \frac{v}{\lambda_1} = \frac{v}{2l}$	$n_2 = \frac{v}{\lambda_2} = \frac{v}{L} = 2n_1$	$n_3 = \frac{v}{\lambda_3} = \frac{3v}{2l} = 3n_1$

(1) Closed organ pipe

(i) In closed organ pipe only odd harmonic are present. Ratio of harmonic is $n : n_1 : n_2 \dots = 1 : 3 : 5 \dots$

(ii) p overtone = $(2p + 1)$ harmonics

(iii) Ratio of overtones = $3 : 5 : 7 \dots$

(iv) The maximum possible wavelength is $4l$

(v) General formula for wavelength is $\lambda = \frac{4l}{(2N - 1)}$; where $N = 1, 2, 3, \dots$ corresponds to order of mode of vibration.

(vi) General formula for frequency $n = \frac{(2N - 1)v}{4l}$

(vii) Position of nodes from closed end $x = 0, \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2} \dots$

(viii) Position of antinodes from closed end $x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4} \dots$

(2) Open organ pipe

(i) In open organ pipe all (even and odd) harmonic are present. Ratio of harmonic is $n : n_1 : n_2 \dots = 1 : 2 : 3 \dots$

(ii) p overtone = $(p + 1)$ harmonics

(iii) Ratio of overtones = $2 : 3 : 5 \dots$

(iv) The maximum possible wavelength is $2l$

(v) General formula for wavelength is $\lambda = \frac{2l}{N}$; where $N = 1, 2, 3, \dots$ corresponds to order of mode of vibration.

(vi) General formula for frequency $n = \frac{Nv}{2l}$

(vii) Position of nodes from one end $x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4} \dots$

(viii) Position of antinodes from one end $x = 0, \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2} \dots$

Tuning Fork

(1) The tuning fork is a metallic device that produces sound of a single frequency.

(2) A tuning fork is really a transversely vibrating rod of rectangular cross-section bent into the shape of U as shown.

(3) The prongs execute transverse vibrations and the stem executes the longitudinal vibration. Both vibrate with the same frequency.

(4) The phase difference between the vibrations produced by both prongs of tuning fork is zero.

(5) Tuning forks are generally taken as the standards of frequency of pure notes.



Fig. 17.28

The frequency of the tuning fork is given by $n \propto \frac{t}{l^2} \sqrt{\frac{Y}{\rho}}$

where t = Thickness of the prongs, l = Length of the prongs, Y = Young's modulus of elasticity and ρ = Density of the material of tuning fork.

(6) If one prong is broken tuning fork does not vibrate.

Effect on frequency of tuning fork

(i) A fork of shorter prongs gives high frequency tone

(ii) The frequency of a tuning fork decreases when it's prongs are loaded (say with wax) near the end.

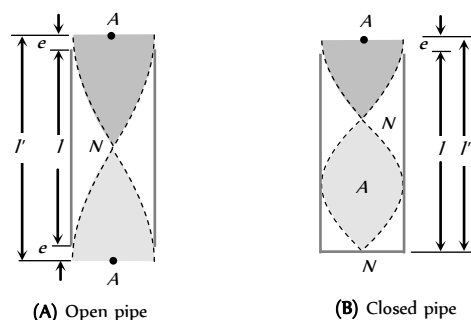
(iii) The frequency of tuning fork increases when prongs are filed near the ends.

(iv) The frequency of a tuning fork decreases if temperature of the fork is increases.

End Correction

Due to finite momentum of air molecules in organ pipe reflection takes place not exactly at open end but some what above it. Hence antinode is not formed exactly at the open end rather it is formed at a little distance away from open end outside it.

The distance of antinode from the open end is known as end correction (e). It is given by $e = 0.6 r$ where r = radius of pipe.

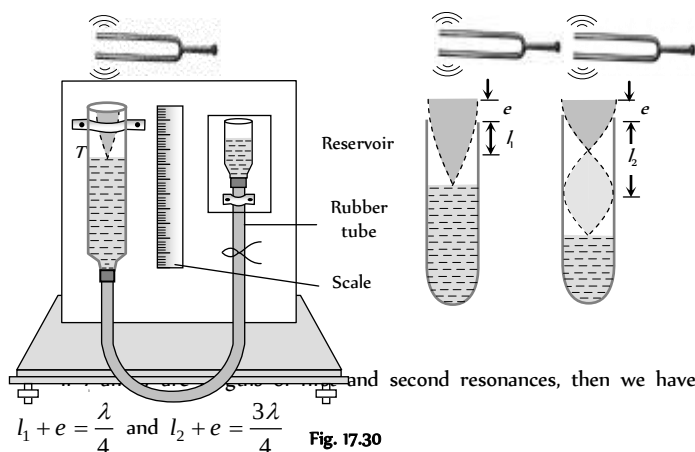
Fig. 17.29
Effect length in open organ pipe $l' = (l + 2e)$

Effect length in closed organ pipe $l = (l + e)$

Resonance Tube

It is used to determine velocity of sound in air by the help of a tuning fork of known frequency.

It is a closed organ pipe having an air column of variable length. When a tuning fork is brought over its mouth. Its air column vibrates with the frequency of the fork. If the length of the air column is varied until its natural frequency equals the frequency of the fork, then the column resonates and emits a loud note.



$$l_1 + e = \frac{\lambda}{4} \text{ and } l_2 + e = \frac{3\lambda}{4}$$

Fig. 17.30

$$\Rightarrow l_2 - l_1 = \frac{\lambda}{2} \Rightarrow \lambda = 2(l_2 - l_1)$$

Speed of sound in air at room (temperature) $v = n\lambda = 2n(l_2 - l_1)$

Also $\frac{l_2 + e}{l_1 + e} = 3 \Rightarrow l_2 = 3l_1 + 2e$ i.e. second resonance is obtained

at length more than thrice the length of first resonance.

Kundt's Tube

The apparatus consists of a long glass tube about 5 cm in diameter, fixed horizontally. A metal rod R clamped firmly at the centre is mounted so that its one end carrying a light disc P_1 (of cork or card board) projects some distance into the glass tube. The other end of the glass tube is closed with a moveable piston P_2 . Any desired length of the air or gas can be enclosed in between the two discs P_1 and P_2 . A small amount of dry lycopodium powder or cork dust is spread along base of the entire length of the tube.

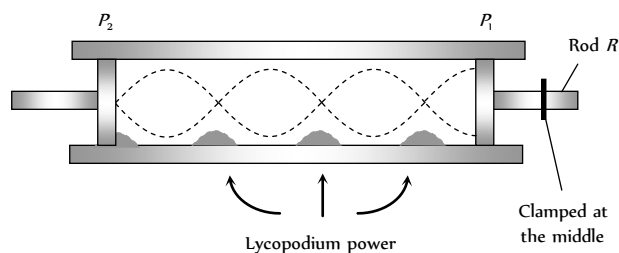


Fig. 17.31

The free end of the metal rod R is rubbed (stroked) along the length with resined cloth. The rod begins to vibrate longitudinally and emits a very high pitched shrill note. These vibrations are impressed upon the air column in the tube through disc P_1 . Let disc P_1 is so adjusted, that the

stationary waves are formed in the air (gas) column in the tube. At antinodes powder is set into oscillations vigorously while it remains unaffected at nodes. Heaps of powder are formed at nodes.

Let n is the frequency of vibration of the rod then, this is also the frequency of sound wave in the air column in the tube.

$$\text{For rod : } \frac{\lambda_{\text{rod}}}{2} = l_{\text{rod}}, \quad \text{For air : } \frac{\lambda_{\text{air}}}{2} = l_{\text{air}}$$

where l is the distance between two heaps of powder in the tube (i.e. distance between two nodes). If v_{air} and v_{rod} are velocity of sound waves in the air and rod respectively, then

$$n = \frac{v_{\text{air}}}{\lambda_{\text{air}}} = \frac{v_{\text{rod}}}{\lambda_{\text{rod}}}. \text{ Therefore } \frac{v_{\text{air}}}{v_{\text{rod}}} = \frac{\lambda_{\text{air}}}{\lambda_{\text{rod}}} = \frac{\lambda_{\text{air}}}{\lambda_{\text{rod}}}$$

Thus knowledge of v_{rod} , determines v_{air}

Kundt's tube may be used for

- Comparison of velocities of sound in different gases.
- Comparison of velocities of sound in different solids
- Comparison of velocities of sound in a solid and in a gas.
- Comparison of density of two gases.
- Determination of γ of a gas.
- Determination of velocity of sound in a liquid.

Beats

When two sound waves of slightly different frequencies, travelling in a medium along the same direction, superimpose on each other, the intensity of the resultant sound at a particular position rises and falls regularly with time. This phenomenon of regular variation in intensity of sound with time at a particular position is called beats.

(1) **Persistence of hearing** : The impression of sound heard by our ears persist in our mind for $1/10^{\text{th}}$ of a second. If another sound is heard before $1/10^{\text{th}}$ second is over, the impression of the two sound mix up and our mind cannot distinguish between the two.

So for the formation of distinct beats, frequencies of two sources of sound should be nearly equal (difference of frequencies less than 10)

(2) **Equation of beats** : If two waves of equal amplitudes ' a ' and slightly different frequencies n_1 and n_2 travelling in a medium in the same direction are.

$$y_1 = a \sin \omega_1 t = a \sin 2\pi n_1 t; \quad y_2 = a \sin \omega_2 t = a \sin 2\pi n_2 t$$

$$\text{By the principle of super position : } \vec{y} = \vec{y}_1 + \vec{y}_2$$

$y = A \sin \pi(n_1 + n_2)t$ where $A = 2a \cos \pi(n_1 - n_2)t$ = Amplitude of resultant wave.

(3) **One beat** : If the intensity of sound is maximum at time $t = 0$, one beat is said to be formed when intensity becomes maximum again after becoming minimum once in between.

(4) **Beat period** : The time interval between two successive beats (i.e. two successive maxima of sound) is called beat period.

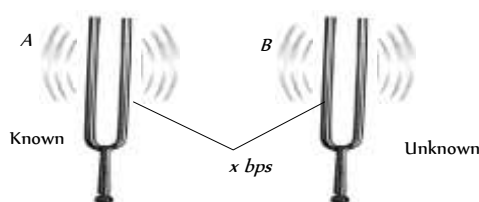
$$n = n_1 \sim n_2$$

(5) **Beat frequency** : The number of beats produced per second is called beat frequency.

$$T = \frac{1}{\text{Beat frequency}} = \frac{1}{n_1 \sim n_2}$$

Determination of Unknown Frequency

Suppose a tuning fork of known frequency (n) is sounded together with another tuning fork of unknown frequency (n) and x beats heard per second.



There are two possibilities Fig. 17.32

$$n_A - n_B = x \quad \dots (i)$$

or $n_B - n_A = x \quad \dots (ii)$

To find the frequency of unknown tuning fork (n) following steps are taken.

(1) Loading or filing of one prong of known or unknown (by wax) tuning fork, so frequency changes (decreases after loading, increases after filing).

(2) Sound them together again, and count the number of heard beats per sec again, let it be x' . These are following four condition arises.

$$(i) \ x' > x \quad (ii) \ x' < x \quad (iii) \ x' = 0 \quad (iv) \ x' = x$$

(3) With the above information, the exact frequency of the unknown tuning fork can be determined as illustrated below.

Suppose two tuning forks A (frequency n_i is known) and B (frequency n_i is unknown) are sounded together and gives x beats/sec. If one prong of unknown tuning fork B is loaded with a little wax (so n_i decreases) and it is sounded again together with known tuning fork A , then in the following four given condition n_i can be determined.

(4) If $x' > x$ than x , then this would happen only when the new frequency of B is more away from n_i . This would happen if originally (before loading), n_i was less than n_i .

$$\text{Thus initially } n_i = n_i - x.$$

(5) If $x' < x$ than x , then this would happen only when the new frequency of B is more nearer to n_i . This would happen if originally (before loading), n_i was more than n_i .

$$\text{Thus initially } n_i = n_i + x.$$

(6) If $x' = x$ then this would means that the new frequency (after loading) differs from n_i by the same amount as was the old frequency (before loading). This means initially $n_i = n_i + x$

$$(\text{and now it has decreased to } n'_i = n_i - x)$$

(7) If $x' = 0$, then this would happen only when the new frequency of B becomes equal to n_i . This would happen if originally n_i was more than n_i .

$$\text{Thus initially } n_i = n_i + x.$$

Table 17.7 ; Frequency of unknown tuning fork for various cases

If remains same $n_B = n_A + x$	If remains same $n_B = n_A - x$
If x becomes zero $n_B = n_A + x$	If x becomes zero $n_B = n_A - x$
By filing	
If B is filed, its frequency increases	If A is filed, its frequency increases
If x increases $n_B = n_A + x$	If x increases $n_B = n_A - x$
If x decrease $n_B = n_A - x$	If x decrease $n_B = n_A + x$
If remains same $n_B = n_A - x$	If remains same $n_B = n_A + x$
If x becomes zero $n_B = n_A - x$	If x becomes zero $n_B = n_A + x$

Doppler's Effect



Whenever there is a relative motion between a source of sound and the observer (listener), the frequency of sound heard by the observer is different from the actual frequency of sound emitted by the source.

The frequency observed by the observer is called the apparent frequency. It may be less than or greater than the actual frequency emitted by the sound source. The difference depends on the relative motion between the source and observer.

(1) When observer and source are stationary

(i) Sound waves propagate in the form of spherical wavefronts (shown as circles)

(ii) The distance between two successive circles is equal to wavelength λ .

(iii) Number of waves crossing the observer = Number of waves emitted by the source

(iv) Thus apparent frequency (n') = actual frequency (n).

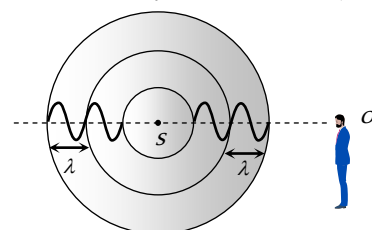


Fig. 17.33

(2) When source is moving but observer is at rest

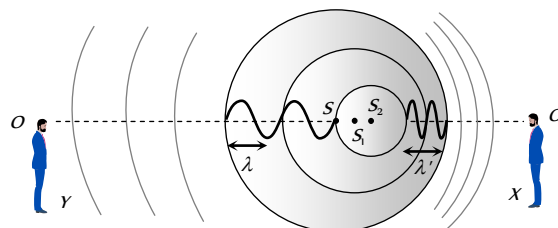


Fig. 17.34

- S, S, S are the positions of the source at three different positions.
- Waves are represented by non-concentric circles, they appear compressed in the forward direction and spread out in backward direction.
- For observer (X)

$$\text{Apparent wavelength } \lambda' < \text{Actual wavelength } \lambda$$

$\Rightarrow$ Apparent frequency $n' >$ Actual frequency n

For observer (Y) : $\lambda' > \lambda \Rightarrow n' < n$

(3) When source is stationary but observer is moving

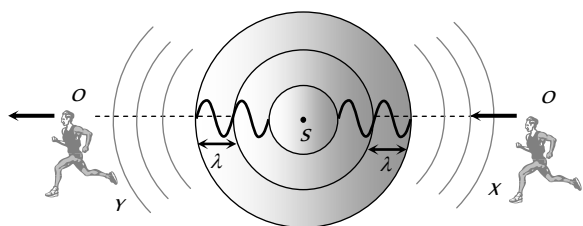


Fig. 17.35

(i) Waves are again represented by concentric circles.

(ii) No change in wavelength received by either observer X or Y.

(iii) Observer X (moving towards) receives wave fronts at shorter interval thus $n' > n$.

(iv) Observer Y receives wavelengths at longer interval thus $n' < n$.

(4) **General expression for apparent frequency** : Suppose observed (O) and source (S) are moving in the same direction along a line with velocities v and v_s respectively. Velocity of sound is v and velocity of medium is v_m then apparent frequency observed by observer is given by

$$n' = \left[\frac{(v + v_m) - v_0}{(v + v_m) - v_s} \right] n$$

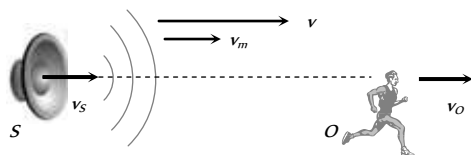


Fig. 17.36

If medium is stationary i.e. $v_m = 0$ then $n' = n \left[\frac{v - v_0}{v - v_s} \right]$

Sign convention for different situation

- (i) The direction of v is always taken from source to observer.
- (ii) All the velocities in the direction of v are taken positive.
- (iii) All the velocities in the opposite direction of v are taken negative.

Common Cases in Doppler's Effect

Case 1 : Source is moving but observer at rest.

(1) Source is moving towards the observer

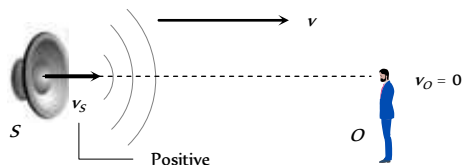


Fig. 17.37

$$\text{Apparent frequency } n' = n \left[\frac{v - 0}{v - (+v_s)} \right] = n \left[\frac{v}{v - v_s} \right]$$

$$\text{Apparent wavelength } \lambda' = \lambda \left(\frac{v - v_s}{v} \right)$$

(2) Source is moving away from the observer.

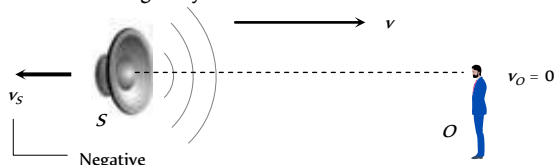


Fig. 17.38

$$\text{Apparent frequency } n' = n \left[\frac{v - 0}{v - (-v_s)} \right] = n \left[\frac{v}{v + v_s} \right]$$

$$\text{Apparent wavelength } \lambda' = \lambda \left(\frac{v + v_s}{v} \right)$$

Case 2: Source is at rest but observer is moving.

(1) Observer is moving towards the source.

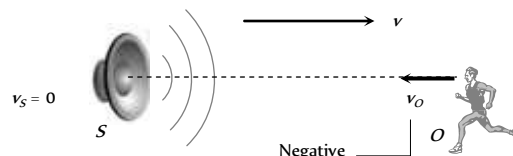


Fig. 17.39

$$\text{Apparent frequency } n' = n \left[\frac{v - (-v_o)}{v - 0} \right] = n \left[\frac{v + v_o}{v} \right]$$

$$\text{Apparent wavelength } \lambda' = \frac{(v + v_o)}{n'} = \frac{(v + v_o)}{n \left(\frac{v + v_o}{v} \right)} = \frac{v}{n} = \lambda$$

(2) Observer is moving away from the source

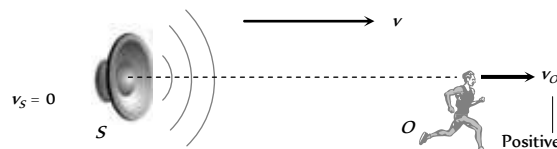


Fig. 17.40

$$\text{Apparent frequency } n' = n \left[\frac{v - (+v_o)}{v - 0} \right] = n \left[\frac{v - v_o}{v} \right]$$

$$\text{Apparent wavelength } \lambda' = \lambda$$

Case 3: When source and observer both are moving

(1) When both are moving towards each other

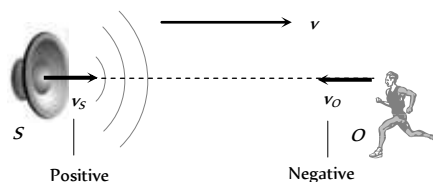


Fig. 17.41

$$(i) \text{ Apparent frequency } n' = n \left[\frac{v - (-v_o)}{v - (+v_s)} \right] = n \left[\frac{v + v_o}{v - v_s} \right]$$

$$(ii) \text{ Apparent wavelength } \lambda' = \lambda \left(\frac{v - v_s}{v} \right)$$

(iii) Velocity of wave with respect to observer $= (v + v_o)$

(2) When both are moving away from each other.

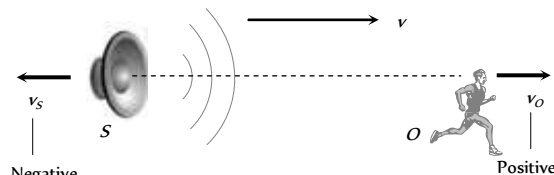


Fig. 17.42

$$(i) \text{ Apparent frequency } n' = n \left[\frac{v - (+v_O)}{v - (-v_S)} \right] = n \left[\frac{v - v_O}{v + v_S} \right]$$

$$(n' < n)$$

$$(ii) \text{ Apparent wavelength } \lambda' = \lambda \left(\frac{v + v_S}{v} \right)$$

$$(\lambda' > \lambda)$$

Velocity of waves with respect to observer = $(v - v)$

(3) When source is moving behind observer

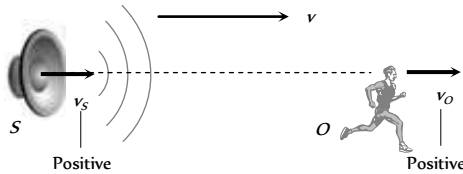


Fig. 17.43

$$(i) \text{ Apparent frequency } n' = n \left(\frac{v - v_O}{v - v_S} \right)$$

(a) If $v_O < v_S$, then $n' > n$

(b) If $v_O > v_S$ then $n' < n$

(c) If $v_O = v_S$ then $n' = n$

$$(ii) \text{ Apparent wavelength } \lambda' = \lambda \left(\frac{v - v_S}{v} \right)$$

(iii) Velocity of waves with respect to observer = $(v - v_O)$

(4) When observer is moving behind the source

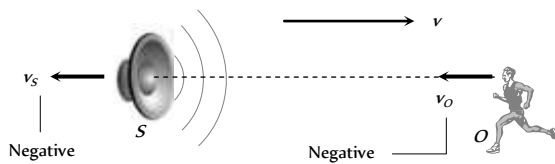


Fig. 17.44

$$(i) \text{ Apparent frequency } n' = n \left(\frac{v - (-v_O)}{v - (-v_S)} \right)$$

(a) If $v_O > v_S$, then $n' > n$

(b) If $v_O < v_S$ then $n' < n$

(c) If $v_O = v_S$ then $n' = n$

$$(ii) \text{ Apparent wavelength } \lambda' = \lambda \left(\frac{v + v_S}{v} \right)$$

(iii) The velocity of waves with respect to observer = $(v + v_O)$

Case 4: Crossing

(1) Moving sound source crosses a stationary observer

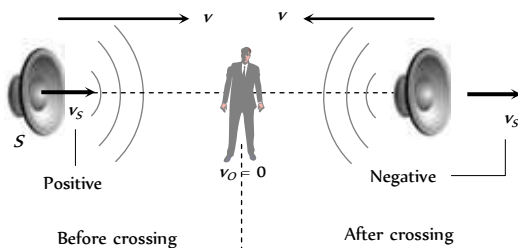


Fig. 17.45

Apparent frequency before crossing

$$n'_{\text{Before}} = n \left[\frac{v - 0}{v - (+v_S)} \right] = n \left[\frac{v}{v - v_S} \right]$$

Apparent frequency

$$n'_{\text{After}} = n \left[\frac{v - 0}{v - (-v_S)} \right] = n \left[\frac{v}{v + v_S} \right]$$

$$\text{Ratio of two frequency } \frac{n'_{\text{Before}}}{n'_{\text{After}}} = \left[\frac{v + v_S}{v - v_S} \right] > 1$$

$$\text{Change in apparent frequency } n'_{\text{Before}} - n'_{\text{After}} = \frac{2nv_S v}{(v^2 - v_S^2)}$$

$$\text{If } v_S \ll v \text{ then } n'_{\text{Before}} - n'_{\text{After}} = \frac{2nv_S}{v}$$

(2) Moving observer crosses a stationary source

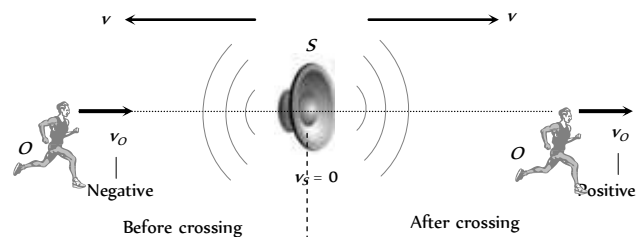


Fig. 17.46

Apparent frequency before crossing

$$n'_{\text{Before}} = n \left[\frac{v - (-v_O)}{v - 0} \right] = n \left[\frac{v + v_O}{v} \right]$$

Apparent frequency

$$n'_{\text{After}} = n \left[\frac{v - (+v_O)}{v - 0} \right] = n \left[\frac{v - v_O}{v} \right]$$

$$\text{Ratio of two frequency } \frac{n'_{\text{Before}}}{n'_{\text{After}}} = \left[\frac{v + v_S}{v - v_S} \right]$$

$$\text{Change in apparent frequency } n'_{\text{Before}} - n'_{\text{After}} = \frac{2nv_O}{v}$$

Case 5: Both moves in the same direction with same velocity $n' = n$, i.e. there will be no Doppler effect because relative motion between source and listener is zero.

Case 6: Source and listener moves at right angle to the direction of wave propagation. $n' = n$

It means there is no change in frequency of sound heard if there is a small displacement of source and listener at right angle to the direction of wave propagation but for a large displacement the frequency decreases because the distance between source of sound and listener increases.

Some Typical Cases of Doppler's Effect

(1) **Moving car towards wall :** When a car is moving towards a stationary wall as shown in figure. If the car sounds a horn, wave travels toward the wall and is reflected from the wall. When the reflected wave is

heard by the driver, it appears to be of relatively high pitch. If we wish to measure the frequency of reflected sound then the problem.

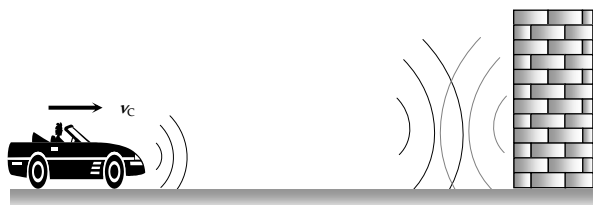


Fig. 17.47

Can be solved in a different manner by using method of sound images. In this procedure we assume the image of the sound source behind the reflector.

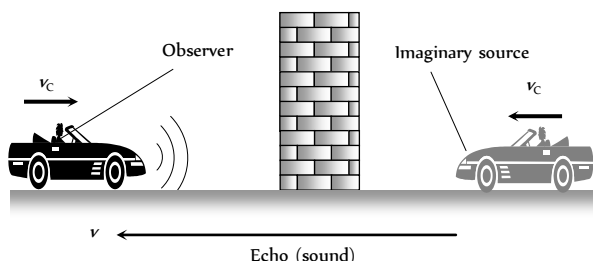


Fig. 17.48

Here we assume that the sound which is reflected by the stationary wall is coming from the image of car which is at the back of it and coming toward it with velocity v . Now the frequency of sound heard by car driver can directly be given as

$$n' = n \left[\frac{v - (-v_c)}{v - (+v_c)} \right] = n \left[\frac{v + v_c}{v - v_c} \right]$$

This method of images for solving problems of Doppler effect is very convenient but is used only for velocities of source and observer which are very small compared to the speed of sound and it should not be used frequently when the reflector of sound is moving.

(2) **Moving target :** Let a sound source S and observer O are at rest (stationary). The frequency of sound emitted by the source is n and velocity of waves is v .

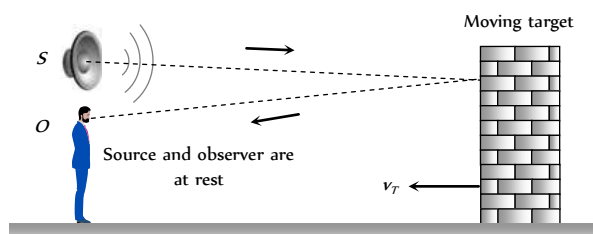


Fig. 17.49

A target is moving towards source and observer, with a velocity v . Our aim is to find out the frequency observed by the observer, for the waves reaching it after reflection from the moving target. The formula is derived by applying Doppler equations twice, first with the target as observer and then with the target as source.

The frequency n' of the waves reaching surface of the moving target (treating it as observer) will be $n' = \left(\frac{v + v_T}{v} \right) n$

Now these waves are reflected by the moving target (which now acts as a source). Therefore the apparent frequency, for the real observer O will

$$\text{be } n'' = \frac{v}{v - v_T} n' \Rightarrow n'' = \frac{v + v_T}{v - v_T} n$$

(i) If the target is moving away from the observer, then

$$n' = \frac{v - v_T}{v + v_T} n$$

(ii) If target velocity is much less than the speed of sound, ($v_T \ll v$),

then $n' = \left(1 + \frac{2v_T}{v} \right) n$, for approaching target

and $n' = \left(1 - \frac{2v_T}{v} \right) n$, for receding target

(3) Transverse Doppler's effect

(i) If a source is moving in a direction making an angle θ w.r.t. the observer

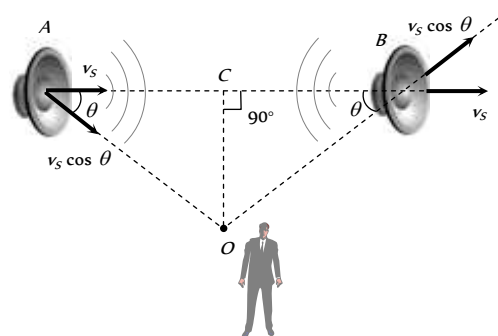


Fig. 17.50

The apparent frequency heard by observer O at rest

$$\text{At point A : } n' = \frac{nv}{v - v_s \cos \theta}$$

As source moves along AB , value of θ increases, $\cos \theta$ decreases, n' goes on decreasing.

At point C : $\theta = 90^\circ$, $\cos \theta = \cos 90^\circ = 0$, $n' = n$.

At point B : the apparent frequency of sound becomes

$$n'' = \frac{nv}{v + v_s \cos \theta}$$

(ii) When two cars are moving on perpendicular roads : When car-1 sounds a horn of frequency n , the apparent frequency of sound heard by car-2 can be given as $n' = n \left[\frac{v + v_2 \cos \theta_2}{v - v_1 \cos \theta_1} \right]$

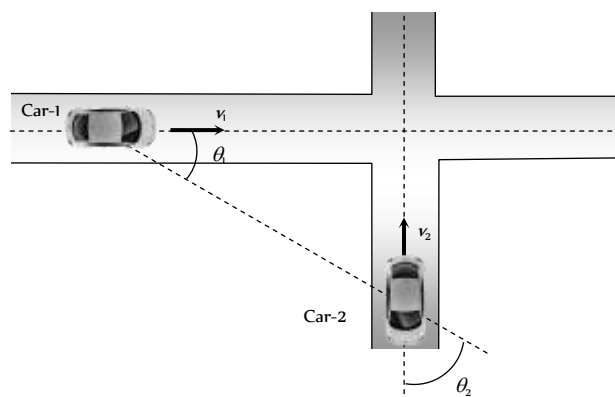


Fig. 17.51

(4) **Rotating source/observer :** Suppose that a source of sound/observer is rotating in a circle of radius r with angular velocity ω (Linear velocity $v = r\omega$)

(i) **When source is rotating**

(a) Towards the observer heard frequency will be maximum

$$\text{i.e. } n_{\max} = \frac{nv}{v - v_s}$$

(b) Away from the observer heard frequency will be minimum

$$\text{and } n_{\min} = \frac{nv}{v + v_s}$$

(c) Ratio of maximum and minimum frequency

$$\frac{n_{\max}}{n_{\min}} = \frac{v + v_s}{v - v_s}$$

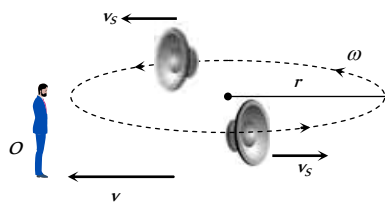


Fig. 17.52

(ii) **When observer is rotating**

(a) Towards the source heard frequency will be maximum

$$\text{i.e. } n_{\max} = \frac{nv}{v - v_o}$$

(b) Away from the source heard frequency will be minimum

$$\text{and } n_{\min} = \frac{nv}{v + v_o}$$

(c) Ratio of maximum and minimum frequency

$$\frac{n_{\max}}{n_{\min}} = \frac{v + v_o}{v - v_o}$$

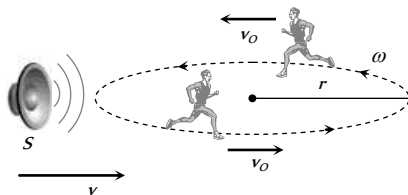


Fig. 17.53

(iii) **Observer is situated at the centre of circle** : There will be no change in frequency of sound heard, if the source is situated at the centre of the circle along which listener is moving..

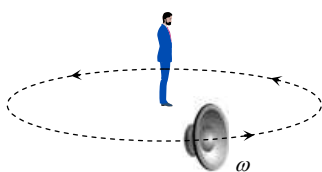


Fig. 17.54

(5) **SONAR** : Sonar means Sound Navigation and Ranging.

(i) Ultrasonic waves are used to detect the presence of big rocks, submarines *etc* in the sea.

(ii) The waves emitted by a source are reflected by the target and received back at the SONAR station.

(iii) If v is velocity of sound waves in water and v_s is velocity of target (submarine), the apparent frequency of reflected waves will be

$$n' = \left(1 \pm \frac{2v_{\text{sub}}}{v} \right) n$$

+ sign is for target approaching the receiver and – sign for target moving away.

Conditions for No Doppler's Effect

(1) When source (S) and listener (L) both are at rest.

(2) When medium alone is moving.

(3) When S and L move in such a way that distance between S and L remains constant.

(4) When source S and listener L , are moving in mutually perpendicular directions.

(5) If the velocity of source and listener is equal to or greater than the sound velocity then Doppler effect is not seen.

Musical Sound



other rapidly at regular interval of time without a sudden change in amplitude.

(1) **Noise** : A noise consists of a series of waves following each other at irregular intervals of time with sudden changes in amplitude.

(2) **Pitch** : The pitch of a sound is the characteristic which distinguishes between a shrill (or sharp) sound and a grave (or flat) sound.

(i) A sound of high pitch is said to be shrill and its frequency is high.

(ii) A sound of low pitch is said to be grave and its frequency is low.

(iii) The pitch of female voice is higher than the pitch of male voice.

(iv) The pitch of sound produced by roaring of lion is lower where as the pitch of sound produced by mosquito whisper is high.

(3) **Quality (or timbre)** : A musical instrument vibrates with many frequencies at the same time. The quality of any musical sound is determined by the number of overtones and their relative intensities.

(i) The quality of sound enables us to distinguish between two sounds having same intensity and pitch.

(ii) The sounds of different instruments (such as Tabla and Mridang) are said to differ in quality.

(iii) Due to quality of sound one can recognise the voice of his friend without seeing him.

(4) **Loudness** : Characteristic of sound, on account of which the sound appears to be intense or slow.

(i) The loudness that we sense is related to the intensity of sound though it is not directly proportional

(ii) The loudness depends on intensity as well as upon the sensitiveness of ear.

(iii) Our perception of loudness is better co-related with the sound level measured in decible (dB) and defined as follows $\beta = 10 \log_{10} \left(\frac{I}{I_0} \right)$;

where I = The minimum intensity that can be heard called threshold of hearing = 10^{-12} W/m^2 at 1 KHz .

(iv) At the threshold of hearing $\beta = 0$. At the threshold of pain $\beta = 10 \log_{10} \frac{1}{10^{-2}} = 120 \text{ dB}$.

(v) When the intensity doubles, the intensity level changes by 3 dB .

(vi) When the intensity increases 10 times the level increases by 10 dB .

Table 17.8 ; Different sound intensity level

Source of sound	dB
-----------------	------

Rustling leaves	10
Whisper	20
Quiet room	30
Normal level of speech (inside)	30
Street traffic (inside car)	65
Riveting tool	80
Thunder	100
Indoor rock concert	120

Interval : The ratio of the frequencies of the two notes is called the interval between them e.g. interval between two notes of frequencies 256 Hz and 512 Hz is 1 : 2.

Table 17.9 : Different interval

Name of interval	Frequency ratio
Unison	1 : 1
Octave	2 : 1
Major tone	9 : 8
Minor tone	10 : 9
Semi tone	16 : 15

Musical scale : It consists of a series of notes of successively increasing frequency, having constant intervals. The note of the lowest frequency is called the key note.

These are many kinds of musical scales. The most commonly used scale is called major diatonic scale. It is formed by introducing six more notes between a given note and its octave, so that these are eight notes in all.

Table 17.10 : Major diatonic scale

Symbol	Indian name	Western name	Frequency in the base of 256 Hz	Interval between successive notes
C	SA	DO	256	
D	RE	RE	288	9/8
E	GA	MI	320	10/9
F	MA	FA	341	16/15
G	PA	SOL	384	9/8
A	DHA	LA	427	10/9
B	Ni	SI	480	9/8
C ₁	SA	DO	512	16/15

Acoustics of Buildings

Acoustics is the branch of physics that deals with the generation, propagation and reception of sound.

W.C. Sabine was the first to carry out the scientific study of architectural acoustics by laying down following rules.

The sound must be loud enough.

The quality of sound must be unaltered.

The successive sounds of speech or music must remain distinct.

These should not be unnecessary interference or resonance of sound in the auditorium.

These should be no echoes in the auditorium.

(1) **Reverberation :** Phenomenon of persistence or prolongation of sound in the auditorium is called reverberation.

(2) **Reverberation time :** The time gap between the initial direct note and the reflected note upto the minimum audibility level is called reverberation time.

(3) **Sabine law :** Sabine derived an expression of the reverberation time which is $t = K \cdot \frac{V}{\alpha S}$; where K is constant, V = Volume of the hall, S =

Surface area exposed to the sound α = Co-efficient of absorption.

(4) **Controlling the reverberation time :** It may be controlled as follows :

By hanging heavy curtains on the doors.

By having few open windows in the hall.

By having large audience.

By using absorbing materials in the walls and roofs of the hall.

Tips & Tricks

✍ In an open pipe all harmonics are present whereas in a closed organ pipe, only alternate harmonics of frequencies $n_1, 3n_1, 5n_1, \dots$ etc. are present.

The harmonics of frequencies

$$2n_1, 4n_1, 6n_1, \dots \text{ are missing.}$$

Hence musical sound produces

by an open organ pipe is

sweeter than that produced by

a closed organ pipe.



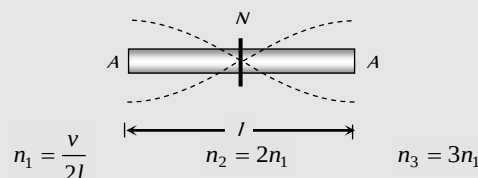
✍ If an open pipe is half submerged in water, it will become a closed organ pipe of length half that of an open pipe. Its fundamental frequency

$$\text{will become } n' = \frac{v}{4\left(\frac{l}{2}\right)} = \frac{v}{2l} = n_1 \text{ i.e., equal to that of open pipe.}$$

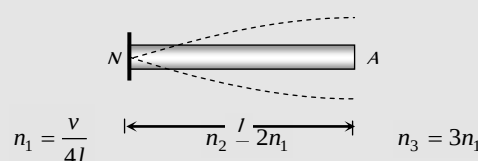
i.e., frequency remains unchanged.

✍ **Vibrating clamped rod :** Frequency of vibration of clamped rod are same as that of organ pipes

Middle clamping : Similar to open organ pipe



End clamping : Similar to closed organ pipe



✍ Sound produced in air is not heard by the diver deep inside the water because most of the sound is reflected from the surface of water

in comparison to the refraction.

✍ If the difference between the apparent frequency of a source of sound as perceived by an observer during its approach and recession is

$$x\% \text{ of the natural frequency of source then speed of } v_s = \frac{v_{\text{sound}}}{200} \times (v^2 \gg v_s^2)$$

✍ In a Tabla, the membrane is loaded about the centre why? a note is musical when it is rich

in harmonics and not partial

overtones. Ordinarily a stretched

membrane vibrates with such

overtones. But when the stretched

membrane is loaded at the centre, its overtones become nearly harmonics, so its sound becomes fairly musical.



✍ All harmonics are overtones but all overtones are not harmonics.

✍ Stethoscope work on the principle of reflection of sound.

✍ Ultrasonic waves can be produced by utilizing piezoelectric effect.

✍ There is no atmosphere on moon, therefore propagation of sound is not possible there. To do conversation on moon, the astronaut uses an instrument which can transmit and detect electromagnetic waves.

✍ Doppler effect gives information regarding the change in frequency only. It does not say about intensity of sound.

✍ Doppler effect in sound is asymmetric but in light it is symmetric.

✍ If three tuning forks of frequencies n , $n + x$ and $n + 2x$ are sounded together to produce waves of equal amplitude these three waves produce beats with beat frequency = x beats/sec

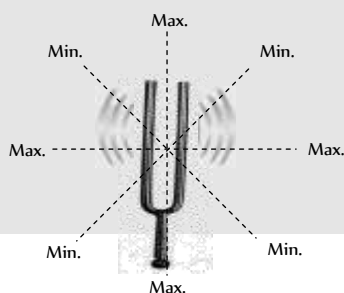


✍ If N tuning forks are so arranged that every fork gives x beats per sec with the next then the frequency of last fork will be

$$n_{\text{last}} = n_{\text{first}} + (N - 1) x$$



✍ If a vibrating tuning fork is rotated about its stem, maximum and minimum number of beats heard by an observer in one revolution of tuning fork are 4.



✍ The tuning of radio receiving set to a particular station is based on forced vibration.

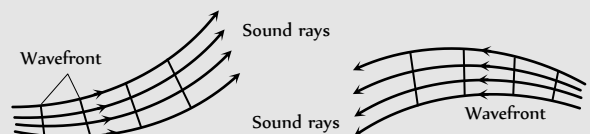
✍ To avoid resonant vibration of the bridge, soldiers are ordered to break steps while crossing a bridge.

✍ **Confusion :** So many students often confuse whether the equation of a plan progressive wave should be

$$y = a \sin (\omega t - kx) \text{ or } y = a \sin (kx - \omega t)$$

Both the equations represent a travelling wave but these two are not same. These waves differ by a phase difference of π .

✍ **Audibility of sound in day/night :** During the day temperature of air is maximum and it diminishes upwards. Therefore velocity of sound is also decreases upwards ($v \propto \sqrt{T}$). The plane wavefronts, initially vertical are turned, upwards so sound rays curl up during the day. At night the conditions are reversed hence audibility of sound is better in night as compared to day.



Ordinary Thinking

Objective Questions

Basics of Mechanical Waves

- Which of the following statements is wrong [NCERT 1976, 79]
 - Sound travels in straight line
 - Sound is a form of energy
 - Sound travels in the form of waves
 - Sound travels faster in vacuum than in air
- The relation between frequency ' n ' wavelength ' λ ' and velocity of propagation ' v ' of wave is [EAMCET 1979; CPMT 1976, 85]

(a) $n = v\lambda$	(b) $n = \lambda / v$
(c) $n = v / \lambda$	(d) $n = 1 / v$

3. Ultrasonic, Infrasonic and audible waves travel through a medium with speeds V_u , V_i and V_a respectively, then
[CPMT 1989]
- (a) V_u , V_i and V_a are nearly equal
(b) $V_u \geq V_a \geq V_i$
(c) $V_u \leq V_a \leq V_i$
(d) $V_a \leq V_u$ and $V_u \approx V_i$
4. The distance between two consecutive crests in a wave train produced in a string is 5 cm. If 2 complete waves pass through any point per second, the velocity of the wave is
[CPMT 1990]
- (a) 10 cm/sec (b) 2.5 cm/sec
(c) 5 cm/sec (d) 15 cm/sec
5. A tuning fork makes 256 vibrations per second in air. When the velocity of sound is 330 m/s, then wavelength of the tone emitted is
[MH CET 1999; CBSE PMT 1999]
- (a) 0.56 m (b) 0.89 m
(c) 1.11 m (d) 1.29 m
6. A man sets his watch by a whistle that is 2 km away. How much will his watch be in error. (speed of sound in air 330 m/sec)
(a) 3 seconds fast (b) 3 seconds slow
(c) 6 seconds fast (d) 6 seconds slow
7. When a sound wave of frequency 300 Hz passes through a medium the maximum displacement of a particle of the medium is 0.1 cm. The maximum velocity of the particle is equal to [MNR 1992; UPSEAT 1998, 2000; RPMT 2002; Pb. PET 2004]
- (a) 60π cm/sec (b) 30π cm/sec
(c) 30 cm/sec (d) 60 cm/sec
8. Sound waves have the following frequencies that are audible to human beings [CPMT 1975]
- (a) 5 c/s (b) 27000 c/s
(c) 5000 c/s (d) 50,000 c/s
9. Velocity of sound waves in air is 330 m/sec. For a particular sound in air, a path difference of 40 cm is equivalent to a phase difference of 1.6π . The frequency of this wave is
[CBSE PMT 1990]
- (a) 165 Hz (b) 150 Hz
(c) 660 Hz (d) 330 Hz
10. The wavelength of ultrasonic waves in air is of the order of
[EAMCET 1989]
- (a) 5×10^{-5} cm (b) 5×10^{-8} cm
(c) 5×10^5 cm (d) 5×10^8 cm
11. The relation between phase difference ($\Delta\phi$) and path difference (Δx) is [MNR 1995; UPSEAT 1999, 2000]
- (a) $\Delta\phi = \frac{2\pi}{\lambda} \Delta x$ (b) $\Delta\phi = 2\pi\lambda\Delta x$
(c) $\Delta\phi = \frac{2\pi\lambda}{\Delta x}$ (d) $\Delta\phi = \frac{2\Delta x}{\lambda}$
12. A hospital uses an ultrasonic scanner to locate tumours in a tissue. The operating frequency of the scanner is 4.2 MHz. The speed of sound in a tissue is 1.7 km/s. The wavelength of sound in the tissue is close to
[CBSE PMT 1995]
- (a) 4×10^{-4} m (b) 8×10^{-3} m
(c) 4×10^{-3} m (d) 8×10^{-4} m
13. The minimum audible wavelength at room temperature is about
(a) 0.2 Å (b) 5 Å
(c) 5 cm to 2 metre (d) 20 mm
14. The ratio of the speed of sound in nitrogen gas to that in helium gas, at 300 K is [IIT 1999]
- (a) $\sqrt{2/7}$ (b) $\sqrt{1/7}$
(c) $\sqrt{3/5}$ (d) $\sqrt{6/5}$
15. In a sinusoidal wave, the time required for a particular point to move from its maximum displacement to zero displacement is 0.170 second. The frequency of the wave is
[K CET 1994; AFMC 1998; CBSE PMT 1998; AIIMS 2001; AFMC 2002; CPMT 2004]
- (a) 1.47 Hz (b) 0.36 Hz
(c) 0.73 Hz (d) 2.94 Hz
16. The number of waves contained in unit length of the medium is called [MP PET 1991]
- (a) Elastic wave (b) Wave number
(c) Wave pulse (d) Electromagnetic wave
17. The frequency of a rod is 200 Hz. If the velocity of sound in air is 340 ms^{-1} , the wavelength of the sound produced is
[EAMCET (Med.) 1995; Pb. PMT 1999; CPMT 2000]
- (a) 1.7 cm (b) 6.8 cm
(c) 1.7 m (d) 6.8 m
18. Frequency range of the audible sounds is
[EAMCET (Med.) 1995; RPMT 1997]
- (a) 0 Hz – 30 Hz (b) 20 Hz – 20 kHz
(c) 20 kHz – 20,000 kHz (d) 20 kHz – 20 MHz
19. In a medium sound travels 2 km in 3 sec and in air, it travels 3 km in 10 sec. The ratio of the wavelengths of sound in the two media is
(a) 1 : 8 (b) 1 : 18
(c) 8 : 1 (d) 20 : 9
20. A stone is dropped into a lake from a tower 500 metre high. The sound of the splash will be heard by the man approximately after [CPMT 1992; J Kerala PMT 2005]
- (a) 11.5 seconds (b) 21 seconds
(c) 10 seconds (d) 14 seconds
21. When sound waves travel from air to water, which of the following remains constant
[AFMC 1993; DCE 1999; CPMT 2004]
- (a) Velocity (b) Frequency
(c) Wavelength (d) All the above
22. A stone is dropped in a well which is 19.6m deep. Echo sound is heard after 2.06 sec (after dropping) then the velocity of sound is

- (a) 332.6 m/sec (b) 326.7 m/sec
(c) 300.4 m/sec (d) 290.5 m/sec
23. At what temperature velocity of sound is double than that of at 0°C
(a) 819 K (b) 819°C
(c) 600°C (d) 600 K
24. Velocity of sound is maximum in
[AFMC 1998; BCECE 2001; RPMT 1999, 02]
(a) Air (b) Water
(c) Vacuum (d) Steel
25. If velocity of sound in a gas is 360 m/s and the distance between a compression and the nearest rarefaction is 1m, then the frequency of sound is [KCET 1999]
(a) 90 Hz (b) 180 Hz
(c) 360 Hz (d) 720 Hz
26. If the density of oxygen is 16 times that of hydrogen, what will be the ratio of their corresponding velocities of sound waves [KCET 1999]
(a) 1 : 4 (b) 4 : 1
(c) 16 : 1 (d) 1 : 16
27. At which temperature the speed of sound in hydrogen will be same as that of speed of sound in oxygen at 100°C
[UPSEAT 1999]
(a) -148°C (b) -212.5°C
(c) -317.5°C (d) -249.7°C
28. A tuning fork produces waves in a medium. If the temperature of the medium changes, then which of the following will change [EAMCET (Med.) 1998; Pb. PMT 1999; MH CET 2001]
(a) Amplitude (b) Frequency
(c) Wavelength (d) Time-period
29. The wave length of light in visible part (λ_V) and for sound (λ_S) are related as [RPMT 1999]
(a) $\lambda_V > \lambda_S$ (b) $\lambda_S > \lambda_V$
(c) $\lambda_S = \lambda_V$ (d) None of these
30. Which of the following is different from others
[AFMC 1994; CPMT 1999; Pb. PMT 2004]
(a) Velocity (b) Wavelength
(c) Frequency (d) Amplitude
31. The phase difference between two points separated by 1m in a wave of frequency 120 Hz is 90° . The wave velocity is [KCET 1999]
(a) 180 m/s (b) 240 m/s
(c) 480 m/s (d) 720 m/s
32. The echo of a gun shot is heard 8 sec. after the gun is fired. How far from him is the surface that reflects the sound (velocity of sound in air = 350 m/s) [JIPMER 1999]
(a) 1400 m (b) 2800 m
(c) 700 m (d) 350 m
33. A man sets his watch by the sound of a siren placed at a distance 1 km away. If the velocity of sound is 330 m/s [JIPMER 1999]
(a) His watch is set 3 sec. faster
(b) His watch is set 3 sec. slower
(c) His watch is set correctly
(d) None of the above
34. Velocity of sound in air is [Pb. PMT 1999; UPSEAT 2000]
(a) Faster in dry air than in moist air
(b) Directly proportional to pressure
(c) Directly proportional to temperature
(d) Independent of pressure of air
35. Two monoatomic ideal gases 1 and 2 of molecular masses m and m_2 respectively are enclosed in separate containers kept at the same temperature. The ratio of the speed of sound in gas 1 to that in gas 2 is given by [IIT-JEE Screening 2000]
(a) $\sqrt{\frac{m_1}{m_2}}$ (b) $\sqrt{\frac{m_2}{m_1}}$
(c) $\frac{m_1}{m_2}$ (d) $\frac{m_2}{m_1}$
36. A man is standing between two parallel cliffs and fires a gun. If he hears first and second echoes after 1.5 s and 3.5s respectively, the distance between the cliffs is (Velocity of sound in air = 340 ms) [EAMCET (Med.) 2000]
(a) 1190 m (b) 850 m
(c) 595 m (d) 510 m
37. When the temperature of an ideal gas is increased by 600 K, the velocity of sound in the gas becomes $\sqrt{3}$ times the initial velocity in it. The initial temperature of the gas is [EAMCET (Med.) 2000]
(a) -73°C (b) 27°C
(c) 127°C (d) 327°C
38. The frequency of a sound wave is n and its velocity is v . If the frequency is increased to $4n$, the velocity of the wave will be
(a) v (b) $2v$
(c) $4v$ (d) $v/4$
39. The temperature at which the speed of sound in air becomes double of its value at 27°C is [CPMT 1997; UPSEAT 2000; DPMT 2003]
(a) 54°C (b) 327°C
(c) 927°C (d) -123°C
40. The speed of a wave in a certain medium is 960 m/s. If 3600 waves pass over a certain point of the medium in 1 minute, the wavelength is [MP PMT 2000]
(a) 2 metres (b) 4 metres
(c) 8 metres (d) 16 metres
41. Speed of sound at constant temperature depends on [RPET 2000; AIIMS 1998]
(a) Pressure (b) Density of gas
(c) Above both (d) None of the above

42. A man standing on a cliff claps his hand hears its echo after 1 sec. If sound is reflected from another mountain and velocity of sound in air is 340 m/sec. Then the distance between the man and reflection point is [RPET 2000]
 (a) 680 m (b) 340 m
 (c) 85 m (d) 170 m
43. What will be the wave velocity, if the radar gives 54 waves per min and wavelength of the given wave is 10 m [RPET 2000]
 (a) 4 m/sec (b) 6 m/sec
 (c) 9 m/sec (d) 5 m/sec
44. Sound velocity is maximum in [Pb. CET 2000; RPMT 2000]
 (a) H_2 (b) N_2
 (c) He (d) O_2
45. The minimum distance of reflector surface from the source for listening the echo of sound is [CPMT 1997; RPMT 1999; KCET 2000]
 (a) 28 m (b) 18 m
 (c) 19 m (d) 16.5 m
46. The type of waves that can be propagated through solid is [CPMT 2000]
 (a) Transverse (b) Longitudinal
 (c) Both (a) and (b) (d) None of these
47. A man stands in front of a hillock and fires a gun. He hears an echo after 1.5 sec. The distance of the hillock from the man is (velocity of sound in air is 330 m/s) [EAMCET (Eng.) 1998; CPMT 2000]
 (a) 220 m (b) 247.5 m
 (c) 268.5 m (d) 292.5 m
48. Velocity of sound in air
 I. Increases with temperature
 II. Decreases with temperature
 III. Increase with pressure
 IV. Is independent of pressure
 V. Is independent of temperature
 Choose the correct answer. [Kerala (Engg.) 2001]
 (a) Only I and II are true (b) Only I and III are true
 (c) Only II and III are true (d) Only I and IV are true
49. The speed of a wave in a medium is 760 m/s. If 3600 waves are passing through a point, in the medium in 2 minutes, then its wavelength is [AFMC 1998; CPMT 2001]
 (a) 13.8 m (b) 25.3 m
 (c) 41.5 m (d) 57.2 m
50. If at same temperature and pressure, the densities for two diatomic gases are respectively d_1 and d_2 , then the ratio of velocities of sound in these gases will be [CPMT 2001]
 (a) $\sqrt{\frac{d_2}{d_1}}$ (b) $\sqrt{\frac{d_1}{d_2}}$
 (c) $d_1 d_2$ (d) $\sqrt{d_1 d_2}$
51. The frequency of a tuning fork is 384 per second and velocity of sound in air is 352 m/s. How far the sound has traversed while fork completes 36 vibration [KCET 2001]
 (a) 3 m (b) 13 m
 (c) 23 m (d) 33 m
52. v_1 and v_2 are the velocities of sound at the same temperature in two monoatomic gases of densities ρ_1 and ρ_2 respectively. If $\rho_1 / \rho_2 = \frac{1}{4}$ then the ratio of velocities v_1 and v_2 will be [KCET 2000; AIIMS 2001]
 (a) 1 : 2 (b) 4 : 1
 (c) 2 : 1 (d) 1 : 4
53. The temperature at which the speed of sound in air becomes double of its value at $0^\circ C$ is [AIIEE 2002]
 (a) 273K (b) 546K
 (c) 1092K (d) 0K
54. If wavelength of a wave is $\lambda = 6000 \text{ \AA}$. Then wave number will be [MH CET 2002]
 (a) $166 \times 10^3 \text{ m}^{-1}$ (b) $16.6 \times 10^{-1} \text{ m}^{-1}$
 (c) $1.66 \times 10^6 \text{ m}^{-1}$ (d) $1.66 \times 10^7 \text{ m}^{-1}$
55. Velocity of sound measured in hydrogen and oxygen gas at a given temperature will be in the ratio [RPET 2001; UPSEAT 2001; KCET 2002, 05]
 (a) 1 : 4 (b) 4 : 1
 (c) 2 : 1 (d) 1 : 1
56. Find the frequency of minimum distance between compression & rarefaction of a wire. If the length of the wire is 1m & velocity of sound in air is 360 m/s [CPMT 2003]
 (a) 90 sec (b) 180s
 (c) 120 sec (d) 360 sec
57. The velocity of sound is v in air. If the density of air is increased to 4 times, then the new velocity of sound will be [BHU 2003]
 (a) $\frac{v_s}{2}$ (b) $\frac{v_s}{12}$
 (c) $12v_s$ (d) $\frac{3}{2}v_s^2$
58. It takes 2.0 seconds for a sound wave to travel between two fixed points when the day temperature is $10^\circ C$. If the temperature rise to $30^\circ C$ the sound wave travels between the same fixed parts in [Orissa JEE 2002]
 (a) 1.9 sec (b) 2.0 sec
 (c) 2.1 sec (d) 2.2 sec
59. If v is the velocity of sound in moist air, v_d is the velocity of sound in dry air, under identical conditions of pressure and temperature [KCET 2002, 2003]
 (a) $v > v_d$ (b) $v < v_d$
 (c) $v = v_d$ (d) $v > v_d$

60. A man, standing between two cliffs, claps his hands and starts hearing a series of echoes at intervals of one second. If the speed of sound in air is 340 ms , the distance between the cliffs is
 (a) 340 m (b) 1620 m
 (c) 680 m (d) 1700 m [AFMC 1997]
61. A source of sound of frequency 600 Hz is placed inside water. The speed of sound in water is 1500 m/s and in air is 300 m/s . The frequency of sound recorded by an observer who is standing in air is
 (a) 200 Hz (b) 3000 Hz
 (c) 120 Hz (d) 600 Hz
62. If the temperature of the atmosphere is increased the following character of the sound wave is effected
 [AFMC 2004]
 (a) Amplitude (b) Frequency
 (c) Velocity (d) Wavelength
63. An underwater sonar source operating at a frequency of 60 KHz directs its beam towards the surface. If the velocity of sound in air is 330 m/s , the wavelength and frequency of waves in air are:
 (a) 5.5 mm , 60 KHz (b) 330 m , 60 KHz
 (c) 5.5 mm , 20 KHz (d) 5.5 mm , 80 KHz
64. Two sound waves having a phase difference of 60° have path difference of
 [CBSE PMT 1996; AIIMS 2001]
 (a) 2λ (b) $\lambda/2$
 (c) $\lambda/6$ (d) $\lambda/3$
65. It is possible to distinguish between the transverse and longitudinal waves by studying the property of
 [CPMT 1976; EAMCET 1994]
 (a) Interference (b) Diffraction
 (c) Reflection (d) Polarisation
66. Water waves are [EAMCET 1979; AIIMS 2004]
 (a) Longitudinal
 (b) Transverse
 (c) Both longitudinal and transverse
 (d) Neither longitudinal nor transverse
67. Sound travels in rocks in the form of
 [NCERT 1968]
 (a) Longitudinal elastic waves only
 (b) Transverse elastic waves only
 (c) Both longitudinal and transverse elastic waves
 (d) Non-elastic waves
68. The waves in which the particles of the medium vibrate in a direction perpendicular to the direction of wave motion is known as
 [EAMCT 1981; AIIMS 1998; DPMT 2000]
 (a) Transverse wave (b) Longitudinal waves
 (c) Propagated waves (d) None of these
69. A medium can carry a longitudinal wave because it has the property of
 [KCET 1994]
 (a) Mass (b) Density
 (c) Compressibility (d) Elasticity
70. Which of the following is the longitudinal wave
 (a) Sound waves (b) Waves on plucked string
 (c) Water waves (d) Light waves [KCET 2004]
71. The nature of sound waves in gases is
 [RPMT 1999; RPET 2000; J & K CET 2004]
 (a) Transverse (b) Longitudinal
 (c) Stationary (d) Electromagnetic [IIT-JEE Screening 2004]
72. Transverse waves can propagate in
 [CPMT 1984; KCET 2000; RPET 2001]
 (a) Liquids (b) Solids
 (c) Gases (d) None of these
73. Sound waves in air are [RPET 2000; AFMC 2001]
 (a) Transverse (b) Longitudinal
 (c) De-Broglie waves (d) All the above
74. Which of the following is not the transverse wave
 [AFMC 1999; BHU 2001]
 (a) X-rays (b) γ -rays [DPMT 2004]
 (c) Visible light wave (d) Sound wave in a gas
75. What is the phase difference between two successive crests in the wave [RPMT 2001, 02; MH CET 2004]
 (a) π (b) $\pi/2$
 (c) 2π (d) 4π
76. A wave of frequency 500 Hz has velocity 360 m/sec . The distance between two nearest points 60° out of phase, is
 [NCERT 1979; MP PET 1989; JIPMER 1997; RPMT 2002, 03; CPMT 1979, 90, 2003; BCECE 2005]
 (a) 0.6 cm (b) 12 cm
 (c) 60 cm (d) 120 cm
77. The following phenomenon cannot be observed for sound waves [NCERT 1982; C
 AFMC 2002; RPMT 2003]
 (a) Refraction (b) Interference
 (c) Diffraction (d) Polarisation
78. When an aeroplane attains a speed higher than the velocity of sound in air, a loud bang is heard. This is because
 [NCERT 1972; J & K CET 2002]
 (a) It explodes
 (b) It produces a shock wave which is received as the bang
 (c) Its wings vibrate so violently that the bang is heard
 (d) The normal engine noises undergo a Doppler shift to generate the bang
79. Ultrasonic waves are those waves [CPMT 1979]
 (a) To which man can hear (b) Man can't hear
 (c) Are of high velocity (d) Of high amplitude
80. A big explosion on the moon cannot be heard on the earth because
 (a) The explosion produces high frequency sound waves which are inaudible
 (b) Sound waves required a material medium for propagation
 (c) Sound waves are absorbed in the moon's atmosphere
 (d) Sound waves are absorbed in the earth's atmosphere
81. Sound waves of wavelength greater than that of audible sound are called [KCET 1999]
 (a) Seismic waves (b) Sonic waves
 (c) Ultrasonic waves (d) Infrasonic waves

82. 'SONAR' emits which of the following waves
[AIIMS 1999]
- (a) Radio waves (b) Ultrasonic waves
(c) Light waves (d) Magnetic waves
83. Which of the following do not require medium for transmission
- (a) Cathode ray (b) Electromagnetic wave
(c) Sound wave (d) None of the above
84. Consider the following
- Waves created on the surfaces of a water pond by a vibrating sources.
 - Wave created by an oscillating electric field in air.
 - Sound waves travelling under water.
- Which of these can be polarized [AMU 2001]
- (a) I and II (b) II only
(c) II and III (d) I, II and III
85. Mechanical waves on the surface of a liquid are
[SCRA 1996]
- (a) Transverse
(b) Longitudinal
(c) Torsional
(d) Both transverse and longitudinal
86. The ratio of densities of nitrogen and oxygen is 14:16. The temperature at which the speed of sound in nitrogen will be same at that in oxygen at 55°C is
[EAMCET (Engg.) 1999]
- (a) 35°C (b) 48°C
(c) 65°C (d) 14°C
87. The intensity of sound increases at night due to
[CPMT 2000]
- (a) Increase in density of air (b) Decreases in density of air
(c) Low temperature (d) None of these
88. A wavelength 0.60 cm is produced in air and it travels at a speed of 300 ms. It will be an
[UPSEAT 2000]
- (a) Audible wave (b) Infrasonic wave
(c) Ultrasonic wave (d) None of the above
89. Speed of sound in mercury at a certain temperature is 1450 m/s. Given the density of mercury as $13.6 \times 10^3 \text{ kg/m}^3$, the bulk modulus for mercury is
[JIPMER 2000]
- (a) $2.86 \times 10^9 \text{ N/m}^2$ (b) $3.86 \times 10^9 \text{ N/m}^2$
(c) $4.86 \times 10^9 \text{ N/m}^2$ (d) $5.86 \times 10^9 \text{ N/m}^2$
90. A micro-wave and an ultrasonic sound wave have the same wavelength. Their frequencies are in the ratio (approximately)
- (a) $10^4 : 1$ (b) $10^5 : 1$
(c) $10^6 : 1$ (d) $10^7 : 1$
91. A point source emits sound equally in all directions in a non-absorbing medium. Two points P and Q are at distance of 2m and 3m respectively from the source. The ratio of the intensities of the waves at P and Q is
[CBSE PMT 2005]
- (a) 9 : 4 (b) 2 : 3
(c) 3 : 2 (d) 4 : 9

92. A wave has velocity v in medium P and velocity $2v$ in medium Q. If the wave is incident in medium P at an angle of 30° then the angle of refraction will be
[J & K CET 2005]
- (a) 30° (b) 45°
(c) 60° (d) 90°
93. An observer standing near the sea shore observes 54 waves per minute. If the wavelength of the water wave is 10m then the velocity of water wave is
[RPMT 2000]
[Kerala (Engg.) 2005]
- (a) 540 ms (b) 5.4 ms
(c) 0.184 ms (d) 9 ms
94. Ultrasonic signal sent from SONAR returns to it after reflection from a rock after a lapse of 1 sec. If the velocity of ultrasound in water is 1600 ms, the depth of the rock in water is
- (a) 300 m (b) 400 m
(c) 500 m (d) 800 m

Progressive Waves

1. The equation of a wave is $y = 2 \sin \pi(0.5x - 200t)$, where x and y are expressed in cm and t in sec. The wave velocity is
[MP PMT 1986]
- (a) 100 cm/sec (b) 200 cm/sec
(c) 300 cm/sec (d) 400 cm/sec
2. Equation of a progressive wave is given by
- $$y = 0.2 \cos \pi \left(0.04t + 0.02x - \frac{\pi}{6} \right)$$
- The distance is expressed in cm and time in second. What will be the minimum distance between two particles having the phase difference of $\pi/2$
- (a) 4 cm (b) 8 cm
(c) 25 cm (d) 12.5 cm
3. A travelling wave passes a point of observation. At this point, the time interval between successive crests is 0.2 seconds and
- (a) The wavelength is 5 m
(b) The frequency is 5 Hz
(c) The velocity of propagation is 5 m/s
(d) The wavelength is 0.2 m
4. The equation of a transverse wave is given by
- $$y = 10 \sin \pi(0.01x - 2t)$$
- where x and y are in cm and t is in second. Its frequency is
[MP PET 1990; MNR 1986; RPET 2003]
- (a) 10 sec^{-1} (b) 2 sec^{-1}
(c) 1 sec^{-1} (d) 0.01 sec^{-1}
5. At a moment in a progressive wave, the phase of a particle executing S.H.M. is $\frac{\pi}{3}$. Then the phase of the particle 15 cm ahead and at the time $\frac{T}{2}$ will be, if the wavelength is 60 cm
- (a) $\frac{\pi}{2}$ (b) $\frac{2\pi}{3}$
(c) Zero (d) $\frac{5\pi}{6}$

6. The equation of a wave travelling on a string is $y = 4 \sin \frac{\pi}{2} \left(8t - \frac{x}{8} \right)$. If x and y are in cm , then velocity of wave is [MP PET 1990]
- 64 cm/sec in $-x$ direction
 - 32 cm/sec in $-x$ direction
 - 32 cm/sec in $+x$ direction
 - 64 cm/sec in $+x$ direction
7. The equation of a progressive wave is given by $y = a \sin(628t - 31.4x)$
- If the distances are expressed in cms and time in seconds, then the wave velocity will be [DPMT 1999]
- 314 cm/sec
 - 628 cm/sec
 - 20 cm/sec
 - 400 cm/sec
8. Two waves are given by $y_1 = a \sin(\omega t - kx)$ and $y_2 = a \cos(\omega t - kx)$. The phase difference between the two waves is [MP PMT 1993; SCRA 1996; CET 1998; EAMCET 1991; Orissa JEE 2002]
- $\frac{\pi}{4}$
 - π
 - $\frac{\pi}{8}$
 - $\frac{\pi}{2}$
9. If amplitude of waves at distance r from a point source is A , the amplitude at a distance $2r$ will be [MP PMT 1985]
- $2A$
 - A
 - $A/2$
 - $A/4$
10. The relation between time and displacement for two particles is given by $y_1 = 0.06 \sin 2\pi(0.04t + \phi_1)$, $y_2 = 0.03 \sin 2\pi(1.04t + \phi_2)$
- The ratio of the intensity of the waves produced by the vibrations of the two particles will be [MP PMT 1991]
- 2 : 1
 - 1 : 2
 - 4 : 1
 - 1 : 4
11. A wave is reflected from a rigid support. The change in phase on reflection will be [MP PMT 1990; RPMT 2002]
- $\pi/4$
 - $\pi/2$
 - π
 - 2π
12. A plane wave is represented by $x = 1.2 \sin(314t + 12.56y)$
- Where x and y are distances measured along in x and y direction in meters and t is time in seconds. This wave has [MP PET 1991]
- A wavelength of 0.25 m and travels in $+ve x$ direction
 - A wavelength of 0.25 m and travels in $+ve y$ direction
 - A wavelength of 0.5 m and travels in $-ve y$ direction
 - A wavelength of 0.5 m and travels in $-ve x$ direction
13. The displacement y (in cm) produced by a simple harmonic wave is $y = \frac{10}{\pi} \sin \left(2000\pi t - \frac{\pi x}{17} \right)$. The periodic time and maximum velocity of the particles in the medium will respectively be
- 10^{-3} sec and 330 m/sec
 - 10^{-4} sec and 20 m/sec
 - 10^{-3} sec and 200 m/sec
 - 10^{-2} sec and 2000 m/sec
14. The equation of a wave travelling in a string can be written as $y = 3 \cos \pi(100t - x)$. Its wavelength is [MNR 1985; CPMT 1991; MP PMT 1994, 97; Pb. PET 2004]
- 100 cm
 - 2 cm
 - 5 cm
 - None of the above
15. A transverse wave is described by the equation $Y = Y_0 \sin 2\pi \left(ft - \frac{x}{\lambda} \right)$. The maximum particle velocity is four times the wave velocity if [IIT 1984; MP PMT 1997; EAMCET; 1998; CBSE PMT 2000; AFMC 2000; MP PMT/PET 1998; 01; KCET 1999, 04; Pb. PET 2001; DPMT 2005]
- $\lambda = \frac{\pi Y_0}{4}$
 - $\lambda = \frac{\pi Y_0}{2}$
 - $\lambda = \pi Y_0$
 - $\lambda = 2\pi Y_0$
16. A wave equation which gives the displacement along the Y direction is given by the equation $y = 10^4 \sin(60t + 2x)$, where x and y are in $metres$ and t is time in seconds. This represents a wave [MNR 1983; IIT 1982; RPMT 1998; MP PET 2001]
- Travelling with a velocity of 30 m/sec in the negative X direction
 - Of wavelength π metre
 - Of frequency 30/ π Hz
 - Of amplitude 10^4 metre travelling along the negative X direction
17. A transverse wave of amplitude 0.5 m and wavelength 1 m and frequency 2 Hz is propagating in a string in the negative x -direction. The expression for this wave is [AIIMS 1980]
- $y(x, t) = 0.5 \sin(2\pi x - 4\pi t)$
 - $y(x, t) = 0.5 \cos(2\pi x + 4\pi t)$
 - $y(x, t) = 0.5 \sin(\pi x - 2\pi t)$
 - $y(x, t) = 0.5 \cos(2\pi x + 2\pi t)$
18. The displacement of a particle is given by $y = 5 \times 10^{-4} \sin(100t - 50x)$, where x is in meter and t in sec, find out the velocity of the wave [CPMT 1982]
- 5000 m/sec
 - 2 m/sec
 - 0.5 m/sec
 - 300 m/sec
19. Which one of the following does not represent a travelling wave
- $y = \sin(x - vt)$
 - $y = y_m \sin k(x + vt)$
 - $y = y_m \log(x - vt)$
 - $y = f(x^2 - vt^2)$
20. A wave represented by the given equation $Y = A \sin \left(10\pi x + 15\pi t + \frac{\pi}{3} \right)$, where x is in meter and t is in second. The expression represents [IIT 1990]
- A wave travelling in the positive X direction with a velocity of 1.5 m/sec
 - A wave travelling in the negative X direction with a velocity of 1.5 m/sec

- (c) A wave travelling in the negative X direction with a wavelength of 0.2 m
- (d) A wave travelling in the positive X direction with a wavelength of 0.2 m
21. A plane wave is described by the equation $y = 3 \cos\left(\frac{x}{4} - 10t - \frac{\pi}{2}\right)$. The maximum velocity of the particles of the medium due to this wave is [MP PMT 1994]
- (a) 30 (b) $\frac{3\pi}{2}$
- (c) $3/4$ (d) 40
22. The path difference between the two waves $y_1 = a_1 \sin\left(\omega t - \frac{2\pi x}{\lambda}\right)$ and $y_2 = a_2 \cos\left(\omega t - \frac{2\pi x}{\lambda} + \phi\right)$ is [MP PMT 1994]
- (a) $\frac{\lambda}{2\pi} \phi$ (b) $\frac{\lambda}{2\pi} \left(\phi + \frac{\pi}{2}\right)$
- (c) $\frac{2\pi}{\lambda} \left(\phi - \frac{\pi}{2}\right)$ (d) $\frac{2\pi}{\lambda} \phi$
23. Wave equations of two particles are given by $y_1 = a \sin(\omega t - kx)$, $y_2 = a \sin(kx + \omega t)$, then [BHU 1995]
- (a) They are moving in opposite direction
- (b) Phase between them is 90°
- (c) Phase between them is 180°
- (d) Phase between them is 0°
24. A wave is represented by the equation $y = 0.5 \sin(10t - x)\text{m}$. It is a travelling wave propagating along the $+x$ direction with velocity [Roorkee 1995]
- (a) 10 m/s (b) 20 m/s
- (c) 5 m/s (d) None of these
25. A wave is represented by the equation $y = 7 \sin\left(7\pi t - 0.04\pi x + \frac{\pi}{3}\right)$
- x is in *metres* and t is in seconds. The speed of the wave is [MP PET 1996; AMU (Engg.) 1999]
- (a) 175 m/sec (b) $49\pi\text{ m/sec}$
- (c) $49\pi\text{ m/sec}$ (d) $0.28\pi\text{ m/sec}$
26. The equation of a transverse wave travelling on a rope is given by $y = 10 \sin\pi(0.01x - 2.00t)$ where y and x are in *cm* and t in seconds. The maximum transverse speed of a particle in the rope is about [MP PET 1999; AIIMS 2000]
- (a) 63 cm/s (b) 75 cm/s
- (c) 100 cm/s (d) 121 cm/s
27. As a wave propagates [IIT-JEE 1999]
- (a) The wave intensity remains constant for a plane wave
- (b) The wave intensity decreases as the inverse of the distance from the source for a spherical wave
- (c) The wave intensity decreases as the inverse square of the distance from the source for a spherical wave
- (d) Total intensity of the spherical wave over the spherical surface centered at the source remains constant at all times
28. A transverse wave is represented by the equation $y = y_0 \sin \frac{2\pi}{\lambda}(vt - x)$
- For what value of λ , the maximum particle velocity equal to two times the wave velocity [CBSE PMT 1998; JIPMER 2001, 02; AFMC 2002]
- (a) $\lambda = 2\pi y_0$ (b) $\lambda = \pi y_0 / 3$
- (c) $\lambda = \pi y_0 / 2$ (d) $\lambda = \pi y_0$
29. A travelling wave in a stretched string is described by the equation $y = A \sin(kx - \omega t)$. The maximum particle velocity is [IIT 1997 Re-Exam; UPSEAT 2004]
- (a) $A\omega$ (b) ω/k
- (c) $d\omega/dk$ (d) x/t
30. A wave travels in a medium according to the equation of displacement given by $y(x, t) = 0.03 \sin\pi(2t - 0.01x)$
- where y and x are in *metres* and t in seconds. The wavelength of the wave is [EAMCET 1994; CPMT 2004]
- (a) 200 m (b) 100 m
- (c) 20 m (d) 10 m
31. The particles of a medium vibrate about their mean positions whenever a wave travels through that medium. The phase difference between the vibrations of two such particles
- (a) Varies with time
- (b) Varies with distance separating them
- (c) Varies with time as well as distance
- (d) Is always zero
32. A wave is given by $y = 3 \sin 2\pi\left(\frac{t}{0.04} - \frac{x}{0.01}\right)$, where y is in *cm*. Frequency of wave and maximum acceleration of particle will be
- (a) 100 Hz , $4.7 \times 10^3\text{ cm/s}^2$ (b) 50 Hz , $7.5 \times 10^3\text{ cm/s}^2$
- (c) 25 Hz , $4.7 \times 10^4\text{ cm/s}^2$ (d) 25 Hz , $7.4 \times 10^4\text{ cm/s}^2$
33. Equation of a progressive wave is given by $y = 4 \sin\left\{\pi\left(\frac{t}{5} - \frac{x}{9}\right) + \frac{\pi}{6}\right\}$
- Then which of the following is correct [CBSE PMT 1993]
- (a) $v = 5\text{ m/sec}$ (b) $\lambda = 18\text{ m}$
- (c) $a = 0.04\text{ m}$ (d) $n = 50\text{ Hz}$
34. With the propagation of a longitudinal wave through a material medium, the quantities transmitted in the propagation direction are [CBSE PMT 1993]
- (a) Energy, momentum and mass
- (b) Energy
- (c) Energy and mass
- (d) Energy and linear momentum
35. The frequency of the sinusoidal wave $y = 0.40 \cos[2000\pi t + 0.80\pi x]$ would be [CBSE PMT 1992]
- (a) $1000\pi\text{ Hz}$ (b) 2000 Hz
- (c) 20 Hz (d) $\frac{1000}{\pi}\text{ Hz}$
36. Which of the following equations represents a wave [CBSE PMT 1994; JIPMER 2000]

- (a) $Y = A(\omega t - kx)$ (b) $Y = A \sin \omega t$
(c) $Y = A \cos kx$ (d) $Y = A \sin(at - bx + c)$
37. The equation of a transverse wave is given by
 $y = 100 \sin \pi(0.04z - 2t)$
 where y and z are in cm and t is in seconds. The frequency of the wave in Hz is [SCRA 1998]
 (a) 1 (b) 2
 (c) 25 (d) 100
38. The equation of a plane progressive wave is given by
 $y = 0.025 \sin(100t + 0.25x)$. The frequency of this wave would be [CPMT 1993; JIPMER 2001, 02]
 (a) $\frac{50}{\pi} Hz$ (b) $\frac{100}{\pi} Hz$
 (c) 100 Hz (d) 50 Hz
39. The equation of a sound wave is
 $y = 0.0015 \sin(62.4x + 316t)$
 The wavelength of this wave is [CBSE PMT 1996; AFMC 2002; AIIMS 2002]
 (a) 0.2 unit (b) 0.1 unit
 (c) 0.3 unit (d) Cannot be calculated
40. In the given progressive wave equation, what is the maximum velocity of particle $Y = 0.5 \sin(10\pi t - 5x)$ cm [BHU 1997]
 (a) 5 cm/s (b) 5 $\pi cm/s$
 (c) 10 cm/s (d) 10.5 cm/s
41. A pulse or a wave train travels along a stretched string and reaches the fixed end of the string. It will be reflected back with
 (a) The same phase as the incident pulse but with velocity reversed
 (b) A phase change of 180° with no reversal of velocity
 (c) The same phase as the incident pulse with no reversal of velocity
 (d) A phase change of 180° with velocity reversed
42. The equation of a travelling wave is
 $y = 60 \cos(1800t - 6x)$
 where y is in microns, t in seconds and x in metres. The ratio of maximum particle velocity to velocity of wave propagation is [CBSE PMT 1997; JIPMER 2001, 02]
 (a) 3.6×10^{-11} (b) 3.6×10^{-6}
 (c) 3.6×10^{-4} (d) 3.6
43. The wave equation is $y = 0.30 \sin(314t - 1.57x)$ where t , x and y are in second, meter and centimeter respectively. The speed of the wave is [CPMT 1997; AFMC 1999; CPMT 2001]
 (a) 100 m/s (b) 200 m/s
 (c) 300 m/s (d) 400 m/s
44. Equation of the progressive wave is given by : $y = a \sin \pi(40t - x)$ where a and x are in $metre$ and t in second. The velocity of the wave is [KCET 1999]
 (a) 80 m/s (b) 10 m/s
 (c) 40 m/s (d) 20 m/s
45. Progressive wave of sound is represented by
 $y = a \sin[400\pi t - \pi x / 6.85]$ where x is in m and t is in sec. Frequency of the wave will be [RPMT 1999]
 (a) 200 Hz (b) 400 Hz
 (c) 500 Hz (d) 600 Hz
46. Two waves of frequencies 20 Hz and 30 Hz . Travels out from a common point. The phase difference between them after 0.6 sec is
 (a) Zero (b) $\frac{\pi}{2}$
 (c) π (d) $\frac{3\pi}{4}$
47. The phase difference between two points separated by 0.8 m in a wave of frequency 120 Hz is 90° . Then the velocity of wave will be
 (a) 192 m/s (b) 360 m/s
 (c) 710 m/s (d) 384 m/s
48. The equation of progressive wave is $y = 0.2 \sin 2\pi \left[\frac{t}{0.01} - \frac{x}{0.3} \right]$, where x and y are in metre and t is in second. The velocity of propagation of the wave is
 (a) 30 m/s (b) 40 m/s
 (c) 300 m/s (d) 400 m/s
49. If the equation of transverse wave is $y = 5 \sin 2\pi \left[\frac{t}{0.04} - \frac{x}{40} \right]$, where distance is in cm and time in second, then the wavelength of the wave is [MH CET 2000; DPMT 2003]
 (a) 60 cm (b) 40 cm
 (c) 35 cm (d) 25 cm
50. A wave is represented by the equation : $y = a \sin(0.01x - 2t)$ where a and x are in cm . velocity of propagation of wave is [CBSE PMT 1997]
 (a) 10 cm/s (b) 50 cm/s
 (c) 100 cm/s (d) 200 cm/s
51. A simple harmonic progressive wave is represented by the equation : $y = 8 \sin 2\pi(0.1x - 2t)$ where x and y are in cm and t is in seconds. At any instant the phase difference between two particles separated by 2.0 cm in the x -direction is [MP PMT 2000]
 (a) 18 (b) 36
 (c) 54 (d) 72
52. The intensity of a progressing plane wave in loss-free medium is
 (a) Directly proportional to the square of amplitude of the wave
 (b) Directly proportional to the velocity of the wave
 (c) Directly proportional to the square of frequency of the wave
 (d) Inversely proportional to the density of the medium
53. The equation of progressive wave is $y = a \sin(200t - x)$. where x is in meter and t is in second. The velocity of wave is
 (a) 200 m/sec (b) 100 m/sec
 (c) 50 m/sec (d) None of these
54. A wave is represented by the equation $y = 7 \sin \{\pi(2t - 2x)\}$ where x is in $metres$ and t in seconds. The velocity of the wave is [CPMT 2000; CBSE PMT 2000; Pb. PET 2000]
 (a) 1 m/s (b) 2 m/s
 (c) 5 m/s (d) 10 m/s
55. The equation of a longitudinal wave is represented as
 $y = 20 \cos \pi(50t - x)$. Its wavelength is

[UPSEAT 2001; Orissa PMT 2004]

- (a) 5 cm (b) 2 cm
(c) 50 cm (d) 20 cm

56. A wave equation which gives the displacement along y -direction is given by $y = 0.001 \sin(100t + x)$ where x and y are in meter and t is time in second. This represented a wave

[UPSEAT 2001]

- (a) Of frequency $\frac{100}{\pi}$ Hz
(b) Of wavelength one metre
(c) Travelling with a velocity of $\frac{50}{\pi}$ ms in the positive X -direction
(d) Travelling with a velocity of 100 ms in the negative X -direction

57. A transverse wave is given by $y = A \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right)$. The maximum particle velocity is equal to 4 times the wave velocity when

[MP PMT 2001]

- (a) $\lambda = 2\pi A$ (b) $\lambda = \frac{1}{2}\pi A$
(c) $\lambda = \pi A$ (d) $\lambda = \frac{1}{4}\pi A$

58. The equation of a wave is represented by $y = 10^{-4} \sin \left[100t - \frac{x}{10} \right]$. The velocity of the wave will be

[CBSE PMT 2001]

- (a) 100 m/s (b) 250 m/s
(c) 750 m/s (d) 1000 m/s

59. A wave travelling in positive X -direction with $A = 0.2m$ has a velocity of 360 m/sec. if $\lambda = 60m$, then correct expression for the wave is

[CBSE PMT 2002; KCET 2003]

- (a) $y = 0.2 \sin \left[2\pi \left(6t + \frac{x}{60} \right) \right]$ (b) $y = 0.2 \sin \left[\pi \left(6t + \frac{x}{60} \right) \right]$
(c) $y = 0.2 \sin \left[2\pi \left(6t - \frac{x}{60} \right) \right]$ (d) $y = 0.2 \sin \left[\pi \left(6t - \frac{x}{60} \right) \right]$

60. The equation of a wave motion (with t in seconds and x in metres) is given by $y = 7 \sin \left[7\pi t - 0.4\pi x + \frac{\pi}{3} \right]$. The velocity of the wave will be

[BHU 2002]

- (a) 17.5 m/s (b) 49π m/s
(c) $\frac{49}{2\pi}$ m/s (d) $\frac{2\pi}{49}$ m/s

61. Two waves represented by the following equations are travelling in the same medium $y_1 = 5 \sin 2\pi(75t - 0.25x)$, $y_2 = 10 \sin 2\pi(150t - 0.50x)$

The intensity ratio I_1 / I_2 of the two waves is

[UPSEAT 2002]

- (a) 1 : 2 (b) 1 : 4

- (c) 1 : 8 (d) 1 : 16

62. The equation of a progressive wave is $y = 8 \sin \left[\pi \left(\frac{t}{10} - \frac{x}{4} \right) + \frac{\pi}{3} \right]$.

The wavelength of the wave is

[MH CET 2002]

- (a) 8 m (b) 4 m
(c) 2 m (d) 10 m

63. Which of the following is not true for this progressive wave $y = 4 \sin 2\pi \left(\frac{t}{0.02} - \frac{x}{100} \right)$ where y and x are in cm & t in

sec

[CPMT 2003]

- (a) Its amplitude is 4 cm
(b) Its wavelength is 100 cm
(c) Its frequency is 50 cycles/sec
(d) Its propagation velocity is 50×10^3 cm/sec

64. The equation of a wave is given as $y = 0.07 \sin(2\pi x - 3000\pi t)$. Where x is in metre and t in sec, then the correct statement is

- (a) $\lambda = 1/6m$, $v = 250m/s$ (b) $a = 0.07m$, $v = 300m/s$
(c) $n = 1500$, $v = 200m/s$ (d) None

65. The equation of the propagating wave is $y = 25 \sin(20t + 5x)$, where y is displacement. Which of the following statement is not true

[MP PET 2003]

- (a) The amplitude of the wave is 25 units
(b) The wave is propagating in positive x -direction
(c) The velocity of the wave is 4 units
(d) The maximum velocity of the particles is 500 units

66. In a plane progressive wave given by $y = 25 \cos(2\pi t - \pi x)$, the amplitude and frequency are respectively

[BCECE 2003]

- (a) 25, 100 (b) 25, 1
(c) 25, 2 (d) 50π , 2

67. The displacement y of a wave travelling in the x -direction is given by $y = 10^{-4} \sin \left(600t - 2x + \frac{\pi}{3} \right)$ metres, where x is expressed in

metres and t in seconds. The speed of the wave-motion, in ms, is

- (a) 200 (b) 300
(c) 600 (d) 1200

68. The displacement y of a particle in a medium can be expressed as: $y = 10^{-6} \sin(100t + 20x + \pi/4)m$, where t is in second and x in meter. The speed of wave is

[AIEEE 2004]

- (a) 2000 m/s (b) 5 m/s
(c) 20 m/s (d) 5π m/s

69. If the wave equation $y = 0.08 \sin \frac{2\pi}{\lambda} (200t - x)$ then the velocity of the wave will be

[BCECE 2004]

- (a) $400\sqrt{2}$ (b) $200\sqrt{2}$



- (c) 400 (d) 200
70. The phase difference between two points separated by 0.8 m in a wave of frequency is 120 Hz is $\frac{\pi}{2}$. The velocity of wave is
- (a) 720 m/s (b) 384 m/s
(c) 250 m/s (d) 1 m/s
71. A plane progressive wave is represented by the equation
- $$y = 0.1 \sin \left(200\pi t - \frac{20\pi x}{17} \right)$$
- where y is displacement in m, t in second and x is distance from a fixed origin in meter. The frequency, wavelength and speed of the wave respectively are
- (a) 100 Hz, 1.7 m, 170 m/s (b) 150 Hz, 2.4 m, 200 m/s
(c) 80 Hz, 1.1 m, 90 m/s (d) 120 Hz, 1.25 m, 207 m/s
72. The equation of a travelling wave is given by
- $$y = 0.5 \sin(20x - 400t)$$
- where x and y are in meter and t is in second. The velocity of the wave is [UPSEAT 2004]
- (a) 10 m/s (b) 20 m/s
(c) 200 m/s (d) 400 m/s
73. A transverse progressive wave on a stretched string has a velocity of 10 ms^{-1} and a frequency of 100 Hz. The phase difference between two particles of the string which are 2.5 cm apart will be
- (a) $\frac{\pi}{8}$ (b) $\frac{\pi}{4}$
(c) $\frac{3\pi}{8}$ (d) $\frac{\pi}{2}$
74. A transverse sinusoidal wave of amplitude a , wavelength λ and frequency n is travelling on a stretched string. The maximum speed of any point on the string is $v/10$, where v is the speed of propagation of the wave. If $a = 10^{-3} \text{ m}$ and $v = 10 \text{ ms}^{-1}$, then λ and n are given by [IIT 1998]
- (a) $\lambda = 2\pi \times 10^{-2} \text{ m}$ (b) $\lambda = 10^{-3} \text{ m}$
(c) $n = \frac{10^3}{2\pi} \text{ Hz}$ (d) $n = 10^4 \text{ Hz}$
75. When a longitudinal wave propagates through a medium, the particles of the medium execute simple harmonic oscillations about their mean positions. These oscillations of a particle are characterised by an invariant [SCRA 1998]
- (a) Kinetic energy
(b) Potential energy
(c) Sum of kinetic energy and potential energy
(d) Difference between kinetic energy and potential energy
76. Equation of a progressive wave is given by $y = a \sin \pi \left[\frac{t}{2} - \frac{x}{4} \right]$, where t is in seconds and x is in meters. The distance through which the wave moves in 8 sec is (in meter)
- (a) 8 (b) 16
(c) 2 (d) 4
77. The phase difference between two waves represented by $y_1 = 10^{-6} \sin[100\pi t + (x/50)] \text{ m}$ and $y_2 = 10^{-6} \cos[100\pi t + (x/50)] \text{ m}$

where x is expressed in metres and t is expressed in seconds, is approximately [CBSE PMT 2004]

- (a) 1.5 rad (b) 1.07 rad
(c) 2.07 rad (d) 0.5 rad

78. Equation of motion in the same direction are given by
- $$y_1 = 2a \sin(\omega t - kx) \text{ and } y_2 = 2a \sin(\omega t - kx - \theta)$$
- The amplitude of the medium particle will be [CPMT 2004]
- (a) $2a \cos \theta$ (b) $\sqrt{2} a \cos \theta$
(c) $4a \cos \theta / 2$ (d) $\sqrt{2} a \cos \theta / 2$
79. A particle on the trough of a wave at any instant will come to the mean position after a time (one time period)
- (a) $T/2$ (b) $T/4$
(c) T (d) $2T$
80. If the equation of transverse wave is $Y = 2 \sin(kx - 2t)$, then the maximum particle velocity is [Orissa JEE 2005]
- (a) 4 units (b) 2 units
(c) 0 (d) 6 units

Interference and Superposition of Waves

1. There is constructive interference between the two waves of wavelength λ coming from two different paths at a point. To get maximum sound or constructive interference at that point, the path of one wave is to be increased by [MP PMT 1994]
- (a) $\frac{\lambda}{4}$ (b) $\frac{\lambda}{2}$
(c) $\frac{3\lambda}{4}$ (d) λ
2. When two sound waves with a phase difference of $\pi/2$, and each having amplitude A and frequency ω , are superimposed on each other, then the maximum amplitude and frequency of resultant wave is [MP PMT 1989]
- (a) $\frac{A}{\sqrt{2}} : \frac{\omega}{2}$ (b) $\frac{A}{\sqrt{2}} : \omega$
(c) $\sqrt{2} A : \frac{\omega}{2}$ (d) $\sqrt{2} A : \omega$
3. If the phase difference between the two wave is 2π during superposition, then the resultant amplitude is [DPMT 2001]
- (a) Maximum (b) Minimum
(c) Maximum or minimum (d) None of the above
4. The superposition takes place between two waves of frequency f and amplitude a . The total intensity is directly proportional to [KCE 1998]
- (a) a (b) $2a$
(c) a^2 (d) $4a^2$
5. If two waves of same frequency and same amplitude respectively, on superposition produced a resultant disturbance of the same amplitude, the waves differ in phase by [KCE 1998]
- (a) π (b) $2\pi/3$

- (c) $\pi/2$ (d) Zero
6. Two sources of sound A and B produces the wave of 350 Hz , they vibrate in the same phase. The particle P is vibrating under the influence of these two waves, if the amplitudes at the point P produced by the two waves is 0.3 mm and 0.4 mm , then the resultant amplitude of the point P will be when $AP - BP = 25\text{ cm}$ and the velocity of sound is 350 m/sec
- (a) 0.7 mm (b) 0.1 mm
(c) 0.2 mm (d) 0.5 mm
7. Two waves are propagating to the point P along a straight line produced by two sources A and B of simple harmonic and of equal frequency. The amplitude of every wave at P is ' a ' and the phase of A is ahead by $\frac{\pi}{3}$ than that of B and the distance AP is greater than BP by 50 cm . Then the resultant amplitude at the point P will be, if the wavelength is 1 meter
- [BVP 2003]
- (a) $2a$ (b) $a\sqrt{3}$
(c) $a\sqrt{2}$ (d) a
8. Coherent sources are characterized by the same
- [KCET 1993]
- (a) Phase and phase velocity
(b) Wavelength, amplitude and phase velocity
(c) Wavelength, amplitude and frequency
(d) Wavelength and phase
9. The minimum intensity of sound is zero at a point due to two sources of nearly equal frequencies, when
- (a) Two sources are vibrating in opposite phase
(b) The amplitude of two sources are equal
(c) At the point of observation, the amplitudes of two S.H.M. produced by two sources are equal and both the S.H.M. are along the same straight line
(d) Both the sources are in the same phase
10. Two sound waves (expressed in CGS units) given by $y_1 = 0.3 \sin \frac{2\pi}{\lambda}(vt - x)$ and $y_2 = 0.4 \sin \frac{2\pi}{\lambda}(vt - x + \theta)$ interfere. The resultant amplitude at a place where phase difference is $\pi/2$ will be
- [MP PET 1991]
- (a) 0.7 cm (b) 0.1 cm
(c) 0.5 cm (d) $\frac{1}{10}\sqrt{7}\text{ cm}$
11. If two waves having amplitudes $2A$ and A and same frequency and velocity, propagate in the same direction in the same phase, the resulting amplitude will be
- [MP PET 1991; DPMT 1999]
- (a) $3A$ (b) $\sqrt{5}A$
(c) $\sqrt{2}A$ (d) A
12. The intensity ratio of two waves is $1 : 16$. The ratio of their amplitudes is
- [EAMCET 1983]
- (a) $1 : 16$ (b) $1 : 4$
(c) $4 : 1$ (d) $2 : 1$
13. Out of the given four waves (1), (2), (3) and (4)

$$y = a \sin(kx + \omega t) \quad \dots(1)$$

$$y = a \sin(\omega t - kx) \quad \dots(2)$$

$$y = a \cos(kx + \omega t) \quad \dots(3)$$

$$y = a \cos(\omega t - kx) \quad \dots(4)$$

emitted by four different sources S_1, S_2, S_3 and S_4 respectively, interference phenomena would be observed in space under appropriate conditions when [CPMT 1988]

- (a) Source S_1 emits wave (1) and S_2 emits wave (2)
(b) Source S_3 emits wave (3) and S_4 emits wave (4)
(c) Source S_2 emits wave (2) and S_4 emits wave (4)
(d) S_4 emits waves (4) and S_3 emits waves (3)
14. Two waves of same frequency and intensity superimpose with each other in opposite phases, then after superposition the
- (a) Intensity increases by 4 times
(b) Intensity increases by two times
(c) Frequency increases by 4 times
(d) None of these
15. The superposing waves are represented by the following equations : $y_1 = 5 \sin 2\pi(10t - 0.1x)$, $y_2 = 10 \sin 2\pi(20t - 0.2x)$
- Ratio of intensities $\frac{I_{\max}}{I_{\min}}$ will be
- [AIIMS 1995; KCET 2001]
- (a) 1 (b) 9
(c) 4 (d) 16
16. The displacement of a particle is given by $x = 3 \sin(5\pi t) + 4 \cos(5\pi t)$
- The amplitude of the particle is [MP PMT 1999]
- (a) 3 (b) 4
(c) 5 (d) 7
17. Two waves $y_1 = A_1 \sin(\omega t - \beta_1)$, $y_2 = A_2 \sin(\omega t - \beta_2)$ Superimpose to form a resultant wave whose amplitude is
- [CPMT 1999]
- (a) $\sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\beta_1 - \beta_2)}$
(b) $\sqrt{A_1^2 + A_2^2 + 2A_1A_2 \sin(\beta_1 - \beta_2)}$
(c) $A_1 + A_2$
(d) $|A_1 + A_2|$
18. If the ratio of amplitude of wave is $2 : 1$, then the ratio of maximum and minimum intensity is [MH CET 1999]
- (a) $9 : 1$ (b) $1 : 9$
(c) $4 : 1$ (d) $1 : 4$
19. The two interfering waves have intensities in the ratio $9 : 4$. The ratio of intensities of maxima and minima in the interference pattern will be [AMU 2000]
- (a) $1 : 25$ (b) $25 : 1$
(c) $9 : 4$ (d) $4 : 9$
20. If the ratio of amplitude of two waves is $4 : 3$. Then the ratio of maximum and minimum intensity will be [MHCET 2000]

- (a) 16 : 18 (b) 18 : 16
(c) 49 : 1 (d) 1 : 49
21. Equation of motion in the same direction is given by $y_1 = A \sin(\omega t - kx)$, $y_2 = A \sin(\omega t - kx - \theta)$. The amplitude of the medium particle will be [BHU 2003]
- (a) $2A \cos \frac{\theta}{2}$ (b) $2A \cos \theta$
(c) $\sqrt{2}A \cos \frac{\theta}{2}$ (d) $1.2f, 1.2\lambda$
22. Two waves having the intensities in the ratio of 9 : 1 produce interference. The ratio of maximum to the minimum intensity, is equal to [CPMT 2001; Pb. PET 2004]
- (a) 2 : 1 (b) 4 : 1
(c) 9 : 1 (d) 10 : 8
23. The displacement of the interfering light waves are $y_1 = 4 \sin \omega t$ and $y_2 = 3 \sin\left(\omega t + \frac{\pi}{2}\right)$. What is the amplitude of the resultant wave [RPMT 1996; Orissa JEE 2005]
- (a) 5 (b) 7
(c) 1 (d) 0
24. Two waves are represented by $y_1 = a \sin\left(\omega t + \frac{\pi}{6}\right)$ and $y_2 = a \cos \omega t$. What will be their resultant amplitude [RPMT 1996]
- (a) a (b) $\sqrt{2}a$
(c) $\sqrt{3}a$ (d) $2a$
25. The amplitude of a wave represented by displacement equation $y = \frac{1}{\sqrt{a}} \sin \omega t \pm \frac{1}{\sqrt{b}} \cos \omega t$ will be [BVP 2003]
- (a) $\frac{a+b}{ab}$ (b) $\frac{\sqrt{a} + \sqrt{b}}{ab}$
(c) $\frac{\sqrt{a} \pm \sqrt{b}}{ab}$ (d) $\sqrt{\frac{a+b}{ab}}$
26. Two waves having equations $x_1 = a \sin(\omega t + \phi_1)$, $x_2 = a \sin(\omega t + \phi_2)$
If in the resultant wave the frequency and amplitude remain equal to those of superimposing waves. Then phase difference between them is [CBSE PMT 2001]
- (a) $\frac{\pi}{6}$ (b) $\frac{2\pi}{3}$
(c) $\frac{\pi}{4}$ (d) $\frac{\pi}{3}$

Beats

1. Two tuning forks when sounded together produced 4 beats/sec. The frequency of one fork is 256. The number of beats heard increases when the fork of frequency 256 is loaded with wax. The frequency of the other fork is [CPMT 1976; MP PMT 1993]
- (a) 504 (b) 520
(c) 260 (d) 252
2. Beats are the result of [CPMT 1971; J & K CET 2002]
- (a) Diffraction
(b) Destructive interference
(c) Constructive and destructive interference
(d) Superposition of two waves of nearly equal frequency
3. Two adjacent piano keys are struck simultaneously. The notes emitted by them have frequencies n_1 and n_2 . The number of beats heard per second is [CPMT 1974, 78; CBSE PMT 1993]
- (a) $\frac{1}{2}(n_1 - n_2)$ (b) $\frac{1}{2}(n_1 + n_2)$
(c) $n_1 \sim n_2$ (d) $2(n_1 - n_2)$
4. A tuning fork of frequency 100 when sounded together with another tuning fork of unknown frequency produces 2 beats per second. On loading the tuning fork whose frequency is not known and sounded together with a tuning fork of frequency 100 produces one beat, then the frequency of the other tuning fork is
- (a) 102 (b) 98
(c) 99 (d) 101
5. A tuning fork sounded together with a tuning fork of frequency 256 emits two beats. On loading the tuning fork of frequency 256, the number of beats heard are 1 per second. The frequency of tuning fork is [NCERT 1975, 81; MP PET 1985]
- (a) 257 (b) 258
(c) 256 (d) 254
6. If two tuning forks A and B are sounded together, they produce 4 beats per second. A is then slightly loaded with wax, they produce 2 beats when sounded again. The frequency of A is 256. The frequency of B will be [CPMT 1976; RPET 1998]
- (a) 250 (b) 252
(c) 260 (d) 262
7. The frequencies of two sound sources are 256 Hz and 260 Hz. At $t = 0$, the intensity of sound is maximum. Then the phase difference at the time $t = 1/16$ sec will be
- (a) Zero (b) π
(c) $\pi/2$ (d) $\pi/4$
8. Two tuning forks have frequencies 450 Hz and 454 Hz respectively. On sounding these forks together, the time interval between successive maximum intensities will be [MP PET 1989; MP PMT 2003]
- (a) $1/4$ sec (b) $1/2$ sec
(c) 1 sec (d) 2 sec

9. When a tuning fork of frequency 341 is sounded with another tuning fork, six beats per second are heard. When the second tuning fork is loaded with wax and sounded with the first tuning fork, the number of beats is two per second. The natural frequency of the second tuning fork is
[MP PET 1989]
- (a) 334 (b) 339
(c) 343 (d) 347
10. Two tuning forks of frequencies 256 and 258 vibrations/sec are sounded together, then time interval between consecutive maxima heard by the observer is
[MP PET/PMT 1988]
- (a) 2 sec (b) 0.5 sec
(c) 250 sec (d) 252 sec
11. A tuning fork gives 5 beats with another tuning fork of frequency 100 Hz. When the first tuning fork is loaded with wax, then the number of beats remains unchanged, then what will be the frequency of the first tuning fork
[MP PMT 1985]
- (a) 95 Hz (b) 100 Hz
(c) 105 Hz (d) 110 Hz
12. Tuning fork F_1 has a frequency of 256 Hz and it is observed to produce 6 beats/second with another tuning fork F_2 . When F_2 is loaded with wax, it still produces 6 beats/second with F_1 . The frequency of F_2 before loading was
[MP PET 1990]
- (a) 253 Hz (b) 262 Hz
(c) 250 Hz (d) 259 Hz
13. A tuning fork and a sonometer wire were sounded together and produce 4 beats per second. When the length of sonometer wire is 95 cm or 100 cm, the frequency of the tuning fork is
(a) 156 Hz (b) 152 Hz
(c) 148 Hz (d) 160 Hz
14. Two tuning forks A and B vibrating simultaneously produce 5 beats. Frequency of B is 512. It is seen that if one arm of A is filed, then the number of beats increases. Frequency of A will be
(a) 502 (b) 507
(c) 517 (d) 522
15. The beats are produced by two sound sources of same amplitude and of nearly equal frequencies. The maximum intensity of beats will be that of one source
[CPMT 1999]
- (a) Same (b) Double
(c) Four times (d) Eight times
16. Beats are produced by two waves given by $y_1 = a \sin 2000\pi t$ and $y_2 = a \sin 2008\pi t$. The number of beats heard per second is
(a) Zero (b) One
(c) Four (d) Eight
17. A tuning fork whose frequency as given by manufacturer is 512 Hz is being tested with an accurate oscillator. It is found that the fork produces a beat of 2 Hz when oscillator reads 514 Hz but produces a beat of 6 Hz when oscillator reads 510 Hz. The actual frequency of fork is
[MNR 1979; RPMT 1999]
- (a) 508 Hz (b) 512 Hz
(c) 516 Hz (d) 518 Hz
18. A tuning fork of frequency 480 Hz produces 10 beats per second when sounded with a vibrating sonometer string. What must have been the frequency of the string if a slight increase in tension produces lesser beats per second than before
(a) 460 Hz (b) 470 Hz
(c) 480 Hz (d) 490 Hz
19. When a tuning fork A of unknown frequency is sounded with another tuning fork B of frequency 256 Hz, then 3 beats per second are observed. After that A is loaded with wax and sounded, the again 3 beats per second are observed. The frequency of the tuning fork A is
[MP PMT 1994]
- (a) 250 Hz (b) 253 Hz
(c) 259 Hz (d) 262 Hz
20. A source of sound gives five beats per second when sounded with another source of frequency 100 s^{-1} . The second harmonic of the source together with a source of frequency 205 s^{-1} gives five beats per second. What is the frequency of the source
(a) 105 s^{-1} (b) 205 s^{-1}
(c) 95 s^{-1} (d) 100 s^{-1}
21. When two sound waves are superimposed, beats are produced when they have
[MP PET 1995; CBSE PMT 1992, 99; DCE 2000; DPMT 2000, 01]
- (a) Different amplitudes and phases
(b) Different velocities
(c) Different phases
(d) Different frequencies
22. Two tuning forks A and B give 4 beats per second. The frequency of A is 256 Hz. On loading B slightly, we get 5 beats in 2 seconds. The frequency of B after loading is
[MP PMT 1990; Haryana CEE 1996]
- (a) 253.5 Hz (b) 258.5 Hz
(c) 260 Hz (d) 252 Hz
23. A tuning fork A of frequency 200 Hz is sounded with fork B, the number of beats per second is 5. By putting some wax on A, the number of beats increases to 8. The frequency of fork B is
(a) 200 Hz (b) 195 Hz
(c) 192 Hz (d) 205 Hz
24. Two tuning forks, A and B, give 4 beats per second when sounded together. The frequency of A is 320 Hz. When some wax is added to B and it is sounded with A, 4 beats per second are again heard. The frequency of B is
[MP PMT 1997]
- (a) 312 Hz (b) 316 Hz
(c) 324 Hz (d) 328 Hz
25. Two tuning forks have frequencies 380 and 384 Hz respectively. When they are sounded together, they produce 4 beats. After hearing the maximum sound, how long will it take to hear the minimum sound
[MP PMT/PET 1998]
- (a) $\frac{1}{2} \text{ sec}$ (b) $\frac{1}{4} \text{ sec}$

- (c) $\frac{1}{8} \text{ sec}$ (d) $\frac{1}{16} \text{ sec}$
26. Beats are produced with the help of two sound waves of amplitudes 3 and 5 *units*. The ratio of maximum to minimum intensity in the beats is [MP PMT 1999]
- (a) 2 : 1 (b) 5 : 3
(c) 4 : 1 (d) 16 : 1
27. Two waves of lengths 50 *cm* and 51 *cm* produced 12 beats per second. The velocity of sound is [CBSE PMT 1999; Pb. PET 2001; AFMC 2003]
- (a) 306 *m/s* (b) 331 *m/s*
(c) 340 *m/s* (d) 360 *m/s*
28. Two waves $y = 0.25 \sin 316 t$ and $y = 0.25 \sin 310 t$ are travelling in same direction. The number of beats produced per second will be [CPMT 1993; JIPMER 2000]
- (a) 6 (b) 3
(c) $3/\pi$ (d) 3π
29. The couple of tuning forks produces 2 beats in the time interval of 0.4 seconds. So the beat frequency is [CPMT 1996]
- (a) 8 *Hz* (b) 5 *Hz*
(c) 2 *Hz* (d) 10 *Hz*
30. An unknown frequency x produces 8 beats per seconds with a frequency of 250 *Hz* and 12 beats with 270 *Hz* source, then x is
- (a) 258 *Hz* (b) 242 *Hz*
(c) 262 *Hz* (d) 282 *Hz*
31. Beats are produced by two waves
 $y_1 = a \sin 1000 \pi t$, $y_2 = a \sin 998 \pi t$
The number of beats heard/sec is [KCET 1998]
- (a) 0 (b) 2
(c) 1 (d) 4
32. The wavelengths of two waves are 50 and 51 *cm* respectively. If the temperature of the room is 20°C, then what will be the number of beats produced per second by these waves, when the speed of sound at 0°C is 332 *m/sec* [UPSEAT 1999]
- (a) 14 (b) 10
(c) 24 (d) None of these
33. Maximum number of beats frequency heard by a human being is
- (a) 10 (b) 4
(c) 20 (d) 6
34. Two sound waves of slightly different frequencies propagating in the same direction produce beats due to [MP PET 2000]
- (a) Interference (b) Diffraction
(c) Polarization (d) Refraction
35. On sounding tuning fork *A* with another tuning fork *B* of frequency 384 *Hz*, 6 *beats* are produced per second. After loading the prongs of *A* with some wax and then sounding it again with *B*, 4 *beats* are produced per second. What is the frequency of the tuning fork *A* [MP PMT 2000]
- (a) 388 *Hz* (b) 380 *Hz*
(c) 378 *Hz* (d) 390 *Hz*
36. It is possible to hear beats from the two vibrating sources of frequency [UPSEAT 2001]
- (a) 100 *Hz* and 150 *Hz* (b) 20 *Hz* and 25 *Hz*
(c) 400 *Hz* and 500 *Hz* (d) 1000 *Hz* and 1500 *Hz*
37. A tuning fork gives 4 beats with 50 *cm* length of a sonometer wire. If the length of the wire is shortened by 1 *cm*, the number of beats is still the same. The frequency of the fork is
- (a) 396 (b) 400
(c) 404 (d) 384
38. Two sound waves of wavelengths 5 *m* and 6 *m* formed 30 beats in 3 seconds. The velocity of sound is [EAMCET 2001]
- (a) 300 *ms* (b) 310 *ms*
(c) 320 *ms* (d) 330 *ms*
39. The wavelength of a particle is 99 *cm* and that of other is 100 *cm*. Speed of sound is 396 *m/s*. The number of beats heard is
- (a) 4 (b) 5
(c) 1 (d) 8
40. A tuning fork arrangement (pair) produces 4 *beats/sec* with one fork of frequency 288 *cps*. A little wax is placed on the unknown fork and it then produces 2 *beats/sec*. The frequency of the unknown fork is [CPMT 1997; KCET 2000] [KCET 1998; AIEEE 2002]
- (a) 286 *cps* (b) 292 *cps*
(c) 294 *cps* (d) 288 *cps*
41. A tuning fork vibrates with 2 beats in 0.04 second. The frequency of the fork is [AFMC 2003]
- (a) 50 *Hz* (b) 100 *Hz*
(c) 80 *Hz* (d) None of these
42. Two sound sources when sounded simultaneously produce four beats in 0.25 second. the difference in their frequencies must be
- (a) 4 (b) 8
(c) 16 (d) 1
43. A tuning fork of known frequency 256 *Hz* makes 5 beats per second with the vibrating string of a piano. The beat frequency decreases to 2 beats per second when the tension in the piano string is slightly increased. The frequency of the piano string before increasing the tension was [RPMT 2000] [AIEEE 2003]
- (a) 256 + 5 *Hz* (b) 256 + 2 *Hz*
(c) 256 - 2 *Hz* (d) 256 - 5 *Hz*
44. When temperature increases, the frequency of a tuning fork [AIEEE 2002]
- (a) Increases
(b) Decreases
(c) Remains same
(d) Increases or decreases depending on the material
45. Two strings *X* and *Y* of a sitar produce a beat frequency 4 *Hz*. When the tension of the string *Y* is slightly increased the beat frequency is

found to be 2 Hz. If the frequency of X is 300 Hz, then the original frequency of Y was

[UPSEAT 2000]

- (a) 296 Hz (b) 298 Hz
(c) 302 Hz (d) 304 Hz

46. The frequency of tuning forks A and B are respectively 3% more and 2% less than the frequency of tuning fork C . When A and B are simultaneously excited, 5 beats per second are produced. Then the frequency of the tuning fork ' A ' (in Hz) is

- (a) 98 (b) 100
(c) 103 (d) 105

47. When a tuning fork vibrates, the waves produced in the fork are

- (a) Longitudinal (b) Transverse
(c) Progressive (d) Stationary

48. Two vibrating tuning forks produce progressive waves given by $Y_1 = 4 \sin 500\pi t$ and $Y_2 = 2 \sin 506\pi t$. Number of beats produced per minute is [CBSE PMT 2005]

- (a) 360 (b) 180
(c) 3 (d) 60

49. When a tuning fork produces sound waves in air, which one of the following is same in the material of tuning fork as well as in air

- (a) Wavelength (b) Frequency
(c) Velocity (d) Amplitude

50. The disc of a siren containing 60 holes rotates at a constant speed of 360 rpm. The emitted sound is in unison with a tuning fork of frequency [KCET 2005]

- (a) 10 Hz (b) 360 Hz
(c) 216 Hz (d) 6 Hz

51. A sound source of frequency 170 Hz is placed near a wall. A man walking from a source towards the wall finds that there is a periodic rise and fall of sound intensity. If the speed of sound in air is 340 m/s the distance (in metres) separating the two adjacent positions of minimum intensity is

[MNR 1992; UPSEAT 2000; CPMT 2002]

- (a) 1/2 (b) 1
(c) 3/2 (d) 2

Stationary Waves

1. The distance between the nearest node and antinode in a stationary wave is

[MP PET 1984; CBSE PMT 1993; AFMC 1996; RPET 2002]

- (a) λ (b) $\frac{\lambda}{2}$
(c) $\frac{\lambda}{4}$ (d) 2λ

2. In stationary wave [MP PET 1987; BHU 1995]

- (a) Strain is maximum at nodes
(b) Strain is maximum at antinodes
(c) Strain is minimum at nodes
(d) Amplitude is zero at all the points

3. The phase difference between the two particles situated on both the sides of a node is [MP PET 2002]

- (a) 0° (b) 90°

- (c) 180° (d) 360°

4. Which of the property makes difference between progressive and stationary waves [MP PMT 1987]

- (a) Amplitude (b) Frequency
(c) Propagation of energy (d) Phase of the wave

5. Stationary waves are formed when

[NCERT 1983]

- (a) Two waves of equal amplitude and equal frequency travel along the same path in opposite directions
(b) Two waves of equal wavelength and equal amplitude travel along the same path with equal speeds in opposite directions
(c) Two waves of equal wavelength and equal phase travel along the same path with equal speed
(d) Two waves of equal amplitude and equal speed travel along the same path in opposite direction [AFMC 2001]

6. For the stationary wave $y = 4 \sin\left(\frac{\pi x}{15}\right) \cos(96\pi t)$, the distance between a node and the next antinode is [MP PMT 1987]

- (a) 7.5 (b) 15
(c) 22.5 (d) 30

7. The equation of stationary wave along a stretched string is given by

$$y = 5 \sin \frac{\pi x}{4} \cos 40\pi t, \text{ where } x \text{ and } y \text{ are in cm and } t \text{ in second. [AFMC 2005]}$$

The separation between two adjacent nodes is [CPMT 1990; MP PET 1999; AMU 1999; DPMT 2004; BHU 2005]

- (a) 1.5 cm (b) 3 cm
(c) 6 cm (d) 4 cm

8. The equation $\vec{\phi}(x, t) = \vec{j} \sin\left(\frac{2\pi}{\lambda} v t\right) \cos\left(\frac{2\pi}{\lambda} x\right)$ represents

[MNR 1994]

- (a) Transverse progressive wave
(b) Longitudinal progressive wave
(c) Longitudinal stationary wave
(d) Transverse stationary wave

9. The equation of a stationary wave is $y = 0.8 \cos\left(\frac{\pi x}{20}\right) \sin 200\pi t$, where x is in cm and t is in sec. The separation between consecutive nodes will be

[MP PET 1994]

- (a) 20 cm (b) 10 cm
(c) 40 cm (d) 30 cm

10. In a stationary wave, all particles are

[MP PMT 1994]

- (a) At rest at the same time twice in every period of oscillation
(b) At rest at the same time only once in every period of oscillation
(c) Never at rest at the same time
(d) Never at rest at all

11. A wave represented by the given equation $y = a \cos(kx - \omega t)$ is superposed with another wave to form a stationary wave such that the point $x = 0$ is a node. The equation for the other wave is

AIIMS 1998; SCRA 1998; MP PET 2001; KCET 2001;

AIEEE 2002; UPSEAT 2004]

- (a) $y = a \sin(kx + \omega t)$ (b) $y = -a \cos(kx + \omega t)$
(c) $y = -a \cos(kx - \omega t)$ (d) $y = -a \sin(kx - \omega t)$

12. At a certain instant a stationary transverse wave is found to have maximum kinetic energy. The appearance of string at that instant is

(a) Sinusoidal shape with amplitude $A/3$
 (b) Sinusoidal shape with amplitude $A/2$
 (c) Sinusoidal shape with amplitude A
 (d) Straight line

13. The equation $y = 0.15 \sin 5x \cos 300t$, describes a stationary wave. The wavelength of the stationary wave is

[MP PMT 1995]

(a) Zero (b) 1.256 metres
 (c) 2.512 metres (d) 0.628 metre

14. In stationary waves, antinodes are the points where there is

[MP PMT 1996]

(a) Minimum displacement and minimum pressure change
 (b) Minimum displacement and maximum pressure change
 (c) Maximum displacement and maximum pressure change
 (d) Maximum displacement and minimum pressure change

15. In stationary waves all particles between two nodes pass through the mean position

[MP PMT 1999; KCET 2001]

(a) At different times with different velocities
 (b) At different times with the same velocity
 (c) At the same time with equal velocity
 (d) At the same time with different velocities

16. Standing waves can be produced [IIT-JEE 1999]

(a) On a string clamped at both the ends
 (b) On a string clamped at one end and free at the other
 (c) When incident wave gets reflected from a wall
 (d) When two identical waves with a phase difference of π are moving in the same direction

17. A standing wave having 3 nodes and 2 antinodes is formed between two atoms having a distance 1.21 Å between them. The wavelength of the standing wave is

[CBSE PMT 1998; MH CET 2002; AIIMS 2000; BHU 2001]

(a) 1.21 Å (b) 2.42 Å
 (c) 6.05 Å (d) 3.63 Å

18. In stationary waves, distance between a node and its nearest antinode is 20 cm. The phase difference between two particles having a separation of 60 cm will be

[CMEET Bihar 1995]

(a) Zero (b) $\pi/2$
 (c) π (d) $3\pi/2$

19. Stationary waves of frequency 300 Hz are formed in a medium in which the velocity of sound is 1200 metre/sec. The distance between a node and the neighbouring antinode is

(a) 1 m (b) 2 m
 (c) 3 m (d) 4 m

20. Which two of the given transverse waves will give stationary waves when get superimposed

[RPET 1997; MP PET 1993]

$$z_1 = a \cos(kx - \omega t) \quad \dots (A)$$

[AIIMS 1995]

$$z_2 = a \cos(kx + \omega t) \quad \dots (B)$$

$$z_3 = a \cos(ky - \omega t) \quad \dots (C)$$

(a) A and B (b) A and C
 (c) B and C (d) Any two

21. A standing wave is represented by

$$Y = A \sin(100t) \cos(0.01x)$$

where Y and A are in millimetre, t is in seconds and x is in metre. The velocity of wave is

[CBSE PMT 1994; AFMC 2002]

(a) 10^4 m/s
 (b) 1 m/s
 (c) 10^{-4} m/s
 (d) Not derivable from above data

22. A wave of frequency 100 Hz is sent along a string towards a fixed end. When this wave travels back after reflection, a node is formed at a distance of 10 cm from the fixed end of the string. The speed of incident (and reflected) wave are

[CBSE PMT 1994]

(a) 40 m/s (b) 20 m/s
 (c) 10 m/s (d) 5 m/s

23. $y = a \cos(kx + \omega t)$ superimposes on another wave giving a stationary wave having node at $x = 0$. What is the equation of the other wave [BHU 1998; DPMT 2000]

(a) $-a \cos(kx + \omega t)$ (b) $a \cos(kx - \omega t)$
 (c) $-a \cos(kx - \omega t)$ (d) $-a \sin(kx + \omega t)$

24. Two waves are approaching each other with a velocity of 20 m/s and frequency n . The distance between two consecutive nodes is

(a) $\frac{20}{n}$ (b) $\frac{10}{n}$
 (c) $\frac{5}{n}$ (d) $\frac{n}{10}$

25. Energy is not carried by which of the following waves

[RPMT 1998; AIIMS 1998, 99]

(a) Stationary (b) Progressive
 (c) Transverse (d) Electromagnetic

26. The stationary wave produced on a string is represented by the equation $y = 5 \cos(\pi x / 3) \sin 40\pi t$. Where x and y are in cm and t is in seconds. The distance between consecutive nodes is

(a) 5 cm (b) π cm
 (c) 3 cm [SCRA 1994] (d) 40 cm

27. Two sinusoidal waves with same wavelengths and amplitudes travel in opposite directions along a string with a speed 10 ms. If the minimum time interval between two instants when the string is flat is 0.5 s, the wavelength of the waves is

(a) 25 m (b) 20 m
 (c) 15 m (d) 10 m

28. "Stationary waves" are so called because in them [MP PMT 2001]
- The particles of the medium are not disturbed at all
 - The particles of the medium do not execute SHM
 - There occurs no flow of energy along the wave
 - The interference effect can't be observed
29. Two waves are approaching each other with a velocity of 16 m/s and frequency n . The distance between two consecutive nodes is
- $\frac{16}{n}$
 - $\frac{8}{n}$
 - $\frac{n}{16}$
 - $\frac{n}{8}$
30. Stationary waves [Kerala (Med.) 2002]
- Transport energy
 - Does not transport energy
 - Have nodes and antinodes
 - Both (b) and (c)
31. In a stationary wave all the particles [KCET 2002]
- On either side of a node vibrate in same phase
 - In the region between two nodes vibrate in same phase
 - In the region between two antinodes vibrate in same phase
 - Of the medium vibrate in same phase
32. When a stationary wave is formed then its frequency is [Kerala (Engg.) 2002]
- Same as that of the individual waves
 - Twice that of the individual waves
 - Half that of the individual waves
 - None of the above
33. In stationary waves [RPMT 1998; JIPMER 2002]
- Energy is uniformly distributed
 - Energy is minimum at nodes and maximum at antinodes
 - Energy is maximum at nodes and minimum at antinodes
 - Alternating maximum and minimum energy producing at nodes and antinodes
34. Equation of a stationary wave is $y = 10 \sin \frac{\pi x}{4} \cos 20\pi t$. Distance between two consecutive nodes is [MP PMT 2002]
- 4
 - 2
 - 1
 - 8
35. At nodes in stationary waves [SCRA 1994; UPSEAT 2000; MP PET 2003; RPET 2003]
- Change in pressure and density are maximum
 - Change in pressure and density are minimum
 - Strain is zero
 - Energy is minimum
36. Consider the three waves z_1, z_2 and z_3 as
- $$z_1 = A \sin(kx - \omega t), \quad z_2 = A \sin(kx + \omega t)$$
- and $z_3 = A \sin(ky - \omega t)$. Which of the following represents a standing wave [DCE 2004]
- $z_1 + z_2$
 - $z_2 + z_3$
 - $z_3 + z_1$
 - $z_1 + z_2 + z_3$
37. The following equations represent progressive transverse waves $Z_1 = A \cos(\omega t - kx)$, $Z_2 = A \cos(\omega t + kx)$, $Z_3 = A \cos(\omega t + ky)$ and $Z_4 = A \cos(2\omega t - 2ky)$. A stationary wave will be formed by superposing [MP PET 1993]
- Z_1 and Z_2
 - Z_1 and Z_4
 - Z_2 and Z_3
 - Z_3 and Z_4
38. [CPMT 2001; Pb. PMT 1999] Two travelling waves $y_1 = A \sin[k(x - ct)]$ and $y_2 = A \sin[k(x + ct)]$ are superimposed on string. The distance between adjacent nodes is [IIT 1992]
- ct / π
 - $ct / 2\pi$
 - $\pi / 2k$
 - π / k
39. A string vibrates according to the equation $y = 5 \sin\left(\frac{2\pi x}{3}\right) \cos 20\pi t$, where x and y are in cm and t in sec. The distance between two adjacent nodes is [UPSEAT 2005]
- 3 cm
 - 4.5 cm
 - 6 cm
 - 1.5 cm

Vibration of String

1. A string fixed at both the ends is vibrating in two segments. The wavelength of the corresponding wave is [SCRA 1994]
- $\frac{l}{4}$
 - $\frac{l}{2}$
 - l
 - $2l$
2. A 1 cm long string vibrates with fundamental frequency of 256 Hz. If the length is reduced to $\frac{1}{4}$ cm keeping the tension unaltered, the new fundamental frequency will be [BHU 1997]
- 64
 - 256
 - 512
 - 1024
3. Standing waves are produced in a 10 m long stretched string. If the string vibrates in 5 segments and the wave velocity is 20 m/s, the frequency is [CBSE PMT 1997; AIIMS 1998; JIPMER 2000]
- 2 Hz
 - 4 Hz
 - 5 Hz
 - 10 Hz
4. The velocity of waves in a string fixed at both ends is 2 m/s. The string forms standing waves with nodes 5.0 cm apart. The frequency of vibration of the string in Hz is [SCRA 1998]
- 40
 - 30
 - 20
 - 10
5. Which of the following is the example of transverse wave [CPMT 1999]
- Sound waves
 - Compressional waves in a spring
 - Vibration of string
 - All of these
6. A stretched string of 1m length and mass 5×10^{-4} kg is having tension of 20N. If it is plucked at 25cm from one end then it will vibrate with frequency

[RPET 1999; RPMT 2002]

- (a) 100 Hz (b) 200 Hz
(c) 256 Hz (d) 400 Hz
7. Two similar sonometer wires given fundamental frequencies of 500 Hz. These have same tensions. By what amount the tension be increased in one wire so that the two wires produce 5 beats/sec [RPET 1999]
- (a) 1% (b) 2%
(c) 3% (d) 4%
8. A string is producing transverse vibration whose equation is $y = 0.021 \sin(\pi x + 30t)$, Where x and y are in meters and t is in seconds. If the linear density of the string is $1.3 \times 10^{-4} \text{ kg/m}$, then the tension in the string in N will be

[RPET 1999; RPMT 2002]

- (a) 10 (b) 0.5
(c) 1 (d) 0.117
9. If the tension of sonometer's wire increases four times then the fundamental frequency of the wire will increase by [RPMT 1999]
- (a) 2 times (b) 4 times
(c) 1/2 times (d) None of the above
10. If vibrations of a string are to be increased by a factor of two, then tension in the string must be made

[AIIMS 1999; Pb. PET 2000]

- (a) Half (b) Twice
(c) Four times (d) Eight times
11. Four wires of identical length, diameters and of the same material are stretched on a sonometre wire. If the ratio of their tensions is 1 : 4 : 9 : 16 then the ratio of their fundamental frequencies are [KCET 2000]
- (a) 16 : 9 : 4 : 1 (b) 4 : 3 : 2 : 1
(c) 1 : 4 : 2 : 16 (d) 1 : 2 : 3 : 4
12. A tuning fork vibrating with a sonometer having 20 cm wire produces 5 beats per second. The beat frequency does not change if the length of the wire is changed to 21 cm. the frequency of the tuning fork (in Hertz) must be

[UPSEAT 2000; Pb. PET 2004]

- (a) 200 (b) 210
(c) 205 (d) 215
13. A stretched string of length l , fixed at both ends can sustain stationary waves of wavelength λ , given by

[UPSEAT 2000; Pb. PET 2004; CPMT 2005]

- (a) $\lambda = \frac{n^2}{2l}$ (b) $\lambda = \frac{l^2}{2n}$
(c) $\lambda = \frac{2l}{n}$ (d) $\lambda = 2ln$
14. If you set up the seventh harmonic on a string fixed at both ends, how many nodes and antinodes are set up in it [AMU 2000]
- (a) 8, 7 (b) 7, 7
(c) 8, 9 (d) 9, 8
15. If you set up the ninth harmonic on a string fixed at both ends, its frequency compared to the seventh harmonic

[AMU (Engg.) 2000]

- (a) Higher (b) Lower

- (c) Equal (d) None of the above

16. Frequency of a sonometer wire is n . Now its tension is increased 4 times and its length is doubled then new frequency will be
- (a) $n/2$ (b) $4n$
(c) $2n$ (d) n

A device used for investigating the vibration of a fixed string or wire is [BHU 2000]

- (a) Sonometer (b) barometer
(c) Hydrometer (d) None of these
18. A string on a musical instrument is 50 cm long and its fundamental frequency is 270 Hz. If the desired frequency of 1000 Hz is to be produced, the required length of the string is

[EAMCET (Engg.) 1998; CPMT 2000; Pb. PET 2001]

- (a) 13.5 cm (b) 2.7 cm
(c) 5.4 cm (d) 10.3 cm
19. The tension in a piano wire is 10N. What should be the tension in the wire to produce a note of double the frequency
- (a) 5 N (b) 20 N
(c) 40 N (d) 80 N
20. To increase the frequency from 100 Hz to 400 Hz the tension in the string has to be changed by [RPET 2001]
- (a) 4 times (b) 16 times
(c) 20 times (d) None of these
21. In order to double the frequency of the fundamental note emitted by a stretched string, the length is reduced to $\frac{3}{4}$ th of the original length and the tension is changed. The factor by which the tension is to be changed, is [EAMCET 2001]

- (a) $\frac{3}{8}$ (b) $\frac{2}{3}$
(c) $\frac{8}{9}$ (d) $\frac{9}{4}$

22. A string of 7 m length has a mass of 0.035 kg. If tension in the string is 60.5 N, then speed of a wave on the string is

[CBSE PMT 2001]

- (a) 77 m/s (b) 102 m/s
(c) 110 m/s (d) 165 m/s

23. A second harmonic has to be generated in a string of length l stretched between two rigid supports. The point where the string has to be plucked and touched are

[KCET 2001]

- (a) Plucked at $\frac{l}{4}$ and touch at $\frac{l}{2}$
(b) Plucked at $\frac{l}{4}$ and touch at $\frac{3l}{4}$
(c) Plucked at $\frac{l}{2}$ and touched at $\frac{l}{4}$
(d) Plucked at $\frac{l}{2}$ and touched at $\frac{3l}{4}$

24. Transverse waves of same frequency are generated in two steel wires A and B. The diameter of A is twice of B and the tension in A is half that in B. The ratio of velocities of wave in A and B is

- (a) $1 : 3\sqrt{2}$ (b) $1 : 2\sqrt{2}$

- (c) 1 : 2 (d) $\sqrt{2} : 1$
25. A sonometer wire resonates with a given tuning fork forming standing waves with five antinodes between the two bridges when a mass of 9 kg is suspended from the wire. When this mass is replaced by a mass M , the wire resonates with the same tuning fork forming three antinodes for the same positions of the bridges. The value of M is
[IIT-JEE (Screening) 2002]
- (a) 25 kg (b) 5 kg
(c) 12.5 kg (d) $1/25$ kg
26. The tension of a stretched string is increased by 69%. In order to keep its frequency of vibration constant, its length must be increased by
[KCET 2002]
- (a) 20% (b) 30%
(c) $\sqrt{69}\%$ (d) 69%
27. The length of a sonometer wire tuned to a frequency of 250 Hz is 0.60 metre. The frequency of tuning fork with which the vibrating wire will be in tune when the length is made 0.40 metre is
- (a) 250 Hz (b) 375 Hz
(c) 256 Hz (d) 384 Hz
28. Length of a string tied to two rigid supports is 40 cm. Maximum length (wavelength in cm) of a stationary wave produced on it is
- (a) 20 (b) 80
(c) 40 (d) 120
29. A string in musical instrument is 50 cm long and its fundamental frequency is 800 Hz. If a frequency of 1000 Hz is to be produced, then required length of string is
[AIIMS 2002]
- (a) 62.5 cm (b) 50 cm
(c) 40 cm (d) 37.5 cm
30. Two wires are in unison. If the tension in one of the wires is increased by 2%, 5 beats are produced per second. The initial frequency of each wire is
[MP PET 2002]
- (a) 200 Hz (b) 400 Hz
(c) 500 Hz (d) 1000 Hz
31. Two uniform strings A and B made of steel are made to vibrate under the same tension. If the first overtone of A is equal to the second overtone of B and if the radius of A is twice that of B , the ratio of the lengths of the strings is
[EAMCET 2003]
- (a) 1 : 2 (b) 1 : 3
(c) 1 : 4 (d) 1 : 6
32. If the length of a stretched string is shortened by 40% and the tension is increased by 44%, then the ratio of the final and initial fundamental frequencies is
[EAMCET 2003]
- (a) 2 : 1 (b) 3 : 2
(c) 3 : 4 (d) 1 : 3
33. Two wires are fixed in a sonometer. Their tensions are in the ratio 8 : 1. The lengths are in the ratio 36 : 35. The diameters are in the ratio 4 : 1. Densities of the materials are in the ratio 1 : 2. If the lower frequency in the setting is 360 Hz, the beat frequency when the two wires are sounded together is
- (a) 5 (b) 8 (c) 6 (d) 10
34. The first overtone of a stretched wire of given length is 320 Hz. The first harmonic is :
[DPMT 2004]
- (a) 320 Hz (b) 160 Hz
(c) 480 Hz (d) 640 Hz
35. Two perfectly identical wires are in unison. When the tension in one wire is increased by 1%, then on sounding them together 3 beats are heard in 2 sec. The initial frequency of each wire is :
- (a) 220s^{-1} (b) 320s^{-1}
(c) 150s^{-1} (d) 300s^{-1}
36. A tuning fork of frequency 392 Hz, resonates with 50 cm length of a string under tension (T). If length of the string is decreased by 2%, keeping the tension constant, the number of beats heard when the string and the tuning fork made to vibrate simultaneously is
- (a) 4 (b) 6
(c) 8 (d) 12
37. The sound carried by air from a sitar to a listener is a wave of the following type:
[BHU 2002] [MP PMT 1987; RPET 2001]
- (a) Longitudinal stationary (b) Transverse progressive
(c) Transverse stationary (d) Longitudinal progressive
38. In Melde's experiment in the transverse mode, the frequency of the tuning fork and the frequency of the waves in the strings are in the ratio
[AIIEEE 2002] [KCET 2004]
- (a) 1 : 1 (b) 1 : 2
(c) 2 : 1 (d) 4 : 1
39. The frequency of transverse vibrations in a stretched string is 200 Hz. If the tension is increased four times and the length is reduced to one-fourth the original value, the frequency of vibration will be
- (a) 25 Hz (b) 200 Hz
(c) 400 Hz (d) 1600 Hz
40. Three similar wires of frequency n , n and n are joined to make one wire. Its frequency will be
[CBSE PMT 2000]
- (a) $n = n_1 + n_2 + n_3$ (b) $\frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$
(c) $\frac{1}{\sqrt{n}} = \frac{1}{\sqrt{n_1}} + \frac{1}{\sqrt{n_2}} + \frac{1}{\sqrt{n_3}}$ (d) $\frac{1}{n^2} = \frac{1}{n_1^2} + \frac{1}{n_2^2} + \frac{1}{n_3^2}$
41. A steel rod 100 cm long is clamped at its mid-point. The fundamental frequency of longitudinal vibrations of the rod is given to be 2.53 kHz. What is the speed of sound in steel
[AFMC 2000]
- (a) 5.06 km/s (b) 6.06 km/s
(c) 7.06 km/s (d) 8.06 km/s
42. Two wires are producing fundamental notes of the same frequency. Change in which of the following factors of one wire will not produce beats between them
[BHU (Med.) 1999]
- (a) Amplitude of the vibrations
(b) Material of the wire
(c) Tension
(d) Diameter of the wires
- [KCET 2005]



43. Calculate the frequency of the second harmonic formed on a string of length 0.5 m and mass $2 \times 10^{-3}\text{ kg}$ when stretched with a tension of 20 N [BHU (Med.) 2000]

(a) 274.4 Hz (b) 744.2 Hz
(c) 44.72 Hz (d) 447.2 Hz

44. The fundamental frequency of a string stretched with a weight of 4 kg is 256 Hz . The weight required to produce its octave is

(a) 4 kg wt (b) 8 kg wt
(c) 12 kg wt (d) 16 kg wt

45. Two vibrating strings of the same material but lengths L and $2L$ have radii $2r$ and r respectively. They are stretched under the same tension. Both the strings vibrate in their fundamental modes, the one of length L with frequency n and the other with frequency n' . The ratio n/n' is given by

[IIT-JEE (Screening) 2000]

(a) 2 (b) 4
(c) 8 (d) 1

46. If the tension and diameter of a sonometer wire of fundamental frequency n are doubled and density is halved then its fundamental frequency will become

[CBSE PMT 2001]

(a) $\frac{n}{4}$ (b) $\sqrt{2}n$
(c) n (d) $\frac{n}{\sqrt{2}}$

47. In a sonometer wire, the tension is maintained by suspending a 50.7 kg mass from the free end of the wire. The suspended mass has a volume of 0.0075 m^3 . The fundamental frequency of the wire is 260 Hz . If the suspended mass is completely submerged in water, the fundamental frequency will become (take $g = 10\text{ ms}^{-2}$)

[KCET 2001]

(a) 240 Hz (b) 230 Hz
(c) 220 Hz (d) 200 Hz

48. A string is rigidly tied at two ends and its equation of vibration is given by $y = \cos 2\pi t \sin \pi x$. Then minimum length of string is

[RPM 2001]

(a) 1 m (b) $\frac{1}{2}\text{ m}$
(c) 5 m (d) $2\pi\text{ m}$

49. Fundamental frequency of sonometer wire is n . If the length, tension and diameter of wire are tripled, the new fundamental frequency is

(a) $\frac{n}{\sqrt{3}}$ (b) $\frac{n}{3}$
(c) $n\sqrt{3}$ (d) $\frac{n}{3\sqrt{3}}$

50. A string of length 2 m is fixed at both ends. If this string vibrates in its fourth normal mode with a frequency of 500 Hz then the waves would travel on it with a velocity of

[BCECE 2005]

(a) 125 m/s (b) 250 m/s
(c) 500 m/s (d) 1000 m/s

51. The fundamental frequency of a sonometre wire is n . If its radius is doubled and its tension becomes half, the material of the wire remains same, the new fundamental frequency will be

(a) n (b) $\frac{n}{\sqrt{2}}$
(c) $\frac{n}{2}$ (d) $\frac{n}{2\sqrt{2}}$

52. In an experiment with sonometer a tuning fork of frequency 256 Hz resonates with a length of 25 cm and another tuning fork resonates with a length of 100 cm . Tension of the string remaining constant the frequency of the second tuning fork is

(a) 163.84 Hz (b) 400 Hz
(c) 320 Hz (d) 204.8 Hz

Organ Pipe (Vibration of Air Column)

1. The length of two open organ pipes are l and $(l + \Delta l)$ respectively. Neglecting end correction, the frequency of beats between them will be approximately

[MP PET 1994; BHU 1995]

(a) $\frac{v}{2l}$ (b) $\frac{v}{4l}$
(c) $\frac{v\Delta l}{2l^2}$ (d) $\frac{v\Delta l}{l}$

(Here v is the speed of sound)

2. A tube closed at one end and containing air is excited. It produces the fundamental note of frequency 512 Hz . If the same tube is open at both the ends the fundamental frequency that can be produced is

(a) 1024 Hz (b) 512 Hz
(c) 256 Hz (d) 128 Hz

3. A closed pipe and an open pipe have their first overtones identical in frequency. Their lengths are in the ratio

[Roorkee 1999]

(a) $1 : 2$ (b) $2 : 3$
(c) $3 : 4$ (d) $4 : 5$

4. The first overtone in a closed pipe has a frequency

[JIPMER 1999]

(a) Same as the fundamental frequency of an open tube of same length
(b) Twice the fundamental frequency of an open tube of same length
(c) Same as that of the first overtone of an open tube of same length
(d) None of the above

5. An empty vessel is partially filled with water, then the frequency of vibration of air column in the vessel

[KCET 2000]

(a) Remains same
(b) Decreases
(c) Increases
(d) First increases then decreases

6. It is desired to increase the fundamental resonance frequency in a tube which is closed at one end. This can be achieved by

(a) Replacing the air in the tube by hydrogen gas
(b) Increasing the length of the tube
(c) Decreasing the length of the tube
(d) Opening the closed end of the tube

7. An air column in a pipe, which is closed at one end, will be in resonance with a vibrating body of frequency 166 Hz , if the length of the air column is

[UPSEAT 2001]

- (a) 2.00 m (b) 1.50 m (c) Third (d) Fourth
- (c) 1.00 m (d) 0.50 m
8. If the velocity of sound in air is 350 m/s. Then the fundamental frequency of an open organ pipe of length 50 cm, will be [CPMT 1997; MH CET 2001; Pb. PMT 2001] [Pb. PMT 2002]
- (a) 350 Hz (b) 175 Hz
- (c) 900 Hz (d) 750 Hz
9. If the length of a closed organ pipe is 1m and velocity of sound is 330 m/s, then the frequency for the second note is [AFMC 2001]
- (a) $4 \times \frac{330}{4}$ Hz (b) $3 \times \frac{330}{4}$ Hz
- (c) $2 \times \frac{330}{4}$ Hz (d) $2 \times \frac{4}{330}$ Hz
10. The fundamental note produced by a closed organ pipe is of frequency f . The fundamental note produced by an open organ pipe of same length will be of frequency [BHU 2001]
- (a) $\frac{f}{2}$ (b) f
- (c) $2f$ (d) $4f$
11. If the velocity of sound in air is 336 m/s. The maximum length of a closed pipe that would produce a just audible sound will be [KCET 2001]
- (a) 3.2 cm (b) 4.2 m
- (c) 4.2 cm (d) 3.2 m
12. An organ pipe P_1 closed at one end vibrating in its first overtone and another pipe P_2 open at both ends vibrating in its third overtone are in resonance with a given tuning fork. The ratio of lengths of P_1 and P_2 is [EAMCET 1997; MH CET 1999; AFMC 2001]
- (a) 1 : 2 (b) 1 : 3
- (c) 3 : 8 (d) 3 : 4
13. A resonance air column of length 20 cm resonates with a tuning fork of frequency 250 Hz. The speed of sound in air is [AFMC 1999; BHU 2000; CPMT 2001]
- (a) 300 m/s (b) 200 m/s
- (c) 150 m/s (d) 75 m/s
14. A cylindrical tube, open at both ends, has a fundamental frequency f_0 in air. The tube is dipped vertically into water such that half of its length is inside water. The fundamental frequency of the air column now is [RPET 1999; RPMT 1998, 2000; J & K CET 2000; KCET 2002; BHU 2002; BCECE 2003]
- (a) $3f_0/4$ (b) f_0
- (c) $f_0/2$ (d) $2f_0$
15. If the length of a closed organ pipe is 1.5 m and velocity of sound is 330 m/s, then the frequency for the second note is [CBSE PMT 2002]
- (a) 220 Hz (b) 165 Hz
- (c) 110 Hz (d) 55 Hz
16. A pipe 30 cm long is open at both ends. Which harmonic mode of the pipe is resonantly excited by a 1.1 kHz source? (Take speed of sound in air = 330 ms) [AMU 2002]
- (a) First (b) Second
17. Two closed organ pipes, when sounded simultaneously gave 4 beats per sec. If longer pipe has a length of 1m. Then length of shorter pipe will be, ($v = 300$ m/s)
- (a) 185.5 cm (b) 94.9 cm
- (c) 90 cm (d) 80 cm
18. A source of sound placed at the open end of a resonance column sends an acoustic wave of pressure amplitude ρ_0 inside the tube. If the atmospheric pressure is ρ_A , then the ratio of maximum and minimum pressure at the closed end of the tube will be
- (a) $\frac{(\rho_A + \rho_0)}{(\rho_A - \rho_0)}$ (b) $\frac{(\rho_A + 2\rho_0)}{(\rho_A - 2\rho_0)}$
- (c) $\frac{\rho_A}{\rho_A}$ (d) $\frac{(\rho_A + \frac{1}{2}\rho_0)}{(\rho_A - \frac{1}{2}\rho_0)}$
19. Two closed pipe produce 10 beats per second when emitting their fundamental nodes. If their length are in ratio of 25 : 26. Then their fundamental frequency in Hz, are [MH CET 2002]
- (a) 270, 280 (b) 260, 270
- (c) 260, 250 (d) 260, 280
20. A closed organ pipe and an open organ pipe are tuned to the same fundamental frequency. What is the ratio of lengths
- (a) 1 : 2 (b) 2 : 1
- (c) 2 : 3 (d) 4 : 3
21. An open pipe resonates with a tuning fork of frequency 500 Hz. it is observed that two successive nodes are formed at distances 16 and 46 cm from the open end. The speed of sound in air in the pipe is
- (a) 230 m/s (b) 300 m/s
- (c) 320 m/s (d) 360 m/s
22. Find the fundamental frequency of a closed pipe, if the length of the air column is 42 m. (speed of sound in air = 332 m/sec)
- (a) 2 Hz (b) 4 Hz
- (c) 7 Hz (d) 9 Hz
23. If v is the speed of sound in air then the shortest length of the closed pipe which resonates to a frequency n [KCET 2003]
- (a) $\frac{v}{4n}$ (b) $\frac{v}{2n}$
- (c) $\frac{2n}{v}$ (d) $\frac{4n}{v}$
24. The frequency of fundamental tone in an open organ pipe of length 0.48 m is 320 Hz. Speed of sound is 320 m/sec. Frequency of fundamental tone in closed organ pipe will be [MP PMT 2003]
- (a) 153.8 Hz (b) 160.0 Hz
- (c) 320.0 Hz (d) 143.2 Hz
25. If fundamental frequency of closed pipe is 50 Hz then frequency of 2nd overtone is [AFMC 2004]
- (a) 100 Hz (b) 50 Hz
- (c) 250 Hz (d) 150 Hz



26. Two open organ pipes of length 25 cm and 25.5 cm produce 10 beat/sec. The velocity of sound will be [Pb. PMT 2004]
- (a) 255 m/s (b) 250 m/s
(c) 350 m/s (d) None of these
27. What is minimum length of a tube, open at both ends, that resonates with tuning fork of frequency 350 Hz? [velocity of sound in air = 350 m/s] [DPMT 2004]
- (a) 50 cm (b) 100 cm
(c) 75 cm (d) 25 cm
28. Two open organ pipes give 4 beats/sec when sounded together in their fundamental nodes. If the length of the pipe are 100 cm and 102.5 cm respectively, then the velocity of sound is :
- (a) 496 m/s (b) 328 m/s
(c) 240 m/s (d) 160 m/s
29. The harmonics which are present in a pipe open at one end are [UPSEAT 2000; MHCET 2004]
- (a) Odd harmonics
(b) Even harmonics
(c) Even as well as odd harmonics
(d) None of these
30. An open pipe is suddenly closed at one end with the result that the frequency of third harmonic of the closed pipe is found to be higher by 100 Hz, then the fundamental frequency of open pipe is: [UPSEAT 2001; RPET 2004]
- (a) 480 Hz (b) 300 Hz
(c) 240 Hz (d) 200 Hz
31. Tube A has both ends open while tube B has one end closed, otherwise they are identical. The ratio of fundamental frequency of tube A and B is [AIEEE 2002; CPMT 2004]
- (a) 1 : 2 (b) 1 : 4
(c) 2 : 1 (d) 4 : 1
32. If the temperature increases, then what happens to the frequency of the sound produced by the organ pipe [RPET 1996; DPMT 2000; RPMT 2001]
- (a) Increases (b) Decreases
(c) Unchanged (d) Not definite
33. Apparatus used to find out the velocity of sound in gas is [AFMC 2004]
- (a) Melde's apparatus (b) Kundt's tube
(c) Quincke's tube (d) None of these
34. Standing stationary waves can be obtained in an air column even if the interfering waves are [CPMT 1972]
- (a) Of different pitches
(b) Of different amplitudes
(c) Of different qualities
(d) Moving with different velocities
35. The stationary wave $y = 2a \sin kx \cos \omega t$ in a closed organ pipe is the result of the superposition of $y = a \sin(\omega t - kx)$ and
- (a) $y = -a \cos(\omega t + kx)$ (b) $y = -a \sin(\omega t + kx)$
(c) $y = a \sin(\omega t + kx)$ (d) $y = a \cos(\omega t + kx)$
36. Stationary waves are set up in air column. Velocity of sound in air is 330 m/s and frequency is 165 Hz. Then distance between the nodes is [EAMCET (Engg.) 1995; CPMT 1999]
- (a) 2 m (b) 1 m
(c) 0.5 m (d) 4 m
37. An open pipe of length l vibrates in fundamental mode. The pressure variation is maximum at [EAMCET (Med.) 1999]
- (a) $1/4$ from ends
(b) The middle of pipe
(c) The ends of pipe
(d) At $1/8$ from ends of pipe middle of the pipe
38. Fundamental frequency of pipe is 100 Hz and other two frequencies are 300 Hz and 500 Hz then [RPMT 1998, 2003; CPMT 2001]
- (a) Pipe is open at both the ends
(b) Pipe is closed at both the ends
(c) One end open and another end is closed
(d) None of the above
39. Fundamental frequency of an open pipe of length 0.5 m is equal to the frequency of the first overtone of a closed pipe of length l . The value of l is (m) [KCET 1999]
- (a) 1.5 (b) 0.75
(c) 2 (d) 1
40. In a closed organ pipe the frequency of fundamental note is 50 Hz. The note of which of the following frequencies will not be emitted by it [J & K CET 2000]
- (a) 50 Hz (b) 100 Hz
(c) 150 Hz (d) None of the above
41. On producing the waves of frequency 1000 Hz in a Kundt's tube, the total distance between 6 successive nodes is 85 cm. Speed of sound in the gas filled in the tube is [AFMC 1999]
- (a) 330 m/s (b) 340 m/s
(c) 350 m/s (d) 300 m/s
42. What is the base frequency if a pipe gives notes of frequencies 425, 255 and 595 and decide whether it is closed at one end or open at both ends [UPSEAT 2001]
- (a) 17, closed (b) 85, closed
(c) 17, open (d) 85, open
43. A student determines the velocity of sound with the help of a closed organ pipe. If the observed length for fundamental frequency is 24.7 m, the length for third harmonic will be [RPET 2002]
- (a) 74.1 m (b) 72.7 cm
(c) 75.4 cm (d) 73.1 cm
44. An open pipe of length 33 cm resonates with frequency of 100 Hz. If the speed of sound is 330 m/s, then this frequency is
- (a) Fundamental frequency of the pipe
(b) Third harmonic of the pipe

- (c) Second harmonic of the pipe
(d) Fourth harmonic of the pipe
45. In a resonance tube the first resonance with a tuning fork occurs at 16 cm and second at 49 cm. If the velocity of sound is 330 m/s, the frequency of tuning fork is [DPMT 2002]
- (a) 500 (b) 300
(c) 330 (d) 165
46. Two closed organ pipes of length 100 cm and 101 cm 16 beats in 20 sec. When each pipe is sounded in its fundamental mode calculate the velocity of sound [AFMC 2003]
- (a) 303 ms (b) 332 ms
(c) 323.2 ms (d) 300 ms
47. In open organ pipe, if fundamental frequency is n then the other frequencies are [BCECE 2005]
- (a) $n, 2n, 3n, 4n$ (b) $n, 3n, 5n$
(c) $n, 2n, 4n, 8n$ (d) None of these
48. If in an experiment for determination of velocity of sound by resonance tube method using a tuning fork of 512 Hz, first resonance was observed at 30.7 cm and second was obtained at 63.2 cm, then maximum possible error in velocity of sound is (consider actual speed of sound in air is 332 m/s)
- (a) 204 cm/sec (b) 110 cm/sec
(c) 58 cm/sec (d) 80 cm/sec
49. An organ pipe, open from both end produces 5 beats per second when vibrated with a source of frequency 200 Hz. The second harmonic of the same pipes produces 10 beats per second with a source of frequency 420 Hz. The frequency of source is
- (a) 195 Hz (b) 205 Hz
(c) 190 Hz (d) 210 Hz
50. In one metre long open pipe what is the harmonic of resonance obtained with a tuning fork of frequency 480 Hz [J & K CET 2005]
- (a) First (b) Second
(c) Third (d) Fourth
51. An organ pipe open at one end is vibrating in first overtone and is in resonance with another pipe open at both ends and vibrating in third harmonic. The ratio of length of two pipes is
- (a) 1 : 2 (b) 4 : 1
(c) 8 : 3 (d) 3 : 8
52. In a resonance pipe the first and second resonances are obtained at depths 22.7 cm and 70.2 cm respectively. What will be the end correction [J & K CET 2005]
- (a) 1.05 cm (b) 115.5 cm
(c) 92.5 cm (d) 113.5 cm
53. An open tube is in resonance with string (frequency of vibration of tube is n). If tube is dipped in water so that 75% of length of tube is inside water, then the ratio of the frequency of tube to string now will be [J & K CET 2005]
- (a) 1 (b) 2

(c) $\frac{2}{3}$

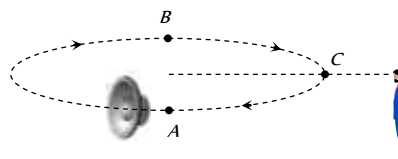
(d) $\frac{3}{2}$

Doppler's Effect

1. Doppler shift in frequency does not depend upon [MP PMT 1993; DPMT 2000]
- (a) The frequency of the wave produced
(b) The velocity of the source
(c) The velocity of the observer
(d) Distance from the source to the listener
2. A source of sound of frequency 450 cycles/sec is moving towards a stationary observer with 34 m/sec speed. If the speed of sound is 340 m/sec, then the apparent frequency will be
- (a) 410 cycles/sec (b) 500 cycles/sec
(c) 550 cycles/sec (d) 450 cycles/sec
3. The wavelength is 120 cm when the source is stationary. If the source is moving with relative velocity of 60 m/sec towards the observer, then the wavelength of the sound wave reaching to the observer will be (velocity of sound = 330 m/s)
- (a) 98 cm (b) 140 cm
(c) 120 cm (d) 144 cm
4. The frequency of a whistle of an engine is 600 cycles/sec is moving with the speed of 30 m/sec towards an observer. The apparent frequency will be (velocity of sound = 330 m/s) [IIT-JEE (Screening) 2005]
- (a) 600 cps (b) 660 cps
(c) 990 cps (d) 330 cps
5. A source of sound emits waves with frequency f Hz and speed V m/sec. Two observers move away from this source in opposite directions with a speed $0.2 V$ relative to the source. The ratio of frequencies heard by the two observers will be [DCE 2005]
- (a) 3 : 2 (b) 2 : 3
(c) 1 : 1 (d) 4 : 10
6. The source producing sound and an observer both are moving along the direction of propagation of sound waves. If the respective velocities of sound, source and an observer are v , v_s and v_o , then the apparent frequency heard by the observer will be (n = frequency of sound) [MP PMT 1989]
- (a) $\frac{n(v+v_o)}{v-v_s}$ (b) $\frac{n(v-v_o)}{v-v_s}$
(c) $\frac{n(v-v_o)}{v+v_s}$ (d) $\frac{n(v+v_o)}{v+v_s}$
7. An observer moves towards a stationary source of sound of frequency n . The apparent frequency heard by him is $2n$. If the velocity of sound in air is 332 m/sec, then the velocity of the observer is [MP PET 1990]
- (a) 166 m/sec (b) 664 m/sec
(c) 332 m/sec (d) 1328 m/sec
8. An observer is moving towards the stationary source of sound, then
- (a) Apparent frequency will be less than the real frequency
(b) Apparent frequency will be greater than the real frequency
(c) Apparent frequency will be equal to real frequency

- (d) Only the quality of sound will change
9. A whistle sends out 256 waves in a second. If the whistle approaches the observer with velocity $1/3$ of the velocity of sound in air, the number of waves per second the observer will receive [MP PET 1990; DPMT 2002]
- (a) 384 (b) 192
(c) 300 (d) 200
10. A person feels 2.5% difference of frequency of a motor-car horn. If the motor-car is moving to the person and the velocity of sound is 320 m/sec, then the velocity of car will be
- (a) 8 m/s (approx.) (b) 800 m/s
(c) 7 m/s (d) 6 m/s (approx.)
11. Two passenger trains moving with a speed of 108 km/hour cross each other. One of them blows a whistle whose frequency is 750 Hz. If sound speed is 330 m/s, then passengers sitting in the other train, after trains cross each other will hear sound whose frequency will be [MP PMT 1991]
- (a) 900 Hz (b) 625 Hz
(c) 750 Hz (d) 800 Hz
12. With what velocity an observer should move relative to a stationary source so that he hears a sound of double the frequency of source
- (a) Velocity of sound towards the source
(b) Velocity of sound away from the source
(c) Half the velocity of sound towards the source
(d) Double the velocity of sound towards the source
13. A source of sound emitting a note of frequency 200 Hz moves towards an observer with a velocity v equal to the velocity of sound. If the observer also moves away from the source with the same velocity v , the apparent frequency heard by the observer is
- (a) 50 Hz (b) 100 Hz
(c) 150 Hz (d) 200 Hz
14. Doppler's effect will not be applicable when the velocity of sound source is
- (a) Equal to that of the sound velocity
(b) Less than the velocity of sound
(c) Greater than the velocity of sound
(d) Zero
15. An observer while going on scooter hears sound of two sirens of same frequencies from two opposite directions. If he travels along the direction of one of the siren, then he
- (a) Listens resonance
(b) Listens beats
(c) Will not listen sound due to destructive interference
(d) Will listen intensive sound due to constructive interference
16. A source of sound is travelling towards a stationary observer. The frequency of sound heard by the observer is of three times the original frequency. The velocity of sound is v m/sec. The speed of source will be [MP PET 1991]
- (a) $\frac{2}{3}v$ (b) v
(c) $\frac{3}{2}v$ (d) $3v$
17. A sound source is moving towards a stationary observer with $1/10$ of the speed of sound. The ratio of apparent to real frequency is [CPMT 1977; NCERT 1977; KCET 2001, 03]
- (a) 10/9 (b) 11/10
(c) $(11/10)^2$ (d) $(9/10)^2$
18. The speed of sound in air at a given temperature is 350 m/s. An engine blows whistle at a frequency of 1200 cps. It is approaching the observer with velocity 50 m/s. The apparent frequency in cps heard by the observer will be [CPMT 1988; MP PET 1989]
- (a) 600 (b) 1050
(c) 1400 (d) 2400
19. Suppose that the speed of sound in air at a given temperature is 400 m/sec. An engine blows a whistle at 1200 Hz frequency. It is approaching an observer at the speed of 100 m/sec. What is the apparent frequency as heard by the observer
- (a) 600 Hz (b) 1200 Hz
(c) 1500 Hz (d) 1600 Hz
20. A source of frequency 150 Hz is moving in the direction of a person with a velocity of 110 m/s. The frequency heard by the person will be (speed of sound in medium = 330 m/s) [MP PMT 1991]
- (a) 225 Hz (b) 200 Hz
(c) 150 Hz (d) 100 Hz
21. The Doppler's effect is applicable for [AFMC 1998]
- (a) Light waves (b) Sound waves
(c) Space waves (d) Both (a) and (b)
22. A source of sound is moving with constant velocity of 20 m/s emitting a note of frequency 1000 Hz. The ratio of frequencies observed by a stationary observer while the source is approaching him and after it crosses him will be [MP PET 1994]
- (a) 9 : 8 (b) 8 : 9
(c) 1 : 1 (d) 9 : 10
(Speed of sound $v = 340$ m/s)
23. A source of sound S is moving with a velocity 50 m/s towards a stationary observer. The observer measures the frequency of the source as 1000 Hz. What will be the apparent frequency of the source when it is moving away from the observer after crossing him? The velocity of sound in the medium is 350 m/s
- (a) 750 Hz (b) 857 Hz
(c) 1143 Hz (d) 1333 Hz
24. A source and listener are both moving towards each other with speed $v/10$, where v is the speed of sound. If the frequency of the note emitted by the source is f , the frequency heard by the listener would be nearly [MP PMT 1994; MP PET 2001]
- (a) $1.11 f$ (b) $1.22 f$
(c) f (d) $1.27 f$
25. A table is revolving on its axis at 5 revolutions per second. A sound source of frequency 1000 Hz is fixed on the table at 70 cm from the axis. The minimum frequency heard by a listener standing at a distance from the table will be (speed of sound = 352 m/s)

- (a) 1000 Hz (b) 1066 Hz
(c) 941 Hz (d) 352 Hz
26. A source of sound S of frequency 500 Hz situated between a stationary observer O and a wall W , moves towards the wall with a speed of 2 m/s. If the velocity of sound is 332 m/s, then the number of beats per second heard by the observer is (approximately)
(a) 8 (b) 6
(c) 4 (d) 2
27. A motor car blowing a horn of frequency 124 vib/sec moves with a velocity 72 km/hr towards a tall wall. The frequency of the reflected sound heard by the driver will be (velocity of sound in air is 330 m/s) [MP PET 1997]
(a) 109 vib/sec (b) 132 vib/sec
(c) 140 vib/sec (d) 248 vib/sec
28. A source of sound of frequency n is moving towards a stationary observer with a speed S . If the speed of sound in air is V and the frequency heard by the observer is n_1 , the value of n_1/n is
(a) $(V+S)/V$ (b) $V/(V+S)$
(c) $(V-S)/V$ (d) $V/(V-S)$
29. A vehicle with a horn of frequency n is moving with a velocity of 30 m/s in a direction perpendicular to the straight line joining the observer and the vehicle. The observer perceives the sound to have a frequency $n + n_1$. Then (if the sound velocity in air is 300 m/s) [CBSE PMT 1998; AIIMS 2000]
(a) $n_1 = 10n$ (b) $n_1 = 0$
(c) $n_1 = 0.1n$ (d) $n_1 = -0.1n$
30. A whistle giving out 450 Hz approaches a stationary observer at a speed of 33 m/s. The frequency heard by the observer in Hz is
(a) 409 (b) 429
(c) 517 (d) 500
31. An observer is moving away from source of sound of frequency 100 Hz. His speed is 33 m/s. If speed of sound is 330 m/s, then the observed frequency is [EAMCET (Engg.) 1995; CPMT 1999]
(a) 90 Hz (b) 100 Hz
(c) 91 Hz (d) 110 Hz
32. An observer standing at station observes frequency 219 Hz when a train approaches and 184 Hz when train goes away from him. If velocity of sound in air is 340 m/s, then velocity of train and actual frequency of whistle will be [RPET 1997]
(a) 15.5 ms^{-1} , 200 Hz (b) 19.5 ms^{-1} , 205 Hz
(c) 29.5 ms^{-1} , 200 Hz (d) 32.5 ms^{-1} , 205 Hz
33. At what speed should a source of sound move so that stationary observer finds the apparent frequency equal to half of the original frequency [RPMT 1996]
(a) $\frac{v}{2}$ (b) $2v$
(c) $\frac{v}{4}$ (d) v
34. A boy is walking away from a wall towards an observer at a speed of 1 metre/sec and blows a whistle whose frequency is 680 Hz. The number of beats heard by the observer per second is (Velocity of sound in air = 340 metres/sec) [MP PMT 1995]
(a) Zero (b) 2
(c) 8 (d) 4
35. The driver of a car travelling with speed 30 metres per second towards a hill sounds a horn of frequency 600 Hz. If the velocity of sound in air is 330 metres per second, the frequency of the reflected sound as heard by the driver is [MP PMT 1996]
(a) 720 Hz (b) 555.5 Hz
(c) 550 Hz (d) 500 Hz
36. Two sirens situated one kilometer apart are producing sound of frequency 330 Hz. An observer starts moving from one siren to the other with a speed of 2 m/s. If the speed of sound be 330 m/s, what will be the beat frequency heard by the observer [RPMT 1996; CPMT 2002]
(a) 8 (b) 4
(c) 6 (d) 1
37. A source of sound is travelling with a velocity 40 km/hour towards observer and emits sound of frequency 2000 Hz. If velocity of sound is 1220 km/hour, then what is the apparent frequency heard by an observer [AFMC 1997]
(a) 2210 Hz (b) 1920 Hz
(c) 2068 Hz (d) 2086 Hz
38. A source of sound and listener are approaching each other with a speed of 40 m/s. The apparent frequency of note produced by the source is 400 cps. Then, its true frequency (in cps) is (velocity of sound in air = 360 m/s) [JIPMER 1997, Cancelled]
(a) 420 (b) 360
(c) 400 (d) 320
39. A siren emitting sound of frequency 500 Hz is going away from a static listener with a speed of 50 m/sec. The frequency of sound to be heard, directly from the siren, is [AIIMS 1999; Pb. PMT 2003]
(a) 434.2 Hz (b) 589.3 Hz
(c) 481.2 Hz (d) 286.5 Hz
40. A man sitting in a moving train hears the whistle of the engine. The frequency of the whistle is 600 Hz [JIPMER 1999]
(a) The apparent frequency as heard by him is smaller than 600 Hz
(b) The apparent frequency is larger than 600 Hz
(c) The frequency as heard by him is 600 Hz
(d) None of the above
41. A source of sound of frequency 500 Hz is moving towards an observer with velocity 30 m/s. The speed of sound is 330 m/s. the frequency heard by the observer will be [MP PET 2000; Kerala PMT 2005; UPSEAT 2005]
(a) 550 Hz (b) 458.3 Hz

- (c) 530 Hz (d) 545.5 Hz
42. A source of sound of frequency 90 vibrations/sec is approaching a stationary observer with a speed equal to $1/10$ the speed of sound. What will be the frequency heard by the observer
(a) 80 vibrations/sec (b) 90 vibrations/sec
(c) 100 vibrations/sec (d) 120 vibrations/sec
43. A whistle of frequency 500 Hz tied to the end of a string of length 1.2 m revolves at 400 rev/min. A listener standing some distance away in the plane of rotation of whistle hears frequencies in the range (speed of sound = 340 m/s)
[KCET 2000; AMU 1999; Pb. PET 2003]
(a) 436 to 586 (b) 426 to 574
(c) 426 to 584 (d) 436 to 674
44. A train moves towards a stationary observer with speed 34 m/s. The train sounds a whistle and its frequency registered by the observer is f_1 . If the train's speed is reduced to 17 m/s, the frequency registered is f_2 . If the speed of sound is 340 m/s then the ratio f_1 / f_2 is
[IIT-JEE (Screening) 2000]
(a) 18/19 (b) 1/2
(c) 2 (d) 19/18
45. If source and observer both are relatively at rest and if speed of sound is increased then frequency heard by observer will
[RPET 2000; J & K CET 2004]
(a) Increases (b) Decreases
(c) Can not be predicted (d) Will not change
46. A source and an observer move away from each other with a velocity of 10 m/s with respect to ground. If the observer finds the frequency of sound coming from the source as 1950 Hz, then actual frequency of the source is (velocity of sound in air = 340 m/s)
[MH CET 2000; AFMC 2000; CBSE PMT 2001]
(a) 1950 Hz (b) 2068 Hz
(c) 2132 Hz (d) 2486 Hz
47. A source is moving towards an observer with a speed of 20 m/s and having frequency of 240 Hz. The observer is now moving towards the source with a speed of 20 m/s. Apparent frequency heard by observer, if velocity of sound is 340 m/s, is [CPMT 2000; KCET 2001; MH CET 2004]
(a) 240 Hz (b) 270 Hz
(c) 280 Hz (d) 360 Hz
48. A siren placed at a railway platform is emitting sound of frequency 5 kHz. A passenger sitting in a moving train A records a frequency of 5.5 kHz while the train approaches the siren. During his return journey in a different train B he records a frequency of 6.0 kHz while approaching the same siren. The ratio of the velocity of train B to that of train A is
[IIT-JEE (Screening) 2002]
(a) 242/252 (b) 2
(c) 5/6 (d) 11/6
49. A whistle revolves in a circle with an angular speed of 20 rad/sec using a string of length 50 cm. If the frequency of sound from the whistle is 385 Hz, then what is the minimum frequency heard by an observer, which is far away from the centre in the same plane? ($v = 340$ m/s)
[MP PMT 2000]
[CBSE PMT 2002]
(a) 333 Hz (b) 374 Hz
(c) 385 Hz (d) 394 Hz
50. A Siren emitting sound of frequency 800 Hz is going away from a static listener with a speed of 30 m/s, frequency of the sound to be heard by the listener is (take velocity of sound as 330 m/s)
[CPMT 1996; AIIMS 2002; Pb. PMT 2001]
(a) 733.3 Hz (b) 644.8 Hz
(c) 481.2 Hz (d) 286.5 Hz
51. A car sounding a horn of frequency 1000 Hz passes an observer. The ratio of frequencies of the horn noted by the observer before and after passing of the car is 11 : 9. If the speed of sound is v , the speed of the car is
[MP PET 2002]
(a) $\frac{1}{10}v$ (b) $\frac{1}{2}v$
(c) $\frac{1}{5}v$ (d) v
52. What should be the velocity of a sound source moving towards a stationary observer so that apparent frequency is double the actual frequency (Velocity of sound is v)
[MP PMT 2002]
(a) v (b) $2v$
(c) $\frac{v}{2}$ (d) $\frac{v}{4}$
[EAMCET 1997;
53. Two trains are moving towards each other at speeds of 20 m/s and 15 m/s relative to the ground. The first train sounds a whistle of frequency 600 Hz. the frequency of the whistle heard by a passenger in the second train before the train meets is (the speed of sound in air is 340 m/s)
[UPSEAT 2002]
(a) 600 Hz (b) 585 Hz
(c) 645 Hz (d) 666 Hz
54. A small source of sound moves on a circle as shown in the figure and an observer is standing on O. Let n_1, n_2 and n_3 be the frequencies heard when the source is at A, B and C respectively. Then
[UPSEAT 2002]
(a) $n_1 > n_2 > n_3$
(b) $n_2 > n_3 > n_1$
(c) $n_1 = n_2 > n_3$
(d) $n_2 > n_1 > n_3$
- 
55. A source and an observer approach each other with same velocity 50 m/s. If the apparent frequency is 435 sec, then the real frequency is
[CPMT 2003]

- (a) 320 s (b) 360 sec
(c) 390 sec (d) 420 sec
56. A source emits a sound of frequency of 400 Hz, but the listener hears it to be 390 Hz. Then [Orissa JEE 2003]
- (a) The listener is moving towards the source
(b) The source is moving towards the listener
(c) The listener is moving away from the source
(d) The listener has a defective ear
57. Doppler effect is applicable for [AFMC 2003]
- (a) Moving bodies
(b) One is moving and other are stationary
(c) For relative motion
(d) None of these
58. A source and an observer are moving towards each other with a speed equal to $\frac{v}{2}$ where v is the speed of sound. The source is emitting sound of frequency n . The frequency heard by the observer will be [MP PET 2003]
- (a) Zero (b) n
(c) $\frac{n}{3}$ (d) $3n$
59. When an engine passes near to a stationary observer then its apparent frequencies occurs in the ratio 5/3. If the velocity of engine is [MP PMT 2003]
- (a) 540 m/s (b) 270 m/s
(c) 85 m/s (d) 52.5 m/s
60. A police car horn emits a sound at a frequency 240 Hz when the car is at rest. If the speed of the sound is 330 m/s, the frequency heard by an observer who is approaching the car at a speed of 11 m/s, is :
- (a) 248 Hz (b) 244 Hz
(c) 240 Hz (d) 230 Hz
61. A person carrying a whistle emitting continuously a note of 272 Hz is running towards a reflecting surface with a speed of 18 km/hour. The speed of sound in air is 345ms^{-1} . The number of beats heard by him is [Kerala (Engg.) 2002]
- (a) 4 (b) 6
(c) 8 (d) 3
62. A bus is moving with a velocity of 5 m/s towards a huge wall. the driver sounds a horn of frequency 165 Hz. If the speed of sound in air is 355 m/s, the number of beats heard per second by a passenger on the bus will be [KCET 2001; BHU 2002]
- (a) 6 (b) 5
(c) 3 (d) 4
63. A source of sound of frequency 256 Hz is moving rapidly towards a wall with a velocity of 5 m/s. The speed of sound is 330 m/s. If the observer is between the wall and the source, then beats per second heard will be [UPSEAT 2002]
- (a) 7.8 Hz (b) 7.7 Hz

- (c) 3.9 Hz (d) Zero

64. The apparent frequency of a note, when a listener moves towards a stationary source, with velocity of 40 m/s is 200 Hz. When he moves away from the same source with the same speed, the apparent frequency of the same note is 160 Hz. The velocity of sound in air is (in m/s) [KCET 1998]
- (a) 360 (b) 330
(c) 320 (d) 340
65. An observer moves towards a stationary source of sound, with a velocity one-fifth of the velocity of sound. What is the percentage increase in the apparent frequency [AIEEE 2005]
- (a) 5% (b) 20%
(c) Zero (d) 0.5%

Musical Sound

1. The walls of the halls built for music concerts should [NCERT 1979]
- (a) Amplify sound (b) Transmit sound
(c) Reflect sound (d) Absorb sound
2. A spherical source of power 4 W and frequency 800 Hz is emitting sound waves. The intensity of waves at a distance 200 m is [CPMT 1999; JIPMER 2000]
- (a) $8 \times 10^{-6} \text{ W/m}^2$ (b) $2 \times 10^{-4} \text{ W/m}^2$
(c) $1 \times 10^{-4} \text{ W/m}^2$ (d) 4 W/m^2
3. If the pressure amplitude in a sound wave is tripled, then the intensity of sound is increased by a factor of [CPMT 1992; JIPMER 2000]
- (a) 9 (b) 3
(c) 6 (d) $\sqrt{3}$
4. If the amplitude of sound is doubled and the frequency reduced to one-fourth, the intensity of sound at the same point will be [UPSEAT 2004]
- (a) Increased by a factor of 2
(b) Decreased by a factor of 2
(c) Decreased by a factor of 4
(d) Unchanged
5. Intensity level of a sound of intensity I is 30 dB. The ratio $\frac{I}{I_0}$ is (Where I_0 is the threshold of hearing) [KCET 1999; J & K CET 2005]
- (a) 3000 (b) 1000
(c) 300 (d) 30
6. Decibel is unit of [RPMT 2000]
- (a) Intensity of light (b) X-rays radiation capacity
(c) Sound loudness (d) Energy of radiation
7. Quality of a musical note depends on [MP PMT 1998; KCET 1999; RPET 2000]
- (a) Harmonics present
(b) Amplitude of the wave
(c) Fundamental frequency
(d) Velocity of sound in the medium
8. When we hear a sound, we can identify its source from

[KCET (Med.) 2001]

- (a) Amplitude of sound
- (b) Intensity of sound
- (c) Wavelength of sound
- (d) Overtones present in the sound

9. A man X can hear only upto 10 kHz and another man Y upto 20 kHz. A note of frequency 500 Hz is produced before them from a stretched string. Then

[KCET 2002]

- (a) Both will hear sounds of same pitch but different quality
- (b) Both will hear sounds of different pitch but same quality
- (c) Both will hear sounds of different pitch and different quality
- (d) Both will hear sounds of same pitch and same quality

10. The amplitude of two waves are in ratio 5 : 2. If all other conditions for the two waves are same, then what is the ratio of their energy densities

[MH CET 2004]

- (a) 5 : 2
- (b) 10 : 4
- (c) 2.5 : 1
- (d) 25 : 4

11. A is singing a note and at the same time B is singing a note with exactly one-eighth the frequency of the note of A . The energies of two sounds are equal, the amplitude of the note of B is [NCERT 1981; AIIMS 2001]

- (a) Same that of A
- (b) Twice as that of A
- (c) Four times as that of A
- (d) Eight times as that of A

12. The loudness and pitch of a sound depends on

[KCET 2004; Pb. PET 2003]

- (a) Intensity and velocity
- (b) Frequency and velocity
- (c) Intensity and frequency
- (d) Frequency and number of harmonics

13. If T is the reverberation time of an auditorium of volume V then

- (a) $T \propto \frac{1}{V}$
- (b) $T \propto \frac{1}{V^2}$
- (c) $T \propto V^2$
- (d) $T \propto V$

14. The intensity of sound from a radio at a distance of 2 metres from its speaker is $1 \times 10^{-2} \mu W/m^2$. The intensity at a distance of 10 meters would be

[CPMT 2005]

- (a) $0.2 \times 10^{-2} \mu W/m^2$
- (b) $1 \times 10^{-2} \mu W/m^2$
- (c) $4 \times 10^{-4} \mu W/m^2$
- (d) $5 \times 10^{-2} \mu W/m^2$

15. The intensity of sound wave while passing through an elastic medium falls down by 10% as it covers one metre distance through the medium. If the initial intensity of the sound wave was 100 decibels, its value after it has passed through 3 metre thickness of the medium will be

[CPMT 1988]

- (a) 70 decibel
- (b) 72.9 decibel
- (c) 81 decibel
- (d) 60 decibel

16. A musical scale is constructed by providing intermediate frequencies between a note and its octave which

[CPMT 1972; NCERT 1980]

- (a) Form an arithmetic progression
- (b) Form a geometric progression

- (c) Bear a simple ratio with their neighbours

- (d) Form a harmonic progression

17. In a harmonium the intermediate notes between a note and its octave form

[CPMT 1973]

- (a) An arithmetic progression
- (b) A geometric progression
- (c) A harmonic progression
- (d) An exponential progression

18. The power of a sound from the speaker of a radio is 20 mW. By turning the knob of the volume control, the power of the sound is increased to 400 mW. The power increase in decibels as compared to the original power is

- (a) 13 dB
- (b) 10 dB
- (c) 20 dB
- (d) 800 dB

19. If separation between screen and source is increased by 2% what would be the effect on the intensity

[CPMT 2003]

- (a) Increases by 4%
- (b) Increases by 2%
- (c) Decreases by 2%
- (d) Decreases by 4%

20. The musical interval between two tones of frequencies 320 Hz and 240 Hz is

[MP PMT 1992; AFMC 1992]

- (a) 80
- (b) $\left(\frac{4}{3}\right)$
- (c) 560
- (d) 320×240

21. In an orchestra, the musical sounds of different instruments are distinguished from one another by which of the following characteristics

[CBSE PMT 1993]

- (a) Pitch
- (b) Loudness
- (c) Quality
- (d) Overtones

22. The intensity level due to two waves of the same frequency in a given medium are 4 bel and 5 bel. Then the ratio of amplitudes is

[KCET 2003]

- (a) 1 : 4
- (b) 1 : 2
- (c) 1 : 10
- (d) 1 : 10

23. It is possible to recognise a person by hearing his voice even if he is hidden behind a wall. This is due to the fact that his voice

- (a) Has a definite pitch
- (b) Has a definite quality
- (c) Has a definite loudness
- (d) Can penetrate the wall

24. Of the following the one which emits sound of higher pitch is

- (a) Mosquito
- (b) Lion
- (c) Man
- (d) Woman

25. In the musical octave 'Sa', 'Re', 'Ga'

- (a) The frequency of the note 'Sa' is greater than that of 'Re', 'Ga'
- (b) The frequency of the note 'Sa' is smaller than that of 'Re', 'Ga'
- (c) The frequency of all the notes 'Sa', 'Re', 'Ga' is the same
- (d) The frequency decreases in the sequence 'Sa', 'Re', 'Ga'

26. Tone A has frequency of 240 Hz. Of the following tones, the one which will sound least harmonious with A is

- (a) 240
- (b) 480
- (c) 360
- (d) 450

27. Learned Indian classical vocalists do not like the accompaniment of a harmonium because

[MP PMT 1992]

- (a) Intensity of the notes of the harmonium is too large

- (b) Notes of the harmonium are too shrill
(c) Diatonic scale is used in the harmonium
(d) Tempered scale is used in the harmonium
28. Each of the properties of sound listed in column A primarily depends on one of the quantities in column B. Choose the matching pairs from two columns
- | | |
|----------|-----------|
| Column A | Column B |
| Pitch | Waveform |
| Quality | Frequency |
| Loudness | Intensity |
- [IIT 1980]
- (a) Pitch-waveform, Quality-frequency; Loudness-intensity
(b) Pitch-frequency, Quality-waveform; Loudness-intensity
(c) Pitch-intensity, Quality-waveform; Loudness- frequency
(d) Pitch-waveform, Quality- intensity; Loudness-frequency
29. Intensity level 200 *cm* from a source of sound is 80 *dB*. If there is no loss of acoustic power in air and intensity of threshold hearing is 10^{-12} Wm^{-2} then, what is the intensity level at a distance of 400 *cm* from source
- (a) Zero (b) 54 *dB*
(c) 64 *dB* (d) 44 *dB*
30. A point source emits sound equally in all directions in a non-absorbing medium. Two points *P* and *Q* are at distances of 2*m* and 3*m* respectively from the source. The ratio of the intensities of the waves at *P* and *Q* is [CBSE PMT 2005]
- (a) 9 : 4 (b) 2 : 3
(c) 3 : 2 (d) 4 : 9
31. Quality depends on [AFMC 2003]
- (a) Intensity (b) Loudness
(c) Timbre (d) Frequency
32. Two waves having sinusoidal waveforms have different wavelengths and different amplitude. They will be having [BHU 2005]
- (a) Same pitch and different intensity
(b) Same quality and different intensity
(c) Different quality and different intensity
(d) Same quality and different pitch

Critical Thinking

Objective Questions

1. A wave disturbance in a medium is described by $y(x, t) = 0.02 \cos\left(50\pi t + \frac{\pi}{2}\right) \cos(10\pi x)$, where *x* and *y* are in metres and *t* in seconds [IIT 1995]
- (a) A displacement node occurs at *x* = 0.15 *m*
(b) An antinode occurs at *x* = 0.3 *m*
(c) The wavelength of the wave is 0.2 *m*
(d) The speed of the wave is 5.0 *m/s*
2. The (*x*, *y*) coordinates of the corners of a square plate are (0, 0), (*L*, 0), (*L*, *L*) and (0, *L*). The edges of the plate are clamped and

transverse standing waves are set up in it. If *u*(*x*, *y*) denotes the displacement of the plate at the point (*x*, *y*) at some instant of time, the possible expression(s) for *u* is(are) (*a* = positive constant)

[IIT 1998; Orissa PMT 2004]

- (a) $a \cos \frac{\pi x}{2L} \cos \frac{\pi y}{2L}$ (b) $a \sin \frac{\pi x}{L} \sin \frac{\pi y}{L}$
(c) $a \sin \frac{\pi x}{L} \sin \frac{2\pi y}{L}$ (d) $a \cos \frac{2\pi x}{L} \cos \frac{\pi y}{L}$

3. The ends of a stretched wire of length *L* are fixed at *x* = 0 and *x* = *L*. In one experiment, the displacement of the wire is $y_1 = A \sin(\pi x / L) \sin \omega t$ and energy is *E*₁, and in another experiment its displacement is $y_2 = A \sin(2\pi x / L) \sin 2\omega t$ and energy is *E*₂. Then

[IIT-JEE (Screening) 2001]

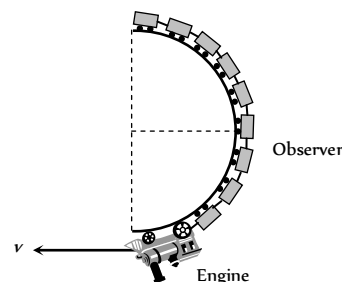
- (a) $E_2 = E_1$ (b) $E_2 = 2E_1$
(c) $E_2 = 4E_1$ (d) $E_2 = 16E_1$

4. In a large room, a person receives direct sound waves from a source 120 metres away from him. He also receives waves from the same source which reach him, being reflected from the 25 metre high ceiling at a point halfway between them. The two waves interfere constructively for wavelength of

[Roorkee 1982]

- (a) 20, 20/3, 20/5 etc (b) 10, 5, 2.5 etc
(c) 10, 20, 30 etc (d) 15, 25, 35 etc

5. A train has just completed a *U*-curve in a track which is a semicircle. The engine is at the forward end of the semi circular part of the track while the last carriage is at the rear end of the semicircular track. The driver blows a whistle of frequency 200 *Hz*. Velocity of sound is 340 *m/sec*. Then the apparent frequency as observed by a passenger in the middle of a train when the speed of the train is 30 *m/sec* is



- (a) 209 *Hz* (b) 288 *Hz*
(c) 200 *Hz* (d) 181 *Hz*

6. Two identical flutes produce fundamental notes of frequency 300 *Hz* at 27° *C*. If the temperature of air in one flute is increased to 31° *C*, the number of the beats heard per second will be

- (a) 1 (b) 2
(c) 3 (d) 4

7. In the experiment for the determination of the speed of sound in air using the resonance column method, the length of the air column that resonates in the fundamental mode, with a tuning fork is 0.1 *m*. when this length is changed to 0.35 *m*, the same tuning fork resonates with the first overtone. Calculate the end correction

[IIT-JEE (Screening) 2003]

- (a) 0.012 *m* (b) 0.025 *m*
(c) 0.05 *m* (d) 0.024 *m*



8. A closed organ pipe of length L and an open organ pipe contain gases of densities ρ_1 and ρ_2 respectively. The compressibility of gases are equal in both the pipes. Both the pipes are vibrating in their first overtone with same frequency. The length of the open organ pipe is

[IIT-JEE (Screening) 2004]

- (a) $\frac{L}{3}$ (b) $\frac{4L}{3}$
(c) $\frac{4L}{3} \sqrt{\frac{\rho_1}{\rho_2}}$ (d) $\frac{4L}{3} \sqrt{\frac{\rho_2}{\rho_1}}$

9. A string of length 0.4 m and mass 10^{-2} kg is tightly clamped at its ends. The tension in the string is 1.6 N . Identical wave pulses are produced at one end at equal intervals of time Δt . The minimum value of Δt which allows constructive interference between successive pulses is [IIT 1998]

(a) 0.05 s (b) 0.10 s
(c) 0.20 s (d) 0.40 s

10. Two identical stringed instruments have frequency 100 Hz . If tension in one of them is increased by 4% and they are sounded together then the number of beats in one second is [EAMCET (Engg.) 1995]

(a) 1 (b) 8
(c) 4 (d) 2

11. The difference between the apparent frequency of a source of sound as perceived by an observer during its approach and recession is 2% of the natural frequency of the source. If the velocity of sound in air is 300 m/sec , the velocity of the source is (It is given that velocity of source $\ll$ velocity of sound) [CPMT 1982; RPET 1998]

(a) 6 m/sec (b) 3 m/sec
(c) 1.5 m/sec (d) 12 m/sec

12. A sound wave of frequency ν travels horizontally to the right. It is reflected from a large vertical plane surface moving to the left with a speed v . The speed of sound in the medium is c , then

- (a) The frequency of the reflected wave is $\frac{\nu(c+v)}{c-v}$
(b) The wavelength of the reflected wave is $\frac{c(c-v)}{\nu(c+v)}$
(c) The number of waves striking the surface per second is $\frac{\nu(c+v)}{c}$
(d) The number of beats heard by a stationary listener to the left of the reflecting surface is $\frac{\nu v}{c-v}$

13. Two cars are moving on two perpendicular roads towards a crossing with uniform speeds of 72 km/hr and 36 km/hr . If first car blows horn of frequency 280 Hz , then the frequency of horn heard by the driver of second car when line joining the cars make 45° angle with the roads; will be [RPET 1997]

(a) 321 Hz (b) 298 Hz
(c) 289 Hz (d) 280 Hz

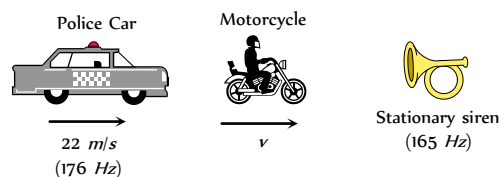
14. Two whistles A and B produces notes of frequencies 660 Hz and 596 Hz respectively. There is a listener at the mid-point of the line joining them. Now the whistle B and the listener start moving with speed 30 m/s away from the whistle A . If speed of sound be 330 m/s , how many beats will be heard by the listener [RPET 1996]

(a) 2 (b) 4
(c) 6 (d) 8

15. A source producing sound of frequency 170 Hz is approaching a stationary observer with a velocity 17 ms . The apparent change in the wavelength of sound heard by the observer is (speed of sound in air = 340 ms) [EAMCET (Engg.) 2000]

(a) 0.1 m (b) 0.2 m
(c) 0.4 m (d) 0.5 m

16. A police car moving at 22 m/s , chases a motorcyclist. The police man sounds his horn at 176 Hz , while both of them move towards a stationary siren of frequency 165 Hz . Calculate the speed of the motorcycle, if it is given that he does not observe any beats [IIT-JEE (Screening) 2003]



- (a) 33 m/s (b) 22 m/s
(c) Zero (d) 11 m/s

17. An observer moves towards a stationary source of sound with a speed $1/5$ of the speed of sound. The wavelength and frequency of the source emitted are λ and f respectively. The apparent frequency and wavelength recorded by the observer are respectively [CBSE PMT 1999; BCECE 2005]

- (a) $1.2 f, \lambda$ (b) $f, 1.2 \lambda$
(c) $0.8 f, 0.8 \lambda$ (d) $1.2 f, 1.2 \lambda$

18. A light pointer fixed to one prong of a tuning fork touches a vertical plate. The fork is set vibrating and the plate is allowed to fall freely. If eight oscillations are counted when the plate falls through 10 cm , the frequency of the tuning fork is [IIT 1977; KCET 2002]

- (a) 360 Hz (b) 280 Hz
(c) 560 Hz (d) 56 Hz

19. Oxygen is 16 times heavier than hydrogen. Equal volumes of hydrogen and oxygen are mixed. The ratio of speed of sound in the mixture to that in hydrogen is [KCET 2004]

- (a) $\sqrt{\frac{1}{8}}$ (b) $\sqrt{\frac{32}{17}}$
(c) $\sqrt{8}$ (d) $\sqrt{\frac{2}{17}}$

20. The equation of displacement of two waves are given as $y_1 = 10 \sin\left(3\pi t + \frac{\pi}{3}\right)$; $y_2 = 5(\sin 3\pi t + \sqrt{3} \cos 3\pi t)$. Then what is the ratio of their amplitudes [AIIMS 1997; Haryana PMT 2000]

- (a) $1 : 2$ (b) $2 : 1$
(c) $1 : 1$ (d) None of these

21. The equation $y = A \cos^2\left(2\pi nt - 2\pi \frac{x}{\lambda}\right)$ represents a wave with [RPET 1996]

- (a) Amplitude $A/2$, frequency $2n$ and wavelength $\lambda/2$
(b) Amplitude $A/2$, frequency $2n$ and wavelength λ
(c) Amplitude A , frequency $2n$ and wavelength 2λ
(d) Amplitude A , frequency n and wavelength λ

22. In a wave motion $y = a \sin(kx - \omega t)$, y can represent [IIT-JEE 1999]

- (a) Electric field (b) Magnetic field
(c) Displacement (d) Pressure

23. Consider ten identical sources of sound all giving the same frequency but having phase angles which are random. If the average

intensity of each source is I_0 , the average of resultant intensity I due to all these ten sources will be

[MP PMT 1990]

- (a) $I = 100 I_0$ (b) $I = 10 I_0$
(c) $I = I_0$ (d) $I = \sqrt{10} I_0$

24. Ten tuning forks are arranged in increasing order of frequency in such a way that any two nearest tuning forks produce 4 beats/sec. The highest frequency is twice of the lowest. Possible highest and the lowest frequencies are

[MP PMT 1990; MHCET 2002]

- (a) 80 and 40 (b) 100 and 50
(c) 44 and 22 (d) 72 and 36

25. 41 forks are so arranged that each produces 5 beats per sec when sounded with its near fork. If the frequency of last fork is double the frequency of first fork, then the frequencies of the first and last fork are respectively

[MP PET 1997; KCET 2002]

- (a) 200, 400 (b) 205, 410
(c) 195, 390 (d) 100, 200

26. Two identical wires have the same fundamental frequency of 400 Hz when kept under the same tension. If the tension in one wire is increased by 2% the number of beats produced will be

- (a) 4 (b) 2
(c) 8 (d) 1

27. 25 tuning forks are arranged in series in the order of decreasing frequency. Any two successive forks produce 3 beats/sec. If the frequency of the first tuning fork is the octave of the last fork, then the frequency of the 21st fork is

[Kerala (Engg.) 2001]

- (a) 72 Hz (b) 288 Hz
(c) 84 Hz (d) 87 Hz

28. 16 tuning forks are arranged in the order of increasing frequencies. Any two successive forks give 8 beats per sec when sounded together. If the frequency of the last fork is twice the first, then the frequency of the first fork is

[CBSE PMT 2000; MP PET 2001]

- (a) 120 (b) 160
(c) 180 (d) 220

29. Two identical straight wires are stretched so as to produce 6 beats per second when vibrating simultaneously. On changing the tension in one of them, the beat frequency remains unchanged. Denoting by T_1, T_2 , the higher and the lower initial tensions in the strings, then it could be said that while making the above change in tension [IIT 1991]

- (a) T_2 was decreased (b) T_2 was increased
(c) T_1 was increased (d) T_1 was kept constant

30. The frequency of a stretched uniform wire under tension is in resonance with the fundamental frequency of a closed tube. If the tension in the wire is increased by 8 N, it is in resonance with the first overtone of the closed tube. The initial tension in the wire is

- (a) 1 N (b) 4 N

- (c) 8 N (d) 16 N

31. A metal wire of linear mass density of 9.8 g/m is stretched with a tension of 10 kg weight between two rigid supports 1 metre apart. The wire passes at its middle point between the poles of a permanent magnet, and it vibrates in resonance when carrying an alternating current of frequency n . The frequency n of the alternating source is

[AIEEE 2003]

- (a) 25 Hz (b) 50 Hz
(c) 100 Hz (d) 200 Hz

32. A wire of density 9×10^{-3} kg/m is stretched between two clamps 1 m apart and is subjected to an extension of 4.9×10^{-2} m. The lowest frequency of transverse vibration in the wire is ($Y = 9 \times 10^{10}$ N/m) [UPSEAT 2000]

- (a) 40 Hz (b) 35 Hz
(c) 30 Hz (d) 25 Hz

33. A man is watching two trains, one leaving and the other coming in with equal speeds of 4 m/sec. If they sound their whistles, each of frequency 240 Hz, the number of beats heard by the man (velocity of sound in air = 320 m/sec) will be equal to

[MP PET 1999; RPMT 2000; BHU 2004, 05]

- (a) 6 (b) 3
(c) 0 [JIPMER 1999] (d) 12

34. An open pipe is in resonance in its 2nd harmonic with tuning fork of frequency f_1 . Now it is closed at one end. If the frequency of the tuning fork is increased slowly from f_1 then again a resonance is obtained with a frequency f_2 . If in this case the pipe vibrates n^{th} harmonics then

[IIT-JEE (Screening) 2005]

- (a) $n = 3, f_2 = \frac{3}{4} f_1$ (b) $n = 3, f_2 = \frac{5}{4} f_1$
(c) $n = 5, f_2 = \frac{5}{4} f_1$ (d) $n = 5, f_2 = \frac{3}{4} f_1$

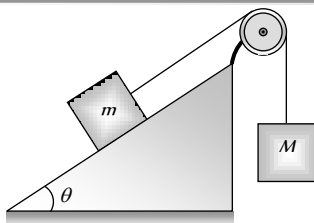
35. Two speakers connected to the same source of fixed frequency are placed 2.0 m apart in a box. A sensitive microphone placed at a distance of 4.0 m from their midpoint along the perpendicular bisector shows maximum response. The box is slowly rotated until the speakers are in line with the microphone. The distance between the midpoint of the speakers and the microphone remains unchanged. Exactly five maximum responses are observed in the microphone in doing this. The wavelength of the sound wave is

- (a) 0.2 m (b) 0.4 m
(c) 0.6 m (d) 0.8 m

36. A wire of 9.8×10^{-3} kgm⁻¹ passes over a frictionless light pulley fixed on the top of a frictionless inclined plane which makes an angle of 30° with the horizontal. Masses m and M are tied at the two ends of wire such that m rests on the plane and M hangs freely vertically downwards. The entire system is in equilibrium and a transverse wave propagates along the wire with a velocity of 100 ms⁻¹.

[EAMCET (Engg.) 2000] option

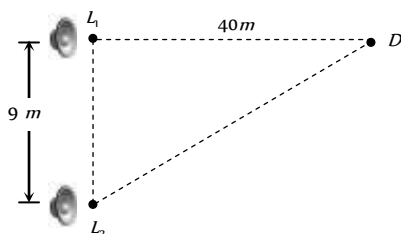
- (a) $m = 20 \text{ kg}$
 (b) $m = 5 \text{ kg}$
 (c) $m = 2 \text{ kg}$
 (d) $m = 7 \text{ kg}$



37. A man standing in front of a mountain beats a drum at regular intervals. The rate of drumming is generally increased and he finds that the echo is not heard distinctly when the rate becomes 40 per minute. He then moves nearer to the mountain by 90 m and finds that echo is again not heard when the drumming rate becomes 60 per minute. The distance between the mountain and the initial position of the man is

- (a) 205 m (b) 300 m
 (c) 180 m (d) 270 m

38. Two loudspeakers L_1 and L_2 driven by a common oscillator and amplifier, are arranged as shown. The frequency of the oscillator is gradually increased from zero and the detector at D records a series of maxima and minima. If the speed of sound is 330 m/s then the frequency at which the first maximum is observed is



- (a) 165 Hz (b) 330 Hz
 (c) 496 Hz (d) 660 Hz

39. The displacement due to a wave moving in the positive x -direction is given by $y = \frac{1}{(1+x^2)}$ at time $t = 0$ and by $y = \frac{1}{[1+(x-1)^2]}$

at $t = 2$ seconds, where x and y are in metres. The velocity of the wave in m/s is

- (a) 0.5 (b) 1
 (c) 2 (d) 4

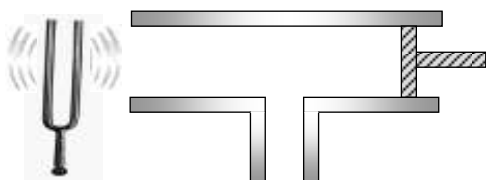
40. A person speaking normally produces a sound intensity of 40 dB at a distance of 1 m. If the threshold intensity for reasonable audibility is 20 dB, the maximum distance at which he can be heard clearly is

- (a) 4 m (b) 5 m
 (c) 10 m (d) 20 m

41. A string of length L and mass M hangs freely from a fixed point. Then the velocity of transverse waves along the string at a distance x from the free end is

- (a) $\sqrt{gL}$ (b) $\sqrt{gx}$
 (c) gL (d) gx

42. Vibrating tuning fork of frequency n is placed near the open end of a long cylindrical tube. The tube has a side opening and is fitted with a movable reflecting piston. As the piston is moved through 8.75 cm, the intensity of sound changes from a maximum to minimum. If the speed of sound is 350 m/s. Then n is



- (a) 500 Hz (b) 1000 Hz
 (c) 2000 Hz (d) 4000 Hz

43. A stone is hung in air from a wire which is stretched over a sonometer. The bridges of the sonometer are L cm apart when the wire is in unison with a tuning fork of frequency N . When the stone is completely immersed in water, the length between the bridges is l cm for re-establishing unison, the specific gravity of the material of the stone is

- (a) $\frac{L^2}{L^2 + l^2}$ (b) $\frac{L^2 - l^2}{L^2}$
 (c) $\frac{L^2}{L^2 - l^2}$ (d) $\frac{L^2 - l^2}{L^2}$

44. The displacement of a particle in string stretched in X direction is represented by y . Among the following expressions for y , those describing wave motions are

[IIT 1987]

- (a) $\cos kx \sin \omega t$ (b) $k^2 x^2 - \omega^2 t^2$
 (c) $\cos(kx + \omega t)$ (d) $\cos(k^2 x^2 - \omega^2 t^2)$

45. Three waves of equal frequency having amplitudes 10 μm , 4 μm and 7 μm arrive at a given point with successive phase difference of $\frac{\pi}{2}$.

The amplitude of the resulting wave in μm is given by

- (a) 7 (b) 6
 (c) 5 (d) 4

46. There are three sources of sound of equal intensity with frequencies 400, 401 and 402 vib/sec. The number of beats heard per second is

[MNR 1980; J & K CET 2005]

- (a) 0 (b) 1
 (c) 2 (d) 3

47. A tuning fork of frequency 340 Hz is vibrated just above the tube of 120 cm height. Water is poured slowly in the tube. What is the minimum height of water necessary for the resonance (speed of sound in the air = 340 m/sec)

[CBSE PMT 1999; UPSEAT 1999]

- (a) 15 cm (b) 25 cm
 (c) 30 cm (d) 45 cm

48. An organ pipe is closed at one end has fundamental frequency of 1500 Hz. The maximum number of overtones generated by this pipe which a normal person can hear is :

[AIIMS 2004]

- (a) 14 (b) 13
 (c) 6 (d) 9

49. In Melde's experiment, the string vibrates in 4 loops when a 50 gram weight is placed in the pan of weight 15 gram. To make the string to vibrate in 6 loops the weight that has to be removed from the pan is

[MH CET 2004]

- (a) 0.0007 kg wt (b) 0.0021 kg wt

- (c) 0.036 kg wt (d) 0.0029 kg wt

50. A racing car moving towards a cliff, sounds its horn. The driver observes that the sound reflected from the cliff has a pitch one octave higher than the actual sound of the horn. If v is the velocity of sound, then the velocity of the car is

[KCET 2002; CBSE PMT 2004]

- (a) $v/\sqrt{2}$ (b) $v/2$
(c) $v/3$ (d) $v/4$

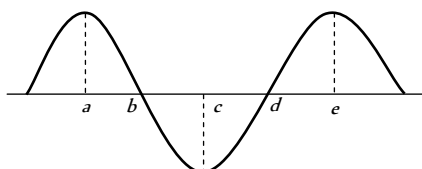
51. An earthquake generates both transverse (S) and longitudinal (P) sound waves in the earth. The speed of S waves is about 4.5 km/s and that of P waves is about 8.0 km/s . A seismograph records P and S waves from an earthquake. The first P wave arrives 4.0 min before the first S wave. The epicenter of the earthquake is located at a distance about

[AIIMS 2003]

- (a) 25 km (b) 250 km
(c) 2500 km (d) 5000 km

Graphical Questions

1. The rope shown at an instant is carrying a wave travelling towards right, created by a source vibrating at a frequency n . Consider the following statements



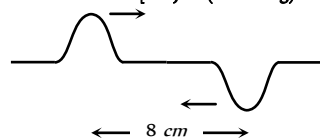
- I. The speed of the wave is $4n \times ab$
II. The medium at a will be in the same phase as d after $\frac{4}{3n} \text{ s}$
III. The phase difference between b and e is $\frac{3\pi}{2}$

Which of these statements are correct [AMU 2001]

- (a) I, II and III (b) II only
(c) I and III (d) III only

2. Two pulses in a stretched string whose centres are initially 8 cm apart are moving towards each other as shown in the figure. The speed of each pulse is 2 cm/s . After 2 seconds, the total energy of the pulses will be

[IIT-JEE (Screening) 2001]



- (a) Zero
(b) Purely kinetic
(c) Purely potential
(d) Partly kinetic and partly potential

A Assertion & Reason

For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below:

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 (b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 (c) If assertion is true but reason is false.
 (d) If the assertion and reason both are false.
 (e) If assertion is false but reason is true.

1. Assertion : Two persons on the surface of moon cannot talk to each other.
Reason : There is no atmosphere on moon.
2. Assertion : Transverse waves are not produced in liquids and gases.
Reason : Light waves are transverse waves.
3. Assertion : Sound waves cannot propagate through vacuum but light waves can.
Reason : Sound waves cannot be polarised but light waves can be polarised. [AIIMS 1998]
4. Assertion : The velocity of sound increases with increase in humidity.
Reason : Velocity of sound does not depend upon the medium.
5. Assertion : Ocean waves hitting a beach are always found to be nearly normal to the shore.
Reason : Ocean waves are longitudinal waves.
6. Assertion : Compression and rarefaction involve changes in density and pressure.
Reason : When particles are compressed, density of medium increases and when they are rarefied, density of medium decreases.
7. Assertion : Transverse waves travel through air in an organ pipe.
Reason : Air possesses only volume elasticity.
8. Assertion : Sound would travel faster on a hot summer day than on a cold winter day.
Reason : Velocity of sound is directly proportional to the square of its absolute temperature.
9. Assertion : The basic of Laplace correction was that, exchange of heat between the region of compression and rarefaction in air is not possible.
Reason : Air is a bad conductor of heat and velocity of sound in air is large.
10. Assertion : Particle velocity and wave velocity both are independent of time.
Reason : For the propagation of wave motion, the medium must have the properties of elasticity and inertia.
11. Assertion : When we start filling an empty bucket with water, the pitch of sound produced goes on decreasing.
Reason : The frequency of man voice is usually higher than that of woman.
12. Assertion : A tuning fork is made of an alloy of steel, nickel and chromium.
Reason : The alloy of steel, nickel and chromium is called elinvar.
13. Assertion : The change in air pressure effect the speed of sound.
Reason : The speed of sound in a gas is proportional to square root of pressure.
14. Assertion : Solids can support both longitudinal and transverse waves but only longitudinal waves can propagate in gases.
Reason : For the propagation of transverse waves, medium must also necessarily have the property of rigidity.
15. Assertion : Under given conditions of pressure and temperature, sound travels faster in a monoatomic gas than in diatomic gas.
Reason : Opposition for wave to travel is more in diatomic gas than monoatomic gas.
16. Assertion : The speed of sound in solids is maximum though their density is large.
Reason : The coefficient of elasticity of solid is large.
17. Assertion : On a rainy day sound travel slower than on a dry day.
Reason : When moisture is present in air the density of air increases.
18. Assertion : To hear distinct beats, difference in frequencies of two sources should be less than 10.
Reason : More the number of beats per sec more difficult to hear them.
19. Assertion : Sound produced by an open organ pipe is richer than the sound produced by a closed organ pipe.
Reason : Outside air can enter the pipe from both ends, in case of open organ pipe.
20. Assertion : It is not possible to have interference between the waves produced by two violins.
Reason : For interference of two waves the phase difference between the waves must remain constant.
21. Assertion : Beats can also be observed by two light sources as in sound.
Reason : Light sources have constant phase deference.
22. Assertion : In the case of a stationary wave, a person hear a loud sound at the nodes as compared to the antinodes.
Reason : In a stationary wave all the particles of the medium vibrate in phase.
23. Assertion : Velocity of particles, while crossing mean position (in stationary waves) varies from maximum at antinodes to zero at nodes.
Reason : Amplitude of vibration at antinodes is maximum and at nodes, the amplitude is zero, And all particles between two successive nodes cross the mean position together.
24. Assertion : Where two vibrating tuning forks having frequencies 256 Hz and 512 Hz are held near each other, beats cannot be heard.
Reason : The principle of superposition is valid only if the frequencies of the oscillators are nearly equal.
25. Assertion : The fundamental frequency of an open organ pipe increases as the temperature is increased.
Reason : As the temperature increases, the velocity of sound increases more rapidly than length of the pipe.

26. Assertion : Sound travel faster in solids than gases.
Reason : Solid possess greater density than gases.
[AIIMS 2000]
27. Assertion : Like sound, light can not propagate in vacuum.
Reason : Sound is a square wave. It propagates in a medium by a virtue of damping oscillation.
[AIIMS 2000]
28. Assertion : Speed of wave = $\frac{\text{Wave length}}{\text{Time period}}$
Reason : Wavelength is the distance between two nearest particles in phase.
[AIIMS 2002]
29. Assertion : The flash of lightening is seen before the sound of thunder is heard.
Reason : Speed of sound is greater than speed of light
[AIIMS 2002]
30. Assertion : When a beetle moves along the sand with in a few tens of centimeters of a sand scorpion the scorpion immediately turn towards the beetle and dashes to it
Reason : When a beetle disturbs the sand, it sends pulses along the sands surface one set of pulses is longitudinal while other set is transverse. [AIIMS 2003]
31. Assertion : The reverberation time dependent on the shape of enclosure, position of source and observer.
Reason : The unit of absorption coefficient in *mks* system is metric sabine.
[EAMCET 2004]

86	d	87	a	88	c	89	a	90	a
91	a	92	d	93	d	94	d		

Progressive Waves

1	d	2	c	3	b	4	c	5	d
6	d	7	c	8	d	9	c	10	c
11	c	12	c	13	c	14	b	15	b
16	abcd	17	b	18	b	19	d	20	bc
21	a	22	b	23	a	24	a	25	a
26	a	27	acd	28	d	29	a	30	a
31	b	32	d	33	b	34	d	35	d
36	d	37	a	38	a	39	b	40	b
41	d	42	c	43	b	44	c	45	a
46	a	47	d	48	a	49	b	50	d
51	d	52	abc	53	a	54	a	55	b
56	d	57	b	58	d	59	c	60	a
61	b	62	a	63	d	64	a	65	b
66	b	67	b	68	b	69	d	70	b
71	a	72	b	73	d	74	ac	75	c
76	b	77	b	78	c	79	b	80	a

Answers

Basics of Mechanical Waves

1	d	2	c	3	a	4	a	5	d
6	d	7	a	8	c	9	c	10	a
11	a	12	a	13	d	14	c	15	a
16	b	17	c	18	b	19	d	20	a
21	b	22	b	23	b	24	d	25	b
26	a	27	d	28	c	29	b	30	d
31	c	32	a	33	b	34	d	35	b
36	b	37	b	38	a	39	c	40	d
41	d	42	d	43	c	44	a	45	d
46	c	47	b	48	d	49	b	50	a
51	d	52	c	53	c	54	c	55	b
56	a	57	a	58	a	59	a	60	a
61	d	62	c	63	a	64	c	65	d
66	c	67	c	68	a	69	d	70	a
71	b	72	b	73	b	74	d	75	c
76	b	77	d	78	b	79	b	80	b
81	d	82	b	83	b	84	b	85	d

Interference and Superposition of Waves

1	b	2	d	3	a	4	d	5	b
6	d	7	d	8	bc	9	c	10	c
11	a	12	b	13	c	14	d	15	b
16	c	17	a	18	a	19	b	20	c
21	a	22	b	23	a	24	c	25	d
26	b								

Beats

1	c	2	d	3	c	4	a	5	d
6	b	7	c	8	a	9	d	10	b
11	c	12	b	13	a	14	c	15	c
16	c	17	c	18	b	19	c	20	a
21	d	22	c	23	d	24	c	25	c
26	d	27	a	28	c	29	b	30	a
31	c	32	a	33	a	34	a	35	d
36	b	37	a	38	a	39	a	40	b
41	a	42	c	43	d	44	b	45	a
46	c	47	a	48	b	49	b	50	b
51	b								

Stationary Waves

1	c	2	c	3	c	4	c	5	b
6	a	7	b	8	d	9	a	10	a
11	b	12	d	13	b	14	d	15	d
16	abc	17	a	18	d	19	a	20	a
21	a	22	b	23	c	24	b	25	a
26	c	27	d	28	c	29	b	30	d
31	b	32	a	33	b	34	a	35	a
36	a	37	a	38	d	39	d		

Vibration of String

1	c	2	d	3	c	4	c	5	c
6	b	7	b	8	d	9	a	10	c
11	d	12	c	13	c	14	a	15	a
16	d	17	a	18	a	19	c	20	b
21	d	22	c	23	a	24	b	25	a
26	b	27	b	28	b	29	c	30	c
31	b	32	a	33	d	34	b	35	d
36	c	37	d	38	a	39	d	40	b
41	a	42	a	43	d	44	d	45	d
46	c	47	a	48	b	49	d	50	c
51	d	52	b						

Organ Pipe (Vibration of Air Column)

1	c	2	a	3	c	4	d	5	c
6	acd	7	d	8	a	9	b	10	c
11	b	12	c	13	b	14	b	15	b
16	a	17	b	18	a	19	c	20	a
21	b	22	a	23	a	24	b	25	c
26	a	27	a	28	b	29	a	30	d
31	c	32	a	33	b	34	b	35	b
36	b	37	b	38	c	39	b	40	b
41	b	42	b	43	a	44	c	45	a
46	c	47	a	48	d	49	b	50	c
51	a	52	a	53	b				

Doppler's Effect

1	d	2	b	3	a	4	b	5	c
6	b	7	c	8	b	9	a	10	a
11	b	12	a	13	d	14	c	15	b
16	a	17	a	18	c	19	d	20	a
21	d	22	a	23	a	24	b	25	c
26	b	27	c	28	d	29	b	30	d
31	a	32	c	33	d	34	d	35	a
36	b	37	c	38	d	39	a	40	c
41	a	42	c	43	a	44	d	45	d

46	b	47	b	48	b	49	b	50	a
51	a	52	c	53	d	54	b	55	a
56	c	57	c	58	d	59	c	60	a
61	c	62	b	63	a	64	a	65	b

Musical Sound

1	d	2	a	3	a	4	c	5	b
6	c	7	a	8	d	9	d	10	d
11	d	12	c	13	d	14	c	15	b
16	c	17	b	18	a	19	d	20	b
21	c	22	d	23	b	24	a	25	b
26	d	27	d	28	b	29	b	30	a
31	d	32	a						

Critical Thinking Questions

1	abcd	2	bc	3	c	4	a	5	c
6	b	7	b	8	c	9	b	10	d
11	b	12	abc	13	b	14	b	15	a
16	b	17	a	18	d	19	a	20	c
21	a	22	abcd	23	b	24	d	25	a
26	a	27	c	28	a	29	b	30	a
31	b	32	b	33	a	34	c	35	b
36	a	37	d	38	b	39	a	40	c
41	b	42	b	43	c	44	ac	45	c
46	b	47	d	48	c	49	c	50	c
51	c								

Graphical Questions

1	c	2	b	3	a	4	b	5	d
6	c	7	d	8	d	9	c	10	c
11	c	12	c	13	c	14	b	15	bd
16	d	17	b	18	d				

Assertion and Reason

1	a	2	b	3	b	4	c	5	c
6	a	7	e	8	c	9	c	10	e
11	d	12	b	13	e	14	a	15	c
16	a	17	d	18	b	19	b	20	a
21	d	22	c	23	a	24	c	25	a
26	b	27	d	28	b	29	c	30	a
31	e								

AS

Answers and Solutions

Basics of Mechanical Waves

1. (d) Air is more rarer for sound to travel as compared to vacuum.
2. (c)
3. (a)
4. (a) $v = n\lambda = 2 \times 5 = 10 \text{ cm/sec}$
5. (d) $v = n\lambda \Rightarrow \lambda = \frac{v}{n} = \frac{330}{256} = 1.29 \text{ m}$
6. (d) Time lost in covering the distance of 2 km by the sound waves $t = \frac{d}{v} = \frac{2000}{330} = 6.06 \text{ sec} \approx 6 \text{ sec}$
7. (a) $v_{\max} = a\omega = a \times 2\pi n = 0.1 \times 2\pi \times 300 = 60\pi \text{ cm/sec}$
8. (c) Audible range of frequency is 20 Hz to 20 kHz
9. (c) Phase difference $= \frac{2\pi}{\lambda} \times \text{path difference}$
 $\Rightarrow 1.6\pi = \frac{2\pi}{\lambda} \times 40 \Rightarrow \lambda = 50 \text{ cm} = 0.5 \text{ m}$
 $\Rightarrow v = n\lambda \Rightarrow 330 = 0.5 \times n \Rightarrow n = 660 \text{ Hz}$
10. (a) $\lambda = \frac{v}{n}; n \approx 50,000 \text{ Hz}, v = 330 \text{ m/sec} \Rightarrow \lambda = \frac{330}{50000} \text{ m}$
 $= 6.6 \times 10^{-5} \text{ cm} \approx 5 \times 10^{-5} \text{ cm}$
11. (a)
12. (a) $\lambda = \frac{v}{n} = \frac{1.7 \times 1000}{4.2 \times 10^6} = 4 \times 10^{-4} \text{ m}$
13. (d) Since maximum audible frequency is 20,000 Hz, hence
 $\lambda_{\min} = \frac{v}{n_{\max}} = \frac{340}{20,000} \approx 20 \text{ mm}$
14. (c) Velocity of sound in gas $v = \sqrt{\frac{\gamma RT}{M}} \Rightarrow v \propto \sqrt{\frac{\gamma T}{M}}$
 $\Rightarrow \frac{v_{N_2}}{v_{He}} = \sqrt{\frac{\gamma_{N_2}}{\gamma_{He}} \times \frac{M_{He}}{M_{H_2}}} = \sqrt{\frac{\frac{7}{5} R \times 4}{\frac{5}{3} R \times 28}} = \frac{\sqrt{3}}{5}$
15. (a) Time required for a point to move from maximum displacement to zero displacement is $t = \frac{T}{4} = \frac{1}{4n}$
 $\Rightarrow n = \frac{1}{4t} = \frac{1}{4 \times 0.170} = 1.47 \text{ Hz}$
16. (b) Wave number is the reciprocal of wavelength and is written as
 $\bar{n} = \frac{1}{\lambda}$
17. (c) $\lambda = \frac{v}{n} = \frac{340}{200} = 1.7 \text{ m}$
18. (b)
19. (d) $v \propto \lambda \Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{v_1}{v_2} = \frac{2/3}{3/10} = \frac{20}{9}$

20. (a) The time taken by the stone to reach the lake

$$t_1 = \sqrt{\left(\frac{2h}{g}\right)} = \sqrt{\left(\frac{2 \times 500}{10}\right)} = 10 \text{ sec} \quad (\text{Using } h = ut + \frac{1}{2}gt^2)$$

Now time taken by sound from lake to the man

$$t_2 = \frac{h}{v} = \frac{500}{340} \approx 1.5 \text{ sec}$$

$$\Rightarrow \text{Total time} = t_1 + t_2 = 10 + 1.5 = 11.5 \text{ sec.}$$

21. (b) When medium changes, velocity and wavelength changes but frequency remains constant.
22. (b) $t = \sqrt{\frac{2h}{g}} + \frac{h}{v} = \sqrt{\frac{2 \times 19.6}{9.8}} + \frac{19.6}{v} = 2.06$
 $\Rightarrow v = 326.7 \text{ m/s}$
23. (b) $v \propto \sqrt{T} \Rightarrow \frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}} \Rightarrow 2 = \sqrt{\frac{T_2}{(273+0)}}$
 $\Rightarrow T_2 = 273 \times 4 = 1092 \text{ K} = 819^\circ \text{C}$
24. (d) Velocity of sound in steel is maximum out of the given materials water and air. In vacuum sound cannot travel, its speed is zero.

25. b) Distance between a compression and the nearest rarefaction is

$$\frac{\lambda}{2} = 1 \text{ m}. \text{ Hence } n = \frac{v}{\lambda} = \frac{360}{2} = 180 \text{ Hz}.$$

26. (a) $v = \sqrt{\frac{\gamma P}{\rho}} \Rightarrow \frac{v_{O_2}}{v_{H_2}} = \sqrt{\frac{\rho_{H_2}}{\rho_{O_2}}} = \sqrt{\frac{1}{16}} = \frac{1}{4}$

27. (d) Speed of sound in gases is $v = \sqrt{\frac{\gamma RT}{M}} \Rightarrow T \propto M$

(Because γ , constant). Hence $\frac{T_{H_2}}{T_{O_2}} = \frac{M_{H_2}}{M_{O_2}}$

$$\Rightarrow \frac{T_{H_2}}{(273 + 100)} = \frac{2}{32} \Rightarrow T_{H_2} = 23.2 \text{ K} = -249.7^\circ \text{C}$$

28. (c) If the temperature changes then velocity of wave and its wavelength changes. Frequency amplitude and time period remains constant.

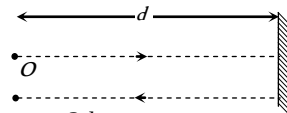
29. (b)

30. (d)

31. (c) Path difference $\Delta = \frac{\lambda}{2\pi} \times \phi \Rightarrow 1 = \frac{\lambda}{2\pi} \times \frac{\pi}{2} \Rightarrow \lambda = 4 \text{ m}$

Hence $v = n\lambda = 120 \times 4 = 480 \text{ m/s}$

32. (a) Suppose the distance between shooter and reflecting surface is d . Hence time interval for hearing echo is



$$t = \frac{2d}{v} \Rightarrow 8 = \frac{2d}{350} \Rightarrow d = 1400 \text{ m}.$$

33. (b) Time = $\frac{\text{Distance}}{\text{Velocity}} = \frac{1000}{330} = 3.03 \text{ sec}.$

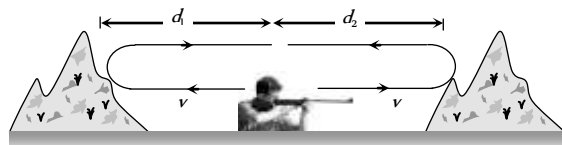
Sound will be heard after 3.03 sec. So his watch is set 3 sec, slower.

34. (d) $v = \sqrt{\frac{\gamma P}{\rho}}$; as P changes, ρ also changes. Hence $\frac{P}{\rho}$ remains constant so speed remains constant.

35. (b) Speed of sound in gases is given by

$$v = \sqrt{\frac{\gamma RT}{M}} \Rightarrow v \propto \frac{1}{\sqrt{M}} \Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{m_2}{m_1}}$$

36. (b)



$$2d_1 + 2d_2 = v \times t_1 + v \times t_2 \Rightarrow 2(d_1 + d_2) = v(t_1 + t_2)$$

$$d_1 + d_2 = \frac{v(t_1 + t_2)}{2} = \frac{340 \times (1.5 + 3.5)}{2} = 850 \text{ m}.$$

37. (b) By using $v = \sqrt{\frac{\gamma RT}{M}} \Rightarrow v \propto \sqrt{T}$

$$\frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}} = \sqrt{\frac{T + 600}{T}} = \sqrt{3} \Rightarrow T = 300 \text{ K} = 27^\circ \text{C}$$

38. (a) Velocity of sound is independent of frequency. Therefore it is same (v) for frequency n and $4n$.

39. (c) $v = \sqrt{\frac{\gamma RT}{M}} \Rightarrow v \propto \sqrt{T}$

i.e. if v is doubled then T becomes four times,

hence $T_2 = 4T_1 = 4(273 + 27) = 1200 \text{ K} = 927^\circ \text{C}$

40. (d) $n = \frac{3600}{60} = 60 \text{ Hz} \Rightarrow \lambda = \frac{v}{n} = \frac{960}{60} = 16 \text{ m}$

41. (d) Speed of sound, doesn't depend up on pressure and density medium.

42. (d) If d is the distance between man and reflecting surface of sound then for hearing echo

$$2d = v \times t \Rightarrow d = \frac{340 \times 1}{2} = 170 \text{ m}$$

43. (c) $n = \frac{54}{60} \text{ Hz}, \lambda = 10 \text{ m} \Rightarrow v = n\lambda = 9 \text{ m/s}.$

44. (a) $v = \sqrt{\frac{\gamma RT}{M}} \Rightarrow v \propto \frac{1}{\sqrt{M}}$. Since M is minimum for H_2 so sound velocity is maximum in H_2 .

45. (d) $2d = v \times t$, where v = velocity of sound = 332 m/s

$$t = \text{Persistence of hearing} = \frac{1}{10} \text{ sec}.$$

$$\Rightarrow d = \frac{v \times t}{2} = \frac{332 \times \frac{1}{10}}{2} = 16.5 \text{ m}$$

46. (c) Since solid has both the properties (rigidity and elasticity)

47. (b) If d is the distance between man and reflecting surface of sound then for hearing echo

$$2d = v \times t \Rightarrow d = \frac{330 \times 1.5}{2} = 247.5 \text{ m}$$

48. (d) Speed of sound $v \propto \sqrt{T}$ and it is independent of pressure.

49. (b) Frequency of wave is

$$n = \frac{3600}{2 \times 60} \text{ Hz} \Rightarrow \lambda = \frac{v}{n} = \frac{760}{30} = 25.3 \text{ m}.$$

50. (a) Speed of sound $v = \sqrt{\frac{\gamma P}{\rho}} \Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{d_2}{d_1}} (\because P - \text{constant})$

51. (d) $\lambda = \frac{v}{n} = \frac{352}{384}$; during 1 vibration of fork sound will travel $\frac{352}{384} \text{ m}$ during 36 vibration of fork sound will travel $\frac{352}{384} \times 36 = 33 \text{ m}$

52. (c) At given temperature and pressure

$$v \propto \frac{1}{\sqrt{\rho}} \Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{\rho_2}{\rho_1}} = \sqrt{\frac{4}{1}} = 2:1$$

53. (c) $v \propto \sqrt{T} \Rightarrow \sqrt{\frac{T_2}{T_1}} = \frac{v_2}{v_1} \Rightarrow T_2 = T_1 \left(\frac{v_2}{v_1} \right)^2$
 $\Rightarrow T_2 = 273 \times 4 = 1092 \text{ K}$

54. (c) $\bar{n} = \frac{1}{\lambda} = \frac{1}{6000 \times 10^{-10}} = 1.66 \times 10^6 \text{ m}^{-1}$
55. (b) $v \propto \frac{1}{\sqrt{M}} \Rightarrow \frac{v_{H_2}}{v_{O_2}} = \sqrt{\frac{M_{O_2}}{M_{H_2}}} = \sqrt{\frac{32}{2}} \Rightarrow \frac{v_{H_2}}{v_{O_2}} = \frac{4}{1}$
56. (a) The minimum distance between compression and rarefaction of the wire $l = \frac{\lambda}{4} \therefore$ Wave length $\lambda = 4l$
Now by $v = n\lambda \Rightarrow n = \frac{360}{4 \times 1} = 90 \text{ sec}^{-1}$.
57. (a) $v_{\text{sound}} \propto \frac{1}{\sqrt{\rho}} \Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{\rho_2}{\rho_1}} = \sqrt{\frac{4}{1}} = 2 \Rightarrow v_2 = \frac{v_1}{2} = \frac{v_s}{2}$
58. (a) Suppose the distance between two fixed points is d then $t = \frac{d}{v}$ also $v \propto \sqrt{T} \Rightarrow \frac{t_1}{t_2} = \frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}}$
 $\Rightarrow \frac{2}{t_2} = \sqrt{\frac{303}{283}} \Rightarrow t_2 = 1.9 \text{ sec.}$
59. (a) The density of moist air (i.e. air mixed with water vapours) is less than the density of dry air
Hence from $v = \sqrt{\frac{\gamma P}{\rho}} \Rightarrow v_{\text{moist air}} > v_{\text{dry air}}$
60. (a) Total time taken for both the echoes $t = t_1 + t_2 = 2 \text{ sec}$
-
- but $t = \frac{2d_1}{v} + \frac{2d_2}{v} \Rightarrow t = \frac{2}{v}(d_1 + d_2)$
 $\Rightarrow (d_1 + d_2) = \frac{v \times t}{2} = \frac{340 \times 2}{2} = 340 \text{ m.}$
61. (d) Frequency of sound does not change with medium, because it is characteristics of source.
62. (c) Since $v = \sqrt{\frac{\gamma RT}{M}}$ i.e., $v \propto \sqrt{T}$
63. (a) Frequency of waves remains same, i.e. 60 Hz
and wavelength $\lambda = \frac{v}{n} = \frac{330}{60 \times 10^3} = 5.5 \text{ mm.}$
64. (c) Path difference $\Delta = \frac{\lambda}{2\pi} \times \phi = \frac{\lambda}{2\pi} \times \frac{\pi}{3} = \frac{\lambda}{6}$
65. (d) Interference, diffraction and reflection occurs in both transverse and longitudinal waves. Polarisation occurs only in transverse waves.
66. (c) Water waves are transverse as well as longitudinal in nature.
67. (c)
68. (a) In transverse waves medium particles vibrate perpendicular to the direction of propagation of wave.
69. (d)
70. (a) Wave on a plucked string is stationary wave. Light waves are EM waves. Water waves are transverse as well as longitudinal.
71. (b)

72. (b) Transverse wave can propagate in solids but not in liquids and gases.
73. (b) Because sound waves in gases are longitudinal.
74. (d)
75. (c) Since distance between two consecutive crests is λ , so
 $\phi = \frac{2\pi}{\lambda} \times \lambda = 2\pi$.
76. (b) The distance between two points i.e. path difference between them $\Delta = \frac{\lambda}{2\pi} \times \phi = \frac{\lambda}{2\pi} \times \frac{\pi}{3} = \frac{\lambda}{6} = \frac{v}{6n}$ ($\because v = n\lambda$) $\Rightarrow$
 $\Delta = \frac{360}{6 \times 500} = 0.12 \text{ m} = 12 \text{ cm}$
77. (d) Sound waves are longitudinal in nature so they can not be polarised
78. (b)
79. (b) Ultrasonic waves are those of higher frequencies than maximum audible range frequencies (audible range of frequencies is 20 Hz to 20000 Hz)
80. (b)
81. (d) Infrasonic waves have frequency less than (20 Hz) audible sound and wavelength more than audible sound.
82. (b) SONAR emits ultrasonic waves.
83. (b) EM waves do not requires medium for their propagation.
84. (b)
85. (d)
86. (d) $v = \sqrt{\frac{\gamma RT}{M}} \Rightarrow \frac{T_N}{T_0} = \frac{M_N}{M_0} \Rightarrow \frac{T_N}{273 + 55} = \frac{14}{16} = \frac{7}{8}$
 $\Rightarrow T_N = 287 \text{ K} = 14^\circ \text{C}$
87. (a) We know that at night amount of carbon dioxide in atmosphere increases which raises the density of atmosphere. Since intensity is directly proportional to density, intensity of sound is more at night.
88. (c) $n = \frac{v}{\lambda} = \frac{300}{0.6 \times 10^{-2}} \text{ Hz} = \frac{3}{6} \times 10^4 \text{ Hz} = 50,000 \text{ Hz}$
 $\Rightarrow$ Wave is ultrasonic.
89. (a) $v = \sqrt{\frac{K}{\rho}} \therefore K = v^2 \rho = 2.86 \times 10^{10} \text{ N/m}^3$
90. (a) $n = \frac{v}{\lambda} \propto v \Rightarrow \frac{n_{MW}}{n_{US}} \approx \frac{3 \times 10^8}{3 \times 10^2} \approx 10^6 : 1$
91. (a) Intensity $\propto \frac{1}{(\text{Distance})^2} \Rightarrow \frac{I_1}{I_2} = \left(\frac{d_2}{d_1}\right)^2 = \left(\frac{3}{2}\right)^2 = \frac{9}{4}$
92. (d) $v = \frac{\sin i}{\sin r} = \frac{v_1}{v_2}$
 $\Rightarrow \sin r = \sin 30^\circ \times \frac{2u}{u} \Rightarrow \sin r = \frac{1}{2} \times 2 \times 1 \Rightarrow r = 90^\circ$
93. (d) Number of waves per minute = 54
 $\therefore$ Number of waves per second = 54/60
Now $v = n\lambda \Rightarrow n = \frac{54}{60} \times 10 = 9 \text{ m/s.}$

94. (d) If d is the distance of rock from SONAR then

$$2d = vt \Rightarrow d = \frac{v \times t}{2} = \frac{1600 \times 1}{2} = 800 \text{ m}$$

Progressive Waves

1. (d) Comparing given equation with standard equation of progressive wave. The velocity of wave

$$v = \frac{\omega (\text{Co-efficient of } t)}{k (\text{Co-efficient of } x)} = \frac{200\pi}{0.5\pi} = 400 \text{ cm/s}$$

2. (c) Comparing with $y = a \cos(\omega t + kx - \phi)$,

$$\text{We get } k = \frac{2\pi}{\lambda} = 0.02 \Rightarrow \lambda = 100 \text{ cm}$$

Also, it is given that phase difference between particles

$$\Delta\phi = \frac{\pi}{2}. \text{ Hence path difference between them}$$

$$\Delta = \frac{\lambda}{2\pi} \times \Delta\phi = \frac{\lambda}{2\pi} \times \frac{\pi}{2} = \frac{\lambda}{4} = \frac{100}{4} = 25 \text{ cm}$$

3. (b) Phase difference between two successive crest is 2π . Also, phase difference $(\Delta\phi) = \frac{2\pi}{T}$ time interval (Δt)

$$\Rightarrow 2\pi = \frac{2\pi}{T} \times 0.2 \Rightarrow \frac{1}{T} = 5 \text{ sec}^{-1} \Rightarrow n = 5 \text{ Hz}$$

4. (c) Comparing with the standard equation,

$$y = A \sin \frac{2\pi}{\lambda} (vt - x), \text{ we have}$$

$$v = 200 \text{ cm/sec}, \lambda = 200 \text{ cm}; \therefore n = \frac{v}{\lambda} = 1 \text{ sec}^{-1}$$

5. (d) Let the phase of second particle be ϕ . Hence phase difference between two particles is $\Delta\phi = \frac{2\pi}{\lambda} \Delta x$

$$\Rightarrow \left(\phi - \frac{\pi}{3} \right) = \frac{2\pi}{60} \times 15 \Rightarrow \phi - \frac{\pi}{3} = \frac{\pi}{2} \Rightarrow \phi = \frac{5\pi}{6}$$

6. (d) The given equation can be written as $y = 4 \sin \left(4\pi t - \frac{\pi x}{16} \right)$

$$\Rightarrow (v) = \frac{\text{Co-efficient of } t(\omega)}{\text{Co-efficient of } x(K)}$$

$$\Rightarrow v = \frac{4\pi}{\pi/16} = 64 \text{ cm/sec along } +x \text{ direction.}$$

7. (c) $v = \frac{\text{Co-efficient of } t}{\text{Co-efficient of } x} = \frac{628}{31.4} = 20 \text{ cm/sec}$

8. (d) $y_1 = a \sin(\omega t - kx)$

$$\text{and } y_2 = a \cos(\omega t - kx) = a \sin \left(\omega t - kx + \frac{\pi}{2} \right)$$

$$\text{Hence phase difference between these two is } \frac{\pi}{2}.$$

9. (c) $I \propto a^2 \propto \frac{1}{d^2} \Rightarrow a \propto \frac{1}{d}$

10. (c) $\frac{I_1}{I_2} = \frac{a_1^2}{a_2^2} = \left(\frac{0.06}{0.03} \right)^2 = 4$

11. (c) After reflection from rigid support, a wave suffers a phase change of π .

12. (c) The given equation representing a wave travelling along $-y$ direction (because '+' sign is given between t term and x term).
On comparing it with $x = A \sin(\omega t + ky)$

$$\text{We get } k = \frac{2\pi}{\lambda} = 12.56 \Rightarrow \lambda = \frac{2 \times 3.14}{12.56} = 0.5 \text{ m}$$

13. (c) Comparing with $y = a \sin(\omega t - kx) \Rightarrow a = \frac{10}{\pi}, \omega = 200\pi$

$$\therefore v_{\max} = a\omega = \frac{10}{\pi} \times 2000\pi = 2000 \text{ m/sec}$$

$$\text{and } \omega = \frac{2\pi}{T} \Rightarrow 200\pi = \frac{2\pi}{T} \Rightarrow T = 10^{-3} \text{ sec}$$

14. (b) Comparing the given equation with $y = a \cos(\omega t - kx)$

$$\text{We get } k = \frac{2\pi}{\lambda} = \pi \Rightarrow \lambda = 2 \text{ cm}$$

15. (b) Comparing the given equation with $y = a \sin(\omega t - kx)$, We get $a = Y, \omega = 2\pi f, k = \frac{2\pi}{\lambda}$. Hence maximum particle velocity

$$(v_{\max})_{\text{particle}} = a\omega = Y_0 \times 2\pi f \quad \text{and} \quad \text{wave velocity}$$

$$(v)_{\text{wave}} = \frac{\omega}{k} = \frac{2\pi f}{2\pi/\lambda} = f\lambda$$

$$\therefore (v_{\max})_{\text{particle}} = 4v_{\text{wave}} \Rightarrow Y_0 \times 2\pi f = 4f\lambda \Rightarrow \lambda = \frac{\pi Y_0}{2}.$$

16. (a,b,c,d) On comparing the given equation with

$$y = a \sin(\omega t + kx), \text{ it is clear that wave is travelling in negative } x\text{-direction.}$$

It's amplitude $a = 10 \text{ m}$ and $\omega = 60, k = 2$. Hence frequency

$$n = \frac{\omega}{2\pi} = \frac{60}{2\pi} = \frac{30}{\pi} \text{ Hz}$$

$$k = \frac{2\pi}{\lambda} = 2 \Rightarrow \lambda = \pi \text{ m and } v = \frac{\omega}{k} = \frac{60}{2} = 30 \text{ m/s}$$

17. (b) $\therefore y = a \cos \left(\frac{2\pi}{\lambda} vt + \frac{2\pi x}{\lambda} \right) = 0.5 \cos(4\pi t + 2\pi x)$

18. (b) $v = \frac{\text{Co-efficient of } t}{\text{Co-efficient of } x} = \frac{100}{50} = 2 \text{ m/sec.}$

19. (d) $y = f(x^2 - vt^2)$ doesn't follow the standard wave equation.

20. (b,c) Standard wave equation which travel in negative x -direction is $y = A \sin(\omega t + kx + \phi_0)$

$$\text{For the given wave } \omega = 2\pi m = 15\pi, k = \frac{2\pi}{\lambda} = 10\pi$$

$$\text{Now } v = \frac{\text{Co-efficient of } t}{\text{Co-efficient of } x} = \frac{\omega}{k} = \frac{15\pi}{10\pi} = 1.5 \text{ m/sec}$$

$$\text{and } \lambda = \frac{2\pi}{k} = \frac{2\pi}{10\pi} = 0.2 \text{ m.}$$

21. (a) $v_{\max} = a\omega = 3 \times 10 = 30$

22. (b) $y_1 = a_1 \sin \left(\omega t - \frac{2\pi x}{\lambda} \right)$ and

$$y_2 = a_2 \cos\left(\omega t - \frac{2\pi x}{\lambda} + \phi\right) = a_2 \sin\left(\omega t - \frac{2\pi x}{\lambda} + \phi + \frac{\pi}{2}\right)$$

$$\text{So phase difference} = \phi + \frac{\pi}{2} \text{ and } \Delta = \frac{\lambda}{2\pi} \left(\phi + \frac{\pi}{2}\right)$$

23. (a) Both waves are moving opposite to each other.

24. (a) The velocity of wave

$$v = \frac{\omega (\text{Co-efficient of } t)}{k (\text{Co-efficient of } x)} = \frac{10}{1} = 10 \text{ m/s}$$

25. (a) $v = \frac{\text{Co-efficient of } t}{\text{Co-efficient of } x} = \frac{7\pi}{0.04} = 175 \text{ m/s}.$

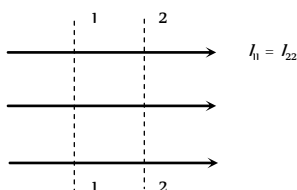
26. (a) The given equation is $y = 10 \sin(0.01\pi x - 2\pi t)$

Hence $\omega = \text{coefficient of } t = 2\pi$

$$\Rightarrow \text{Maximum speed of the particle } v_{\max} = a\omega = 10 \times 2\pi$$

$$= 10 \times 2 \times 3.14 = 62.8 \approx 63 \text{ cm/s}$$

27. (a,c,d) For a travelling wave, the intensity of wave remains constant if it is a plane wave.



Intensity of wave is inversely proportional to the square of the distance from the source if the wave is spherical

$$\left(I = \frac{P}{4\pi r^2}\right)$$

Intensity of spherical wave on the spherical surface centred at source always remains same. Here total intensity means power P .

28. (d) On comparing the given equation with standard equation $y = a \sin \frac{2\pi}{\lambda}(vt - x)$. It is clear that wave speed $(v)_{\text{wave}} = v$ and maximum particle velocity $(v_{\max})_{\text{particle}} = a\omega = y_0 \times \text{co-efficient of } t = y_0 \times \frac{2\pi v}{\lambda}$

$$\therefore (v_{\max})_{\text{particle}} = 2(\omega)_{\text{wave}} \Rightarrow \frac{a \times 2\pi v}{\lambda} = 2v \Rightarrow \lambda = \pi y_0$$

29. (a) Given $y = A \sin(kx - \omega t)$

$$\Rightarrow v = \frac{dy}{dt} = -A\omega \cos(kx - \omega t): \Rightarrow v_{\max} = A\omega$$

30. (a) Comparing with $y(x, t) = a \sin(\omega t - kx)$

$$k = \frac{2\pi}{\lambda} = 0.01\pi \Rightarrow \lambda = 200 \text{ m}.$$

31. (b)

32. (d) Comparing the given equation with standard equation $y = a \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda}\right) \Rightarrow T = 0.04 \text{ sec} \Rightarrow v = \frac{1}{T} = 25 \text{ Hz}$

$$\text{Also } (A)_{\max} = \omega^2 a = \left(\frac{2\pi}{T}\right)^2 \times a = \left(\frac{2\pi}{0.04}\right)^2 \times 3$$

$$= 7.4 \times 10^4 \text{ cm/sec.}$$

33. (b) From the given equation amplitude $a = 0.04 \text{ m}$

$$\text{Frequency} = \frac{\text{Co-efficient of } t}{2\pi} = \frac{\pi/5}{2\pi} = \frac{1}{10} \text{ Hz}$$

$$\text{Wave length } \lambda = \frac{2\pi}{\text{Co-efficient of } x} = \frac{2\pi}{\pi/9} = 18 \text{ m}.$$

$$\text{Wave speed } v = \frac{\text{Co-efficient of } t}{\text{Co-efficient of } x} = \frac{\pi/5}{\pi/9} = 1.8 \text{ m/s}.$$

34. (d)

35. (d) Compare the given equation with $y = a \cos(\omega t + k\phi)$

$$\Rightarrow \omega = 2\pi n = 2000 \Rightarrow n = \frac{1000}{\pi} \text{ Hz}$$

36. (d) $y = A \sin(at - bx + c)$ represents equation of simple harmonic progressive wave as it describes displacement of any particle (x) at any time (t). or It represents a wave because it satisfies wave equation $\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}$.

37. (a) Here $\omega = 2\pi n = 2\pi \Rightarrow n = 1$

38. (a) Compare the given equation with $y = a \sin(\omega t + kx)$. We get $\omega = 2\pi n = 100 \Rightarrow n = \frac{50}{\pi} \text{ Hz}$

39. (b) Compare with $y = a \sin(\omega t - kx)$

$$\text{We have } k = \frac{2\pi}{\lambda} = 62.4 \Rightarrow \lambda = \frac{2\pi}{62.4} = 0.1$$

40. (b) Maximum velocity of the particle

$$v_{\max} = a\omega = 0.5 \times 10\pi = 5\pi \text{ cm/sec}$$

41. (d) On reflection from fixed end (denser medium) a phase difference of π is introduced.

42. (c) Maximum particle velocity $v_{\max} = \omega a$ and wave velocity

$$v = \frac{\omega}{k} \Rightarrow \frac{v_{\max}}{v} = \frac{\omega a}{\omega/k} = ka. \text{ From the given}$$

$$\text{equation } k = \text{Co-efficient of } x = 6 \text{ micron} = 6 \times 10^{-6} \text{ m}$$

$$\Rightarrow \frac{v_{\max}}{v} = ka = 6 \times 10^{-6} \times 60 = 3.6 \times 10^{-4}$$

43. (b) $\omega = 314$, $k = 1.57$ and $v = \frac{\omega}{k} = \frac{314}{1.57} = 200 \text{ m/s}.$

44. (c) $v = \frac{\text{Co-efficient of } t}{\text{Co-efficient of } x} = \frac{40}{1} = 40 \text{ m/s}$

45. (a) $n = \frac{\omega}{2\pi} = \frac{400\pi}{2\pi} = 200 \text{ Hz}$ (As $\omega = 400\pi$)

46. (a) Beats period = $\frac{1}{30-20} = 0.1 \text{ sec}$

$$\Delta\phi = \frac{2\pi}{T} \Delta t = \frac{2\pi}{0.1} \times 0.6 = 2\pi \times 6 = 12\pi \text{ or Zero.}$$

47. (d) Path difference $\Delta = \frac{\lambda}{2\pi} \times \phi = \frac{\lambda}{2\pi} \times \frac{\pi}{2} = \frac{\lambda}{4}$

$$\therefore \Delta = 0.8 \text{ m} \Rightarrow \frac{\lambda}{4} = 0.8 \Rightarrow \lambda = 3.2 \text{ m}.$$

$$\therefore v = n\lambda = 120 \times 3.2 = 384 \text{ m/s}$$

48. (a) $v = \frac{\text{co-efficient of } t}{\text{co-efficient of } x} = \frac{2\pi/0.01}{2\pi/0.3} = 30 \text{ m/s}$
49. (b) Comparing with $y = a \sin 2\pi \left[\frac{t}{T} - \frac{x}{\lambda} \right] \Rightarrow \lambda = 40 \text{ cm}$
50. (d) $v = \frac{\omega}{k} = \frac{\text{Co-efficient of } t}{\text{Co-efficient of } x} = \frac{2}{0.01} = 200 \text{ cm/sec.}$
51. (d) From the given equation $k = 0.2\pi$
 $\Rightarrow \frac{2\pi}{\lambda} = 0.2\pi \Rightarrow \lambda = 10 \text{ cm}$
 $\Delta\phi = \frac{2\pi}{\lambda} \Delta x = \frac{2\pi}{10} \times 2 = \frac{2\pi}{5} = 72^\circ$
52. (a,b,c) $I = 2\pi n^2 a^2 \rho v \Rightarrow I \propto n^2 a^2 v$
53. (a) comparing the given equation with $y = a \sin(\omega t - kx)$
 $\omega = 200, k = 1$ so $v = \frac{\omega}{k} = 200 \text{ m/s}$
54. (a) $v = \frac{\omega}{k} = \frac{2\pi}{2\pi} = 1 \text{ m/s}$
55. (b) By comparing it with standard equation
 $y = a \cos(\omega t - kx) \Rightarrow k = \frac{2\pi}{\lambda} = \pi \Rightarrow \lambda = 2 \text{ cm}$
56. (d) Compare the given equation with
 $y = a \sin(\omega t + kx) \Rightarrow \omega = 2\pi n = 100 \Rightarrow n = \frac{50}{\pi} \text{ Hz}$
 $k = \frac{2\pi}{\lambda} = 1 \Rightarrow \lambda = 2\pi$ and $v = \omega/k = 100 \text{ m/s}$
 Since '+' is given between t terms and x term, so wave is travelling in negative x -direction.
57. (b) Given $A\omega = 4v \Rightarrow A2\pi n = 4n\lambda \Rightarrow \lambda = \frac{\pi A}{2}$
58. (d) $v = \frac{\omega}{k} = \frac{100}{1/10} = 1000 \text{ m/s}$
59. (c) A wave travelling in positive x -direction may be represented as
 $y = A \sin \frac{2\pi}{\lambda} (vt - x)$. On putting values
 $y = 0.2 \sin \frac{2\pi}{60} (360t - x) \Rightarrow y = 0.2 \sin 2\pi \left(6t - \frac{x}{60} \right)$
60. (a) $v = \frac{\omega}{k} = \frac{7\pi}{0.4\pi} = 17.5 \text{ m/s}$
61. (b) $\frac{I_1}{I_2} = \frac{a_1^2}{a_2^2} \Rightarrow \frac{I_1}{I_2} = \frac{25}{100} = \frac{1}{4}$
62. (a) From the given equation $k = \frac{2\pi}{\lambda} = \text{Co-efficient of } x$
 $= \frac{\pi}{4} \Rightarrow \lambda = 8 \text{ m}$
63. (d) $y = 4 \sin 2\pi \left(\frac{t}{0.02} - \frac{x}{100} \right)$
 Comparing this equation with $y = a \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right)$
 $v = \frac{\text{Co-efficient of } t}{\text{Co-efficient of } x} = \frac{1/0.02}{1/100}$
64. (a) Comparing the given equation with $y = a \sin(\omega t - kx)$
 We get $\omega = 3000\pi \Rightarrow n = \frac{\omega}{2\pi} = 1500 \text{ Hz}$
 and $k = \frac{2\pi}{\lambda} = 12\pi \Rightarrow \lambda = \frac{1}{6} \text{ m}$
 So, $v = n\lambda \Rightarrow v = 1500 \times \frac{1}{6} = 250 \text{ m/s}$
65. (b) Positive sign in the argument of $\sin$ indicating that wave is travelling in negative x -direction.
66. (b) Comparing the given equation with $y = a \cos(\omega t - kx)$
 $a = 25, \omega = 2\pi n = 2\pi \Rightarrow n = 1 \text{ Hz}$
67. (b) $v = \frac{\omega}{k} = \frac{600}{2} = 300 \text{ m/sec.}$
68. (b) $v = \frac{\text{Co-efficient of } t}{\text{Co-efficient of } x} = \frac{\omega}{k} = \frac{100}{20} = 5 \text{ m/s.}$
69. (d) Comparing with standard wave equation
 $y = a \sin \frac{2\pi}{\lambda} (vt - x)$, we get, $v = 200 \text{ m/s.}$
70. (b) Phase difference $= \frac{2\pi}{\lambda} \times \text{path difference}$
 $\Rightarrow \frac{\pi}{2} = \frac{2\pi}{\lambda} \times 0.8 \Rightarrow \lambda = 4 \times 0.8 = 3.2 \text{ m}$
 Velocity $v = n\lambda = 120 \times 3.2 = 384 \text{ m/s.}$
71. (a) Comparing the given equation with standard equation
 We get $\omega = 2\pi n = 200\pi \Rightarrow n = 100 \text{ Hz}$
 $k = \frac{20\pi}{17} \Rightarrow \lambda = \frac{2\pi}{k} = \frac{2\pi}{20\pi/17} = 1.7 \text{ m}$
 and $v = \frac{\omega}{k} = \frac{200\pi}{20\pi/17} = 170 \text{ m/s.}$
72. (b) Given, $y = 0.5 \sin(20x - 400t)$
 Comparing with $y = a \sin(\omega t - kx)$
 Gives velocity of wave $v = \frac{\omega}{k} = \frac{400}{20} = 20 \text{ m/s.}$
73. (d) $v = n\lambda \Rightarrow \lambda = 10 \text{ cm}$
 Phase difference $= \frac{2\pi}{\lambda} \times \text{Path difference} = \frac{2\pi}{10} \times 2.5 = \frac{\pi}{2}$
74. (a, c) $v_{\max} = a\omega = \frac{v}{10} = \frac{10}{10} = \text{m/sec}$
 $\Rightarrow a\omega = a \times 2\pi n = 1 \Rightarrow n = \frac{10^3}{2\pi} \quad (\because a = 10^{-3} \text{ m})$
 Since $v = n\lambda \Rightarrow \lambda = \frac{v}{n} = \frac{10}{10^3/2\pi} = 2\pi \times 10^{-2} \text{ m}$
75. (c) Total energy is conserved.
76. (b) $v = \frac{\text{Co-efficient of } t}{\text{Co-efficient of } x} = \frac{1/2}{1/4} = 2 \text{ m/s}$
 Hence $d = vt = 2 \times 8 = 16 \text{ m}$
77. (b) $y_1 = 10^{-6} \sin[100t + (x/50) + 0.5]$

$$y_2 = 10^{-6} \sin \left[100t + \left(\frac{x}{50} \right) + \left(\frac{\pi}{2} \right) \right]$$

Phase difference ϕ

$$= [100t + (x/50) + 1.57] - [100t + (x/50) + 0.5] \\ = 1.07 \text{ radians.}$$

78. (c) Resultant amplitude

$$A_R = 2A \cos \left(\frac{\theta}{2} \right) = 2 \times (2a) \cos \left(\frac{\theta}{2} \right) = 4a \cos \left(\frac{\theta}{2} \right)$$

79. (b) The particle will come after a time $\frac{T}{4}$ to its mean position.

80. (a) Maximum particle velocity $= a\omega = 2 \times 2 = 4$ units.

Interference and Superposition of Waves

1. (b) With path difference $\frac{\lambda}{2}$, waves are out of phase at the point of observation.

2. (d) $A_{\max} = \sqrt{A^2 + A^2} = A\sqrt{2}$, frequency will remain same i.e. ω

3. (a) Phase difference is 2π means constructive interference so resultant amplitude will be maximum.

4. (d) Resultant amplitude

$$A = \sqrt{a^2 + a^2 + 2aa \cos \phi} = \sqrt{4a^2 \cos^2 \left(\frac{\phi}{2} \right)}$$

$$\therefore I \propto A^2 \Rightarrow I \propto 4a^2$$

5. (b) $A^2 = a^2 = a^2 + a^2 + 2a^2 \cos \theta \Rightarrow \cos \theta = -\frac{1}{2} \Rightarrow \theta = \frac{2\pi}{3}$

6. (d) $\lambda = \frac{v}{n} = \frac{350}{350} = 1 \text{ m} = 100 \text{ cm}$

Also path difference (Δx) between the waves at the point of observation is $AP - BP = 25 \text{ cm}$. Hence

$$\Rightarrow \Delta \phi = \frac{2\pi}{\lambda} (\Delta x) = \frac{2\pi}{1} \times \left(\frac{25}{100} \right) = \frac{\pi}{2}$$

$$\Rightarrow A = \sqrt{(a_1)^2 + (a_2)^2} = \sqrt{(0.3)^2 + (0.4)^2} = 0.5 \text{ mm}$$

7. (d) Path difference (Δx) $= 50 \text{ cm} = \frac{1}{2} \text{ m}$

$$\therefore \text{Phase difference } \Delta \phi = \frac{2\pi}{\lambda} \times \Delta x \Rightarrow \phi = \frac{2\pi}{1} \times \frac{1}{2} = \pi$$

$$\text{Total phase difference} = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

$$\Rightarrow A = \sqrt{a^2 + a^2 + 2a^2 \cos(2\pi/3)} = a$$

8. (b,c) Because in general phase velocity = wave velocity. But in case of complex waves (many waves together) phase velocity $\neq$ wave velocity.

$\therefore$ If two waves have same λ, v ; then they have same frequency too

9. (c) If two waves of nearly equal frequency superpose, they

give beats if they both travel in straight line and $I_{\min} = 0$ if they have equal amplitudes.

10. (c) Resultant amplitude $= \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \phi}$

$$= \sqrt{0.3^2 + 0.4^2 + 2 \times 0.3 \times 0.4 \times \cos \frac{\pi}{2}} = 0.5 \text{ cm}$$

11. (a) In the same phase $\phi = 0$ so resultant amplitude $= a_1 + a_2 = 2A + A = 3A$

12. (b) $\frac{I_1}{I_2} = \left(\frac{a_1}{a_2} \right)^2 = \frac{1}{16} \Rightarrow \frac{a_1}{a_2} = \frac{1}{4}$

13. (c) For interference, two waves must have a constant phase relation ship. Equation '1' and '3' and '2' and '4' have a constant phase relationship of $\frac{\pi}{2}$ out of two choices. Only one S_1 emitting '2' and S_2 emitting '4' is given so only (c) option is correct.

14. (d) This is a case of destructive interference.

15. (b) $a_1 = 5, a_2 = 10 \Rightarrow \frac{I_{\max}}{I_{\min}} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2} = \left(\frac{5 + 10}{5 - 10} \right)^2 = \frac{9}{1}$

16. (c) For the given super imposing waves

$$a_1 = 3, a_2 = 4 \text{ and phase difference } \phi = \frac{\pi}{2}$$

$$\Rightarrow A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \pi/2} = \sqrt{(3)^2 + (4)^2} = 5$$

17. (a) Phase difference between the two waves is $\phi = (\omega t - \beta_2) - (\omega t - \beta_1) = (\beta_1 - \beta_2)$

$$\therefore \text{Resultant amplitude } A = \sqrt{A_1^2 + A_2^2 + 2A_1A_2 \cos(\beta_1 - \beta_2)}$$

18. (a) $\frac{I_{\max}}{I_{\min}} = \left(\frac{\frac{a_1}{a_2} + 1}{\frac{a_1}{a_2} - 1} \right)^2 = \left(\frac{2 + 1}{2 - 1} \right)^2 = 9/1$

19. (b) $\frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{\frac{I_1}{I_2}} + 1}{\sqrt{\frac{I_1}{I_2}} - 1} \right)^2 = \left(\frac{\sqrt{\frac{9}{4}} + 1}{\sqrt{\frac{9}{4}} - 1} \right)^2 = \frac{25}{1}$

20. (c) $\frac{I_{\max}}{I_{\min}} = \left(\frac{\frac{a_1}{a_2} + 1}{\frac{a_1}{a_2} - 1} \right)^2 = \left(\frac{\frac{4}{3} + 1}{\frac{4}{3} - 1} \right)^2 = \frac{49}{1}$

21. (a) The resultant amplitude is given by

$$A_R = \sqrt{A^2 + A^2 + 2AA \cos \theta} = \sqrt{2A^2(1 + \cos \theta)} \\ = 2A \cos \theta/2 \quad (\because H \cos \theta = 2 \cos^2 \theta/2)$$

22. (b) $\frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{\frac{I_1}{I_2}} + 1}{\sqrt{\frac{I_1}{I_2}} - 1} \right)^2 = \left(\frac{\sqrt{\frac{9}{1}} + 1}{\sqrt{\frac{9}{1}} - 1} \right)^2 = \frac{4}{1}$

23. (a) Since $\phi = \frac{\pi}{2} \Rightarrow A = \sqrt{a_1^2 + a_2^2} = \sqrt{(4)^2 + (3)^2} = 5$

24. (c) $A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \phi}$

Putting $a_1 = a_2 = a$ and $\phi = \frac{\pi}{3}$, we get $A = \sqrt{3}a$

25. (d) $y = \frac{1}{\sqrt{a}} \sin \omega t \pm \frac{1}{\sqrt{b}} \sin \left(\omega t + \frac{\pi}{2} \right)$

Here phase difference $= \frac{\pi}{2} \therefore$ The resultant amplitude

$$= \sqrt{\left(\frac{1}{\sqrt{a}}\right)^2 + \left(\frac{1}{\sqrt{b}}\right)^2} = \sqrt{\frac{1}{a} + \frac{1}{b}} = \sqrt{\frac{a+b}{ab}}$$

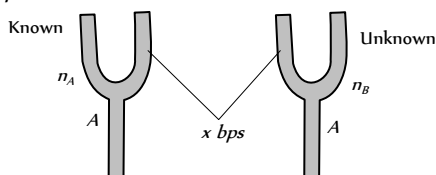
26. (b) Superposition of waves does not alter the frequency of resultant wave and resultant amplitude

$$\Rightarrow a^2 = a^2 + a^2 + 2a^2 \cos \phi = 2a^2(1 + \cos \phi)$$

$$\Rightarrow \cos \phi = -1/2 = \cos 2\pi/3 \therefore \phi = 2\pi/3$$

Beats

1. (c) Suppose two tuning forks are named A and B with frequencies $n_A = 256 \text{ Hz}$ (known), $n_i = ?$ (unknown), and beat frequency $x = 4 \text{ bps}$.



Frequency of unknown tuning fork may be

$$n_B = 256 + 4 = 260 \text{ Hz}$$

$$\text{or } = 256 - 4 = 252 \text{ Hz}$$

It is given that on sounding waxed fork A (fork of frequency 256 Hz) and fork B , number of beats (beat frequency) increases. It means that with decrease in frequency of A , the difference in new frequency of A and the frequency of B has increased. This is possible only when the frequency of A while decreasing is moving away from the frequency of B .

This is possible only if $n_i = 260 \text{ Hz}$.

Alternate method : It is given $n_A = 256 \text{ Hz}$, $n_B = ?$ and $x = 4 \text{ bps}$

Also after loading A (i.e. $n_i \downarrow$), beat frequency (i.e. x) increases ($\uparrow$).

Apply these informations in two possibilities to know the frequency of unknown tuning fork.

$$n_i \downarrow - n_i = x \uparrow \quad \dots (i)$$

$$n_i - n_i \downarrow = x \uparrow \quad \dots (ii)$$

It is obvious that equation (i) is wrong (ii) is correct so

$$n_i = n_i + x = 256 + 4 = 260 \text{ Hz}$$

2. (d)

3. (c)

4. (a) Suppose n_i = known frequency = 100 Hz , $n_i = ?$

$x = 2 = \text{Beat frequency}$, which is decreasing after loading (i.e. $x \downarrow$)

Unknown tuning fork is loaded so $n_i \downarrow$

$$\text{Hence } n_i - n_i \downarrow = x \downarrow \quad \dots (i) \quad \text{--- Wrong}$$

$$n_i \downarrow - n_i = x \downarrow \quad \dots (ii) \quad \text{--- Correct}$$

$$\Rightarrow n_i = n_i + x = 100 + 2 = 102 \text{ Hz}$$

5. (d) n_i = Known frequency = 256 Hz , $n_i = ?$

$x = 2 \text{ bps}$, which is decreasing after loading (i.e. $x \downarrow$) known tuning fork is loaded so $n_i \downarrow$

$$\text{Hence } n_i \downarrow - n_i = x \downarrow \quad \dots (i) \quad \text{--- Correct}$$

$$n_i - n_i \downarrow = x \downarrow \quad \dots (ii) \quad \text{--- Wrong}$$

$$\Rightarrow n_i = n_i - x = 256 - 2 = 254 \text{ Hz}$$

6. (b) n_i = Known frequency = 256 Hz , $n_i = ?$

$x = 4 \text{ bps}$, which is decreasing after loading (i.e. $x \downarrow$) also known tuning fork is loaded so $n_i \downarrow$

$$\text{Hence } n_i \downarrow - n_i = x \downarrow \quad \dots (i) \quad \text{--- Correct}$$

$$n_i - n_i \downarrow = x \downarrow \quad \dots (ii) \quad \text{--- Wrong}$$

$$\Rightarrow n_i = n_i - x = 256 - 4 = 252 \text{ Hz}$$

7. (c) Time interval between two consecutive beats

$$T = \frac{1}{n_1 - n_2} = \frac{1}{260 - 256} = \frac{1}{4} \text{ sec so, } t = \frac{1}{16} = \frac{T}{4} \text{ sec}$$

$$\text{By using time difference} = \frac{T}{2\pi} \times \text{Phase difference}$$

$$\Rightarrow \frac{T}{4} = \frac{T}{2\pi} \times \phi \Rightarrow \phi = \frac{\pi}{2}$$

8. (a) The time interval between successive maximum intensities will

$$\text{be } \frac{1}{n_1 - n_2} = \frac{1}{454 - 450} = \frac{1}{4} \text{ sec.}$$

9. (d) n_i = Known frequency = 341 Hz , $n_i = ?$

$x = 6 \text{ bps}$, which is decreasing (i.e. $x \downarrow$) after loading (from 6 to 1 bps)

Unknown tuning fork is loaded so $n_i \downarrow$

$$\text{Hence } n_i - n_i \downarrow = x \downarrow \quad \dots (i) \quad \text{--- Wrong}$$

$$n_i \downarrow - n_i = x \downarrow \quad \dots (ii) \quad \text{--- Correct}$$

$$\Rightarrow n_i = n_i + x = 341 + 6 = 347 \text{ Hz}$$

10. (b) $T = \frac{1}{258 - 256} = 0.5 \text{ sec}$

11. (c) Suppose n_i = known frequency = 100 Hz , $n_i = ?$

$x = 5 \text{ bps}$, which remains unchanged after loading

Unknown tuning fork is loaded so $n_i \downarrow$

$$\text{Hence } n_i - n_i \downarrow = x \quad \dots (i)$$

$$n_i \downarrow - n_i = x \quad \dots (ii)$$

From equation (i), it is clear that as n_i decreases, beat frequency (i.e. $n_i - (n_i)_-$) can never be x again.

From equation (ii), as $n_i \downarrow$, beat frequency (i.e. $(n_i)_- - n_i$) decreases as long as $(n_i)_-$ remains greater than n_i . If $(n_i)_-$ become lesser than n_i the beat frequency will increase again and will be x . Hence this is correct.

$$\text{So, } n_i = n_i + x = 100 + 5 = 105 \text{ Hz}$$

12. (b) n_i = Known frequency = 256 Hz , $n_i = ?$

$x = 6 \text{ bps}$, which remains the same after loading.

Unknown tuning fork F_i is loaded so $n_i \downarrow$

$$\text{Hence } n_i - n_i \downarrow = x \quad \dots (i) \quad \text{--- Wrong}$$

$$n_1 \downarrow - n_2 = x \quad \dots (ii) \quad \longrightarrow \text{Correct}$$

$$\Rightarrow n_1 = n_2 + x = 256 + 6 = 262 \text{ Hz}$$

13. (a) Probable frequencies of tuning fork be $n + 4$ or $n - 4$

$$\text{Frequency of sonometer wire } n \propto \frac{1}{l}$$

$$\therefore \frac{n+4}{n-4} = \frac{100}{95} \text{ or } 95(n+4) = 100(n-4)$$

$$\text{or } 95n + 380 = 100n - 400 \text{ or } 5n = 780 \text{ or } n = 156$$

14. (c) After filling frequency increases, so n_A decreases ($\downarrow$). Also it is given that beat frequency increases (i.e., $x \uparrow$)

$$\text{Hence } n_1 \downarrow - n_2 = x \uparrow \quad \dots (i) \quad \longrightarrow \text{Correct}$$

$$n_1 - n_2 \uparrow = x \uparrow \quad \dots (ii) \quad \longrightarrow \text{Wrong}$$

$$\Rightarrow n_1 = n_2 + x = 512 + 5 = 517 \text{ Hz}$$

15. (c) Intensity $\propto$ (amplitude)

$$\text{as } A_{\max} = 2a_0 \text{ (} a_0 = \text{amplitude of one source) so } I_{\max} = 4I_0$$

16. (c) Number of beats per second = $n_1 \sim n_2$

$$\omega_1 = 2000\pi = 2\pi n_1 \Rightarrow n_1 = 1000$$

$$\text{and } \omega_2 = 2008\pi = 2\pi n_2 \Rightarrow n_2 = 1004$$

$$\text{Number of beats heard per sec} = 1004 - 1000 = 4$$

17. (c) The tuning fork whose frequency is being tested produces 2 beats with oscillator at 514 Hz, therefore, frequency of tuning fork may either be 512 or 516. With oscillator frequency 510 it gives 6 beats/sec, therefore frequency of tuning fork may be either 516 or 504.

Therefore, the actual frequency is 516 Hz which gives 2 beats/sec with 514 Hz and 6 beats/sec with 510 Hz.

18. (b) If suppose n_1 = frequency of string = $\frac{1}{2l} \sqrt{\frac{T}{m}}$

$$n_1 = \text{Frequency of tuning fork} = 480 \text{ Hz}$$

$$x = \text{Beats heard per second} = 10$$

as tension T increases, so n_1 increases ($\uparrow$)

Also it is given that number of beats per sec decreases (i.e., $x \downarrow$)

$$\text{Hence } n_1 \uparrow - n_2 = x \downarrow \quad \dots (i) \quad \longrightarrow \text{Wrong}$$

$$n_1 - n_2 \uparrow = x \downarrow \quad \dots (ii) \quad \longrightarrow \text{Correct}$$

$$\Rightarrow n_1 = n_2 + x = 480 - 10 = 470 \text{ Hz}$$

19. (c) It is given that

$$n_1 = \text{Unknown frequency} = ?$$

$$n_2 = \text{Known frequency} = 256 \text{ Hz}$$

$$x = 3 \text{ bps, which remains same after loading}$$

Unknown tuning fork A is loaded so $n_1 \downarrow$

$$\text{Hence } n_1 \downarrow - n_2 = x \quad \dots (i) \quad \longrightarrow \text{Correct}$$

$$n_1 - n_2 \downarrow = x \quad \dots (ii) \quad \longrightarrow \text{Wrong}$$

$$\Rightarrow n_1 = n_2 + x = 256 + 3 = 259 \text{ Hz}$$

20. (a) Frequency of the source = $100 \pm 5 = 105 \text{ Hz}$ or 95 Hz .

Second harmonic of the source = 210 Hz or 190 Hz.

As the second harmonic gives 5 beats/sec with sound of frequency 205 Hz, the second harmonic should be 210 Hz.

$$\Rightarrow \text{Frequency of the source} = 105 \text{ Hz}$$

21. (d) For producing beats, there must be small difference in frequency.

22. (c) n_1 = Known frequency = 256 Hz, n_2 = ?

$x = 4$ beats per sec which is decreasing (4 bps to $\frac{5}{2}$ bps) after loading (i.e., $x \downarrow$)

Unknown tuning fork B, is loaded so $n_2 \downarrow$

$$\text{Hence } n_1 - n_2 \downarrow = x \downarrow \quad \dots (i) \quad \longrightarrow \text{Wrong}$$

$$n_1 \downarrow - n_2 = x \downarrow \quad \dots (ii) \quad \longrightarrow \text{Correct}$$

$$\Rightarrow n_1 = n_2 + x = 256 + 4 = 260 \text{ Hz}$$

23. (d) $n_1 \downarrow - n_2 = x \uparrow \quad \dots (i) \quad \longrightarrow \text{Wrong}$

$$n_1 - n_2 \downarrow = x \uparrow \quad \dots (ii) \quad \longrightarrow \text{Correct}$$

$$\Rightarrow n_1 = n_2 + x = 200 + 5 = 205 \text{ Hz}$$

24. (c) $n_1 - n_2 \downarrow = x$ (same) $\dots (i) \quad \longrightarrow \text{Wrong}$

$$n_1 \downarrow - n_2 = x \text{ (same)} \quad \dots (ii) \quad \longrightarrow \text{Correct}$$

$$\Rightarrow n_1 = n_2 + x = 320 + 4 = 324 \text{ Hz}$$

25. (c) Beat period $T = \frac{1}{n_1 \sim n_2} = \frac{1}{384 - 380} = \frac{1}{4} \text{ sec}$. Hence minimum time interval between maxima and minima $t = \frac{T}{2} = \frac{1}{8} \text{ sec}$.

26. (d) $\frac{I_{\max}}{I_{\min}} = \left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2 = \frac{(5+3)^2}{(5-3)^2} = \frac{16}{1}$

27. (a) $n_1 = \frac{v}{\lambda_1} = \frac{v}{0.50}$ and $n_2 = \frac{v}{\lambda_2} = \frac{v}{0.51}$

$$\Delta n = n_1 - n_2 = v \left[\frac{1}{0.05} - \frac{1}{0.51} \right] = 12$$

$$\Rightarrow v = \frac{12 \times 0.51 \times 0.50}{0.01} = 306 \text{ m/s}$$

28. (c) $n_1 = \frac{316}{2\pi}$ and $n_2 = \frac{310}{2\pi}$ Number of beats heard per second = $n_1 - n_2 = \frac{316}{2\pi} - \frac{310}{2\pi} = \frac{3}{\pi}$

29. (b) Beat frequency = $\frac{2}{0.4} = 5 \text{ Hz}$

30. (a) Since source of frequency x gives 8 beats per second with frequency 250 Hz, its possible frequency are 258 or 242. As source of frequency x gives 12 beats per second with a frequency 270 Hz, its possible frequencies 282 or 258 Hz. The only possible frequency of x which gives 8 beats with frequency 250 Hz also 12 beats per second with 258 Hz.

31. (c) $n_1 = \frac{1000\pi}{2\pi} = 500 \text{ Hz}$ and $n_2 = \frac{998\pi}{2\pi} = 499 \text{ Hz}$

$$\text{Hence beat frequency} = n_1 - n_2 = 1$$

32. (a) $v_0 = 332 \text{ m/s}$. Velocity sound at t is $v_t = (v_0 + 0.61t)$

$$\Rightarrow v_{20} = v_0 + 0.61 \times 20 = 344.2 \text{ m/s}$$

$$\Rightarrow \Delta n = v_{20} \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right) = 344.2 \left(\frac{100}{50} - \frac{100}{51} \right) = 14$$

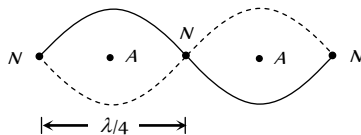
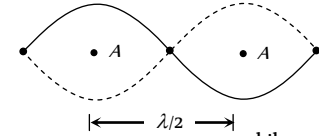
33. (a) Persistence of hearing is 10 sec.
34. (a)
35. (d) $n_1 = ?$, $n_2 = 384 \text{ Hz}$
 $x = 6 \text{ bps}$, which is decreasing (from 6 to 4) i.e. $x \downarrow$
 Tuning fork A is loaded so $n_1 \downarrow$
 Hence $n_1 \downarrow - n_2 = x \downarrow$ —————> Correct
 $n_1 - n_2 = x \downarrow$ —————> Wrong
 $\Rightarrow n_1 = n_2 + x = 384 + 6 = 390 \text{ Hz}$
36. (b) For hearing beats, difference of frequencies should be approximately 10 Hz.
37. (a) $n \propto \frac{1}{l} \Rightarrow n_1 l_1 = n_2 l_2 \Rightarrow (n+4)49 = (n-4)50 \Rightarrow n = 396$
38. (a) No of beats, $x = \Delta n = \frac{30}{3} = 10 \text{ Hz}$
 $\Rightarrow$ Also $\Delta n = v \left[\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right] = v \left[\frac{1}{5} - \frac{1}{6} \right] = 10 \Rightarrow v = 300 \text{ m/s}$
39. (a) $\Delta n = v \left[\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right] = 396 \left[\frac{1}{0.99} - \frac{1}{1} \right] = 3.96 \approx 4$.
40. (b) n_1 = Known frequency = 288 cps, $n_2 = ?$
 $x = 4 \text{ bps}$, which is decreasing (from 4 to 2) after loading i.e. $x \downarrow$
 Unknown fork is loaded so $n_2 \downarrow$
 Hence $n_1 - n_2 \downarrow = x \downarrow$ —————> Wrong
 $n_1 \downarrow - n_2 \downarrow = x \downarrow$ —————> Correct
 $\Rightarrow n_2 = n_1 + x = 288 + 4 = 292 \text{ Hz}$
41. (a) Frequency = $\frac{\text{Number of beats}}{\text{Time}} = \frac{2}{0.04} = 50 \text{ Hz}$
42. (c) No. of beats = frequency difference = $\frac{4}{0.25} = 16$
43. (d) Suppose n_p = frequency of piano = ? ($n_p \propto \sqrt{T}$)
 n_f = Frequency of tuning fork = 256 Hz
 x = Beat frequency = 5 bps, which is decreasing (5 $\rightarrow$ 2) after clanging the tension of piano wire
 Also, tension of piano wire is increasing so $n_p \downarrow$
 Hence $n_f \uparrow - n_p = x \downarrow$ —————> Wrong
 $n_f - n_p \uparrow = x \downarrow$ —————> Correct
 $\Rightarrow n_p = n_f - x = 256 - 5 = 251 \text{ Hz}$
44. (b) With temperature rise frequency of tuning fork decreases. Because, the elastic properties are modified when temperature is changed
 also, $n_t = n_0(1 - 0.00011t)$
 where n_t = frequency at $t^\circ\text{C}$, n_0 = frequency at 0°C
45. (a) $n_x = 300 \text{ Hz}$, $n_y = ?$
 x = beat frequency = 4 Hz, which is decreasing (4 $\rightarrow$ 2) after increasing the tension of the string y .
 Also tension of wire y increasing so $n_y \uparrow$ ($\because n \propto \sqrt{T}$)
 Hence $n_x - n_y \uparrow = x \downarrow$ —————> Correct

$$n_y \uparrow - n_x = x \downarrow \longrightarrow \text{Wrong}$$

$$\Rightarrow n_y = n_x - x = 300 - 4 = 296 \text{ Hz}$$

46. (c) Let n be the frequency of fork C then
 $n_A = n + \frac{3n}{100} = \frac{103n}{100}$ and $n_B = n - \frac{2n}{100} = \frac{98n}{100}$
 but $n_A - n_B = 5 \Rightarrow \frac{5n}{100} = 5 \Rightarrow n = 100 \text{ Hz}$
 $\therefore n_A = \frac{(103)(100)}{100} = 103 \text{ Hz}$
47. (a)
48. (b) From the given equations of progressive waves $\omega_1 = 500\pi$ and $\omega_2 = 506\pi \therefore n_1 = 250$ and $n_2 = 253$
 So beat frequency = $n_2 - n_1 = 253 - 250 = 3 \text{ beats per sec}$
 $\therefore$ Number of beats per min = 180.
49. (b)
50. (b) Frequency = $\frac{360}{60} \times 60 = 360 \text{ Hz}$.
51. (b) $v = n\lambda \Rightarrow \lambda = \frac{v}{n} = \frac{340}{170} \Rightarrow \lambda = 2$
 Distance separating the position of minimum intensity = $\frac{\lambda}{2} = \frac{2}{2} = 1 \text{ m}$

Stationary Waves

1. (c) The distance between the nearest node and antinode in a stationary wave is $\frac{\lambda}{4}$
- 
2. (c) At nodes pressure change (strain) is maximum
3. (c) Both the sides of a node, two antinodes are present with separation $\frac{\lambda}{2}$
- So phase difference between them $\phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{2} = \pi$
- 
4. (c) Progressive wave propagate energy while no propagation of energy takes place in stationary waves.
5. (b)
6. (a) Comparing given equation with standard equation
 $y = 2a \sin \frac{2\pi x}{\lambda} \cos \frac{2\pi vt}{\lambda}$ gives us $\frac{2\pi}{\lambda} = \frac{\pi}{15} \Rightarrow \lambda = 30$
 Distance between nearest node and antinodes = $\frac{\lambda}{4} = \frac{30}{4} = 7.5$

7. (b) On comparing the given equation with standard equation

$$y = 2a \sin \frac{2\pi x}{\lambda} \cos \frac{2\pi vt}{\lambda} \Rightarrow \frac{2\pi x}{\lambda} = \frac{\pi x}{3} \Rightarrow \lambda = 6$$

$$\text{Separation between two adjacent nodes} = \frac{\lambda}{2} = 3 \text{ cm}$$

8. (d)

9. (a) On comparing the given equation with standard equation

$$y = 2a \sin \frac{2\pi x}{\lambda} \cos \frac{2\pi vt}{\lambda}$$

$$\text{We get } \frac{2\pi}{\lambda} = \frac{\pi}{20} \Rightarrow \lambda = 40$$

Separation between two consecutive nodes =

$$\frac{\lambda}{2} = \frac{40}{2} = 20 \text{ cm}$$

10. (a)

11. (b) Since the point $x = 0$ is a node and reflection is taking place from point $x = 0$. This means that reflection must be taking place from the fixed end and hence the reflected ray must

suffer an additional phase change of π or a path change of $\frac{\lambda}{2}$.

$$\text{So, if } y_{\text{incident}} = a \cos(kx - \omega t)$$

$$\Rightarrow y_{\text{reflected}} = a \cos(-kx - \omega t + \pi) = -a \cos(\omega t + kx)$$

12. (d) Particles have kinetic energy maximum at mean position.

13. (b) On comparing the given equation with standard equation

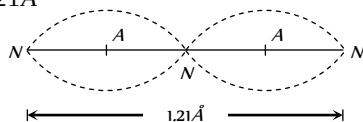
$$\frac{2\pi}{\lambda} = 5 \Rightarrow \lambda = \frac{6.28}{5} = 1.256 \text{ m}$$

14. (d)

15. (d)

16. (a,b,c) Standing waves can be produced only when two similar type of waves (same frequency and speed, but amplitude may be different) travel in opposite directions.

17. (a) $\lambda = 1.21 \text{ Å}$



18. (d) $\frac{\lambda}{4} = 20 \Rightarrow \lambda = 80 \text{ cm}$, also $\Delta\phi = \frac{\lambda}{2\pi} \cdot \Delta x$

$$\Rightarrow \Delta\phi = \frac{60}{80} \times 2\pi = \frac{3\pi}{2}$$

19. (a) Required distance $= \frac{\lambda}{4} = \frac{v/n}{4} = \frac{1200}{4 \times 300} = 1 \text{ m}$

20. (a) Waves A and B satisfied the conditions required for a standing wave.

21. (a) By comparing given equation with $y = a \sin(\omega t) \cos kx$

$$\Rightarrow v = \frac{\omega}{k} = \frac{100}{0.01} = 10^4 \text{ m/s}$$

22. (b) At fixed end node is formed and distance between two consecutive nodes $\frac{\lambda}{2} = 10 \text{ cm} \Rightarrow \lambda = 20 \text{ cm}$

$$\Rightarrow v = n\lambda = 20 \text{ m/sec}$$

23. (c) $a \cos(kx + \omega t)$

$$\text{hence } y_{\text{reflected}} = a \cos(-kx + \omega t + \pi) = -a \cos(kx - \omega t)$$

24. (b) Distance between the consecutive node $= \frac{\lambda}{2}$,

$$\text{but } \lambda = \frac{v}{n} = \frac{20}{n} \text{ so } \frac{\lambda}{2} = \frac{10}{n}$$

25. (a) Energy is not carried by stationary waves

26. (c) On comparing the given equation with standard equation

$$\Rightarrow \frac{2\pi}{\lambda} = \frac{\pi}{3} \Rightarrow \lambda = 6 \text{ cm}. \text{ Hence, distance between two consecutive nodes } \Rightarrow \lambda = 3 \text{ cm}$$

27. (d) Minimum time interval between two instants when the string is

$$\text{flat} = \frac{T}{2} = 0.5 \text{ sec} \Rightarrow T = 1 \text{ sec}$$

$$\text{Hence } \lambda = v \times T = 10 \times 1 = 10 \text{ m}.$$

28. (c)

29. (b) Distance between two nodes $= \frac{\lambda}{2} = \frac{v}{2n} = \frac{16}{2n} = \frac{8}{n}$

30. (d)

31. (b) In stationary wave all the particles in one particular segment (i.e., between two nodes) vibrates in the same phase.

32. (a) If $y_{\text{incident}} = a \sin(\omega t - kx)$ and $y_{\text{stationary}} = a \sin(\omega t) \cos kx$

then it is clear that frequency of both is same (ω)

33. (b)

34. (a) On comparing the given equation with standard equation

$$\frac{2\pi}{\lambda} = \frac{\pi}{4} \Rightarrow \lambda = 8$$

$$\text{Hence distance between two consecutive nodes } \frac{\lambda}{2} = 4$$

35. (a)

36. (a) Waves $Z_1 = A \sin(kx - \omega t)$ is travelling towards positive x -direction.

Wave $Z_2 = A \sin(kx + \omega t)$, is travelling towards negative x -direction.

Wave $Z_3 = A \sin(ky - \omega t)$ is travelling towards positive y direction.

Since waves Z_1 and Z_2 are travelling along the same line so they will produce stationary wave.

37. (a) When two waves of equal frequency and travelling in opposite direction superimpose, then the stationary wave is produced. Hence Z_1 and Z_2 produces stationary wave.

38. (d) The distance between adjacent nodes $x = \frac{\lambda}{2}$

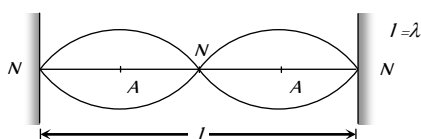
$$\text{Also } k = \frac{2\pi}{\lambda}. \text{ Hence } x = \frac{\pi}{k}.$$

39. (d) $y = 5 \sin\left(\frac{2\pi x}{3}\right) \cos 20\pi t$, comparing with equation

$$y = 2a \sin \frac{2\pi x}{\lambda} \cos \frac{2\pi vt}{\lambda} \Rightarrow \lambda = 3, \text{ distance between two adjacent nodes} = \lambda/2 = 1.5 \text{ cm}.$$

Vibration of String

1. (c)



2. (d) $n \propto \frac{1}{l} \Rightarrow \frac{n_2}{n_1} = \frac{l_1}{l_2} \Rightarrow n_2 = \frac{l_1}{l_2} n_1 = \frac{1 \times 256}{1/4} = 1024 \text{ Hz}$

3. (c) String vibrates in five segments so $\frac{5}{2} \lambda = l \Rightarrow \lambda = \frac{2l}{5}$

Hence $n = \frac{v}{\lambda} = 5 \times \frac{v}{2l} = 5 \times \frac{20}{2 \times 10} = 5 \text{ Hz}$

4. (c) Here $\frac{\lambda}{2} = 5.0 \text{ cm} \Rightarrow \lambda = 10 \text{ cm}$

Hence $n = \frac{v}{\lambda} = \frac{200}{10} = 20 \text{ Hz}$

5. (c)

6. (b) As we know plucking distance from one end $= \frac{l}{2p}$

$\Rightarrow 25 = \frac{100}{2p} \Rightarrow p = 2$. Hence frequency of vibration

$n = \frac{p}{2l} \sqrt{\frac{T}{m}} = \frac{2}{2 \times 1} \sqrt{\frac{20}{5 \times 10^{-4}}} = 200 \text{ Hz}.$

7. (b) To produce 5 beats/sec. Frequency of one wire should be increase up to 505 Hz. i.e. increment of 1% in basic frequency.

$n \propto \sqrt{T}$ or $T \propto n^2 \Rightarrow \frac{\Delta T}{T} = 2 \frac{\Delta n}{n}$

$\Rightarrow$ percentage change in Tension = 2(1%) = 2%

8. (d) $y = 0.021 \sin(\kappa + 30t) \Rightarrow v = \frac{\omega}{k} = \frac{30}{1} = 30 \text{ m/s}.$

Using, $v = \sqrt{\frac{T}{m}} \Rightarrow 30 = \sqrt{\frac{T}{1.3 \times 10^{-4}}} \Rightarrow T = 0.117 \text{ N}$

9. (a) $n \propto \sqrt{T}$

10. (c) $n \propto \sqrt{T}$

11. (d) $n \propto \sqrt{T}$

$\Rightarrow n_1 : n_2 : n_3 : n_4 = \sqrt{1} : \sqrt{4} : \sqrt{9} : \sqrt{16} = 1 : 2 : 3 : 4$

12. (c) Let the frequency of tuning fork be N

As the frequency of vibration string $\propto \frac{1}{\text{length of string}}$

For sonometer wire of length 20 cm, frequency must be $(N + 5)$ and that for the sonometer wire of length 21 cm, the frequency must be $(N - 5)$ as in each case the tuning fork produces 5 beats/sec with sonometer wire

Hence $n_1 l_1 = n_2 l_2 \Rightarrow (N + 5) \times 20 = (N - 5) \times 21$

$\Rightarrow N = 205 \text{ Hz}.$

13. (c) $\lambda = \frac{2l}{p}$ (p = Number of loops)

14. (a) String will vibrate in 7 loops so it will have 8 nodes 7 antinodes.

Number of harmonics = Number of loops = Number of antinodes $\Rightarrow$ Number of antinodes = 7

Hence number of nodes = Number of antinodes + 1
 $= 7 + 1 = 8$

15. (a)

16. (d) $n \propto \frac{1}{l} \sqrt{T} \Rightarrow \frac{n'}{n} = \sqrt{\frac{T'}{T}} \times \frac{l}{l'} = \sqrt{4} \times \frac{1}{2} = 1 \Rightarrow n' = n$

17. (a) Sonometer is used to produce resonance of sound source with stretched vibrating string.

18. (a) $n \propto \frac{1}{l} \Rightarrow \frac{l_2}{l_1} = \frac{n_1}{n_2} \Rightarrow l_2 = l_1 \left(\frac{n_1}{n_2} \right) = 50 \times \frac{270}{1000} = 13.5 \text{ cm}$

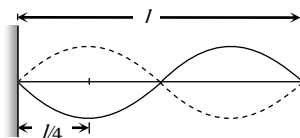
19. (c) $n \propto \sqrt{T} \Rightarrow \frac{n_1}{n_2} = \sqrt{\frac{T_1}{T_2}} \Rightarrow \frac{n}{2n} = \sqrt{\frac{10}{T_2}} \Rightarrow T_2 = 40 \text{ N}$

20. (b) $n \propto \sqrt{T}$

21. (d) $n = \frac{1}{2l} \sqrt{\frac{T}{m}} \Rightarrow n \propto \sqrt{T}$
 $\Rightarrow \frac{T_2}{T_1} = \left(\frac{n_2}{n_1} \right)^2 \left(\frac{l_2}{l_1} \right)^2 = (2)^2 \left(\frac{3}{4} \right)^2 = \frac{9}{4}$

22. (c) $v = \sqrt{\frac{T}{m}} \Rightarrow v = \sqrt{\frac{60.5}{(0.035/7)}} = 110 \text{ m/s}$

23. (a) Second harmonic means 2 loops in a total length



Hence plucking distance from one end $= \frac{l}{2p} = \frac{l}{2 \times 2} = \frac{l}{4}.$

24. (b) $v = \sqrt{\frac{T}{m}} = \sqrt{\frac{T}{\pi^2 \rho}}$

$v \propto \frac{\sqrt{T}}{r} \Rightarrow \frac{v_A}{v_B} = \sqrt{\frac{T_A}{T_B}} \cdot \frac{r_B}{r_A} = \sqrt{\frac{1}{2}} \cdot \frac{1}{2} = \frac{1}{2\sqrt{2}}$

25. (a) The frequency of vibration of a string $n = \frac{p}{2l} \sqrt{\frac{T}{m}}$

Also number of loops = Number of antinodes.

Hence, with 5 antinodes and hanging mass of 9 kg.

We have $p = 5$ and $T = 9g \Rightarrow n_1 = \frac{5}{2l} \sqrt{\frac{9g}{m}}$

With 3 antinodes and hanging mass M

We have $p = 3$ and $T = Mg \Rightarrow n_2 = \frac{3}{2l} \sqrt{\frac{Mg}{m}}$

$\therefore n_1 = n_2 \Rightarrow \frac{5}{2l} \sqrt{\frac{9g}{m}} = \frac{3}{2l} \sqrt{\frac{Mg}{m}} \Rightarrow M = 25 \text{ kg}.$

26. (b) $n \propto \frac{\sqrt{T}}{l} \Rightarrow l \propto \sqrt{T}$ (As $n = \text{constant}$)

$$\Rightarrow \frac{l_2}{l_1} = \sqrt{\frac{T_2}{T_1}} = l_1 \sqrt{\frac{169}{100}} \Rightarrow l_2 = 1.3l_1 = l_1 + 30\% \text{ of } l_1$$

27. (b) $n_1 l_1 = n_2 l_2 \Rightarrow 250 \times 0.6 = n_2 \times 0.4 \Rightarrow n_2 = 375 \text{ Hz}$

28. (b) In fundamental mode of vibration wavelength is maximum $\Rightarrow$
 $l = \frac{\lambda}{2} = 40 \text{ cm} \Rightarrow \lambda = 80 \text{ cm}$

29. (c) $n_1 l_1 = n_2 l_2 \Rightarrow 800 \times 50 = 1000 \times l_2 \Rightarrow l_2 = 40 \text{ cm}$

30. (c) $n \propto \sqrt{T} \Rightarrow \frac{\Delta n}{n} = \frac{\Delta T}{2T}$

If tension increases by 2%, then frequency must increase by 1%.

If initial frequency $n_1 = n$ then final frequency $n_2 - n = 5$

$$\Rightarrow \frac{101}{100} n - n = 5 \Rightarrow n = 500 \text{ Hz.}$$

Short trick : If you can remember then apply following formula to solve such type of problems.

Initial frequency of each wire (n)

$$= \frac{(\text{Number of beats heard per sec}) \times 200}{(\text{per centage change in tension of the wire})}$$

$$\text{Here } n = \frac{5 \times 200}{2} = 500 \text{ Hz}$$

31. (b) First overtone of string A = Second overtone of string B.
 $\Rightarrow$ Second harmonic of A = Third harmonic of B

$$\Rightarrow n_2 = n_3 \Rightarrow [2(n_1)]_A = [3(n_1)]_B \quad (\because n_1 = \frac{1}{2l} \sqrt{\frac{T}{\pi^2 \rho}})$$

$$\Rightarrow 2 \left[\frac{1}{2l_A r_A} \sqrt{\frac{T}{\pi \rho}} \right] = 3 \left[\frac{1}{2l_B r_B} \sqrt{\frac{T}{\pi \rho}} \right]$$

$$\frac{l_A}{l_B} = \frac{2}{3} \frac{r_B}{r_A} \Rightarrow \frac{l_A}{l_B} = \frac{2}{3} \times \frac{r_B}{(2r_B)} = \frac{1}{3}$$

32. (a) Fundamental frequency in case of string is

$$n = \frac{1}{2l} \sqrt{\frac{T}{m}} \Rightarrow n \propto \frac{\sqrt{T}}{l} \Rightarrow \frac{n'}{n} = \sqrt{\frac{T'}{T}} \times \frac{l}{l'}$$

$$\text{putting } T' = T + 0.44T = \frac{144}{100} T \text{ and } l' = l - 0.4l = \frac{3}{5} l$$

$$\text{We get } \frac{n'}{n} = \frac{2}{1}.$$

33. (d) Frequency in a stretched string is given by

$$n = \frac{1}{2l} \sqrt{\frac{T}{\pi^2 \rho}} = \frac{1}{l} \sqrt{\frac{T}{\pi d^2 \rho}} \quad (d = \text{Diameter of string})$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{l_2}{l_1} \sqrt{\frac{T_1}{T_2}} \times \left(\frac{d_2}{d_1} \right)^2 \times \left(\frac{\rho_2}{\rho_1} \right)$$

$$= \frac{35}{36} \sqrt{\frac{8}{1} \times \left(\frac{1}{4} \right)^2 \times \frac{2}{1}} = \frac{35}{36} \Rightarrow n_2 = \frac{36}{35} \times 360 = 370$$

$$\text{Hence beat frequency} = n_2 - n_1 = 10$$

34. (b) Frequency of first overtone or second harmonic (n)
 $= 320 \text{ Hz}$. So, frequency of first harmonic

$$n_1 = \frac{n_2}{2} = \frac{320}{2} = 160 \text{ Hz}$$

35. (d) Similar to Q. 30

Initial frequency of each wire (n)

$$= \frac{(\text{Number of beats heard per sec}) \times 200}{(\text{per centage change in tension of the wire})}$$

$$= \frac{(3/2) \times 200}{1} = 300 \text{ sec}^{-1}$$

36. (c) $n \propto \frac{1}{l} \Rightarrow \frac{\Delta n}{n} = -\frac{\Delta l}{l}$

If length is decreased by 2% then frequency increases by 2%

$$\text{i.e., } \frac{n_2 - n_1}{n_1} = \frac{2}{100}$$

$$\Rightarrow n_2 - n_1 = \frac{2}{100} \times n_1 = \frac{2}{100} \times 392 = 7.8 \approx 8.$$

37. (d) Observer receives sound waves (music) which are longitudinal progressive waves.

38. (a) Because both tuning fork and string are in resonance condition.

39. (d) $n = \frac{1}{2l} \sqrt{\frac{T}{m}} \Rightarrow \frac{n_1}{n_2} = \frac{l_2}{l_1} \sqrt{\frac{T_1}{T_2}} = \frac{1}{4} \sqrt{\frac{1}{4}} = \frac{1}{8}$

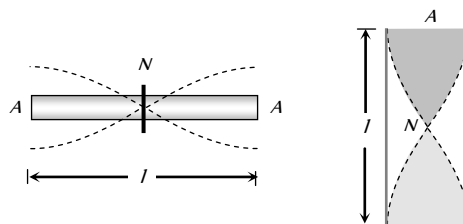
$$\Rightarrow n_2 = 8n_1 = 8 \times 200 = 1600 \text{ Hz}$$

40. (b) $n = \frac{1}{2l} \sqrt{\frac{T}{m}} \Rightarrow n_1 l_1 = n_2 l_2 = n_3 l_3 = k$

$$l_1 + l_2 + l_3 = l \Rightarrow \frac{k}{n_1} + \frac{k}{n_2} + \frac{k}{n_3} = \frac{k}{n}$$

$$\Rightarrow \frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3} + \dots$$

41. (a) If a rod clamped at middle, then it vibrates with similar fashion as open organ pipe vibrates as shown.



Hence, fundamental frequency of vibrating rod is given by

$$n_1 = \frac{v}{2l} \Rightarrow 2.53 = \frac{v}{4 \times 1} \Rightarrow v = 5.06 \text{ km/sec.}$$

42. (a) Change in amplitude does not produce change in frequency,
 $\left(n = \frac{1}{2l} \sqrt{\frac{T}{\pi^2 \rho}} \right).$

43. (d) Mass per unit length $m = \frac{2 \times 10^{-4}}{0.5} \text{ kg/m} = 4 \times 10^{-4} \text{ kg/m}$

Frequency of 2nd harmonic $n_2 = 2n_1$

$$= 2 \times \frac{1}{2l} \sqrt{\frac{T}{m}} = \frac{1}{0.5} \sqrt{\frac{20}{4 \times 10^{-4}}} = 447.2 \text{ Hz}$$

44. (d) $n = \frac{1}{2l} \sqrt{\frac{T}{m}} \Rightarrow n \propto \sqrt{T}$ For octave, $n' = 2n$

$$\Rightarrow \frac{n'}{n} = \sqrt{\frac{T'}{T}} = 2 \Rightarrow T' = 4T = 16 \text{ kg} \cdot \text{wt}$$

$$45. \quad (d) \quad \text{Fundamental frequency } n = \frac{1}{2l} \sqrt{\frac{T}{\pi^2 \rho}}$$

where m = Mass per unit length of wire

$$\Rightarrow n \propto \frac{1}{l} \Rightarrow \frac{n_1}{n_2} = \frac{r_2}{r_1} \times \frac{l_2}{l_1} = \frac{r}{2r} \times \frac{2L}{L} = 1$$

$$46. \quad (c) \quad n = \frac{1}{2l} \sqrt{\frac{T}{\pi^2 \rho}} \propto \sqrt{\frac{T}{r^2 \rho}}$$

$$\Rightarrow \frac{n_1}{n_2} = \sqrt{\left(\frac{T_1}{T_2}\right) \left(\frac{r_2}{r_1}\right)^2 \left(\frac{\rho_2}{\rho_1}\right)} = \sqrt{\left(\frac{1}{2}\right) \left(\frac{2}{1}\right)^2 \left(\frac{1}{2}\right)} = 1$$

$$\therefore n_1 = n_2$$

$$47. \quad (a) \quad n = \frac{p}{2l} \sqrt{\frac{T}{m}} \propto \sqrt{T} \Rightarrow \frac{n_1}{n_2} = \sqrt{\frac{T_1}{T_2}}$$

$$\Rightarrow \frac{260}{n_2} = \sqrt{\frac{50.7g}{(50.7 - 0.0075 \times 10^3)g}} \Rightarrow n_2 \approx 240$$

$$48. \quad (b) \quad \text{Given equation of stationary wave is}$$

$$y = \sin 2\pi x \cos 2\pi t, \text{ comparing it with standard equation}$$

$$y = 2A \sin \frac{2\pi x}{\lambda} \cos \frac{2\pi t}{\lambda}$$

$$\text{We have } \frac{2\pi x}{\lambda} = 2\pi x \Rightarrow \lambda = 1 \text{ m}$$

$$\text{Minimum distance of string (first mode) } L_{\min} = \frac{\lambda}{2} = \frac{1}{2} \text{ m}$$

$$49. \quad (d) \quad n = \frac{1}{2l} \sqrt{\frac{T}{\pi^2 \rho}} \Rightarrow n \propto \frac{\sqrt{T}}{l} \Rightarrow \frac{n_1}{n_2} = \sqrt{\frac{T_1}{T_2}} \times \frac{l_2}{l_1} \times \frac{r_2}{r_1}$$

$$= \sqrt{\frac{T}{3T}} \times \frac{3l}{l} \times \frac{2r}{r} = 3\sqrt{3} \Rightarrow n_2 = \frac{n}{3\sqrt{3}}$$

$$50. \quad (c) \quad \text{For string } \lambda = \frac{2l}{p}$$

where p = No. of loops = Order of vibration

Hence for forth mode $p = 4 \Rightarrow \lambda = \frac{l}{2}$

$$\text{Hence } v = n\lambda = 500 \times \frac{2}{2} = 500 \text{ Hz}$$

$$51. \quad (d) \quad n = \frac{1}{2l} \sqrt{\frac{T}{\pi^2 \rho}} \Rightarrow n \propto \frac{\sqrt{T}}{l}$$

$$\Rightarrow \frac{n_2}{n_1} = \frac{r_1}{r_2} \sqrt{\frac{T_2}{T_1}} = \frac{1}{2} \times \sqrt{\frac{1}{2}} = \frac{1}{2\sqrt{2}}$$

52. (b) In case of sonometer frequency is given by

$$n = \frac{p}{2l} \sqrt{\frac{T}{m}} \Rightarrow \frac{n_2}{n_1} = \frac{l_1}{l_2} \Rightarrow n_2 = \frac{25}{16} \times 256 = 400 \text{ Hz}$$

$$1. \quad (c) \quad \lambda_1 = 2l, \lambda_2 = 2l + 2\Delta l \Rightarrow n_1 = \frac{v}{2l} \text{ and } n_2 = \frac{v}{2l + 2\Delta l}$$

$$\Rightarrow \text{No. of beats} = n_1 - n_2 = \frac{v}{2} \left(\frac{1}{l} - \frac{1}{l + \Delta l} \right) = \frac{v\Delta l}{2l^2}$$

2. (a) Fundamental frequency of open pipe is double that of the closed pipe.

3. (c) If is given that
First overtone of closed pipe = First overtone of open pipe $\Rightarrow$
 $3\left(\frac{v}{4l_1}\right) = 2\left(\frac{v}{2l_2}\right)$; where l_1 and l_2 are the lengths of closed and open organ pipes hence $\frac{l_1}{l_2} = \frac{3}{4}$

$$4. \quad (d) \quad \text{First overtone for closed pipe} = \frac{3v}{4l}$$

Fundamental frequency for open pipe = $\frac{v}{2l}$

$$\text{First overtone for open pipe} = \frac{2v}{2l}$$

5. (c) For closed pipe in general $n = \frac{v}{4l}(2N - 1) \Rightarrow n \propto \frac{1}{l}$
i.e. if length of air column decreases frequency increases.

6. (a,c,d) Fundamental frequency of closed pipe $n = \frac{v}{4l}$

$$\text{where } v = \sqrt{\frac{\gamma RT}{M}} \Rightarrow v \propto \frac{1}{\sqrt{M}}$$

$$\therefore M_{H_2} < M_{air} \Rightarrow v_{H_2} > v_{air}$$

Hence fundamental frequency with H_2 will be more as compared to air. So option (a) is correct.

Also $n \propto \frac{1}{l}$, hence if l decreases n increases so option (c) is correct.

It is well known that $(n)_{\text{open}} = 2(n)_{\text{closed}}$ hence option (d) is correct.

$$7. \quad (d) \quad \text{For closed pipe } n_1 = \frac{v}{4l} \Rightarrow l = \frac{v}{4n} = \frac{332}{4 \times 166} = 0.5 \text{ m}$$

$$8. \quad (a) \quad \text{Fundamental frequency of open pipe}$$

$$n_1 = \frac{v}{2l} = \frac{350}{2 \times 0.5} = 350 \text{ Hz}$$

$$9. \quad (b) \quad \text{For closed pipe } n_1 = \frac{v}{4l} = \frac{330}{4} \text{ Hz}$$

Second note = $3n_1 = \frac{3 \times 330}{4} \text{ Hz}$

$$10. \quad (c) \quad n_{\text{closed}} = \frac{v}{4l}, n_{\text{open}} = \frac{v}{2l} \Rightarrow n_{\text{open}} = 2n_{\text{closed}} = 2f$$

$$11. \quad (b) \quad \text{Minimum audible frequency} = 20 \text{ Hz}$$

$$\Rightarrow \frac{v}{4l} = 20 \Rightarrow l = \frac{336}{4 \times 20} = 4.2 \text{ m}$$

$$12. \quad (c) \quad \text{First overtone of closed organ pipe } n_1 = \frac{3v}{4l_1}$$

$$\text{Third overtone of open organ pipe } n_2 = \frac{4v}{2l_2}$$

Organ Pipe (Vibration of Air Column)

$$n_1 = n_2 \text{ (Given)} \Rightarrow \frac{3v}{4l_1} = \frac{4v}{2l_2} \Rightarrow \frac{l_1}{l_2} = \frac{3}{8}$$

13. (b) For closed pipe $n_1 = \frac{v}{4l} \Rightarrow 250 = \frac{v}{4 \times 0.2} \Rightarrow v = 200 \text{ m/s}$

14. (b) $n_{\text{open}} = \frac{v}{2l_{\text{open}}}$

$$n_{\text{closed}} = \frac{v}{4l_{\text{closed}}} = \frac{v}{4l_{\text{open}}/2} = \frac{v}{2l_{\text{open}}}$$

$$\left(\text{As } l_{\text{closed}} = \frac{l_{\text{open}}}{2} \right), \text{ i.e. frequency remains unchanged.}$$

15. (b) For closed pipe second note = $\frac{3v}{4l} = \frac{3 \times 330}{4 \times 1.5} = 165 \text{ Hz}$.

16. (a) Fundamental frequency of open pipe

$$n_1 = \frac{v}{2l} = \frac{330}{2 \times 0.3} = 550 \text{ Hz}$$

$$\text{First harmonic} = 2 \times n_1 = 1100 \text{ Hz} = 1.1 \text{ kHz}$$

17. (b) For first pipe $n_1 = \frac{v}{4l_1}$ and for second pipe $n_2 = \frac{v}{4l_2}$

$$\text{So, number of beats} = n_2 - n_1 = 4$$

$$\Rightarrow 4 = \frac{v}{4} \left(\frac{1}{l_2} - \frac{1}{l_1} \right) \Rightarrow 16 = 300 \left(\frac{1}{l_2} - \frac{1}{1} \right) \Rightarrow l_2 = 94.9 \text{ cm}$$

18. (a) Maximum pressure at closed end will be atmospheric pressure adding with acoustic wave pressure

$$\text{So } \rho_{\text{max}} = \rho_A + \rho_0 \text{ and } \rho_{\text{min}} = \rho_A - \rho_0$$

$$\text{Thus } \frac{\rho_{\text{max}}}{\rho_{\text{min}}} = \frac{\rho_A + \rho_0}{\rho_A - \rho_0}$$

19. (c) $n_1 - n_2 = 10$ (i)

$$\text{Using } n_1 = \frac{v}{4l_1} \text{ and } n_2 = \frac{v}{4l_2}$$

$$\Rightarrow \frac{n_1}{n_2} = \frac{l_2}{l_1} = \frac{26}{25} \text{(ii)}$$

$$\text{After solving these equation } n_1 = 260 \text{ Hz, } n_2 = 250 \text{ Hz}$$

20. (a) Let l_1 and l_2 be the length's of closed and open pipes respectively. (Neglecting end correction)

$$l_1 = \frac{\lambda_1}{4} \Rightarrow \lambda_1 = 4l_1 \text{ and } l_2 = \frac{\lambda_2}{2} \Rightarrow \lambda_2 = 2l_2$$

$$\text{Given } n_1 = n_2 \text{ so } \frac{v}{\lambda_1} = \frac{v}{\lambda_2} \Rightarrow \frac{v}{4l_1} = \frac{v}{2l_2} = \frac{l_1}{l_2} = \frac{1}{2}$$

21. (b) Distance between two consecutive nodes

$$= \frac{\lambda}{2} = 46 - 16 = 30 \Rightarrow \lambda = 60 \text{ cm} = 0.6 \text{ m}$$

$$\therefore v = n\lambda = 500 \times 0.6 = 300 \text{ m/s}$$

22. (a) For closed pipe $n = \frac{v}{4l} \Rightarrow n = \frac{332}{4 \times 42} = 2 \text{ Hz}$.

23. (a) For shortest length of pipe mode of vibration must be fundamental i.e., $n = \frac{v}{4l} \Rightarrow l = \frac{v}{4n}$.

24. (b) $n_{\text{Closed}} = \frac{1}{2}(n_{\text{Open}}) = \frac{1}{2} \times 320 = 160 \text{ Hz}$

25. (c) Frequency of 2^{nd} overtone $n_3 = 5n_1 = 5 \times 50 = 250 \text{ Hz}$.

26. (a) $\Delta n = n_1 - n_2 \Rightarrow 10 = \frac{v}{2l_1} - \frac{v}{2l_2} = \frac{v}{2} \left[\frac{1}{l_1} - \frac{1}{l_2} \right]$

$$\Rightarrow 10 = \frac{v}{2} \left[\frac{1}{0.25} - \frac{1}{0.255} \right] \Rightarrow v = 255 \text{ m/s}$$

27. (a) Fundamental frequency $n = \frac{v}{2l}$

$$\Rightarrow 350 = \frac{350}{2l} \Rightarrow l = \frac{1}{2} \text{ m} = 50 \text{ cm}$$

28. (b) $\Delta n = n_1 - n_2 \Rightarrow 4 = \frac{v}{2l_1} - \frac{v}{2l_2} = \frac{v}{2} \left[\frac{1}{1.00} - \frac{1}{1.025} \right]$

$$\Rightarrow 8 = [1 - 0.975] \Rightarrow v = \frac{8}{0.025} \approx 328 \text{ m/s}$$

29. (a) In closed pipe only odd harmonics are present

30. (d) Fundamental frequency of open organ pipe = $\frac{v}{2l}$

$$\text{Frequency of third harmonic of closed pipe} = \frac{3v}{4l}$$

$$\therefore \frac{3v}{4l} = 100 + \frac{v}{2l} \Rightarrow \frac{3v}{4l} - \frac{2v}{4l} = \frac{v}{4l} = 100 \Rightarrow \frac{v}{2l} = 200 \text{ Hz}$$

31. (c) $n_A = \frac{v}{2l}; n_B = \frac{v}{4l} \Rightarrow n_A / n_B = 2 : 1$

32. (a) Due to rise in temperature, the speed of sound increases. Since

$$n = \frac{v}{\lambda} \text{ and } \lambda \text{ remains unchanged, hence } n \text{ increases.}$$

33. (b)

34. (b)

35. (b) In closed organ pipe. If $y_{\text{incident}} = a \sin(\omega t - kx)$

$$\text{then } y_{\text{reflected}} = a \sin(\omega t + kx + \pi) = -a \sin(\omega t + kx)$$

Superimposition of these two waves give the required stationary wave.

36. (b) $v = 330 \text{ m/s}; n = 165 \text{ Hz}$. Distance between two

$$\text{successive nodes} = \frac{\lambda}{2} = \frac{v}{2n} = \frac{330}{2 \times 165} = 1 \text{ m}$$

37. (b) At the middle of pipe, node is formed.

38. (c) For closed organ pipe $n_1 : n_2 : n_3 \dots = 1 : 3 : 5 : \dots$

39. (b) First tone of open pipe = first overtone of closed pipe

$$\Rightarrow \frac{v}{2l_0} = \frac{3v}{4l_c} \Rightarrow l_c = \frac{3 \times 2 \times 0.5}{4} = 0.75 \text{ m}$$

40. (b) Only odd harmonics are present.

41. (b) Distance between six successive node

$$= \frac{5\lambda}{2} = 85 \text{ cm} \Rightarrow \lambda = \frac{2 \times 85}{5} = 34 \text{ cm} = 0.34 \text{ m}$$

$$\text{Therefore speed of sound in gas}$$

$$= n\lambda = 1000 \times 0.34 = 340 \text{ m/s}$$

42. (b) Let the base frequency be n for closed pipe then notes are $n, 3n, 5n, \dots$

$$\therefore \text{note } 3n = 255 \Rightarrow n = 85, \text{ note } 5n = 85 \times 5 = 425$$

$$\text{note } 7n = 7 \times 85 = 595$$

43. (a) $l_2 = 3l_1 = 3 \times 24.7 = 74.1 \text{ cm}$

44. (c) Frequency of p^{th} harmonic

$$n = \frac{pv}{2l} \Rightarrow p = \frac{2ln}{v} = \frac{2 \times 0.33 \times 1000}{330} = 2$$

45. (a) For closed pipe $l_1 = \frac{v}{4n}$; $l_2 = \frac{3v}{4n} \Rightarrow v = 2n(l_2 - l_1)$

$$\Rightarrow n = \frac{v}{2(l_2 - l_1)} = \frac{330}{2 \times (0.49 - 0.16)} = 500 \text{ Hz}$$

46. (c) Number of beats per second,

$$n = \frac{16}{20} = \frac{4}{5} \Rightarrow n = n_1 - n_2 = \frac{v}{4} \left(\frac{1}{l_1} - \frac{1}{l_2} \right)$$

$$\Rightarrow \frac{4}{5} = \frac{v}{4} \left(\frac{1}{1} - \frac{1}{1.01} \right) = \frac{0.01v}{4 \times 1.01}$$

$$v = \frac{16 \times 101}{5} = 323.2 \text{ ms}^{-1}$$

47. (a) In open organ pipe both even and odd harmonics are produced.

48. (d) Using $\lambda = 2(l_2 - l_1) \Rightarrow v = 2n(l_2 - l_1)$

$$\Rightarrow 2 \times 512(63.2 - 30.7) = 33280 \text{ cm/s}$$

$$\text{Actual speed of sound } v_0 = 332 \text{ m/s} = 33200 \text{ cm/s}$$

$$\text{Hence error} = 33280 - 33200 = 80 \text{ cm/s}$$

49. (b) Initially number of beats per second = 5

$$\therefore \text{Frequency of pipe} = 200 \pm 5 = 195 \text{ Hz or } 205 \text{ Hz} \dots(i)$$

$$\text{Frequency of second harmonics of the pipe} = 2n \text{ and number of beats in this case} = 10$$

$$\therefore 2n = 420 \pm 10 \Rightarrow 410 \text{ Hz or } 430 \text{ Hz}$$

$$\Rightarrow n = 205 \text{ Hz or } 215 \text{ Hz} \dots(ii)$$

$$\text{From equation (i) and (ii) it is clear that } n = 205 \text{ Hz}$$

50. (c) In case of open pipe, $n = \frac{N}{2l}$ where N = order of harmonics =

$$\text{order of mode of vibration} \Rightarrow N = \frac{n \times 2l}{v}$$

$$= \frac{480}{330} \times 2 \times 1 = 3 \quad (\text{Here } v = 330 \text{ m/s})$$

51. (a) In first overtone of organ pipe open at one end,

$$\text{end, } n_c = \frac{3v}{4l_c} \dots(i)$$

$$\text{Third harmonic or second overtone of organ pipe open at both}$$

$$\text{end, } n_0 = \frac{3v}{2l_0} \dots(ii)$$

$$\text{given } n_c = n_0 \Rightarrow \frac{3v}{4l_c} = \frac{3v_0}{2l_0} \Rightarrow \frac{l_c}{l_0} = \frac{1}{2}$$

52. (a) For end correction x , $\frac{l_2 + x}{l_1 + x} = \frac{3\lambda/4}{\lambda/4} = 3$

$$x = \frac{l_2 - 3l_1}{2} = \frac{70.2 - 3 \times 22.7}{2} = 1.05 \text{ cm}$$

53. (b) For open tube, $n_0 = \frac{v}{2l}$

$$\text{For closed tube length available for resonance is}$$

$$l' = l \times \frac{25}{100} = \frac{l}{4} \therefore \text{Fundamental frequency of water filled}$$

$$\text{tube } n = \frac{v}{4l'} = \frac{v}{4 \times (l/4)} = \frac{v}{l} = 2n_0 \Rightarrow \frac{n}{n_0} = 2$$

Doppler's Effect

1. (d)

2. (b) $n' = n \left(\frac{v}{v - v_o} \right) = 450 \left(\frac{340}{340 - 34} \right) = 500 \text{ cycles/sec}$

3. (a) $n' = n \left(\frac{v}{v - v_s} \right) \Rightarrow \lambda' = \lambda \left(\frac{v - v_s}{v} \right)$

$$\Rightarrow \lambda' = 120 \left(\frac{330 - 60}{330} \right) = 98 \text{ cm.}$$

4. (b) $n' = n \left(\frac{v}{v - v_s} \right) = 600 \left(\frac{330}{300} \right) = 660 \text{ cps}$

5. (c) Both listeners, hears the same frequencies.

6. (b)

7. (c) $n' = n \left(\frac{v + v_o}{v} \right) \Rightarrow 2n = n \left(\frac{v + v_o}{v} \right) \Rightarrow \frac{v + v_o}{v} = 2$

$$\Rightarrow v_o = v = 332 \text{ m/sec}$$

8. (b) Apparent frequency in this case $n' = \frac{n(v + v_o)}{v}$

$$\therefore \frac{v + v_o}{v} > 1 \Rightarrow \frac{n'}{n} > 1 \text{ i.e. } n' > n.$$

9. (a) Wave number = $\frac{1}{\lambda}$ but $\frac{1}{\lambda'} = \frac{1}{\lambda} \left(\frac{v}{v - v_s} \right)$ and $v_s = \frac{v}{3}$

$$\therefore (\text{W.N.})' = (\text{W.N.}) \left(\frac{v}{v - v/3} \right) = 256 \times \frac{v}{2v/3}$$

$$= \frac{3}{2} \times 256 = 384$$

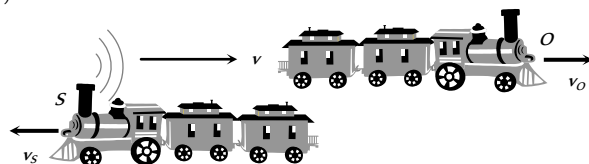
10. (a) By Doppler's formula $n' = \frac{nv}{(v - v_s)}$

$$\text{Since, source is moving towards the listener so } n' > n.$$

$$\text{If } n = 100 \text{ then } n' = 102.5$$

$$\Rightarrow 102.5 = \frac{100 \times 320}{(320 - v_s)} \Rightarrow v_s = 8 \text{ m/sec}$$

11. (b)

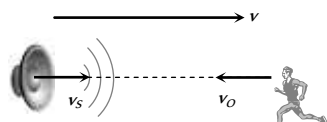


$$n' = n \left(\frac{v - v_O}{v + v_S} \right) = 750 \left(\frac{330 - 180 \times \frac{5}{18}}{330 + 108 \times \frac{5}{18}} \right) = 625 \text{ Hz}$$

12. (a) By using $n' = n \left(\frac{v}{v - v_S} \right)$

$$2n = n \left(\frac{v - v_O}{v - 0} \right) \Rightarrow v_O = -v = -(\text{Speed of sound})$$

Negative sign indicates that observer is moving opposite to the direction of velocity of sound, as shown



13. (d) Since there is no relative motion between observer and source, therefore there is no apparent change in frequency.

14. (c)
15. (b)

16. (a) $n' = n \left(\frac{v}{v - v_S} \right) \Rightarrow \frac{n'}{n} = \frac{v}{v - v_S} \Rightarrow \frac{v}{v - v_S} = 3 \Rightarrow v_S = \frac{2v}{3}$

17. (a) $n' = n \left(\frac{v}{v - v_S} \right) = n \left(\frac{v}{v - v/10} \right) \Rightarrow \frac{n'}{n} = \frac{10}{9}$

18. (c) $n' = n \left(\frac{v}{v - v_S} \right) = 1200 \times \left(\frac{350}{350 - 50} \right) = 1400 \text{ cps}$

19. (d) $n' = n \left(\frac{v}{v - v_S} \right) = 1200 \left(\frac{400}{400 - 100} \right) = 1600 \text{ Hz}$

20. (a) $n' = \frac{v}{v - v_S} \times n = \left(\frac{330}{330 - 110} \right) \times 150 = 225 \text{ Hz}$

21. (d) Doppler's effect is applicable for both light and sound waves.

22. (a) When source is approaching the observer, the frequency heard

$$n_a = \left(\frac{v}{v - v_S} \right) \times n = \left(\frac{340}{340 - 20} \right) \times 1000 = 1063 \text{ Hz}$$

When source is receding, the frequency heard

$$n_r = \left(\frac{v}{v + v_S} \right) \times n = \frac{340}{340 + 20} \times 1000 = 944$$

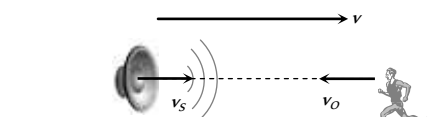
$$\Rightarrow n_a : n_r = 9 : 8$$

Short tricks : $\frac{n_a}{n_r} = \frac{v + v_S}{v - v_S} = \frac{340 + 20}{340 - 20} = \frac{9}{8}$

23. (a) By using $\frac{n_{\text{approaching}}}{n_{\text{receding}}} = \frac{v + v_S}{v - v_S}$

$$\Rightarrow \frac{1000}{n_r} = \frac{350 + 50}{350 - 50} \Rightarrow n_r = 750 \text{ Hz}$$

24. (b) When source and listener both are moving towards each other then, the frequency heard

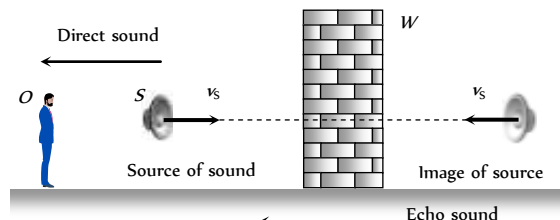


$$n' = n \left(\frac{v + v_O}{v - v_S} \right) \Rightarrow n' = f \left(\frac{v + v/10}{v - v/10} \right) = 1.22 f$$

25. (c) For source $v_s = r\omega = 0.70 \times 2\pi \times 5 = 22 \text{ m/sec}$
Minimum frequency is heard when the source is receding the man. It is given by $n_{\min} = n \frac{v}{v + v_S}$

$$= 1000 \times \frac{352}{352 + 22} = 941 \text{ Hz}$$

26. (b) For direct sound source is moving away from the observer so frequency heard in this case



$$n_1 = n \left(\frac{v}{v + v_S} \right) = 500 \left(\frac{332}{332 + 2} \right) = 500 \left(\frac{332}{334} \right) \text{ Hz}$$

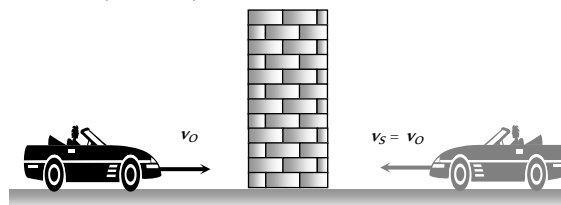
The other sound is echo, reaching the observer from the wall and can be regarded as coming from the image of source formed by reflection at the wall. This image is approaching the observer in the direction of sound.

Hence for reflected sound, frequency heard by the observer is

$$n_2 = n \left(\frac{v}{v - v_S} \right) = 500 \left(\frac{332}{332 - 2} \right) = 500 \left(\frac{332}{330} \right) \text{ Hz}$$

$$\text{Beats frequency} = n_2 - n_1 = 500 \times 332 \left(\frac{1}{330} - \frac{1}{334} \right) = 6$$

27. (c) Similar to previous question



The frequency of reflected sound heard by the driver

$$n' = n \left(\frac{v - (-v_O)}{v - v_S} \right) = n \left(\frac{v + v_O}{v - v_S} \right)$$

$$= 124 \left[\frac{330 + (72 \times 5/18)}{330 - (72 \times 5/18)} \right] = 140 \text{ vibration/sec}$$

28. (d) By using $n' = n \frac{v}{v - v_S} \Rightarrow \frac{n_1}{n} = \left(\frac{V}{V - S} \right)$

29. (b) In this case Doppler's effect is not applicable.

30. (d) The apparent frequency heard by the observer is given by

$$n' = \frac{v}{v - v_S} n = \frac{330}{330 - 33} \times 450 = \frac{330}{297} \times 450 = 500 \text{ Hz}$$

31. (a) $n' = n \left(\frac{v - v_O}{v} \right) = \left(\frac{330 - 33}{330} \right) \times 100 = 90 \text{ Hz}$

32. (c) When train is approaching frequency heard by the observer is

$$n_a = n \left(\frac{v}{v - v_S} \right) \Rightarrow 219 = n \left(\frac{340}{340 - v_S} \right) \dots(i)$$

when train is receding (goes away), frequency heard by the observer is

$$n_r = n \left(\frac{v}{v + v_s} \right) \Rightarrow 184 = n \left(\frac{340}{340 + v_s} \right) \quad \dots(ii)$$

On solving equation (i) and (ii) we get $n = 200\text{Hz}$

and $v_s = 29.5\text{m/s}$.

33. (d) Frequency is decreasing (becomes half), it means source is going away from the observer. In this case frequency observed by the observer is

$$n' = n \left(\frac{v}{v + v_s} \right) \Rightarrow \frac{n}{2} = n \left(\frac{v}{v + v_s} \right) \Rightarrow v_s = v$$

34. (d) Observer hears two frequencies

(i) n_1 which is coming from the source directly

(ii) n_2 which is coming from the reflection image of source

$$\text{so, } n_1 = 680 \left(\frac{340}{340 - 1} \right) \text{ and } n_2 = 680 \left(\frac{340}{340 + 1} \right)$$

$$\Rightarrow n_1 - n_2 = 4 \text{ beats}$$

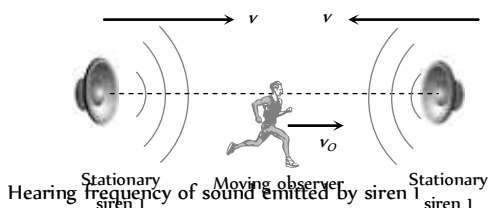
35. (a) From the figure, it is clear that

Frequency of reflected sound heard by the driver.

$$n' = n \left[\frac{v - (-v_o)}{v - v_s} \right] = n \left[\frac{v + v_o}{v - v_s} \right] = n \left[\frac{v + v_{car}}{v - v_{car}} \right]$$

$$= 600 \left[\frac{330 + 30}{330 - 30} \right] = 720 \text{ Hz.}$$

36. (b) Observer is moving away from siren 1 and towards the siren 2.



Hearing frequency of sound emitted by siren 1

$$n_1 = n \left(\frac{v - v_o}{v} \right) = 330 \left(\frac{330 - 2}{330} \right) = 328 \text{ Hz}$$

Hearing frequency of sound emitted by siren 2

$$n_2 = n \left(\frac{v + v_o}{v} \right) = 330 \left(\frac{330 + 2}{330} \right) = 332 \text{ Hz}$$

Hence, beat frequency $= n_2 - n_1 = 332 - 328 = 4$.

37. (c) $n' = n \left(\frac{v}{v - v_s} \right) = \frac{2000 \times 1220}{(1220 - 40)} = 2068 \text{ Hz}$

38. (d) $n' = n \left(\frac{v + v_o}{v - v_s} \right) \Rightarrow 400 = n \left(\frac{360 + 40}{360 - 40} \right) \Rightarrow n = 320 \text{ cps}$

39. (a) $n' = n \left(\frac{v}{v + v_s} \right) = 500 \times \left(\frac{330}{300 + 50} \right) = 434.2 \text{ Hz}$

40. (c) Since there is no relative motion between the listener and source, hence actual frequency will be heard by listener.

41. (a) $n' = n \left(\frac{v}{v - v_s} \right) \Rightarrow n' = 500 \left(\frac{330}{330 - 30} \right) = 550 \text{ Hz.}$

42. (c) $n' = n \left(\frac{v}{v - v_s} \right) = 90 \left(\frac{v}{v - \frac{v}{10}} \right) = 100 \frac{\text{Vibration}}{\text{sec}}$

43. (a) The linear velocity of Whistle

$$v_s = r\omega = 1.2 \times 2\pi \frac{400}{60} = 50 \text{ m/s}$$

When Whistle approaches the listener, heard frequency will be maximum and when listener recedes away, heard frequency will be minimum

$$\text{So, } n_{\max} = n \left(\frac{v}{v - v_s} \right) = 500 \left(\frac{340}{290} \right) = 586 \text{ Hz}$$

$$n_{\min} = n \left(\frac{v}{v + v_s} \right) = 500 \left(\frac{340}{390} \right) = 436 \text{ Hz}$$

44. (d) By using $n' = n \left(\frac{v}{v - v_s} \right)$

$$\Rightarrow f_1 = n \left(\frac{v}{v - v_s} \right) = n \left(\frac{340}{340 - 34} \right) = \frac{340}{306} n$$

$$\text{and } f_2 = n \left(\frac{340}{340 - 17} \right) = n \left(\frac{340}{323} \right) \Rightarrow \frac{f_1}{f_2} = \frac{323}{306} = \frac{19}{18}$$

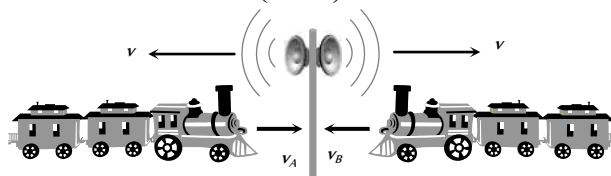
45. (d) No change in frequency.

46. (b) $n' = n \left(\frac{v - v_o}{v + v_s} \right) = n \left(\frac{340 - 10}{340 + 10} \right) = 1950 \Rightarrow n = 2068 \text{ Hz}$

47. (b) $n' = n \left(\frac{v + v_o}{v - v_s} \right) = 240 \left(\frac{340 + 20}{340 - 20} \right) = 270 \text{ Hz.}$

48. (b) In both the cases observer is moving towards, the source.

$$\text{Hence by using } n' = n \left(\frac{v + v_o}{v} \right)$$



When passenger is sitting in train A, then

$$5.5 = 5 \left(\frac{v + v_A}{v} \right) \quad \dots(i)$$

when passenger is sitting in train B, then

$$6 = 5 \left(\frac{v + v_B}{v} \right) \quad \dots(ii)$$

On solving equation (i) and (ii) we get $\frac{v_B}{v_A} = 2$

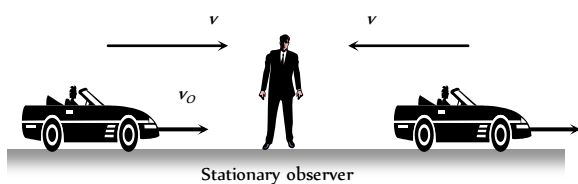
49. (b) Minimum frequency will be heard, when whistle moves away from the listener.

$$n_{\min} = n \left(\frac{v}{v + v_s} \right) \text{ where } v = r\omega = 0.5 \times 10 = 1 \text{ m/s}$$

$$\Rightarrow n_{\min} = 385 \left(\frac{340}{340 + 10} \right) = 374 \text{ Hz.}$$

50. (a) $n' = n \left(\frac{v}{v + v_s} \right) = 800 \left(\frac{330}{330 + 30} \right) = 733.33 \text{ Hz}$

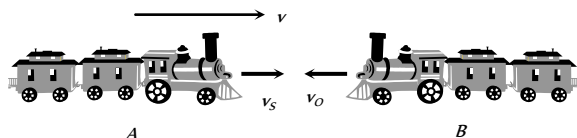
51. (a) $n_{\text{Before}} = \frac{v}{v - v_c} n$ and $n_{\text{After}} = \frac{v}{v + v_c} n$



$$\frac{n_{\text{Before}}}{n_{\text{After}}} = \frac{11}{9} = \left(\frac{v + v_c}{v - v_c} \right) \Rightarrow v_c \Rightarrow \frac{v}{10}$$

52. (c) By using $n' = \left(\frac{v}{v - v_s} \right) \Rightarrow 2n = n \left(\frac{v}{v - v_s} \right) \Rightarrow v_s = \frac{v}{2}$

53. (d) The frequency of whistle heard by passenger in the train B, is



$$n' = n \left(\frac{v + v_0}{v - v_s} \right) = 600 \left(\frac{340 + 15}{340 - 20} \right) \approx 666 \text{ Hz}$$

54. (b) At point A, source is moving away from observer so apparent frequency $n_1 < n$ (actual frequency) At point B source is coming towards observer so apparent frequency $n_2 > n$ and point C source is moving perpendicular to observer so $n_3 = n$

Hence $n_2 > n_3 > n_1$

55. (a) $n' = n \left[\frac{v + v_O}{v - v_s} \right]$; Here $v = 332 \text{ m/s}$ and $v_0 = v_s = 50 \text{ m/s}$

$$\Rightarrow 435 = n \left[\frac{332 + 50}{332 - 50} \right] \Rightarrow n = 321.12 \text{ sec}^{-1} \approx 320 \text{ sec}^{-1}$$

56. (c) Since apparent frequency is lesser than the actual frequency, hence the relative separation between source and listener should be increasing.

57. (c)

58. (d) $n' = n \left(\frac{v + v_0}{v - v_s} \right) = n \left(\frac{v + v/2}{v - v/2} \right) = 3n$

59. (c) When engine approaches towards observer $n' = n \left(\frac{v}{v - v_s} \right)$

when engine going away from observer $n'' = \left(\frac{v}{v + v_s} \right) n$

$$\therefore \frac{n'}{n''} = \frac{v + v_s}{v - v_s} \Rightarrow \frac{5}{3} = \frac{340 + v_s}{340 - v_s} \Rightarrow v_s = 85 \text{ m/s}$$

60. (a) Frequency heard by the observer

$$n' = n \left(\frac{v + v_0}{v} \right) = 240 \left(\frac{330 + 11}{330} \right) = 248 \text{ Hz}$$

61. (c) According to the concept of sound image

$$n' = \frac{v + v_{\text{person}}}{v - v_{\text{person}}} \cdot 272 = \frac{345 + 5}{345 - 5} \times 272 = 280 \text{ Hz}$$

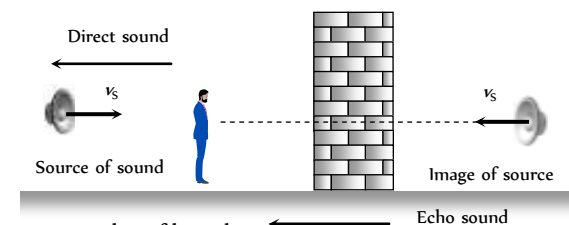
$$\Delta n = \text{Number of beats} = 280 - 272 = 8 \text{ Hz}$$

62. (b) According to the concept of sound image

$$n' = \frac{v + v_B}{v - v_B} \times n = \frac{355 + 5}{355 - 5} \times 165 = 170 \text{ Hz}$$

$$\text{Number of beats} = n' - n = 170 - 165 = 5$$

63. (a) The observer will hear two sound, one directly from source and other from reflected image of sound



Hence number of beats heard per second

$$= \left(\frac{v}{v - v_s} \right) n - \left(\frac{v}{v + v_s} \right) n$$

$$= \frac{2nvv_s}{v^2 - v_s^2} = \frac{2 \times 256 \times 330 \times 5}{335 \times 325} = 7.8 \text{ Hz}$$

64. (a) When a listener moves towards a stationary source apparent frequency

$$n' = \left(\frac{v + v_O}{v} \right) n = 200 \quad \dots (i)$$

When listener moves away from the same source

$$n'' = \left(\frac{v - v_O}{v} \right) n = 160 \quad \dots (ii)$$

From (i) and (ii)

$$\frac{v + v_O}{v - v_O} = \frac{200}{160} \Rightarrow \frac{v + v_O}{v - v_O} = \frac{5}{4} \Rightarrow v = 360 \text{ m/sec}$$

65. (b) When observer moves towards stationary source then apparent frequency

$$n' = \left[\frac{v + v_O}{v} \right] n = \left[\frac{v + v/5}{v} \right] n = \frac{6}{5} n = 1.2n$$

Increment in frequency = $0.2n$ so percentage change in

$$\text{frequency} = \frac{0.2n}{n} \times 100 = 20\%$$

Musical Sound

1. (d)

2. (a) $\text{Intensity} = \frac{\text{Power}}{\text{Area}} = \frac{4}{4\pi \times (200)^2} = 7.9 \times 10^{-6} \text{ W/m}^2$

3. (a) $\text{Intensity} \propto (\text{Amplitude})^2$

4. (c) $I = 2\pi^2 a^2 n^2 v \rho \Rightarrow I \propto a^2 n^2 \Rightarrow \frac{I_1}{I_2} = \left(\frac{a_1}{a_2} \right)^2 \times \left(\frac{n_1}{n_2} \right)^2$

$$= \left(\frac{1}{2} \right)^2 \times \left(\frac{1}{1/4} \right)^2 \Rightarrow I_2 = \frac{I_1}{4}$$

5. (b) $L = 10 \log_{10} \left(\frac{I}{I_0} \right) = 30 \Rightarrow \frac{I}{I_0} = 10^3$
6. (c)
7. (a) The quality of sound depends upon the number of harmonics present. Due to different number of harmonics present in two sounds, the shape of the resultant wave is also different.
8. (d) The sounds of different source are said to differ in quality. The number of overtones and their relative intensities determines the quality of any musical sound.

9. (d)
10. (d) Energy density $\propto (\text{amplitude})^2$
11. (d) Energy $\propto a^2 n^2 \Rightarrow \frac{a_B}{a_A} = \frac{n_A}{n_B}$ ($\because$ energy is same)

$$\Rightarrow \frac{a_B}{a_A} = \frac{8}{1}$$

12. (c) Loudness depends upon intensity while pitch depends upon frequency.

13. (d) Reverberation time $T = \frac{kV}{\alpha S} \Rightarrow T \propto V$.

14. (c) $I \propto \frac{1}{r^2} \Rightarrow \frac{I_2}{I_1} = \frac{r_1^2}{r_2^2} \Rightarrow \frac{I_2}{1 \times 10^{-2}} = \frac{2^2}{10^2} = \frac{4}{100}$
- $$\Rightarrow I_2 = \frac{4 \times 10^{-2}}{100} = 4 \times 10^{-4} \text{ } \mu \text{ W/m}^2$$

15. (b) After passing the 3 meter intensity is given by

$$I_3 = \frac{90}{100} \times \frac{90}{100} \times \frac{90}{100} \times I = 72.9\% \text{ of } I$$

So, the intensity is 72.9 decibel.

16. (c)

17. (b)

18. (a) $P \propto I$

$$L_1 = 10 \log_{10} \left(\frac{I_1}{I_0} \right) \text{ and } L_2 = 10 \log_{10} \left(\frac{I_2}{I_0} \right)$$

$$\text{So } L_2 - L_1 = 10 \log_{10} \left(\frac{I_2}{I_1} \right)$$

$$= 10 \log_{10} \left(\frac{P_2}{P_1} \right) = 10 \log_{10} \left(\frac{400}{20} \right) = 10 \log_{10} 20$$

$$= 10 \log(2 \times 10) = 10(0.301 + 1) = 13 \text{ dB}$$

19. (d) $I \propto \frac{1}{r^2} \Rightarrow \frac{\Delta I}{I} = -2 \frac{\Delta r}{r} = -2 \times 2 = -4\%$

Hence intensity is decreased by 4%.

20. (b) Musical interval is the ratio of frequencies = $\frac{320}{240} = \frac{4}{3}$

21. (c)

22. (d) By using $L = \log_{10} \frac{I}{I_0}$

$$L_2 - L_1 = \log_{10} \frac{I_2}{I_0} - \log_{10} \frac{I_1}{I_0}$$

$$5 - 1 = \log_{10} \frac{I_2}{I_1} \Rightarrow 4 = \log_{10} \frac{I_2}{I_1} \Rightarrow \frac{I_2}{I_1} = 10^4$$

$$\Rightarrow \frac{a_2^2}{a_1^2} = 10^4 \Rightarrow \frac{a_2}{a_1} = \frac{10^2}{1} \Rightarrow \frac{a_1}{a_2} = \frac{1}{10^2}$$

23. (b)

24. (a) Pitch of mosquito is higher among all given options.

25. (b) The frequency of note 'Sa' is 256 Hz while that of note 'Re' and 'Ga' respectively are 288 Hz and 320 Hz.

26. (d)

27. (d) Indian classical vocalists don't like harmonium because it uses tempered scale.

28. (b)

29. (b) $I \propto \frac{1}{r^2} \Rightarrow \frac{I_2}{I_1} = \frac{r_1^2}{r_2^2} = \frac{2^2}{(40)^2} = \frac{1}{400} \Rightarrow I_1 = 400 I_2$

$$\text{Intensity level at point 1, } L_1 = 10 \log_{10} \left(\frac{I_1}{I_0} \right)$$

$$\text{and intensity at point 2, } L_2 = 10 \log_{10} \left(\frac{I_2}{I_0} \right)$$

$$\therefore L_1 - L_2 = 10 \log \frac{I_1}{I_2} = 10 \log_{10} (400)$$

$$\Rightarrow L_1 - L_2 = 10 \times 2.602 = 26$$

$$L_2 = L_1 - 26 = 80 - 26 = 54 \text{ dB}$$

30. (a) Intensity $\propto \frac{1}{(\text{Distance})^2} \Rightarrow \frac{I_1}{I_2} = \left(\frac{d_2}{d_1} \right)^2 = \left(\frac{3}{2} \right)^2 = \frac{9}{4}$.

31. (d)

32. (a) The pitch depends upon the frequency of the source. As the two waves have different amplitude therefore they having different intensity. While quality depends on number of harmonics/overtone produced and their relative intensity. Assuming that their frequencies are the same.

Critical Thinking Questions

1. (a,b,c,d) $y = 0.02 \cos(10\pi x) \cos \left(50\pi t + \frac{\pi}{2} \right)$

At node, amplitude = 0

$$\Rightarrow \cos(10\pi x) = 0 \Rightarrow 10\pi x = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$\Rightarrow x = \frac{1}{20} = 0.05 \text{ m, } 0.15 \text{ m} \dots$$

At antinode, amplitude is maximum

$$\Rightarrow \cos(10\pi x) = \pm 1 \Rightarrow x = 0, \pi, 2\pi \dots$$

$$\Rightarrow x = 0, 0.1 \text{ m, } 0.2 \text{ m} \dots$$

Now $\lambda = 2 \times \text{Distance between two nodes or antinodes}$

$$= 2 \times 0.1 = 0.2 \text{ m and } \frac{2\pi vt}{\lambda} = 50\pi$$

$$v = 25\lambda = 25 \times 0.2 = 5 \text{ m/sec.}$$

2. (b,c) Since the edges are clamped, displacement of the edges

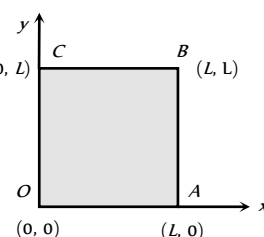
$u(x, y) = 0$ for line –

OA i.e. $y = 0$, $0 \leq x \leq L$ (0, L)

AB i.e. $x = L$, $0 \leq y \leq L$

BC i.e. $y = L$, $0 \leq x \leq L$

OC i.e. $x = 0$, $0 \leq y \leq L$



The above conditions are satisfied only in alternatives (b) and (c).

Note that $u(x, y) = 0$, for all four values e.g. in alternative (d), $u(x, y) = 0$ for $y = 0, y = L$ but it is not zero for $x = 0$ or $x = L$. Similarly in option (a). $u(x, y) = 0$ at $x = L, y = L$ but it is not zero for $x = 0$ or $y = 0$, while in options (b) and (c), $u(x, y) = 0$ for $x = 0, y = 0, x = L$ and $y = L$

3. (c) Energy (E) $\propto$ (Amplitude) (Frequency)

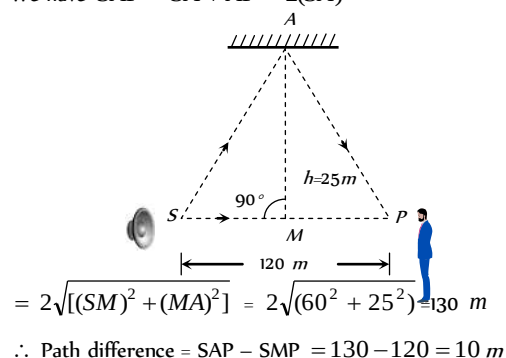
Amplitude is same in both the cases, but frequency 2ω in the second case is two times the frequency (ω) in the first case. Hence $E_2 = 4E_1$.

4. (a) Let S be source of sound and P the person or listener.

The waves from S reach point P directly following the path SMP and being reflected from the ceiling at point A following the path SAP . M is mid-point of SP (i.e. $SM = MP$) and $\angle SMA = 90^\circ$

Path difference between waves $\Delta x = SAP - SMP$

We have $SAP = SA + AP = 2(SA)$



Path difference due to reflection from ceiling = $\frac{\lambda}{2}$

$\therefore$ Effective path difference $\Delta x = 10 + \frac{\lambda}{2}$

For constructive interference

$$\Delta x = 10 + \frac{\lambda}{2} = n\lambda \Rightarrow (2n-1)\frac{\lambda}{2} = 10(n=1, 2, 3, \dots)$$

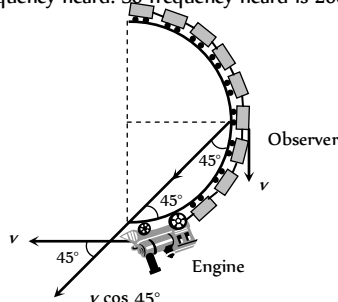
$$\therefore \text{Wavelength } \lambda = \frac{2 \times 10}{(2n-1)} = \frac{20}{2n-1}. \text{ The possible}$$

$$\text{wavelength are } \lambda = 20, \frac{20}{3}, \frac{20}{5}, \frac{20}{7}, \frac{20}{9}, \dots$$

$$= 20 \text{ m}, 6.67 \text{ m}, 4 \text{ m}, 2.85 \text{ m}, 2.22 \text{ m}, \dots$$

5. (c) The situation is shown in the fig.

Both the source (engine) and the observer (Person in the middle of the train) have the same speed, but their direction of motion is right angles to each other. The component of velocity of observer towards source is $v \cos 45^\circ$ and that of source along the time joining the observer and source is also $v \cos 45^\circ$. There is number relative motion between them, so there is no change in frequency heard. So frequency heard is 200 Hz.



6. (b) Velocity of sound increases if the temperature increases. So with $v = n\lambda$, if v increases n will increase

$$\text{at } 27^\circ \text{C}, v_1 = n\lambda, \text{ at } 31^\circ \text{C}, v_2 = (n+x)\lambda$$

$$\text{Now using } v \propto \sqrt{T} \quad \left(\because v = \sqrt{\frac{\gamma RT}{M}} \right)$$

$$\frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}} = \frac{n+x}{n}$$

$$\Rightarrow \frac{300+x}{300} = \sqrt{\frac{273+31}{273+27}} = \sqrt{\frac{304}{300}} = \sqrt{\frac{300+4}{300}}$$

$$\Rightarrow 1 + \frac{x}{300} = \left(1 + \frac{4}{300}\right)^{1/2} = \left(1 + \frac{1}{2} \times \frac{4}{300}\right) \Rightarrow x = 2.$$

$$[\because (1+x)^n = 1+nx]$$

7. (b) Let x be the end correction then according to question.

$$\frac{v}{4(l_1+x)} = \frac{3v}{4(l_2+x)} \Rightarrow x = 2.5 \text{ cm} = 0.025 \text{ m}.$$

8. (c) Frequency of first overtone of closed pipe = Frequency of first overtone of open pipe

$$\Rightarrow \frac{3v}{4L_1} = \frac{v}{L_2} \Rightarrow \frac{3}{4L_1} \sqrt{\frac{\gamma P}{\rho_1}} = \frac{1}{L_2} \sqrt{\frac{\gamma P}{\rho_2}} \quad \left[\because v = \sqrt{\frac{\gamma P}{\rho}} \right]$$

$$\Rightarrow L_2 = \frac{4L_1}{3} \sqrt{\frac{\rho_1}{\rho_2}} = \frac{4L}{3} \sqrt{\frac{\rho_1}{\rho_2}}$$

9. (b) For string, $\frac{\text{Mass}}{\text{Length}} = m = \frac{10^{-2}}{0.4} = 2.5 \times 10^{-2} \text{ kg/m}$

$$\therefore \text{Velocity } v = \sqrt{\frac{T}{m}} = \sqrt{\frac{16}{2.5 \times 10^{-2}}} = 8 \text{ m/s}$$

For constructive interference between successive pulses.

$$\Delta t_{\min} = \frac{2l}{v} = \frac{2(0.4)}{8} = 0.1 \text{ sec}$$

(After two reflections, the wave pulse is in same phase as it was produced since in one reflection it's phase changes by π , and if at this moment next identical pulse is produced, then constructive interference will be obtained.)

10. (d) Frequency of vibration in tight string

$$n = \frac{p}{2l} \sqrt{\frac{T}{m}} \Rightarrow n \propto \sqrt{T} \Rightarrow \frac{\Delta n}{n} = \frac{\Delta T}{2T} = \frac{1}{2} \times (4\%) = 2\%$$

$$\Rightarrow \text{Number of beats} = \Delta n = \frac{2}{100} \times n = \frac{2}{100} \times 100 = 2$$

11. (b) When the source approaches the observer

$$\text{Apparent frequency } n' = \frac{v}{v-v_s} \cdot n = n \left[\frac{1}{1 - \frac{v_s}{v}} \right]$$

$$= n \left[1 - \frac{v_s}{v} \right]^{-1} = n \left[1 + \frac{v_s}{v} \right]$$

(Neglecting higher powers because $v_s \ll v$)

When the source recedes the observed apparent frequency

$$n'' = n \left[1 - \frac{v_s}{v} \right]$$

$$\text{Given } n' - n'' = \frac{2}{100} n, v = 300 \text{ m/sec}$$

$$\therefore \frac{2}{100} n = n \left[1 + \frac{v_s}{v} \right] - n \left[1 - \frac{v_s}{v} \right] = n \left[2 \frac{v_s}{v} \right]$$

$$\Rightarrow \frac{2}{100} = 2 \frac{v_s}{v} \Rightarrow v_s = \frac{v}{100} = \frac{300}{100} = 3 \text{ m/sec}.$$

12. (a,b,c) Number of waves striking the surface per second (or the frequency of the waves reaching surface of the moving target)

$$n' = \frac{(c+v)}{\lambda} = \frac{v(c+v)}{c}$$

Now these waves are reflected by the moving target

(Which now act as a source). Therefore apparent frequency of

$$\text{reflected sound } n'' = \left(\frac{c}{c-v} \right) n' = v \left(\frac{c+v}{c-v} \right)$$

$$\text{The wavelength of reflected wave} = \frac{c}{n''} = \frac{c(c-v)}{v(c+v)}$$

The number of beats heard by stationary listener

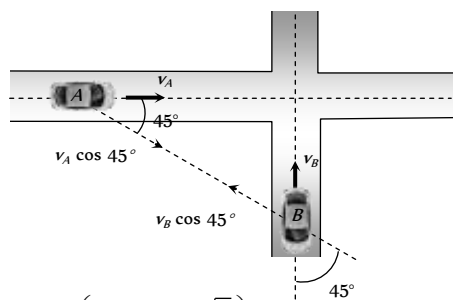
$$= n'' - v = v \left(\frac{c+v}{c-v} \right) - v = \frac{2v}{(c-v)}$$

Hence option (a) (b) and (c) are correct.

13. (b) Here $v_A = 72 \text{ km/hr} = 20 \text{ m/sec}$

$$v_B = 36 \text{ km/hr} = 10 \text{ m/sec}$$

$$n' = n \left(\frac{v + v_B \cos 45^\circ}{v - v_A \cos 45^\circ} \right)$$



$$\Rightarrow n' = 280 \left(\frac{340 + 10/\sqrt{2}}{340 - 20/\sqrt{2}} \right) = 298 \text{ Hz}$$

14. (b) For observer note of B will not change due to zero relative motion.

Observed frequency of sound produced by A

$$= 660 \frac{(330 - 30)}{330} = 600 \text{ Hz}$$

$$\therefore \text{No. of beats} = 600 - 596 = 4$$

15. (a) $\lambda = \frac{v}{n} = \frac{340}{170} = 2\text{m}$, $n' = \frac{340}{340 - 17} \times 170 \Rightarrow n' = 178.9 \text{ Hz}$

$$\text{Now } \lambda' = \frac{v}{n'} = \frac{340}{178.9} = 1.9$$

$$\Rightarrow \lambda - \lambda' = 2 - 1.9 = 0.1$$

16. (b) n_1 = Frequency of the police car horn observer heard by motorcyclist

n_2 = Frequency of the siren heard by motorcyclist.

v = Speed of motor cyclist

$$n_1 = \frac{330 - v}{330 - 22} \times 176; n_2 = \frac{330 + v}{330} \times 165$$

$$\therefore n_1 - n_2 = 0 \Rightarrow v = 22 \text{ m/s}$$

17. (a) $n' = \frac{v + v_0}{v} \cdot n = \frac{v + \frac{v}{5}}{v} \cdot f = \frac{6}{5} f = 1.2f$ and since the source is stationary, so wave length remains unchanged for observer.

18. (d) Time of fall = $\sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 10}{1000}} = \frac{1}{\sqrt{50}}$

In this time number of oscillations are eight.

$$\text{So time for 1 oscillation} = \frac{1}{8\sqrt{50}}$$

$$\text{Frequency} = 8\sqrt{50} \text{ Hz} = 56 \text{ Hz}$$

19. (a) Density of mixture = $\rho_{\text{mix}} = \frac{V_{O_2} \rho_{O_2} + V_{H_2} \rho_{H_2}}{V_{O_2} + V_{H_2}}$
- $$= \frac{V(\rho_{O_2} + \rho_{H_2})}{2V} = \frac{\rho_{O_2} + \rho_{H_2}}{2} \text{ (since } V_{O_2} = V_{H_2} = V)$$

$$= \frac{\rho_{H_2} + 16\rho_{H_2}}{2} = 8.5\rho_{H_2} \Rightarrow v \propto \frac{1}{\sqrt{\rho}}$$

$$\Rightarrow \frac{V_{\text{mix}}}{V_{H_2}} = \sqrt{\frac{\rho_{H_2}}{\rho_{\text{mix}}}} = \sqrt{\frac{\rho_{H_2}}{8.5\rho_{H_2}}} \approx \sqrt{\frac{1}{8}}$$

20. (c) $y_1 = 10 \sin\left(3\pi t + \frac{\pi}{3}\right)$... (i)

$$\text{and } y_2 = 5[\sin 3\pi t + \sqrt{3} \cos 3\pi t]$$

$$= 5 \times 2 \left[\frac{1}{2} \times \sin 3\pi t + \frac{\sqrt{3}}{2} \times \cos 3\pi t \right]$$

$$= 10 \left[\cos \frac{\pi}{3} \sin 3\pi t + \sin \frac{\pi}{3} \cos 3\pi t \right]$$

$$= 10 \left[\sin\left(3\pi t + \frac{\pi}{3}\right) \right] \quad \dots \text{ (ii)}$$

$$(\because \sin(A + B) = \sin A \cos B + \cos A \sin B)$$

Comparing equation (i) and (ii) we get ratio of amplitude 1 : 1.

21. (a) The given equation can be written as

$$y = \frac{A}{2} \cos\left(4\pi t - \frac{4\pi x}{\lambda}\right) + \frac{A}{2} \quad \left(\because \cos^2 \theta = \frac{1 + \cos 2\theta}{2}\right)$$

$$\text{Hence amplitude} = \frac{A}{2} \text{ and frequency} = \frac{\omega}{2\pi} = \frac{4\pi}{2\pi} = 2n$$

$$\text{and wave length} = \frac{2\pi}{k} = \frac{2\pi}{4\pi/\lambda} = \frac{\lambda}{2}$$

22. (a,b,c,d) In case of sound wave, y can represent pressure and displacement, while in case of an electromagnetic wave it represents electric and magnetic fields.

(In general y is any general physical quantity which is made to oscillate at one place and these oscillations are propagated to other places also).

23. (b) In case of interference of two waves resultant intensity

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$$

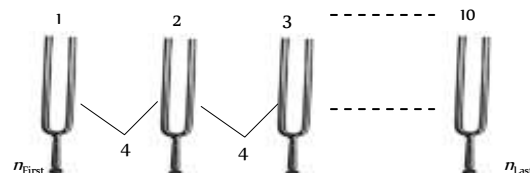
If ϕ varies randomly with time, so $(\cos \phi)_{\text{av}} = 0$

$$\Rightarrow I = I_1 + I_2$$

For n identical waves, $I = I_0 + I_0 + \dots = nI_0$

here $I = 10I_0$.

24. (d)



$$\text{Using } n_{\text{last}} = n_{\text{first}} + (N - 1)x$$

where N = Number of tuning fork in series

x = beat frequency between two successive forks

$$\Rightarrow 2n = n + (10 - 1) \times 4 \Rightarrow n = 36 \text{ Hz}$$

$$\therefore n_{\text{last}} = 36 \text{ Hz and } n_{\text{last}} = 2 \times n_{\text{first}} = 72 \text{ Hz}$$

25. (a) Similar to previous question

$$n_{\text{last}} = n_{\text{first}} + (N - 1)x$$

$$2n = n_{\text{first}} + (41 - 1) \times 5$$

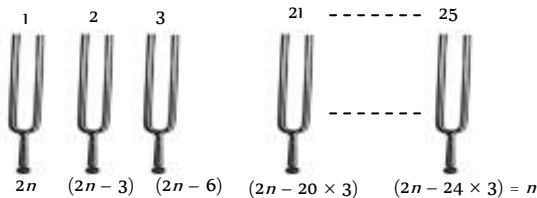
$$\Rightarrow n_{\text{max}} = 200 \text{ Hz and } n_{\text{min}} = 400 \text{ Hz}$$

26. (a) $n \propto \sqrt{T} \Rightarrow \frac{\Delta n}{n} = \frac{1}{2} \frac{\Delta T}{T}$

$$\text{Beat frequency} = \Delta n = \left(\frac{1}{2} \frac{\Delta T}{T} \right) n = \frac{1}{2} \times \frac{2}{100} \times 400 = 4$$

27. (c) According to the question frequencies of first and last tuning forks are $2n$ and n respectively.

Hence frequency in given arrangement are as follows



$$\Rightarrow 2n - 24 \times 3 = n \Rightarrow n = 72 \text{ Hz}$$

So, frequency of 21st tuning fork

$$n_{21} = (2 \times 72 - 20 \times 3) = 84 \text{ Hz}$$

28. (a) Using $n_{\text{max}} = n_{\text{min}} + (N-1)\lambda$

$$\Rightarrow 2n = n + (16-1) \times 8 \Rightarrow n = 120 \text{ Hz}$$

29. (b) Using $n = \frac{1}{2l} \sqrt{\frac{T}{m}}$;

$$\text{As } T_1 > T_2 \Rightarrow n_1 > n_2 \text{ giving } n_1 - n_2 = 6$$

The beat frequency of 6 will remain fixed when

- (i) n_1 remains same but n_2 is increased to a new value ($n_2' - n_2 = 12$) by increasing tension T_2 .
 (ii) n_1 remains same but n_1 is decreased to a new value ($n_1 - n_1' = 12$) by decreasing tension T_1 .

30. (a) According to problem

$$\frac{1}{2L} \sqrt{\frac{T}{m}} = \frac{v}{4L} \quad \dots (i)$$

$$\text{and } \frac{1}{2L} \sqrt{\frac{T+8}{m}} = \frac{3v}{4L} \quad \dots (ii)$$

$$\text{Dividing equation (i) and (ii), } \sqrt{\frac{T}{T+8}} = \frac{1}{3} \Rightarrow T = 1 \text{ N}$$

31. (b) In condition of resonance, frequency of a.c. will be equal to natural frequency of wire

$$n = \frac{1}{2l} \sqrt{\frac{T}{m}} = \frac{1}{2 \times 1} \sqrt{\frac{10 \times 9.8}{9.8 \times 10^{-3}}} = \frac{100}{2} = 50 \text{ Hz}$$

32. (b) For wire if

M = mass, ρ = density, A = Area of cross section

V = volume, l = length, Δl = change in length

$$\text{Then mass per unit length } m = \frac{M}{l} = \frac{Al\rho}{l} = A\rho$$

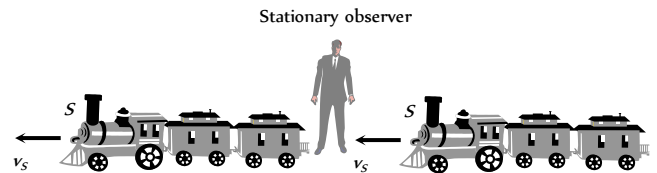
$$\text{And Young's modulus of elasticity } y = \frac{T/A}{\Delta l/l}$$

$$\Rightarrow T = \frac{Y\Delta l A}{l}. \text{ Hence lowest frequency of vibration}$$

$$n = \frac{1}{2l} \sqrt{\frac{T}{m}} = \frac{1}{2l} \sqrt{\frac{y \left(\frac{\Delta l}{l} \right) A}{A\rho}} = \frac{1}{2l} \sqrt{\frac{y\Delta l}{l\rho}}$$

$$\Rightarrow n = \frac{1}{2 \times 1} \sqrt{\frac{9 \times 10^{10} \times 4.9 \times 10^{-4}}{1 \times 9 \times 10^3}} = 35 \text{ Hz}$$

33. (a)



Receding train

Approaching train

Frequency of sound heard by the man from approaching train

$$n_a = n \left(\frac{v}{v - v_s} \right) = 240 \left(\frac{320}{320 - 4} \right) = 243 \text{ Hz}$$

Frequency of sound heard by the man from receding train

$$n_r = n \left(\frac{v}{v + v_s} \right) = 240 \left(\frac{320}{320 + 4} \right) = 237 \text{ Hz}$$

Hence, number of beats heard by man per sec

$$= n_a - n_r = 243 - 237 = 6$$

Short trick : Number of beats heard per sec

$$= \frac{2nvv_s}{v^2 - v_s^2} = \frac{2nvv_s}{(v - v_s)(v + v_s)} = \frac{2 \times 240 \times 320 \times 4}{(320 - 4)(320 + 4)} = 6$$

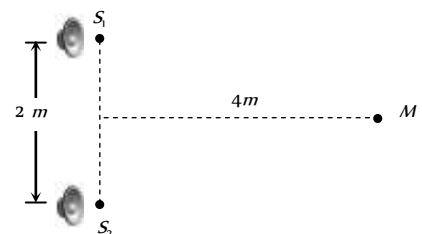
34. (c) Open pipe resonance frequency $f_1 = \frac{2v}{2L}$

$$\text{Closed pipe resonance frequency } f_2 = \frac{nv}{4L}$$

$$f_2 = \frac{n}{4} f_1 \text{ (where } n \text{ is odd and } f_2 > f_1) \therefore n = 5$$

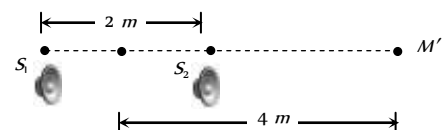
35. (b) Initially $SM = SM$

$$\Rightarrow \text{Path Difference } (\Delta x) = S_1M - S_2M = 0.$$



Finally when the box is rotated

$$\text{Path Difference} = S_1M' - S_2M' \Rightarrow \Delta x = 5 - 3 = 2m$$



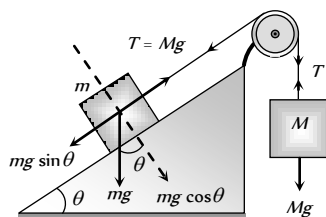
For maxima

$$\text{Path Difference} = (\text{Even multiple}) \frac{\lambda}{2} \Rightarrow \Delta x = (2n) \frac{\lambda}{2}$$

For 5 maximum responses

$$\Rightarrow 2 = 2(5) \frac{\lambda}{2} \left\{ \because \Delta x = (2n) \frac{\lambda}{2} \right\} \Rightarrow \lambda = \frac{2}{5} = 0.4m.$$

36. (a) $v = \sqrt{\frac{T}{\mu}}$



For equilibrium $Mg = mg \sin 30 = T$

$$\Rightarrow M = \frac{m}{2} \Rightarrow 100 = \sqrt{\frac{Mg}{9.8 \times 10^{-3}}} = \sqrt{\frac{M(9.8)}{9.8 \times 10^{-3}}}$$

$$\Rightarrow 100 = \sqrt{M(1000)} \Rightarrow M = 10kg \text{ and } m = 20kg$$

37. (d) For not hearing the echo the time interval between the beats of drum must be equal to time of echo.

$$\Rightarrow t_1 = \frac{2d}{v} = \frac{60}{40} = \frac{3}{2} \quad \dots(i)$$

$$\text{and } t_2 = \frac{2(d-90)}{v} = \frac{60}{60} = 1$$

$$\Rightarrow 2d - 180 = v \quad \dots(ii)$$

Form (i), we get $2d = \frac{3}{2}v$. Substituting in (ii), we get

$$\Rightarrow \frac{3}{2}v - 180 = v \Rightarrow 180 = \frac{v}{2} \Rightarrow v = 360ms^{-1}$$

$$\Rightarrow \frac{2(d)}{360} = \frac{3}{2} \Rightarrow d = 270m.$$

38. (b) Path difference between the wave reaching at D

$$\Delta x = L_2P - L_1P = \sqrt{40^2 + 9^2} - 40 = 41 - 40 = 1m$$

$$\text{For maximum } \Delta x = (2n) \frac{\lambda}{2}$$

$$\text{For first maximum } (n=1) \Rightarrow 1 = 2(1) \frac{\lambda}{2} \Rightarrow \lambda = 1m$$

$$\Rightarrow n = \frac{v}{\lambda} = 330Hz.$$

39. (a) In a wave equation, x and t must be related in the form $(x-vt)$.

$$\text{We rewrite the given equations } y = \frac{1}{1+(x-vt)^2}$$

$$\text{For } t=0, \text{ this becomes } y = \frac{1}{(1+x^2)}, \text{ as given}$$

$$\text{For } t=2, \text{ this becomes } y = \frac{1}{[1+(x-2v)^2]} = \frac{1}{[1+(x-1)^2]}$$

$$\Rightarrow 2v=1 \text{ or } v=0.5m/s.$$

40. (c) $dB = 10 \log_{10} \left(\frac{I}{I_0} \right)$; where $I_0 = 10^{-12} Wm^{-2}$

$$\text{Since } 40 = 10 \log_{10} \left(\frac{I_1}{I_0} \right) \Rightarrow \frac{I_1}{I_0} = 10^4 \quad \dots(i)$$

$$\text{Also } 20 = 10 \log_{10} \left(\frac{I_2}{I_0} \right) \Rightarrow \frac{I_2}{I_0} = 10^2 \quad \dots(ii)$$

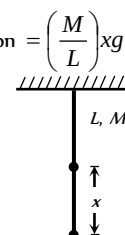
$$\Rightarrow \frac{I_2}{I_1} = 10^{-2} = \frac{r_1^2}{r_2^2} \Rightarrow r_2^2 = 100r_1^2 \Rightarrow r_2 = 10m$$

$$\{\because r_1 = 1m\}$$

41. (b) Velocity $v = \sqrt{\frac{T}{m}}$; where $T = \text{weight of part of rope hanging}$

$$\text{below the point under consideration} = \left(\frac{M}{L} \right) xg$$

$$\Rightarrow v = \sqrt{\frac{\left(\frac{M}{L} \right) xg}{\left(\frac{M}{L} \right)}} = \sqrt{xg}.$$



42. (b) When the piston is moved through a distance of $8.75cm$, the path difference produced is $2 \times 8.75cm = 17.5cm$. This must be equal to $\frac{\lambda}{2}$ for maximum to change to minimum. $\therefore$

$$\frac{\lambda}{2} = 17.5cm \Rightarrow \lambda = 35cm = 0.35m$$

$$\text{So, } v = n\lambda \Rightarrow n = \frac{v}{\lambda} = \frac{350}{0.35} = 1000Hz$$

43. (c) Frequency of vib. is stretched string $n = \frac{1}{2(\text{Length})} \sqrt{\frac{T}{m}}$

When the stone is completely immersed in water, length changes but frequency doesn't ($\because$ unison reestablished)

$$\text{Hence length} \propto \sqrt{T} \Rightarrow \frac{L}{l} = \sqrt{\frac{T_{air}}{T_{water}}} = \sqrt{\frac{V\rho g}{V(\rho-1)g}}$$

(Density of stone = ρ and density of water = 1)

$$\Rightarrow \frac{L}{l} = \sqrt{\frac{\rho}{\rho-1}} \Rightarrow \rho = \frac{L^2}{L^2 - l^2}$$

44. (a,c) $y = \cos kx \sin \omega t$ and $y = \cos(kx + \omega t)$ represent wave motion, because they satisfies the wave equation $\frac{\partial^2}{\partial t^2} = v^2 \frac{\partial^2}{\partial x^2}$.

45. (c) The wave 1 and 3 reach out of phase. Hence resultant phase difference between them is π .

$$\therefore \text{Resultant amplitude of 1 and 3} = 10 - 7 = 3 \mu m$$

$$\text{This wave has phase difference of } \frac{\pi}{2} \text{ with } 4 \mu m$$

$$\therefore \text{Resultant amplitude} = \sqrt{3^2 + 4^2} = 5 \mu m$$

46. (b) Let $n-1$ ($= 400$), n ($= 401$) and $n+1$ ($= 402$) be the frequencies of the three waves. If a be the amplitude of each then

$$y = a \sin 2\pi(n-1)t, \quad y = a \sin 2\pi nt \quad \text{and}$$

$$y_3 = a \sin 2\pi(n+1)t$$

Resultant displacement due to all three waves is

$$Y = y_1 + y_2 + y_3$$

$$= a \sin 2\pi nt + a [\sin 2\pi(n-1)t + \sin 2\pi(n+1)t]$$

$$= a \sin 2\pi nt + a [2 \sin 2\pi nt \cos 2\pi t]$$

$$\left[\text{Using } \sin C + \sin D = 2 \sin \frac{C+D}{2} \cos \frac{C-D}{2} \right]$$

$$\Rightarrow Y = a(1 + \cos 2\pi t) \sin 2\pi nt$$

This is the resultant wave having amplitude $= (1 + \cos 2\pi t)$

For maximum amplitude $\cos 2\pi t = 1 \Rightarrow 2\pi t = 2m\pi$ where $m = 0, 1, 2, 3, \dots$

$$\Rightarrow t = 0, 1, 2, 3, \dots$$

Hence time interval between two successive maximum is 1 sec. So beat frequency = 1

Also for minimum amplitude $(2 \cos 2\pi t) = 0$

$$\Rightarrow \cos 2\pi t = -\frac{1}{2}$$

$$\Rightarrow 2\pi t = 2m\pi + \frac{2\pi}{3} \Rightarrow t = \frac{1}{3}$$

$$\Rightarrow t = \frac{1}{3}, \frac{4}{3}, \frac{7}{3}, \frac{10}{3}, \dots \quad (\text{for } m = 0, 1, 2, \dots)$$

Hence time interval between two successive minima is 1 sec so, number of beats per second = 1

Note : PET/PMT Aspirants can remember result only.

47. (d) Because the tuning fork is in resonance with air column in the pipe closed at one end, the frequency is $n = \frac{(2N-1)v}{4l}$ where

$N = 1, 2, 3, \dots$ corresponds to different mode of vibration

putting $n = 340 \text{ Hz}$, $v = 340 \text{ m/s}$, the length of air column in the pipe can be

$$l = \frac{(2N-1)340}{4 \times 340} = \frac{(2N-1)}{4} m = \frac{(2N-1) \times 100}{4} \text{ cm}$$

For $N = 1, 2, 3, \dots$ we get $l = 25 \text{ cm}, 75 \text{ cm}, 125 \text{ cm} \dots$

As the tube is only 120 cm long, length of air column after water is poured in it may be 25 cm or 75 cm only, 125 cm is not possible, the corresponding length of water column in the tube will be $(120 - 25) \text{ cm} = 95 \text{ cm}$ or $(120 - 75) \text{ cm} = 45 \text{ cm}$.

Thus minimum length of water column is 45 cm.

48. (c) Critical hearing frequency for a person is 20,000 Hz.

If a closed pipe vibration in N^{th} mode then frequency of

$$\text{vibration } n = \frac{(2N-1)v}{4l} = (2N-1)n_1$$

(where n_1 = fundamental frequency of vibration)

$$\text{Hence } 20,000 = (2N-1) \times 1500 \Rightarrow N = 7.1 \approx 7$$

Also, in closed pipe

Number of over tones = (No. of mode of vibration) - 1

$$= 7 - 1 = 6.$$

49. (c) Frequency of vibration of string is given by

$$n = \frac{p}{2l} \sqrt{\frac{T}{m}} \Rightarrow p\sqrt{T} = \text{constant} \Rightarrow \frac{p_1}{p_2} = \sqrt{\frac{T_2}{T_1}}$$

$$\text{Hence } \frac{4}{6} = \sqrt{\frac{T_2}{(50+15)gm - \text{force}}} \Rightarrow T_2 = 28.8 \text{ gm} - f$$

Hence weight removed from the pan

$$= T_1 - T_2 = 65 - 28.8 = 3.62 \text{ gm-force} = 0.036 \text{ kg-f}$$

50. (c) Frequency of reflected sound heard by driver $n' = n \left(\frac{v + v_O}{v - v_S} \right)$



It is given that $n' = 2n$

$$\text{Hence, } 2n = n \left(\frac{v + v_{car}}{v - v_{car}} \right) \Rightarrow v_{car} = v/3.$$

51. (c) Suppose d = distance of epicenter of Earth quake from point of observation

v_s = Speed of S-wave and v_p = Speed of P-wave then

$$d = v_p t_p = v_s t_s \quad \text{or} \quad 8 t_p = 4.5 t_s$$

$$\Rightarrow t_p = \frac{4.5}{8} t_s, \quad \text{given that } t_s - t_p = 240$$

$$\Rightarrow t_s - \frac{4.5}{8} t_s = 240 \Rightarrow t_s = \frac{240 \times 8}{3.5} = 548.5 \text{ s}$$

$$\therefore d = v_s t_s = 4.5 \times 548.5 = 2468.6 \approx 2500 \text{ km}$$

Graphical Questions

1. (c) Speed = $n\lambda = n(4ab) = 4n \times ab$ $\left(Asab = \frac{\lambda}{4}\right)$
 Path difference between b and e is $\frac{3\lambda}{4}$
 So the phase difference = $\frac{2\pi}{\lambda} \cdot \text{Path difference}$

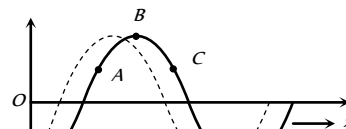
$$= \frac{2\pi}{\lambda} \cdot \frac{3\lambda}{4} = \frac{3\pi}{2}$$
2. (b) After 2 sec the pulses will overlap completely. The string becomes straight and therefore does not have any potential energy and its entire energy must be kinetic.
3. (a) When the train is approaching the stationary observer frequency heard by the observer $n' = \frac{v + v_0}{v} n$
 when the train is moving away from the observer then frequency heard by the observer $n'' = \frac{v - v_0}{v} n$
 it is clear that n' and n'' are constant and independent of time. Also $n' > n''$.
4. (b) Equation of A , B , C and D are
 $y_A = A \sin \omega t$, $y_B = A \sin(\omega t + \pi/2)$
 $y_C = A \sin(\omega t - \pi/2)$, $y_D = A \sin(\omega t - \pi)$
 It is clear that wave C lags behind by a phase angle of $\pi/2$ and the wave B is ahead by a phase angle at $\pi/2$.
5. (d) Points B and F are in same phase as they are λ distance apart.
6. (c) The particle velocity is maximum at B and is given by
 $\frac{dy}{dt} = (v_p)_{\max} = \omega A$
 Also wave velocity is $\frac{dx}{dt} = v = \frac{\omega}{k}$
 So slope $\frac{dy}{dx} = \frac{(v_p)_{\max}}{v} = kA$
7. (d) When pulse is reflected from a rigid support, the pulse is inverted both lengthwise and sideways
8. (d) Given equation $y = y_0 \sin(\omega t - \phi)$
 at $t = 0$, $y = -y_0 \sin \phi$
 this is the case with curve marked D .
9. (c) We know frequency $n = \frac{P}{2l} \sqrt{\frac{T}{\pi^2 \rho}} \Rightarrow n \propto \frac{1}{\sqrt{\rho}}$
i.e., graph between n and $\sqrt{\rho}$ will be hyperbola.
10. (c) Energy density (E) = $\frac{I}{v} = 2\pi^2 \rho n^2 A^2$
 $v_{\max} = \omega A = 2\pi n A \Rightarrow E \propto (v_{\max})^2$
i.e., graph between E and $v_{\max}$ will be a parabola symmetrical about E axis.
11. (c) Here $A = 0.05m$, $\frac{5\lambda}{2} = 0.025 \Rightarrow \lambda = 0.1m$
 Now standard equation of wave

$$y = A \sin \frac{2\pi}{\lambda} (vt - x) \Rightarrow y = 0.05 \sin 2\pi(33t - 10x)$$

12. (c) After two seconds each wave travel a distance of $2.5 \times 2 = 5$ cm *i.e.* the two pulses will meet in mutually opposite phase and hence the amplitude of resultant will be zero.



13. (c) $n_Q = 341 \pm 3 = 344 \text{ Hz}$ or 338 Hz
 on waxing Q , the number of beats decreases hence $n_Q = 344 \text{ Hz}$
14. (b) For observer approaching a stationary source
 $n' = \frac{v + v_0}{v} n$ and given $v_0 = at \Rightarrow n' = \left(\frac{an}{v}\right)t + n$
 this is the equation of straight line with positive intercept n and positive slope $\left(\frac{n}{v}\right)$.
15. (b,d) Since A is moving upwards, therefore, after an elemental time interval the wave will be as shown dotted in following fig. It means, the wave is travelling leftward. Therefore, (a) is wrong.



Displacement amplitude of the wave means maximum possible displacement of medium particles due to propagation of the wave, which is equal to the displacement at B at the instant shown in fig. Hence (b) is correct.

From figure, it is clear that C is moving downwards at this instant. Hence (c) is wrong.

The phase difference between two points will be equal to $\frac{\pi}{2}$ if distance between them is equal to $\frac{\lambda}{4}$. Between A and C , the distance is less than $\frac{\lambda}{2}$. It may be equal to $\frac{\lambda}{4}$. Hence, phase difference between these two points may be equal to $\frac{\pi}{2}$.

16. (d) Intensity $\propto a^2 \omega^2$
 here $\frac{a_A}{a_B} = \frac{2}{1}$ and $\frac{\omega_A}{\omega_B} = \frac{1}{2} \Rightarrow \frac{I_A}{I_B} = \left(\frac{2}{1}\right)^2 \times \left(\frac{1}{2}\right)^2 = \frac{1}{1}$
17. (b) At $t = 0$ and $x = \frac{\pi}{2k}$. The displacement
 $y = a_0 \sin\left(\omega x_0 - k \times \frac{\pi}{2k}\right) = -a_0 \sin \frac{\pi}{2} = -a_0$
 from graph. Point of maximum displacement (a) in negative direction is Q .
18. (d) Particle velocity (v_p) = $-v \times \text{Slope of the graph at that point}$
 At point 1 : Slope of the curve is positive, hence particle velocity is negative or downward ($\downarrow$)

At point 2 : Slope negative, hence particle velocity is positive or upwards ($\uparrow$)

At point 3 : Again slope of the curve is positive, hence particle velocity is negative or downward ($\downarrow$)

Assertion and Reason

1. (a) Sound waves require material medium to travel. As there is no atmosphere (vacuum) on the surface of moon, therefore the sound waves cannot reach from one person to another.
2. (b) Transverse waves travel in the form of crests and troughs involving change in shape of the medium. As liquids and gases do not possess the elasticity of shape, therefore, transverse waves cannot be produced in liquid and gases. Also light wave is one example of transverse wave.
3. (b) Sound waves cannot propagate through vacuum because sound waves are mechanical waves. Light waves can propagate through vacuum because light waves are electromagnetic waves. Since sound waves are longitudinal waves, the particles moves in the direction of propagation, therefore these waves cannot be polarised.
4. (c) Velocity of sound in gas medium is $v = \sqrt{\frac{K}{\rho}} = \sqrt{\frac{\gamma P}{\rho}}$
 γ is ratio of its principal heat capacities (C_p / C_v). For moist air ρ is less than that for dry air and γ is slightly greater.
 $\therefore$ velocity of sound increases with increase in humidity.
5. (c) Ocean waves are transverse waves travelling in concentric circles of ever increasing radius. When they hit the shore, their radius of curvature is so large that they can be treated as plane waves. Hence they hit the shore nearly normal to the shore.
6. (a) A compression is a region of medium in which particles come closer *i.e.*, distance between the particles becomes less than the normal distance between them. Thus there is a temporary decrease in volume and a consequent increase in density of medium. Similarly in rarefaction, particle get farther apart and a consequent decrease in density.
7. (e) Since transverse wave can propagate through medium which posses elasticity of shape. Air posses only volume elasticity therefore transverse wave cannot propagate through air.
8. (c) The velocity of sound in a gas is directly proportional to the square root of its absolute temperature (as $v = \sqrt{\frac{\gamma RT}{M}}$). Since temperature of a hot day is more than cold winter day, therefore sound would travel faster on a hot summer day than on a cold winter day.
9. (c) According to Laplace, the changes in pressure and volume of a gas, when sound waves propagated through it, are not isothermal, but adiabatic. A gas is a bad conductor of heat. It does not allow the free exchange of heat between compressed layer, rarefied layer and surrounding.
10. (e) The velocity of every oscillating particle of the medium is different of its different positions in one oscillation but the velocity of wave motion is always constant *i.e.*, particle velocity vary with respect to time, while the wave velocity is independent of time.
 Also for wave propagation medium must have the properties of elasticity and inertia.
11. (d) A bucket can be treated as a pipe closed at one end. The frequency of the note produced $= \frac{v}{4L}$, here L equal to depth

of water level from the open end. As the bucket is filled with water L decreases, hence frequency increases. Therefore, frequency or pitch of sound produced goes on increasing.

Also, the frequency of woman voice is usually higher than that of man.

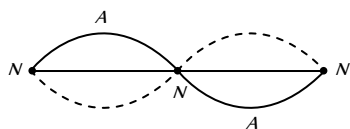
12. (b) A tuning fork is made of a material for which elasticity does not change. Since the alloy of nickel, steel and chromium (elinvar) has constant elasticity, therefore it is used for the preparation of tuning fork.
13. (e) Speed of sound in cases in independent of pressure because $v = \sqrt{\frac{\gamma P}{\rho}}$. At constant temperature, if P changes then ρ also changes in such a way that the ratio $\frac{P}{\rho}$ remains constant
 hence there is no effect of the pressure change on the speed of sound.
14. (a) For the propagation of transverse waves, medium must have the property of rigidity. Because gases have no rigidity, (they do not posses shear elasticity), hence transverse waves cannot be produced is gases. On the other hand, the solids possess both volume and shear elasticity and likewise both the longitudinal and transverse waves can be transmitted through them.
15. (c) Velocity of sound in a gas $v = \sqrt{\frac{\gamma P}{\rho}}$. For monoatomic gas $\gamma = 1.67$; for diatomic $\gamma = 1.40$. Therefore v is larger in case of monoatomic gas compared to its values in diatomic gas.
16. (a) The velocity of sound in solid is given by, $v = \sqrt{E / \rho}$. Though ρ is large for solids, but their coefficient of elasticity E is much larger (compared to that of liquids and gases). That is why v is maximum in case of solid.
17. (d) When moisture is present in air, the density of air decreases. It is because the density of water vapours is less than that of dry air. The velocity of sound is inversely proportional to the square root of density, hence sound travel faster in moist air than in the dry air. Therefore, on a rainy day sound travels faster than on a dry day.
18. (b) According to the property of persistence of hearing, the impression of a sound heard persists on our mind for $\frac{1}{10}$ sec.
 Therefore, number of beats per sec should be less than 10. Hence difference in frequencies of two sources must be less than 10.
19. (b) Sound produced by an open organ pipe is richer because it contains all harmonics and frequency of fundamental note in an open organ pipe is twice the fundamental frequency in a closed organ pipe of same length.
 Reason is also correct, but it is not explaining the assertion.
20. (a) Since the initial phase difference between the two waves coming from different violins changes, therefore, the waves produced by two different violins does not interfere because two waves interfere only when the phase difference between them remain constant throughout.
21. (d) As emission of light from atom is a random and rapid phenomenon. The phase at a point due to two independent light source will change rapidly and randomly. Therefore,

instead of beats, we shall get uniform intensity. However if light sources are LASER beams of nearly equal frequencies, it may possible to observe the phenomenon of beats in light.

22. (c) The person will hear the loud sound at nodes than at antinodes. We know that at anti-nodes the displacement is maximum and pressure change is minimum while at nodes the displacement is zero and pressure change is maximum. The sound is heard due to variation of pressure.

Also in stationary waves particles in two different segment vibrates in opposite phase.

23. (a) Stationary wave



A node is a place of zero amplitude and an antinode is a place of maximum amplitude.

24. (c) The principle of superposition does not state that the frequencies of the oscillation should be nearly equal. For beats to be heard the condition is that difference in frequencies of the two oscillations should not be more than 10 times per seconds for a normal human ear to recognise it. Hence we cannot hear beats in the case of two tuning forks vibrating at frequencies 256 Hz and 512 Hz respectively.

25. (a) The fundamental frequency of an open organ pipe is $n = \frac{v}{2l}$.

As temperature increases, both v and l increase but v increases more rapidly than l . Hence, the fundamental frequency increases as the temperature increases.

26. (b) Since, velocity of sound,

$$v = \sqrt{\frac{E}{\rho}}$$

As, the elasticity of solid is large than that of gases. Hence, it is obvious that velocity of sound is greater in solids than in gases.

27. (d)

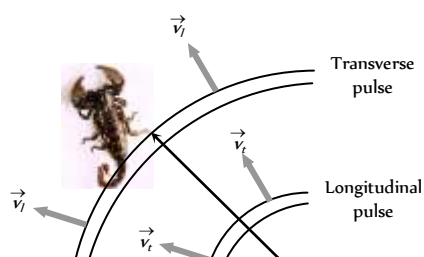
28. (b) Velocity of wave = $\frac{\text{Distance travelled by wave } (\lambda)}{\text{Time period } (T)}$

Wavelength is also defined as the distance between two nearest points in phase.

29. (c) Speed of light is greater than that of sound, hence flash of lightening is seen before the sound of thunder.

30. (a) A beetle motion sends fast longitudinal pulses and slower transverse waves along the sand surface. The sand scorpion first intercept the longitudinal pulses and learns the direction of the beetle; it is in the direction of which ever leg is disturbed earliest by the pulses. The scorpion then senses the time interval (Δt) between that first interception and the interception of slower transverse waves and uses it to determine the distance of the beetle. The distance is given by

$$\Delta t = \frac{d}{v_t} - \frac{d}{v_l}$$



31. (e)

Waves and Sound

Self Evaluation Test -17

1. An engine is moving on a circular track with a constant speed. It is blowing a whistle of frequency 500 Hz . The frequency received by an observer standing stationary at the centre of the track is

- (a) 500 Hz
 (b) More than 500 Hz
 (c) Less than 500 Hz
 (d) More or less than 500 Hz depending on the actual speed of the engine



2. In a resonance tube, the first resonance is obtained when the level of water in the tube is at 16 cm from the open end. Neglecting end correction, the next resonance will be obtained when the level of water from the open end is

- (a) 24 cm (b) 32 cm
 (c) 48 cm (d) 64 cm

3. To raise the pitch of a stringed musical instrument the player can

- (a) Loosen the string (b) Tighten the string
 (c) Shorten the string (d) Both (b) and (c)

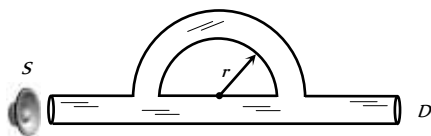
4. A wave travelling along positive x -axis is given by $y = A \sin(\omega t - kx)$. If it is reflected from rigid boundary such that 80% amplitude is reflected, then equation of reflected wave is

- (a) $y = A \sin(\omega t + kx)$ (b) $y = -0.8A \sin(\omega t + kx)$
 (c) $y = 0.8A \sin(\omega t + kx)$ (d) $y = A \sin(\omega t + 0.8 kx)$

5. The frequency of the first harmonic of a string stretched between two points is 100 Hz . The frequency of the third overtone is

- (a) 200 Hz (b) 300 Hz
 (c) 400 Hz (d) 600 Hz

6. A sound wave of wavelength 32 cm enters the tube at S as shown in the figure. Then the smallest radius r so that a minimum of sound is heard at detector D is



- (a) 7 cm (b) 14 cm
 (c) 21 cm (d) 28 cm

7. A stretched wire of length 110 cm is divided into three segments whose frequencies are in ratio $1 : 2 : 3$. Their lengths must be

- (a) $20\text{ cm} ; 30\text{ cm} ; 60\text{ cm}$
 (b) $60\text{ cm} ; 30\text{ cm} ; 20\text{ cm}$
 (c) $60\text{ cm} ; 20\text{ cm} ; 30\text{ cm}$
 (d) $30\text{ cm} ; 60\text{ cm} ; 20\text{ cm}$

8. Unlike a laboratory sonometer, a stringed instrument is seldom plucked in the middle. Supposing a sitar string is plucked at about

$\frac{1}{4}$ th of its length from the end. The most prominent harmonic would be

- (a) Eighth (b) Fourth
 (c) Third (d) Second

9. If $n_1, n_2, n_3, \dots$ are the frequencies of segments of a stretched string, the frequency n of the string is given by

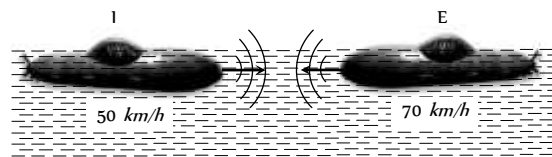
- (a) $n = n_1 + n_2 + n_3 + \dots$ (b) $n = \sqrt{n_1 \times n_2 \times n_3 \times \dots}$
 (c) $\frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3} + \dots$ (d) None of these

10. The equation of stationary wave along a stretched string is given by

$y = 5 \sin \frac{\pi x}{3} \cos 40\pi t$ where x and y are in centimetre and t in second. The separation between two adjacent nodes is :

- (a) 6 cm (b) 4 cm
 (c) 3 cm (d) 1.5 cm

11. An Indian submarine and an enemy submarine move towards each other during maneuvers in motionless water in the Indian ocean. The Indian submarine moves at 50 km/h , and the enemy submarine at 70 km/h . The Indian sub sends out a sonar signal (sound wave in water) at 1000 Hz . Sonar waves travel at 5500 km/h . What is the frequency detected by the Indian submarine



- (a) 1.02 kHz (b) 2 kHz
 (c) 2.5 kHz (d) 4.7 kHz

12. Two trains, one coming towards and another going away from an observer both at 4 m/s produce whistle simultaneously of frequency 300 Hz . Find the number of beats produced

- (a) 5 (b) 6
 (c) 7 (d) 12

13. A source of sound emits $200\pi W$ power which is uniformly distributed over a sphere of 10 m radius. What is the loudness of sound on the surface of a sphere

- (a) 200 dB (b) $200\pi\text{ dB}$
 (c) 120 dB (d) $120\pi\text{ dB}$

14. When a wave travels in a medium, the particle displacement is given by $y(x, t) = 0.03 \sin \pi(2t - 0.01x)$ where y and x are meters and t in seconds. The phase difference, at a given instant of time between two particle 25 m apart in the medium, is

- (a) $\frac{\pi}{8}$ (b) $\frac{\pi}{4}$

- (c) $\frac{\pi}{2}$ (d) π
15. A sine wave has an amplitude A and wavelength λ . Let V be the wave velocity and v be the maximum velocity of a particle in the medium. Then [KCET 2001]
- (a) $V = v$ if $\lambda = \frac{3A}{2\pi}$ (b) $V = v$ if $A = 2\pi\lambda$
- (c) $V = v$ if $A = \frac{\lambda}{2\pi}$ (d) V can not be equal to v
16. A pipe open at both ends produces a note of frequency f . When the pipe is kept with $\frac{3}{4}$ th of its length in water, it produced a note of frequency f_1 . The ratio $\frac{f_1}{f}$ is [KCET 1998]
- (a) $\frac{3}{4}$ (b) $\frac{4}{3}$
- (c) $\frac{1}{2}$ (d) 2
17. A man fires a bullet standing between two cliffs. First echo is heard after 3 seconds and second echo is heard after 5 seconds. If the velocity of sound is 330 m/s, then the distance between the cliffs is
- (a) 1650 m (b) 1320 m
- (c) 990 m (d) 660 m
18. The equation for spherical progressive wave is (where r is the distance from the source) [CPMT 2002]
- (a) $y = a \sin(\omega t - kx)$ (b) $y = \frac{a}{\sqrt{r}} \sin(\omega t - kx)$
- (c) $y = \frac{a}{2} \sin(\omega t - kx)$ (d) $y = \frac{a}{r} \sin(\omega t - kx)$
19. A tuning fork A produces 4 beats/sec with another tuning fork B of frequency 320 Hz. On filing the fork A , 4 beats/sec are again heard. The frequency of fork A , after filing is [KCET (Engg./Med.) 1999]
- (a) 324 Hz (b) 320 Hz
- (c) 316 Hz (d) 314 Hz
20. The number of beats produced per second by two vibrations: $x_1 = \sin 646\pi t$ and $x_2 = \sin 652\pi t$ is
- (a) 2 (b) 3
- (c) 4 (d) 6
21. 50 tuning forks are arranged in increasing order of their frequencies such that each gives 4 beats/sec with its previous tuning fork. If the frequency of the last fork is octave of the first, then the frequency of the first tuning fork is [DPMT 2005]
- (a) 200 Hz (b) 204 Hz
- (c) 196 Hz (d) None of these
22. The fundamental frequency of a closed pipe is 220 Hz. If $\frac{1}{4}$ of the pipe is filled with water, the frequency of the first overtone of the pipe now is [EAMCET (Med.) 2000]
- (a) 220 Hz (b) 440 Hz
- (c) 880 Hz (d) 1760 Hz
23. A glass tube 1.5 m long and open at both ends, is immersed vertically in a water tank completely. A tuning fork of 660 Hz is vibrated and kept at the upper end of the tube and the tube is gradually raised out of water. The total number of resonances heard before the tube comes out of water, taking velocity of sound air 330 m/sec is [EAMCET (Engg.) 1999]
- (a) 12 [AFMC 2000] (b) 6
- (c) 8 (d) 4
24. In the 5th overtone of an open organ pipe, these are (N -stands for nodes and A -for antinodes)
- (a) 2N, 3A (b) 3N, 4A
- (c) 4N, 5A (d) 5N, 4A
25. An engine approaches a hill with a constant speed. When it is at a distance of 0.9 km it blows a whistle, whose echo is heard by the driver after 5 sec. If speed of sound in air is 330 m/s, the speed of engine is



- (a) 10 m/s (b) 20 m/s
- (c) 30 m/s (d) 40 m/s

AS Answers and Solutions

(SET -17)

1. (a) Since there is no relative motion between the source and listener, So apparent frequency equals original frequency.
2. (c) Next resonance length after the fundamental is $3l_1 = 3 \times 16 = 48 \text{ cm}$.
3. (d) Higher pitch means higher frequency
Frequency of a stringed system is given by

$$n = \frac{p}{2l} \sqrt{\frac{T}{m}} \Rightarrow n \propto \frac{\sqrt{T}}{l}$$

Hence, to get higher frequency (higher pitch) tension should be increase and length should be shorten.

4. (b) On getting reflected from a rigid boundary the wave suffers
Hence if $y_{\text{incident}} = A \sin(\omega t - kx)$

then $y_{\text{reflected}} = (0.8A) \sin\{\omega t - k(-x) + \pi\}$

$= -0.8A \sin(\omega t + kx)$ an additional phase change of π .

5. (c) Third overtone is the fourth harmonic i.e.,
 $n_4 = 4n_1 = 4 \times 100 = 400 \text{ Hz}$

6. (b) Path difference $(\pi - 2r) = \frac{\lambda}{2} = \frac{32}{2} = 16$,

$$r = \frac{16}{\pi - 2} = 14 \text{ cm}.$$

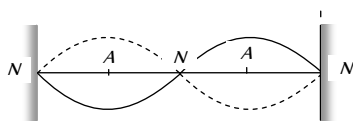
7. (b) $l_1 + l_2 + l_3 = 110 \text{ cm}$ and $n_1 l_1 = n_2 l_2 = n_3 l_3$

$$n_1 : n_2 : n_3 :: 1 : 2 : 3$$

$$\therefore \frac{n_1}{n_2} = \frac{1}{2} = \frac{l_2}{l_1} \Rightarrow l_2 = \frac{l_1}{2} \text{ and } \frac{n_1}{n_3} = \frac{1}{3} = \frac{l_3}{l_1} \Rightarrow l_3 = \frac{l_1}{3}$$

$$\therefore l_1 + \frac{l_1}{2} + \frac{l_1}{3} = 110 \text{ so } l_1 = 60 \text{ cm}, l_2 = 30 \text{ cm}, l_3 = 20 \text{ cm}.$$

8. (d) When plucked at one fourth it gives two loops, and hence 2nd harmonic is produced.



9. (c) For a vibrating string

$$n_1 l_1 = n_2 l_2 = n_3 l_3, \dots = \text{constant} = k \text{ (say)} = n l$$

$$\text{Also } l_1 + l_2 + l_3 + l_4 + \dots = 1$$

$$\frac{k}{n_1} + \frac{k}{n_2} + \frac{k}{n_3} + \frac{k}{n_4} + \dots = \frac{k}{n} \Rightarrow \frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3} + \dots$$

10. (c) Given $y = 5 \sin \frac{\pi x}{3} \cos 40 \pi t$

$$\text{Comparing with } y = 2a \cos \frac{2\pi x}{\lambda} \sin \frac{2\pi t}{T} \Rightarrow \lambda = 6 \text{ cm}.$$

$$\therefore \text{The separation between adjacent nodes} = \frac{\lambda}{2} = 3 \text{ cm}.$$

11. (a) Frequency detected by Indian submarine

$$n' = n \left[\frac{v + v_{\text{sub}}}{v - v_{\text{sub}}} \right] = 1000 \left[\frac{5500 + 50}{5500 - 50} \right] \approx 1.02 \text{ kHz}.$$

12. (c) $\Delta n = \left[\frac{v}{v-u} - \frac{v}{v+u} \right] n = \frac{2uv}{v^2 - u^2} n$

$$= \frac{2 \times 4 \times 332}{(332)^2 - (4)^2} \times 300 \approx 7$$

13. (c) Intensity = $\frac{\text{power}}{\text{area}} = \frac{200\pi}{2\pi \times 10^{-2}} = 1 \text{ Watt/m}$

$$\text{Now } L = 10 \log_{10} \frac{I}{I_0} = 10 \log_{10} \left(\frac{1}{10^{-12}} \right)$$

$$= 10 \log_{10} 10^{12} = 120 \text{ dB}$$

14. (b) $y(x, t) = 0.03 \sin \pi(2t - 0.01x) = 0.03 \sin(2\pi t - 0.01\pi x)$

$$k = 0.01\pi = \frac{2\pi}{\lambda} \Rightarrow \Delta \phi = \frac{2\pi}{\lambda} \Delta x = 0.01\pi \times 25 = \frac{\pi}{4}$$

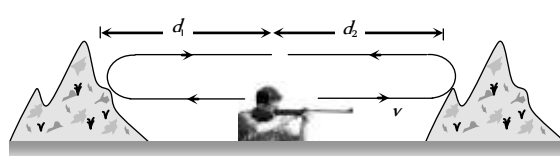
15. (c) Let wave velocity (v) = maximum particle velocity $\Rightarrow$

$$n\lambda = \omega A = 2\pi nA \Rightarrow A = \frac{\lambda}{2\pi}$$

16. c) For open pipe $f_1 = \frac{v}{2l}$ and for closed pipe

$$f_2 = \frac{v}{4 \times \left(\frac{l}{4}\right)} = \frac{v}{l} = 2f_1 \Rightarrow \frac{f_1}{f_2} = \frac{1}{2}$$

17. (b)



$$2(d_1 + d_2) = v(t_1 + t_2) \Rightarrow d_1 + d_2 = \frac{330 \times (3 + 5)}{2} = 1320 \text{ m}$$

18. (d) For spherical wave intensity (I) $\propto \frac{1}{(\text{Distance})^2}$

$$\text{also } I \propto a^2 \Rightarrow a \propto \frac{1}{r}. \text{ Hence equation of a cylindrical wave}$$

$$\text{is } y = \frac{1}{r} \sin(\omega t - kx)$$

19. (a) $n_i = ?$, n_e = Known frequency = 320 Hz

$$x = 4 \text{ bps, which remains same after filing.}$$

$$\text{Unknown fork A is filed so } n_i \uparrow$$

$$\text{Hence } n_i \uparrow - n_e = x \quad \text{--- Wrong}$$

$$n_e - n_i \uparrow = x \quad \text{--- Correct}$$

$$\Rightarrow n_i = n_e - x = 320 - 4 = 316 \text{ Hz}.$$

$$\text{This is the frequency before filing.}$$

$$\text{But in question frequency after filing is asked which must be greater than } 316 \text{ Hz, such that it produces } 4 \text{ beats per sec. Hence it is } 324 \text{ Hz.}$$

20. (b) From the given equation $\omega_1 = 2\pi n_1 = 646\pi \Rightarrow n_1 = 323$

$$\text{and } \omega_2 = 2\pi n_2 = 652\pi \Rightarrow n_2 = 326$$

$$\text{Hence, beat frequency} = 326 - 323 = 3$$

21. (c) Frequencies of tuning forks is given by

$$n_{\text{last}} = n_{\text{first}} + (N - 1)x$$

$$2n = n + (50 - 1) \times 4 \Rightarrow n = 196 \text{ Hz}.$$

22. (c) Fundamental frequency of closed pipe

$$n = \frac{v}{4l} = 220 \text{ Hz} \Rightarrow v = 220 \times 4l$$

If $\frac{1}{4}$ of the pipe is filled with water then remaining length of

air column is $\frac{3l}{4}$

$$\text{Now fundamental frequency} = \frac{v}{4\left(\frac{3l}{4}\right)} = \frac{v}{3l} \text{ and}$$

First overtone = $3 \times$ fundamental frequency

$$= \frac{3v}{3l} = \frac{v}{l} = \frac{220 \times 4l}{l} = 880 \text{ Hz}.$$

23. (b) Suppose N resonance occurred before tube coming out.

$$\text{Hence by using } l = \frac{(2N-1)v}{4n}$$

$$\Rightarrow 1.5 = \frac{(2N-1) \times 330}{4 \times 660} \Rightarrow N \approx 6.$$

24. (c) In open organ pipe 5th overtone corresponds to 4th harmonic mode.

Also in open pipe, Number of nodes = Order of mode of vibration and number of antinodes = (Number of nodes + 1). Here number of nodes = 4, Number of antinodes = 4 + 1 = 5.

25. (c) If the speed of engine is v , the distance traveled by engine in 5 sec will be $5v$, and hence the distance traveled by sound in reaching the hill and coming back to the moving driver = $900 + (900 - 5v) = 1800 - 5v$

So the time interval between original sound and its echo

$$t = \frac{(1800 - 5v)}{330} = 5 \Rightarrow v = 30 \text{ m/s}.$$



Chapter 19

Current Electricity

Electric Current

(1) The time rate of flow of charge through any cross-section is called current. $i = \lim_{\Delta t \rightarrow 0} \frac{\Delta Q}{\Delta t} = \frac{dQ}{dt}$. If flow is uniform then $i = \frac{Q}{t}$. Current is a scalar quantity. It's S.I. unit is *ampere* (A) and C.G.S. unit is *emu* and is called *biot* (Bi), or *ab ampere*. $1A = (1/10) Bi(ab amp.)$

(2) *Ampere* of current means the flow of 6.25×10^{18} *electrons/sec* through any cross-section of the conductor.

(3) The conventional direction of current is taken to be the direction of flow of positive charge, i.e. field and is opposite to the direction of flow of negative charge.

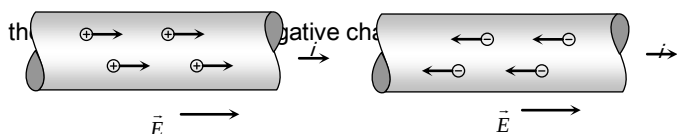


Fig. 19.1

(4) The net charge in a current carrying conductor is zero.

(5) For a given conductor current does not change with change in cross-sectional area. In the following figure $i_1 = i_2 = i_3$

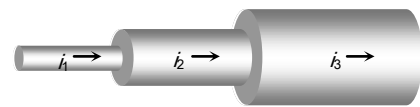


Fig. 19.2

(6) **Current due to translatory motion of charge :** If n particle each having a charge q , pass through a given area in time t then $i = \frac{nq}{t}$

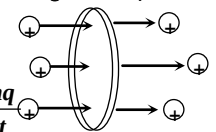


Fig. 19.3

If n particles each having a charge q pass per second per unit area, the current associated with cross-sectional area A is

$$i = nqA$$

If there are n particle per unit volume each having a charge q and moving with velocity v , the current thorough, cross section A is $i = nqvA$

Table : 19.1 Types of current

Alternating current (ac)	Direct current (dc)	
(i)	(i) (Pulsating dc)	(Constant dc)

2 Current Electricity

Magnitude and direction both varies with time ac $\rightarrow$ Rectifier $\rightarrow$ dc	dc $\rightarrow$ Inverter $\rightarrow$ ac
(ii) Shows heating effect only	(ii) Shows heating effect, chemical effect and magnetic effect of current
(iii) It's symbol is 	(iii) It's symbol is 

(7) **Current due to rotatory motion of charge** : If a point charge q is moving in a circle of radius r with speed v (frequency ν , angular speed ω and time period T) then corresponding current $i = qv = \frac{q}{T} = \frac{qv}{2\pi r} = \frac{q\omega}{2\pi}$

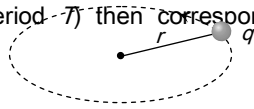


Fig. 19.4

(8) **Current carriers** : The charged particles whose flow in a definite direction constitutes the electric current are called current carriers. In different situation current carriers are different.

- (i) Solids : In solid conductors like metals current carriers are free electrons.
- (ii) Liquids : In liquids current carriers are positive and negative ions.
- (iii) Gases : In gases current carriers are positive ions and free electrons.
- (iv) Semi conductor : In semi conductors current carriers are holes and free electrons.

Current Density (J)

Current density at any point inside a conductor is defined as a vector having magnitude equal to current per unit area surrounding that point. Remember area is normal to the direction of charge flow (or current passes) through that point.

(1) Current density at point P is given by $\vec{J} = \frac{di}{dA} \vec{n}$

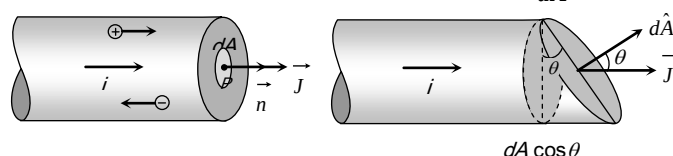


Fig. 19.5

(2) If the cross-sectional area is not normal to the current, but makes an angle θ with the direction of current then

$$J = \frac{di}{dA \cos \theta} \Rightarrow di = J dA \cos \theta = \vec{J} \cdot d\vec{A} \Rightarrow i = \int \vec{J} \cdot d\vec{A}$$

(3) If current density $\vec{J}$ is uniform for a normal cross-section $\vec{A}$ then $J = \frac{i}{A}$

(4) Current density $\vec{J}$ is a vector quantity. It's direction is same as that of $\vec{E}$. It's S.I. unit is amp/m^2 and dimension $[L^{-2}A]$.

(5) In case of uniform flow of charge through a cross-section normal to it as $i = nqvA \Rightarrow J = \frac{i}{A} = nqv$.

(6) Current density relates with electric field as $\vec{J} = \sigma \vec{E} = \frac{\vec{E}}{\rho}$; where σ = conductivity and ρ = resistivity or specific resistance of substance.

Drift Velocity

Drift velocity is the average uniform velocity acquired by free electrons inside a metal by the application of an electric field which is responsible for current through it. Drift velocity is very small it is of the order of 10^{-4} m/s as compared to thermal speed ($\approx 10^5 \text{ m/s}$) of electrons at room temperature.

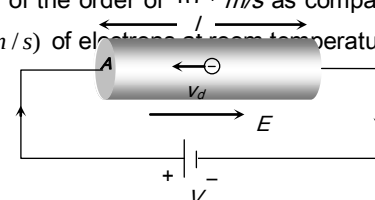


Fig. 19.6

If suppose for a conductor

n = Number of electron per unit volume of the conductor

A = Area of cross-section

V = potential difference across the conductor

E = electric field inside the conductor

i = current, J = current density, ρ = specific resistance, σ = conductivity ($\sigma = \frac{1}{\rho}$) then current relates with drift velocity as

$i = neAv_d$ we can also write

$$v_d = \frac{i}{neA} = \frac{J}{ne} = \frac{\sigma E}{ne} = \frac{E}{\rho ne} = \frac{V}{\rho l ne}$$

(1) The direction of drift velocity for electron in a metal is opposite to that of applied electric field (*i.e.* current density $\vec{J}$).

$v_d \propto E$ *i.e.*, greater the electric field, larger will be the drift velocity.

(2) When a steady current flows through a conductor of non-uniform cross-section drift velocity varies inversely with area of cross-section ($v_d \propto \frac{1}{A}$)

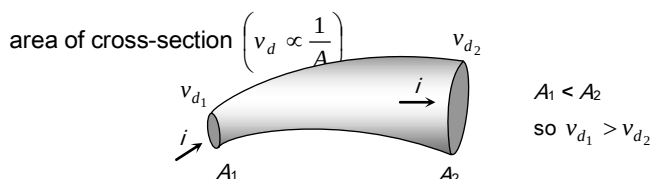


Fig. 19.7

(3) If diameter (d) of a conductor is doubled, then drift velocity of electrons inside it will not change.

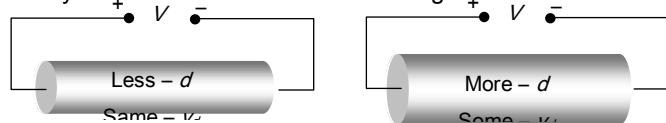


Fig. 19.8

(1) **Relaxation time (τ)** : The time interval between two successive collisions of electrons with the positive ions in the

metallic lattice is defined as relaxation time $\tau = \frac{\text{mean free path}}{\text{r.m.s. velocity of electrons}} = \frac{\lambda}{v_{rms}}$. With rise in temperature v_{rms} increases consequently τ decreases.

(2) **Mobility** : Drift velocity per unit electric field is called mobility of electron *i.e.* $\mu = \frac{v_d}{E}$. Its unit is $\frac{m^2}{\text{volt} - \text{sec}}$.

Ohm's Law

If the physical conditions of the conductor (length, temperature, mechanical strain etc.) remains same, then the current flowing through the conductor is directly proportional to the potential difference across its two ends *i.e.* $i \propto V \Rightarrow V = iR$ where R is a proportionality constant, known as electric resistance.

(1) Ohm's law is not a universal law, the substances, which obey Ohm's law are known as ohmic substance.

(2) Graph between V and i for a metallic conductor is a straight line as shown. At different temperatures V - i curves are different.

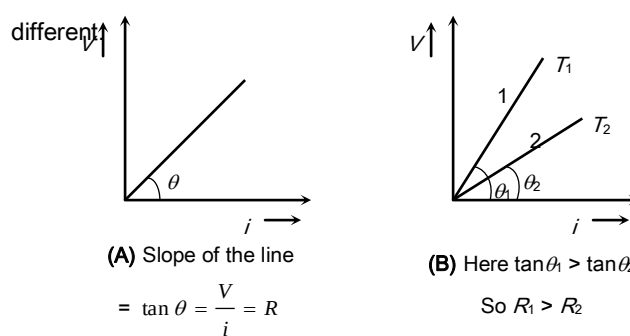


Fig. 19.9

(3) The device or substances which don't obey Ohm's law *e.g.* gases, crystal rectifiers, thermionic valve, transistors etc. are known as non-ohmic or non-linear conductors. For these V - i curve is not linear.

$$\text{Static resistance } R_{st} = \frac{V}{i} = \frac{1}{\tan \theta}$$

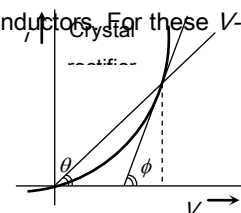


Fig. 19.10

$$\text{Dynamic resistance } R_{dyn} = \frac{\Delta V}{\Delta I} = \frac{1}{\tan \phi}$$

Resistance

(1) The property of substance by virtue of which it opposes the flow of current through it, is known as the resistance.

(2) **Formula of resistance** : For a conductor if l = length of a conductor A = Area of cross-section of conductor, n = No. of free electrons per unit volume in conductor, τ = relaxation time then resistance of conductor $R = \rho \frac{l}{A} = \frac{m}{ne^2 \tau} \cdot \frac{l}{A}$; where ρ = resistivity of the material of conductor

(3) **Unit and dimension** : It's S.I. unit is *Volt/Amp.* or *Ohm* (Ω).

$$\text{Also } 1 \text{ ohm} = \frac{1 \text{ volt}}{1 \text{ Amp}} = \frac{10^8 \text{ emu of potential}}{10^{-1} \text{ emu of current}} = 10^9 \text{ emu of resistance.}$$

It's dimension is $[ML^2T^{-3}A^{-2}]$.

(4) **Dependence of resistance** : Resistance of a conductor depends upon the following factors.

(i) **Length of the conductor** : Resistance of a conductor is directly proportional to its length *i.e.* $R \propto l$ and inversely proportional to its area of cross-section *i.e.* $R \propto \frac{1}{A}$

(ii) **Temperature** : For a conductor

Resistance $\propto$ temperature

If R_0 = resistance of conductor at $0^\circ C$

R_t = resistance of conductor at $t^\circ C$

and α, β = temperature co-efficient of resistance

then $R_t = R_0(1 + \alpha t + \beta t^2)$ for $t > 300^\circ C$ and

$$R_t = R_0(1 + \alpha t) \text{ for } t \leq 300^\circ C \text{ or } \alpha = \frac{R_t - R_0}{R_0 \times t}$$

If R_1 and R_2 are the resistances at $t_1^\circ C$ and $t_2^\circ C$ respectively then $\frac{R_1}{R_2} = \frac{1 + \alpha t_1}{1 + \alpha t_2}$.

The value of α is different at different temperature. Temperature coefficient of resistance averaged over the temperature range $t_1^\circ C$ to $t_2^\circ C$ is given by $\alpha = \frac{R_2 - R_1}{R_1(t_2 - t_1)}$ which

gives $R_2 = R_1 [1 + \alpha (t_2 - t_1)]$. This formula gives an approximate value.

Table 19.2 : Variation of resistance of some electrical material

with temperature		
Material	Temp. coefficient of resistance (α)	Variation of resistance with temperature rise
Metals	Positive	Increases
Solid non-metal	Zero	Independent
Semi-conductor	Negative	Decreases
Electrolyte	Negative	Decreases
Ionised gases	Negative	Decreases
Alloys	Small positive value	Almost constant

Resistivity (ρ), Conductivity (σ) and Conductance (C)

(1) **Resistivity** : From $R = \rho \frac{l}{A}$; If $l = 1m$, $A = 1 m^2$ then

$R = \rho$ *i.e.* resistivity is numerically equal to the resistance of a substance having unit area of cross-section and unit length.

(i) **Unit and dimension** : It's S.I. unit is *ohm $\times m$* and dimension is $[ML^3T^{-3}A^{-2}]$

(ii) It's formula : $\rho = \frac{m}{ne^2 \tau}$

(iii) Resistivity is the intrinsic property of the substance. It is independent of shape and size of the body (*i.e.* l and A).

(iv) For different substances their resistivity is also different *e.g.* $\rho_{\text{silver}} = \text{minimum} = 1.6 \times 10^{-8} \Omega\text{-m}$ and $\rho_{\text{fused quartz}} = \text{maximum} \approx 10^{16} \Omega\text{-m}$

$$\rho_{\text{insulator (Maximum for fused quartz)}} > \rho_{\text{alloy}} > \rho_{\text{semi-conductor}} > \rho_{\text{conductor (Minimum for silver)}}$$

(v) Resistivity depends on the temperature. For metals $\rho_t = \rho_0(1 + \alpha \Delta t)$ *i.e.* resistivity increases with temperature.

(vi) Resistivity increases with impurity and mechanical stress.

(vii) Magnetic field increases the resistivity of all metals except iron, cobalt and nickel.

(viii) Resistivity of certain substances like selenium, cadmium, sulphides is inversely proportional to intensity of light falling upon them.

(2) **Conductivity** : Reciprocal of resistivity is called conductivity (σ) i.e. $\sigma = \frac{1}{\rho}$ with unit mho/m and dimensions $[M^{-1}L^{-3}T^3A^2]$.

(3) **Conductance** : Reciprocal of resistance is known as conductance. $C = \frac{1}{R}$ Its unit is $\frac{1}{\Omega}$ or Ω^{-1} or "Siemen".

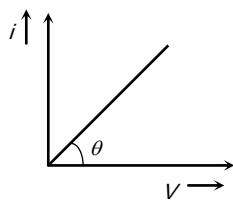


Fig. 19.11

Stretching of Wire

If a conducting wire stretches, its length increases, area of cross-section decreases so resistance increases but volume remain constant.

Suppose for a conducting wire before stretching its length = l_1 , area of cross-section = A_1 , radius = r_1 , diameter = d_1 , and resistance $R_1 = \rho \frac{l_1}{A_1}$

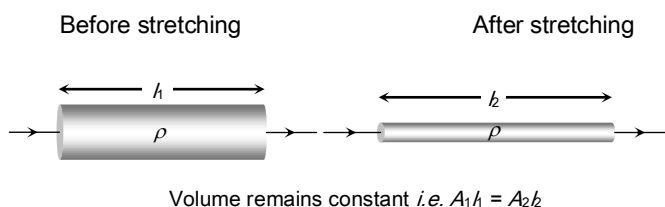


Fig. 19.12

After stretching length = l_2 , area of cross-section = A_2 , radius = r_2 , diameter = d_2 and resistance = $R_2 = \rho \frac{l_2}{A_2}$

Ratio of resistances before and after stretching

$$\frac{R_1}{R_2} = \frac{l_1}{l_2} \times \frac{A_2}{A_1} = \left(\frac{l_1}{l_2}\right)^2 = \left(\frac{A_2}{A_1}\right)^2 = \left(\frac{r_2}{r_1}\right)^4 = \left(\frac{d_2}{d_1}\right)^4$$

$$(1) \text{ If length is given then } R \propto l^2 \Rightarrow \frac{R_1}{R_2} = \left(\frac{l_1}{l_2}\right)^2$$

$$(2) \text{ If radius is given then } R \propto \frac{1}{r^4} \Rightarrow \frac{R_1}{R_2} = \left(\frac{r_2}{r_1}\right)^4$$

Electrical Conducting Materials For Specific Use

(1) **Filament of electric bulb** : Is made up of tungsten which has high resistivity, high melting point.

(2) **Element of heating devices (such as heater, geyser or press)** : Is made up of nichrome which has high resistivity and high melting point.

(3) **Resistances of resistance boxes (standard resistances)** : Are made up of alloys (manganin, constantan or nichrome) these materials have moderate resistivity which is practically independent of temperature so that the specified value of resistance does not alter with minor changes in temperature.

(4) **Fuse-wire** : Is made up of tin-lead alloy (63% tin + 37% lead). It should have low melting point and high resistivity. It is used in series as a safety device in an electric circuit and is designed so as to melt and thereby open the circuit if the current exceeds a predetermined value due to some fault. The function of a fuse is independent of its length.

Safe current of fuse wire relates with its radius as $i \propto r^{3/2}$.

(5) **Thermistors** : A thermistor is a heat sensitive resistor usually prepared from oxides of various metals such as nickel, copper, cobalt, iron etc. These compounds are also semiconductor. For thermistors α is very high which may be positive or negative. The resistance of thermistors changes very rapidly with change of temperature.

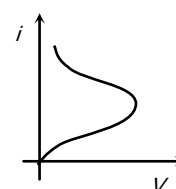


Fig. 19.13

Thermistors are used to detect small temperature change and to measure very low temperature.

Colour Coding of Resistance

To know the value of resistance colour code is used. These code are printed in form of set of rings or strips. By reading the values of colour bands, we can estimate the value of resistance.

The carbon resistance has normally four coloured rings or bands as shown in following figure.

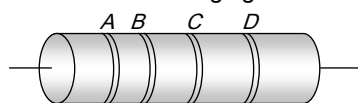


Fig. 19.14

Colour band A and B : Indicate the first two significant figures of resistance in *ohm*.

Band C : Indicates the decimal multiplier *i.e.* the number of zeros that follows the two significant figures A and B.

Band D : Indicates the tolerance in percent about the indicated value or in other words it represents the percentage accuracy of the indicated value.

The tolerance in the case of gold is $\pm 5\%$ and in silver is $\pm 10\%$. If only three bands are marked on carbon resistance, then it indicate a tolerance of 20%.

Table 19.3 : Colour code for carbon resistance

Letters as an aid to memory	Colour	Figure (A, B)	Multiplier (C)
B	Black	0	10^0
B	Brown	1	10^1
R	Red	2	10^2

O	Orange	3	10^3
Y	Yellow	4	10^4
G	Green	5	10^5
B	Blue	6	10^6
V	Violet	7	10^7
G	Grey	8	10^8
W	White	9	10^9

To remember the sequence of colour code following sentence should kept in memory.

B B R O Y Great Britain Very Good Wife.

Grouping of Resistance

(1) Series grouping

(i) Same current flows through each resistance but potential difference distributes in the ratio of resistance *i.e.* $V \propto R$

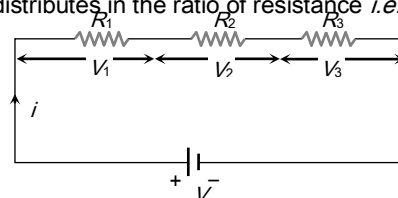


Fig. 19.15

(ii) $R_{eq} = R_1 + R_2 + R_3$ equivalent resistance is greater than the maximum value of resistance in the combination.

(iii) If n identical resistance are connected in series $R_{eq} = nR$ and potential difference across each resistance

$$V' = \frac{V}{n}$$

(2) Parallel grouping

(i) Same potential difference appeared across each resistance but current distributes in the reverse ratio of their resistance

$$i.e. i \propto \frac{1}{R}$$

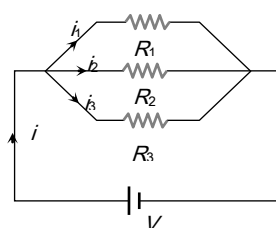


Fig. 19.16

(ii) Equivalent resistance is given by $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

$$\text{or } R_{eq} = (R_1^{-1} + R_2^{-1} + R_3^{-1})^{-1} \text{ or } R_{eq} = \frac{R_1 R_2 R_3}{R_1 R_2 + R_2 R_3 + R_3 R_1}$$

Equivalent resistance is smaller than the minimum value of resistance in the combination.

(iv) If two resistance in parallel

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} = \frac{\text{Multiplication}}{\text{Addition}}$$

(v) Current through any resistance

$$i' = i \times \left[\frac{\text{Resistance of opposite branch}}{\text{Total resistance}} \right]$$

Where i = required current (branch current),

i = main current

$$i_1 = i \left(\frac{R_2}{R_1 + R_2} \right)$$

$$\text{and } i_2 = i \left(\frac{R_1}{R_1 + R_2} \right)$$

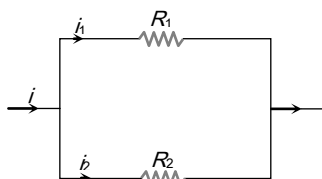


Fig. 19.17

(vi) In n identical resistance are connected in parallel

$$R_{eq} = \frac{R}{n} \text{ and current through each resistance } i' = \frac{i}{n}$$

Cell



The device which converts chemical energy into electrical energy is known as electric cell. Cell is a source of constant emf but not constant current.

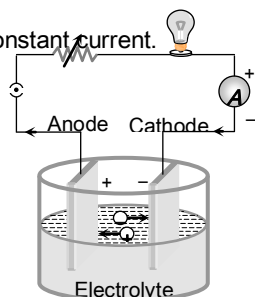
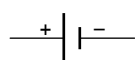


Fig. 19.18



Symbol of cell

(1) **Emf of cell ($\mathcal{E}$)** : The potential difference across the terminals of a cell when it is not supplying any current is called its emf.

(2) **Potential difference (V)** : The voltage across the terminals of a cell when it is supplying current to external resistance is called potential difference or terminal voltage. Potential difference is equal to the product of current and resistance of that given part i.e. $V = iR$.

(3) **Internal resistance (r)** : In case of a cell the opposition of electrolyte to the flow of current through it is called internal resistance of the cell. The internal resistance of a cell depends on the distance between electrodes ($r \propto d$), area of electrodes [$r \propto (1/A)$] and nature, concentration ($r \propto C$) and temperature of electrolyte [$r \propto (1/\text{temp.})$].

A cell is said to be ideal, if it has zero internal resistance.

Cell in Various Positions

(1) **Closed circuit** : Cell supplies a constant current in the circuit.

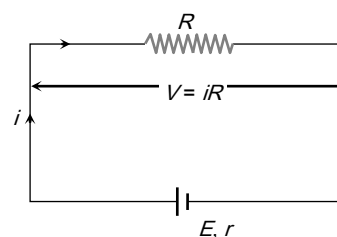


Fig. 19.19

(i) Current given by the cell $i = \frac{E}{R + r}$

(ii) Potential difference across the resistance $V = iR$

(iii) Potential drop inside the cell $= ir$

(iv) Equation of cell $E = V + ir$ ($E > V$)

(v) Internal resistance of the cell $r = \left(\frac{E}{V} - 1 \right) \cdot R$

(vi) Power dissipated in external resistance (load)

$$P = Vi = i^2 R = \frac{V^2}{R} = \left(\frac{E}{R+r} \right)^2 \cdot R$$

Power delivered will be maximum when $R = r$ so

$$P_{\max} = \frac{E^2}{4r}$$

This statement in generalised form is called "*maximum power transfer theorem*".

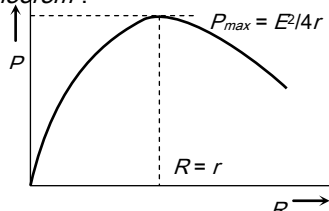


Fig. 19.20

(vii) When the cell is being charged *i.e.* current is given to the cell then $E = V - ir$ and $E < V$.

(2) **Open circuit** : When no current is taken from the cell it is said to be in open circuit

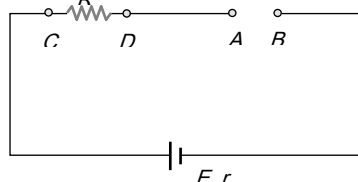


Fig. 19.21

(i) Current through the circuit $i = 0$

(ii) Potential difference between A and B, $V_{AB} = E$

(iii) Potential difference between C and D, $V_{CD} = 0$

(3) **Short circuit** : If two terminals of cell are join together by a thick conducting wire

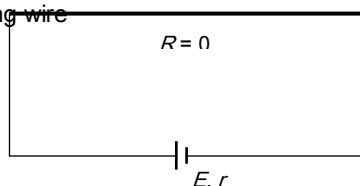


Fig. 19.22

(i) Maximum current (called short circuit current) flows momentarily $i_{sc} = \frac{E}{r}$

(ii) Potential difference $V = 0$

Grouping of Cells



Group of cell is called a battery.

In series grouping of cell's their emf's are additive or subtractive while their internal resistances are always additive. If dissimilar plates of cells are connected together their emf's are added to each other while if their similar plates are connected together their emf's are subtractive.



Fig. 19.23

(1) **Series grouping** : In series grouping anode of one cell is connected to cathode of other cell and so on. If n identical cells are connected in series

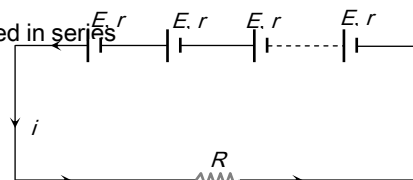


Fig. 19.24

(i) Equivalent emf of the combination $E_{eq} = nE$

(ii) Equivalent internal resistance $r_{eq} = nr$

(iii) Main current = Current from each cell $= i = \frac{nE}{R + nr}$

(iv) Potential difference across external resistance $V = iR$

(v) Potential difference across each cell $V' = \frac{V}{n}$

(vi) Power dissipated in the external circuit $= \left(\frac{nE}{R + nr} \right)^2 \cdot R$

(vii) Condition for maximum power $R = nr$ and

$$P_{\max} = n \left(\frac{E^2}{4r} \right)$$

(viii) This type of combination is used when $nr \ll R$.

(2) **Parallel grouping** : In parallel grouping all anodes are connected at one point and all cathode are connected together at other point. If n identical cells are connected in parallel

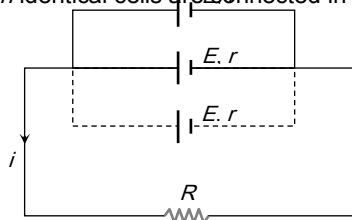


Fig. 19.25

(i) Equivalent emf $E_{eq} = E$

(ii) Equivalent internal resistance $R_{eq} = r/n$

(iii) Main current $i = \frac{E}{R + r/n}$

(iv) potential difference across external resistance = p.d. across each cell $= V = iR$

(v) Current from each cell $i' = \frac{i}{n}$

(vi) Power dissipated in the circuit $P = \left(\frac{E}{R + r/n} \right)^2 \cdot R$

(vii) Condition for max. power is $R = r/n$ and

$$P_{\max} = n \left(\frac{E^2}{4r} \right)$$

(viii) This type of combination is used when $nr \gg R$

(3) **Mixed Grouping** : If n identical cell's are connected in a row and such m row's are connected in parallel as shown.

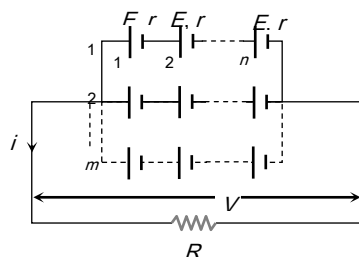


Fig. 19.26

(i) Equivalent emf of the combination $E_{eq} = nE$

(ii) Equivalent internal resistance of the combination

$$r_{eq} = \frac{nr}{m}$$

(iii) Main current flowing through the load

$$i = \frac{nE}{R + \frac{nr}{m}} = \frac{mnE}{mR + nr}$$

(iv) Potential difference across load $V = iR$

(v) Potential difference across each cell $V' = \frac{V}{n}$

(vi) Current from each cell $i' = \frac{i}{n}$

(vii) Condition for maximum power $R = \frac{nr}{m}$

$$\text{and } P_{\max} = (mn) \frac{E^2}{4r}$$

(viii) Total number of cell $= mn$

Kirchoff's Laws

(1) **Kirchoff's first law** : This law is also known as junction rule or current law (KCL). According to it the algebraic sum of currents meeting at a junction is zero i.e. $\sum i = 0$.

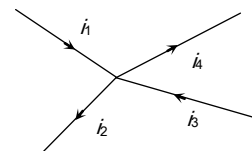


Fig. 19.27

In a circuit, at any junction the sum of the currents entering the junction must equal the sum of the currents leaving the junction. $i_1 + i_3 = i_2 + i_4$

(ii) This law is simply a statement of "conservation of charge".

(2) **Kirchoff's second law** : This law is also known as loop rule or voltage law (KVL) and according to it "the algebraic sum of the changes in potential in complete traversal of a mesh (closed loop) is zero", i.e. $\sum V = 0$

(i) This law represents "*conservation of energy*".

(ii) If there are n meshes in a circuit, the number of independent equations in accordance with loop rule will be $(n - 1)$.

(3) **Sign convention for the application of Kirchoff's law** :

For the application of Kirchoff's laws following sign convention are to be considered

(i) The change in potential in traversing a resistance in the direction of current is $-iR$ while in the opposite direction $+iR$

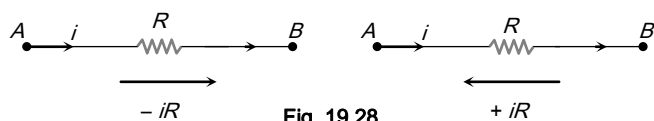


Fig. 19.28

(ii) The change in potential in traversing an emf source from negative to positive terminal is $+E$ while in the opposite direction $-E$ irrespective of the direction of current in the circuit.

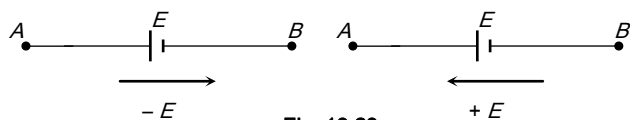


Fig. 19.29

(iii) The change in potential in traversing a capacitor from the negative terminal to the positive terminal is $+\frac{q}{C}$ while in opposite direction $-\frac{q}{C}$.

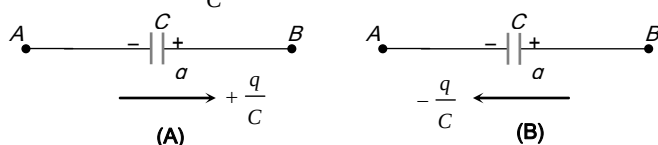


Fig. 19.30

(iv) The change in voltage in traversing an inductor in the direction of current is $-L \frac{di}{dt}$ while in opposite direction it is $+L \frac{di}{dt}$

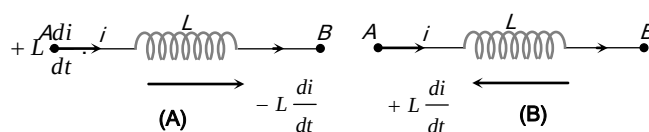


Fig. 19.31

Different Measuring Instruments



(1) **Galvanometer** : It is an instrument used to detect small current passing through it by showing deflection. Galvanometers are of different types e.g. moving coil galvanometer, moving magnet galvanometer, hot wire galvanometer. In dc circuit usually moving coil galvanometer are used.

(i) **It's symbol** : ; where G is the total internal resistance of the galvanometer.

(ii) **Full scale deflection current** : The current required for full scale deflection in a galvanometer is called full scale deflection current and is represented by i_g .

(iii) **Shunt** : The small resistance connected in parallel to galvanometer coil, in order to control current flowing through the galvanometer is known as shunt.

Table 19.4 : Merits and demerits of shunt

Merits of shunt	Demerits of shunt

To protect the galvanometer coil from burning	Shunt resistance decreases the sensitivity of galvanometer.
It can be used to convert any galvanometer into ammeter of desired range.	

(2) **Ammeter** : It is a device used to measure current and is always connected in series with the 'element' through which current is to be measured.

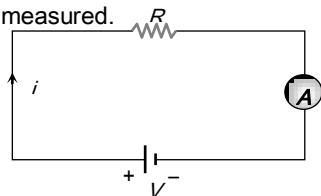


Fig. 19.32

(i) The reading of an ammeter is always lesser than actual current in the circuit.

(ii) Smaller the resistance of an ammeter more accurate will be its reading. An ammeter is said to be ideal if its resistance r is zero.

(iii) **Conversion of galvanometer into ammeter** : A galvanometer may be converted into an ammeter by connecting a low resistance (called shunt S) in parallel to the galvanometer G as shown in figure.

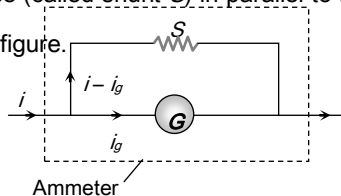


Fig. 19.33

(a) Equivalent resistance of the combination $= \frac{GS}{G+S}$

(b) G and S are parallel to each other hence both will have equal potential difference *i.e.* $i_g G = (i - i_g)S$; which gives

Required shunt $S = \frac{i_g}{(i - i_g)} G$

(c) To pass n th part of main current (*i.e.* $i_g = \frac{i}{n}$) through the galvanometer, required shunt $S = \frac{G}{(n-1)}$.

(3) **Voltmeter** : It is a device used to measure potential difference and is always put in parallel with the 'circuit element' across which potential difference is to be measured.

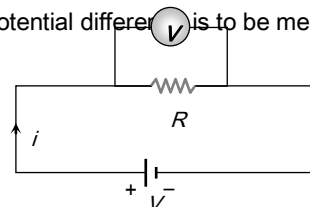


Fig. 19.34

(i) The reading of a voltmeter is always lesser than true value.

(ii) Greater the resistance of voltmeter, more accurate will be its reading. A voltmeter is said to be ideal if its resistance is infinite, *i.e.*, it draws no current from the circuit element for its operation.

(iii) **Conversion of galvanometer into voltmeter** : A galvanometer may be converted into a voltmeter by connecting a large resistance R in series with the galvanometer as shown in the figure.

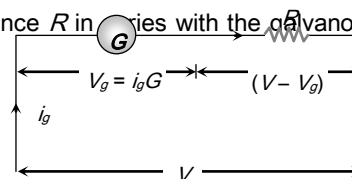


Fig. 19.35

(a) Equivalent resistance of the combination $= G + R$

(b) According to ohm's law $V = i_g (G + R)$; which gives

Required series resistance $R = \frac{V}{i_g} - G = \left(\frac{V}{V_g} - 1 \right) G$

(c) If n th part of applied voltage appeared across galvanometer (*i.e.* $V_g = \frac{V}{n}$) then required series resistance

$R = (n-1) G$.

(4) Wheatstone bridge

: Wheatstone bridge is an arrangement of four resistance which can be used to measure one of them in terms of rest. Here arms AB and BC are called ratio arm and arms AC and BD are called conjugate arms

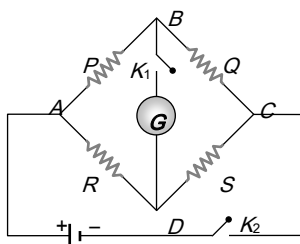


Fig. 19.36

(i) **Balanced bridge** : The bridge is said to be balanced when deflection in galvanometer is zero *i.e.* no current flows through the galvanometer or in other words $V_B = V_D$. In the balanced condition $\frac{P}{Q} = \frac{R}{S}$, on mutually changing the position of cell and galvanometer this condition will not change.

(ii) **Unbalanced bridge** : If the bridge is not balanced current will flow from D to B if $V_D > V_B$ *i.e.* $(V_A - V_D) < (V_A - V_B)$ which gives $PS > RQ$.

(iii) **Applications of wheatstone bridge** : Meter bridge, post office box and Carey Foster bridge are instruments based on the principle of wheatstone bridge and are used to measure unknown resistance.

(5) **Meter bridge** : In case of meter bridge, the resistance wire AC is 100 cm long. Varying the position of tapping point B , bridge is balanced. If in balanced position of bridge $AB = l$, $BC = (100 - l)$ so that $\frac{P}{Q} = \frac{l}{(100 - l)}$. Also $\frac{P}{Q} = \frac{R}{S} \Rightarrow S = \frac{(100 - l)}{l} R$

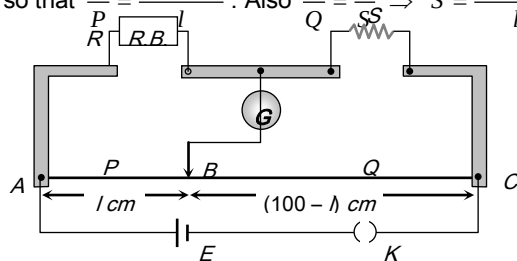


Fig. 19.37

Potentiometer is a device mainly used to measure emf of a given cell and to compare emf's of cells. It is also used to measure internal resistance of a given cell.

(1) **Circuit diagram** : Potentiometer consists of a long resistive wire AB of length L (about 6 m to 10 m long) made up of manganin or constantan and a battery of known voltage e and internal resistance r called supplier battery or driver cell. Connection of these two forms primary circuit.

One terminal of another cell (whose emf E is to be measured) is connected at one end of the main circuit and the other terminal at any point on the resistive wire through a galvanometer G . This forms the secondary circuit. Other details are as follows

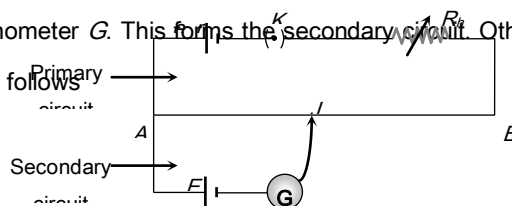


Fig. 19.38

J = Jockey

K = Key

R = Resistance of potentiometer wire,

ρ = Specific resistance of potentiometer wire.

R_h = Variable resistance which controls the current through the wire AB

(i) The specific resistance (ρ) of potentiometer wire must be high but its temperature coefficient of resistance (α) must be low.

(ii) All higher potential points (terminals) of primary and secondary circuits must be connected together at point A and all lower potential points must be connected to point B or jockey.

Potentiometer

(iii) The value of known potential difference must be greater than the value of unknown potential difference to be measured.

(iv) The potential gradient must remain constant. For this the current in the primary circuit must remain constant and the jockey must not be slid in contact with the wire.

(v) The diameter of potentiometer wire must be uniform everywhere.

(2) **Potential gradient (x)** : Potential difference (or fall in potential) per unit length of wire is called potential gradient *i.e.*

$$x = \frac{V \text{ volt}}{L \text{ m}} \quad \text{where } V = iR = \left(\frac{e}{R + R_h + r} \right) \cdot R.$$

$$\text{So } x = \frac{V}{L} = \frac{iR}{L} = \frac{i\rho}{(R + R_h + r)} \cdot \frac{R}{L}$$

(i) Potential gradient directly depends upon

(a) The resistance per unit length (R/L) of potentiometer wire.

(b) The radius of potentiometer wire (*i.e.* Area of cross-section)

(c) The specific resistance of the material of potentiometer wire (*i.e.* ρ)

(d) The current flowing through potentiometer wire (i)

(ii) potential gradient indirectly depends upon

(a) The emf of battery in the primary circuit (*i.e.* e)

(b) The resistance of rheostat in the primary circuit (*i.e.* R_h)

(3) **Working** : Suppose jockey is made to touch a point J on wire then potential difference between A and J will be $V = x l$

At this length (l) two potential difference are obtained

(i) V due to battery e and

(ii) E due to unknown cell

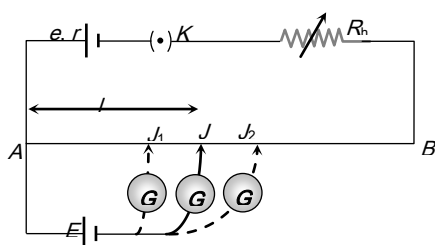


Fig. 19.39

If $V > E$ then current will flow in galvanometer circuit in one direction

If $V < E$ then current will flow in galvanometer circuit in opposite direction

If $V = E$ then no current will flow in galvanometer circuit this condition is known as null deflection position, length l is known as balancing length.

In balanced condition $E = x l$

$$\text{or } E = x l = \frac{V}{L} l = \frac{iR}{L} l = \left(\frac{e}{R + R_h + r} \right) \cdot \frac{R}{L} \times l$$

$$\text{If } V \text{ is constant then } L \propto l \Rightarrow \frac{x_1}{x_2} = \frac{L_1}{L_2} = \frac{l_1}{l_2}$$

(6) **Standardization of potentiometer** : The process of determining potential gradient experimentally is known as standardization of potentiometer.

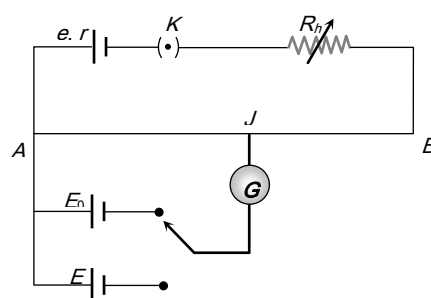


Fig. 19.40

Let the balancing length for the standard emf E_0 is l_0 then

$$\text{by the principle of potentiometer } E_0 = x l_0 \Rightarrow x = \frac{E_0}{l_0}$$

(7) **Sensitivity of potentiometer** : A potentiometer is said to be more sensitive, if it measures a small potential difference more accurately.

(i) The sensitivity of potentiometer is assessed by its potential gradient. The sensitivity is inversely proportional to the potential gradient.

(ii) In order to increase the sensitivity of potentiometer

(a) The resistance in primary circuit will have to be decreased.

(b) The length of potentiometer wire will have to be increased so that the length may be measured more accuracy.

Table 19.5 : Difference between voltmeter and potentiometer

Voltmeter	Potentiometer
It's resistance is high but finite	Its resistance is infinite
It draws some current from source of emf	It does not draw any current from the source of unknown emf
The potential difference measured by it is lesser than the actual potential difference	The potential difference measured by it is equal to actual potential difference
Its sensitivity is low	Its sensitivity is high
It is a versatile instrument	It measures only emf or potential difference
It is based on deflection method	It is based on zero deflection method

Application of Potentiometer

(1) **To determine the internal resistance of a primary cell**

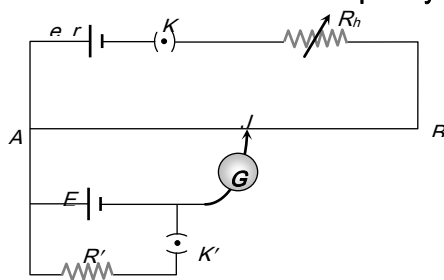


Fig. 19.41

(i) Initially in secondary circuit key K' remains open and balancing length (l_1) is obtained. Since cell E is in open circuit so it's emf balances on length l_1 i.e. $E = x l_1$ (i)

(ii) Now key K' is closed so cell E comes in closed circuit. If the process of balancing repeated again then potential difference V balances on length l_2 i.e. $V = x l_2$

.... (ii)

(iii) By using formula internal resistance $r = \left(\frac{E}{V} - 1 \right) \cdot R'$

$$r = \left(\frac{l_1 - l_2}{l_2} \right) \cdot R'$$

(2) **Comparison of emf's of two cell** : Let l_1 and l_2 be the balancing lengths with the cells E_1 and E_2 respectively then $E_1 = x l_1$ and $E_2 = x l_2 \Rightarrow \frac{E_1}{E_2} = \frac{l_1}{l_2}$

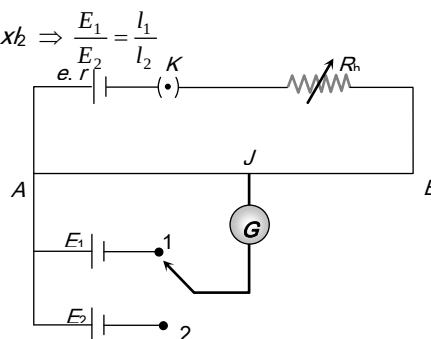


Fig. 19.42

Let $E_1 > E_2$ and both are connected in series. If balancing length is l_1 when cell assist each other and it is l_2 when they oppose each other as shown then :



$$(E_1 + E_2) = x l_1$$

$$(E_1 - E_2) = x l_2$$

$$\Rightarrow \frac{E_1 + E_2}{E_1 - E_2} = \frac{l_1}{l_2} \quad \text{or} \quad \frac{E_1}{E_2} = \frac{l_1 + l_2}{l_1 - l_2}$$

(3) **Comparison of resistances** : Let the balancing length for resistance R_1 (when XY is connected) is l_1 and let balancing length for resistance $R_1 + R_2$ (when YZ is connected) is l_2 .

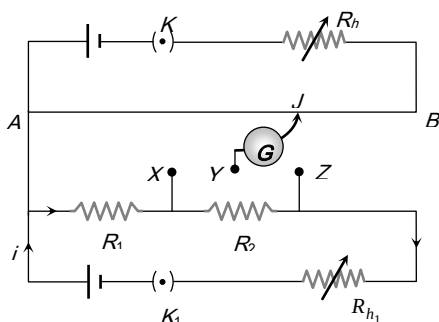


Fig. 19.43

$$\text{Then } iR_1 = xl_1 \text{ and } i(R_1 + R_2) = xl_2 \Rightarrow \frac{R_2}{R_1} = \frac{l_2 - l_1}{l_1}$$

(4) **To determine thermo emf**

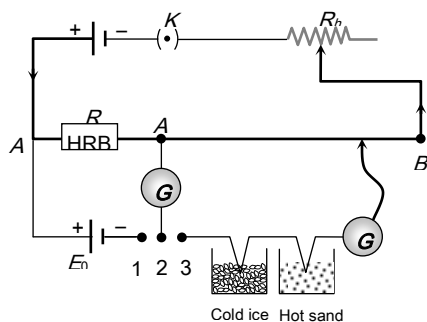


Fig. 19.44

(i) The value of thermo-emf in a thermocouple for ordinary temperature difference is very low (10^{-6} volt). For this the potential gradient x must be also very low (10^{-4} V/m). Hence a high resistance (R) is connected in series with the potentiometer wire in order to reduce current.

(ii) The potential difference across R must be equal to the emf of standard cell *i.e.* $iR = E_0 \therefore i = \frac{E_0}{R}$

(iii) The small thermo emf produced in the thermocouple $e = xl$

(iv) $x = i\rho = \frac{iR}{L} \therefore e = \frac{iRl}{L}$ where L = length of potentiometer wire, ρ = resistance per unit length, l = balancing length for e

(5) **Calibration of ammeter** : Checking the correctness of ammeter readings with the help of potentiometer is called calibration of ammeter.

(i) In the process of calibration of an ammeter the current flowing in a circuit is measured by an ammeter and the same current is also measured with the help of potentiometer. By comparing both the values, the errors in the ammeter readings are determined.

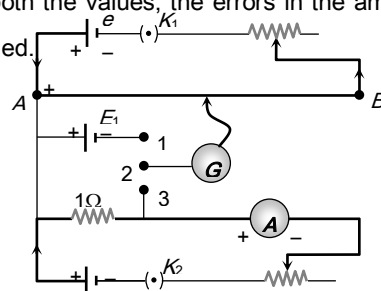


Fig. 19.45

(ii) For the calibration of an ammeter, 1Ω standard resistance coil is specifically used in the secondary circuit of the potentiometer, because the potential difference across 1Ω is equal to the current flowing through it *i.e.* $V = i$.

(iii) If the balancing length for the emf E_0 is l_0 then $E_0 = xl_0 \Rightarrow x = \frac{E_0}{l_0}$ (Process of standardisation)

(iv) Let i' current flows through 1Ω resistance giving potential difference as $V' = i'(1) = xl_1$ where l_1 is the balancing length. So error can be found as $\Delta i = i - i' = i - xl_1 = i - \frac{E_0}{l_0} \times l_1$

(6) **Calibration of voltmeter**

(i) Practical voltmeters are not ideal, because these do not have infinite resistance. The error of such practical voltmeter

can be found by comparing the voltmeter reading with calculated value of p.d. by potentiometer.

(ii) If h is balancing length for E_0 the emf of standard cell by connecting 1 and 2 of bi-directional key, then $x = E_0/h$.

(iii) The balancing length l_1 for unknown potential difference V is given by (by closing 2 and 3) $V = x l_1 = (E_0 / l_0) l_1$.

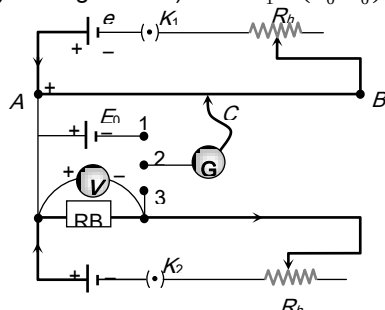


Fig. 19.46

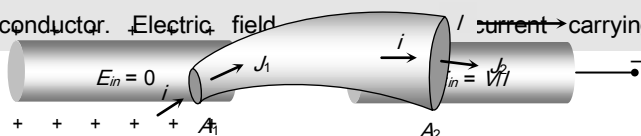
If the voltmeter reading is V then the error will be $(V - V')$ which may be $+ve$, $-ve$ or zero.

Tips & Tricks

✍ Human body, though has a large resistance of the order of $k\Omega$ (say $10 k\Omega$), is very sensitive to minute currents even as low as a few mA . Electrocutation, excites and disorders the nervous system of the body and hence one fails to control the activity of the body.

✍ dc flows uniformly throughout the cross-section of conductor while ac mainly flows through the outer surface area of the conductor. This is known as skin effect.

✍ It is worth noting that electric field inside a charged conductor is zero, but it is non zero inside a current carrying conductor and is given by $E = \frac{V}{l}$ where V = potential difference across the conductor and l = length of the conductor. Electric field



conductor is zero.

✍ For a given conductor $JA = i = \text{constant}$ so that $J \propto \frac{1}{A}$ i.e., $J_1 A_1 = J_2 A_2$; this is called equation of continuity

✍ The drift velocity of electrons is small because of the frequent collisions suffered by electrons.

✍ The small value of drift velocity produces a large amount of electric current, due to the presence of extremely large number of free electrons in a conductor.

The propagation of current is almost at the speed of light and involves electromagnetic process. It is due to this reason that the electric bulb glows immediately when switch is on.

✍ In the absence of electric field, the paths of electrons between successive collisions are straight line while in presence of electric field the paths are generally curved.

✍ Free electron density in a metal is given by $n = \frac{N_A x d}{A}$ where N_A = Avogadro number, x = number of free electrons per atom, d = density of metal and A = Atomic weight of metal.

✍ In the absence of radiation loss, the time in which a fuse will melt does not depend on its length but varies with radius as $t \propto r^4$.

✍ If length (l) and mass (m) of a conducting wire is given then $R \propto \frac{l^2}{m}$.

✍ Macroscopic form of Ohm's law is $R = \frac{V}{i}$, while its

microscopic form is $J = \sigma E$.

✍ After stretching if length increases by n times then resistance will increase by n^2 times *i.e.* $R_2 = n^2 R_1$. Similarly if radius be reduced to $\frac{1}{n}$ times then area of cross-section decreases $\frac{1}{n^2}$ times so the resistance becomes n^4 times *i.e.* $R_2 = n^4 R_1$.

✍ After stretching if length of a conductor increases by $x\%$ then resistance will increase by $2x\%$ (valid only if $x < 10\%$)

✍ Decoration of lightning in festivals is an example of series grouping whereas all household appliances connected in parallel grouping.

✍ Using n conductors of equal resistance, the number of possible combinations is 2^{n-1} .

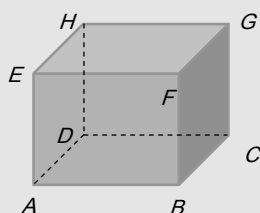
✍ If the resistance of n conductors are totally different, then the number of possible combinations will be 2^n .

✍ If n identical resistances are first connected in series and then in parallel, the ratio of the equivalent resistance is given by $\frac{R_p}{R_s} = \frac{n^2}{1}$.

✍ If a wire of resistance R , cut in n equal parts and then these parts are collected to form a bundle then equivalent resistance of combination will be $\frac{R}{n^2}$.

✍ If equivalent resistance of R_1 and R_2 in series and parallel be R_s and R_p respectively then $R_1 = \frac{1}{2} \left[R_s + \sqrt{R_s^2 - 4R_s R_p} \right]$ and $R_2 = \frac{1}{2} \left[R_s - \sqrt{R_s^2 - 4R_s R_p} \right]$.

✍ If a skeleton cube is made with 12 equal resistance each having resistance R then the net resistance across

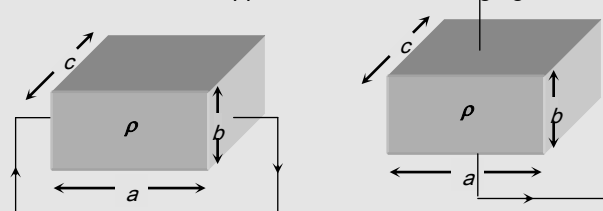


The longest diagonal (EC or AG) $= \frac{5}{6} R$

The diagonal of face (*e.g.* AC , ED ,) $= \frac{3}{4} R$

A side (*e.g.* AB , BC) $= \frac{7}{12} R$

✍ Resistance of a conducting body is not unique but depends on its length and area of cross-section *i.e.* how the potential difference is applied. See the following figures



Length = a

Area of cross-section = $b \times c$

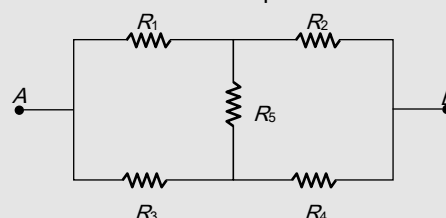
Resistance $R = \rho \left(\frac{a}{b \times c} \right)$

Length = b

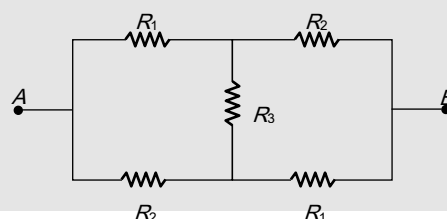
Area of cross-section = $a \times c$

Resistance $R = \rho \left(\frac{b}{a \times c} \right)$

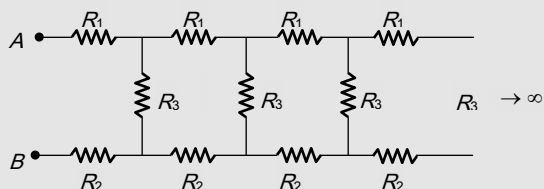
✍ Some standard results for equivalent resistance



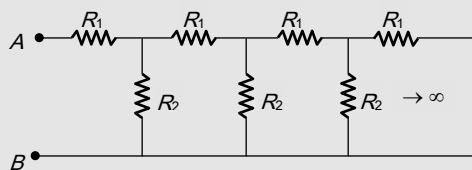
$$R_{AB} = \frac{R_1 R_2 (R_3 + R_4) + (R_1 + R_2) R_3 R_4 + R_5 (R_1 + R_2) (R_3 + R_4)}{R_5 (R_1 + R_2 + R_3 + R_4) + (R_1 + R_3) (R_2 + R_4)}$$



$$R_{AB} = \frac{2R_1R_2 + R_3(R_1 + R_2)}{2R_3 + R_1 + R_2}$$



$$R_{AB} = \frac{1}{2}(R_1 + R_2) + \frac{1}{2}[(R_1 + R_2)^2 + 4R_3(R_1 + R_2)]^{1/2}$$



$$R_{AB} = \frac{1}{2}R_1 \left[1 + \sqrt{1 + 4\left(\frac{R_2}{R_1}\right)} \right]$$

✍ It is a common misconception that “current in the circuit will be maximum when power consumed by the load is maximum.”

Actually current $i = E/(R+r)$ is maximum ($= E/r$) when $R = \min = 0$ with $P_L = (E/r)^2 \times 0 = 0 \min$. while power consumed by the load $E^2R/(R+r)^2$ is maximum ($= E^2/4r$) when $R = r$ and $i = (E/2r) \neq \max (= E/r)$.

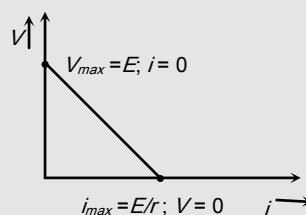
✍ Emf is independent of the resistance of the circuit and depends upon the nature of electrolyte of the cell while potential difference depends upon the resistance between the two points of the circuit and current flowing through the circuit.

✍ Whenever a cell or battery is present in a branch there must be some resistance (internal or external or both)

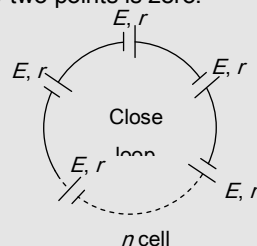
present in that branch. In practical situation it always happen because we can never have an ideal cell or battery with zero resistance.

✍ In series grouping of identical cells. If one cell is wrongly connected then it will cancel out the effect of two cells *e.g.* If in the combination of n identical cells (each having emf E and internal resistance r) if x cell are wrongly connected then equivalent emf $E_{eq} = (n - 2x)E$ and equivalent internal resistance $r_{eq} = nr$.

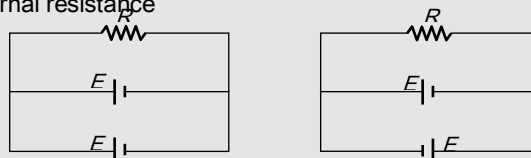
✍ Graphical view of open circuit and closed circuit of a cell.



✍ If n identical cells are connected in a loop in order, then emf between any two points is zero.



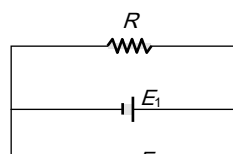
✍ In parallel grouping of two identical cell having no internal resistance



$$E_{eq} = E$$

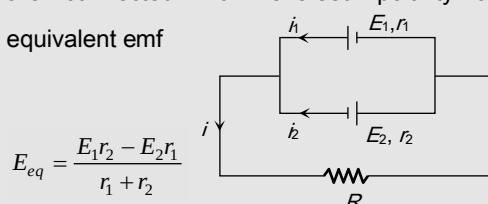
$$E_{eq} = 0$$

✍ When two cell's of different emf and no internal resistance are connected in parallel then equivalent emf is



indeterminate, note that connecting a wire with a cell with no resistance is equivalent to short circuiting. Therefore the total current that will be flowing will be infinity.

✍ In the parallel combination of non-identical cell's if they are connected with reversed polarity as shown then equivalent emf



✍ Wheatstone bridge is most sensitive if all the arms of bridge have equal resistances *i.e.* $P = Q = R = S$

✍ If the temperature of the conductor placed in the right gap of metre bridge is increased, then the balancing length decreases and the jockey moves towards left.

✍ In Wheatstone bridge to avoid inductive effects the battery key should be pressed first and the galvanometer key afterwards.

✍ The measurement of resistance by Wheatstone bridge is not affected by the internal resistance of the cell.

✍ In case of zero deflection in the galvanometer current flows in the primary circuit of the potentiometer, not in the galvanometer circuit.

✍ A potentiometer can act as an ideal voltmeter.



Ordinary Thinking

Objective Questions

Electric Conduction, Ohm's Law and Resistance

- Current of 4.8 amperes is flowing through a conductor. The number of electrons per second will be [CPMT 1986]
 - 3×10^{19}
 - 7.68×10^{21}
 - 7.68×10^{20}
 - 3×10^{20}
- When the current i is flowing through a conductor, the drift velocity is v . If $2i$ current is flowed through the same metal but having double the area of cross-section, then the drift velocity will be
 - $v/4$
 - $v/2$
 - v
 - $4v$
- When current flows through a conductor, then the order of drift velocity of electrons will be [CPMT 1986]
 - 10^{10} m/sec
 - 10^{-2} cm/sec
 - 10^4 cm/sec
 - 10^{-1} cm/sec
- Every atom makes one free electron in copper. If 1.1 ampere current is flowing in the wire of copper having 1 mm diameter, then the drift velocity (approx.) will be (Density of copper $= 9 \times 10^3 \text{ kg m}^{-3}$ and atomic weight = 63) [CPMT 1989]
 - 0.3 mm/sec
 - 0.1 mm/sec
 - 0.2 mm/sec
 - 0.2 cm/sec
- Which one is not the correct statement [NCERT 1978]
 - $1 \text{ volt} \times 1 \text{ coulomb} = 1 \text{ joule}$
 - $1 \text{ volt} \times 1 \text{ ampere} = 1 \text{ joule/second}$
 - $1 \text{ volt} \times 1 \text{ watt} = 1 \text{ H.P.}$
 - Watt-hour can be expressed in eV
- If a 0.1 % increase in length due to stretching, the percentage increase in its resistance will be [MNR 1990; MP PMT 1996; UPSEAT 1999; MP PMT 2000]
 - 0.2 %
 - 2 %
 - 1 %
 - 0.1 %



1056 Current Electricity

7. The specific resistance of manganin is $50 \times 10^{-8} \text{ ohm} \times \text{m}$.

The resistance of a cube of length 50 cm will be

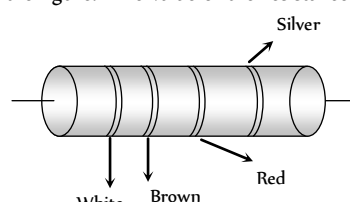
- (a) 10^{-6} ohm (b) $2.5 \times 10^{-5} \text{ ohm}$
(c) 10^{-8} ohm (d) $5 \times 10^{-4} \text{ ohm}$

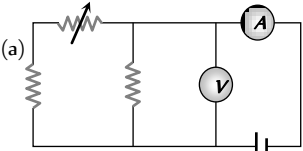
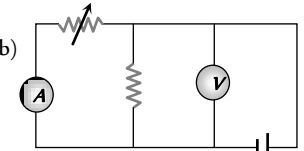
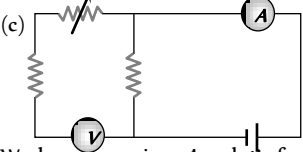
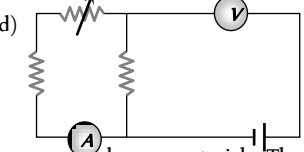
8. The resistivity of iron is $1 \times 10^{-7} \text{ ohm-m}$. The resistance of a iron wire of particular length and thickness is 1 ohm . If the length and the diameter of wire both are doubled, then the resistivity in ohm-m will be [CPMT 1983; DPMT 1999]
- (a) 1×10^{-7} (b) 2×10^{-7}
(c) 4×10^{-7} (d) 8×10^{-7}
9. The temperature coefficient of resistance for a wire is $0.00125/^{\circ}\text{C}$. At 300K its resistance is 1 ohm . The temperature at which the resistance becomes 2 ohm is [IIT 1980; MP PET 2002; KCET 2003; MP PMT 2001; Orissa JEE 2002]
- (a) 1154 K (b) 1100 K
(c) 1400 K (d) 1127 K
10. When the length and area of cross-section both are doubled, then its resistance [MP PET 1989]
- (a) Will become half (b) Will be doubled
(c) Will remain the same (d) Will become four times
11. The resistance of a wire is 20 ohms . It is so stretched that the length becomes three times, then the new resistance of the wire will be
- (a) 6.67 ohms (b) 60.0 ohms
(c) 120 ohms (d) 180.0 ohms
12. The resistivity of a wire [MP PMT 1984; DPMT 1982]
- (a) Increases with the length of the wire
(b) Decreases with the area of cross-section
(c) Decreases with the length and increases with the cross-section of wire
(d) None of the above statement is correct
13. Ohm's law is true
- (a) For metallic conductors at low temperature
(b) For metallic conductors at high temperature
(c) For electrolytes when current passes through them
(d) For diode when current flows
14. The example for non-ohmic resistance is [MP PMT 1978]
- (a) Copper wire (b) Carbon resistance
(c) Diode (d) Tungston wire
15. Drift velocity v_d varies with the intensity of electric field as per the relation [CPMT 1981; BVP 2003]
- (a) $v_d \propto E$ (b) $v_d \propto \frac{1}{E}$
(c) $v_d = \text{constant}$ (d) $v_d \propto E^2$
16. On increasing the temperature of a conductor, its resistance increases because [CPMT 1982]
- (a) Relaxation time decreases
(b) Mass of the electrons increases
(c) Electron density decreases
(d) None of the above
17. In a conductor 4 coulombs of charge flows for 2 seconds . The value of electric current will be [CPMT 1984]
- (a) 4 volts (b) 4 amperes
(c) 2 amperes (d) 2 volts
18. The specific resistance of a wire is ρ , its volume is 3 m^3 and its resistance is 3 ohms , then its length will be [CPMT 1984]
- (a) $\sqrt{\frac{1}{\rho}}$ (b) $\frac{3}{\sqrt{\rho}}$
(c) $\frac{1}{\rho} \sqrt{3}$ (d) $\rho \sqrt{\frac{1}{3}}$
19. 62.5×10^{18} electrons per second are flowing through a wire of area of cross-section 0.1 m^2 , the value of current flowing will be
- (a) 1 A (b) 0.1 A
(c) 10 A (d) 0.11 A
20. A piece of wire of resistance 4 ohms is bent through 180° at its mid point and the two halves are twisted together, then the resistance is [CPMT 1971]
- (a) 8 ohms (b) 1 ohm
(c) 2 ohms (d) 5 ohms
21. When a piece of aluminium wire of finite length is drawn through a series of dies to reduce its diameter to half its original value, its resistance will become [MP PET 1989]
- [NCERT 1974; AIIMS 1997; MH CET 2000; UPSEAT 2001; CBSE PMT 2002]
- (a) Two times (b) Four times
(c) Eight times (d) Sixteen times
22. A wire 100 cm long and 2.0 mm diameter has a resistance of 0.7 ohm , the electrical resistivity of the material is
- (a) $4.4 \times 10^{-6} \text{ ohm} \times \text{m}$ (b) $2.2 \times 10^{-6} \text{ ohm} \times \text{m}$
(c) $1.1 \times 10^{-6} \text{ ohm} \times \text{m}$ (d) $0.22 \times 10^{-6} \text{ ohm} \times \text{m}$
23. A certain wire has a resistance R . The resistance of another wire identical with the first except having twice its diameter is
- (a) $2R$ (b) $0.25R$
(c) $4R$ (d) $0.5R$
24. In hydrogen atom, the electron makes 6.6×10^{15} revolutions per second around the nucleus in an orbit of radius $0.5 \times 10^{-10} \text{ m}$. It is equivalent to a current nearly
- (a) 1 A (b) 1 mA
(c) $1 \mu\text{A}$ (d) $1.6 \times 10^{-19} \text{ A}$
25. A wire of length 5 m and radius 1 mm has a resistance of 1 ohm . What length of the wire of the same material at the same temperature and of radius 2 mm will also have a resistance of 1 ohm
- (a) 1.25 m (b) 2.5 m
(c) 10 m (d) 20 m
26. When there is an electric current through a conducting wire along its length, then an electric field must exist
- (a) Outside the wire but normal to it
(b) Outside the wire but parallel to it
(c) Inside the wire but parallel to it
(d) Inside the wire but normal to it
27. Through a semiconductor, an electric current is due to drift of
- (a) Free electrons
(b) Free electrons and holes
(c) Positive and negative ions
(d) Protons

28. In an electrolyte 3.2×10^{18} bivalent positive ions drift to the right per second while 3.6×10^{18} monovalent negative ions drift to the left per second. Then the current is
 (a) 1.6 amp to the left (b) 1.6 amp to the right
 (c) 0.45 amp to the right (d) 0.45 amp to the left
29. A metallic block has no potential difference applied across it, then the mean velocity of free electrons is T = absolute temperature of the block)
 (a) Proportional to T
 (b) Proportional to $\sqrt{T}$
 (c) Zero
 (d) Finite but independent of temperature
30. The specific resistance of all metals is most affected by
 (a) Temperature (b) Pressure
 (c) Degree of illumination (d) Applied magnetic field
31. The positive temperature coefficient of resistance is for
 (a) Carbon (b) Germanium
 (c) Copper (d) An electrolyte
32. The fact that the conductance of some metals rises to infinity at some temperature below a few Kelvin is called
 (a) Thermal conductivity (b) Optical conductivity
 (c) Magnetic conductivity (d) Superconductivity
33. Dimensions of a block are $1\text{ cm} \times 1\text{ cm} \times 100\text{ cm}$. If specific resistance of its material is $3 \times 10^{-7}\text{ ohm-m}$, then the resistance between the opposite rectangular faces is
 [MP PET 1993]
 (a) $3 \times 10^{-9}\text{ ohm}$ (b) $3 \times 10^{-7}\text{ ohm}$
 (c) $3 \times 10^{-5}\text{ ohm}$ (d) $3 \times 10^{-3}\text{ ohm}$
34. In the above question, the resistance between the square faces is
 (a) $3 \times 10^{-9}\text{ ohm}$ (b) $3 \times 10^{-7}\text{ ohm}$
 (c) $3 \times 10^{-5}\text{ ohm}$ (d) $3 \times 10^{-3}\text{ ohm}$
35. There is a current of 20 amperes in a copper wire of 10^{-6} square metre area of cross-section. If the number of free electrons per cubic metre is 10^{29} , then the drift velocity is
 (a) $125 \times 10^{-3}\text{ m/sec}$ (b) $12.5 \times 10^{-3}\text{ m/sec}$
 (c) $1.25 \times 10^{-3}\text{ m/sec}$ (d) $1.25 \times 10^{-4}\text{ m/sec}$
36. The electric intensity E , current density j and specific resistance k are related to each other by the relation
 [DPMT 2001]
 (a) $E = j/k$ (b) $E = jk$
 (c) $E = k/j$ (d) $k = jE$
37. The resistance of a wire of uniform diameter d and length L is R . The resistance of another wire of the same material but diameter $2d$ and length $4L$ will be
 [CPMT 1984; MP PET 2002]
 (a) $2R$ (b) R
 (c) $R/2$ (d) $R/4$
38. There is a current of 1.344 amp in a copper wire whose area of cross-section normal to the length of the wire is 1 mm^2 . If the number of free electrons per cm^3 is 8.4×10^{22} , then the drift velocity would be
 [CPMT 1990]
 (a) 1.0 mm/sec (b) 1.0 m/sec
 (c) 0.1 mm/sec (d) 0.01 mm/sec
39. It is easier to start a car engine on a hot day than on a cold day. This is because the internal resistance of the car battery
 (a) Decreases with rise in temperature
 (b) Increases with rise in temperature
 (c) Decreases with a fall in temperature
 (d) Does not change with a change in temperature
40. 5 amperes of current is passed through a metallic conductor. The charge flowing in one minute in coulombs will be
 [MP PET 1984]
 (a) 5 (b) 12
 (c) $1/12$ (d) 300
41. Two wires of the same material are given. The first wire is twice as long as the second and has twice the diameter of the second. The resistance of the first will be
 [MP PMT 1993]
 (a) Twice of the second (b) Half of the second
 (c) Equal to the second (d) Four times of the second
42. An electric wire is connected across a cell of e.m.f. E . The current I is measured by an ammeter of resistance R . According to ohm's law
 (a) $E = I^2 R$ (b) $E = IR$
 (c) $E = R/I$ (d) $E = I/R$
43. The resistances of a wire at temperatures $t^\circ\text{C}$ and 0°C are related by
 [MP PMT 1993]
 (a) $R_t = R_0(1 + \alpha t)$ (b) $R_t = R_0(1 - \alpha t)$
 (c) $R_t = R_0^2(1 + \alpha t)$ (d) $R_t = R_0^2(1 - \alpha t)$
44. An electric wire of length ' l ' and area of cross-section a has a resistance $R\text{ ohms}$. Another wire of the same material having same length and area of cross-section $4a$ has a resistance of
 [MP PET 1993]
 (a) $4R$ (b) $R/4$
 (c) $R/16$ (d) $16R$
45. For which of the following the resistance decreases on increasing the temperature
 [MP PET 1993]
 (a) Copper (b) Tungsten
 (c) Germanium (d) Aluminium
46. If n, e, τ and m respectively represent the density, charge relaxation time and mass of the electron, then the resistance of a wire of length l and area of cross-section A will be
 [CPMT 1992]
 (a) $\frac{ml}{ne^2 \tau A}$ (b) $\frac{m \tau^2 A}{ne^2 l}$
 (c) $\frac{ne^2 \tau A}{2ml}$ (d) $\frac{ne^2 A}{2m \tau l}$
47. The relaxation time in conductors
 [DPMT 2003]
 (a) Increases with the increase of temperature
 (b) Decreases with the increase of temperature
 (c) It does not depend on temperature
 (d) All of sudden changes at 400 K
48. Which of the following statement is correct
 (a) Liquids obey fully the ohm's law
 (b) Liquids obey partially the ohm's law
 (c) There is no relation between current and p.d. for liquids

- (d) None of the above
49. A certain piece of silver of given mass is to be made like a wire. Which of the following combination of length (L) and the area of cross-sectional (A) will lead to the smallest resistance [MP PMT 1995; CBSE PMT 1997]
- (a) L and A
 (b) $2L$ and $A/2$
 (c) $L/2$ and $2A$
 (d) Any of the above, because volume of silver remains same
50. The resistance of a wire is 10Ω . Its length is increased by 10% by stretching. The new resistance will now be [CPMT 2000; Pb PET 2004]
- (a) 12Ω (b) 1.2Ω
 (c) 13Ω (d) 11Ω
51. Resistance of tungsten wire at 150°C is 133Ω . Its resistance temperature coefficient is $0.0045/^\circ\text{C}$. The resistance of this wire at 500°C will be [DPMT 2004]
- (a) 180Ω (b) 225Ω
 (c) 258Ω (d) 317Ω
52. A metal wire of specific resistance $64 \times 10^{-6} \text{ ohm-cm}$ and length 198 cm has a resistance of 7 ohm , the radius of the wire will be [MP PET 1994]
- (a) 2.4 cm (b) 0.24 cm
 (c) 0.024 cm (d) 24 cm
53. A copper wire of length 1 m and radius 1 mm is joined in series with an iron wire of length 2 m and radius 3 mm and a current is passed through the wires. The ratio of the current density in the copper and iron wires is [MP PMT 1994]
- (a) $18 : 1$ (b) $9 : 1$
 (c) $6 : 1$ (d) $2 : 3$
54. For a metallic wire, the ratio V/i (V = the applied potential difference, i = current flowing) is [MP PMT 1994; BVP 2003]
- (a) Independent of temperature
 (b) Increases as the temperature rises
 (c) Decreases as the temperature rises
 (d) Increases or decreases as temperature rises, depending upon the metal
55. The resistance of a wire is R . If the length of the wire is doubled by stretching, then the new resistance will be [Roorkee 1992; AFMC 1995; KCET 1993; AMU (Med.) 1999; CBSE PMT 1999; MP PET 2001; UPSEAT 2001]
- (a) $2R$ (b) $4R$
 (c) R (d) $\frac{R}{4}$
56. Which of the following has a negative temperature coefficient
- (a) C (b) Fe
 (c) Mn (d) Ag
57. The reciprocal of resistance is [AFMC 1995]
- (a) Conductance (b) Resistivity
- (c) Voltage (d) None of the above
58. A solenoid is at potential difference 60 V and current flows through it is 15 ampere , then the resistance of coil will be [AFMC 1995]
- (a) 4Ω (b) 8Ω
 (c) 0.25Ω (d) 2Ω
59. All of the following statements are true except [Manipal MEE 1995]
- (a) Conductance is the reciprocal of resistance and is measured in *Siemens*
 (b) *Ohm's* law is not applicable at very low and very high temperatures
 (c) *Ohm's* law is applicable to semiconductors
 (d) *Ohm's* law is not applicable to electron tubes, discharge tubes and electrolytes
60. A potential difference of V is applied at the ends of a copper wire of length l and diameter d . On doubling only d , drift velocity
- (a) Becomes two times (b) Becomes half
 (c) Does not change (d) Becomes one fourth
61. If the resistance of a conductor is 5Ω at 50°C and 7Ω at 100°C then the mean temperature coefficient of resistance of the material is
- (a) $0.008/^\circ\text{C}$ (b) $0.006/^\circ\text{C}$
 (c) $0.004/^\circ\text{C}$ (d) $0.001/^\circ\text{C}$
62. The resistance of a discharge tube is [AFMC 1996; CBSE PMT 1999]
- (a) *Ohmic* (b) *Non-ohmic*
 (c) Both (a) and (b) (d) Zero
63. We are able to obtain fairly large currents in a conductor because
- (a) The electron drift speed is usually very large
 (b) The number density of free electrons is very high and this can compensate for the low values of the electron drift speed and the very small magnitude of the electron charge
 (c) The number density of free electrons as well as the electron drift speeds are very large and these compensate for the very small magnitude of the electron charge
 (d) The very small magnitude of the electron charge has to be divided by the still smaller product of the number density and drift speed to get the electric current
64. A platinum resistance thermometer has a resistance of 50Ω at 20°C . When dipped in a liquid the resistance becomes 76.8Ω . The temperature coefficient of resistance for platinum is $\alpha = 3.92 \times 10^{-3} / ^\circ\text{C}$. The temperature of the liquid is
- (a) 100°C (b) 137°C
 (c) 167°C (d) 200°C
65. In a wire of circular cross-section with radius r , free electrons travel with a drift velocity V when a current I flows through the wire. What is the current in another wire of half the radius and of the same material when the drift velocity is $2V$ [AFMC 1995]
- [MP PET 1997]

- (a) $2I$ (b) I
(c) $I/2$ (d) $I/4$
66. The resistivity of a wire depends on its [MP PMT/PET 1998]
(a) Length (b) Area of cross-section
(c) Shape (d) Material
67. The conductivity of a superconductor is [Similar to KCET 1993; MP PMT/PET 1998]
(a) Infinite (b) Very large
(c) Very small (d) Zero
68. In a neon discharge tube 2.9×10^{18} Ne^+ ions move to the right each second while 1.2×10^{18} electrons move to the left per second. Electron charge is $1.6 \times 10^{-19} C$. The current in the discharge tube
(a) $1 A$ towards right (b) $0.66 A$ towards right
(c) $0.66 A$ towards left (d) Zero
69. A steady current flows in a metallic conductor of non-uniform cross-section. The quantity/ quantities constant along the length of the conductor is/are [KCET 1994, IIT 1997 Cancelled; CBSE PMT 2001]
(a) Current, electric field and drift speed
(b) Drift speed only
(c) Current and drift speed
(d) Current only
70. The resistivity of alloys $= R_{\text{alloy}}$; the resistivity of constituent metals R_{metal} . Then, usually [KCET 1994]
(a) $R_{\text{alloy}} = R_{\text{metal}}$
(b) $R_{\text{alloy}} < R_{\text{metal}}$
(c) There is no simple relation between R_{alloy} and R_{metal}
(d) $R_{\text{alloy}} > R_{\text{metal}}$
71. Two wires A and B of same material and same mass have radius $2r$ and r . If resistance of wire A is 34Ω , then resistance of B will be
(a) 544Ω (b) 272Ω
(c) 68Ω (d) 17Ω
72. Two rods of same material and length have their electric resistance in ratio $1:2$. When both rods are dipped in water, the correct statement will be [RPMT 1997]
(a) A has more loss of weight
(b) B has more loss of weight
(c) Both have same loss of weight
(d) Loss of weight will be in the ratio $1:2$
73. $20 \mu A$ current flows for 30 seconds in a wire, transfer of charge will be [RPMT 1997]
(a) $2 \times 10^{-4} C$ (b) $4 \times 10^{-4} C$
(c) $6 \times 10^{-4} C$ (d) $8 \times 10^{-4} C$
74. σ_1 and σ_2 are the electrical conductivities of Ge and Na respectively. If these substances are heated, then
(a) Both σ_1 and σ_2 increase
(b) σ_1 increases and σ_2 decreases
(c) σ_1 decreases and σ_2 increases
(d) Both σ_1 and σ_2 decrease
75. $1.6 mA$ current is flowing in conducting wire then the number of electrons flowing per second is [RPMT 1999]
(a) 10^6 (b) 10^7
(c) 10^8 (d) 10^9
76. A current I is passing through a wire having two sections P and Q of uniform diameters d and $d/2$ respectively. If the mean drift velocity of electrons in sections P and Q is denoted by v_1 and v_2 respectively, then [Roorkee 1999]
(a) $v_1 = v_2$ (b) $v_1 = \frac{1}{2} v_2$
(c) $v_1 = \frac{1}{4} v_2$ (d) $v_1 = 2 v_2$
77. If an electric current is passed through a nerve of a man, then man
(a) Begins to laugh
(b) Begins to weep
(c) Is excited
(d) Becomes insensitive to pain
78. The resistance of a coil is 4.2Ω at $100^\circ C$ and the temperature coefficient of resistance of its material is $0.004/^\circ C$. Its resistance at $0^\circ C$ is [KCET 1999]
(a) 6.5Ω (b) 5Ω
(c) 3Ω (d) 4Ω
79. Masses of three wires of copper are in the ratio of $1:3:5$ and their lengths are in the ratio of $5:3:1$. The ratio of their electrical resistances are [AFMC 2000]
(a) $1:3:5$ (b) $5:3:1$
(c) $1:15:125$ (d) $125:15:1$
80. Conductivity increases in the order of [AFMC 2000]
(a) Al, Ag, Cu (b) Al, Cu, Ag
(c) Cu, Al, Ag (d) Ag, Cu, Al
81. A uniform wire of resistance R is uniformly compressed along its length, until its radius becomes n times the original radius. Now resistance of the wire becomes [KCET 2000]
(a) $\frac{R}{n^4}$ (b) $\frac{R}{n^2}$
(c) $\frac{R}{n}$ (d) nR
82. The resistance of a conductor is 5 ohm at $50^\circ C$ and 6 ohm at $100^\circ C$. Its resistance at $0^\circ C$ is [KCET 2000]
(a) 1 ohm (b) 2 ohm
(c) 3 ohm (d) 4 ohm
83. If an electron revolves in the path of a circle of radius of $0.5 \times 10^{-8} m$ at frequency of 5×10^{14} cycles/s the electric current in the circle is (Charge of an electron $= 1.6 \times 10^{-19} C$) [EAMCET 2000]

- (a) 0.4 mA (b) 0.8 mA
(c) 1.2 mA (d) 1.6 mA
84. Equal potentials are applied on an iron and copper wire of same length. In order to have the same current flow in the two wires, the ratio $r(\text{iron})/r(\text{copper})$ of their radii must be (Given that specific resistance of iron = $1.0 \times 10^{-7} \text{ ohm-m}$ and specific resistance of copper = $1.7 \times 10^{-8} \text{ ohm-m}$)
[MP PMT 2000]
- (a) About 1.2 (b) About 2.4
(c) About 3.6 (d) About 4.8
85. An electron (charge = $1.6 \times 10^{-19} \text{ coulomb}$) is moving in a circle of radius $5.1 \times 10^{-11} \text{ m}$ at a frequency of $6.8 \times 10^{14} \text{ revolutions/sec}$. The equivalent current is approximately
[MP PET 2000]
- (a) $5.1 \times 10^{-3} \text{ amp}$ (b) $6.8 \times 10^{-3} \text{ amp}$
(c) $1.1 \times 10^{-3} \text{ amp}$ (d) $2.2 \times 10^{-3} \text{ amp}$
86. A rod of a certain metal is 1.0 m long and 0.6 cm in diameter. Its resistance is $3.0 \times 10^{-3} \text{ ohm}$. Another disc made of the same metal is 2.0 cm in diameter and 1.0 mm thick. What is the resistance between the round faces of the disc
[MP PET 2000]
- (a) $1.35 \times 10^{-8} \text{ ohm}$ (b) $2.70 \times 10^{-7} \text{ ohm}$
(c) $4.05 \times 10^{-6} \text{ ohm}$ (d) $8.10 \times 10^{-5} \text{ ohm}$
87. At what temperature will the resistance of a copper wire become three times its value at 0°C (Temperature coefficient of resistance for copper = $4 \times 10^{-5} \text{ per } ^\circ \text{C}$)
[MP PET 2000]
- (a) 400°C (b) 450°C
(c) 500°C (d) 550°C
88. An electron revolves $6 \times 10^{14} \text{ times/sec}$ in circular loop. The current in the loop is
[MNR 1995; UPSEAT 2000]
- (a) 0.96 mA (b) $0.96 \mu \text{ A}$
(c) 28.8 A (d) None of these
89. The charge of an electron is $1.6 \times 10^{-19} \text{ C}$. How many electrons strike the screen of a cathode ray tube each second when the beam current is 16 mA
[AMU (Med.) 2000]
- (a) 10^7 (b) 10^8
(c) 10^9 (d) 10^{10}
90. If potential $V = 100 \pm 0.5 \text{ Volt}$ and current $I = 10 \pm 0.2 \text{ amp}$ are given to us. Then what will be the value of resistance
[EAMCET 2001]
- (a) $10 \pm 0.7 \text{ ohm}$ (b) $5 \pm 2 \text{ ohm}$
(c) $0.1 \pm 0.2 \text{ ohm}$ (d) None of these
91. A nichrome wire 50 cm long and one square millimetre cross-section carries a current of 4 A when connected to a 2 V battery. The resistivity of nichrome wire in ohm metre is
[EAMCET 2001]
- (a) 1×10^{-6} (b) 4×10^{-7}
(c) 3×10^{-7} (d) 2×10^{-7}
92. If an observer is moving with respect to a stationary electron, then he observes
[DCE 2001]
- (a) Only magnetic field (b) Only electric field
(c) Both (a) and (b) (d) None of the above
93. Calculate the amount of charge flowing in 2 minutes in a wire of resistance 10Ω when a potential difference of 20 V is applied between its ends
[Kerala (Engg.) 2001]
- (a) 120 C (b) 240 C
(c) 20 C (d) 4 C
94. If a wire of resistance R is melted and recasted to half of its length, then the new resistance of the wire will be
[KCET (Med.) 2001]
- (a) $R/4$ (b) $R/2$
(c) R (d) $2R$
95. The drift velocity does not depend upon
[BHU 2001]
- (a) Cross-section of the wire (b) Length of the wire
(c) Number of free electrons (d) Magnitude of the current
96. There is a current of 40 ampere in a wire of 10^{-6} m^2 area of cross-section. If the number of free electron per m^3 is 10^{29} , then the drift velocity will be
[Pb. PMT 2001]
- (a) $1.25 \times 10^3 \text{ m/s}$ (b) $2.50 \times 10^{-3} \text{ m/s}$
(c) $25.0 \times 10^{-3} \text{ m/s}$ (d) $250 \times 10^{-3} \text{ m/s}$
97. At room temperature, copper has free electron density of $8.4 \times 10^{28} \text{ per m}^3$. The copper conductor has a cross-section of 10^{-6} m^2 and carries a current of 5.4 A . The electron drift velocity in copper is
[MP PET 2000]
[UPSEAT 2002]
- (a) 400 m/s (b) 0.4 m/s
(c) 0.4 mm/s (d) 72 m/s
98. The resistance of a 5 cm long wire is 10Ω . It is uniformly stretched so that its length becomes 20 cm . The resistance of the wire is
- (a) 160Ω (b) 80Ω
(c) 40Ω (d) 20Ω
99. The resistance of an incandescent lamp is
[KCET 2002]
- (a) Greater when switched off
(b) Smaller when switched on
(c) Greater when switched on
(d) The same whether it is switched off or switched on
100. In the figure a carbon resistor has bands of different colours on its body as mentioned in the figure. The value of the resistance is
[RPET 2001]
- (a) $2.2 \text{ k}\Omega$
(b) $3.3 \text{ k}\Omega$
(c) $5.6 \text{ k}\Omega$
(d) $9.1 \text{ k}\Omega$
- 
101. By increasing the temperature, the specific resistance of a conductor and a semiconductor
[AIEEE 2002]
- (a) Increases for both
(b) Decreases for both
(c) Increases, decreases
(d) Decreases, increases
102. Which of the following is vector quantity
[AFMC 2002]
- (a) Current density (b) Current
(c) Wattless current (d) Power

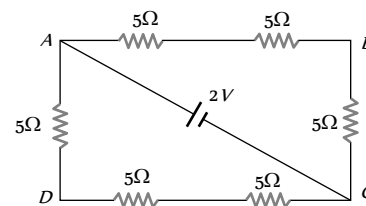
103. Masses of 3 wires of same metal are in the ratio 1 : 2 : 3 and their lengths are in the ratio 3 : 2 : 1. The electrical resistances are in ratio [CPMT 2002]
(a) 1 : 4 : 9 (b) 9 : 4 : 1
(c) 1 : 2 : 3 (d) 27 : 6 : 1
104. A current of 1 mA is flowing through a copper wire. How many electrons will pass a given point in one second [$e = 1.6 \times 10^{-19}$ Coulomb] [RPMT 2000; MP PMT 2002]
(a) 6.25×10^{19} (b) 6.25×10^{15}
(c) 6.25×10^{31} (d) 6.25×10^8
105. The drift velocity of free electrons in a conductor is ' v ' when a current ' I ' is flowing in it. If both the radius and current are doubled, then drift velocity will be [BHU 2002]
(a) v (b) $\frac{v}{2}$
(c) $\frac{v}{4}$ (d) $\frac{v}{8}$
106. A wire of radius r has resistance R . If it is stretched to a radius of $\frac{3r}{4}$, its resistance becomes [BHU 2002]
(a) $\frac{9R}{16}$ (b) $\frac{16R}{9}$
(c) $\frac{81R}{256}$ (d) $\frac{256R}{81}$
107. The resistance of a conductor increases with [CBSE PMT 2002]
(a) Increase in length
(b) Increase in temperature
(c) Decrease in cross-sectional area
(d) All of these
108. A copper wire has a square cross-section, 2.0 mm on a side. It carries a current of 8 A and the density of free electrons is $8 \times 10^{28} \text{ m}^{-3}$. The drift speed of electrons is equal to [AMU (Med.) 2002]
(a) $0.156 \times 10^{-3} \text{ m.s}$ (b) $0.156 \times 10^{-2} \text{ m.s}$
(c) $3.12 \times 10^{-3} \text{ m.s}$ (d) $3.12 \times 10^{-2} \text{ m.s}$
109. Two wires of same material have length L and $2L$ and cross-sectional areas $4A$ and A respectively. The ratio of their specific resistance would be [MHCET 2002]
(a) 1 : 2 (b) 8 : 1
(c) 1 : 8 (d) 1 : 1
110. When a current flows through a conductor its temperature [MHCET 2002]
(a) May increase or decrease
(b) Remains same
(c) Decreases
(d) Increases
111. What length of the wire of specific resistance $48 \times 10^{-8} \Omega \text{ m}$ is needed to make a resistance of 4.2Ω (diameter of wire = 0.4 mm) [CBSE PMT 2000; Pb. PMT 2002]
(a) 4.1 m (b) 3.1 m
(c) 2.1 m (d) 1.1 m
112. A strip of copper and another of germanium are cooled from room temperature to 80 K. The resistance of [AIEEE 2003]
(a) Each of these increases
(b) Each of these decreases
(c) Copper strip increases and that of germanium decreases
(d) Copper strip decreases and that of germanium increases
113. The length of a given cylindrical wire is increased by 100 %. Due to the consequent decrease in diameter the change in the resistance of the wire will be [AIEEE 2003]
(a) 300 % (b) 200 %
(c) 100 % (d) 50 %
114. Express which of the following setups can be used to verify Ohm's law [IIT-JEE (Screening) 2003]
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115. We have two wires A and B of same mass and same material. The diameter of the wire A is half of that B . If the resistance of wire A is 24 ohm then the resistance of wire B will be
(a) 12 Ohm (b) 3.0 Ohm
(c) 1.5 Ohm (d) None of the above
116. In a hydrogen discharge tube it is observed that through a given cross-section 3.13×10^{15} electrons are moving from right to left and 3.12×10^{15} protons are moving from left to right. What is the electric current in the discharge tube and what is its direction
(a) 1 mA towards right (b) 1 mA towards left
(c) 2 mA towards left (d) 2 mA towards right
117. A steady current i is flowing through a conductor of uniform cross-section. Any segment of the conductor has [MP PET 1996]
(a) Zero charge
(b) Only positive charge
(c) Only negative charge
(d) Charge proportional to current i
118. The length of the wire is doubled. Its conductance will be [Kerala PMT 2004]
(a) Unchanged (b) Halved
(c) Quadrupled (d) 1/4 of the original value
119. A source of e.m.f. $E = 15 \text{ V}$ and having negligible internal resistance is connected to a variable resistance so that the current in the circuit increases with time as $i = 1.2 t + 3$. Then, the total charge that will flow in first five second will be
(a) 10 C (b) 20 C
(c) 30 C (d) 40 C
120. The new resistance of wire of $R \Omega$, whose radius is reduced half, is [J & K CET]
(a) 16 R (b) 3 R
(c) 2R (d) R
121. A resistance R is stretched to four times its length. Its new resistance will be [ISM Dhanbad 1994; UPSEAT 2003]
(a) 4 R (b) 64 R

- (c) $R/4$ (d) $16R$
122. What is the resistance of a carbon resistance which has bands of colours brown, black and brown [DCE 1999]
(a) $100\ \Omega$ (b) $1000\ \Omega$
(c) $10\ \Omega$ (d) $1\ \Omega$
123. The lead wires should have [Pb. PMT 2000]
(a) Larger diameter and low resistance
(b) Smaller diameter and high resistance
(c) Smaller diameter and low resistance
(d) Larger diameter and high resistance
124. The alloys constantan and manganin are used to make standard resistance due to they have [MH CET 2000; NCERT 1990]
(a) Low resistivity
(b) High resistivity
(c) Low temperature coefficient of resistance
(d) Both (b) and (c)
125. When a potential difference is applied across the ends of a linear metallic conductor [MP PET 1997]
(a) The free electrons are accelerated continuously from the lower potential end to the higher potential end of the conductor
(b) The free electrons are accelerated continuously from the higher potential end to the lower potential end of the conductor
(c) The free electrons acquire a constant drift velocity from the lower potential end to the higher potential end of the conductor
(d) The free electrons are set in motion from their position of rest
126. The electric resistance of a certain wire of iron is R . If its length and radius are both doubled, then [CBSE PMT 2004]
(a) The resistance will be doubled and the specific resistance will be halved
(b) The resistance will be halved and the specific resistance will remain unchanged
(c) The resistance will be halved and the specific resistance will be doubled
(d) The resistance and the specific resistance, will both remain unchanged
127. A wire of diameter 0.02 metre contains 10^{24} free electrons per cubic metre. For an electrical current of 100 A , the drift velocity of the free electrons in the wire is nearly [UPSEAT 2004]
(a) $1 \times 10^{-4}\text{ m/s}$ (b) $5 \times 10^{-4}\text{ m/s}$
(c) $2 \times 10^{-4}\text{ m/s}$ (d) $8 \times 10^{-4}\text{ m/s}$
128. The following four wires are made of the same material and are at the same temperature. Which one of them has highest electrical resistance [UPSEAT 2004]
(a) Length = 50 cm , diameter = 0.5 mm
(b) Length = 100 cm , diameter = 1 mm
(c) Length = 200 cm , diameter = 2 mm
(d) Length = 300 cm , diameter = 3 mm
129. The colour sequence in a carbon resistor is red, brown, orange and silver. The resistance of the resistor is [DCE 2004]
(a) $21 \times 10^{\pm 10}\%$ (b) $23 \times 10^{\pm 10}\%$
(c) $21 \times 10^{\pm 5}\%$ (d) $12 \times 10^{\pm 5}\%$
130. A thick wire is stretched so that its length become two times. Assuming that there is no change in its density, then what is the ratio of change in resistance of wire to the initial resistance of wire
(a) $2:1$ (b) $4:1$
(c) $3:1$ (d) $1:4$
131. The length of the resistance wire is increased by 10% . What is the corresponding change in the resistance of wire [MH CET 2004]
(a) 10% (b) 25%
(c) 21% (d) 9%
132. The electric field E , current density J and conductivity σ of a conductor are related as [Kerala PMT 2005]
(a) $\sigma = E/j$ (b) $\sigma = j/E$
(c) $\sigma = jE$ (d) $\sigma = 1/jE$
133. Two wires that are made up of two different materials whose specific resistance are in the ratio $2:3$, length $3:4$ and area $4:5$. The ratio of their resistances is [Kerala PMT 2005]
(a) $6:5$ (b) $6:8$
(c) $5:8$ (d) $1:2$

Grouping of Resistances

1. The potential difference between points A and B of adjoining figure is [CPMT 1991]

- (a) $\frac{2}{3}V$
(b) $\frac{8}{9}V$
(c) $\frac{4}{3}V$
(d) $2V$



2. Two resistors of resistance R_1 and R_2 having $R_1 > R_2$ are connected in parallel. For equivalent resistance R , the correct statement is [CPMT 1978; KCET (Med.) 2000]

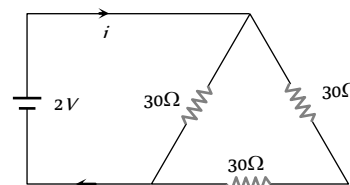
- (a) $R > R_1 + R_2$ (b) $R_1 < R < R_2$
(c) $R_2 < R < (R_1 + R_2)$ (d) $R < R_1$

3. A wire of resistance R is divided in 10 equal parts. These parts are connected in parallel, the equivalent resistance of such connection will be [CPMT 1973, 91]

- (a) $0.01R$ (b) $0.1R$
(c) $10R$ (d) $100R$

4. The current in the adjoining circuit will be [IIT 1983; CPMT 1991, 92; MH CET 2002; Pb. PMT 2001; Kerala PMT 2004]

- (a) $\frac{1}{45}\text{ ampere}$
(b) $\frac{1}{15}\text{ ampere}$
(c) $\frac{1}{10}\text{ ampere}$



(d) $\frac{1}{5}$ ampere

5. There are 8 equal resistances R . Two are connected in parallel, such four groups are connected in series, the total resistance of the system will be [MP PMT 1987]

(a) $R/2$ (b) $2R$
(c) $4R$ (d) $8R$

6. Three resistances of one ohm each are connected in parallel. Such connection is again connected with $2/3\Omega$ resistor in series. The resultant resistance will be [MP PMT 1985]

(a) $\frac{5}{3}\Omega$ (b) $\frac{3}{2}\Omega$
(c) 1Ω (d) $\frac{2}{3}\Omega$

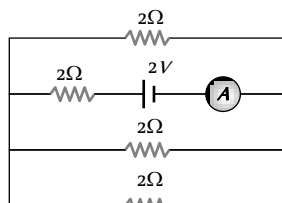
7. The lowest resistance which can be obtained by connecting 10 resistors each of $1/10$ ohm is

[MP PMT 1984; EAMCET 1994]

(a) $1/250\Omega$ (b) $1/200\Omega$
(c) $1/100\Omega$ (d) $1/10\Omega$

8. The reading of the ammeter as per figure shown is

(a) $\frac{1}{8}A$
(b) $\frac{3}{4}A$
(c) $\frac{1}{2}A$
(d) $2A$



9. Three resistors each of 2 ohm are connected together in a triangular shape. The resistance between any two vertices will be

[CPMT 1983; MP PET 1990; MP PMT 1993; DCE 2004]

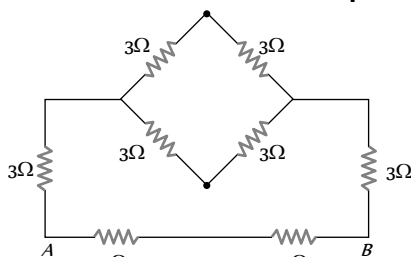
(a) $4/3$ ohm (b) $3/4$ ohm
(c) 3 ohm (d) 6 ohm

10. There are n similar conductors each of resistance R . The resultant resistance comes out to be x when connected in parallel. If they are connected in series, the resistance comes out to be

(a) x/n^2 (b) n^2x
(c) x/n (d) nx

11. Equivalent resistance between A and B will be [CPMT 1981]

(a) 2 ohm
(b) 18 ohm
(c) 6 ohm
(d) 3.6 ohm

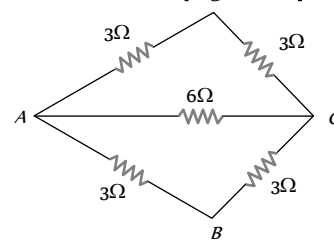


12. A wire has a resistance of 12 ohm. It is bent in the form of equilateral triangle. The effective resistance between any two corners of the triangle is

(a) 9 ohms (b) 12 ohms
(c) 6 ohms (d) $8/3$ ohms

13. The effective resistance between the points A and B in the figure is [MP PET 1994]

(a) 5Ω
(b) 2Ω
(c) 3Ω
(d) 4Ω

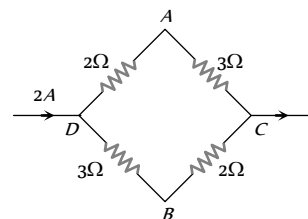


14. Three resistances of magnitude 2, 3 and 5 ohm are connected in parallel to a battery of 10 volts and of negligible resistance. The potential difference across 3Ω resistance will be

(a) 2 volts (b) 3 volts
(c) 5 volts (d) 10 volts

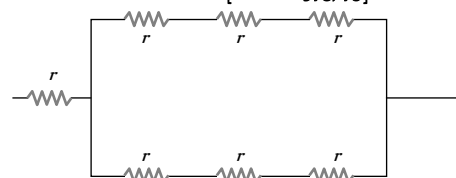
15. A current of 2 A flows in a system of conductors as shown. The potential difference ($V_A - V_B$) will be [CPMT 1975, 76]

(a) +2 V
(b) +1 V
(c) -1 V
(d) -2 V



16. Referring to the figure below, the effective resistance of the network is [NCERT 1973, 75]

(a) $2r$
(b) $4r$
(c) $10r$
(d) $5r/2$



17. Two resistances are joined in parallel whose resultant is $\frac{6}{8}$ ohm.

One of the resistance wire is broken and the effective resistance becomes 2Ω . Then the resistance in ohm of the wire that got broken was

[DPMT 2004]

[CPMT 1976; DPMT 1982]

(a) $3/5$ (b) 2
(c) $6/5$ (d) 3

18. Given three equal resistors, how many different combination of all the three resistors can be made [NCERT 1970]

(a) Six (b) Five
(c) Four (d) Three

19. Lamps used for household lighting are connected in

(a) Series (b) Parallel
(c) Mixed circuit (d) None of the above

20. The equivalent resistance of resistors connected in series is always [CPMT 1984;

(a) Equal to the mean of component resistors
(b) Less than the lowest of component resistors
(c) In between the lowest and the highest of component resistors
(d) Equal to sum of component resistors

21. A cell of negligible resistance and e.m.f. 2 volts is connected to series combination of 2, 3 and 5 ohm. The potential difference in volts between the terminals of 3 ohm resistance will be

(a) 0.6 (b) $2/3$

- (c) 3 (d) 6

22. Four wires of equal length and of resistances $10\ \text{ohms}$ each are connected in the form of a square. The equivalent resistance between two opposite corners of the square is

[NCERT 1977]

- (a) $10\ \text{ohm}$ (b) $40\ \text{ohm}$
(c) $20\ \text{ohm}$ (d) $10/4\ \text{ohm}$

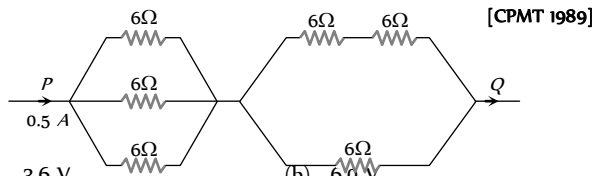
23. Two resistors are connected (a) in series (b) in parallel. The equivalent resistance in the two cases are $9\ \text{ohm}$ and $2\ \text{ohm}$ respectively. Then the resistances of the component resistors are

- (a) $2\ \text{ohm}$ and $7\ \text{ohm}$ (b) $3\ \text{ohm}$ and $6\ \text{ohm}$
(c) $3\ \text{ohm}$ and $9\ \text{ohm}$ (d) $5\ \text{ohm}$ and $4\ \text{ohm}$

24. Resistors of $1, 2, 3\ \text{ohm}$ are connected in the form of a triangle. If a $1.5\ \text{volt}$ cell of negligible internal resistance is connected across $3\ \text{ohm}$ resistor, the current flowing through this resistance will be

- (a) $0.25\ \text{amp}$ (b) $0.5\ \text{amp}$
(c) $1.0\ \text{amp}$ (d) $1.5\ \text{amp}$

25. Resistances of $6\ \text{ohm}$ each are connected in the manner shown in adjoining figure. With the current $0.5\ \text{ampere}$ as shown in figure, the potential difference $V_P - V_Q$ is

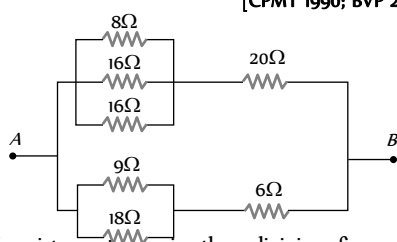


- (a) $3.6\ \text{V}$ (b) $6.0\ \text{V}$
(c) $3.0\ \text{V}$ (d) $7.2\ \text{V}$

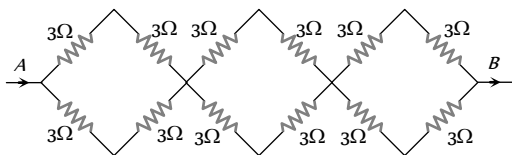
26. The equivalent resistance of the arrangement of resistances shown in adjoining figure between the points A and B is

[CPMT 1990; BVP 2003]

- (a) $6\ \text{ohm}$
(b) $8\ \text{ohm}$
(c) $16\ \text{ohm}$
(d) $24\ \text{ohm}$



27. In the network of resistors shown in the adjoining figure, the equivalent resistance between A and B is



- (a) $54\ \text{ohm}$ (b) $18\ \text{ohm}$
(c) $36\ \text{ohm}$ (d) $9\ \text{ohm}$

28. A wire is broken in four equal parts. A packet is formed by keeping the four wires together. The resistance of the packet in comparison to the resistance of the wire will be

[MP PET 1985; AFMC 2005]

- (a) Equal (b) One fourth

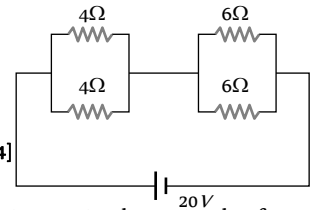
- (c) One eight

- (d) $\frac{1}{16}\ \text{th}$

29. Four resistances are connected in a circuit in the given figure. The electric current flowing through $4\ \text{ohm}$ and $6\ \text{ohm}$ resistance is respectively

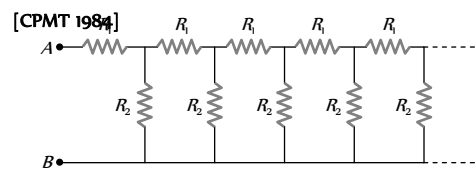
[MP PET 1993]

- (a) $2\ \text{amp}$ and $4\ \text{amp}$
(b) $1\ \text{amp}$ and $2\ \text{amp}$
(c) $1\ \text{amp}$ and $1\ \text{amp}$ [CPMT 1984]
(d) $2\ \text{amp}$ and $2\ \text{amp}$



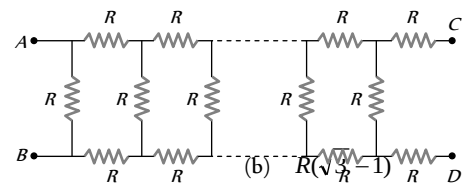
30. An infinite sequence of resistance is shown in the figure. The resultant resistance between A and B will be, when $R_1 = 1\ \text{ohm}$ and $R_2 = 2\ \text{ohm}$

[MP PET 1993]



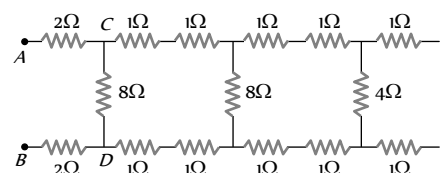
- (a) Infinity (b) $1\ \Omega$
(c) $2\ \Omega$ (d) $1.5\ \Omega$

31. In the figure, the value of resistors to be connected between C and D so that the resistance of the entire circuit between A and B does not change with the number of elementary sets used is



- (a) R (b) $R(\sqrt{3}-1)$
(c) $3R$ (d) $R(\sqrt{3}+1)$

32. In the figure shown, the total resistance between A and B is

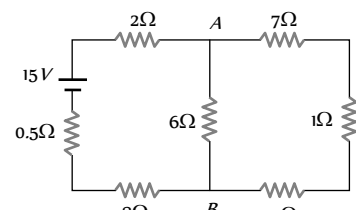


- (a) $12\ \Omega$ (b) $4\ \Omega$
(c) $6\ \Omega$ (d) $8\ \Omega$

33. The current from the battery in circuit diagram shown is

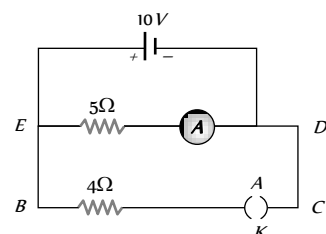
[IIT 1989]

- (a) $1\ \text{A}$
(b) $2\ \text{A}$
(c) $1.5\ \text{A}$
(d) $3\ \text{A}$



34. In the given figure, when key K is opened, the reading of the ammeter A will be

- (a) $50\ \text{A}$

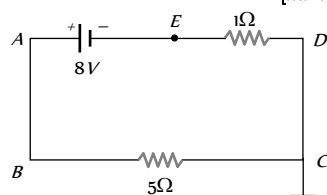


- (b) 2 A
(c) 0.5 A
(d) $\frac{10}{9}\text{ A}$

35. In the given circuit, the potential of the point E is

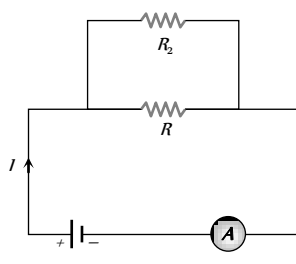
[MP PMT 2003]

- (a) Zero
(b) -8 V
(c) $-4/3\text{ V}$
(d) $4/3\text{ V}$



36. If a resistance R_2 is connected in parallel with the resistance R in the circuit shown, then possible value of current through R and the possible value of R_2 will be

- (a) $\frac{I}{3}, R$
(b) $I, 2R$
(c) $\frac{I}{3}, 2R$
(d) $\frac{I}{2}, R$



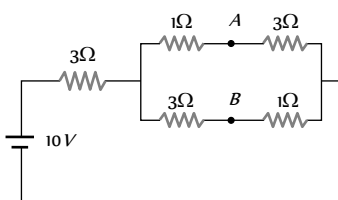
37. Four wires AB, BC, CD, DA of resistance 4 ohm each and a fifth wire BD of resistance 8 ohm are joined to form a rectangle $ABCD$ of which BD is a diagonal. The effective resistance between the points A and B is

[MP PMT 1994]

- (a) 24 ohm
(b) 16 ohm
(c) $\frac{4}{3}\text{ ohm}$
(d) $\frac{8}{3}\text{ ohm}$

38. A battery of e.m.f. 10 V is connected to resistance as shown in figure. The potential difference $V_A - V_B$ between the points A and B is

- (a) -2 V
(b) 2 V
(c) 5 V
(d) $\frac{20}{11}\text{ V}$



39. Three resistances, each of 1 ohm , are joined in parallel. Three such combinations are put in series, then the resultant resistance will be

- (a) 9 ohm
(b) 3 ohm
(c) 1 ohm
(d) $\frac{1}{3}\text{ ohm}$

40. A student has 10 resistors of resistance ' r '. The minimum resistance made by him from given resistors is

[AFMC 1995]

- (a) 10 r
(b) $\frac{r}{10}$
(c) $\frac{r}{100}$
(d) $\frac{r}{5}$

41. Two wires of same metal have the same length but their cross-sections are in the ratio $3:1$. They are joined in series. The

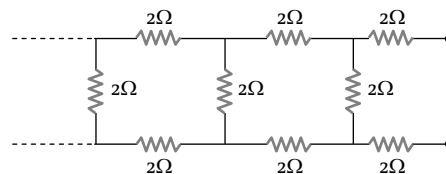
resistance of the thicker wire is 10Ω . The total resistance of the combination will be

[CBSE PMT 1995]

- (a) 40Ω
(b) $\frac{40}{3}\Omega$
(c) $\frac{5}{2}\Omega$
(d) 100Ω

42. The equivalent resistance of the following infinite network of resistances is

[AIIMS 1995]

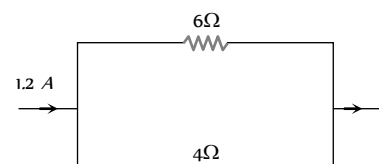


- (a) Less than 4Ω
(b) 4Ω
(c) More than 4Ω but less than 12Ω
(d) 12Ω

43. In the figure given below, the current passing through 6Ω resistor is

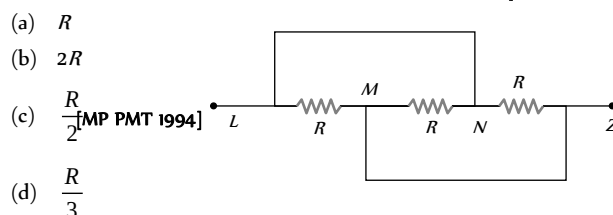
[Manipal MEE 1995]

- (a) 0.40 ampere
(b) 0.48 ampere
(c) 0.72 ampere
(d) 0.80 ampere



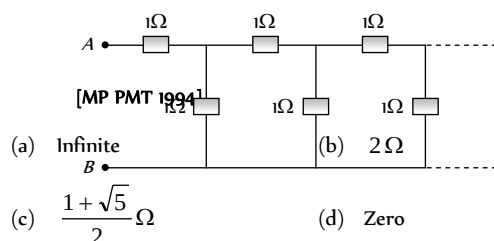
44. Three equal resistances each of value R are joined as shown in the figure. The equivalent resistance between M and N is

[MP PET 1995]



- (a) R
(b) $2R$
(c) $\frac{R}{2}$
(d) $\frac{R}{3}$

45. The equivalent resistance between points A and B of an infinite network of resistances each of 1Ω connected as shown, is



[MP PMT 1994]

- (a) Infinite
(b) 2Ω
(c) $\frac{1+\sqrt{5}}{2}\Omega$
(d) Zero

46. A copper wire of resistance R is cut into ten parts of equal length. Two pieces each are joined in series and then five such combinations are joined in parallel. The new combination will have a resistance

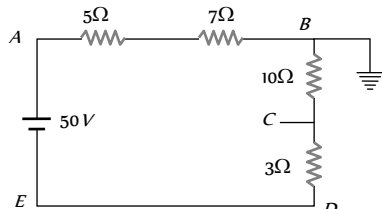
- (a) R
(b) $\frac{R}{4}$
(c) $\frac{R}{5}$
(d) $\frac{R}{25}$

47. A wire has resistance $12\ \Omega$. It is bent in the form of a circle. The effective resistance between the two points on any diameter is equal to
[JIPMER 1999]

(a) $12\ \Omega$ (b) $6\ \Omega$
(c) $3\ \Omega$ (d) $24\ \Omega$

48. In the circuit shown, the point 'B' is earthed. The potential at the point 'A' is

(a) $14\ V$
(b) $24\ V$
(c) $26\ V$
(d) $50\ V$

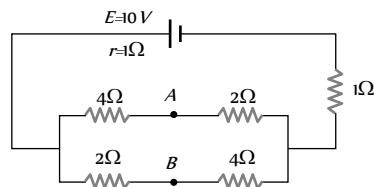


49. Three resistors each of $4\ \Omega$ are connected together to form a network. The equivalent resistance of the network cannot be

(a) $1.33\ \Omega$ (b) $3.0\ \Omega$
(c) $6.0\ \Omega$ (d) $12.0\ \Omega$

50. In the circuit shown below, the cell has an e.m.f. of $10\ V$ and internal resistance of $1\ \Omega$. The other resistances are shown in the figure. The potential difference $V_A - V_B$ is

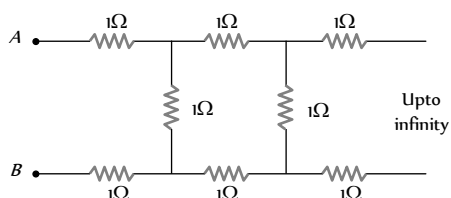
(a) $6\ V$
(b) $4\ V$
(c) $2\ V$
(d) $-2\ V$



51. A wire of resistance R is cut into ' n ' equal parts. These parts are then connected in parallel. The equivalent resistance of the combination will be
[MP PMT/PET 1998; BHU 2005]

(a) nR (b) $\frac{R}{n}$
(c) $\frac{n}{R}$ (d) $\frac{R}{n^2}$

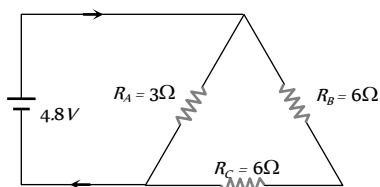
52. The resistance between the terminal points A and B of the given infinitely long circuit will be
[MP PMT/PET 1998]



(a) $(\sqrt{3} - 1)$ (b) $(1 - \sqrt{3})$
(c) $(1 + \sqrt{3})$ (d) $(2 + \sqrt{3})$

53. The current in the given circuit is
[CBSE PMT 1999]

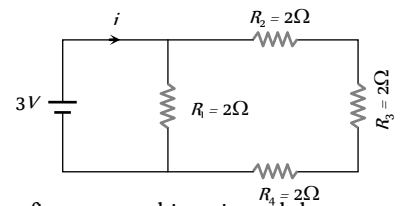
(a) $8.31\ A$
(b) $6.82\ A$



(c) $4.92\ A$
(d) $2\ A$

54. What is the current (i) in the circuit as shown in figure
[AIIMS 1998]

(a) $2\ A$
(b) $1.2\ A$
(c) $1\ A$
(d) $0.5\ A$



55. n equal resistors are first connected in series and then connected in parallel. What is the ratio of the maximum to the minimum resistance
[KCET 1994]

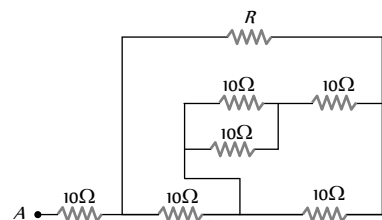
(a) n (b) $\frac{1}{n^2}$
(c) n^2 (d) $\frac{1}{n}$

56. A uniform wire of $16\ \Omega$ is made into the form of a square. Two opposite corners of the square are connected by a wire of resistance $16\ \Omega$. The effective resistance between the other two opposite corners is
[EAMCET (Med.) 1995]

(a) $32\ \Omega$ (b) $20\ \Omega$
(c) $8\ \Omega$ (d) $4\ \Omega$

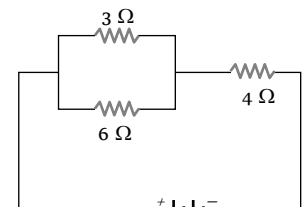
57. For what value of R the net resistance of the circuit will be $18\ \Omega$

(a) $8\ \Omega$
(b) $10\ \Omega$
(c) $16\ \Omega$
(d) $24\ \Omega$



58. In the figure, current through the $3\ \Omega$ resistor is $0.8\ \text{ampere}$, then potential drop through $4\ \Omega$ resistor is
[CBSE PMT 1993; AFMC 1999; MP PMT 2004]

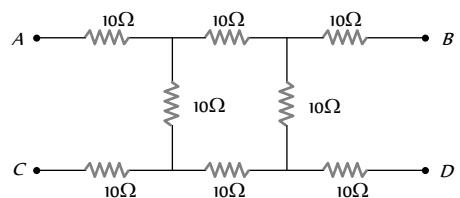
(a) $9.6\ V$
(b) $2.6\ V$
(c) $4.8\ V$
(d) $1.2\ V$



59. Three resistances $4\ \Omega$ each of are connected in the form of an equilateral triangle. The effective resistance between two corners is

(a) $8\ \Omega$ (b) $12\ \Omega$
(c) $\frac{3}{8}\ \Omega$ (d) $\frac{8}{3}\ \Omega$

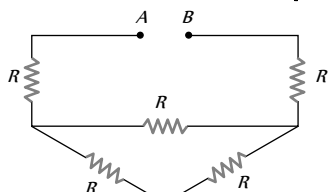
60. What will be the equivalent resistance between the two points A and D
[CBSE PMT 1996]



- (a) $10\ \Omega$ (b) $20\ \Omega$
(c) $30\ \Omega$ (d) $40\ \Omega$

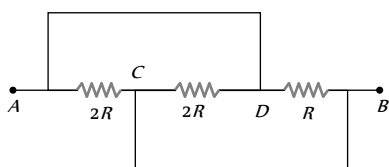
61. What is the equivalent resistance between A and B in the figure below if $R = 3\ \Omega$ [SCRA 1996]

- (a) $9\ \Omega$
(b) $12\ \Omega$
(c) $15\ \Omega$
(d) None of these



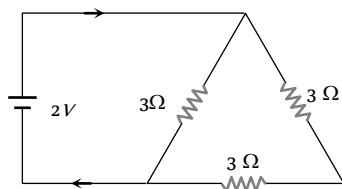
62. What is the equivalent resistance between A and B [BHU 1997; MP PET 2001]

- (a) $\frac{2}{3}R$
(b) $\frac{3}{2}R$
(c) $\frac{R}{2}$
(d) $2R$



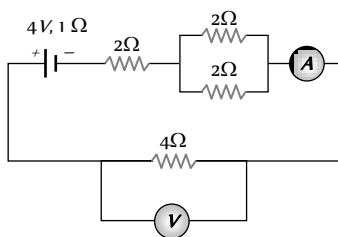
63. The current in the following circuit is [CBSE PMT 1997]

- (a) $\frac{1}{8}A$
(b) $\frac{2}{9}A$
(c) $\frac{2}{3}A$
(d) $1A$



64. What is the equivalent resistance of the circuit [KCET 1998]

- (a) $6\ \Omega$
(b) $7\ \Omega$
(c) $8\ \Omega$
(d) $9\ \Omega$

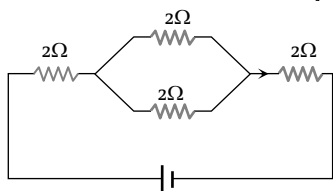


65. 10 wires (same length, same area, same material) are connected in parallel and each has $1\ \Omega$ resistance, then the equivalent resistance will be [RPMT 1999]

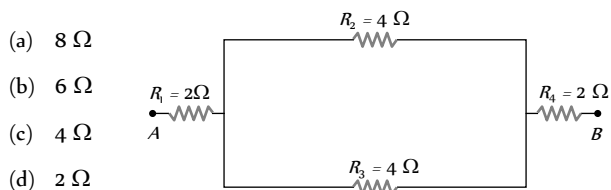
- (a) $10\ \Omega$ (b) $1\ \Omega$
(c) $0.1\ \Omega$ (d) $0.001\ \Omega$

66. The equivalent resistance of the circuit shown in the figure is [CPMT 1999]

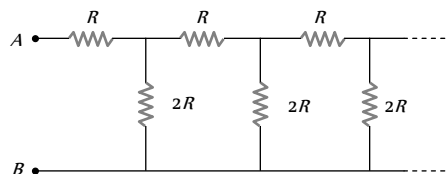
- (a) $8\ \Omega$
(b) $6\ \Omega$
(c) $5\ \Omega$
(d) $4\ \Omega$



67. In the given figure, the equivalent resistance between the points A and B is [AIIMS 1999]



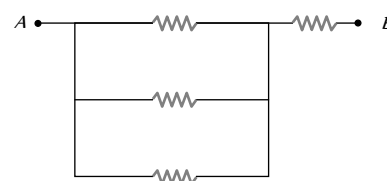
68. An infinite ladder network is arranged with resistances R and $2R$ as shown. The effective resistance between terminals A and B is



- (a) ∞ (b) R
(c) $2R$ (d) $3R$

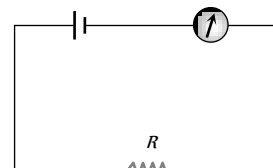
69. If all the resistors shown have the value $2\ \text{ohm}$ each, the equivalent resistance over AB is [JIPMER 1999]

- (a) $2\ \text{ohm}$
(b) $4\ \text{ohm}$
(c) $1\frac{2}{3}\ \text{ohm}$
(d) $2\frac{2}{3}\ \text{ohm}$



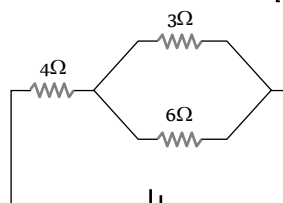
70. A battery of $\text{emf } 10\text{ V}$ and internal resistance $3\ \Omega$ is connected to a resistor as shown in the figure. If the current in the circuit is 0.5 A , then the resistance of the resistor will be

- (a) $19\ \Omega$
(b) $17\ \Omega$
(c) $10\ \Omega$
(d) $12\ \Omega$



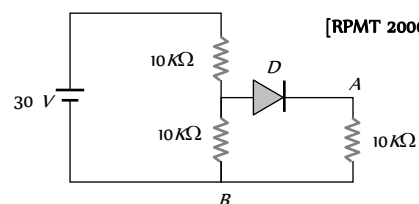
71. The potential drop across the $3\ \Omega$ resistor is [CPMT 2000]

- (a) 1 V
(b) 1.5 V
(c) 2 V
(d) 3 V

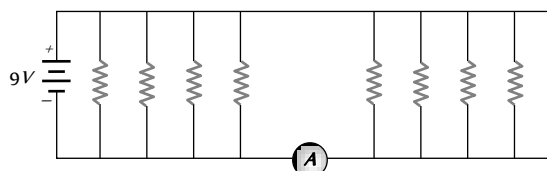


72. In the given figure, potential difference between A and B is [RPMT 2000]

- (a) 0
(b) 5 volt
(c) 10 volt
(d) 15 volt



73. If each resistance in the figure is of $9\ \Omega$ then reading of ammeter is



- (a) 5 A (b) 8 A
(c) 2 A (d) 9 A

74. Four resistances $10\ \Omega$, $5\ \Omega$, $7\ \Omega$ and $3\ \Omega$ are connected so that they form the sides of a rectangle AB , BC , CD and DA respectively. Another resistance of $10\ \Omega$ is connected across the diagonal AC . The equivalent resistance between A and B is

- (a) $2\ \Omega$ (b) $5\ \Omega$
(c) $7\ \Omega$ (d) $10\ \Omega$

75. Two wires of equal diameters, of resistivities ρ_1 and ρ_2 and lengths l and l , respectively, are joined in series. The equivalent resistivity of the combination is

[EAMCET (Engg.) 2000]

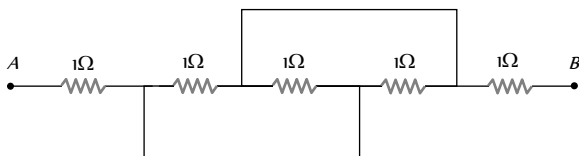
- (a) $\frac{\rho_1 l_1 + \rho_2 l_2}{l_1 + l_2}$ (b) $\frac{\rho_1 l_2 + \rho_2 l_1}{l_1 - l_2}$
(c) $\frac{\rho_1 l_2 + \rho_2 l_1}{l_1 + l_2}$ (d) $\frac{\rho_1 l_1 - \rho_2 l_2}{l_1 - l_2}$

76. Four resistances of $100\ \Omega$ each are connected in the form of square. Then, the effective resistance along the diagonal points is

- (a) $200\ \Omega$ (b) $400\ \Omega$
(c) $100\ \Omega$ (d) $150\ \Omega$

77. Equivalent resistance between the points A and B is (in Ω)

[AMU (Engg.) 2000]



- (a) $\frac{1}{5}$ (b) $1\frac{1}{4}$
(c) $2\frac{1}{3}$ (d) $3\frac{1}{2}$

78. Two wires of the same material and equal length are joined in parallel combination. If one of them has half the thickness of the other and the thinner wire has a resistance of $8\ \text{ohms}$, the resistance of the combination is equal to

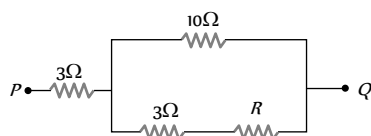
[AMU (Engg.) 2000]

- (a) $\frac{5}{8}\ \text{ohms}$ (b) $\frac{8}{5}\ \text{ohms}$
(c) $\frac{3}{8}\ \text{ohms}$ (d) $\frac{8}{3}\ \text{ohms}$

79. In the circuit shown here, what is the value of the unknown resistor R so that the total resistance of the circuit between points P and Q is also equal to R

[MP PET 2001]

- (a) $3\ \text{ohms}$
(b) $\sqrt{39}\ \text{ohms}$
(c) $\sqrt{69}\ \text{ohms}$



(d) $10\ \text{ohms}$

80. A uniform wire of resistance $9\ \Omega$ is cut into 3 equal parts. They are connected in the form of equilateral triangle ABC . A cell of e.m.f. $2\ \text{V}$ and negligible internal resistance is connected across B and C . Potential difference across AB is

[Kerala (Engg.) 2001]

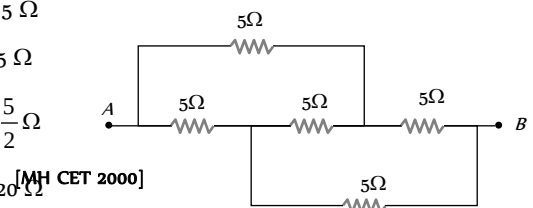
- (a) $1\ \text{V}$ (b) $2\ \text{V}$
(c) $3\ \text{V}$ (d) $0.5\ \text{V}$

81. [EAMCET (Med.) 2000] Three resistors of resistances $2\ \Omega$, $4\ \Omega$ and $8\ \Omega$ are connected in parallel, then the equivalent resistance of the combination will be [KCET 2001]

- (a) $\frac{8}{7}\ \Omega$ (b) $\frac{7}{8}\ \Omega$
(c) $\frac{7}{4}\ \Omega$ (d) $\frac{4}{9}\ \Omega$

82. Effective resistance between A and B is [UPSEAT 2001]

- (a) $15\ \Omega$
(b) $5\ \Omega$
(c) $\frac{5}{2}\ \Omega$
(d) $20\ \Omega$



83. The effective resistance of two resistors in parallel is $\frac{12}{7}\ \Omega$. If one of the resistors is disconnected the resistance becomes $4\ \Omega$. The resistance of the other resistor is [MH CET 2002]

- (a) $4\ \Omega$ (b) $3\ \Omega$
(c) $\frac{12}{7}\ \Omega$ (d) $\frac{7}{12}\ \Omega$

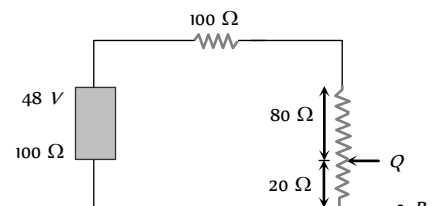
84. Two resistance wires on joining in parallel the resultant resistance is $\frac{6}{5}\ \text{ohms}$. One of the wire breaks, the effective resistance is $2\ \text{ohms}$. The resistance of the broken wire is

[MP PET 2001, 2002]

- (a) $\frac{3}{5}\ \text{ohm}$ (b) $2\ \text{ohm}$
(c) $\frac{6}{5}\ \text{ohm}$ (d) $3\ \text{ohm}$

85. In the circuit, the potential difference across PQ will be nearest to

- (a) $9.6\ \text{V}$
(b) $6.6\ \text{V}$
(c) $4.8\ \text{V}$
(d) $3.2\ \text{V}$



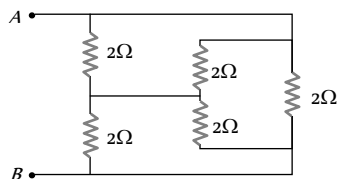
86. Three resistors are connected to form the sides of a triangle ABC , the resistance of the sides AB , BC and CA are $40\ \text{ohms}$, $60\ \text{ohms}$ and $100\ \text{ohms}$ respectively. The effective resistance between the points A and B in ohms will be

[JIPMER 2002]

- (a) 32 (b) 64
(c) 50 (d) 200

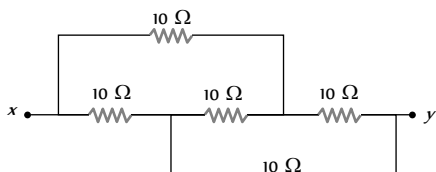
87. Find the equivalent resistance across AB [Orissa JEE 2002]

- (a) $1\ \Omega$
(b) $2\ \Omega$
(c) $3\ \Omega$
(d) $4\ \Omega$



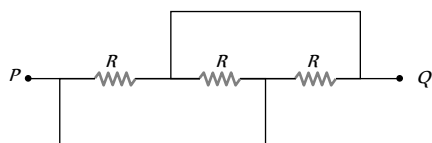
88. The equivalent resistance between x and y in the circuit shown is

- (a) $10\ \Omega$
(b) $40\ \Omega$
(c) $20\ \Omega$
(d) $\frac{5}{2}\ \Omega$



89. The equivalent resistance between the points P and Q of the circuit given is [Pb. PMT 2002]

- (a) $\frac{R}{4}$
(b) $\frac{R}{3}$
(c) $4R$
(d) $2R$



90. Two wires of the same dimensions but resistivities ρ_1 and ρ_2 are connected in series. The equivalent resistivity of the combination is

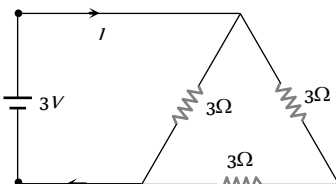
- (a) $\rho_1 + \rho_2$ (b) $\frac{\rho_1 + \rho_2}{2}$
(c) $\sqrt{\rho_1 \rho_2}$ (d) $2(\rho_1 + \rho_2)$

91. Three unequal resistors in parallel are equivalent to a resistance $1\ \text{ohm}$. If two of them are in the ratio $1 : 2$ and if no resistance value is fractional, the largest of the three resistances in ohms is

- (a) 4 (b) 6
(c) 8 (d) 12

92. A 3 volt battery with negligible internal resistance is connected in a circuit as shown in the figure. The current I , in the circuit will be

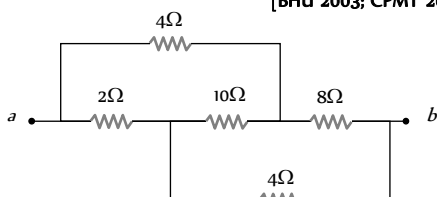
- (a) $\frac{1}{3}\text{ A}$
(b) 1 A
(c) 1.5 A
(d) 2 A



93. Find the equivalent resistance between the points a and b

[BHU 2003; CPMT 2004]

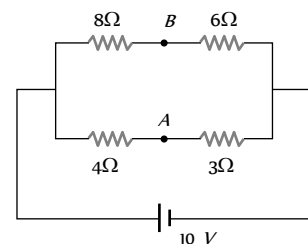
- (a) $2\ \Omega$
(b) $4\ \Omega$
(c) $8\ \Omega$
(d) $16\ \Omega$



94. The potential difference between point A & B is

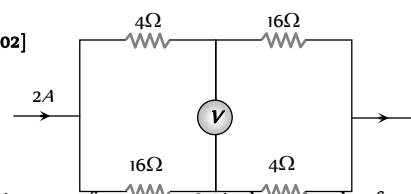
[BHU 2003; CPMT 2004; MP PMT 2005]

- (a) $\frac{20}{7}\text{ V}$
(b) $\frac{40}{7}\text{ V}$
(c) $\frac{10}{7}\text{ V}$
(d) 0



95. In the circuit shown below, The reading of the voltmeter V is

- (a) 12 [MP PMT 2002]
(b) 8 V
(c) 20 V
(d) 16 V



96. A wire has a resistance of $12\ \text{ohm}$. It is bent in the form of equilateral triangle. The effective resistance between any two corners of the triangle is

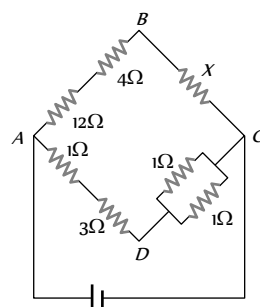
- (a) 9 ohms (b) 12 ohms
(c) 6 ohms (d) $\frac{8}{3}\ \text{ohms}$

97. A series combination of two resistors $1\ \Omega$ each is connected to a 12 V battery of internal resistance $0.4\ \Omega$. The current flowing through it will be [MH CET (Med.) 1999]

- (a) 3.5 A (b) 5 A
(c) 6 A (d) 10 A

98. In the circuit shown in the adjoining figure, the current between B and C is zero, the unknown resistance is of [KCET 2003]

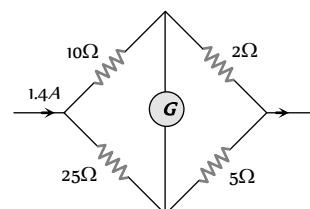
[CPMT 1986]



- (a) $4\ \Omega$ [EAMCET 2003]
(b) $2\ \Omega$
(c) $3\ \Omega$
(d) emf of a cell is required to find the value of X [AIEEE 2003]

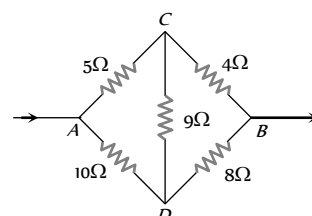
99. In the circuit shown in the figure, the current flowing in $2\ \Omega$ resistance [CPMT 1989; MP PMT 2004]

- (a) 1.4 A
(b) 1.2 A
(c) 0.4 A
(d) 1.0 A



100. Five resistors are connected as shown in the diagram. The equivalent resistance between A and B is

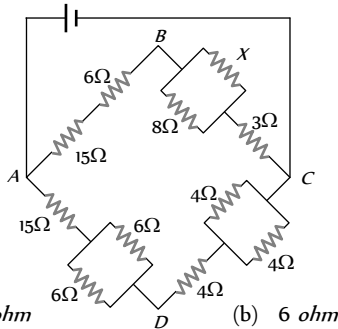
- (a) 6 ohm
(b) 9 ohm



[MP PMT 1996]

(c) $12\ \text{ohm}$ (d) $15\ \text{ohm}$

101. In the figure given the value of X resistance will be, when the p.d. between B and D is zero [MP PET 1993]

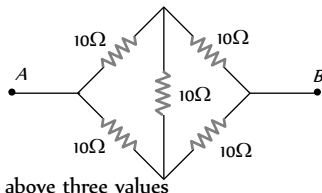


- (a) $4\ \text{ohm}$ (b) $6\ \text{ohm}$
(c) $8\ \text{ohm}$ (d) $9\ \text{ohm}$

102. The effective resistance between points A and B is

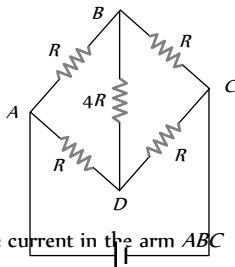
[NCERT 1974; MP PMT 2000]

- (a) $10\ \Omega$
(b) $20\ \Omega$
(c) $40\ \Omega$
(d) None of the above three values



103. Five resistors of given values are connected together as shown in the figure. The current in the arm BD will be

[MP PMT 1995]

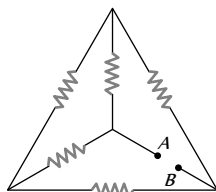


- (a) Half the current in the arm ABC
(b) Zero
(c) Twice the current in the arm ABC
(d) Four times the current in the arm ABC

104. In the network shown in the figure, each of the resistance is equal to $2\ \Omega$. The resistance between the points A and B is

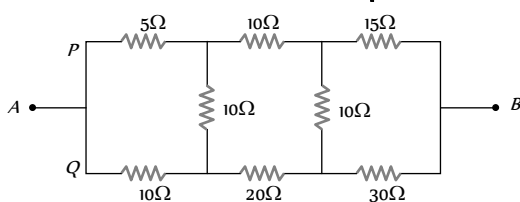
[CBSE PMT 1995]

- (a) $1\ \Omega$
(b) $4\ \Omega$
(c) $3\ \Omega$
(d) $2\ \Omega$



105. In the arrangement of resistances shown below, the effective resistance between points A and B is

[MP PMT 1997; RPET 2001]



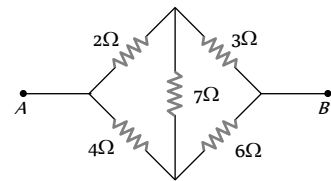
- (a) $20\ \Omega$ (b) $30\ \Omega$

(c) $90\ \Omega$ (d) $110\ \Omega$

106. Five resistances are connected as shown in the figure. The effective resistance between the points A and B is

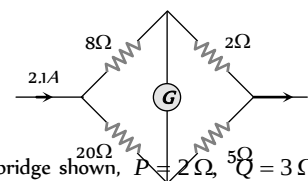
[MP PMT 1999; KCET 2001; BHU 2001, 05]

- (a) $\frac{10}{3}\ \Omega$
(b) $\frac{20}{3}\ \Omega$
(c) $15\ \Omega$
(d) $6\ \Omega$



107. In the given figure, when galvanometer shows no deflection, the current (in ampere) flowing through $5\ \Omega$ resistance will be

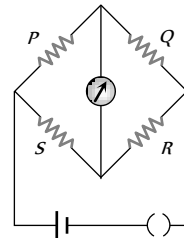
- (a) 0.5
(b) 0.6
(c) 0.9
(d) 1.5



108. In the Wheatstone's bridge shown, $P = 2\ \Omega$, $Q = 3\ \Omega$, $R = 6\ \Omega$ and $S = 8\ \Omega$. In order to obtain balance, shunt resistance across 'S' must be

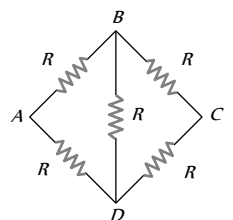
[SCRA 1998]

- (a) $2\ \Omega$
(b) $3\ \Omega$
(c) $6\ \Omega$
(d) $8\ \Omega$



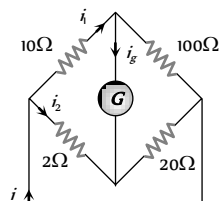
109. Five equal resistances each of value R are connected in a form shown alongside. The equivalent resistance of the network

- (a) Between the points B and D is R
(b) Between the points B and D is $\frac{R}{2}$
(c) Between the points A and C is R
(d) Between the points A and C is $\frac{R}{2}$



110. In the circuit shown below the resistance of the galvanometer is $20\ \Omega$. In which case of the following alternatives are the currents arranged strictly in the decreasing order

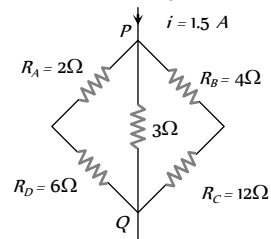
- (a) i, i, i, i
(b) i, i, i, i
(c) i, i, i, i
(d) i, i, i, i



111. Potential difference between the points P and Q in the electric circuit shown is

[KCET 1999]

- (a) $4.5\ \text{V}$
(b) $1.2\ \text{V}$
(c) $2.4\ \text{V}$

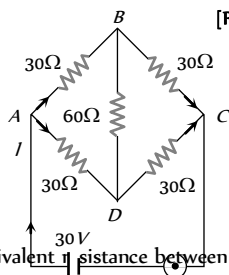


(d) 2.88 V

112. The current between B and D in the given figure is

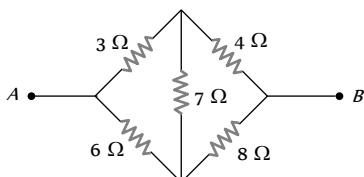
[RPET 2000; DCE 2001]

- (a) 1 amp
(b) 2 amp
(c) Zero
(d) 0.5 amp



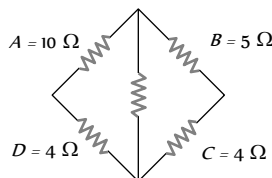
113. In the given figure, equivalent resistance between A and B will be

- (a) $\frac{14}{3} \Omega$
(b) $\frac{3}{14} \Omega$
(c) $\frac{9}{14} \Omega$
(d) $\frac{14}{9} \Omega$



114. In a typical Wheatstone network, the resistances in cyclic order are $A = 10 \Omega$, $B = 5 \Omega$, $C = 4 \Omega$ and $D = 4 \Omega$ for the bridge to be balanced

[KCET 2000]

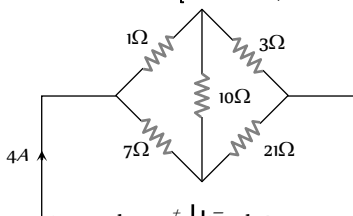


- (a) 10Ω should be connected in parallel with A
(b) 10Ω should be connected in series with A
(c) 5Ω should be connected in series with B
(d) 5Ω should be connected in parallel with B

115. In the circuit shown in figure, the current drawn from the battery is $4A$. If 10Ω resistor is replaced by 20Ω resistor, then current drawn from the circuit will be

[KCET 2000; CBSE PMT 2001]

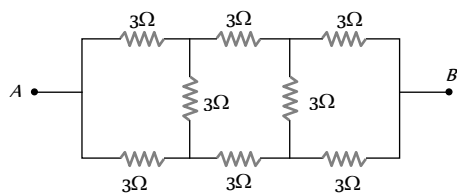
- (a) 1 A
(b) 2 A
(c) 3 A
(d) 0 A



116. Calculate the equivalent resistance between A and B

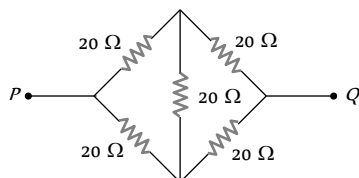
[UPSEAT 2001]

- (a) $\frac{9}{2} \Omega$
(b) 3Ω
(c) 6Ω
(d) $\frac{5}{3} \Omega$



117. The equivalent resistance between P and Q in the given figure, is

- (a) 50Ω



(b) 40Ω

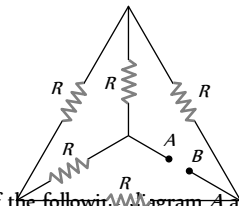
(c) 30Ω

(d) 20Ω

118. If each of the resistance of the network shown in the figure is R , the equivalent resistance between A and B is

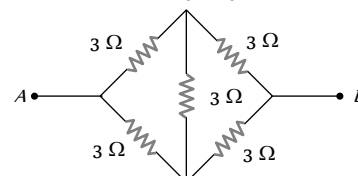
[KCET 2002]

- (a) $5 R$
(b) $3 R$
(c) R [CBSE PMT 2000]
(d) $R/2$



119. The equivalent resistance of the following diagram A and B is

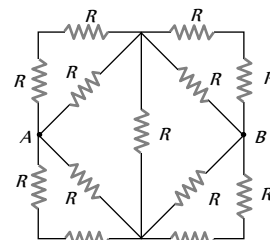
- (a) $\frac{2}{3} \Omega$
(b) 9Ω
(c) 6Ω
(d) None of these



120. Thirteen resistances each of resistance R ohm are connected in the circuit as shown in the figure below. The effective resistance between A and B is

[KCET 2003]

- (a) $2R \Omega$
(b) $\frac{4R}{3} \Omega$
(c) $\frac{2R}{3} \Omega$
(d) $R \Omega$

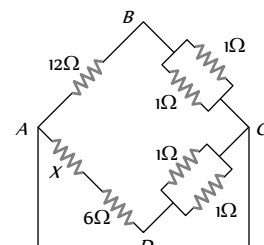


121. In a Wheatstone's bridge all the four arms have equal resistance R . If the resistance of the galvanometer arm is also R , the equivalent resistance of the combination as seen by the battery is

- (a) $\frac{R}{2}$ (b) R
(c) $2R$ (d) $\frac{R}{4}$

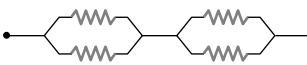
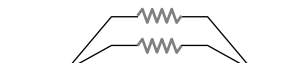
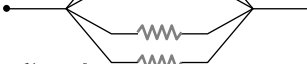
122. For what value of unknown resistance X , the potential difference between B and D will be zero in the circuit shown in the figure

- (a) 4Ω
(b) 6Ω
(c) 2Ω
(d) 5Ω

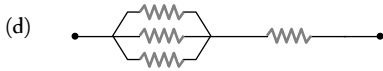


123. Which arrangement of four identical resistances should be used to draw maximum energy from a cell of voltage V

[MP PMT 2004]

- (a) 
(b) 
(c) 

[MH CET (Med.) 2001]



124. An unknown resistance R is connected in series with a resistance of $10\ \Omega$. This combination is connected to one gap of a metre bridge while a resistance R_1 is connected in the other gap. The balance point is at 50 cm . Now, when the $10\ \Omega$ resistance is removed the balance point shifts to 40 cm . The value of R is (in ohm)

(a) 60 (b) 40
(c) 20 (d) 10

125. A wire has a resistance of $6\ \Omega$. It is cut into two parts and both half values are connected in parallel. The new resistance is

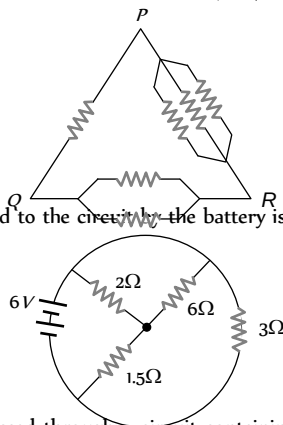
(a) $12\ \Omega$ (b) $1.5\ \Omega$
(c) $3\ \Omega$ (d) $6\ \Omega$

126. Six equal resistances are connected between points P , Q and R as shown in the figure. Then the net resistance will be maximum between [IIT-JEE (Screening) 2004]

(a) P and Q
(b) Q and R
(c) P and R
(d) Any two points

127. The total current supplied to the circuit by the battery is [AIEEE 2004]

(a) 1 A
(b) 2 A
(c) 4 A
(d) 6 A



128. An electric current is passed through a circuit containing two wires of the same material, connected in parallel. If the lengths and radii of the wires are in the ratio of $4/3$ and $2/3$, then the ratio of the currents passing through the wire will be [AIEEE 2004]

(a) 3 (b) $1/3$
(c) $8/9$ (d) 2

129. If a rod has resistance $4\ \Omega$ and if rod is turned as half cycle then the resistance along diameter [BCECE 2004]

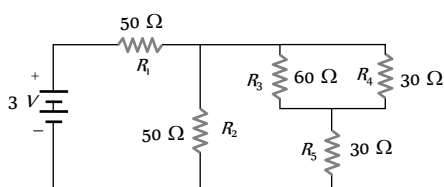
(a) $1.56\ \Omega$ (b) $2.44\ \Omega$
(c) $4\ \Omega$ (d) $2\ \Omega$

130. If three resistors of resistance $2\ \Omega$, $4\ \Omega$ and $5\ \Omega$ are connected in parallel then the total resistance of the combination will be [Pb. PMT 2004]

(a) $\frac{20}{19}\ \Omega$ (b) $\frac{19}{20}\ \Omega$
(c) $\frac{19}{10}\ \Omega$ (d) $\frac{10}{19}\ \Omega$

131. In circuit shown below, the resistances are given in ohms and the battery is assumed ideal with emf equal to 3 volt . The voltage across the resistance R is [UPSEAT 2004; Kerala PMT 2004]

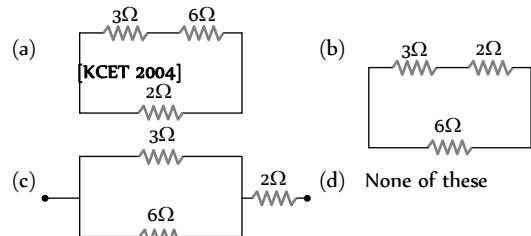
(a) 0.4 V
(b) 0.6 V
(c) 1.2 V
(d) 1.5 V



132. A parallel combination of two resistors, of $1\ \Omega$ each, is connected in series with a $1.5\ \Omega$ resistor. The total combination is connected across a 10 V battery. The current flowing in the circuit is

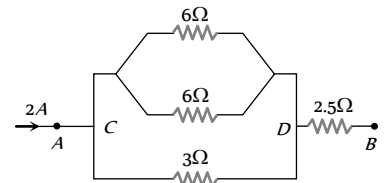
(a) 5 A (b) 20 A
(c) 0.2 A (d) 0.4 A

133. [KCET 2004] are provided three resistances $2\ \Omega$, $3\ \Omega$ and $6\ \Omega$. How will you connect them so as to obtain the equivalent resistance of $4\ \Omega$



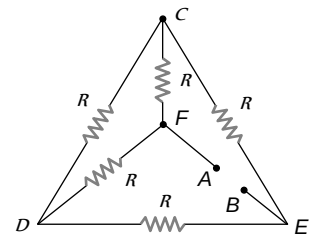
134. The equivalent resistance and potential difference between A and B for the circuit is respectively [Pb. PMT 2003]

(a) $4\ \Omega$, 8 V
(b) $8\ \Omega$, 4 V
(c) $2\ \Omega$, 2 V
(d) $16\ \Omega$, 8 V



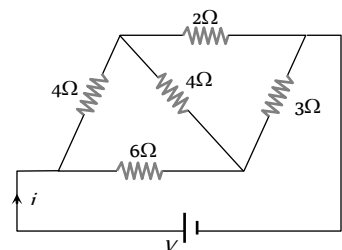
135. Five equal resistances each of resistance R are connected as shown in the figure. A battery of V volts is connected between A and B . The current flowing in $AFCEB$ will be [CBSE PMT 2004]

(a) $\frac{3V}{R}$
(b) $\frac{V}{R}$
(c) $\frac{V}{2R}$
(d) $\frac{2V}{R}$



136. For the network shown in the figure the value of the current i is

(a) $\frac{9V}{35}$
(b) $\frac{5V}{18}$
(c) $\frac{5V}{9}$
(d) $\frac{18V}{5}$

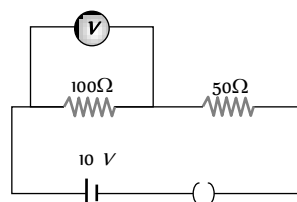
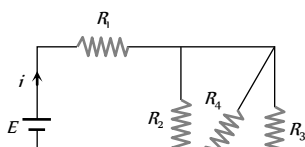


137. When a wire of uniform cross-section a , length l and resistance R is bent into a complete circle, resistance between any two of diametrically opposite points will be [CBSE PMT 2005]

(a) $\frac{R}{4}$ (b) $\frac{R}{8}$
(c) $4R$ (d) $\frac{R}{2}$

138. The current in a simple series circuit is 5.0 amp . When an additional resistance of 2.0 ohms is inserted, the current drops to 4.0 amp . The original resistance of the circuit in ohms was

139. In the circuit given $E = 6.0 \text{ V}$, $R_1 = 100 \text{ ohms}$, $R_2 = R_3 = 50 \text{ ohms}$, $R_4 = 75 \text{ ohms}$. The equivalent resistance of the circuit, in ohms, is
- (a) 11.875
(b) 26.31
(c) 118.75
(d) None of these
140. By using only two resistance coils singly, in series, or in parallel one should be able to obtain resistances of 3, 4, 12 and 16 ohms. The separate resistances of the coil are
- [KCET 2005]
- (a) 3 and 4
(b) 4 and 12
(c) 12 and 16
(d) 16 and 3
141. In the given circuit, the voltmeter records 5 volts. The resistance of the voltmeter in ohms is
- [KCET 2005]



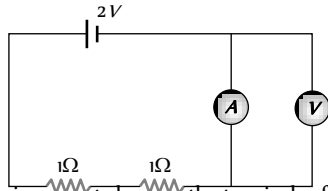
Kirchhoff's Law, Cells

1. In the adjoining circuit, the battery E_1 has an *e.m.f.* of 12 volt and zero internal resistance while the battery E has an *e.m.f.* of 2 volt. If the galvanometer G reads zero, then the value of the resistance X in ohm is
- [NCERT 1990; AIEEE 2005]
- (a) 10
(b) 100
(c) 500
(d) 200
2. The magnitude and direction of the current in the circuit shown will be
- [CPMT 1986, 88]
-
- (a) $\frac{7}{3} \text{ A}$ from a to b through e
(b) $\frac{7}{3} \text{ A}$ from b to a through e
(c) 1 A from b to a through e
(d) 1 A from a to b through e
3. A cell of *e.m.f.* 1.5 V having a finite internal resistance is connected to a load resistance of 2Ω . For maximum power transfer the internal resistance of the cell should be
- [BIT 1988]
- (a) 4 ohm
(b) 0.5 ohm
(c) 2 ohm
(d) None of these

4. By a cell a current of 0.9 A flows through 2 ohm resistor and 0.3 A through 7 ohm resistor. The internal resistance of the cell is [KCET 2003]
- (a) 0.5Ω
(b) 1.0Ω
(c) 1 [KCET 2005]
(d) 2.0Ω
5. The *e.m.f.* of a cell is E volts and internal resistance is r ohm. The resistance in external circuit is also r ohm. The p.d. across the cell will be [CPMT 1985; NCERT 1973]
- (a) $E/2$
(b) $2E$
(c) $4E$
(d) $E/4$
6. A cell of *e.m.f.* E is connected with an external resistance R , then p.d. across cell is V . The internal resistance of cell will be [MNR 1987; Kerala PET 2003]
- (a) $\frac{(E - V)R}{E}$
(b) $\frac{(E - V)R}{V}$
(c) $\frac{(V - E)R}{V}$
(d) $\frac{(V - E)R}{E}$
7. Two cells, *e.m.f.* of each is E and internal resistance r are connected in parallel between the resistance R . The maximum energy given to the resistor will be, only when [MNR 1988; MP PET 2000; UPSEAT 2001]
- (a) $R = r/2$
(b) $R = r$
(c) $R = 2r$
(d) $R = 0$
8. Kirchhoff's first law i.e. $\sum i = 0$ at a junction is based on the law of conservation of [CBSE PMT 1997; AIIMS 2000; MP PMT 2002; RPMT 2001; DPMT 2005]
- (a) Charge
(b) Energy
(c) Momentum
(d) Angular momentum
9. Kirchhoff's second law is based on the law of conservation of [RPET 2003; MH CET 2001]
- (a) Charge
(b) Energy
(c) Momentum
(d) Sum of mass and energy
10. The figure below shows currents in a part of electric circuit. The current i is [CPMT 1981; RPET 1999]
- (a) 1.7 amp
(b) 3.7 amp
(c) 1.3 amp
(d) 1 amp
-
11. The terminal potential difference of a cell is greater than its *e.m.f.* when it is
- (a) Being discharged
(b) In open circuit
(c) Being charged
(d) Being either charged or discharged
12. In the circuit shown, potential difference between X and Y will be
- (a) Zero
(b) 20 V
(c) 60 V
(d) 120 V
-

13. In the above question, potential difference across the $40\ \Omega$ resistance will be
 (a) Zero (b) $80\ V$
 (c) $40\ V$ (d) $120\ V$
14. In the circuit shown, A and V are ideal ammeter and voltmeter respectively. Reading of the voltmeter will be

- (a) $2\ V$
 (b) $1\ V$
 (c) $0.5\ V$
 (d) Zero



15. When a resistance of $2\ \text{ohm}$ is connected across the terminals of a cell, the current is $0.5\ \text{amperes}$. When the resistance is increased to $5\ \text{ohm}$, the current is $0.25\ \text{amperes}$. The internal resistance of the cell is
 (a) $0.5\ \text{ohm}$ (b) $1.0\ \text{ohm}$
 (c) $1.5\ \text{ohm}$ (d) $2.0\ \text{ohm}$
16. The terminal potential difference of a cell when short-circuited is (E = E.M.F. of the cell)
 (a) E (b) $E/2$
 (c) Zero (d) $E/3$
17. A primary cell has an e.m.f. of $1.5\ \text{volts}$, when short-circuited it gives a current of $3\ \text{amperes}$. The internal resistance of the cell is
 (a) $4.5\ \text{ohm}$ (b) $2\ \text{ohm}$
 (c) $0.5\ \text{ohm}$ (d) $1/4.5\ \text{ohm}$
18. A $50\ V$ battery is connected across a $10\ \text{ohm}$ resistor. The current is $4.5\ \text{amperes}$. The internal resistance of the battery is
 (a) Zero (b) $0.5\ \text{ohm}$
 (c) $1.1\ \text{ohm}$ (d) $5.0\ \text{ohm}$
19. The potential difference in open circuit for a cell is $2.2\ \text{volts}$. When a $4\ \text{ohm}$ resistor is connected between its two electrodes the potential difference becomes $2\ \text{volts}$. The internal resistance of the cell will be
 [MP PMT 1984; SCRA 1994; CBSE PMT 2002]
 (a) $1\ \text{ohm}$ (b) $0.2\ \text{ohm}$
 (c) $2.5\ \text{ohm}$ (d) $0.4\ \text{ohm}$
20. A new flashlight cell of e.m.f. $1.5\ \text{volts}$ gives a current of $15\ \text{amps}$, when connected directly to an ammeter of resistance $0.04\ \Omega$. The internal resistance of cell is [MP PET 1994]
 (a) $0.04\ \Omega$ (b) $0.06\ \Omega$
 (c) $0.10\ \Omega$ (d) $10\ \Omega$
21. A cell whose e.m.f. is $2\ V$ and internal resistance is $0.1\ \Omega$, is connected with a resistance of $3.9\ \Omega$. The voltage across the cell terminal will be
 [CPMT 1990; MP PET 1993; CBSE PMT 1999; AFMC 1999; Pb. PMT 2000; AIIMS 2001]
 (a) $0.50\ V$ (b) $1.90\ V$
 (c) $1.95\ V$ (d) $2.00\ V$
22. The reading of a high resistance voltmeter when a cell is connected across it is $2.2\ V$. When the terminals of the cell are also connected to a resistance of $5\ \Omega$ the voltmeter reading drops to $1.8\ V$. Find the internal resistance of the cell
 (a) $1.2\ \Omega$ (b) $1.3\ \Omega$
 (c) $1.1\ \Omega$ (d) $1.4\ \Omega$
23. When cells are connected in parallel, then

- (a) The current decreases (b) The current increases
 (c) The e.m.f. increases (d) The e.m.f. decreases

24. The internal resistance of a cell depends on
 (a) The distance between the plates
 (b) The area of the plates immersed
 (c) The concentration of the electrolyte
 (d) All the above
25. n identical cells each of e.m.f. E and internal resistance r are connected in series. An external resistance R is connected in series to this combination. The current through R is
 [DPMT 2002]

- (a) $\frac{nE}{R + nr}$ (b) $\frac{nE}{nR + r}$
 [MP PMT 1996]
 (c) $\frac{E}{R + nr}$ (d) $\frac{nE}{R + r}$

26. A cell of internal resistance r is connected to an external resistance R . The current will be maximum in R , if

- (a) $R = r$ (b) $R < r$
 (c) $R > r$ (d) $R = r/2$

27. To get the maximum current from a parallel combination of n identical cells each of internal resistance r in an external resistance R , when [CPMT 1976, 83]
 [DPMT 1999]

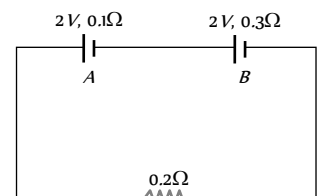
- (a) $R \gg r$ (b) $R \ll r$
 (c) $R = r$ (d) None of these

28. Two identical cells send the same current in $2\ \Omega$ resistance, whether connected in series or in parallel. The internal resistance of the cell should be
 [CPMT 1985; BHU 1997; Pb. PMT 2001]

[NCERT 1982; Kerala PMT 2002]

- (a) $1\ \Omega$ (b) $2\ \Omega$
 (c) $\frac{1}{2}\ \Omega$ (d) $2.5\ \Omega$

29. The internal resistances of two cells shown are $0.1\ \Omega$ and $0.3\ \Omega$. If $R = 0.2\ \Omega$, the potential difference across the cell



- (a) B will be zero
 (b) A will be zero
 (c) A and B will be $2\ V$
 (d) A will be $> 2\ V$ and B will be $< 2\ V$

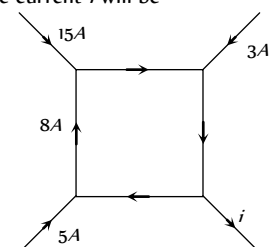
30. A torch battery consisting of two cells of $1.45\ \text{volts}$ and an internal resistance $0.15\ \Omega$, each cell sending currents through the filament of the lamps having resistance $1.5\ \text{ohms}$. The value of current will be [MP PET 1995]
 (a) $16.11\ \text{amp}$ (b) $1.611\ \text{amp}$
 (c) $0.1611\ \text{amp}$ (d) $2.6\ \text{amp}$

31. The electromotive force of a primary cell is $2\ \text{volts}$. When it is short-circuited it gives a current of $4\ \text{amperes}$. Its internal resistance in ohms is
 [MP PET 1995]

- (a) 0.5 (b) 5.0
 (c) 2.0 (d) 8.0

32. [KCET 2003; MP PMT 2003]
 The figure shows a network of currents. The magnitude of currents is shown here. The current i will be

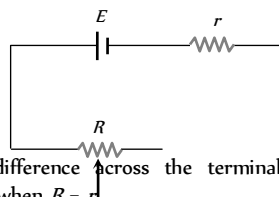
- (a) $3\ A$



[MP PMT 1995]

- (b) 13 A
(c) 23 A
(d) -3 A

33. A battery of e.m.f. E and internal resistance r is connected to a variable resistor R as shown here. Which one of the following is true

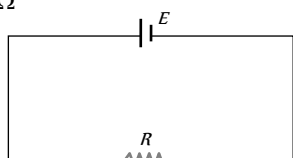


- (a) Potential difference across the terminals of the battery is maximum when $R = r$
(b) Power delivered to the resistor is maximum when $R = r$
(c) Current in the circuit is maximum when $R = r$
(d) Current in the circuit is maximum when $R \gg r$

34. A dry cell has an e.m.f. of 1.5 V and an internal resistance of $0.05\ \Omega$. The maximum current obtainable from this cell for a very short time interval is [Haryana CEE 1996]

- (a) 30 A (b) 300 A
(c) 3 A (d) 0.3 A

35. Consider the circuit given here with the following parameters
E.M.F. of the cell = 12 V . Internal resistance of the cell = $2\ \Omega$.
Resistance $R = 4\ \Omega$



Which one of the following statements is true

- (a) Rate of energy loss in the source is 8 W
(b) Rate of energy conversion in the source is 16 W
(c) Power output in is 8 W
(d) Potential drop across R is 16 V
36. A current of two amperes is flowing through a cell of e.m.f. 5 volts and internal resistance $0.5\ \Omega$ from negative to positive electrode. If the potential of negative electrode is 10 V , the potential of positive electrode will be

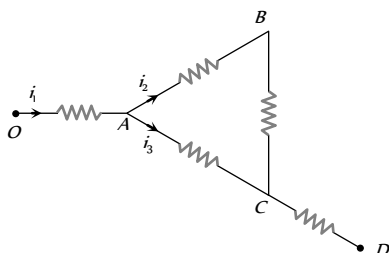
- (a) 5 V (b) 14 V
(c) 15 V (d) 16 V

37. 100 cells each of e.m.f. 5 V and internal resistance $1\ \Omega$ are to be arranged so as to produce maximum current in a $25\ \Omega$ resistance. Each row is to contain equal number of cells. The number of rows should be [MP PMT 1997]

- (a) 2 (b) 4
(c) 5 (d) 10

38. The current in the arm CD of the circuit will be [MP PMT/PET 1998; MP PMT 2000; DPMT 2000]

- (a) $i_1 + i_2$
(b) $i_2 + i_3$
(c) $i_1 + i_3$



- (d) $i_1 - i_2 + i_3$

39. When a resistance of $2\ \Omega$ is connected across the terminals of a cell, the current is 0.5 A . When the resistance is increased to $5\ \Omega$, the current is 0.25 A . The e.m.f. of the cell is

[MP PET 1999, 2000; Pb. PMT 2002; MP PMT 2000]

- (a) 1.0 V (b) 1.5 V
(c) 2.0 V (d) 2.5 V

40. Two non-ideal identical batteries are connected in parallel. Consider the following statements [MP PMT 1999]

- (i) The equivalent e.m.f. is smaller than either of the two e.m.f.s
(ii) The equivalent internal resistance is smaller than either of the two internal resistances
(a) Both (i) and (ii) are correct
(b) (i) is correct but (ii) is wrong
(c) (ii) is correct but (i) is wrong
(d) Both (i) and (ii) are wrong

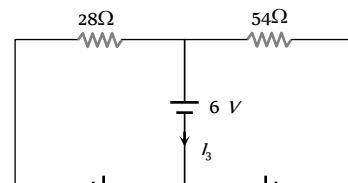
41. If six identical cells each having an e.m.f. of 6 V are connected in parallel, the e.m.f. of the combination is

[EAMCET (Med.) 1995; Pb. PMT 1999; CPMT 2000]

- (a) 1 V (b) 36 V
(c) $\frac{1}{6}\text{ V}$ (d) 6 V

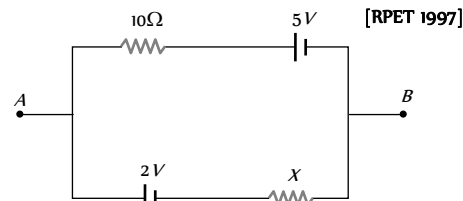
42. Consider the circuit shown in the figure. The current I_3 is equal to

- (a) 5 amp
(b) 3 amp
(c) -3 amp
(d) $-5/6\text{ amp}$



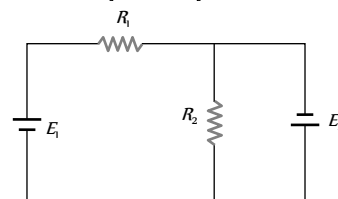
43. If $V_{AB} = 4\text{ V}$ in the given figure, then resistance X will be

- (a) $5\ \Omega$
(b) $10\ \Omega$
(c) $15\ \Omega$
(d) $20\ \Omega$



44. Two resistances R_1 and R_2 are joined as shown in the figure to two batteries of e.m.f. E_1 and E_2 . If E_2 is short-circuited, the current through R_1 is [NDA 1995]

- (a) E_1 / R_1
(b) E_2 / R_1
(c) E_2 / R_2
(d) $E_1 / (R_2 + R_1)$



45. A storage battery has e.m.f. 15 volts and internal resistance $0.05\ \Omega$. Its terminal voltage when it is delivering 10 ampere is

- (a) 30 volts (b) 1.00 volts
(c) 14.5 volts (d) 15.5 volts

46. The number of dry cells, each of e.m.f. 1.5 volt and internal resistance $0.5\ \Omega$ that must be joined in series with a resistance of $20\ \Omega$ so as to send a current of 0.6 ampere through the circuit is

- (a) 2 (b) 8
(c) 10 (d) 12

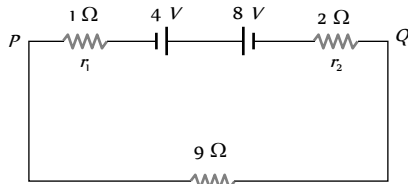
47. Emf is most closely related to [DCE 1999]

- (a) Mechanical force (b) Potential difference

- (c) Electric field (d) Magnetic field
48. For driving a current of 2 A for 6 minutes in a circuit, 1000 J of work is to be done. The e.m.f. of the source in the circuit is
 (a) 1.38 V (b) 1.68 V
 (c) 2.04 V (d) 3.10 V

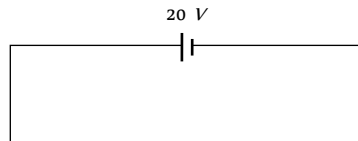
49. Two batteries of e.m.f. 4 V and 8 V with internal resistances $1\ \Omega$ and $2\ \Omega$ are connected in a circuit with a resistance of $9\ \Omega$ as shown in figure. The current and potential difference between the points P and Q are [AFMC 1999]

- (a) $\frac{1}{3}\text{ A}$ and 3 V
 (b) $\frac{1}{6}\text{ A}$ and 4 V
 (c) $\frac{1}{9}\text{ A}$ and 9 V
 (d) $\frac{1}{2}\text{ A}$ and 12 V



50. In the shown circuit, what is the potential difference across A and B

- (a) 50 V
 (b) 45 V
 (c) 30 V
 (d) 20 V



51. Four identical cells each having an electromotive force (e.m.f.) of 12 V , are connected in parallel. The resultant electromotive force (e.m.f.) of the combination is [CPMT 1999]

- (a) 48 V (b) 12 V
 (c) 4 V (d) 3 V

52. Electromotive force is the force, which is able to maintain a constant

- (a) Current (b) Resistance
 (c) Power (d) Potential difference

53. A cell of *emf* 6 V and resistance $0.5\ \Omega$ is short circuited. The current in the cell is [JIPMER 1999]

- (a) 3 amp (b) 12 amp
 (c) 24 amp (d) 6 amp

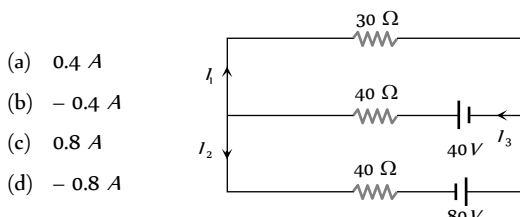
54. A storage cell is charged by 5 amp D.C. for 18 hours . Its strength after charging will be [JIPMER 1999]

- (a) 18 AH (b) 5 AH
 (c) 90 AH (d) 15 AH

55. A battery having e.m.f. 5 V and internal resistance $0.5\ \Omega$ is connected with a resistance of $4.5\ \Omega$ then the voltage at the terminals of battery is [RPMT 2000]

- (a) 4.5 V (b) 4 V
 (c) 0 V (d) 2 V

56. In the given circuit the current I is [DCE 2000]



- (a) 0.4 A
 (b) -0.4 A
 (c) 0.8 A
 (d) -0.8 A

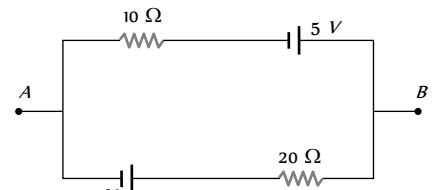
57. The internal resistance of a cell of e.m.f. 12 V is $5 \times 10^{-2}\ \Omega$. It is connected across an unknown resistance. Voltage across the cell, when a current of 60 A is drawn from it, is [CBSE PMT 2000]

- (a) 15 V (b) 12 V
 (c) 9 V (d) 6 V

58. The current in the given circuit is [CPMT 1999]

[AIIMS 2000; MH CET 2003]

- (a) 0.1 A
 (b) 0.2 A
 (c) 0.3 A
 (d) 0.4 A



59. A current of 2.0 ampere passes through a cell of e.m.f. 1.5 volts having internal resistance of $0.15\ \Omega$. The potential difference measured, in *volts*, across both the ends of the cell will be

- (a) 1.35 (b) 1.50
 (c) 1.00 (d) 1.20

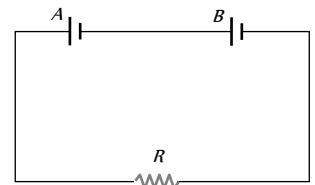
60. A battery has e.m.f. 4 V and internal resistance r . When this battery is connected to an external resistance of $2\ \Omega$, a current of 1 amp. flows in [AIIMS 1999]. How much current will flow if the terminals of the battery are connected directly

[MP PET 2001]

- (a) 1 amp (b) 2 amp
 (c) 4 amp (d) Infinite

61. Two batteries A and B each of e.m.f. 2 V are connected in series to an external resistance $R = 1\ \Omega$. If the internal resistance of battery A is $1.9\ \Omega$ and that of B is $0.9\ \Omega$, what is the potential difference between the terminals of battery A

- (a) 2 V
 (b) 3.8 V [PMT 1999]
 (c) Zero
 (d) None of the above



62. When a resistor of $11\ \Omega$ is connected in series with an electric cell, the current flowing in it is 0.5 A . Instead, when a resistor of $5\ \Omega$ is connected to the same electric cell in series, the current increases by 0.4 A . The internal resistance of the cell is

- (a) $1.5\ \Omega$ (b) $2\ \Omega$
 (c) $2.5\ \Omega$ (d) $3.5\ \Omega$

63. The internal resistance of a cell is the resistance of

[BHU 1999, 2000; AIIMS 2001]

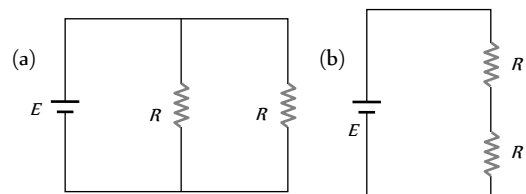
- (a) Electrodes of the cell
 (b) Vessel of the cell
 (c) Electrolyte used in the cell
 (d) Material used in the cell

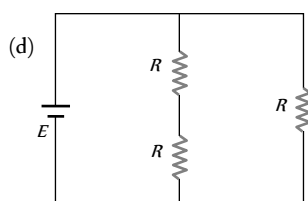
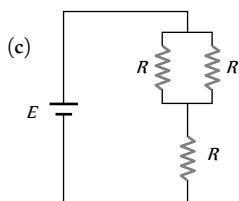
64. How much work is required to carry a $6\ \mu\text{C}$ charge from the negative terminal to the positive terminal of a 9 V battery

[KCET (Med.) 2001]

- (a) $54 \times 10^{-3}\text{ J}$ (b) $54 \times 10^{-6}\text{ J}$
 (c) $54 \times 10^{-9}\text{ J}$ (d) $54 \times 10^{-12}\text{ J}$

65. Consider four circuits shown in the figure below. In which circuit power dissipated is greatest (Neglect the internal resistance of the power supply) [Orissa JEE 2002]





66. The *emf* of a battery is 2 V and its internal resistance is $0.5\ \Omega$. The maximum power which it can deliver to any external circuit will be [AMU (Med.) 2002]

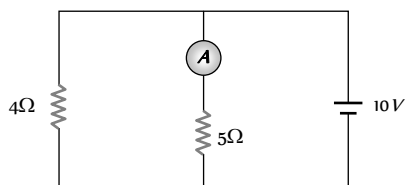
(a) 8 Watt (b) 4 Watt
(c) 2 Watt (d) None of the above

67. Kirchhoff's I law and II law of current, proves the [CBSE PMT 1993; BHU 2002; AFMC 2003]

(a) Conservation of charge and energy
(b) Conservation of current and energy
(c) Conservation of mass and charge
(d) None of these

68. In the circuit, the reading of the ammeter is (assume internal resistance of the battery be zero)

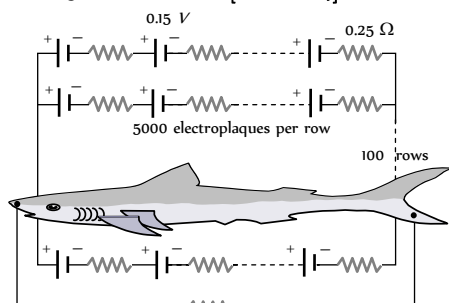
(a) $\frac{40}{29}\text{ A}$
(b) $\frac{10}{9}\text{ A}$
(c) $\frac{5}{3}\text{ A}$
(d) 2 A



69. In the above question, if the internal resistance of the battery is 1 ohm , then what is the reading of ammeter

(a) $5/3\text{ A}$ (b) $40/29\text{ A}$
(c) $10/9\text{ A}$ (d) 1 A

70. Eels are able to generate current with biological cells called electroplaques. The electroplaques in an eel are arranged in 100 rows, each row stretching horizontally along the body of the fish containing 5000 electroplaques. The arrangement is suggestively shown below. Each electroplaques has an *emf* of 0.15 V and internal resistance of $0.25\ \Omega$ [AIIMS 2004]



The water surrounding the eel completes a circuit between the head and its tail. If the water surrounding it has a resistance of $500\ \Omega$, the current an eel can produce in water is about

(a) 1.5 A (b) 3.0 A
(c) 15 A (d) 30 A

71. Current provided by a battery is maximum when [AFMC 2004]

(a) Internal resistance equal to external resistance
(b) Internal resistance is greater than external resistance

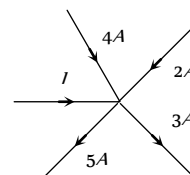
(c) Internal resistance is less than external resistance
(d) None of these

72. A battery is charged at a potential of 15 V for 8 hours when the current flowing is 10 A . The battery on discharge supplies a current of 5 A for 15 hours . The mean terminal voltage during discharge is 14 V . The "Watt-hour" efficiency of the battery is

(a) 82.5% (b) 80%
(c) 90% (d) 87.5%

73. In the given current distribution what is the value of I

[Orissa PMT 2004]



(a) 3 A
(b) 8 A
(c) 2 A
(d) 5 A

74. A capacitor is connected to a cell of *emf* E having some internal resistance r . The potential difference across the

[CPMT 2004; MP PMT 2005]

(a) Cell is $< E$ (b) Cell is E
(c) Capacitor is $> E$ (d) Capacitor is $< E$

75. When the resistance of $9\ \Omega$ is connected at the ends of a battery, its potential difference decreases from 40 volt to 30 volt . The internal resistance of the battery is [DPMT 2003]

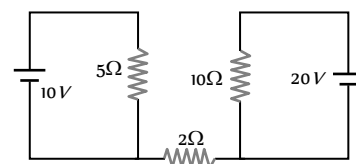
(a) $6\ \Omega$ (b) $3\ \Omega$
(c) $9\ \Omega$ (d) $15\ \Omega$

76. The maximum power drawn out of the cell from a source is given by (where r is internal resistance) [DCE 2002]

(a) $E^2 / 2r$ (b) $E^2 / 4r$
(c) E^2 / r (d) $E^2 / 3r$

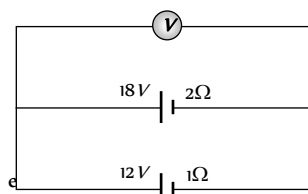
77. Find out the value of current through $2\ \Omega$ resistance for the given circuit [IIT-JEE (Screening) 2005]

(a) 5 A
(b) 2 A
(c) Zero
(d) 4 A



78. Two batteries, one of *emf* 18 volts and internal resistance $2\ \Omega$ and the other of *emf* 12 volt and internal resistance $1\ \Omega$, are connected as shown. The voltmeter V will record a reading of

(a) 15 volt
(b) 30 volt
(c) 14 volt
(d) 18 volt



79. Two sources of equal *emf* E and internal resistance R_1 and R_2 are connected in series with a resistance R . The internal resistances of the two sources are R_1 and

R_2 ($R_2 > R_1$). If the potential difference across the source having internal resistance R_2 is zero, then

[AIEEE 2005]

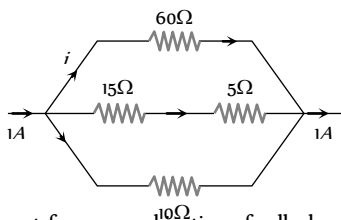
- (a) $R = R_1 R_2 / (R_1 + R_2)$
 (b) $R = R_1 R_2 / (R_2 - R_1)$
 (c) $R = R_2 \times (R_1 + R_2) / (R_2 - R_1)$
 (d) $R = R_2 - R_1$

80. An energy source will supply a constant current into the load if its internal resistance is [AIEEE 2005]

- (a) Zero
 (b) Non-zero but less than the resistance of the load
 (c) Equal to the resistance of the load
 (d) Very large as compared to the load resistance

81. The magnitude of i in ampere unit is [KCET 2005]

- (a) 0.1
 (b) 0.3
 (c) 0.6
 (d) None of these

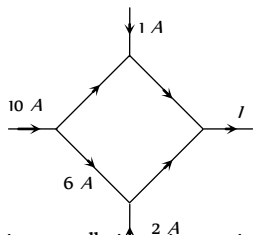


82. To draw maximum current from a combination of cells, how should the cells be grouped [AFMC 2005]

- (a) Series
 (b) Parallel
 (c) Mixed
 (d) Depends upon the relative values of external and internal resistance

83. The figure shows a network of currents. The magnitude of currents is shown here. The current I will be [BCECE 2005]

- (a) 3 A
 (b) 9 A
 (c) 13 A
 (d) 19 A



84. The n rows each containing m cells in series are joined in parallel. Maximum current is taken from this combination across an external resistance of 3Ω resistance. If the total number of cells used are 24 and internal resistance of each cell is 0.5Ω then

- (a) $m = 8, n = 3$ (b) $m = 6, n = 4$
 (c) $m = 12, n = 2$ (d) $m = 2, n = 12$

85. A cell of constant e.m.f. first connected to a resistance R_1 and then connected to a resistance R_2 . If power delivered in both cases is then the internal resistance of the cell is

[Orissa JEE 2005]

- (a) $\sqrt{R_1 R_2}$ (b) $\sqrt{\frac{R_1}{R_2}}$

(c) $\frac{R_1 - R_2}{2}$

(d) $\frac{R_1 + R_2}{2}$

Different Measuring Instruments

1. In meter bridge or Wheatstone bridge for measurement of resistance, the known and the unknown resistances are interchanged. The error so removed is

[MNR 1988; MP PET 1995]

- (a) End correction
 (b) Index error
 (c) Due to temperature effect
 (d) Random error

2. A galvanometer can be converted into an ammeter by connecting

[MP PMT 1987, 93; CPMT 1973, 75, 96, 2000;

MP PET 1994; AFMC 1993, 95; RPET 2000; DCE 2000]

- (a) Low resistance in series
 (b) High resistance in parallel
 (c) Low resistance in parallel
 (d) High resistance in series

3. A cell of internal resistance 1.5Ω and of e.m.f. 1.5 volt balances 500 cm on a potentiometer wire. If a wire of 15Ω is connected between the balance point and the cell, then the balance point will shift [MP PMT 1985]

- (a) To zero (b) By 500 cm
 (c) By 750 cm (d) None of the above

4. 10^{-3} amp is flowing through a resistance of 1000Ω . To measure the correct potential difference, the voltmeter is to be used of which the resistance should be [MP PMT 1985]

- (a) 0Ω (b) 500Ω
 (c) 1000Ω (d) $>> 1000\Omega$

5. A galvanometer of 100Ω resistance gives full scale deflection when 10 mA of current is passed. To convert it into 10 A range ammeter, the resistance of the shunt required will be

- (a) -10Ω (b) 1Ω
 (c) 0.1Ω (d) 0.01Ω

6. 50Ω and 100Ω resistors are connected in series. This connection is connected with a battery of 2.4 volts. When a voltmeter of 100Ω resistance is connected across 100Ω resistor, then the reading of the voltmeter will be

[MP PMT 1985]

- (a) 1.6 V (b) 1.0 V
 (c) 1.2 V (d) 2.0 V

7. A 2 volt battery, a 15Ω resistor and a potentiometer of 100 cm length, all are connected in series. If the resistance of potentiometer wire is 5Ω , then the potential gradient of the potentiometer wire is [AIIMS 1982]

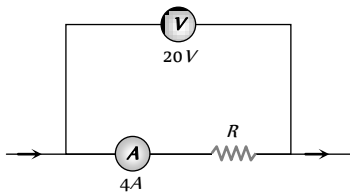
- (a) 0.005 V/cm (b) 0.05 V/cm
 (c) 0.02 V/cm (d) 0.2 V/cm

8. An ammeter gives full scale deflection when current of 1.0 A is passed in it. To convert it into 10 A range ammeter, the ratio of its resistance and the shunt resistance will be

[MP PMT 1985]

- (a) 1 : 9 (b) 1 : 10

- (c) 1 : 11 (d) 9 : 1
9. By ammeter, which of the following can be measured
[MP PET 1981; DPMT 2001]
- (a) Electric potential (b) Potential difference
(c) Current (d) Resistance
10. The resistance of 1 A ammeter is $0.018\ \Omega$. To convert it into 10 A ammeter, the shunt resistance required will be
[MP PET 1982]
- (a) $0.18\ \Omega$ (b) $0.0018\ \Omega$
(c) $0.002\ \Omega$ (d) $0.12\ \Omega$
11. For measurement of potential difference, potentiometer is preferred in comparison to voltmeter because
[MP PET 1983]
- (a) Potentiometer is more sensitive than voltmeter
(b) The resistance of potentiometer is less than voltmeter
(c) Potentiometer is cheaper than voltmeter
(d) Potentiometer does not take current from the circuit
12. In order to pass 10% of main current through a moving coil galvanometer of $99\ \text{ohm}$, the resistance of the required shunt is [MP PET 1990, 99; MP PMT 1994; RPET 2001; KCET 2003, 05]
- (a) $9.9\ \Omega$ (b) $10\ \Omega$
(c) $11\ \Omega$ (d) $9\ \Omega$
13. An ammeter of $5\ \text{ohm}$ resistance can read $5\ \text{mA}$. If it is to be used to read 100 volts, how much resistance is to be connected in series
[MP PET 1991; MP PMT 1996; MP PMT 2000]
- (a) $19.9995\ \Omega$ (b) $199.995\ \Omega$
(c) $1999.95\ \Omega$ (d) $19995\ \Omega$
14. The potential gradient along the length of a uniform wire is $10\ \text{volt/metre}$. B and C are the two points at $30\ \text{cm}$ and $60\ \text{cm}$ point on a meter scale fitted along the wire. The potential difference between B and C will be [CPMT 1986]
- (a) 3 volt (b) 0.4 volt
(c) 7 volt (d) 4 volt
15. $100\ \text{mA}$ current gives a full scale deflection in a galvanometer of $2\ \Omega$ resistance. The resistance connected with the galvanometer to convert it into a voltmeter to measure 5 V is
- (a) $98\ \Omega$ (b) $52\ \Omega$
(c) $50\ \Omega$ (d) $48\ \Omega$
16. When a $12\ \Omega$ resistor is connected with a moving coil galvanometer then its deflection reduces from 50 divisions to 10 divisions. The resistance of the galvanometer is
[CPMT 2002; DPMT 2003]
- (a) $24\ \Omega$ (b) $36\ \Omega$
(c) $48\ \Omega$ (d) $60\ \Omega$
17. A galvanometer can be used as a voltmeter by connecting a
[AFMC 1993; MP PMT 1993, 95; CBSE PMT 2004]
- (a) High resistance in series (b) Low resistance in series
(c) High resistance in parallel (d) Low resistance in parallel
18. The tangent galvanometer, when connected in series with a standard resistance can be used as [MP PET 1994]
- (a) An ammeter
(b) A voltmeter
(c) A wattmeter
(d) Both an ammeter and a voltmeter
19. In Wheatstone's bridge $P = 9\ \text{ohm}$, $Q = 11\ \text{ohm}$, $R = 4\ \text{ohm}$ and $S = 6\ \text{ohm}$. How much resistance must be put in parallel to the resistance S to balance the bridge
[DPMT 1999]
- (a) $24\ \text{ohm}$ (b) $\frac{44}{9}\ \text{ohm}$
(c) $26.4\ \text{ohm}$ (d) $18.7\ \text{ohm}$
20. A Daniel cell is balanced on $125\ \text{cm}$ length of a potentiometer wire. Now the cell is short-circuited by a resistance $2\ \text{ohm}$ and the balance is obtained at $100\ \text{cm}$. The internal resistance of the Daniel cell is [UPSEAT 2002]
- (a) $0.5\ \text{ohm}$ (b) $1.5\ \text{ohm}$
(c) $1.25\ \text{ohm}$ (d) $4/5\ \text{ohm}$
21. Sensitivity of potentiometer can be increased by [MP PET 1994]
- (a) Increasing the e.m.f. of the cell
(b) Increasing the length of the potentiometer wire
(c) Decreasing the length of the potentiometer wire
(d) None of the above
22. A potentiometer is an ideal device of measuring potential difference because
- (a) It uses a sensitive galvanometer
(b) It does not disturb the potential difference it measures
(c) It is an elaborate arrangement
(d) It has a long wire hence heat developed is quickly radiated
23. A battery of 6 volts is connected to the terminals of a three metre long wire of uniform thickness and resistance of the order of $100\ \Omega$. The difference of potential between two points separated by $50\ \text{cm}$ on the wire will be
[CPMT 1984; CBSE PMT 2004]
- (a) 1 V (b) 1.5 V
(c) 2 V (d) 3 V
24. A galvanometer of $10\ \text{ohm}$ resistance gives full scale deflection with 0.01 ampere of current. It is to be converted into an ammeter for measuring 10 ampere current. The value of shunt resistance required will be [MP PET 1984]
- [MNR 1994; UPSEAT 2000]
- (a) $\frac{10}{999}\ \text{ohm}$ (b) $0.1\ \text{ohm}$
(c) $0.5\ \text{ohm}$ (d) $1.0\ \text{ohm}$
25. A potentiometer is used for the comparison of e.m.f. of two cells E_1 and E_2 . For cell E_1 the no deflection point is obtained at $20\ \text{cm}$ and for E_2 the no deflection point is obtained at $30\ \text{cm}$. The ratio of their e.m.f.'s will be [MP PET 1984]
- (a) $2/3$ (b) $1/2$
(c) 1 (d) 2
26. Potential gradient is defined as [MP PET 1994]
- (a) Fall of potential per unit length of the wire
(b) Fall of potential per unit area of the wire
(c) Fall of potential between two ends of the wire

- (d) Potential at any one end of the wire
27. In an experiment of meter bridge, a null point is obtained at the centre of the bridge wire. When a resistance of $10\ \text{ohm}$ is connected in one gap, the value of resistance in other gap is
(a) $10\ \Omega$ (b) $5\ \Omega$
(c) $\frac{1}{5}\ \Omega$ (d) $500\ \Omega$
28. If the length of potentiometer wire is increased, then the length of the previously obtained balance point will
(a) Increase (b) Decrease
(c) Remain unchanged (d) Become two times
29. In potentiometer a balance point is obtained, when
(a) The e.m.f. of the battery becomes equal to the e.m.f. of the experimental cell
(b) The p.d. of the wire between the +ve end to jockey becomes equal to the e.m.f. of the experimental cell
(c) The p.d. of the wire between +ve point and jockey becomes equal to the e.m.f. of the battery
(d) The p.d. across the potentiometer wire becomes equal to the e.m.f. of the battery
30. In the experiment of potentiometer, at balance, there is no current in the
(a) Main circuit
(b) Galvanometer circuit
(c) Potentiometer circuit
(d) Both main and galvanometer circuits
31. If in the experiment of Wheatstone's bridge, the positions of cells and galvanometer are interchanged, then balance points will
(a) Change
(b) Remain unchanged
(c) Depend on the internal resistance of cell and resistance of galvanometer
(d) None of these
32. The resistance of a galvanometer is $90\ \text{ohms}$. If only 10 percent of the main current may flow through the galvanometer, in which way and of what value, a resistor is to be used
(a) $10\ \text{ohms}$ in series (b) $10\ \text{ohms}$ in parallel
(c) $810\ \text{ohms}$ in series (d) $810\ \text{ohms}$ in parallel
33. Two cells when connected in series are balanced on 8 m on a potentiometer. If the cells are connected with polarities of one of the cell is reversed, they balance on 2 m . The ratio of e.m.f.'s of the two cells is
(a) 3 : 5 (b) 5 : 3
(c) 3 : 4 (d) 4 : 3
34. A voltmeter has a resistance of $G\ \text{ohms}$ and range $V\ \text{volts}$. The value of resistance used in series to convert it into a voltmeter of range $nV\ \text{volts}$ is
[MP PMT 1999; MP PET 2002; DPMT 2004; MH CET 2004]
(a) nG (b) $(n-1)G$
(c) $\frac{G}{n}$ (d) $\frac{G}{(n-1)}$
35. Which of the following statement is wrong [MP PET 1994]
(a) Voltmeter should have high resistance
(b) Ammeter should have low resistance
(c) Ammeter is placed in parallel across the conductor in a circuit
(d) Voltmeter is placed in parallel across the conductor in a circuit
36. In the diagram shown, the reading of voltmeter is $20\ \text{V}$ and that of ammeter is $4\ \text{A}$. The value of R should be (Consider given ammeter and voltmeter are not ideal) [RPMT 1997]
[MP PET 1994]
(a) Equal to $5\ \Omega$
(b) Greater from $5\ \Omega$
(c) Less than $5\ \Omega$
(d) Greater or less than $5\ \Omega$ depends on the material of R
- 
37. A moving coil galvanometer has a resistance of $50\ \Omega$ and gives full scale deflection for $10\ \text{mA}$. How could it be converted into an ammeter with a full scale deflection for 1 A
[MP PMT 1996]
(a) $50/99\ \Omega$ in series (b) $50/99\ \Omega$ in parallel
(c) $0.01\ \Omega$ in series (d) $0.01\ \Omega$ in parallel
38. The current flowing through a coil of resistance $900\ \text{ohms}$ is to be reduced by 90%. What value of shunt should be connected across the coil
[Roorkee 1992]
(a) $90\ \Omega$ (b) $100\ \Omega$
(c) $9\ \Omega$ (d) $10\ \Omega$
39. A galvanometer of resistance $25\ \Omega$ gives full scale deflection for a current of $10\ \text{milliampere}$, is to be changed into a voltmeter of range $100\ \text{V}$ by connecting a resistance of ' R ' in series with galvanometer. The value of resistance R in Ω is
[MP PET 1994]
(a) 10000 (b) 10025
(c) 975 (d) 9975
40. In a potentiometer circuit there is a cell of e.m.f. $2\ \text{volt}$, a resistance of $5\ \text{ohm}$ and a wire of uniform thickness of length $1000\ \text{cm}$ and resistance $15\ \text{ohm}$. The potential gradient in the wire is
[MP PET 1996]
(a) $\frac{1}{500}\ \text{V/cm}$ (b) $\frac{3}{2000}\ \text{V/cm}$
(c) $\frac{3}{5000}\ \text{V/cm}$ (d) $\frac{1}{1000}\ \text{V/cm}$
41. The resistance of a galvanometer is $25\ \text{ohm}$ and it requires $50\ \mu\text{A}$ for full deflection. The value of the shunt resistance required to convert it into an ammeter of $5\ \text{amp}$ is
[MP PMT 1994; BHU 1997]
(a) $2.5 \times 10^{-4}\ \text{ohm}$ (b) $1.25 \times 10^{-3}\ \text{ohm}$
(c) $0.05\ \text{ohm}$ (d) $2.5\ \text{ohm}$
42. Which is a *wrong* statement [MP PMT 1994]
(a) The Wheatstone bridge is most sensitive when all the four resistances are of the same order
(b) In a balanced Wheatstone bridge, interchanging the positions of galvanometer and cell affects the balance of the bridge
(c) Kirchhoff's first law (for currents meeting at a junction in an electric circuit) expresses the conservation of charge
(d) The rheostat can be used as a potential divider
43. A voltmeter having a resistance of $998\ \text{ohms}$ is connected to a cell of e.m.f. $2\ \text{volt}$ and internal resistance $2\ \text{ohm}$. The error in the measurement of e.m.f. will be [MP PMT 1994]
(a) $4 \times 10^{-1}\ \text{volt}$ (b) $2 \times 10^{-3}\ \text{volt}$
(c) $4 \times 10^{-3}\ \text{volt}$ (d) $2 \times 10^{-1}\ \text{volt}$

44. For comparing the e.m.f.'s of two cells with a potentiometer, a standard cell is used to develop a potential gradient along the wires. Which of the following possibilities would make the experiment unsuccessful [MP PMT 1994]

(a) The e.m.f. of the standard cell is larger than the E e.m.f.'s of the two cells
(b) The diameter of the wires is the same and uniform throughout
(c) The number of wires is ten
(d) The e.m.f. of the standard cell is smaller than the e.m.f.'s of the two cells

45. Which of the following is correct [BHU 1995]

(a) Ammeter has low resistance and is connected in series
(b) Ammeter has low resistance and is connected in parallel
(c) Voltmeter has low resistance and is connected in parallel
(d) None of the above

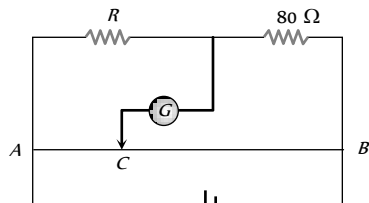
46. An ammeter with internal resistance $90\ \Omega$ reads $1.85\ A$ when connected in a circuit containing a battery and two resistors $700\ \Omega$ and $410\ \Omega$ in series. Actual current will be

[Roorkee 1995]

(a) $1.85\ A$ (b) Greater than $1.85\ A$
(c) Less than $1.85\ A$ (d) None of these

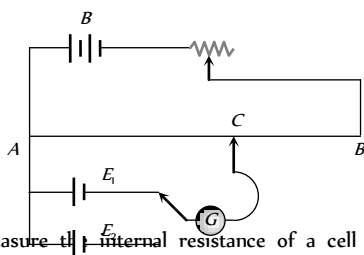
47. AB is a wire of uniform resistance. The galvanometer G shows no current when the length $AC = 20\ cm$ and $CB = 80\ cm$. The resistance R is equal to [MP PMT 1995; RPET 2001]

(a) $2\ \Omega$
(b) $8\ \Omega$
(c) $20\ \Omega$
(d) $40\ \Omega$



48. The circuit shown here is used to compare the e.m.f. of two cells E_1 and E_2 ($E_1 > E_2$). The null point is at C when the galvanometer is connected to E_1 . When the galvanometer is connected to E_2 , the null point will be [MP PMT 1995]

(a) To the left of C
(b) To the right of C
(c) At C itself
(d) Nowhere on AB

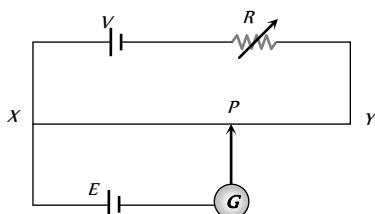


49. In an experiment to measure the internal resistance of a cell by potentiometer, it is found that the balance point is at a length of $2\ m$ when the cell is shunted by a $5\ \Omega$ resistance; and is at a length of $3\ m$ when the cell is shunted by a $10\ \Omega$ resistance. The internal resistance of the cell is, then

[Haryana CEE 1996]

(a) $1.5\ \Omega$ (b) $10\ \Omega$
(c) $15\ \Omega$ (d) $1\ \Omega$

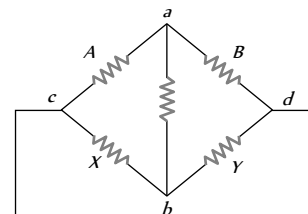
50. A potentiometer circuit shown in the figure is set up to measure e.m.f. of a cell E . As the point P moves from X to Y the galvanometer G shows deflection always in one direction, but the deflection decreases continuously until Y is reached. In order to obtain balance point between X and Y it is necessary to



(a) Decreases the resistance R
(b) Increase the resistance R
(c) Reverse the terminals of battery V
(d) Reverse the terminals of cell E

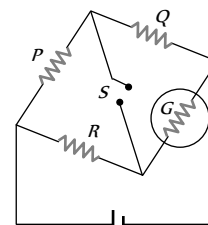
51. In the Wheatstone's bridge (shown in figure) $X = Y$ and $A > B$. The direction of the current between ab will be

(a) From a to b
(b) From b to a
(c) From b to a through c
(d) From a to b through c



52. The figure shows a circuit diagram of a Wheatstone bridge to measure the resistance G of the galvanometer. The relation

$\frac{P}{Q} = \frac{R}{G}$ will be satisfied only when



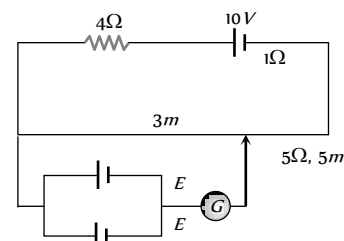
(a) The galvanometer shows a deflection when switch S is closed
(b) The galvanometer shows a deflection when switch S is open
(c) The galvanometer shows no change in deflection whether S is open or closed
(d) The galvanometer shows no deflection

53. The resistance of a galvanometer is $50\ ohms$ and the current required to give full scale deflection is $100\ \mu A$. In order to convert it into an ammeter, reading upto $10\ A$, it is necessary to put a resistance of [MP PMT 1997; AIIMS 1999]

(a) $5 \times 10^{-3}\ \Omega$ in parallel (b) $5 \times 10^{-4}\ \Omega$ in parallel
(c) $10^5\ \Omega$ in series (d) $99,950\ \Omega$ in series

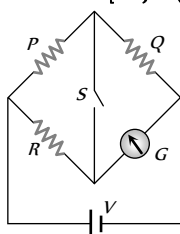
54. A resistance of $4\ \Omega$ and a wire of length $5\ metres$ and resistance $5\ \Omega$ are joined in series and connected to a cell of e.m.f. $10\ V$ and internal resistance $1\ \Omega$. A parallel combination of two identical cells is balanced across $300\ cm$ of the wire. The e.m.f. E of each cell is [MP PMT 1999]

(a) $1.5\ V$
(b) $3.0\ V$
(c) $0.67\ V$
(d) $1.33\ V$



55. The resistivity of a potentiometer wire is $40 \times 10^{-8}\ ohm - m$ and its area of cross-section is $8 \times 10^{-6}\ m^2$. If $0.2\ amp$ current is flowing through the wire, the potential gradient will be

- (a) 10^{-2} volt/m (b) 10^{-1} volt/m
 (c) $3.2 \times 10^{-2} \text{ volt/m}$ (d) 1 volt/m
56. If only 2% of the main current is to be passed through a galvanometer of resistance G , then the resistance of shunt will be
 (a) $\frac{G}{50}$ (b) $\frac{G}{49}$
 (c) $50 G$ (d) $49 G$
57. The resistance of an ideal voltmeter is
 [EAMCET (Med.) 1995; MP PMT/PET 1998; Pb. PMT 1999; CPMT 2000]
 (a) Zero (b) Very low
 (c) Very large (d) Infinite
58. A 100 V voltmeter of internal resistance $20 \text{ k}\Omega$ in series with a high resistance R is connected to a 110 V line. The voltmeter reads 5 V , the value of R is [MP PET 1999]
 (a) $210 \text{ k}\Omega$ (b) $315 \text{ k}\Omega$
 (c) $420 \text{ k}\Omega$ (d) $440 \text{ k}\Omega$
59. Constantan wire is used in making standard resistances because its
 (a) Specific resistance is low
 (b) Density is high
 (c) Temperature coefficient of resistance is negligible
 (d) Melting point is high
60. The net resistance of a voltmeter should be large to ensure that
 (a) It does not get overheated
 (b) It does not draw excessive current
 (c) It can measure large potential difference
 (d) It does not appreciably change the potential difference to be measured
61. A galvanometer has resistance of 7Ω and gives a full scale deflection for a current of 1.0 A . How will you convert it into a voltmeter of range 10 V [MP PMT 1999]
 (a) 3Ω in series (b) 3Ω in parallel
 (c) 17Ω in series (d) 30Ω in series
62. A potentiometer consists of a wire of length 4 m and resistance 10Ω . It is connected to a cell of e.m.f. 2 V . The potential difference per unit length of the wire will be [CBSE PMT 1999; AFMC 2001]
 (a) 0.5 V/m (b) 2 V/m
 (c) 5 V/m (d) 10 V/m
63. In a meter bridge, the balancing length from the left end (standard resistance of one ohm is in the right gap) is found to be 20 cm . The value of the unknown resistance is [CBSE PMT 1999; Pb PMT 2004]
 (a) 0.8Ω (b) 0.5Ω
 (c) 0.4Ω (d) 0.25Ω
64. In the circuit shown $P \neq R$, the reading of the galvanometer is same with switch S open or closed. Then

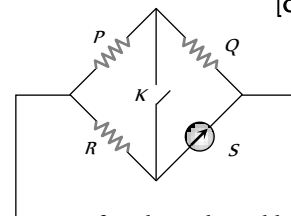


[IIT-JEE (Screening) 1999]

- (a) $I_R = I_G$
 (b) $I_P = I_G$
 (c) $I_Q = I_G$ [MP PMT/PET 1998]
 (d) $I_Q = I_R$

65. In the following Wheatstone bridge $P/Q = R/S$. If key K is closed, then the galvanometer will show deflection [CPMT 1999]

- (a) In left side
 (b) In right side
 (c) No deflection
 (d) In either side



66. A galvanometer having a resistance of 8 ohm is shunted by a wire of resistance 2 ohm . If the total current is 1 amp , the part of it passing through the shunt will be [CBSE PMT 1998]

- (a) 0.25 amp [MP PET 1999] (b) 0.8 amp
 (c) 0.2 amp (d) 0.5 amp

67. A potentiometer wire has length 10 m and resistance 20Ω . A 2.5 V battery of negligible internal resistance is connected across the wire with an 80Ω series resistance. The potential gradient on the wire will be [MP PMT 1999] [KCET 1994]

- (a) $5 \times 10^{-5} \text{ V/mm}$ (b) $2.5 \times 10^{-4} \text{ V/cm}$
 (c) $0.62 \times 10^{-4} \text{ V/mm}$ (d) $1 \times 10^{-5} \text{ V/mm}$

68. An ammeter whose resistance is 180Ω gives full scale deflection when current is 2 mA . The shunt required to convert it into an ammeter reading 20 mA (in ohms) is [EAMCET (Engg.) 1995]

- (a) 18 (b) 20
 (c) 0.1 (d) 10

69. A galvanometer whose resistance is 120Ω gives full scale deflection with a current of 0.05 A so that it can read a maximum current of 10 A . A shunt resistance is added in parallel with it. The resistance of the ammeter so formed is [Bihar MEE 1995]

- (a) 0.06Ω (b) 0.006Ω
 (c) 0.6Ω (d) 6Ω

70. In a potentiometer experiment, the galvanometer shows no deflection when a cell is connected across 60 cm of the potentiometer wire. If the cell is shunted by a resistance of 6Ω , the balance is obtained across 50 cm of the wire. The internal resistance of the cell is [SCRA 1994]

- (a) 0.5Ω (b) 0.6Ω
 (c) 1.2Ω (d) 1.5Ω

71. A voltmeter of resistance 1000Ω gives full scale deflection when a current of 100 mA flow through it. The shunt resistance required across it to enable it to be used as an ammeter reading 1 A at full scale deflection is [SCRA 1994]

- (a) 10000Ω (b) 9000Ω
 (c) 222Ω (d) 111Ω

72. The resistance of 10 metre long potentiometer wire is 10 ohm/metre . A cell of e.m.f. 2.2 volts and a high resistance box are connected in series to this wire. The value of resistance taken from resistance box for getting potential gradient of 2.2 millivolt/metre will be [RPET 1997]

(a) 790 Ω (b) 810 Ω
(c) 990 Ω (d) 1000 Ω

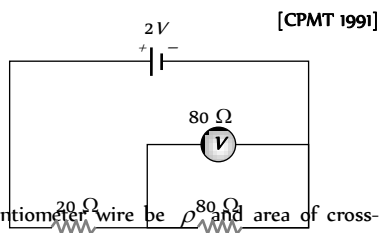
73. We have a galvanometer of resistance $25\text{ }\Omega$. It is shunted by a $2.5\text{ }\Omega$ wire. The part of total current that flows through the galvanometer is given as

[AFMC 1998; MH CET 1999; Pb. PMT 2002]

(a) $\frac{I}{I_0} = \frac{1}{11}$ (b) $\frac{I}{I_0} = \frac{1}{10}$
(c) $\frac{I}{I_0} = \frac{3}{11}$ (d) $\frac{I}{I_0} = \frac{4}{11}$

74. In the adjoining circuit, the e.m.f. of the cell is 2 volt and the internal resistance is negligible. The resistance of the voltmeter is 80 ohm. The reading of the voltmeter will be

(a) 0.80 volt
(b) 1.60 volt
(c) 1.33 volt
(d) 2.00 volt



[CPMT 1991]

75. If the resistivity of a potentiometer wire be ρ and area of cross-section be A , then what will be potential gradient along the wire

(a) $\frac{I\rho}{A}$ (b) $\frac{I}{A\rho}$
(c) $\frac{IA}{\rho}$ (d) $IA\rho$

76. A voltmeter has resistance of 2000 ohms and it can measure upto 2V. If we want to increase its range to 10 V, then the required resistance in series will be

[CPMT 1997, SCRA 1994]

(a) 2000 Ω (b) 4000 Ω
(c) 6000 Ω (d) 8000 Ω

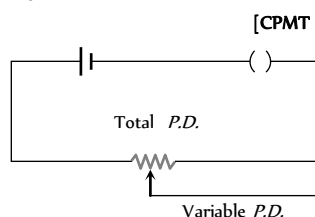
77. For a cell of e.m.f. 2V, a balance is obtained for 50 cm of the potentiometer wire. If the cell is shunted by a $2\text{ }\Omega$ resistor and the balance is obtained across 40 cm of the wire, then the internal resistance of the cell is

[SCRA 1998]

(a) 0.25 Ω (b) 0.50 Ω
(c) 0.80 Ω (d) 1.00 Ω

78. The arrangement as shown in figure is called as

(a) Potential divider
(b) Potential adder
(c) Potential subtractor
(d) Potential multiplier



[CPMT 1999]

79. A potentiometer wire of length 1m and resistance 10 Ω is connected in series with a cell of e.m.f. 2V with internal resistance 1 Ω and a resistance box including a resistance R . If potential difference between the ends of the wire is 1 mV, the value of R is

[KCET 1999]

(a) 20000 Ω (b) 19989 Ω
(c) 10000 Ω (d) 9989 Ω

80. In a balanced Wheatstone's network, the resistances in the arms Q and S are interchanged. As a result of this

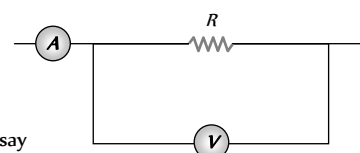
[KCET 1999]

(a) Network is not balanced
(b) Network is still balanced
(c) Galvanometer shows zero deflection
(d) Galvanometer and the cell must be interchanged to balance

81. The ammeter A reads 2 A and the voltmeter V reads 20 V. the value of resistance R is (Assuming finite resistance's of ammeter and voltmeter)

[JIPMER 1999; MP PMT 2004]

(a) Exactly 10 ohm
(b) Less than 10 ohm
(c) More than 10 ohm
(d) We cannot definitely say



82. The resistance of a galvanometer coil is R . What is the shunt resistance required to convert it into an ammeter of range 4 times

(a) $\frac{R}{5}$ (b) $\frac{R}{4}$
(c) $\frac{R}{3}$ (d) 4 R

[RPET 1996]

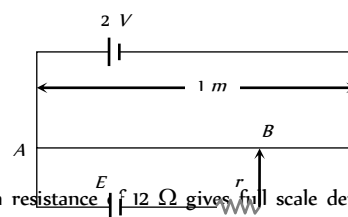
83. If an ammeter is connected in parallel to a circuit, it is likely to be damaged due to excess

[BHU 2000; BCECE 2004]

(a) Current (b) Voltage
(c) Resistance (d) All of these

84. In the given figure, battery E is balanced on 55 cm length of potentiometer wire but when a resistance of 10 Ω is connected in parallel with the battery then it balances on 50 cm length of the potentiometer wire then internal resistance r of the battery is

(a) 1 Ω
(b) 3 Ω
(c) 10 Ω
(d) 5 Ω



85. A galvanometer with a resistance of 12 Ω gives full scale deflection when a current of 3 mA is passed. It is required to convert it into a voltmeter which can read up to 18 V. the resistance to be connected is

[Pb. PMT 2000]

(a) 6000 Ω (b) 5988 Ω
(c) 5000 Ω (d) 4988 Ω

86. The resistance of an ideal ammeter is

[KCET 2000]

(a) Infinite (b) Very high
(c) Small (d) Zero

87. A galvanometer of 25 Ω resistance can read a maximum current of 6mA. It can be used as a voltmeter to measure a maximum of 6 V by connecting a resistance to the galvanometer. Identify the correct choice in the given answers

[EAMCET (Med.) 2000]

(a) 1025 Ω in series (b) 1025 Ω in parallel
(c) 975 Ω in series (d) 975 Ω in parallel

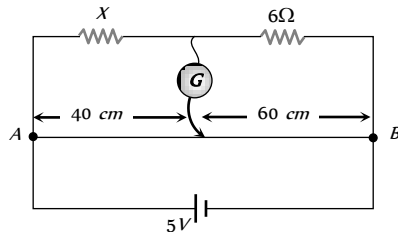
88. A galvanometer has a resistance of $25\ \Omega$ and a maximum of $0.01\ A$ current can be passed through it. In order to change it into an ammeter of range $10\ A$, the shunt resistance required is

(a) $5/999\ \Omega$ (b) $10/999\ \Omega$
(c) $20/999\ \Omega$ (d) $25/999\ \Omega$

89. In the circuit shown, a meter bridge is in its balanced state. The meter bridge wire has a resistance $0.1\ \Omega/cm$. The value of unknown resistance X and the current drawn from the battery of negligible resistance is

[AMU (Engg.) 2000]

(a) $6\ \Omega, 5\ \text{amp}$
(b) $10\ \Omega, 0.1\ \text{amp}$
(c) $4\ \Omega, 1.0\ \text{amp}$
(d) $12\ \Omega, 0.5\ \text{amp}$



90. A galvanometer has 30 divisions and a sensitivity $16\ \mu A/\text{div}$. It can be converted into a voltmeter to read $3\ V$ by connecting

(a) Resistance nearly $6\ k\Omega$ in series
(b) $6\ k\Omega$ in parallel
(c) $500\ \Omega$ in series
(d) It cannot be converted

91. Voltmeters V and V_1 are connected in series across a D.C. line. V reads $80\ \text{volts}$ and has a per volt resistance of $200\ \text{ohms}$. V_1 has a total resistance of $32\ \text{kilo ohms}$. The line voltage is

(a) $120\ \text{volts}$ (b) $160\ \text{volts}$
(c) $220\ \text{volts}$ (d) $240\ \text{volts}$

92. A potentiometer having the potential gradient of $2\ \text{mV/cm}$ is used to measure the difference of potential across a resistance of $10\ \Omega$. If a length of $50\ \text{cm}$ of the potentiometer wire is required to get the null point, the current passing through the $10\ \Omega$ resistor is (in mA)

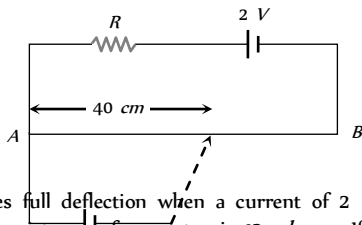
[AMU (Med.) 2000]

(a) 1 (b) 2
(c) 5 (d) 10

93. AB is a potentiometer wire of length $100\ \text{cm}$ and its resistance is $10\ \text{ohms}$. It is connected in series with a resistance $R = 40\ \text{ohms}$ and a battery of e.m.f. $2\ V$ and negligible internal resistance. If a source of unknown e.m.f. E is balanced by $40\ \text{cm}$ length of the potentiometer wire, the value of E is

[MP PET 2001]

(a) $0.8\ V$
(b) $1.6\ V$
(c) $0.08\ V$
(d) $0.16\ V$



94. An ammeter gives full deflection when a current of $2\ \text{amp}$. flows through it. The resistance of ammeter is $12\ \text{ohms}$. If the same ammeter is to be used for measuring a maximum current of $5\ \text{amp}$, then the ammeter must be connected with a resistance of

(a) $8\ \text{ohms}$ in series (b) $18\ \text{ohms}$ in series
(c) $8\ \text{ohms}$ in parallel (d) $18\ \text{ohms}$ in parallel

95. In a circuit 5 percent of total current passes through a galvanometer. If resistance of the galvanometer is G then value of the shunt is

[MP PET 2001]

(a) $19\ G$ (b) $20\ G$

(c) $\frac{G}{20}$ (d) $\frac{G}{19}$

[MP PET 2000]

96. A voltmeter having resistance of $50 \times 10^3\ \Omega$ is used to measure the voltage in a circuit. To increase the range of measurement 3 times the additional series resistance required is

(a) $10\ \text{ohm}$ (b) $150\ \text{kohm}$
(c) $900\ \text{kohm}$ (d) $9 \times 10^4\ \text{ohm}$

97. In a potentiometer experiment two cells of e.m.f. E and E_2 are used in series and in conjunction and the balancing length is found to be $58\ \text{cm}$ of the wire. If the polarity of E_2 is reversed, then the

balancing length becomes $29\ \text{cm}$. The ratio $\frac{E_1}{E_2}$ of the e.m.f. of the

two cells is

[Kerala (Engg.) 2001]

(a) 1 : 1 (b) 2 : 1
(c) 3 : 1 (d) 4 : 1

98. A milliammeter of range $10\ \text{mA}$ has a coil of resistance $1\ \Omega$. To use it as voltmeter of range $10\ \text{volt}$, the resistance that must be connected in series with it, will be

[Kerala PMT 2005]

(a) $999\ \Omega$ (b) $99\ \Omega$
(c) $1000\ \Omega$ (d) None of these

99. A voltmeter has a range $0-V$ with a series resistance R . With a series resistance $2R$, the range is $0-V'$. The correct relation between V and V' is

[CPMT 2001]

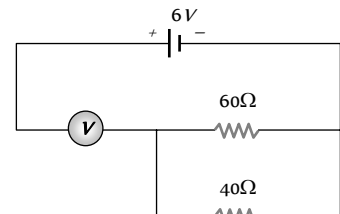
(a) $V' = 2V$ (b) $V' > 2V$
(c) $V' \gg 2V$ (d) $V' < 2V$

[UPSEAT 2000]

100. The measurement of voltmeter in the following circuit is

[AFMC 2001]

(a) $2.4\ V$
(b) $3.4\ V$
(c) $4.0\ V$
(d) $6.0\ V$



101. A $36\ \Omega$ galvanometer is shunted by resistance of $4\ \Omega$. The percentage of the total current, which passes through the galvanometer is

[UPSEAT 2002]

(a) 8 % (b) 9 %
(c) 10 % (d) 91 %

102. An ammeter and a voltmeter of resistance R are connected in series to an electric cell of negligible internal resistance. Their readings are A and V respectively. If another resistance R is connected in parallel with the voltmeter

[EAMCET 2000; KCET 2002]

(a) Both A and V will increase
(b) Both A and V will decrease
(c) A will decrease and V will increase
(d) A will increase and V will decrease

[MP PET 2001]

103. A wire of length $100\ \text{cm}$ is connected to a cell of e.m.f. $2\ V$ and negligible internal resistance. The resistance of the wire is $3\ \Omega$. The additional resistance required to produce a potential drop of $1\ \text{milli volt}$ per cm is

[Kerala PET 2002]

(a) $60\ \Omega$ (b) $47\ \Omega$
(c) $57\ \Omega$ (d) $35\ \Omega$

104. A galvanometer of resistance $20\ \Omega$ is to be converted into an ammeter of range $1\ A$. If a current of $1\ mA$ produces full scale deflection, the shunt required for the purpose is

[Kerala PET 2002]

- (a) $0.01\ \Omega$ (b) $0.05\ \Omega$
(c) $0.02\ \Omega$ (d) $0.04\ \Omega$
105. There are three voltmeters of the same range but of resistances $10000\ \Omega$, $8000\ \Omega$ and $4000\ \Omega$ respectively. The best voltmeter among these is the one whose resistance is
- (a) $10000\ \Omega$ (b) $8000\ \Omega$
(c) $4000\ \Omega$ (d) All are equally good
106. If an ammeter is to be used in place of a voltmeter then we must connect with the ammeter a

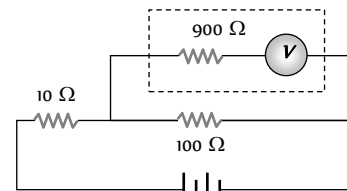
[AIIEE 2002; AFMC 2002]

- (a) Low resistance in parallel
(b) High resistance in parallel
(c) High resistance in series
(d) Low resistance in series
107. A $10\ m$ long wire of $20\ \Omega$ resistance is connected with a battery of $3\ volt$ e.m.f. (negligible internal resistance) and a $10\ \Omega$ resistance is joined to it in series. Potential gradient along wire in volt per meter is
- [MP PMT 2003]
- (a) 0.02 (b) 0.3
(c) 0.2 (d) 1.3
108. A potentiometer has uniform potential gradient across it. Two cells connected in series (i) to support each other and (ii) to oppose each other are balanced over $6\ m$ and $2\ m$ respectively on the potentiometer wire. The e.m.f.'s of the cells are in the ratio of
- (a) $1 : 2$ (b) $1 : 1$
(c) $3 : 1$ (d) $2 : 1$
109. The material of wire of potentiometer is

[MP PMT 2002]

- (a) Copper (b) Steel
(c) Manganin (d) Aluminium
110. To convert a galvanometer into a voltmeter, one should connect a
- (a) High resistance in series with galvanometer
(b) Low resistance in series with galvanometer
(c) High resistance in parallel with galvanometer
(d) Low resistance in parallel with galvanometer
111. To convert a $800\ mV$ range *milli voltmeter* of resistance $40\ \Omega$ into a galvanometer of $100\ mA$ range, the resistance to be connected as shunt is
- [CBSE PMT 2002]
- (a) $10\ \Omega$ (b) $20\ \Omega$
(c) $30\ \Omega$ (d) $40\ \Omega$
112. A $100\ ohm$ galvanometer gives full scale deflection at $10\ mA$. How much shunt is required to read $100\ mA$
- [MP PET 2002]
- (a) $11.11\ ohm$ (b) $9.9\ ohm$
(c) $1.1\ ohm$ (d) $4.4\ ohm$
113. The potential difference across the $100\ \Omega$ resistance in the following circuit is measured by a voltmeter of $900\ \Omega$ resistance. The percentage error made in reading the potential difference is

- (a) $\frac{10}{9}$
(b) 0.1
(c) 1.0
(d) 10.0



114. A cell of internal resistance $3\ ohm$ and *emf* $10\ volt$ is connected to a uniform wire of length $500\ cm$ and resistance $3\ ohm$. The potential gradient in the wire is
- [Kerala PET 2002] [MP PET 2003]
- (a) $30\ mV/cm$ (b) $10\ mV/cm$
(c) $20\ mV/cm$ (d) $4\ mV/cm$

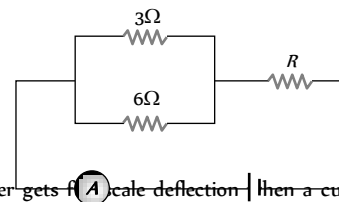
115. An ammeter of $100\ \Omega$ resistance gives full deflection for the current of $10\ amp$. Now the shunt resistance required to convert it into ammeter of $1\ amp$ range, will be

[RPET 2003]

- (a) $10^{-4}\ \Omega$ (b) $10^{-5}\ \Omega$
(c) $10^{-3}\ \Omega$ (d) $10^{-1}\ \Omega$
116. A galvanometer of resistance $36\ \Omega$ is changed into an ammeter by using a shunt of $4\ \Omega$. The fraction f of total current passing through the galvanometer is
- [BCECE 2003]
- (a) $\frac{1}{40}$ (b) $\frac{1}{4}$
(c) $\frac{1}{140}$ (d) $\frac{1}{10}$

117. If the ammeter in the given circuit reads $2\ A$, the resistance R is

- (a) $1\ ohm$ [MP PMT 2002]
(b) $2\ ohm$
(c) $3\ ohm$
(d) $4\ ohm$



118. A $50\ ohm$ galvanometer gets full scale deflection when a current of $0.01\ A$ passes through the coil. When it is converted to a $10\ A$ ammeter, the shunt resistance is

[Orissa JEE 2003]

- [CBSE PMT 2002]
- (a) $0.01\ \Omega$ (b) $0.05\ \Omega$
(c) $2000\ \Omega$ (d) $5000\ \Omega$
119. Resistance in the two gaps of a *meter bridge* are $10\ ohm$ and $30\ ohm$ respectively. If the resistances are interchanged the balance point shifts by
- [Orissa JEE 2003]
- (a) $33.3\ cm$ (b) $66.67\ cm$
(c) $25\ cm$ (d) $50\ cm$

120. A potentiometer has uniform potential gradient. The specific resistance of the material of the potentiometer wire is $10^{-4}\ ohm-meter$ and the current passing through it is $0.1\ ampere$; cross-section of the wire is $10^{-7}\ m$. The potential gradient along the potentiometer wire is
- [KCET 2003]

- (a) $10^{-4}\ V/m$ (b) $10^{-6}\ V/m$
(c) $10^{-2}\ V/m$ (d) $10^{-8}\ V/m$

121. Two resistances of $400\ \Omega$ and $800\ \Omega$ are connected in series with $6\ volt$ battery of negligible internal resistance. A voltmeter of resistance $10,000\ \Omega$ is used to measure the potential difference across $400\ \Omega$. The error in the measurement of potential difference in volts approximately is
- [AMU (Med.) 2002]

- (a) 0.01 (b) 0.02
(c) 0.03 (d) 0.05

122. A galvanometer, having a resistance of $50\ \Omega$ gives a full scale deflection for a current of $0.05\ A$. The length in meter of a resistance wire of area of cross-section $2.97 \times 10^{-7}\ cm$ that can be used to convert the galvanometer into an ammeter which can read a maximum of $5\ A$ current is (Specific resistance of the wire $= 5 \times 10^{-7}\ \Omega m$)

(a) 9 (b) 6
(c) 3 (d) 1.5

123. An ammeter reads upto $1\ ampere$. Its internal resistance is $0.81\ ohm$. To increase the range to $10\ A$ the value of the required shunt is [AIEEE 2003]

(a) $0.09\ \Omega$ (b) $0.03\ \Omega$
(c) $0.3\ \Omega$ (d) $0.9\ \Omega$

124. The length of a wire of a potentiometer is $100\ cm$, and the *emf* of its standard cell is $E\ volt$. It is employed to measure the *e.m.f* of a battery whose internal resistance is $0.5\ \Omega$. If the balance point is obtained at $l = 30\ cm$ from the positive end, the *e.m.f* of the battery is [AIEEE 2003]

(a) $\frac{30E}{100}$
(b) $\frac{30E}{100.5}$
(c) $\frac{30E}{(100 - 0.5)}$
(d) $\frac{30(E - 0.5i)}{100}$, where i is the current in the potentiometer

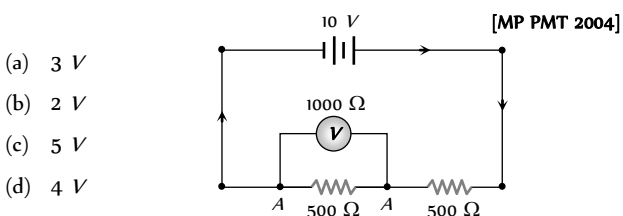
125. Resistance of $100\ cm$ long potentiometer wire is $10\ \Omega$, it is connected to a battery ($2\ volt$) and a resistance R in series. A source of $10\ mV$ gives null point at $40\ cm$ length, then external resistance R is

(a) $490\ \Omega$ (b) $790\ \Omega$
(c) $590\ \Omega$ (d) $990\ \Omega$

126. The *e.m.f* of a standard cell balances across $150\ cm$ length of a wire of potentiometer. When a resistance of $2\ \Omega$ is connected as a shunt with the cell, the balance point is obtained at $100\ cm$. The internal resistance of the cell is [MP PET 1993]

(a) $0.1\ \Omega$ (b) $1\ \Omega$
(c) $2\ \Omega$ (d) $0.5\ \Omega$

127. What is the reading of voltmeter in the following figure



(a) $3\ V$
(b) $2\ V$
(c) $5\ V$
(d) $4\ V$

128. The current flowing in a coil of resistance $90\ \Omega$ is to be reduced by 90% . What value of resistance should be connected in parallel with it [MP PMT 2004]

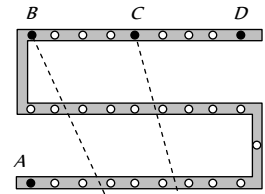
(a) $9\ \Omega$ (b) $90\ \Omega$
(c) $1000\ \Omega$ (d) $10\ \Omega$

129. The maximum current that can be measured by a galvanometer of resistance $40\ \Omega$ is $10\ mA$. It is converted into a voltmeter that can read upto $50\ V$. The resistance to be connected in series with the galvanometer is ... (in *ohm*) [KCET 2004]

(a) 5040 (b) 4960
(c) 2010 (d) 4050

130. For the post office box arrangement to determine the value of unknown resistance the unknown resistance should be connected between [IIT-JEE (Screening) 2004]

(a) B and C
(b) C and D
(c) A and D
(d) B and C



131. A galvanometer of $50\ ohm$ resistance has 25 divisions. A current of $4 \times 10^{-5}\ ampere$ gives a deflection of one division. To convert this galvanometer into a voltmeter having a range of $25\ volts$, it should be connected with a resistance of [CBSE PMT 2004]

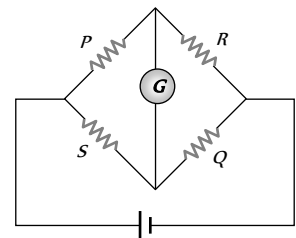
(a) $2500\ \Omega$ as a shunt (b) $2450\ \Omega$ as a shunt
(c) $2550\ \Omega$ in series (d) $2450\ \Omega$ in series

132. In a metre bridge experiment null point is obtained at $20\ cm$ from one end of the wire when resistance X is balanced against another resistance Y . If $X < Y$, then where will be the new position of the null point from the same end, if one decides to balance a resistance of $4X$ against Y [AIEEE 2004]

(a) $50\ cm$ (b) $80\ cm$
(c) $40\ cm$ (d) $70\ cm$

133. In the circuit given, the correct relation to a balanced Wheatstone bridge is [Orissa PMT 2004]

(a) $\frac{P}{Q} = \frac{R}{S}$
(b) $\frac{P}{Q} = \frac{S}{R}$
(c) $\frac{P}{R} = \frac{S}{Q}$



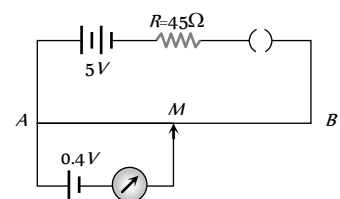
(d) None of these

134. A galvanometer coil of resistance $50\ \Omega$, show full deflection of $100\ \mu A$. The shunt resistance to be added to the galvanometer, to work as an ammeter of range $10\ mA$ is [Pb PET 2000]

(a) $5\ \Omega$ in parallel (b) $0.5\ \Omega$ in series
(c) $5\ \Omega$ in series (d) $0.5\ \Omega$ in parallel

135. In given figure, the potentiometer wire AB has a resistance of $5\ \Omega$ and length $10\ m$. The balancing length AM for the *emf* of $0.4\ V$ is

(a) $0.4\ m$
(b) $4\ m$
(c) $0.8\ m$
(d) $8\ m$



136. A potentiometer consists of a wire of length 4 m and resistance $10\ \Omega$. It is connected to cell of emf 2 V . The potential difference per unit length of the wire will be

[Pb. PET 2002]

- (a) 0.5 V/m (b) 10 V/m
(c) 2 V/m (d) 5 V/m

137. A voltmeter essentially consists of [UPSEAT 2004]

- (a) A high resistance, in series with a galvanometer
(b) A low resistance, in series with a galvanometer
(c) A high resistance in parallel with a galvanometer
(d) A low resistance in parallel with a galvanometer

138. In a potentiometer experiment the balancing with a cell is at length 240 cm . On shunting the cell with a resistance of $2\ \Omega$, the balancing length becomes 120 cm . The internal resistance of the cell is

- (a) $4\ \Omega$ (b) $2\ \Omega$
(c) $1\ \Omega$ (d) $0.5\ \Omega$

139. With a potentiometer null point were obtained at 140 cm and 180 cm with cells of emf 1.1 V and one unknown X volts. Unknown emf is [DCE 2002]

- (a) 1.1 V (b) 1.8 V
(c) 2.4 V (d) 1.41 V

140. A moving coil galvanometer of resistance $100\ \Omega$ is used as an ammeter using a resistance $0.1\ \Omega$. The maximum deflection current in the galvanometer is $100\ \mu\text{A}$. Find the minimum current in the circuit so that the ammeter shows maximum deflection

- (a) 100.1 mA (b) 1000.1 mA
(c) 10.01 mA (d) 1.01 mA

141. Two resistances are connected in two gaps of a metre bridge. The balance point is 20 cm from the zero end. A resistance of 15 ohms is connected in series with the smaller of the two. The null point shifts to 40 cm . The value of the smaller resistance in ohms is

- (a) 3 (b) 6
(c) 9 (d) 12

142. If resistance of voltmeter is $10000\ \Omega$ and resistance of ammeter is $2\ \Omega$ then find R when voltmeter reads 12 V and ammeter reads 0.1 A

- (a) $118\ \Omega$ (b) $120\ \Omega$
(c) $124\ \Omega$ (d) $114\ \Omega$

143. Potentiometer wire of length 1 m is connected in series with $490\ \Omega$ resistance and 2 V battery. If 0.2 mV/cm is the potential gradient, then resistance of the potentiometer wire is

- (a) $4.9\ \Omega$ (b) $7.9\ \Omega$
(c) $5.9\ \Omega$ (d) $6.9\ \Omega$

Critical Thinking

Objective Questions

1. In an electrical cable there is a single wire of radius 9 mm of copper. Its resistance is $5\ \Omega$. The cable is replaced by 6 different insulated copper wires, the radius of each wire is 3 mm . Now the total resistance of the cable will be

[CPMT 1988]

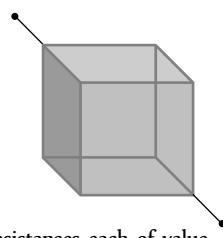
- (a) $7.5\ \Omega$ (b) $45\ \Omega$
(c) $90\ \Omega$ (d) $270\ \Omega$

2. Two uniform wires A and B are of the same metal and have equal masses. The radius of wire A is twice that of wire B . The total resistance of A and B when connected in parallel is

- (a) $4\ \Omega$ when the resistance of wire A is $4.25\ \Omega$
(b) $5\ \Omega$ when the resistance of wire A is $4.25\ \Omega$
(c) $4\ \Omega$ when the resistance of wire B is $4.25\ \Omega$
(d) $4\ \Omega$ when the resistance of wire B is $4.25\ \Omega$

3. Twelve [DCE 2002; AIITB 2005] and same cross-section are connected in the form of a cube. If the resistance of each of the wires is R , then the effective resistance between the two diagonal ends would be [J & K CET 2004]

- (a) $2R$
(b) $12R$
(c) $\frac{5}{6}R$
(d) $8R$

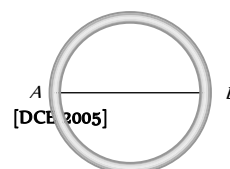


4. You are given several identical resistances each of value $R = 10\ \Omega$ and each capable of carrying maximum current of 1 ampere . It is required to make a suitable combination of these resistances to produce a resistance of $5\ \Omega$ which can carry a current of 4 amperes . The minimum number of resistances of the type R that will be required for this job [IIT JEE (Screening) 2005]

- (a) 4 [KCET 2005] (b) 10
(c) 8 (d) 20

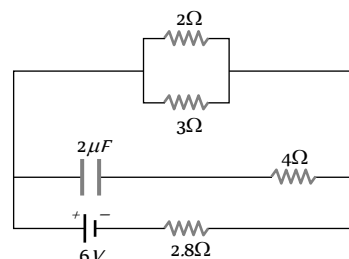
[CBSE PMT 1990]

5. The resistance of a wire is $10^{-6}\ \Omega$ per metre. It is bend in the form of a circle of diameter 2 m . A wire of the same material is connected across its diameter. The total resistance across its diameter AB will be [BCECE 2005]



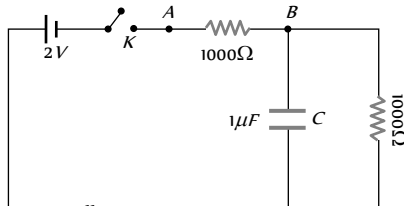
- (a) $\frac{4}{3}\pi \times 10^{-6}\ \Omega$ (b) $\frac{2}{3}\pi \times 10^{-6}\ \Omega$
(c) $0.88 \times 10^{-6}\ \Omega$ (d) $14\pi \times 10^{-6}\ \Omega$

6. In the figure shown, the capacity of the condenser C is $2\ \mu\text{F}$. The current in $2\ \Omega$ resistor is [IIT 1982]



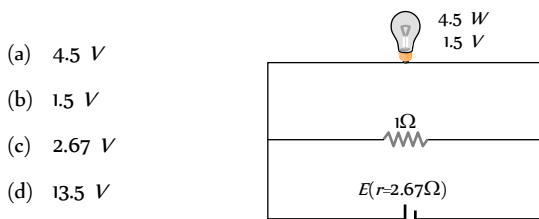
- (a) 9 A (b) 0.9 A
(c) $\frac{1}{9}\text{ A}$ (d) $\frac{1}{0.9}\text{ A}$

7. When the key K is pressed at time $t = 0$, which of the following statements about the current I in the resistor AB of the given circuit is true [CBSE PMT 1995]



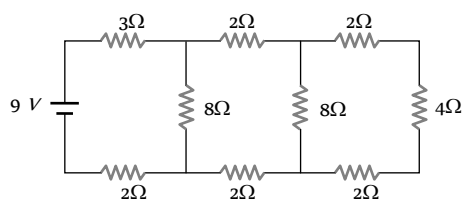
- (a) $I = 2\text{ mA}$ at all t
(b) I oscillates between 1 mA and 2 mA
(c) $I = 1\text{ mA}$ at all t
(d) At $t = 0$, $I = 2\text{ mA}$ and with time it goes to 1 mA

8. A torch bulb rated as 4.5 W , 1.5 V is connected as shown in the figure. The *e.m.f.* of the cell needed to make the bulb glow at full intensity is [MP PMT 1999]



- (a) 4.5 V
(b) 1.5 V
(c) 2.67 V
(d) 13.5 V

9. In the circuit shown in the figure, the current through



[IIT 1998]

- (a) The 3Ω resistor is 0.50 A (b) The 3Ω resistor is 0.25 A
(c) The 4Ω resistor is 0.50 A (d) The 4Ω resistor is 0.25 A

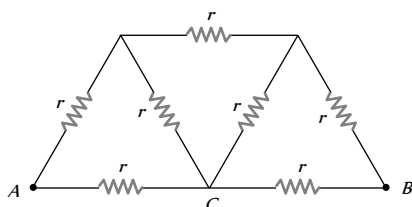
10. There are three resistance coils of equal resistance. The maximum number of resistances you can obtain by connecting them in any manner you choose, being free to use any number of the coils in any way is

[ISM Dhanbad 1994]

- (a) 3 (b) 4
(c) 6 (d) 5

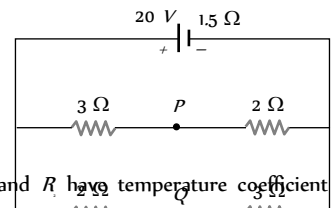
11. In the circuit shown, the value of each resistance is r , then equivalent resistance of circuit between points A and B will be

- (a) $(4/3)r$
(b) $3r/2$
(c) $r/3$
(d) $8r/7$



12. If in the circuit shown below, the internal resistance of the battery is 1.5Ω and V_1 and V_2 are the potentials at P and Q respectively, what is the potential difference between the points P and Q

- (a) Zero
(b) 4 volts ($V_1 > V_2$)
(c) 4 volts ($V_2 > V_1$)
(d) 2.5 volts ($V_2 > V_1$)



13. Two wires of resistance R_1 and R_2 have temperature coefficient of resistance α_1 and α_2 , respectively. These are joined in series. The effective temperature coefficient of resistance is

- (a) $\frac{\alpha_1 + \alpha_2}{2}$ (b) $\sqrt{\alpha_1 \alpha_2}$
(c) $\frac{\alpha_1 R_1 + \alpha_2 R_2}{R_1 + R_2}$ (d) $\frac{\sqrt{R_1 R_2 \alpha_1 \alpha_2}}{\sqrt{R_1^2 + R_2^2}}$

14. Two cells of equal *e.m.f.* and of internal resistances r_1 and r_2 ($r_1 > r_2$) are connected in series. On connecting this combination to an external resistance R , it is observed that the potential difference across the first cell becomes zero. The value of R will be

[MP PET 1985; KCET 2005; Kerala PMT 2005]

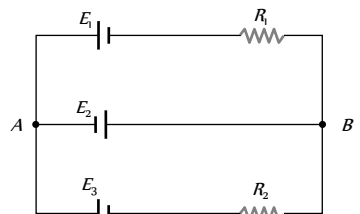
- (a) $r_1 + r_2$ (b) $r_1 - r_2$
(c) $\frac{r_1 + r_2}{2}$ (d) $\frac{r_1 - r_2}{2}$

15. When connected across the terminals of a cell, a voltmeter measures 5 V and a connected ammeter measures 10 A of current. A resistance of 2 ohms is connected across the terminals of the cell. The current flowing through this resistance will be

- (a) 2.5 A (b) 2.0 A
(c) 5.0 A (d) 7.5 A

16. In the circuit shown here, $E_1 = E_2 = E_3 = 2\text{ V}$ and $R_1 = R_2 = 4\text{ ohms}$. The current flowing between points A and B through battery E_1 is

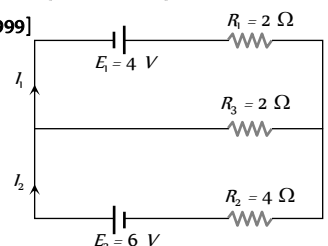
- (a) Zero
(b) 2 amp from A to B
(c) 2 amp from B to A
(d) None of the above



17. In the circuit shown below $E = 4.0\text{ V}$, $R_1 = 2\Omega$, $E_1 = 6.0\text{ V}$, $R_2 = 4\Omega$ and $R_3 = 2\Omega$. The current I is [MP PET 2003]

[Similar to CBSE PMT 1999; RPET 1999]

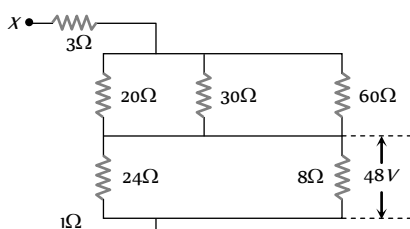
- (a) 1.6 A
(b) 1.8 A
(c) 1.25 A
(d) 1.0 A



18. A microammeter has a resistance of $100\ \Omega$ and full scale range of $50\ \mu\text{A}$. It can be used as a voltmeter or as a higher range ammeter provided a resistance is added to it. Pick the correct range and resistance combination

[SCRA 1996; AMU (Med.) 2001; Roorkee 2000]

- (a) $50\ \text{V}$ range with $10\ \text{k}\Omega$ resistance in series
(b) $10\ \text{V}$ range with $200\ \text{k}\Omega$ resistance in series
(c) $10\ \text{mA}$ range with $1\ \Omega$ resistance in parallel
(d) $10\ \text{mA}$ range with $0.1\ \Omega$ resistance in parallel
19. The potential difference across $8\ \text{ohm}$ resistance is $48\ \text{volt}$ as shown in the figure. The value of potential difference across X and Y points will be [MP PET 1996]



- (a) $160\ \text{volt}$
(b) $128\ \text{volt}$
(c) $80\ \text{volt}$
(d) $62\ \text{volt}$
20. Two resistances R_1 and R_2 are made of different materials. The temperature coefficient of the material of R_1 is α and of the material of R_2 is $-\beta$. The resistance of the series combination of R_1 and R_2 will not change with temperature, if R_1 / R_2 equals

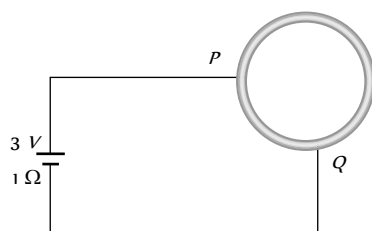
- (a) $\frac{\alpha}{\beta}$
(b) $\frac{\alpha + \beta}{\alpha - \beta}$
(c) $\frac{\alpha^2 + \beta^2}{\alpha\beta}$
(d) $\frac{\beta}{\alpha}$

21. An ionization chamber with parallel conducting plates as anode and cathode has 5×10^7 electrons and the same number of singly-charged positive ions per cm^3 . The electrons are moving at $0.4\ \text{m/s}$. The current density from anode to cathode is $4\ \mu\text{A}/\text{m}^2$. The velocity of positive ions moving towards cathode is

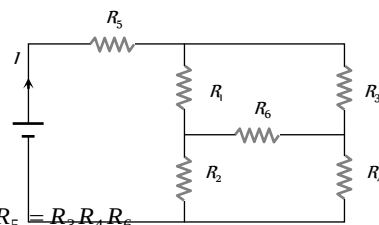
- (a) $0.4\ \text{m/s}$
(b) $16\ \text{m/s}$
(c) Zero
(d) $0.1\ \text{m/s}$

22. A wire of resistance $10\ \Omega$ is bent to form a circle. P and Q are points on the circumference of the circle dividing it into a quadrant and are connected to a Battery of $3\ \text{V}$ and internal resistance $1\ \Omega$ as shown in the figure. The currents in the two parts of the circle are

- (a) $\frac{6}{23}\ \text{A}$ and $\frac{18}{23}\ \text{A}$
(b) $\frac{5}{26}\ \text{A}$ and $\frac{15}{26}\ \text{A}$
(c) $\frac{4}{25}\ \text{A}$ and $\frac{12}{25}\ \text{A}$
(d) $\frac{3}{25}\ \text{A}$ and $\frac{9}{25}\ \text{A}$



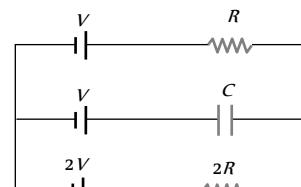
23. In the given circuit, it is observed that the current I is independent of the value of the resistance R . Then the resistance values must satisfy [IIT-JEE (Screening) 2001]



- (a) $R_1 R_2 R_5 = R_3 R_4 R_6$
(b) $\frac{1}{R_5} + \frac{1}{R_6} = \frac{1}{R_1 + R_2} + \frac{1}{R_3 + R_4}$
(c) $R_1 R_4 = R_2 R_3$
(d) $R_1 R_3 = R_2 R_4 = R_5 R_6$

24. In the given circuit, with steady current, the potential drop across the capacitor must be [IIT-JEE (Screening) 2001]

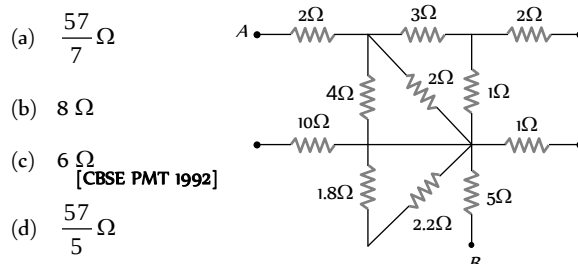
- (a) V
(b) $V/2$
(c) $V/3$
(d) $2V/3$



25. A wire of length L and 3 identical cells of negligible internal resistances are connected in series. Due to current, the temperature of the wire is raised by ΔT in a time t . A number N of similar cells is now connected in series with a wire of the same material and cross-section but of length $2L$. The temperature of the wire is raised by the same amount ΔT in the same time t . the value of N is [MP PMT 1997]

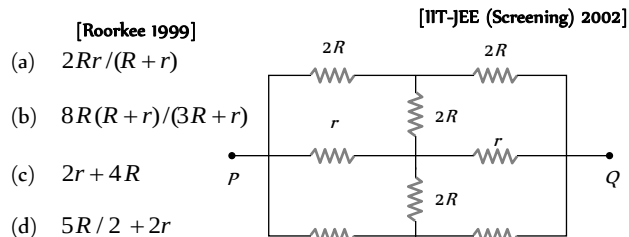
- (a) 4
(b) 6
(c) 8
(d) 9

26. What is the equivalent resistance between the points A and B of the network [AMU (Engg.) 2001]



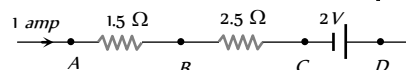
- (a) $\frac{57}{7}\ \Omega$
(b) $8\ \Omega$
(c) $6\ \Omega$
(d) $\frac{57}{5}\ \Omega$

27. The effective resistance between points P and Q of the electrical circuit shown in the figure is [Roorkee 1999]



- (a) $2Rr/(R+r)$
(b) $8R(R+r)/(3R+r)$
(c) $2r+4R$
(d) $5R/2+2r$

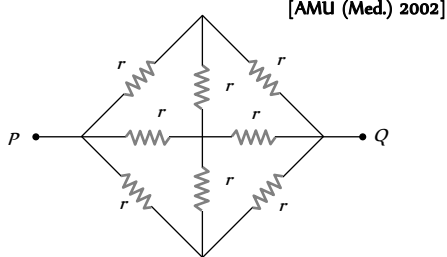
28. In the circuit element given here, if the potential at point B , $V_B = 0$, then the potentials of A and D are given as [AMU (Med.) 2002]



- (a) $V_A = -1.5 \text{ V}, V_D = +2 \text{ V}$ (b) $V_A = +1.5 \text{ V}, V_D = +2 \text{ V}$
 (c) $V_A = +1.5 \text{ V}, V_D = +0.5 \text{ V}$ (d) $V_A = +1.5 \text{ V}, V_D = -0.5 \text{ V}$

29. The equivalent resistance between the points P and Q in the network given here is equal to (given $r = \frac{3}{2} \Omega$)

- (a) $\frac{1}{2} \Omega$
 (b) 1Ω
 (c) $\frac{3}{2} \Omega$
 (d) 2Ω



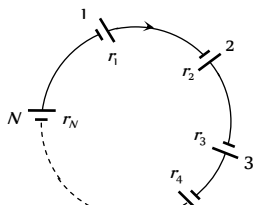
[AMU (Med.) 2002]

30. The current in a conductor varies with time t as $I = 2t + 3t^2$ where I is in ampere and t in seconds. Electric charge flowing through a section of the conductor during $t = 2 \text{ sec}$ to $t = 3 \text{ sec}$ is

- (a) 10 C (b) 24 C
 (c) 33 C (d) 44 C

31. A group of N cells whose emf varies directly with the internal resistance as per the equation $E_i = 1.5 r_i$ are connected as shown in the figure below. The current I in the circuit is

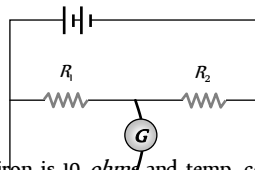
- (a) 0.51 amp
 (b) 5.1 amp
 (c) 0.15 amp
 (d) 1.5 amp



[KCET 2003]

32. In the shown arrangement of the experiment of the meter bridge if AC corresponding to null deflection of galvanometer is x , what would be its value if the radius of the wire AB is doubled

- (a) x
 (b) $x/4$
 (c) $4x$
 (d) $2x$



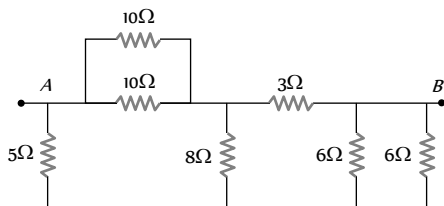
33. The resistance of a wire of iron is 10 ohms and temp. coefficient of resistivity is $5 \times 10^{-3} / ^\circ\text{C}$. At 20°C it carries 30 milliamperes of current. Keeping constant potential difference between its ends, the temperature of the wire is raised to 120°C . The current in milliamperes that flows in the wire is

- (a) 20 (b) 15
 (c) 10 (d) 40

34. Seven resistances are connected as shown in the figure. The equivalent resistance between A and B is

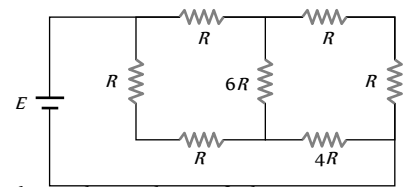
[MP PET 2000]

- (a) 3Ω
 (b) 4Ω
 (c) 4.5Ω
 (d) 5Ω



35. A battery of internal resistance 4Ω is connected to the network of resistances as shown. In order to give the maximum power to the network, the value of R (in Ω) should be

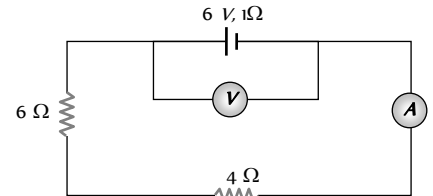
- (a) $4/9$
 (b) $8/9$
 (c) 2
 (d) 18



36. In the circuit shown here, the readings of the ammeter and voltmeter are

[Kerala PMT 2002]

- (a) $6 \text{ A}, 60 \text{ V}$
 (b) $0.6 \text{ A}, 6 \text{ V}$
 (c) $6/11 \text{ A}, 60/11 \text{ V}$
 (d) $11/6 \text{ A}, 11/60 \text{ V}$



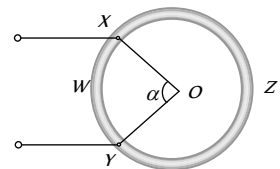
37. Length of a hollow tube is 5 m , its outer diameter is 10 cm and thickness of its wall is 5 mm . If resistivity of the material of the tube is $4.7 \times 10^{-8} \Omega \cdot \text{m}$ then resistance of tube will be

- (a) $5.6 \times 10^{-3} \Omega$ (b) $2 \times 10^{-3} \Omega$
 (c) $4 \times 10^{-3} \Omega$ (d) None of these

38. A wire of resistor R is bent into a circular ring of radius r . Equivalent resistance between two points X and Y on its circumference, when angle XOY is α , can be given by

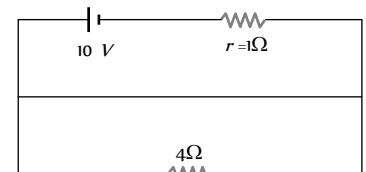
- (a) $\frac{R\alpha}{4\pi^2} (2\pi - \alpha)$
 (b) $\frac{R}{2\pi} (2\pi - \alpha)$
 (c) $R (2\pi - \alpha)$
 (d) $\frac{4\pi}{(2\pi - \alpha)} R$

[IIT-JEE (Screening) 2003]



39. Potential difference across the terminals of the battery shown in figure is (r = internal resistance of battery)

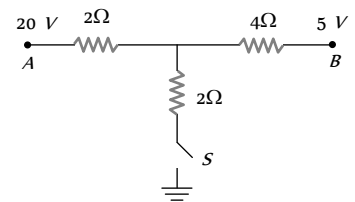
- (a) 8 V
 (b) 10 V
 (c) 6 V
 (d) Zero



40. As the switch S is closed in the circuit shown in figure, current passed through it is

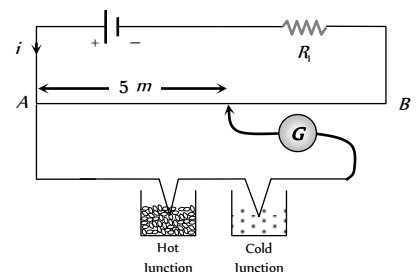
[MP PMT 1994]

- (a) 4.5 A
 (b) 6.0 A
 (c) 3.0 A
 (d) Zero



41. In the following circuit a 10 m long potentiometer wire with resistance 1.2 ohm/m , a resistance R and an accumulator of emf 2 V are connected in series. When the emf of thermocouple is 2.4 mV then the deflection in galvanometer is zero. The current supplied by the accumulator will be

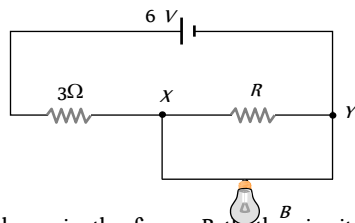
- (a) 4 A



- (b) $8 \times 10^{-4} A$
(c) $4 \times 10^{-4} A$
(d) $8 \times 10^{-4} A$

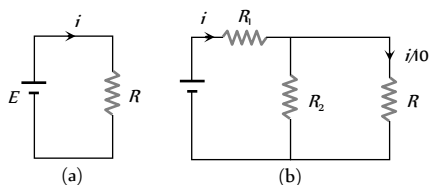
42. In the following circuit, bulb rated as $1.5 V, 0.45 W$. If bulb glows with full intensity then what will be the equivalent resistance between X and Y

- (a) 0.45Ω
(b) 1Ω
(c) 3Ω
(d) 5Ω



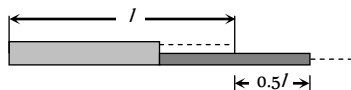
43. Consider the circuits shown in the figure. Both the circuits are taking same current from battery but current through R in the second circuit is $\frac{1}{10}$ th of current through R in the first circuit. If R is 11Ω , the value of R

- (a) 9.9Ω
(b) 11Ω
(c) 8.8Ω
(d) 7.7Ω



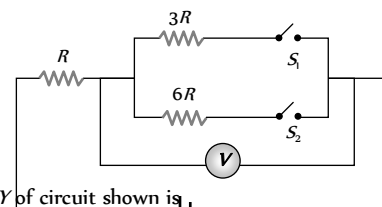
44. In order to quadruple the resistance of a uniform wire, a part of its length was uniformly stretched till the final length of the entire wire was 1.5 times the original length, the part of the wire was fraction equal to

- (a) $1/8$
(b) $1/6$
(c) $1/10$
(d) $1/4$



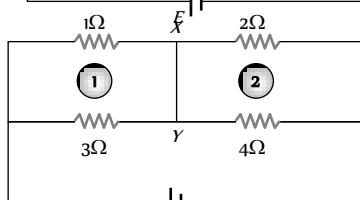
45. In the circuit shown in figure reading of voltmeter is V when only S_1 is closed, reading of voltmeter is V_1 when only S_2 is closed and reading of voltmeter is V_2 when both S_1 and S_2 are closed. Then

- (a) $V_1 > V_2 > V$
(b) $V_1 > V > V_2$
(c) $V_1 > V_2 > V$
(d) $V_1 > V > V_2$



46. Current through wire XY of circuit shown is

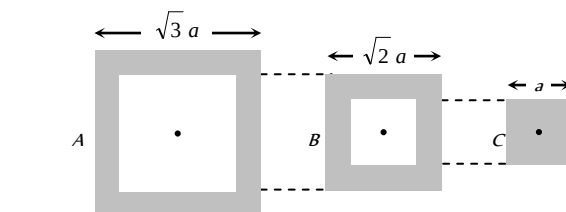
- (a) $1 A$
(b) $4 A$
(c) $2 A$
(d) $3 A$



47. 12 cells each having same emf are connected in series with some cells wrongly connected. The arrangement is connected in series with an ammeter and two cells which are in series. Current is $3 A$ when cells and battery aid each other and is $2 A$ when cells and battery oppose each other. The number of cells wrongly connected is

- (a) 4
(b) 1
(c) 3
(d) 2

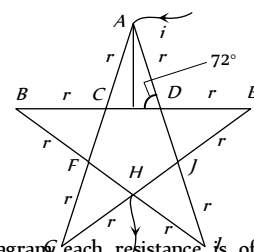
48. Following figure shows cross-sections through three long conductors of the same length and material, with square cross-section of edge lengths as shown. Conductor B will fit snugly within conductor A , and conductor C will fit snugly within conductor B . Relationship between their end to end resistance is



- (a) $R_1 = R_2 = R_3$
(b) $R_1 > R_2 > R_3$
(c) $R_1 < R_2 < R_3$
(d) Information is not sufficient

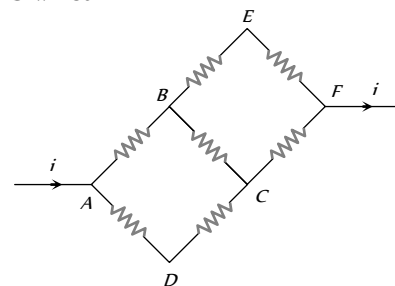
49. In the following star circuit diagram (figure), the equivalent resistance between the points A and H will be

- (a) $1.944 r$
(b) $0.973 r$
(c) $0.486 r$
(d) $0.243 r$



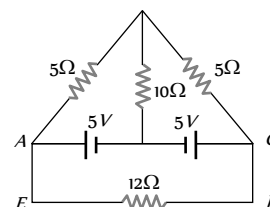
50. In the adjoining circuit diagram each resistance is of 10Ω . The current in the arm AD will be

- (a) $\frac{2i}{5}$
(b) $\frac{3i}{5}$
(c) $\frac{4i}{5}$
(d) $\frac{i}{5}$



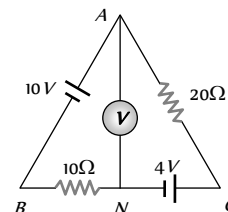
51. In the circuit of adjoining figure the current through 12Ω resistor will be

- (a) $1 A$
(b) $\frac{1}{5} A$
(c) $\frac{2}{5} A$
(d) $0 A$



52. The reading of the ideal voltmeter in the adjoining diagram will be

- (a) $4 V$
(b) $8 V$
(c) $12 V$
(d) $14 V$



53. The resistance of the series combination of two resistance is S . When they are joined in parallel the total resistance is P . If $S = nP$, then the minimum possible value of n is

[AIEEE 2004]

- (a) 4 (b) 3
(c) 2 (d) 1

54. A moving coil galvanometer has 150 equal divisions. Its current sensitivity is 10 divisions per milliamper and voltage sensitivity is 2 divisions per millivolt. In order that each division reads 1 volt, the resistance in ohms needed to be connected in series with the coil will be

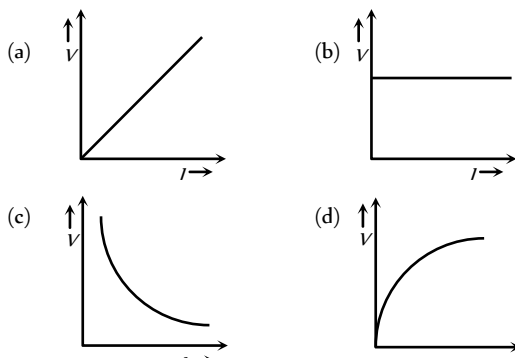
[AIEEE 2005]

- (a) 99995 (b) 9995
(c) 10^3 (d) 10^5

Graphical Questions

1. Which of the adjoining graphs represents ohmic resistance

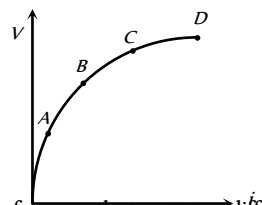
[CPMT 1981; DPMT 2002]



2. Variation of current passing through a conductor I as the voltage applied across its ends as varied is shown in the adjoining diagram. If the resistance (R) is determined at the points A , B , C and D , we will find that

[CPMT 1988]

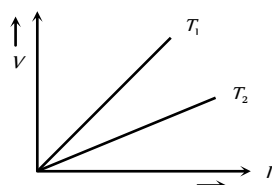
- (a) $R_A = R_B$
(b) $R_A > R_B$
(c) $R_C > R_D$
(d) None of these



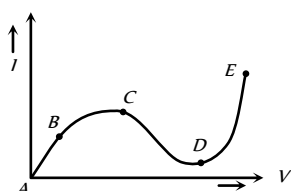
3. The voltage V and current I graph for a conductor at two different temperatures T_1 and T_2 are shown in the figure. The relation between T_1 and T_2 is

[MP PET 1996; KCET 2002]

- (a) $T_1 > T_2$
(b) $T_1 \approx T_2$
(c) $T_1 = T_2$
(d) $T_1 < T_2$

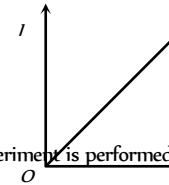


4. From the graph between current I and voltage V shown below, identify the portion corresponding to negative resistance



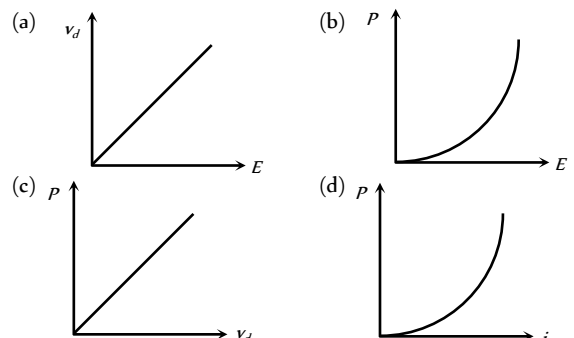
- (a) AB
(b) BC
(c) CD
(d) DE

5. I - V characteristic of a copper wire of length L and area of cross-section A is shown in figure. The slope of the curve becomes

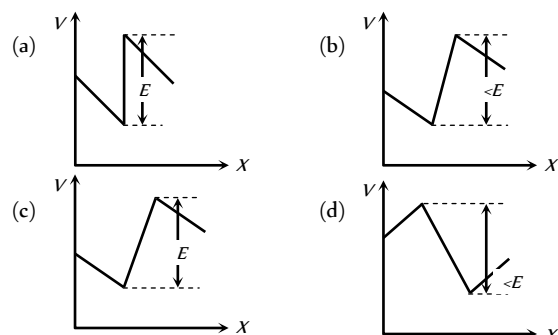


- (a) More if the experiment is performed at higher temperature
(b) More if a wire of steel of same dimension is used
(c) More if the length of the wire is increased
(d) Less if the length of the wire is increased

6. E denotes electric field in a uniform conductor, I corresponding current through it, v_d drift velocity of electrons and P denotes thermal power produced in the conductor, then which of the following graph is incorrect



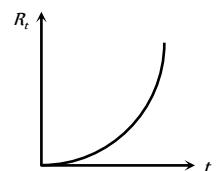
7. The two ends of a uniform conductor are joined to a cell of e.m.f. E and some internal resistance. Starting from the midpoint P of the conductor, we move in the direction of current and return to P . The potential V at every point on the path is plotted against the distance covered (x). Which of the following graphs best represents the resulting curve



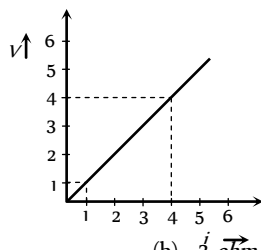
8. The resistance R_t of a conductor varies with temperature t as shown in the figure. If the variation is represented by $R_t = R_0[1 + \alpha t + \beta t^2]$, then

[CPMT 1988]

- (a) α and β are both negative
(b) α and β are both positive
(c) α is positive and β is negative
(d) α is negative and β are positive

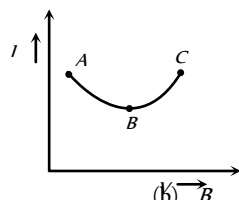


9. Variation of current and voltage in a conductor has been shown in the diagram below. The resistance of the conductor is.



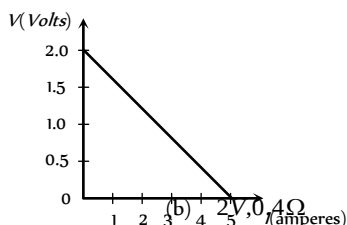
- (a) 4 ohm
(b) 2 ohm
(c) 3 ohm
(d) 1 ohm

10. Resistance as shown in figure is negative at [CPMT 1997]



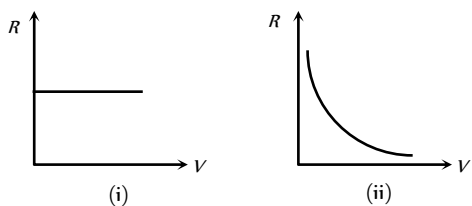
- (a) A
(b) B
(c) C
(d) None of these

11. For a cell, the graph between the potential difference (V) across the terminals of the cell and the current (I) drawn from the cell is shown in the figure. The e.m.f. and the internal resistance of the cell are



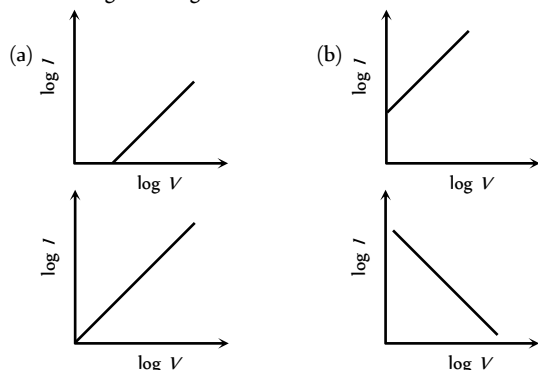
- (a) 2V, 0.5Ω
(b) 2V, 0.4Ω
(c) > 2V, 0.5Ω
(d) > 2V, 0.4Ω

12. The graph which represents the relation between the total resistance R of a multi range moving coil voltmeter and its full scale deflection V is



- (a) (i)
(b) (ii)
(c) (iii)
(d) (iv)

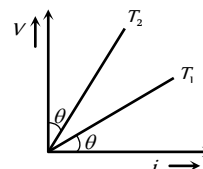
13. When a current I is passed through a wire of constant resistance, it produces a potential difference V across its ends. The graph drawn between $\log I$ and $\log V$ will be



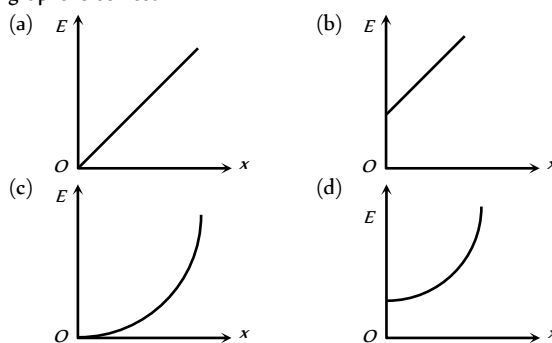
- (c) (d)

14. The $V-i$ graph for a conductor at temperature T_1 and T_2 are as shown in the figure. $(T_2 - T_1)$ is proportional to

- (a) $\cos 2\theta$
(b) $\sin \theta$
(c) $\cot 2\theta$
(d) $\tan \theta$



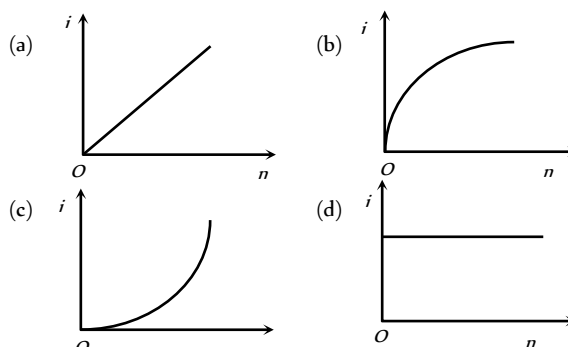
15. A cylindrical conductor has uniform cross-section. Resistivity of its material increase linearly from left end to right end. If a constant current is flowing through it and at a section distance x from left end, magnitude of electric field intensity is E , which of the following graphs is correct



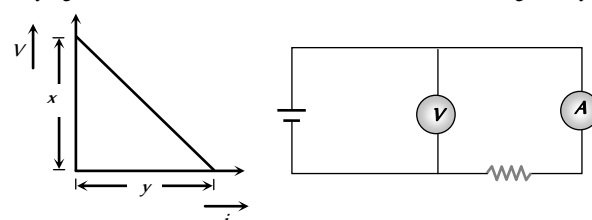
16. The $V-i$ graph for a conductor makes an angle θ with V -axis. Here V denotes the voltage and i denotes current. The resistance of conductor is given by

- (a) $\sin \theta$
(b) $\cos \theta$
(c) $\tan \theta$
(d) $\cot \theta$

17. A battery consists of a variable number ' n ' of identical cells having internal resistances connected in series. The terminals of battery are short circuited and the current i is measured. Which of the graph below shows the relationship between i and n

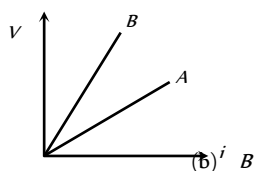


18. In an experiment, a graph was plotted of the potential difference V between the terminals of a cell against the circuit current i by varying load rheostat. Internal conductance of the cell is given by

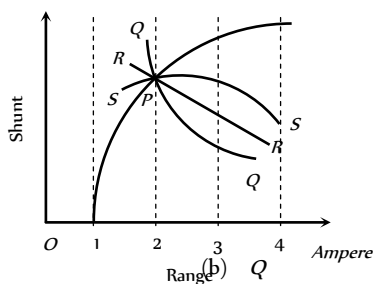


- (a) xy (b) $\frac{y}{x}$
 (c) $\frac{x}{y}$ (d) $(x - y)$

19. $V-i$ graphs for parallel and series combination of two identical resistors are as shown in figure. Which graph represents parallel combination



- (a) A
 (c) A and B both
 (b) Neither A nor B
 (d) Neither A nor B
20. The ammeter has range 1 ampere without shunt. the range can be varied by using different shunt resistances. The graph between shunt resistance and range will have the nature



- (a) P
 (c) R
 (b) Q
 (d) S

Assertion & Reason

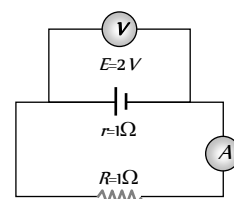
For AIIMS Aspirants

Read the assertion and reason carefully to mark the correct option out of the options given below :

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
 (b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
 (c) If assertion is true but reason is false.
 (d) If the assertion and reason both are false.
 (e) If assertion is false but reason is true.

1. Assertion : The resistivity of a semiconductor increases with temperature.
 Reason : The atoms of a semiconductor vibrate with larger amplitude at higher temperatures thereby increasing its resistivity [AIIMS 2003]
2. Assertion : In a simple battery circuit the point of lowest potential is positive terminal of the battery
 Reason : The current flows towards the point of the higher potential as it flows in such a circuit from the negative to the positive terminal. [AIIMS 2002]
3. Assertion : The temperature coefficient of resistance is positive for metals and negative for p -type semiconductor.
 Reason : The effective charge carriers in metals are negatively charged whereas in p -type semiconductor they are positively charged. [AIIMS 1996]

4. Assertion : In the following circuit emf is $2V$ and internal resistance of the cell is 1Ω and $R = 1\Omega$, then reading of the voltmeter is $1V$.



- Reason : $V = E - ir$ where $E = 2V$, $i = \frac{2}{2} = 1A$ and $R = 1\Omega$ [AIIMS 1995]

5. Assertion : There is no current in the metals in the absence of electric field.

- Reason : Motion of free electron are randomly.

[AIIMS 1994]

6. Assertion : Electric appliances with metallic body have three connections, whereas an electric bulb has a two pin connection.

- Reason : Three pin connections reduce heating of connecting wires.

7. Assertion : The drift velocity of electrons in a metallic wire will decrease, if the temperature of the wire is increased.

- Reason : On increasing temperature, conductivity of metallic wire decreases.

8. Assertion : The electric bulbs glows immediately when switch is on.

- Reason : The drift velocity of electrons in a metallic wire is very high.

9. Assertion : Bending a wire does not effect electrical resistance.

- Reason : Resistance of wire is proportional to resistivity of material.

10. Assertion : In meter bridge experiment, a high resistance is always connected in series with a galvanometer.

- Reason : As resistance increases current through the circuit increases.

11. Assertion : Voltmeter measures current more accurately than ammeter.

- Reason : Relative error will be small if measured from voltmeter.

12. Assertion : Electric field outside the conducting wire which carries a constant current is zero.

- Reason : Net charge on conducting wire is zero.

13. Assertion : The resistance of super-conductor is zero.

- Reason : The super-conductors are used for the transmission of electric power.

14. Assertion : A potentiometer of longer length is used for accurate measurement.

- Reason : The potential gradient for a potentiometer of longer length with a given source of e.m.f. becomes small.

15. Assertion : The e.m.f. of the driver cell in the potentiometer experiment should be greater than the e.m.f. of the cell to be determined.

- Reason : The fall of potential across the potentiometer wire should not be less than the e.m.f. of the cell to be determined.

16. Assertion : A person touching a high power line gets stuck with the line.



1090 Current Electricity

Reason : The current carrying wires attract the man towards it.

17. Assertion : The connecting wires are made of copper.

Reason : The electrical conductivity of copper is high.

Answers

Electric Conduction, Ohm's Law and Resistance

1	a	2	c	3	b	4	b	5	c
6	a	7	a	8	a	9	d	10	c
11	d	12	d	13	a	14	c	15	a
16	a	17	c	18	b	19	c	20	b
21	d	22	b	23	b	24	b	25	d
26	c	27	b	28	b	29	b	30	a
31	c	32	d	33	b	34	d	35	c
36	b	37	b	38	c	39	a	40	d
41	b	42	b	43	a	44	b	45	c
46	a	47	b	48	b	49	c	50	a
51	c	52	c	53	b	54	b	55	b
56	a	57	a	58	a	59	c	60	c
61	a	62	b	63	b	64	c	65	c
66	d	67	a	68	b	69	d	70	d
71	a	72	a	73	c	74	b	75	b
76	c	77	c	78	c	79	d	80	b
81	a	82	d	83	b	84	b	85	c
86	b	87	c	88	a	89	a	90	d
91	a	92	c	93	b	94	a	95	b
96	b	97	c	98	a	99	c	100	d
101	c	102	a	103	d	104	b	105	b
106	d	107	d	108	a	109	d	110	d
111	d	112	d	113	a	114	a	115	c
116	a	117	a	118	b	119	c	120	a
121	d	122	a	123	a	124	d	125	c
126	b	127	c	128	a	129	a	130	c
131	c	132	b	133	c				

Grouping of Resistances

1	c	2	d	3	a	4	c	5	b
6	c	7	c	8	b	9	a	10	b
11	d	12	d	13	b	14	d	15	b
16	d	17	c	18	c	19	b	20	d
21	a	22	a	23	b	24	b	25	c
26	b	27	d	28	d	29	d	30	c
31	b	32	d	33	a	34	b	35	c
36	d	37	d	38	b	39	c	40	b

41	a	42	c	43	b	44	d	45	c
46	d	47	c	48	b	49	b	50	d
51	d	52	c	53	d	54	a	55	c
56	d	57	c	58	c	59	d	60	c
61	d	62	c	63	d	64	c	65	c
66	c	67	b	68	c	69	d	70	b
71	a	72	c	73	a	74	b	75	a
76	c	77	c	78	b	79	c	80	a
81	a	82	b	83	b	84	d	85	d
86	a	87	a	88	a	89	b	90	b
91	b	92	c	93	b	94	d	95	a
96	d	97	b	98	b	99	d	100	a
101	c	102	a	103	b	104	d	105	a
106	a	107	b	108	d	109	bc	110	b
111	d	112	c	113	a	114	a	115	d
116	a	117	d	118	c	119	d	120	c
121	b	122	b	123	b	124	c	125	b
126	a	127	c	128	b	129	c	130	a
131	a	132	a	133	c	134	a	135	b
136	b	137	a	138	b	139	c	140	b
141	b								

Kirchhoff's Law, Cells

1	b	2	d	3	c	4	a	5	a
6	b	7	a	8	a	9	b	10	a
11	c	12	d	13	a	14	d	15	b
16	c	17	c	18	c	19	d	20	b
21	c	22	c	23	b	24	d	25	a
26	a	27	b	28	b	29	a	30	b
31	a	32	c	33	b	34	a	35	a
36	b	37	a	38	b	39	b	40	c
41	d	42	d	43	d	44	a	45	c
46	c	47	b	48	a	49	a	50	d
51	b	52	d	53	b	54	c	55	a
56	b	57	c	58	a	59	d	60	b
61	c	62	c	63	c	64	b	65	a
66	c	67	a	68	d	69	b	70	a
71	a	72	d	73	c	74	b	75	b
76	b	77	c	78	c	79	d	80	d
81	a	82	d	83	c	84	c	85	a

Different Measuring Instruments

1	a	2	c	3	d	4	d	5	c
6	c	7	a	8	d	9	c	10	c
11	d	12	c	13	d	14	a	15	d
16	c	17	a	18	b	19	c	20	a
21	b	22	b	23	a	24	a	25	a
26	a	27	a	28	a	29	b	30	b
31	b	32	b	33	b	34	b	35	c
36	c	37	b	38	b	39	d	40	b

41	a	42	b	43	c	44	d	45	a
46	b	47	c	48	a	49	b	50	a
51	b	52	c	53	b	54	b	55	a
56	b	57	d	58	c	59	c	60	d
61	a	62	a	63	d	64	a	65	d
66	b	67	a	68	b	69	c	70	c
71	d	72	c	73	a	74	c	75	a
76	d	77	b	78	a	79	b	80	a
81	c	82	c	83	a	84	a	85	b
86	d	87	c	88	d	89	c	90	a
91	d	92	d	93	d	94	c	95	d
96	a	97	c	98	a	99	d	100	d
101	c	102	d	103	c	104	c	105	a
106	c	107	c	108	d	109	c	110	a
111	a	112	a	113	c	114	b	115	c
116	d	117	a	118	b	119	d	120	c
121	d	122	c	123	a	124	a	125	b
126	b	127	d	128	d	129	b	130	c
131	d	132	a	133	c	134	d	135	d
136	a	137	a	138	b	139	d	140	a
141	c	142	a	143	a				

Critical Thinking Questions

1	a	2	a	3	c	4	c	5	c
6	b	7	d	8	d	9	d	10	b
11	d	12	d	13	c	14	b	15	b
16	b	17	b	18	b	19	a	20	d
21	d	22	a	23	c	24	c	25	b
26	b	27	a	28	d	29	b	30	b
31	d	32	a	33	a	34	b	35	c
36	c	37	a	38	a	39	d	40	a
41	a	42	b	43	a	44	a	45	b
46	c	47	b	48	a	49	b	50	a
51	d	52	b	53	a	54	b		

Graphical Questions

1	a	2	d	3	a	4	c	5	d
6	c	7	b	8	b	9	d	10	a
11	b	12	d	13	a	14	c	15	b
16	d	17	d	18	b	19	a	20	b

Assertion and Reason

1	d	2	d	3	b	4	a	5	a
6	c	7	b	8	c	9	a	10	c
11	a	12	a	13	b	14	a	15	a
16	d	17	a						

AS Answers and Solutions

Electric Conduction, Ohm's Law and Resistance

1. (a) Number of electrons flowing per second

$$\frac{n}{t} = \frac{i}{e} = 4.8 / 1.6 \times 10^{-19} = 3 \times 10^{19}$$

2. (c) $v_d = \frac{J}{ne} \Rightarrow v_d \propto J$ (current density)

$$J_1 = \frac{i}{A} \text{ and } J_2 = \frac{2i}{2A} = \frac{i}{A} = J_1; \therefore (v_d)_1 = (v_d)_2 = v$$

3. (b) Order of drift velocity $= 10^{-4} \text{ m/sec} = 10^{-2} \text{ cm/sec}$

4. (b) Density of Cu $= 9 \times 10^3 \text{ kg/m}^3$ (mass of 1 m of Cu)

$$\therefore 6.0 \times 10^{23} \text{ atoms has a mass} = 63 \times 10^{-3} \text{ kg}$$

$$\therefore \text{Number of electrons per m are}$$

$$= \frac{6.0 \times 10^{23}}{63 \times 10^{-3}} \times 9 \times 10^3 = 8.5 \times 10^{28}$$

$$\text{Now drift velocity} = v_d = \frac{i}{neA}$$

$$= \frac{1.1}{8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times \pi \times (0.5 \times 10^{-3})^2}$$

$$= 0.1 \times 10^{-3} \text{ m/sec}$$

5. (c) Because 1 H.P. = 746 J/s = 746 watt

6. (a) $R \propto l^2 \Rightarrow \frac{\Delta R}{R} = \frac{2\Delta l}{l} \Rightarrow \frac{\Delta R_0}{R} \% = 2 \times 0.1 = 0.2\%$

7. (a) $R = \frac{\rho l}{A} = 50 \times 10^{-8} \times \frac{50 \times 10^{-2}}{(50 \times 10^{-2})^2} = 10^{-6} \Omega$

8. (a) Resistivity of some material is its intrinsic property and is constant at particular temperature. Resistivity does not depend upon shape.

9. (d) $\frac{\rho_1}{\rho_2} = \frac{(1 + \alpha t_1)}{(1 + \alpha t_2)} \Rightarrow \frac{1}{2} = \frac{(1 + 0.00125 \times 27)}{(1 + 0.00125 \times t)}$

$$\Rightarrow t = 854^\circ\text{C} \Rightarrow T = 1127\text{K}$$

10. (c) $R_1 \propto \frac{l}{A} \Rightarrow R_2 \propto \frac{2l}{2A}$ i.e. $R_2 \propto \frac{l}{A}$

$$\therefore R_1 = R_2$$

11. (d) In case of stretching of wire $R \propto l^2$

$\Rightarrow$ If length becomes 3 times so Resistance becomes 9 times
i.e. $R' = 9 \times 20 = 180\Omega$

12. (d) Resistivity is the property of the material. It does not depend upon size and shape.

13. (a) Because with rise in temperature resistance of conductor increase, so graph between V and i becomes non linear.

14. (c) Because $V-i$ graph of diode is non-linear.

15. (a) $v_d = \frac{e}{m} \times \frac{V}{l} \tau$ or $v_d = \frac{e}{m} \cdot \frac{El}{l} \tau$ (Since $V = El$)

$$\therefore v_d \propto E$$

16. (a) Resistance of conductor depends upon relation as $R \propto \frac{1}{\tau}$.

With rise in temperature rms speed of free electron inside the conductor increase, so relaxation time decrease and hence resistance increases

17. (c) $i = \frac{q}{t} = \frac{4}{2} = 2 \text{ ampere}$

18. (b) Volume $= Al = 3 \Rightarrow A = \frac{3}{l}$

$$\text{Now } R = \rho \frac{l}{A} \Rightarrow 3 = \frac{\rho \times l}{3/l} = \frac{\rho l^2}{3} \Rightarrow l^2 = \frac{9}{\rho} = \frac{3}{\sqrt{\rho}}$$

19. (c) $i = \frac{ne}{t} = \frac{62.5 \times 10^{18} \times 1.6 \times 10^{-19}}{1} = 10 \text{ ampere}$

20. (b) In twisted wire, two halves each of resistance 2Ω are in parallel, so equivalent resistance will be $\frac{2}{2} = 1\Omega$.

21. (d) In stretching of wire $R \propto \frac{1}{r^4}$

22. (b) $R = \frac{\rho L}{A} \Rightarrow 0.7 = \frac{\rho \times 1}{\frac{22}{7} (1 \times 10^{-3})^2}$

$$\rho = 2.2 \times 10^{-6} \text{ ohm-m.}$$

23. (b) $R \propto \frac{1}{A} \Rightarrow R \propto \frac{1}{r^2} \propto \frac{1}{d^2}$ [d = diameter of wire]

24. (b) $i = qv = 1.6 \times 10^{-19} \times 6.6 \times 10^{15} = 10.56 \times 10^{-4} \text{ A} = 1 \text{ mA}$

25. (d) $R \propto \frac{l}{r^2} \Rightarrow \frac{R_1}{R_2} = \frac{l_1}{l_2} \times \frac{r_2^2}{r_1^2} \Rightarrow \frac{1}{1} = \frac{5}{l_2} \times \left(\frac{2}{1}\right)^2 \Rightarrow l_2 = 20 \text{ m}$

26. (c)

27. (b) In semiconductors charge carries are free electrons and holes

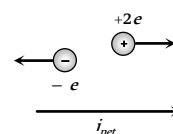
28. (b) Net current $i_{\text{net}} = i_{(+)} + i_{(-)}$

$$= \frac{n_{(+)}q_{(+)}}{t} + \frac{n_{(-)}q_{(-)}}{t}$$

$$= \frac{n_{(+)} \times 2e}{t} + \frac{n_{(-)} \times e}{t}$$

$$= 3.2 \times 10^8 \times 2 \times 1.6 \times 10^{-19} + 3.6 \times 10^8 \times 1.6 \times 10^{-19}$$

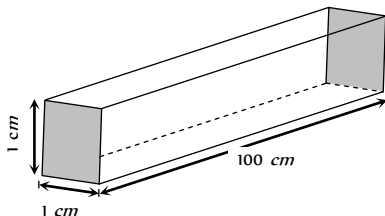
$$= 1.6 \text{ A (towards right)}$$



29. (b) In the absence of external electric field mean velocity of free

$$\text{electron } (V_-) \text{ is given by } V_{rms} = \sqrt{\frac{3KT}{m}} \Rightarrow V_{rms} \propto \sqrt{T}$$

30. (a) With rise in temperature specific resistance increases
 31. (c) For metallic conductors, temperature co-efficient of resistance is positive.
 32. (d)
 33. (b) Length $l = 1 \text{ cm} = 10^{-2} \text{ m}$



Area of cross-section $A = 1 \text{ cm} \times 100 \text{ cm}$

$$= 100 \text{ cm} = 10^{-1} \text{ m}$$

$$\text{Resistance } R = 3 \times 10^{-2} \times \frac{10^{-2}}{10^{-1}} = 3 \times 10^{-3} \Omega$$

34. (d) In the above question for calculating equivalent resistance between two opposite square faces.

$$l = 100 \text{ cm} = 1 \text{ m}, A = 1 \text{ cm} = 10^{-1} \text{ m}, \text{ so resistance } R = 3 \times 10^{-3} \times \frac{1}{10^{-4}} = 3 \times 10^{-7} \Omega$$

35. (c) $v_d = \frac{i}{nAe} = \frac{20}{10^{29} \times 10^{-6} \times 1.6 \times 10^{-19}} = 1.25 \times 10^{-3} \text{ m/s}$

36. (b) Specific resistance $k = \frac{E}{j}$

$$37. (b) R \propto \frac{l}{A} \propto \frac{l}{d^2} \Rightarrow \frac{R_1}{R_2} = \frac{l_1}{l_2} \times \left(\frac{d_2}{d_1}\right)^2 = \frac{L}{4L} \left(\frac{2d}{d}\right)^2 = 1$$

$$\Rightarrow R_1 = R_2 = R$$

$$38. (c) v_d = \frac{i}{nAe} = \frac{1.344}{10^{-6} \times 1.6 \times 10^{-19} \times 8.4 \times 10^{22}} = \frac{1.344}{10 \times 1.6 \times 8.4} = 0.01 \text{ cm/s} = 0.1 \text{ mm/s}$$

39. (a) Internal resistance $\propto \frac{1}{\text{Temperature}}$

$$40. (d) \text{ Charge} = \text{Current} \times \text{Time} = 5 \times 60 = 300 \text{ C}$$

$$41. (b) \text{ By } R = \rho l / A$$

$$42. (b)$$

$$43. (a)$$

$$44. (b) R = \frac{\rho l}{a} \text{ for first wire and } R = \frac{\rho l}{4a} = \frac{R}{4} \text{ for second wire.}$$

45. (c) For semiconductors, resistance decreases on increasing the temperature.

$$46. (a) R = \rho \frac{l}{A} = \frac{n}{ne^2 \tau} \cdot \frac{l}{A}$$

47. (b) Because as temperature increases, the resistivity increases and hence the relaxation time decreases for conductors $\left(\tau \propto \frac{1}{\rho}\right)$.

48. (b) In VI graph, we will not get a straight line in case of liquids.

$$49. (c) R = \rho \frac{l}{A}$$

50. (a) Since $R \propto l^2 \Rightarrow$ If length is increased by 10%, resistance is increases by almost 20%

Hence new resistance $R' = 10 + 20\% \text{ of } 10$

$$= 10 + \frac{20}{100} \times 10 = 12 \Omega.$$

$$51. (c) \frac{R_{150}}{R_{500}} = \frac{[1 + \alpha(150)]}{[1 + \alpha(500)]}. \text{ Putting } R_{150} = 133 \Omega \text{ and } \alpha = 0.0045 / ^\circ\text{C}, \text{ we get } R_{500} = 258 \Omega$$

$$52. (c) R = \rho \frac{l}{A} \Rightarrow 7 = \frac{64 \times 10^{-6} \times 198}{\frac{22}{7} \times r^2} \Rightarrow r = 0.024 \text{ cm}$$

$$53. (b) \text{ Current density } J = \frac{i}{A} = \frac{i}{\pi r^2} \Rightarrow \frac{J_1}{J_2} = \frac{i_1}{i_2} \times \frac{r_2^2}{r_1^2}$$

But the wires are in series, so they have the same current,

$$\text{hence } i_1 = i_2. \text{ So } \frac{J_1}{J_2} = \frac{r_2^2}{r_1^2} = 9 : 1$$

54. (b) As $\frac{V}{i} = R$ and $R \propto \text{temperature}$

55. (b) $R \propto l^2 \Rightarrow$ If l doubled then R becomes 4 times.

56. (a) Temperature coefficient of a semiconductor is negative.

57. (a) The reciprocal of resistance is called conductance

$$58. (a) \text{ Resistance} = \frac{\text{Potential difference}}{\text{Current}}$$

59. (c) Ohm's Law is not obeyed by semiconductors.

60. (c) Drift velocity $v_d = \frac{V}{\rho l n e}$; v_d does not depend upon diameter.

$$61. (a) \text{ Using } R_{T_2} = R_{T_1} [1 + \alpha(T_2 - T_1)]$$

$$\Rightarrow R_{100} = R_{50} [1 + \alpha(100 - 50)]$$

$$\Rightarrow 7 = 5 [1 + (\alpha \times 50)] \Rightarrow \alpha = \frac{(7 - 5)}{250} = 0.008 / ^\circ\text{C}$$

62. (b) This is because of secondary ionisation which is possible in the gas filled in it.

$$63. (b)$$

$$64. (c) \frac{R_1}{R_2} = \frac{(1 + \alpha t_1)}{(1 + \alpha t_2)} \Rightarrow \frac{50}{76.8} = \frac{(1 + 3.92 \times 10^{-3} \times 20)}{(1 + 3.92 \times 10^{-3} t)}$$

$$\Rightarrow t = 167^\circ\text{C}$$

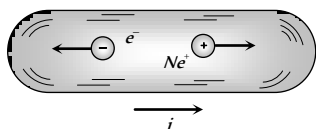
$$65. (c) \text{ From } v_d = \frac{i}{neA} \Rightarrow i \propto v_d A \Rightarrow i \propto v_d r^2$$

66. (d) Resistivity depends only on the material of the conductor.

67. (a) A particular temperature, the resistance of a superconductor is

$$\text{zero} \Rightarrow G = \frac{1}{R} = \frac{1}{0} = \infty$$

68. (b) Net current $i = i_+ + i_- = \frac{(n_+)(q_+)}{t} + \frac{(n_-)(q_-)}{t}$



$$\begin{aligned} \Rightarrow i &= \frac{(n_+)}{t} \times e + \frac{(n_-)}{t} \times e \\ &= 2.9 \times 10^{18} \times 1.6 \times 10^{-19} + 1.2 \times 10^{18} \times 1.6 \times 10^{-19} \\ \Rightarrow i &= 0.66 \text{ A} \end{aligned}$$

69. (d) If E be electric field, then current density $j = \sigma E$

$$\text{Also we know that current density } j = \frac{i}{A}$$

Hence j is different for different area of cross-sections. When j is different, then E is also different. Thus E is not constant. The drift velocity v_d is given by $v_d = \frac{j}{ne}$ = different for different j values. Hence only current i will be constant.

70. (d)

71. (a) $R = \rho \frac{l}{A}$ and mass $m = \text{volume } (V) \times \text{density } (d) = (Al) d$

Since wires have same material so ρ and d is same for both.

$$\text{Also they have same mass} \Rightarrow Al = \text{constant} \Rightarrow l \propto \frac{1}{A}$$

$$\Rightarrow \frac{R_1}{R_2} = \frac{l_1}{l_2} \times \frac{A_2}{A_1} = \left(\frac{A_2}{A_1} \right)^2 = \left(\frac{r_2}{r_1} \right)^4$$

$$\Rightarrow \frac{34}{R_2} = \left(\frac{r}{2r} \right)^4 \Rightarrow R_2 = 544 \Omega$$

72. (a) $R = \rho \frac{l}{A} \Rightarrow \frac{R_1}{R_2} = \frac{A_2}{A_1} (\rho, L \text{ constant}) \Rightarrow \frac{A_1}{A_2} = \frac{R_2}{R_1} = 2$

Now, when a body dipped in water, loss of weight = $V\sigma_L g = AL\sigma_L g$

$$\text{So, } \frac{(\text{Loss of weight})_1}{(\text{Loss of weight})_2} = \frac{A_1}{A_2} = 2; \text{ so } A \text{ has more loss of weight.}$$

73. (c) $Q = it = 20 \times 10^{-3} \times 30 = 6 \times 10^{-2} \text{ C}$

74. (b) Ge is semiconductor and Na is a metal. The conductivity of semiconductor increases and that of the metals decreases with the rise in temperature.

75. (b) $i = \frac{ne}{t} \Rightarrow n = \frac{it}{e} = \frac{1.6 \times 10^{-3} \times 1}{1.6 \times 10^{-19}} = 10^{16}$

76. (c) Drift velocity $v_d = \frac{i}{neA} \Rightarrow v_d \propto \frac{1}{A}$ or $v_d \propto \frac{1}{d^2}$

$$\Rightarrow \frac{v_P}{v_Q} = \left(\frac{d_Q}{d_P} \right)^2 = \left(\frac{d/2}{d} \right)^2 = \frac{1}{4} \Rightarrow v_P = \frac{1}{4} v_Q$$

77. (c) Human body, though has a large resistance of the order, of $K\Omega$ (say $10k\Omega$), is very sensitive to minute currents even as low as a few mA . Electrons, excites and disorders the nervous system of the body and hence one fails to control the activity of the body.

78. (c) $R_t = R_0(1 + \alpha t)$

$$\Rightarrow 4.2 = R_0(1 + 0.004 \times 100) = 1.4R_0 \Rightarrow R_0 = 3\Omega$$

79. (d) $R \propto \frac{l^2}{m} \Rightarrow R_1 : R_2 : R_3 = \left(\frac{l_1}{m_1} \right)^2 : \left(\frac{l_2}{m_2} \right)^2 : \left(\frac{l_3}{m_3} \right)^2$

$$= \frac{25}{1} : \frac{9}{3} : \frac{1}{5} = 25 : 3 : \frac{1}{5} \Rightarrow 125 : 15 : 1$$

80. (b)

81. (a) $\frac{R_1}{R_2} = \left(\frac{r_2}{r_1} \right)^4 \Rightarrow \frac{R}{R_2} = \left(\frac{nr}{r} \right)^4 \Rightarrow R_2 = \frac{R}{n^4}$

82. (d) $\frac{R_1}{R_2} = \frac{(1 + \alpha t_1)}{(1 + \alpha t_2)} \Rightarrow \frac{5}{6} = \frac{(1 + \alpha \times 50)}{(1 + \alpha \times 100)} \Rightarrow \alpha = \frac{1}{200} \text{ per } ^\circ C$

$$\text{Again by } R_t = R_0(1 + \alpha t)$$

$$\Rightarrow 5 = R_0 \left(1 + \frac{1}{200} \times 50 \right) \Rightarrow R_0 = 4\Omega$$

83. (b) $i = \frac{Q}{T} = Qv = 1.6 \times 10^{-19} \times 5 \times 10^{15} = 0.8mA$

84. (b) $\frac{r_{\text{iron}}}{r_{\text{copper}}} = \sqrt{\frac{\rho_{\text{iron}}}{\rho_{\text{copper}}}} = \sqrt{\frac{1 \times 10^{-7}}{1.7 \times 10^{-8}}} \approx 2.4$

85. (c) $i = ev = 1.6 \times 10^{-19} \times 6.8 \times 10^{15} = 1.1 \times 10^{-3} \text{ amp}$

86. (b) Resistivity of the material of the rod

$$\rho = \frac{RA}{l} = \frac{3 \times 10^{-3} \pi (0.3 \times 10^{-2})^2}{1} = 27 \times 10^{-9} \pi \Omega \times m$$

$$\text{Resistance of disc } R = \frac{(\text{Thickness})}{(\text{Area of cross section})}$$

$$= 27 \times 10^{-9} \pi \times \frac{(10^{-3})}{\pi \times (1 \times 10^{-2})^2} = 2.7 \times 10^{-7} \Omega$$

87. (c) By using $R_t = R_0(1 + \alpha t)$

$$3 \times R_0 = R_0(1 + 4 \times 10^{-3} t) \Rightarrow t = 500^\circ C$$

88. (a) $i = 6 \times 10^{15} \times 1.6 \times 10^{-19} = 0.96mA$

89. (a) $i = \frac{ne}{t} \Rightarrow 16 \times 10^{-3} = \frac{n \times 1.6 \times 10^{-19}}{1} \Rightarrow n = 10^{17}$

90. (d) $R = \frac{V}{i} = \frac{100 \pm 0.5}{10 \pm 0.2} = 10 \pm 0.25 \Omega$

91. (a) $R = \frac{V}{i} = \rho \frac{l}{A} \Rightarrow \frac{2}{4} = \rho \frac{50 \times 10^{-2}}{(1 \times 10^{-3})^2} \Rightarrow \rho = 1 \times 10^{-6} \Omega m$

92. (c)

93. (b) $i = \frac{V}{R} = \frac{Q}{t} \Rightarrow Q = \frac{Vt}{R} = \frac{20 \times 2 \times 60}{10} = 240 \text{ C}$

94. (a) $R \propto l^2 \Rightarrow \frac{R_1}{R_2} = \left(\frac{l_1}{l_2}\right)^2 \Rightarrow \frac{R}{R_2} = \left(\frac{l}{l/2}\right)^2 = 4 \Rightarrow R_2 = \frac{R}{4}$.

95. (b)

96. (b) $V_d = \frac{i}{neA} = \frac{40}{10^{29} \times 10^{-6} \times 1.6 \times 10^{-19}}$
 $= 2.5 \times 10^{-3} \text{ m/sec}.$

97. (c) $V_d = \frac{i}{nAe} = \frac{5.4}{8.4 \times 10^{28} \times 10^{-6} \times 1.6 \times 10^{-19}}$
 $= 0.4 \times 10^{-3} \text{ m/sec} = 0.4 \text{ mm/sec}.$

98. (a) $\frac{R_1}{R_2} = \left(\frac{l_1}{l_2}\right)^2 \Rightarrow \frac{10}{R_2} = \left(\frac{5}{20}\right)^2 = \frac{1}{16} = R_2 = 160 \Omega.$

99. (c) $R \propto \frac{1}{\tau}$; where τ = Relaxation time.

When lamp is switched on, temperature of filament increase, hence τ decrease so R increases

100. (d) $R = 91 \times 10^2 \approx 9.1 \text{ k}\Omega.$

101. (c)

102. (a)

103. (d) $R \propto \frac{l^2}{m} \Rightarrow R_1 : R_2 : R_3 = \frac{l_1^2}{m_1} : \frac{l_2^2}{m_2} : \frac{l_3^2}{m_3}$
 $\Rightarrow R_1 : R_2 : R_3 = \frac{9}{1} : \frac{4}{2} : \frac{1}{3} = 27 : 6 : 1.$

104. (b) $n = \frac{1 \times 10^{-3}}{1.6 \times 10^{-19}} = 6.25 \times 10^{15}.$

105. (b) $v_d = \frac{i}{ne\pi r^2} \Rightarrow v_d \propto \frac{i}{r^2} \Rightarrow \frac{v}{v'} = \frac{i}{i'} \times \left(\frac{r_2}{r_1}\right)^2 \Rightarrow v' = \frac{v}{2}.$

106. (d) $\frac{R_1}{R_2} = \left(\frac{r_2}{r_1}\right)^4 \Rightarrow \frac{R}{R_2} = \left(\frac{3r/4}{r}\right)^4 = \frac{81}{256} = R_2 = \frac{256}{81} R$

107. (d)

108. (a) $v_d = \frac{i}{nAe} = \frac{8}{8 \times 10^{28} \times (2 \times 10^{-3})^2 \times 1.6 \times 10^{-19}}$
 $= 0.156 \times 10^{-3} \text{ m/sec}.$

109. (d) Specific resistance doesn't depend upon length and area.

110. (d) Heating effect of current.

111. (d) $l = \frac{R\pi r^2}{\rho} = \frac{4.2 \times 3.14 \times (0.2 \times 10^{-3})^2}{48 \times 10^{-8}} = 1.1 \text{ m}$

112. (d) For conductors, resistance $\propto$ Temperature and for semi-conductor, resistance $\propto \frac{1}{\text{Temperature}}$

113. (a) If suppose initial length $l_1 = 100$ then $l_2 = 100 + 100 = 200$

$$\frac{R_1}{R_2} = \left(\frac{l_1}{l_2}\right)^2 = \left(\frac{100}{200}\right)^2 \Rightarrow R_2 = 4R_1$$

$$\frac{\Delta R}{R} \times 100 = \frac{R_2 - R_1}{R_1} \times 100 = \frac{4R_1 - R_1}{R_1} \times 100 = 300\%.$$

114. (a) Ammeter is always connected in series and Voltmeter is always connected in parallel.

115. (c) Same mass, same material *i.e.* volume is same or $Al = \text{constant}$

$$\text{Also, } R = \rho \frac{l}{A} \Rightarrow \frac{R_1}{R_2} = \frac{l_1}{l_2} \times \frac{A_2}{A_1} = \left(\frac{A_2}{A_1}\right)^2 \left(\frac{d_2}{d_1}\right)^4$$

$$\Rightarrow \frac{24}{R_2} = \left(\frac{d}{d/2}\right)^4 = 16 \Rightarrow R_2 = 1.5 \Omega.$$

116. (a) $I = n_e q_e + n_p q_p = 1 \text{ mA}$ towards right

117. (a) As steady current is flowing through the conductor, hence the number of electrons entering from one end and outgoing from the other end of any segment is equal. Hence charge will be zero.

118. (b) Conductance $C = \frac{1}{R} = \frac{A}{\rho l} \Rightarrow C \propto \frac{1}{l}$

119. (c) $i = \frac{dQ}{dt} \Rightarrow dQ = i dt \Rightarrow Q = \int_{t_1}^{t_2} i dt = \int_0^5 (1.2t + 3) dt$
 $= \left[\frac{1.2t^2}{2} + 3t \right]_0^5 = 30 \text{ C}$

120. (a) In stretching, $\frac{R_2}{R_1} = \left(\frac{r_1}{r_2}\right)^4 \Rightarrow \frac{R_2}{R} = \left(\frac{2}{1}\right)^4 \Rightarrow R_2 = 16R$

121. (d) $R' = n^2 R \Rightarrow R' = 16R$

122. (a)

Significant figures		Multiplier
Brown	Black	Brown
1	0	10

$$\therefore R = 10 \times 10 = 100 \Omega$$

123. (a)

124. (d)

125. (c)

126. (b) $R \propto \frac{l}{r^2} \Rightarrow \frac{R_2}{R_1} = \frac{l_2}{l_1} \times \frac{r_1^2}{r_2^2} = \left(\frac{2}{1}\right) \times \left(\frac{1}{2}\right)^2 = \frac{1}{2}$

$\Rightarrow R_2 = \frac{R_1}{2}$, specific resistance doesn't depend upon length, and radius.

127. (c) By using $v_d = \frac{i}{neA} = \frac{100}{10^{28} \times 1.6 \times 10^{-19} \times \frac{\pi}{4} \times (0.02)^2}$
 $= 2 \times 10^{-4} \text{ m/sec}$

128. (a) $R \propto \frac{l}{r^2}$. For highest resistance $\frac{l}{r^2}$ should be maximum, which is correct for option (a)

129. (a) Red, brown, orange, silver red and brown represents the first two significant figures.

Significant figures		Multiplier	Tolerance
Red	Brown	Orange	Silver
2	1	10	$\pm 10\%$

$$\therefore R = 21 \times 10^3 \pm 10\%$$

130. (c) In stretching $R \propto l^2 \Rightarrow \frac{R_2}{R_1} = \frac{l_2^2}{l_1^2} \Rightarrow \frac{R_2}{R_1} = \left(\frac{2}{1}\right)^2$

$\Rightarrow R_2 = 4R_1$. Change in resistance $= R_2 - R_1 = 3R_1$

Now, $\frac{\text{Change in resistance}}{\text{Original resistance}} = \frac{3R_1}{R_1} = \frac{3}{1}$

131. (c) $\frac{R_1}{R_2} = \left(\frac{l_1}{l_2}\right)^2$, If $l_1 = 100$ then $l_2 = 110$

$\Rightarrow \frac{R_1}{R_2} = \left(\frac{100}{110}\right)^2 \Rightarrow R_2 = 1.21R_1$

% change $\frac{R_2 - R_1}{R_1} \times 100 = 21\%$

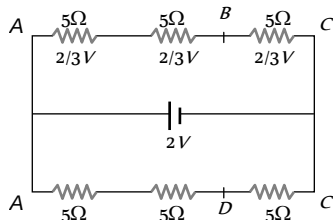
132. (b)

133. (c) Resistance $= \rho \frac{l}{A}$

$\therefore \frac{R_1}{R_2} = \frac{\rho_1}{\rho_2} \times \frac{l_1}{l_2} \times \frac{A_2}{A_1} = \frac{2}{3} \times \frac{3}{4} \times \frac{5}{4} = \frac{5}{8}$

Grouping of Resistances

1. (c) The given circuit can be redrawn as follows



For identical resistances, potential difference distributes equally among all. Hence potential difference across each resistance is $\frac{2}{3}V$, and potential difference between A and B is $\frac{4}{3}V$.

2. (d) Equivalent resistance of parallel resistors is always less than any of the member of the resistance system.

3. (a) Each part will have a resistance $r = R/10$

Let equivalent resistance be r_R , then

$\frac{1}{r_R} = \frac{1}{r} + \frac{1}{r} + \frac{1}{r} \dots \dots \dots 10 \text{ times}$

$\therefore \frac{1}{r_R} = \frac{10}{r} = \frac{10}{R/10} = \frac{100}{R} \Rightarrow r_R = \frac{R}{100} = 0.01R$

4. (c) $R_{\text{equivalent}} = \frac{(30 + 30)30}{(30 + 30) + 30} = \frac{60 \times 30}{90} = 20\Omega$

$\therefore i = \frac{V}{R} = \frac{2}{20} = \frac{1}{10} \text{ ampere}$

5. (b) Resistance of parallel group $= \frac{R}{2}$

$\therefore \text{Total equivalent resistance} = 4 \times \frac{R}{2} = 2R$

6. (c) Resistance of 1 ohm group $= \frac{R}{n} = \frac{1}{3}\Omega$

This is in series with $\frac{2}{3}\Omega$ resistor.

$\therefore \text{Total resistance} = \frac{2}{3} + \frac{1}{3} = \frac{3}{3}\Omega = 1\Omega$

7. (c) Lowest resistance will be in the case when all the resistors are connected in parallel.

$\frac{1}{R} = \frac{1}{0.1} + \frac{1}{0.1} \dots \dots \dots 10 \text{ times}$

$\frac{1}{R} = 10 + 10 \dots \dots \dots 10 \text{ times}$

$\frac{1}{R} = 100 \text{ i.e. } R = \frac{1}{100}\Omega$

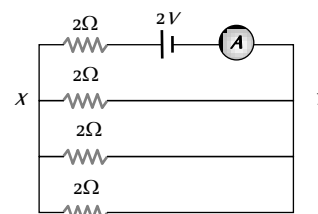
8. (b) Resistance across XY $= \frac{2}{3}\Omega$

Total resistance

$= 2 + \frac{2}{3} = \frac{8}{3}\Omega$

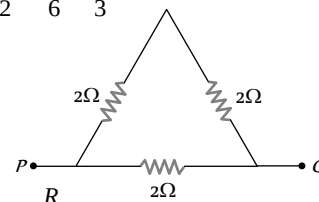
Current through ammeter

$= \frac{2}{8/3} = \frac{6}{8} = \frac{3}{4}A$



9. (a) Equivalent resistance of the combination

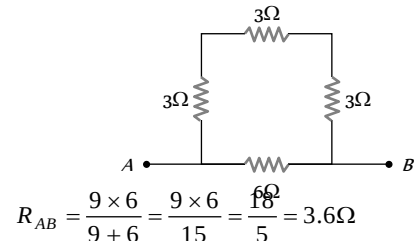
$= \frac{(2+2) \times 2}{2+2+2} = \frac{8}{6} = \frac{4}{3}\Omega$



10. (b) In parallel, $x = \frac{R}{n}$ $R = nx$

In series, $R + R + R \dots n \text{ times} = nR = n(nx) = nx$

11. (d) The circuit reduces to

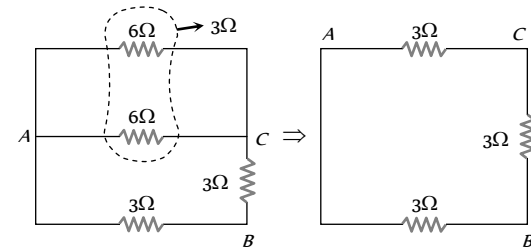


12. (d) As resistance $\propto$ Length

Resistance of each arm $= \frac{12}{3} = 4\Omega$

$\Rightarrow R_{\text{effective}} = \frac{4 \times 8}{4 + 8} = \frac{8}{3}\Omega$

13. (b) Given circuit is equivalent to



So the equivalent resistance between points A and B is equal to

$$R = \frac{6 \times 3}{6 + 3} = 2\Omega$$

14. (d) Potential difference across all resistors in parallel combination is same.

15. (b) Current through each arm DAC and $DBC = 1A$

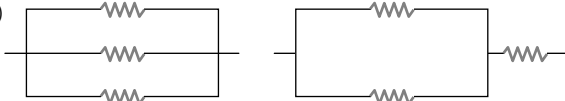
$$V_D - V_A = 2 \text{ and } V_D - V_B = 3 \Rightarrow V_A - V_B = +1V$$

16. (d) $R_{\text{eq}} = r + \frac{3r}{2} = \frac{5r}{2}$

17. (c) If resistances are R_1 and R_2 then $\frac{R_1 R_2}{R_1 + R_2} = \frac{6}{8}$ (i)

Suppose R_2 is broken then $R_1 = 2\Omega$ (ii)

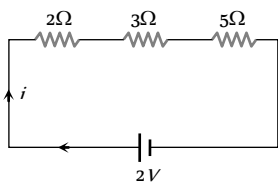
On solving equations (i) and (ii) we get $R_2 = 6/5\Omega$

18. (c) 

19. (b) Because all the lamps have same voltage. 

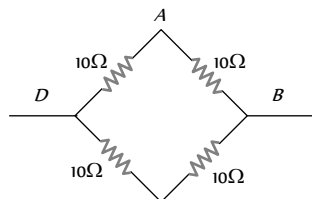
20. (d) $R_{\text{series}} = R_1 + R_2 + R_3 + \dots$

21. (a) Current supplied by cell $i = \frac{2}{2+3+5} = \frac{1}{5} A$



So potential difference across 3 will be $V = \frac{3 \times 1}{5} = 0.6V$

22. (a) According to the problem, we arrange four resistance as follows



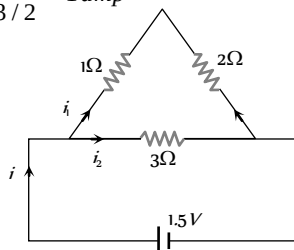
$$\text{Equivalent resistance} = \frac{20 \times 20}{40} = 10\Omega$$

23. (b) $R_1 + R_2 = 9$ and $\frac{R_1 R_2}{R_1 + R_2} = 2 \Rightarrow R_1 R_2 = 18$

$$R_1 - R_2 = \sqrt{(R_1 + R_2)^2 - 4R_1 R_2} = \sqrt{81 - 72} = 3$$

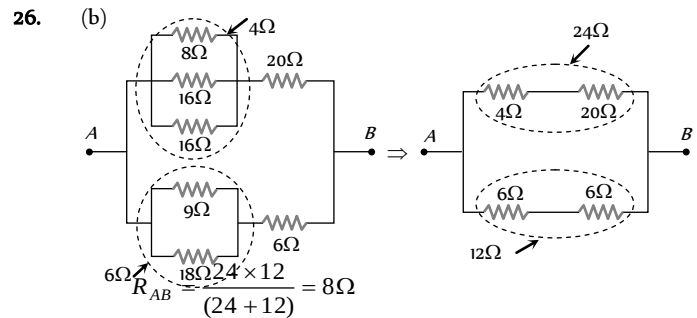
$$R_1 = 6\Omega, R_2 = 3\Omega$$

24. (b) $i_1 + i_2 = \frac{1.5}{3/2} = 1 \text{ amp}$

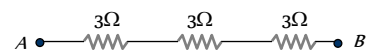


$$\frac{i_1}{i_2} = \frac{3}{3} \Rightarrow i_1 = i_2 \therefore i_2 = 0.5A = i_1$$

25. (c) $V_p - V_q = \left(\frac{6}{3} + \frac{12 \times 6}{12 + 6} \right) (0.5) = (2 + 4)(0.5) = 3V$



27. (d) The network can be redrawn as follows



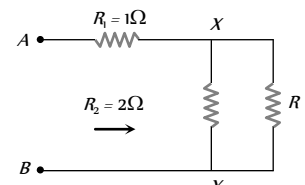
$$\Rightarrow R_{eq} = 9\Omega$$

28. (d) Let the resistance of the wire be R , then we know that resistance is proportional to the length of the wire. So each of the four wires will have $R/4$ resistance and they are connected in parallel. So the effective resistance will be

$$\frac{1}{R_1} = \left(\frac{4}{R} \right) 4 \Rightarrow R_1 = \frac{R}{16}$$

29. (d) Equivalent resistance $= \frac{4 \times 4}{4 + 4} + \frac{6 \times 6}{6 + 6} = 5\Omega$ So the current in the circuit $= \frac{20}{5} = 4 \text{ ampere}$ Hence the current flowing through each resistance $= 2 \text{ ampere}$.

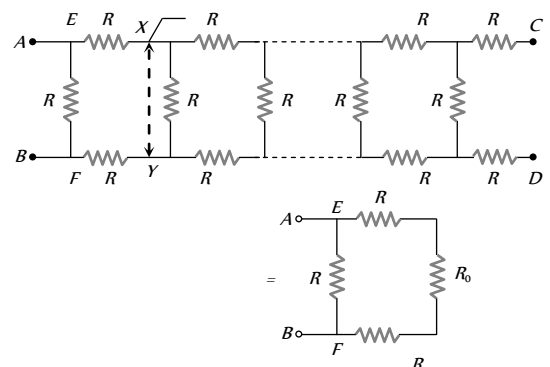
30. (c) Let the resultant resistance be R . If we add one more branch, then the resultant resistance would be the same because this is an infinite sequence.



$$\therefore \frac{R R_2}{R + R_2} + R_1 = R \Rightarrow 2R + R + 2 = R^2 + 2R$$

$$\Rightarrow R^2 - R - 2 = 0 \Rightarrow R = -1 \text{ or } R = 2\Omega$$

31. (b) Cut the series from XY and let the resistance towards right of XY be R_0 whose value should be such that when connected across AB does not change the entire resistance. The combination is reduced to as shown below.



The resistance across $EF, R_{EF} = (R_0 + 2R)$

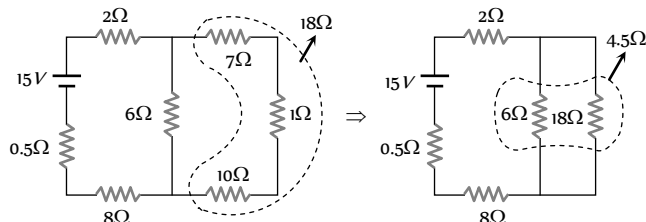
$$\text{Thus } R_{AB} = \frac{(R_0 + 2R)R}{R_0 + 2R + R} = \frac{R_0 R + 2R^2}{R_0 + 3R} = R_0$$

$$\Rightarrow R_0^2 + 2RR_0 - 2R^2 + 0 \Rightarrow R_0 = R(\sqrt{3} - 1)$$

32. (d) The last two resistance are out of circuit. Now 8Ω is in parallel with $(1 + 1 + 4 + 1 + 1)\Omega$.

$$\therefore R = 8\Omega \parallel 8\Omega = \frac{8}{2} = 4\Omega \Rightarrow R_{AB} = 4 + 2 + 2 = 8\Omega$$

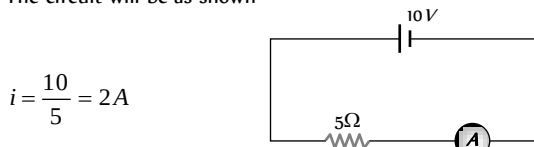
33. (a) The given circuit can be simplified as follows



On further solving equivalent resistance $R = 15\Omega$

$$\text{Hence current from the battery } i = \frac{15}{15} = 1A$$

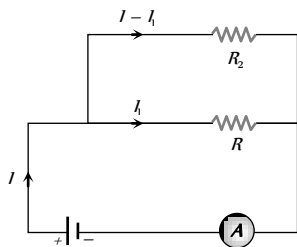
34. (b) The circuit will be as shown



35. (c) The current in the circuit $= \frac{8}{5+1} = \frac{4}{3}$

$$\text{Now } V_C - V_E = \frac{4}{3} \times 1 \Rightarrow V_E = -\frac{4}{3}V$$

36. (d) According to the figure, $(I - I_1)R_2 = I_1 R$

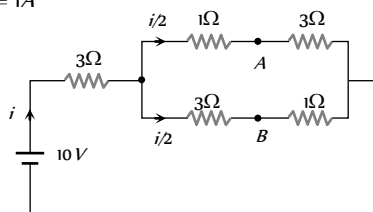


Only two values satisfying the above relation are $\frac{I}{2}$ and R

37. (d) Effective resistance between the points A and B is

$$R = \frac{32}{12} = \frac{8}{3}\Omega$$

38. (b) $R_{eq} = 5\Omega$, Current $i = \frac{10}{5} = 2A$ and current in each branch $= 1A$



Potential difference between C and A,

$$V_C - V_A = 1 \times 1 = 1V \quad \dots\dots(i)$$

Potential difference between C and B,

$$V_C - V_B = 1 \times 3 = 3V \quad \dots\dots(ii)$$

On solving (i) and (ii) $V_A - V_B = 2\text{ volt}$

$$\text{Shot Trick : } (V_A - V_B) = \frac{i}{2}(R_2 - R_1) = \frac{2}{2}(3 - 1) = 2V$$

39. (c) $\frac{1}{R} = \frac{1}{1} + \frac{1}{1} + \frac{1}{1} = \frac{3}{1} \Rightarrow R = \frac{1}{3}\text{ ohm}$

Now such three resistance are joined in series, hence total

$$R = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = 1\text{ohm}$$

40. (b) To obtain minimum resistance, all resistors must be connected in parallel.

$$\text{Hence equivalent resistance of combination} = \frac{r}{10}$$

41. (a) For same material and same length

$$\frac{R_2}{R_1} = \frac{A_1}{A_2} = \frac{3}{2} \Rightarrow R_2 = 3R_1$$

Resistance of thick wire $R_1 = 10\Omega$

$$\therefore \text{Resistance of thin wire } R_2 = 30\Omega$$

Total resistance in series $= 10 + 30 = 40\Omega$

42. (c) Similar to Q. No. 30

$$R = 2 + 2 + \frac{2 \times R}{2 + R} \Rightarrow 2R + R^2 = 8 + 4R + 2R$$

$$\Rightarrow R^2 - 4R - 8 = 0 \Rightarrow R = \frac{4 \pm \sqrt{16 + 32}}{2} = 2 \pm 2\sqrt{3}$$

R cannot be negative, hence $R = 2 + 2\sqrt{3} = 5.46\Omega$

43. (b) P.d. across the circuit $= 1.2 \times \frac{6 \times 4}{6 + 4} = 2.88\text{ volt}$

$$\text{Current through } 6\text{ ohm resistance} = \frac{2.88}{6} = 0.48\text{ A}$$

44. (d) Three resistances are in parallel.

$$\therefore \frac{1}{R'} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R} = \frac{3}{R}$$

$$\text{The equivalent resistance } R' = \frac{R}{3}\Omega$$

45. (c) Similar to Q. No. 30. By formula $R = R_1 + \frac{R_2 \times R}{R_2 + R}$

$$\therefore R = 1 + \frac{1 \times R}{1 + R} \Rightarrow R^2 + R = 1 + R + R$$

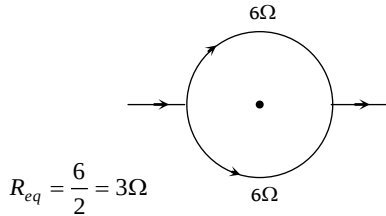
$$\Rightarrow R^2 - R - 1 = 0 \text{ or } R = \frac{1 \pm \sqrt{1 + 4}}{2} = \frac{1 \pm \sqrt{5}}{2}$$

Since R cannot be negative, hence $R = \frac{1 + \sqrt{5}}{2}\Omega$

46. (d) $R \propto l$

Hence every new piece will have a resistance $\frac{R}{10}$. If two pieces are connected in series, then their resistance $= \frac{2R}{10} = \frac{R}{5}$. If 5 such combinations are joined in parallel, then net resistance $= \frac{R}{5 \times 5} = \frac{R}{25}$.

47. (c)

48. (b) Current in the given circuit $i = \frac{50}{(5+7+10+3)} = 2A$

Potential difference between A and B $V_A - V_B = 2 \times 12$
 $\Rightarrow V_A - 0 = 24V \Rightarrow V_A = 24V$

49. (b) If all are in series then $R_{eq} = 12\Omega$

If all are in parallel then $R_{eq} = \frac{4}{3}\Omega = 1.33\Omega$

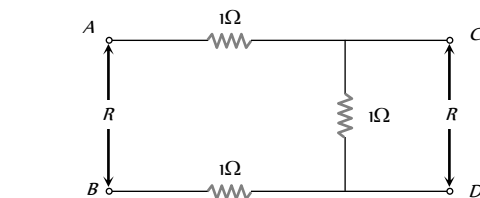
If two are in series then parallel with third, $R_{eq} = \frac{8}{3} = 2.6\Omega$

If two are in parallel then series with third, $R_{eq} = 6\Omega$

50. (d) Equivalent external resistance of the given circuit $R_{eq} = 4\Omega$

Current given by the cell $i = \frac{E}{R_{eq} + r} = \frac{10}{(4+1)} = 2A$

Hence, $(V_A - V_B) = \frac{i}{2} \times (R_2 - R_1) = \frac{2}{2} (2 - 4) = -2V$.

51. (d) Resistance of each part will be $\frac{R}{n}$; such n parts are joined in parallel so $R_{eq} = \frac{R}{n^2}$.52. (c) Let equivalent resistance between A and B be R , then equivalent resistance between C and D will also be R .

$$R' = \frac{R}{R+1} + 2 = R \text{ or } R^2 - 2R - 2 = 0$$

$$\therefore R = \frac{2 \pm \sqrt{4+8}}{2} = \sqrt{3} + 1$$

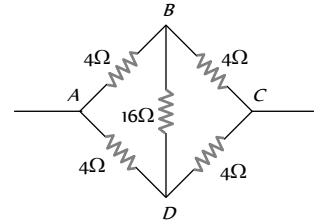
53. (d) 6Ω and 6Ω are in series, so effective resistance is 12Ω which is in parallel with 3Ω , so

$$\frac{1}{R} = \frac{1}{3} + \frac{1}{12} = \frac{15}{36} \Rightarrow R = \frac{36}{15}$$

$$\therefore I = \frac{V}{R} = \frac{4.8 \times 15}{36} = 2A$$

54. (a) Equivalent resistance of the circuit $R = \frac{3}{2}\Omega$

$$\therefore \text{Current through the circuit } i = \frac{V}{R} = \frac{3}{3/2} = 2A$$

55. (c) $R_{\max} = nR$ and $R_{\min} = R/n \Rightarrow \frac{R_{\max}}{R_{\min}} = n^2$ 56. (d) According to the principle of Wheatstone's bridge, the effective resistance between the given points is 4Ω .

57. (c)

58. (c) Current through 6Ω resistance in parallel with 3Ω resistance = $0.4A$

So total current = $0.8 + 0.4 = 1.2A$

Potential drop across $4\Omega = 1.2 \times 4 = 4.8V$

59. (d) Two resistances in series are connected parallel with the third.

$$\text{Hence } \frac{1}{R_p} = \frac{1}{4} + \frac{1}{8} = \frac{3}{8} \Rightarrow R_p = \frac{8}{3}\Omega$$

60. (c) Resistances at C and B are not in the circuit. Use laws of resistances in series and parallel excluding the two resistance.

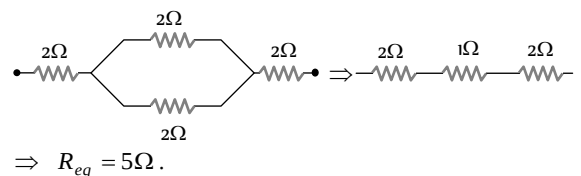
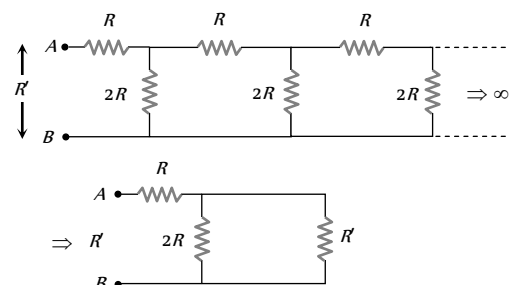
61. (d) After simplifying the network, equivalent resistance obtained between A and B is 8Ω .62. (c) The circuit consists of three resistances ($2R$, $2R$ and R) connected in parallel.

63. (d) Resistance across the battery is

$$\frac{1}{R_p} = \frac{1}{3} + \frac{1}{6} = \frac{2+1}{6} = \frac{3}{6} \Rightarrow R_p = 2\Omega \Rightarrow I = \frac{2}{2} = 1A$$

64. (c) The voltmeter is assumed to have infinite resistance. Hence $(1 + 2 + 1) + 4 = 8\Omega$.65. (c) $R' = \frac{R}{n} = \frac{1}{10} = 0.1\Omega$

66. (c) The given circuit can be redrawn as follows

67. (b) $R_{AB} = R_1 + \frac{R_2 R_3}{R_2 + R_3} + R_4 = 2 + \frac{4 \times 4}{4 + 4} + 2 = 6\Omega$.68. (c) Let equivalent resistance between A and B is R , so given circuit can be reduced as follows

$$R' = R + \frac{2R \times R'}{2R + R'} \Rightarrow R'^2 - RR' - 2R^2 = 0$$

On solving the equation we get $R' = 2R$.

69. (d) $R_{AB} = \frac{R}{3} + R = \frac{2}{3} + 2 = \frac{8}{3} = 2\frac{2}{3} \Omega$.

70. (b) $i = \frac{E}{R+r} \Rightarrow 0.5 = \frac{10}{R+3} \Rightarrow 10 = 0.5R + 1.5 \Rightarrow R = 17 \Omega$.

71. (a) Equivalent resistance $R = 4 + \frac{3 \times 6}{3+6} = 6 \Omega$ and main current

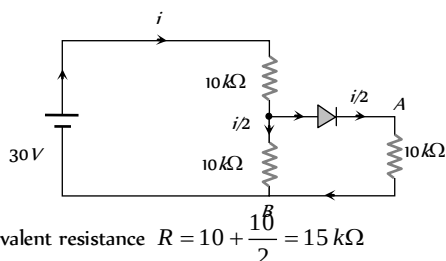
$$i = \frac{E}{R} = \frac{3}{6} = 0.5 A$$

Now potential difference across the combination of 3Ω and

$$6 \Omega, V = 0.5 \times \left(\frac{3 \times 6}{3+6} \right) = 1 \text{ Volt}$$

The same potential difference, also develops across 3Ω resistance.

72. (c)



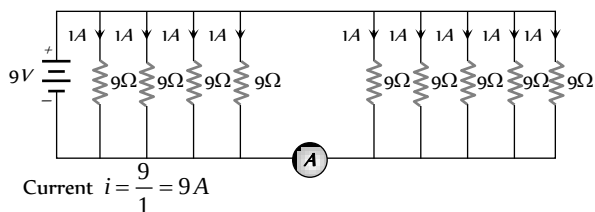
Equivalent resistance $R = 10 + \frac{10}{2} = 15 \text{ k}\Omega$

Current $i = \frac{30}{15} = 2 \times 10^{-3} A$

Hence, potential difference between A and B

$$V = \left(\frac{2 \times 10^{-3}}{2} \right) \times 10 \times 10^3 = 10 \text{ Volt}$$

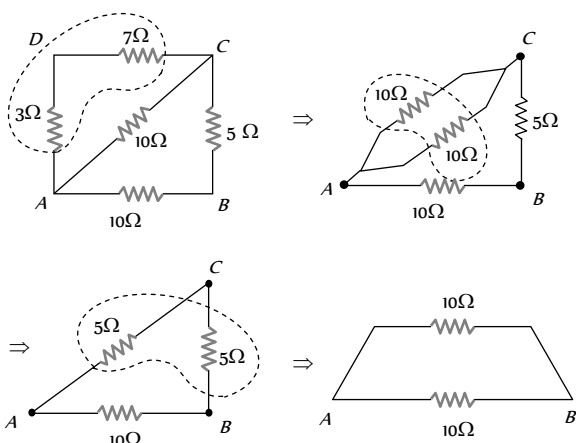
73. (a) Equivalent resistance $R = \frac{9}{9} = 1 \Omega$



Current $i = \frac{9}{1} = 9 A$

Current passes through the ammeter = $5 A$.

74. (b) The figure can be drawn as follows

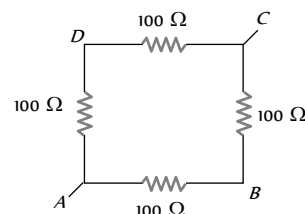


$$\Rightarrow R_{AB} = 5 \Omega.$$

75. (a) $R_1 = \frac{\rho_1 l_1}{A}$ and $R_2 = \frac{\rho_2 l_2}{A}$. In series $R_{eq} = R_1 + R_2$

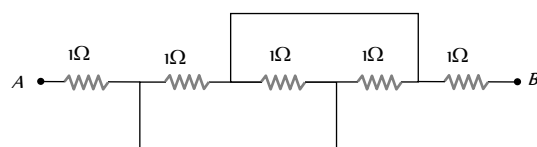
$$\frac{\rho_{eq}(l_1 + l_2)}{A} = \frac{\rho_1 l_1}{A} + \frac{\rho_2 l_2}{A} \Rightarrow \rho_{eq} = \frac{\rho_1 l_1 + \rho_2 l_2}{l_1 + l_2}.$$

76. (c) The figure can be drawn as follows



$$R_{AC} = \frac{200 \times 200}{200 + 200} = 100 \Omega.$$

77. (c)



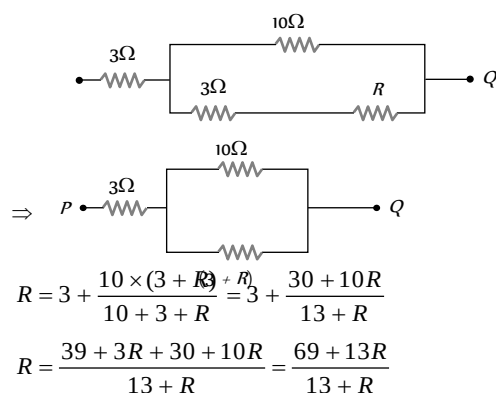
$$R_{AB} = 2 + \frac{1}{3} = 2\frac{1}{3} \Omega.$$

78. (b) ρ - same, l - same, $A_2 = \frac{1}{4} A_1$ (as $r_2 = \frac{r_1}{2}$)

By using $R = \rho \frac{l}{A} \Rightarrow \frac{R_1}{R_2} = \frac{A_2}{A_1} \Rightarrow \frac{R_1}{8} = \frac{1}{4} \Rightarrow R_1 = 2 \Omega$

Hence, $R_{eq} = \frac{R_1 R_2}{R_1 + R_2} = \frac{2 \times 8}{2 + 8} = \frac{8}{5} \Omega$.

79. (c) The given circuit can be simplified as follows

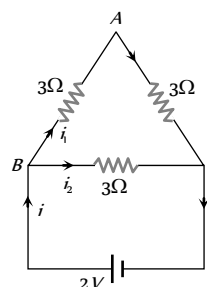


$$R = 3 + \frac{10 \times (3 + R)}{10 + 3 + R} = 3 + \frac{30 + 10R}{13 + R}$$

$$R = \frac{39 + 3R + 30 + 10R}{13 + R} = \frac{69 + 13R}{13 + R}$$

$$13R + R^2 = 69 + 13R \Rightarrow R = \sqrt{69} \Omega.$$

80. (a) The circuit can be drawn as follows



$$\text{Equivalent resistance } R = \frac{3 \times (3 + 3)}{3 + (3 + 3)} = 2\Omega$$

$$\text{Current } i = \frac{2}{2} = 1A. \text{ So, } i_1 = 1 \times \left(\frac{3}{3 + 6} \right) = \frac{1}{3} A.$$

$$\text{Potential difference between } A \text{ and } B = \frac{1}{3} \times 3 = 1\text{volt}.$$

$$81. (a) \frac{1}{R_{eq}} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{4 + 2 + 1}{8} \Rightarrow R_{eq} = \frac{8}{7} \Omega.$$

82. (b) The given figure is balance wheat stone bridge.

$$83. (b) \frac{7}{12} = \frac{1}{4} + \frac{1}{R} \Rightarrow R = 3\Omega$$

$$84. (d) \text{ Suppose resistance of wires are } R_1 \text{ and } R_2 \text{ then}$$

$$\frac{6}{5} = \frac{R_1 R_2}{R_1 + R_2}. \text{ If } R_2 \text{ breaks then } R_1 = 2\Omega$$

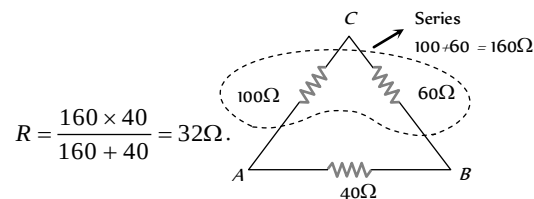
$$\text{Hence, } \frac{6}{5} = \frac{2 \times R_2}{2 + R_2} \Rightarrow R_2 = 3\Omega.$$

85. (d) Potential difference across PQ i.e. p.d. across the resistance of 20Ω , which is $V = i \times 20$

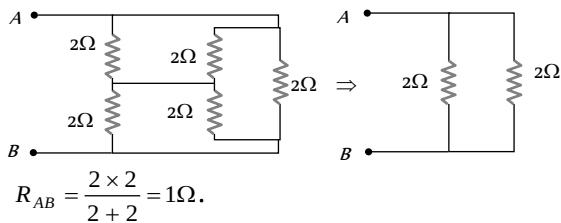
$$\text{and } i = \frac{48}{(100 + 100 + 80 + 20)} = 0.16A$$

$$\therefore V = 0.16 \times 20 = 3.2V.$$

86. (a)



87. (a)



88. (a) Given circuit is a balance Wheatstone bridge circuit.

$$89. (b) \text{ All of three resistance are in parallel So, } R' = R/n = \frac{R}{3}.$$

$$90. (b) R_{eq} = R_1 + R_2 \Rightarrow \frac{\rho_{eff} 2l}{A} = \frac{\rho_1 l}{A} + \frac{\rho_2 l}{A} \Rightarrow \rho_{eff} = \frac{\rho_1 + \rho_2}{2}.$$

91. (b) Two resistance are in ratio 1 : 2 and third resistance is R

$$\text{So, } \frac{1}{x} + \frac{1}{2x} + \frac{1}{R} = 1 \Rightarrow x = \frac{3}{2} \left(\frac{R}{R-1} \right)$$

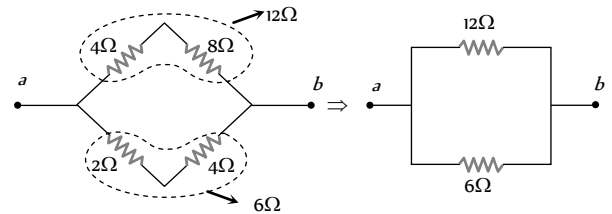
$$\text{As, resistance is not fractional } \Rightarrow \frac{R}{R-1} = 2$$

$$\Rightarrow x = 3, R = 2, 2x = 6$$

Hence, the value of largest resistance = 6Ω .

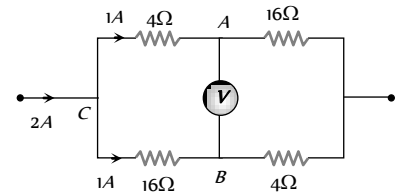
$$92. (c) R = \frac{(3 + 3) \times 3}{(3 + 3) + 3} = 2\Omega \Rightarrow i = \frac{3}{2} = 1.5A.$$

93. (b) Given circuit is a balanced Wheatstone bridge circuit, hence it can be redrawn as follows



94. (d) The given circuit is a balanced wheatstone bridge circuit. Hence potential difference between A and B is zero.

95. (a) In the following circuit potential difference between C and A is $V_C - V_A = 1 \times 4 = 4$ (i)



$$C \text{ and } B \text{ is } V_C - V_B = 1 \times 16 = 16 \text{(ii)}$$

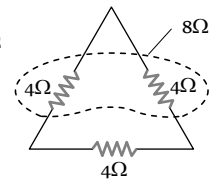
On solving equations (i) and (ii) we get

$$V_A - V_B = 12V.$$

96. (d) As resistance $\propto$ Length

$$\therefore \text{Resistance of each arm} = \frac{12}{3} = 4\Omega$$

$$\therefore R_{effective} = \frac{4 \times 8}{4 + 8} = \frac{8}{3} \Omega$$



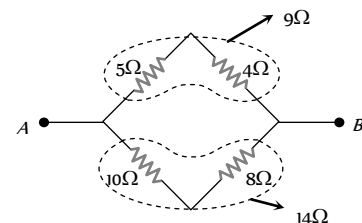
$$97. (b) i = \frac{12}{(1 + 1) + 0.4} = 5A.$$

$$98. (b) \text{ By balanced Wheatstone bridge condition } \frac{16}{X} = \frac{4}{0.5}$$

$$\Rightarrow X = \frac{8}{4} = 2\Omega$$

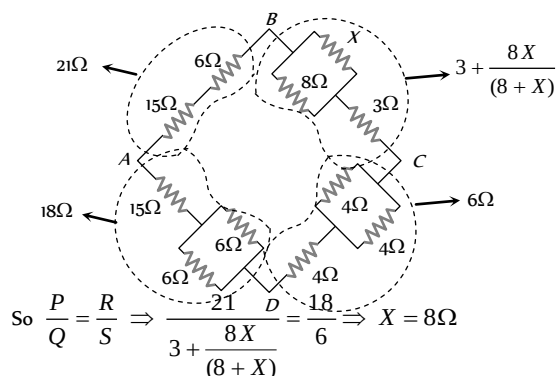
$$99. (d) \text{ Current through } 2\Omega = 1.4 \left\{ \frac{(25 + 5)}{(10 + 2) + (25 + 5)} \right\} = 1A$$

100. (a) Since the given bridge is balanced, hence there will be no current through 9Ω resistance. This resistance has no effect and must be ignored in the calculations.



$$R_{AB} = \frac{9 \times 18}{27} = 6 \Omega$$

101. (c) Potential difference between B and D is zero, it means Wheatstone bridge is in balanced condition

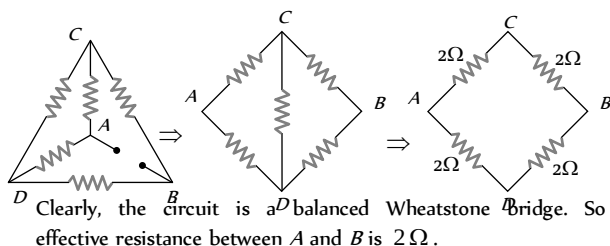


102. (a) This is a balanced Wheatstone bridge. Therefore no current will flow from the diagonal resistance 10Ω

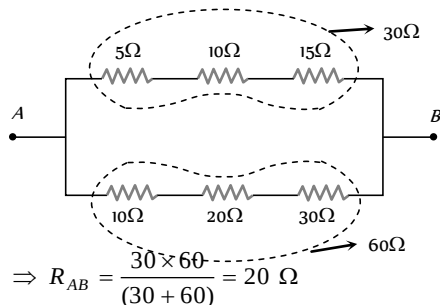
$$\therefore \text{Equivalent resistance} = \frac{(10+10) \times (10+10)}{(10+10) + (10+10)} = 10 \Omega$$

103. (b) This is a balanced Wheatstone bridge circuit. So potential at B and D will be same and no current flows through $4R$ resistance.

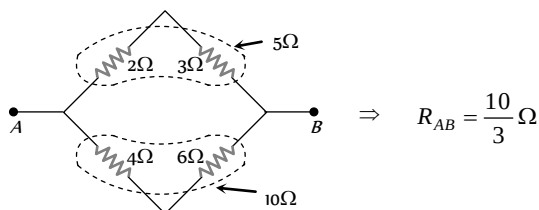
104. (d) The equivalent circuits are as shown below



105. (a) By the concept of balanced Wheatstone bridge, the given circuit can be redrawn as follows



106. (a) The given circuit is a balanced Wheatstone bridge type, hence it can be simplified as follows



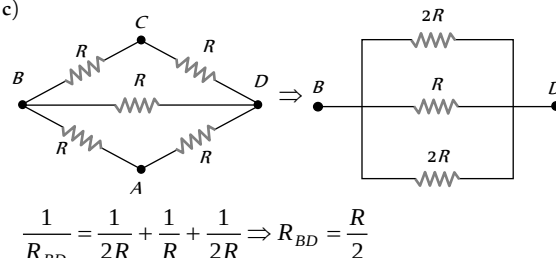
107. (b) Let current through 5Ω resistance be i . Then

$$i \times 25 = (2.1 - i)10 \Rightarrow i = \frac{10}{35} \times 2.1 = 0.6 A$$

108. (d) Let the value of shunt be r . Hence the equivalent resistance of branch containing S will be $\frac{Sr}{S+r}$

$$\text{In balance condition, } \frac{P}{Q} = \frac{Sr/(S+r)}{R}. \text{ This gives } r = 8 \Omega$$

109. (b, c)



Between A and C circuit becomes equivalent to balanced Wheatstone bridge so $R_{AC} = R$.

110. (b) $i \propto \frac{1}{R}$

111. (d) Equivalent resistance between P and Q

$$\frac{1}{R_{PQ}} = \frac{1}{(6+2)} + \frac{1}{3} + \frac{1}{(4+12)} \Rightarrow R_{PQ} = \frac{48}{25}$$

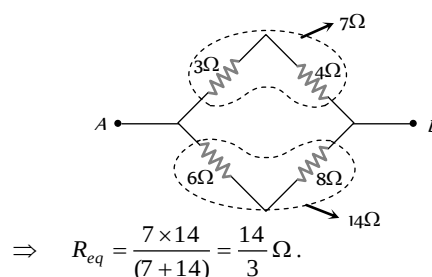
Current between P and Q ; $i = 1.5 A$

So, potential difference between P and Q

$$V_{PQ} = 1.5 \times \frac{48}{25} = 2.88 V.$$

112. (c) Given circuit is a balanced Wheatstone bridge i.e. potential difference between B and D is zero. Hence, no current flows between B and D .

113. (a) The given circuit is a balanced Wheatstone bridge, hence it can be redrawn as follows



114. (a) For a balance Wheatstone bridge.

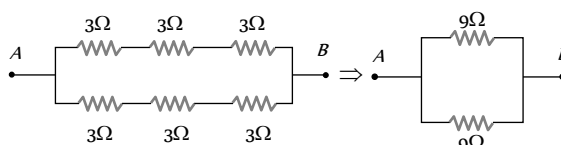
$$\frac{A}{B} = \frac{D}{C} \Rightarrow \frac{10}{5} \neq \frac{4}{4} \text{ (Unbalanced)}$$

$$\frac{A'}{B} = \frac{D}{C} \Rightarrow \frac{A'}{5} = \frac{4}{4} \Rightarrow A' = 5 \Omega$$

$A' (5 \Omega)$ is obtained by connecting a 10Ω resistance in parallel with A .

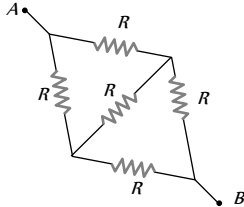
115. (d) Given circuit is a balanced Wheatstone bridge circuit. So there will be no change in equivalent resistance. Hence no further current will be drawn.

116. (a) No current flow through vertical resistances



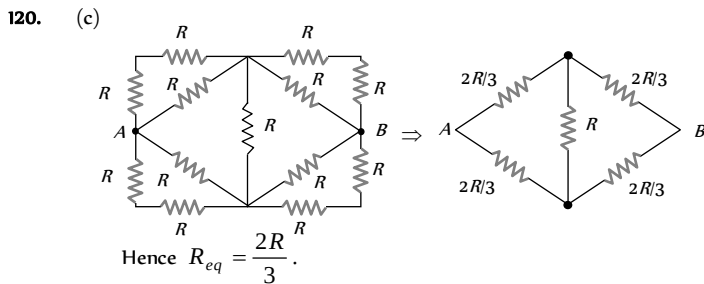
$$R_{AB} = \frac{9}{2} \Omega.$$

117. (d) The given circuit is a balanced Wheatstone bridge.
118. (c) The given circuit can be redrawn as follows



Equivalent resistance between A to B is R .

119. (d) Equivalent resistance of the given circuit is 3Ω .



121. (b)

122. (b) For balanced Wheatstone bridge $\frac{P}{Q} = \frac{R}{S}$

$$\Rightarrow \frac{12}{(1/2)} = \frac{x+6}{(1/2)} \Rightarrow x = 6\Omega$$

123. (b) For maximum energy equivalent resistance of combination should be minimum.

124. (c) For first balancing condition $\frac{10 + R_1}{R_2} = \frac{50}{50}$

$$\Rightarrow R_2 = 10 + R_1. \text{ For second balancing condition}$$

$$\frac{R_1}{R_2} = \frac{40}{60} \Rightarrow \frac{R_1}{10 + R_1} = \frac{2}{3} \Rightarrow R_1 = 20\Omega$$

125. (b) Given $R = 6\Omega$. When resistor is cut into two equal parts and connected in parallel, then

$$R_{eq} = \frac{R/2}{2} = \frac{R}{4} = \frac{6}{4} = 1.5\Omega$$

126. (a) Resistance between P and Q

$$R_{PQ} = R \parallel \left(\frac{R}{3} + \frac{R}{2} \right) = \frac{R \times \frac{5}{6} R}{R + \frac{5}{6} R} = \frac{5}{11} R$$

Resistance between Q and R

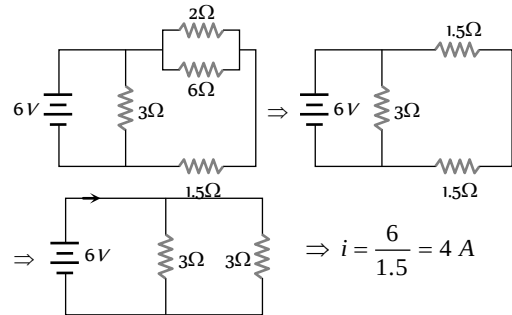
$$R_{QR} = \frac{R}{2} \parallel \left(R + \frac{R}{3} \right) = \frac{\frac{R}{2} \times \frac{4}{3} R}{\frac{R}{2} + \frac{4}{3} R} = \frac{4}{11} R$$

Resistance between P and R

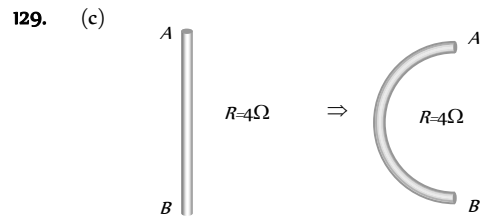
$$R_{PR} = \frac{R}{3} \parallel \left(\frac{R}{2} + R \right) = \frac{\frac{R}{3} \times \frac{3}{2} R}{\frac{R}{3} + \frac{3}{2} R} = \frac{3}{11} R$$

Hence it is clear that R_{PQ} is maximum.

127. (c) Given circuit can be redrawn as follows

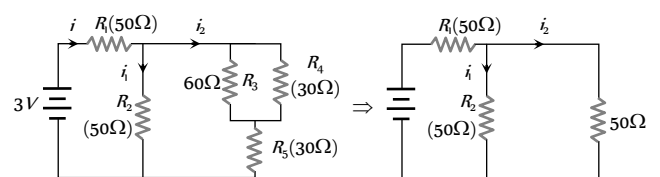


128. (b) $\frac{i_1}{i_2} = \frac{R_2}{R_1} = \frac{l_2}{l_1} \times \left(\frac{r_1}{r_2} \right)^2 = \frac{3}{4} \left(\frac{2}{3} \right)^2 = \frac{1}{3}$



130. (a) $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{2} + \frac{1}{4} + \frac{1}{5} = \frac{19}{20} \Rightarrow R_{eq} = \frac{20}{19} \Omega$

131. (a) Equivalent resistance of the given network $R_{eq} = 75\Omega$



$$\therefore \text{Total current through battery } i = \frac{3}{75}$$

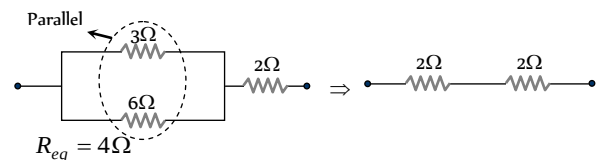
$$i_1 = i_2 = \frac{3}{75 \times 2} = \frac{3}{150}$$

$$\text{Current through } R_4 = \frac{3}{150} \times \frac{60}{(30+60)} = \frac{3}{150} \times \frac{60}{90} = \frac{2}{150} \text{ A}$$

$$V_4 = i_4 \times R_4 = \frac{2}{150} \times 30 = \frac{2}{5} \text{ V} = 0.4 \text{ V}$$

132. (a) $i = \frac{10}{1.5 + (1 \parallel 1)} = \frac{10}{1.5 + 0.5} = 5\text{ A}$

133. (c)



134. (a) The equivalent resistance between C and D is

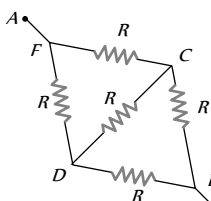
$$\frac{1}{R'} = \frac{1}{6} + \frac{1}{6} + \frac{1}{3} = \frac{2}{3} \quad \text{or} \quad R' = \frac{3}{2} = 1.5 \Omega$$

Now the equivalent resistance between A and B as $R' = 1.5 \Omega$ and 2.5Ω are connected in series, so

$$R'' = 1.5 + 2.5 = 4 \Omega$$

Now by ohm's law, potential difference between A and B is given by $V_A - V_B = iR = 2 \times 4.0 = 8 \text{ volt}$

135. (b) The given circuit can be redrawn as follows



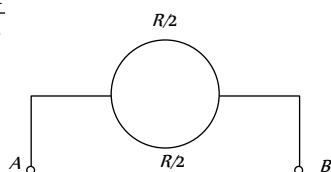
Equivalent resistance between A and B is R and

$$\text{current } i = \frac{V}{R}$$

136. (b) The given network is a balanced Wheatstone bridge. It's equivalent resistance will be $R = \frac{18}{5} \Omega$

$$\text{So current from the battery } i = \frac{V}{R} = \frac{V}{18/5} = \frac{5V}{18}$$

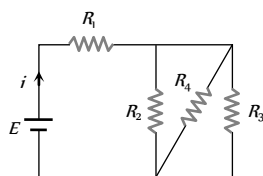
137. (a) $\Rightarrow R_{AB} = \frac{R/2}{2} = \frac{R}{4}$



138. (b) $i \propto \frac{1}{R} \Rightarrow \frac{i_1}{i_2} = \frac{R_2}{R_1} \Rightarrow \frac{5}{4} = \frac{(R+2)}{R} \Rightarrow R = 8 \Omega$

139. (c) In given circuit three resistance R_2 , R_4 and R_3 are parallel.

$$\begin{aligned} \frac{1}{R} &= \frac{1}{R_2} + \frac{1}{R_4} + \frac{1}{R_3} \\ &= \frac{1}{50} + \frac{1}{50} + \frac{1}{75} \\ &= \frac{75+75+50}{50 \times 75} \end{aligned}$$



$$R = \frac{50 \times 75}{75 + 75 + 50} = \frac{50 \times 75}{200} = \frac{75}{4} \Omega = 18.75 \Omega$$

This resistance is in series with R_1

$\therefore$

$$R_{\text{resultant}} = R_1 + R = 100 + 18.75 = 118.75 \Omega$$

140. (b) When resistances 4Ω and 12Ω are connected in series $= 4 + 12 = 16 \Omega$

When these resistance are connected in parallel.

$$\frac{1}{R_p} = \frac{1}{4} + \frac{1}{12} \Rightarrow R_p = \frac{4 \times 12}{4 + 12} = \frac{4 \times 12}{16} = 3 \Omega$$

141. (b) Since voltmeter records 5 V , it means the equivalent. Resistance of voltmeter and 100Ω must be 50 , because in series grouping if resistances are equal, they share equal potential difference. It conclude that resistance of voltmeter must be 100Ω .

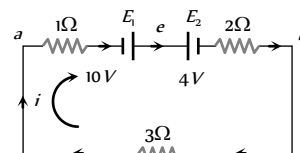
Kirchoff's Law, Cells

1. (b) For no current through galvanometer, we have

$$\left(\frac{E_1}{500 + X} \right) X = E \Rightarrow \left(\frac{12}{500 + X} \right) X = 2 \Rightarrow X = 100 \Omega$$

2. (d) Since $E_1(10 \text{ V}) > E_2(4 \text{ V})$

So current in the circuit will be clockwise.



Applying Kirchoff's voltage law

$$-1 \times i + 10 - 4 - 2 \times i - 3i = 0 \Rightarrow i = 1 \text{ A (a to b via e)}$$

$$\therefore \text{Current} = \frac{V}{R} = \frac{10 - 4}{6} = 1.0 \text{ ampere}$$

3. (c) For maximum power, external resistance = internal resistance.

4. (a) $0.9(2 + r) = 0.3(7 + r) \Rightarrow 6 + 3r = 7 + r \Rightarrow r = 0.5 \Omega$

5. (a) Since both the resistors are same, therefore potential difference $= V + V = E \Rightarrow V = \frac{E}{2}$

6. (b) Let the current in the circuit $= i = \frac{V}{R}$

$$\text{Across the cell, } E = V + ir \Rightarrow r = \frac{E - V}{i} = \frac{E - V}{V/R} = \left(\frac{E - V}{V} \right) R$$

7. (a) For maximum energy, we have
External resistance of the circuit

$$= \text{Equivalent internal resistance of the circuit i.e. } R = \frac{r}{2}$$

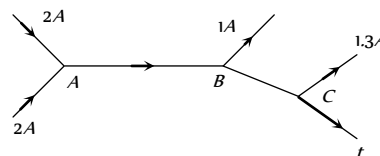
8. (a) Kirchoff's first law is based on the law of conservation of charge.

9. (b) Kirchoff's second law is based on the law of conservation of energy.

10. (a) According to Kirchoff's first law

$$\text{At junction A, } i_{AB} = 2 + 2 = 4 \text{ A}$$

$$\text{At junction B, } i_{AB} = i_{BC} - 1 = 3 \text{ A}$$



$$\text{At junction C, } i = i_{BC} - 1.3 = 3 - 1.3 = 1.7 \text{ amp}$$

11. (c) In charging $V > E$

12. (d) In open circuit of a cell $V = E$

13. (a) Zero (Circuit open means no current and hence no potential difference across resistance).

14. (d) Zero (No potential difference across voltmeter).

15. (b) Let the e.m.f. of cell be E and internal resistance be r . Then
 $0.5 = \frac{E}{(r+2)}$ and $0.25 = \frac{E}{(r+5)}$

$$\text{On dividing, } 2 = \frac{5+r}{2+r} \Rightarrow r = 1\Omega$$

16. (c) In short circuiting $R = 0$, so $V = 0$

17. (c) Short circuit current $i_{SC} = \frac{E}{r} \Rightarrow 3 = \frac{1.5}{r} \Rightarrow r = 0.5\Omega$

18. (c) $i = \frac{50}{R+r} \Rightarrow r = \frac{50}{4.5} - 10 = \frac{5}{4.5} = 1.1\Omega$

19. (d) $(4+r)i = 2.2$ (i)

$$\text{and } 4i = 2 \Rightarrow i = \frac{1}{2}$$

Putting the value of i in (i), we get $r = 0.4 \text{ ohm}$.

20. (b) Let the internal resistance of cell be r , then

$$i = \frac{E}{R+r} \Rightarrow 15 = \frac{1.5}{0.04+r} \Rightarrow r = 0.06\Omega$$

21. (c) The voltage across cell terminal will be given by

$$= \frac{E}{R+r} \times R = \frac{2}{(3.9+0.1)} \times 3.9 = 1.95V$$

22. (c) $E = 2.2 \text{ volt}$, $V = 1.8 \text{ volt}$, $R = 5R$

$$r = \left(\frac{E}{V} - 1 \right) R = \left(\frac{2.2}{1.8} - 1 \right) \times 5 = 1.1\Omega$$

23. (b) In parallel, equivalent resistance is low $\left(i = \frac{E}{R + \frac{r}{n}} \right)$

24. (d) Internal resistance $\propto$ distance $\propto \frac{1}{\text{Area}} \propto$ concentration

25. (a) Total e.m.f. = nE , Total resistance $R + nr \Rightarrow i = \frac{nE}{R + nr}$

26. (a) Current through R is maximum when total internal resistance of the circuit is equal to external resistance.

27. (b) Cells are joined in parallel when internal resistance is higher than a external resistance. ($R \ll r$)

$$i = \frac{E}{R + \frac{r}{n}}$$

28. (b) In series, $i_1 = \frac{2E}{2+2r}$

$$\text{In parallel, } i_2 = \frac{E}{2 + \frac{r}{2}} = \frac{2E}{4+r}$$

$$\text{Since } i_1 = i_2 \Rightarrow \frac{2E}{4+r} = \frac{2E}{2+2r} \Rightarrow r = 2\Omega$$

29. (a) Applying Kirchhoff law

$$(2+2) = (0.1+0.3+0.2)i \Rightarrow i = \frac{20}{3} A$$

Hence potential difference across A

$$= 2 - 0.1 \times \frac{20}{3} = \frac{4}{3} V \quad (\text{less than } 2V)$$

$$\text{Potential difference across } B = 2 - 0.3 \times \frac{20}{3} = 0$$

30. (b) Here two cells are in series.

Therefore total emf = $2E$

Total resistance = $R + 2r$

$$\therefore i = \frac{2E}{R+2r} = \frac{2 \times 1.45}{1.5 + 2 \times 0.15} = \frac{2.9}{1.8} = \frac{29}{18} = 1.611 \text{ amp}$$

31. (a) $E = V + ir$

$$\text{After short-circuiting, } V = 0; \Rightarrow r = \frac{E}{i} = \frac{2}{4} = 0.5\Omega$$

32. (c) By Kirchhoff's current law.

33. (b) For power to be maximum

External resistance = Equivalent internal resistance of the circuit

34. (a) $i = \frac{E}{r} = \frac{1.5}{0.05} = 30A$

35. (a) $i = \frac{12}{(4+2)} = 2A$

$$\text{Energy loss inside the source} = i^2 r = (2)^2 \times 2 = 8W$$

36. (b) $V_2 - V_1 = E - ir = 5 - 2 \times 0.5 = 4 \text{ volt}$

$$\Rightarrow V_2 = 4 + V_1 = 4 + 10 = 14 \text{ volt}$$

37. (a) If m = Number of rows
and n = Number of cells in a row

$$\text{Then } m \times n = 100$$

.....(i)

$$\text{Also condition of maximum current is } R = \frac{nr}{m}$$

$$\Rightarrow 25 = \frac{1 \times n}{m} \Rightarrow n = 25 \text{ } m$$

.....(ii)

On solving (i) and (ii) $m = 2$

38. (b) According to Kirchhoff's law $i_{CD} = i_2 + i_3$

39. (b) Since $i = \left(\frac{E}{R+r} \right)$, we get

$$0.5 = \frac{E}{2+r} \quad \text{.....(i)}$$

$$0.25 = \frac{E}{5+r} \quad \text{.....(ii)}$$

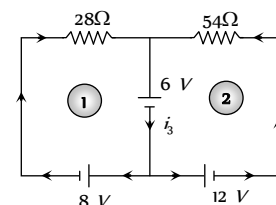
$$\text{Dividing (i) by (ii), we get } 2 = \frac{5+r}{2+r} \Rightarrow r = 1\Omega$$

$$\therefore 0.5 = \frac{E}{2+1} \Rightarrow E = 1.5V$$

40. (c) Because $E_{eq} = E$ and $r_{eq} = \frac{r}{2}$

41. (d) In parallel combination $E_{eq} = E = 6V$

42. (d) Suppose current through different paths of the circuit is as follows.



After applying KVL for loop (1) and loop (2)

$$\text{We get } 28i_1 = -6 - 8 \Rightarrow i_1 = -\frac{1}{2} \text{ A}$$

$$\text{and } 54i_2 = -6 - 12 \Rightarrow i_2 = -\frac{1}{3} \text{ A}$$

$$\text{Hence } i_3 = i_1 + i_2 = -\frac{5}{6} \text{ A}$$

43. (d) $V_{AB} = 4 = \frac{5X + 2 \times 10}{X + 10} \Rightarrow X = 20 \Omega$

44. (a) After short circuiting, R_1 becomes meaningless.

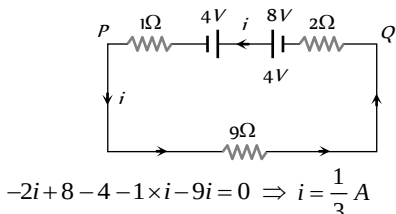
45. (c) $V = E - IR = 15 - 10 \times 0.05 = 14.5 \text{ V}$

46. (c) In series $i = \frac{nE}{nr + R} \Rightarrow 0.6 = \frac{n \times 1.5}{n \times 0.5 \times 20} \Rightarrow n = 10$

47. (b)

48. (a) $P = \frac{W}{t} = Vi \Rightarrow V = \frac{W}{it} = \frac{1000}{2 \times 6 \times 60} = 1.38 \text{ V}$

49. (a) Applying Kirchhoff's voltage law in the given loop.



$$-2i + 8 - 4 - 1 \times i - 9i = 0 \Rightarrow i = \frac{1}{3} \text{ A}$$

$$\text{Potential difference across } PQ = \frac{1}{3} \times 9 = 3 \text{ V}$$

50. (d) Because cell is in open circuit.

51. (b) In parallel combination $E_{eq} = E = 12 \text{ V}$

52. (d)

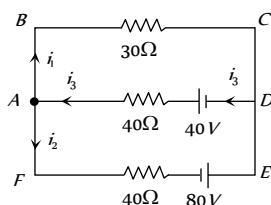
53. (b) $i = \frac{E}{r} = \frac{6}{0.5} = 12 \text{ amp.}$

54. (c) Strength = $5 \times 18 = 90 \text{ AH.}$

55. (a) $i = \frac{E}{R + r} = \frac{5}{4.5 + 0.5} = 1 \text{ A}$

$$V = E - ir = 5 - 1 \times 0.5 = 4.5 \text{ Volt}$$

56. (b) The circuit can be simplified as follows



Applying KCL at junction A

$$i_3 = i_1 + i_2 \quad \dots(i)$$

Applying Kirchhoff's voltage law for the loop ABCDA

$$-30i_1 - 40i_3 + 40 = 0$$

$$\Rightarrow -30i_1 - 40(i_1 + i_2) + 40 = 0$$

$$\Rightarrow 7i_1 + 4i_2 = 4 \quad \dots(ii)$$

Applying Kirchhoff's voltage law for the loop ADEFA.

$$-40i_2 - 40i_3 + 80 + 40 = 0$$

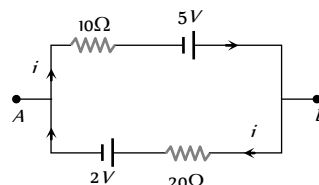
$$\Rightarrow -40i_2 - 40(i_1 + i_2) = -120$$

$$\Rightarrow i_1 + 2i_2 = 3 \quad \dots(iii)$$

On solving equation (ii) and (iii) $i_1 = -0.4 \text{ A.}$

57. (c) $V = E - ir = 12 - 60 \times 5 \times 10^{-2} = 9 \text{ V.}$

58. (a) Applying Kirchhoff's voltage law in the loop



$$-10i + 5 - 20i - 2 = 0 \Rightarrow i = 0.1 \text{ A}$$

59. (d) $V = E - ir = 1.5 - 2 \times 0.15 = 1.20 \text{ Volt.}$

60. (b) $i = \frac{E}{R + r} \Rightarrow 1 = \frac{4}{2 + r} = r = 2 \Omega$

Short circuit when terminals of battery connected directly then

$$\text{current flows which is } i_{SC} = \frac{E}{r} = \frac{4}{2} = 2 \text{ A.}$$

61. (c) $i = \frac{2 + 2}{1 + 1.9 + 0.9} = \frac{4}{3.8} \text{ A}$

$$\text{For cell A } E = V + ir \Rightarrow V = 2 - \frac{4}{3.8} \times 1.9 = 0.$$

62. (c) By using $i = \frac{E}{R + r}$

$$\Rightarrow 0.5 = \frac{E}{11 + r} \Rightarrow E = 5.5 + 0.5r \quad \dots(i)$$

$$\text{and } 0.9 = \frac{E}{5 + r} \Rightarrow E = 4.5 + 0.9r \quad \dots(ii)$$

On solving these equation, we have $r = 2.5 \Omega$

63. (c)

64. (b) $W = qV = 6 \times 10^{-6} \times 9 = 54 \times 10^{-6} \text{ J.}$

65. (a) $P = \frac{V^2}{R_{eq}}$; for P to be maximum R_{eq} should be less. Hence option (a) is correct.

66. (c) $P_{\max} = \frac{E^2}{4r} = \frac{(2)^2}{4 \times 0.5} = 2 \text{ W}$

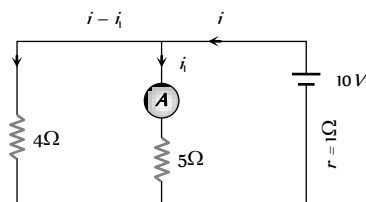
67. (a)

68. (d) Applying Kirchhoff law in the first mesh

$$10 = 5 \times i \Rightarrow i = \frac{10}{5} = 2 \text{ A}$$

69. (b) Applying Kirchhoff law in the first mesh

$$10 = 5i_1 + i \quad \dots(i)$$



Applying in the second mesh

$$5i_1 = 4i - 4i_1 \quad \dots(ii)$$

Solving equation (i) and (ii), we get $i_1 = \frac{40}{29} A$

70. (a) Given problem is the case of mixed grouping of cells

$$\text{So total current produced } i = \frac{nE}{R + \frac{nr}{m}}$$

Here $m = 100$, $n = 5000$, $R = 500 \Omega$

$$E = 0.15 V \text{ and } r = 0.25 \Omega$$

$$\Rightarrow i = \frac{5000 \times 0.15}{500 + \frac{5000 \times 0.25}{100}} = \frac{750}{512.5} \approx 1.5 A$$

71. (a)

72. (d) Watt hour efficiency = $\frac{\text{Discharging energy}}{\text{Charging energy}}$

$$= \frac{14 \times 5 \times 15}{15 \times 8 \times 10} = 0.875 = 87.5\%$$

73. (c) From Kirchoff's junction Law

$$\Rightarrow 4 + 2 + i - 5 - 3 = 0 \Rightarrow i = 2 A$$

74. (b) In the given case cell is in open circuit ($i = 0$) so voltage across the cell is equal to its e.m.f.

75. (b) The internal resistance of battery is given by

$$r = \left(\frac{E}{V} - 1 \right) R = \left(\frac{40}{30} - 1 \right) \times 9 = \frac{9 \times 10}{30} = 3 \Omega$$

76. (b) $i = \frac{E}{r + R} \Rightarrow P = i^2 R \Rightarrow P = \frac{E^2 R}{(r + R)^2}$

Power is maximum when $r = R \Rightarrow P_{\max} = E^2 / 4r$

77. (c) Since the current coming out from the positive terminal is equal to the current entering the negative terminal, therefore, current in the respective loop will remain confined in the loop itself.

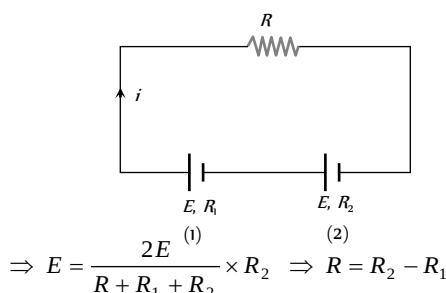
$\therefore$ current through 2Ω resistor = 0

78. (c) Reading of voltmeter

$$= E_{eq} = \frac{E_1 r_2 + E_2 r_1}{r_1 + r_2} = \frac{18 \times 1 + 12 \times 2}{1 + 2} = 14 V$$

79. (d) $i = \frac{2E}{R + R_1 + R_2}$

From cell (2) $E = V + iR_2 = 0 + iR_2$



80. (d)

81. (a) Applying Kirchoff's law in following figure.

At junction A :

$$i + i_1 + i_2 = 1 \quad \dots (i)$$

For Loop (i)

$$-60i + (15 + 5)i_1 = 0$$

$$\Rightarrow i_1 = 3i \quad \dots(ii)$$

For loop (2)

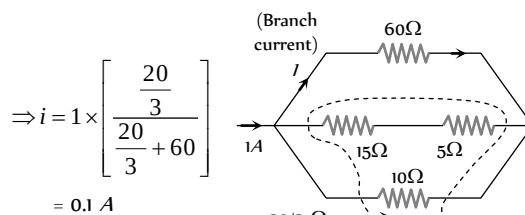
$$-(15 + 5)i + 10i_2 = 0$$

$$\Rightarrow i_2 = i = (3i) = 6i$$

On solving equation (i), (ii) and (iii) we get $i = 0.1 A$

Short Trick : Branch current =

$$\text{main current} \left(\frac{\text{Resistance of opposite branch}}{\text{Total resistance}} \right)$$



82. (d) Maximum current will be drawn from the circuit if resultant resistance of all internal resistances is equal to the value of external resistance if the arrangement is mixed. In series, $R \gg nr$ and in parallel, the external resistance is negligible.

83. (c) On applying Kirchoff's current law $i = 13 A$.

84. (c) Total cells = $m \times n = 24 \quad \dots (i)$

For maximum current in the circuit $R = \frac{mr}{n}$

$$\Rightarrow 3 = \frac{m}{n} \times (0.5) \Rightarrow m = 6n \quad \dots (ii)$$

On solving equation (i) and (ii), we get $m = 12$, $n = 2$

85. (a) Power dissipated = $i^2 R = \left(\frac{E}{R + r} \right)^2 R$

$$\therefore \left(\frac{E}{R_1 + r} \right)^2 R_1 = \left(\frac{E}{R_2 + r} \right)^2 R_2$$

$$\Rightarrow R_1(R_2^2 + r^2 + 2R_2r) = R_2(R_1^2 + r^2 + 2R_1r)$$

$$\Rightarrow R_2^2 R_1 + R_1 r^2 + 2R_2 r = R_1^2 R_2 + R_2 r^2 + 2R_1 R_2 r$$

$$\Rightarrow (R_1 - R_2)r^2 = (R_1 - R_2)r^2 = (R_1 - R_2)R_1 R_2$$

$$\Rightarrow r = \sqrt{R_1 R_2}$$

Different Measuring Instruments

- (a) In meter bridge experiment, it is assumed that the resistance of the L shaped plate is negligible, but actually it is not so. The error created due to this is called, end error. To remove this the resistance box and the unknown resistance must be interchanged and then the mean reading must be taken.
- (c) To convert a galvanometer into an ammeter a low value resistance is to be connected in parallel to it called shunt.
- (d) Balance point has some fixed position on potentiometer wire. It is not affected by the addition of resistance between balance point and cell.

4. (d) Resistance of voltmeter should be greater than the external circuit resistance. An ideal voltmeter has infinite resistance.

5. (c) $S = \frac{i_g G}{i - i_g} = \frac{100 \times 0.01}{(10 - 0.01)} = \frac{1}{10} = 0.1 \Omega$

6. (c) Equivalent resistance of the circuit $R_{eq} = 100 \Omega$

current through the circuit $i = \frac{2.4}{100} A$

P.D. across combination of voltmeter and 100Ω resistance
 $= \frac{2.4}{100} \times 50 = 1.2 V$

Since the voltmeter and 100Ω resistance are in parallel, so the voltmeter reads the same value i.e. $1.2 V$.

7. (a) Potential gradient $= \frac{e}{(R + R_h + r)} \cdot \frac{R}{L}$
 $= \frac{2}{(15 + 5 + 0)} \times \frac{5}{1} = 0.5 \frac{V}{m} = 0.005 \frac{V}{cm}$

8. (d) $S = \frac{i_g G}{(i - i_g)} \Rightarrow \frac{G}{S} = \frac{i - i_g}{i_g} = \frac{10 - 1}{1} = \frac{9}{1}$

9. (c) Ammeter is used to measure the current through the circuit.

10. (c) $S = \frac{i_g G}{(i - i_g)} = \frac{1 \times 0.018}{10 - 1} = \frac{0.018}{9} = 0.002 \Omega$

11. (d) Potentiometer works on null deflection method. In balance condition no current flows in secondary circuit.

12. (c) Shunt resistances $S = \frac{i_g G}{(i - i_g)} = \frac{10 \times 99}{(100 - 10)} = 11 \Omega$

13. (d) By using $R = \frac{V}{i_g} - G \Rightarrow R = \frac{100}{5 \times 10^{-3}} - 5 = 19,995 \Omega$

14. (a) Potential gradient = Change in voltage per unit length
 $\therefore 10 = \frac{V_2 - V_1}{30/100} \Rightarrow V_2 - V_1 = 3 \text{ volt}$

15. (d) $R = \frac{V}{i_g} - G = \frac{5}{100/10^3} - 2 = \frac{5000}{100} - 2 = 48 \Omega$

16. (c) $i_g = \frac{iS}{S + G} \Rightarrow 10 = \frac{50 \times 12}{12 + G} \Rightarrow 12 + G = 60 \Rightarrow G = 48 \Omega$

17. (a) To convert a galvanometer into a voltmeter, a high value resistance is to be connected in series with it.

18. (b)

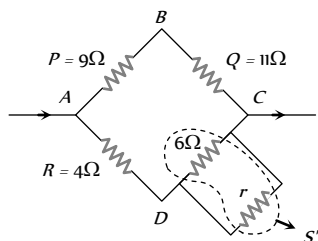
19. (c) $\frac{P}{Q} = \frac{R}{S'}$ (For balancing bridge)

$\Rightarrow S' = \frac{4 \times 11}{9} = \frac{44}{9}$

$\Rightarrow \frac{1}{S'} = \frac{1}{r} + \frac{1}{6}$

$\Rightarrow \frac{9}{44} - \frac{1}{6} = \frac{1}{r}$

$\Rightarrow r = \frac{132}{5} = 26.4 \Omega$



20. (a) $r = \left(\frac{l_1 - l_2}{l_2} \right) R = \left(\frac{25}{100} \right) 2 = 0.5 \Omega$

21. (b) The sensitivity of potentiometer can be increased by decreasing the potential gradient i.e. by increasing the length of potentiometer wire.

(Sensitivity $\propto \frac{1}{P.G.} \propto \text{Length}$)

22. (b) In balance condition, potentiometer doesn't take the current from secondary circuit.

23. (a) Here same current is passing throughout the length of the wire, hence $V \propto R \propto l$

$\Rightarrow \frac{V_1}{V_2} = \frac{l_1}{l_2} \Rightarrow \frac{6}{V_2} = \frac{300}{50} \Rightarrow V_2 = 1 V$

24. (a) $S = \frac{i_g G}{i - i_g} = \frac{10 \times 0.01}{10 - 0.01} = \frac{10}{999} \text{ ohm}$

25. (a) Ratio will be equal to the ratio of no deflection lengths i.e.
 $\frac{E_1}{E_2} = \frac{l_1}{l_2} = \frac{2}{3}$

26. (a) Potential gradient $= \frac{\text{Potential difference}}{\text{Length}}$

27. (a) Wheatstone bridge is balanced, therefore

$\frac{P}{Q} = \frac{R}{S}$ or $1 = \frac{10}{S} \Rightarrow S = 10 \text{ ohm}$

28. (a) When the length of potentiometer wire is increased, the potential gradient decreases and the length of previous balance point is increased.

29. (b)

30. (b)

31. (b) The actual circuit is same.

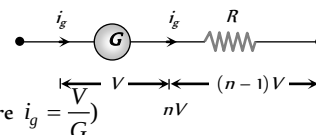
32. (b) $\because i_g = 10\%$ of $i = \frac{i}{10} \Rightarrow S = \frac{G}{(n-1)} = \frac{90}{(10-1)} = 10 \Omega$

33. (b) $\frac{E_1}{E_2} = \frac{l_1 + l_2}{l_1 - l_2} = \frac{(8+2)}{(8-2)} = \frac{5}{3}$

34. (b) Suppose resistance R is connected in series with voltmeter as shown.

By Ohm's law

$i_g \cdot R = (n-1)V$



$\Rightarrow R = (n-1)G$ (where $i_g = \frac{V}{G}$)

35. (c) Ammeter is always connected in series with circuit.

36. (c) If resistance of ammeter is r then

$20 = (R+r)4 \Rightarrow R+r = 5 \Rightarrow R < 5 \Omega$

37. (b) $S = \frac{i_g \times G}{i - i_g} = \frac{10 \times 10^{-3} \times 50}{1 - 10^{-3} \times 10} = \frac{50}{99} \Omega$ in parallel.

38. (b) $\because i_g = (100-90)\%$ of $i = \frac{i}{10}$

$\Rightarrow \text{Required shunt } S = \frac{G}{(n-1)} = \frac{900}{(10-1)} = 100 \Omega$

39. (d) $R = \frac{V}{i_g} - G = \frac{100}{10 \times 10^{-3}} - 25 = 9975 \Omega$

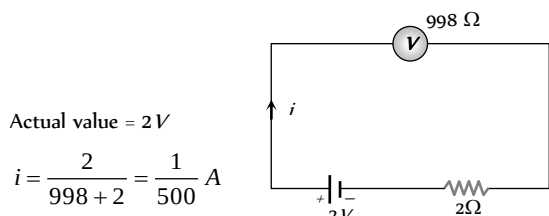
40. (b) Potential gradient $x = \frac{V}{L} = \frac{iR}{L}$

$$\Rightarrow x = \frac{2}{(15+5)} \times \frac{15}{10} = \frac{3}{2000} \text{ volt/cm}$$

$$41. (a) S = \frac{G}{\frac{i}{i_g} - 1} = \frac{25}{\frac{5}{50 \times 10^{-6}} - 1} = \frac{25}{10^5 - 1} = \frac{25}{10^5} = 2.5 \times 10^{-4} \Omega$$

42. (b) In balanced Wheatstone bridge, the arms of galvanometer and cell can be interchanged without affecting the balance of the bridge.

43. (c) Error in measurement = Actual value - Measured value



Actual value = 2 V

$$i = \frac{2}{998 + 2} = \frac{1}{500} \text{ A}$$

$$\text{Since } E = V + ir \Rightarrow V = E - ir = 2 - \frac{1}{500} \times 2 = \frac{998}{500} \text{ V}$$

$$\therefore \text{Measured value} = \frac{998}{500} \text{ V}$$

$$\Rightarrow \text{Error} = 2 - \frac{998}{500} = 4 \times 10^{-3} \text{ volt}$$

44. (d) The emf of the standard cell must be greater than that of experimental cells, otherwise balance point is not obtained.

45. (a)

46. (b) In general, ammeter always reads less than the actual value because of its resistance.

$$47. (c) \text{ By Wheatstone bridge, } \frac{R}{80} = \frac{AC}{BC} = \frac{20}{80} \Rightarrow R = 20 \Omega$$

48. (a) $E \propto l$ (balancing length)

$$49. (b) r = \left(\frac{l_1 - l_2}{l_2} \right) \times R' = \left(\frac{l_1 - 2}{2} \right) \times 5 \quad \dots (i)$$

$$\text{and } r = \left(\frac{l_1 - 3}{3} \right) \times 10 \quad \dots (ii)$$

On solving (i) and (ii) $r = 10 \Omega$

50. (a)

51. (b) In the part $c b d$,

$$V_c - V_b = V_b - V_d \Rightarrow V_b = \frac{V_c + V_d}{2}$$

In the part $c a d$

$$V_c - V_a > V_a - V_d \Rightarrow \frac{V_c + V_d}{2} > V_a \Rightarrow V_b > V_a$$

52. (c) In balance condition, no current will flow through the branch containing S .

$$53. (b) \text{ Resistance in parallel } S = \frac{G i_g}{i - i_g} = \frac{50 \times 100 \times 10^{-6}}{(10 - 100 \times 10^{-6})}$$

$$\Rightarrow S = 5 \times 10^{-4} \Omega$$

$$54. (b) E = x l = \frac{V}{l} = \frac{iR}{L} \times l \Rightarrow E = \frac{e}{(R + R_h + r)} \times \frac{R}{L} \times l$$

$$\Rightarrow E = \frac{10}{(5 + 4 + 1)} \times \frac{5}{5} \times 3 = 3 \text{ V}$$

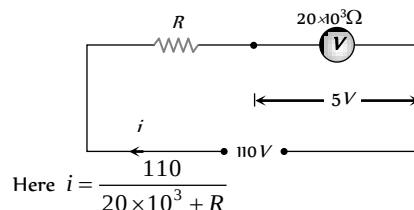
$$55. (a) \text{ Potential gradient} = \frac{V}{L} = \frac{iR}{L} = \frac{i\rho L}{AL} = \frac{i\rho}{A}$$

$$= \frac{0.2 \times 40 \times 10^{-8}}{8 \times 10^{-6}} = 10^{-2} \text{ V/m}$$

$$56. (b) i_g = 2\% \text{ of } i = \frac{i}{50} \Rightarrow S = \frac{G}{(n-1)} = \frac{G}{(50-1)} = \frac{G}{49}$$

57. (d) The resistance of an ideal voltmeter is considered as infinite.

58. (c)



$$\text{Here } i = \frac{110}{20 \times 10^3 + R}$$

$$\therefore V = iR \Rightarrow 5 = \left(\frac{110}{20 \times 10^3 + R} \right) \times 20 \times 10^3$$

$$\Rightarrow 10^5 + 5R = 22 \times 10^5 \Rightarrow R = 21 \times \frac{10^5}{5} = 420 \text{ K}\Omega$$

59. (c) Due to the negligible temperature co-efficient of resistance of constantan wire, there is no change in its resistance value with change in temperature.

60. (d) The resistance of voltmeter is too high, so that it draws negligible current from the circuit, hence potential drop in the external circuit is also negligible.

61. (a) By connecting a series resistance

$$R = \frac{V}{i_g} - G = \frac{10}{1} - 7 = 3 \Omega$$

62. (a) Since potential difference for full length of wire = 2 V

$$\therefore \text{P.D. per unit length of wire} = \frac{2}{4} = 0.5 \frac{\text{V}}{\text{m}}$$

$$63. (d) \frac{X}{1} = \frac{20}{80} \Rightarrow X = \frac{1}{4} \Omega = 0.25 \Omega.$$

64. (a) Reading of galvanometer remains same whether switch S is open or closed, hence no current will flow through the switch i.e. R and G will be in series and same current will flow through them. $I_R = I_G$.

65. (d) Pressing the key does not disturb current in all resistances as the bridge is balanced. Therefore, deflection in the galvanometer in whatever direction it was, will stay.

$$66. (b) i_g S = (i - i_g) G \Rightarrow i_g (S + G) = i G$$

$$\Rightarrow \frac{i_g}{i} = \frac{G}{S + G} = \frac{8}{2 + 8} = 0.8$$

$$67. (a) \text{ Potential gradient } x = \frac{e}{(R + R_h + r)} \cdot \frac{R}{L}$$

$$\Rightarrow x = \frac{2.5}{(20 + 80 + 0)} \times \frac{20}{10} = 5 \times 10^{-5} \frac{\text{V}}{\text{mm}}$$

68. (b) Given $i_g = 2 \text{ mA}$, $i = 20 \text{ mA}$, $G = 180 \Omega$

$$\frac{i_g}{i} = \frac{S}{G + S} \Rightarrow 180 + S = 10S \Rightarrow S = \frac{180}{9} = 20 \Omega$$

$$69. (c) \text{ Resistance of shunted ammeter} = \frac{GS}{G + S}$$

$$\text{Also } \frac{i}{i_g} = 1 + \frac{G}{S} \Rightarrow \frac{GS}{G+S} = \frac{i_g \cdot G}{i}$$

$$\Rightarrow \frac{GS}{G+S} = \frac{0.05 \times 120}{10} = 0.6 \Omega$$

70. (c) $r = \frac{(l_1 - l_2)}{l_2} \times R' = \left(\frac{60 - 50}{50} \right) \times 6 = 1.2 \Omega$

71. (d) By using $\frac{i}{i_g} = 1 + \frac{G}{S}$
 $\Rightarrow \frac{i}{100 \times 10^{-3}} = 1 + \frac{1000}{S} \Rightarrow S = \frac{1000}{9} = 111 \Omega$

72. (c) Potential gradient $x = \frac{V}{L} = \frac{e}{(R + R_h + r)} \cdot \frac{R}{L}$
 $\Rightarrow 2.2 \times 10^{-3} = \frac{2.2}{(10 + R_h)} \times 1 \Rightarrow R' = 990 \Omega$

73. (a) $\frac{i}{i_g} = \frac{G+S}{S} \Rightarrow \frac{i_g}{i} = \frac{S}{G+S} = \frac{2.5}{27.5} = \frac{1}{11}$

74. (c) Total resistance of the circuit $= \frac{80}{2} + 20 = 60 \Omega$
 $\Rightarrow$ Main current $i = \frac{2}{60} = \frac{1}{30} A$

Combination of voltmeter and 80Ω resistance is connected in series with 20Ω , so current through 20Ω and this combination will be same $= \frac{1}{30} A$.

Since the resistance of voltmeter is also 80Ω , so this current is equally distributed in 80Ω resistance and voltmeter (i.e. $\frac{1}{60} A$ through each)

P.D. across 80Ω resistance $= \frac{1}{60} \times 80 = 1.33 V$

75. (a) Potential gradient $x = \frac{V}{L} = \frac{iR}{L} = \frac{i \left(\frac{\rho L}{A} \right)}{L} = \frac{i\rho}{A}$

76. (d) Here $n = \frac{10}{2} = 5$
 $\therefore R = (n-1)G = (5-1)2000 = 8000 \Omega$

77. (b) $r = \left(\frac{l_1 - l_2}{l_1} \right) R = 0.5 \Omega$.

78. (a)

79. (b) $V = i \cdot R = \frac{e}{(R + R_h + r)} \cdot R \Rightarrow 10^{-3} = \frac{2}{(10 + R + r)} \times 10$
 $\Rightarrow R = 19,989 \Omega$.

80. (a)

81. (c) $2R > 20 \Rightarrow R > 10 \Omega$.

82. (c) $\frac{i}{i_g} = 1 + \frac{G}{S} \Rightarrow \frac{4}{1} = 1 + \frac{R}{S} \Rightarrow S = \frac{R}{3}$.

83. (a) When ammeter is connected in parallel to the circuit, net resistance of the circuit decreases. Hence more current is drawn from the battery, which damages the ammeter.

84. (a) $r = \left(\frac{l_1 - l_2}{l_2} \right) \times R' \Rightarrow r = \left(\frac{55 - 50}{50} \right) \times 10 = 1 \Omega$.

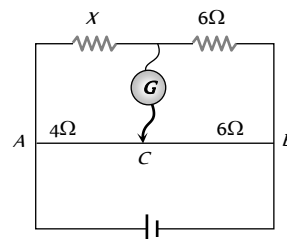
85. (b) $R = \frac{V}{i_g} - G = \frac{18}{3 \times 10^{-3}} - 12 = 5988 \Omega$

86. (d)

87. (c) $R = \frac{V}{i_g} - G = \frac{6}{6 \times 10^{-3}} - 25 = 975 \Omega$ (In series).

88. (d) $i_g = i \frac{S}{G+S} \Rightarrow 0.01 = 10 \frac{S}{25+S}$
 $\Rightarrow 1000S = 25 + S \Rightarrow S = \frac{25}{999} \Omega$.

89. (c)



Resistance of the part $AC = 5V$

$R_{AC} = 0.1 \times 40 = 4 \Omega$ and $R_{CB} = 0.1 \times 60 = 6 \Omega$

In balanced condition $\frac{X}{6} = \frac{4}{6} \Rightarrow X = 4 \Omega$

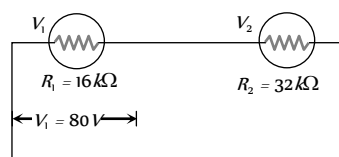
Equivalent resistance $R_{eq} = 5 \Omega$ so current drawn from battery $i = \frac{5}{5} = 1 A$.

90. (a) $(R+G)i_g = V \Rightarrow (R+G) = \frac{V}{i_g}$
 $= \frac{3}{30 \times 16 \times 10^{-6}} = 6.25 k\Omega$

$\therefore$ Value of R is nearly equal to $6 k\Omega$

This is connected in series in a voltmeter.

91. (d)



$R_1 = 80 \times 200 = 16000 \Omega = 16 k\Omega$

Current flowing through V_1 = Current flowing through V_2 = $\frac{80}{16 \times 10^3} = 5 \times 10^{-3} A$.

So, potential differences across V_2 is

$V_2 = 5 \times 10^{-3} \times 32 \times 10^3 = 160 \text{ volt}$

Hence, line voltage $V = V_1 + V_2 = 80 + 160 = 240 V$.

92. (d) $V = xI \Rightarrow iR = xI$
 $\Rightarrow i \times 10 = \left(\frac{2 \times 10^{-3}}{10^{-2}} \right) \times 50 \times 10^{-2} = 0.1$

$\Rightarrow i = 10 \times 10^{-3} A = 10 mA$.

93. (d) $E = \frac{e}{(R + R_h + r)} \frac{R}{L} \times l = \frac{2}{(10 + 40 + 0)} \times \frac{10}{1} \times 0.4 = 0.16 V$.

94. (c) $\frac{i}{i_g} = 1 + \frac{G}{S} \Rightarrow \frac{5}{2} = 1 + \frac{12}{S} \Rightarrow S = 8\Omega$. (In parallel).

95. (d) $\frac{i_g}{i} = \frac{S}{G+S} \Rightarrow \frac{5}{100} = \frac{S}{G+S} \Rightarrow S = \frac{G}{19}$

96. (a) $R = G(n-1) = 50 \times 10^3 (3-1) = 10^5 \Omega$.

97. (c) $\frac{E_1}{E_2} = \frac{l_1 + l_2}{l_1 - l_2} = \frac{58 + 29}{58 - 29} = \frac{3}{1}$

98. (a) $R = \frac{V}{i_g} - G = \frac{10}{10 \times 10^{-3}} - 1 = 999\Omega$.

99. (d) For conversion of galvanometer (of resistances) into voltmeter, a resistance R is connected in series.

$$\therefore i_g = \frac{V_1}{R+G} \text{ and } i_g = \frac{V_2}{2R+G}$$

$$\Rightarrow \frac{V_1}{R+G} = \frac{V_2}{2R+G} \Rightarrow \frac{V_2}{V_1} = \frac{2R+G}{R+G} = \frac{2(R+G)-G}{(R+G)}$$

$$= 2 - \frac{G}{(R+G)} \Rightarrow V_2 = 2V_1 - \frac{V_1 G}{(R+G)} \Rightarrow V_2 < 2V_1$$

100. (d) If the voltmeter is ideal then given circuit is an open circuit, so reading of voltmeter is equal to the e.m.f. of cell i.e., 6V.

101. (c) $\frac{i_g}{i} = \frac{S}{G+S} = \frac{4}{36+4} = \frac{1}{10}$ i.e. 10%.

102. (d) After connecting a resistance R in parallel with voltmeter its effective resistance decreases. Hence less voltage appears across it i.e. V will decrease. Since overall resistance decreases so more current will flow i.e. A will increase.

103. (c) Potential gradient $x = \frac{e}{(R+R_h+r)} \cdot \frac{R}{L}$

$$\Rightarrow \frac{10^{-3}}{10^{-2}} = \frac{2}{(3+R_h+0)} \times \frac{3}{1} \Rightarrow R_h = 57\Omega.$$

104. (c) $\frac{i}{i_g} = 1 + \frac{G}{S} \Rightarrow \frac{1}{10^{-3}} = 1 + \frac{20}{S} \Rightarrow S = \frac{20}{999} \approx 0.02\Omega$.

105. (a) Resistance of voltmeter should be high.

106. (c) If ammeter is used in place of voltmeter (i.e. in parallel) it may damage due to large current in circuit. Hence to control this large amount of current a high resistance must be connected in series.

107. (c) Potential gradient $x = \frac{e}{(R+R_h+r)} \cdot \frac{R}{L}$

$$= \frac{3}{(20+10+0)} \times \frac{20}{10} = 0.2$$

108. (d) $\frac{E_1}{E_2} = \frac{l_1 + l_2}{l_1 - l_2} = \frac{(6+2)}{(6-2)} = \frac{2}{1}$

109. (c) Manganin or constantan are used for making the potentiometer wire.

110. (a)

111. (a) $\frac{i}{i_g} = 1 + \frac{G}{S} \Rightarrow \frac{iG}{V_g} = 1 + \frac{G}{S} \Rightarrow \frac{100 \times 10^{-3} \times 40}{800 \times 10^{-3}} = 1 + \frac{40}{S}$
 $\Rightarrow S = 10\Omega$.

112. (a) $i_g = i \frac{S}{G+S} \Rightarrow 10 \times 10^{-3} = \frac{S}{100+S} \times 100 \times 10^{-3}$

$$90S = 1000 \Rightarrow S = \frac{1000}{90} = 11.11\Omega.$$

113. (c) Before connecting the voltmeter, potential difference across 100Ω resistance

$$V_i = \frac{100}{(100+10)} \times V = \frac{10}{11} V$$

Finally after connecting voltmeter across 100Ω

Equivalent resistance

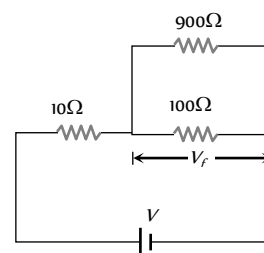
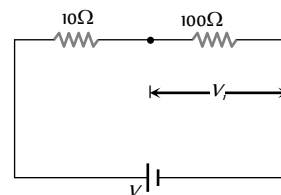
$$\frac{100 \times 900}{(100+900)} = 90\Omega$$

Final potential difference

$$V_f = \frac{90}{(90+10)} \times V = \frac{9}{10} V$$

$$\% \text{ error} = \frac{V_i - V_f}{V_i} \times 100$$

$$= \frac{\frac{10}{11}V - \frac{9}{10}V}{\frac{10}{11}V} \times 100 = 1.0.$$



114. (b) Potential gradient $= \frac{e.R}{(R+r).L} = \frac{10 \times 3}{(3+3) \times 5}$
 $= 1V/m = 10mV/cm$.

115. (c) $\frac{i}{i_g} = 1 + \frac{G}{S} \Rightarrow \frac{1}{10^{-5}} = 1 + \frac{100}{S} \Rightarrow S \approx \frac{100}{10^5} = 10^{-3}\Omega$.

116. (d) $\frac{i_g}{i} = \frac{S}{G+S} = \frac{4}{36+4} = \frac{4}{40} = \frac{1}{10}$

117. (a) $i = \frac{V}{R} \Rightarrow 2 = \frac{6}{\frac{6 \times 3}{6+3} + R} = \frac{6}{2+R} \Rightarrow R = 1\Omega$.

118. (b) $i_g = i \frac{S}{G+S} \Rightarrow \frac{0.01}{10} = \frac{5}{50+S} \Rightarrow S = \frac{50}{999} = 0.05\Omega$.

119. (d) $S = \left(\frac{100-l}{l} \right) \cdot R$

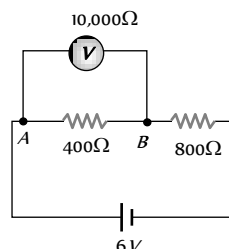
$$\text{Initially, } 30 = \left(\frac{100-l}{l} \right) \times 10 \Rightarrow l = 25cm$$

$$\text{Finally, } 10 = \left(\frac{100-l}{l} \right) \times 30 \Rightarrow l = 75cm$$

So, shift = 50cm.

120. (c) Potential gradient $(x) = \frac{i\rho}{A} = \frac{0.1 \times 10^{-7}}{10^{-6}} = 10^{-2} V/m$

121. (d) Before connecting voltmeter potential difference across 400Ω resistance is



$$V_i = \frac{400}{(400 + 800)} \times 6 = 2V$$

After connecting voltmeter equivalent resistance between A and

$$B = \frac{400 \times 10,000}{(400 + 10,000)} = 384.6\Omega$$

Hence, potential difference measured by voltmeter

$$V_f = \frac{384.6}{(384.6 + 800)} \times 6 = 1.95V$$

$$\text{Error in measurement} = V_i - V_f = 2 - 1.95 = 0.05V$$

122. (c) $\frac{i}{i_g} = 1 + \frac{G}{S} \Rightarrow \frac{5}{0.05} = 1 + \frac{50}{S}$

$$\Rightarrow S = \frac{50}{99} = \frac{\rho \times l}{A} \Rightarrow l = \frac{50}{99} \times \frac{2.97 \times 10^{-2} \times 10^{-4}}{5 \times 10^{-7}} = 3m.$$

123. (a) $\frac{i}{i_g} = 1 + \frac{G}{S} \Rightarrow \frac{10}{1} = 1 + \frac{0.81}{S} \Rightarrow S = 0.09\Omega.$

124. (a) From the principle of potentiometer $V \propto l$

$\Rightarrow \frac{V}{E} = \frac{l}{L}$; where V = emf of battery, E = emf of standard cell, L = Length of potentiometer wire

$$V = \frac{El}{L} = \frac{30E}{100}.$$

125. (b) $E = \frac{e}{(R + R_h + r)} \cdot \frac{R}{L} \times l$

$$\Rightarrow 10 \times 10^{-3} = \frac{2}{(10 + R + 0)} \times \frac{10}{1} \times 0.4 \Rightarrow R = 790\Omega$$

126. (b) Using $r = R \left(\frac{l_1}{l_2} - 1 \right) = 2 \left(\frac{150}{100} - 1 \right) = 1\Omega$

127. (d) Resistance between A and B = $\frac{1000 \times 500}{(1500)} = \frac{1000}{3}$

So, equivalent resistance of the circuit

$$R_{eq} = 500 + \frac{1000}{3} = \frac{2500}{3}$$

$\therefore$ Current drawn from the cell

$$i = \frac{10}{(2500/3)} = \frac{3}{250} A$$

Reading of voltmeter i.e.

$$\text{potential difference across } AB = \frac{3}{250} \times \frac{1000}{3} = 4V$$

128. (d) $i_g = \frac{i}{10} \Rightarrow \text{Required shunt } S = \frac{G}{(n-1)} = \frac{90}{(10-1)} = 10\Omega$

129. (b) $i_g = \frac{50}{10 \times 10^{-3}} - 40 = 4960$

130. (c) Post office box is based on the principle of Wheatstone's bridge

131. (d) Full deflection current $i_g = 25 \times 4 \times 10^{-4} = 100 \times 10^{-4} A$

$$\text{Using } R = \frac{V}{I_g} - G = \frac{25}{100 \times 10^{-4}} - 50 = 2450\Omega \text{ in series.}$$

132. (a) In balancing condition, $\frac{R_1}{R_2} = \frac{l_1}{l_2} = \frac{l_1}{100 - l_1}$

$$\Rightarrow \frac{X}{Y} = \frac{20}{80} = \frac{1}{4} \quad \dots(i)$$

$$\text{and } \frac{4X}{Y} = \frac{l}{100 - l} \quad \dots(ii)$$

$$\Rightarrow \frac{4}{4} = \frac{l}{100 - l} \Rightarrow l = 50cm$$

133. (c)

134. (d) $S = \left(\frac{i_g}{i - i_g} \right) \times G = \frac{100 \times 10^{-6}}{(10 \times 10^{-3} - 100 \times 10^{-6})} \times 50 \approx 0.5\Omega$
(in parallel)

135. (d) $E = \frac{e}{(R + R_h + r)} \cdot \frac{R}{L} \times l \Rightarrow 0.4 = \frac{5}{(5 + 45 + 0)} \times \frac{5}{10} \times l$
 $\Rightarrow l = 8m$

136. (a) Potential difference per unit length = $\frac{V}{L} = \frac{2}{4} = 0.5V/m$

137. (a)

138. (b) $r = R \left(\frac{l_1}{l_2} - 1 \right) = 2 \left(\frac{240}{120} - 1 \right) = 2\Omega$

139. (d) $E = \frac{V}{l}$; E is constant (volt. gradient).

$$\Rightarrow \frac{V_1}{l_1} = \frac{V_2}{l_2} \Rightarrow \frac{1.1}{140} = \frac{V}{180} \Rightarrow V = \frac{180 \times 1.1}{140} = 1.41V$$

140. (a) $I_G \times G = (I - I_G)S \Rightarrow I = \left(1 + \frac{G}{S} \right) I_G \Rightarrow I = 100.1mA$

141. (c) Let S be larger and R be smaller resistance connected in two gaps of meter bridge.

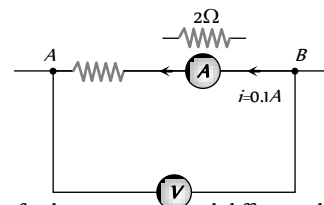
$$\therefore S = \left(\frac{100 - l}{l} \right) R = \frac{100 - 20}{20} R = 4R \quad \dots(i)$$

When 15Ω resistance is added to resistance R , then

$$S = \left(\frac{100 - 40}{40} \right) (R + 15) = \frac{6}{4} (R + 15) \quad \dots(ii)$$

From equations (i) and (ii) $R = 9\Omega$

142. (a) According to following figure



Reading of voltmeter = Potential difference between A and B = $i(R + 2) \Rightarrow 12 = 0.1(R + 2) \Rightarrow R = 118\Omega.$

143. (a) Potential gradient $x = \frac{e}{(R + R_h + r)} \cdot \frac{R}{L}$

$$\Rightarrow \frac{0.2 \times 10^{-3}}{10^{-2}} = \frac{2}{(R + 490 + 0)} \times \frac{R}{1} \Rightarrow R = 4.9\Omega.$$

Critical Thinking Questions

1. (a) Initially : Resistance of given cable

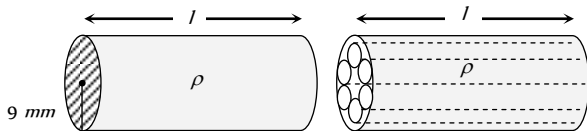
$$R = \rho \frac{l}{\pi \times (9 \times 10^{-3})^2} \quad \dots$$

(i)

Finally : Resistance of each insulated copper wire is

$$R' = \rho \frac{l}{\pi \times (3 \times 10^{-3})^2} \quad \text{Hence equivalent resistance of}$$

$$\text{cable } R_{eq} = \frac{R'}{6} = \frac{1}{6} \times \left(\rho \frac{l}{\pi \times (3 \times 10^{-3})^2} \right) \quad \dots (ii)$$

On solving equation (i) and (ii) we get $R' = 7.5 \Omega$

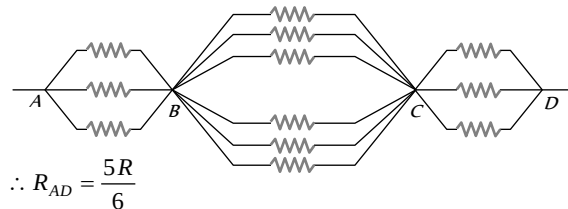
$$2. (a) \frac{R_A}{R_B} = \left(\frac{r_B}{r_A} \right)^4 \Rightarrow \frac{R_A}{R_B} = \left(\frac{1}{2} \right)^4 = \frac{1}{16} \Rightarrow R_B = 16R_A$$

When R_A and R_B are connected in parallel then equivalent

$$\text{resistance } R_{eq} = \frac{R_A R_B}{(R_A + R_B)} = \frac{16}{17} R_A$$

If $R_A = 4.25 \Omega$ then $R_{eq} = 4 \Omega$ i.e. option (a) is correct.

3. (c) The given circuit can be simplified as follows



4. (c) Suppose
- n
- resistors are used for the required job. Suppose equivalent resistance of the combination is
- R'
- and according to energy conservation it's current rating is
- i'
- .

Energy consumed by the combination = $n \times$ (Energy consumed by each resistance)

$$\Rightarrow i'^2 R' = n \times i^2 R \Rightarrow n = \left(\frac{i'}{i} \right)^2 \times \left(\frac{R'}{R} \right) = \left(\frac{4}{1} \right)^2 \times \left(\frac{5}{10} \right) = 8$$

5. (c) Resistance across
- $AB = \frac{1}{R'} = \frac{1}{R} + \frac{1}{R} + \frac{1}{R_1}$

$$R_1 = 2 \times 10^{-6} \Omega$$

and $R = \pi \times 1 \times 10^{-6} \Omega$

On solving,

$$R' = 0.88 \times 10^{-6} \Omega$$

6. (b) No current flows through the capacitor branch in steady state. Total current supplied by the battery

$$i = \frac{6}{2.8 + 1.2} = \frac{3}{2}$$

$$\text{Current through } 2 \Omega \text{ resistor} = \frac{3}{2} \times \frac{3}{5} = 0.9 A$$

7. (d) At time
- $t = 0$
- i.e. when capacitor is charging, current

$$i = \frac{2}{1000} = 2 mA$$

When capacitor is full charged, no current will pass through it,

$$\text{hence current through the circuit } i = \frac{2}{2000} = 1 mA$$

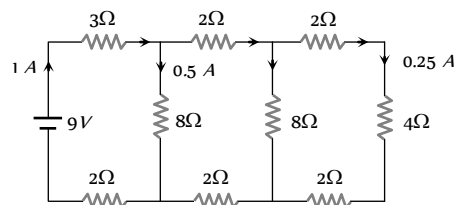
8. (d) Current in the bulb =
- $\frac{P}{V} = \frac{4.5}{1.5} = 3 A$

$$\text{Current in } 1 \Omega \text{ resistance} = \frac{1.5}{1} = 1.5 A$$

Hence total current from the cell $i = 3 + 1.5 = 4.5 A$ By using $E = V + ir \Rightarrow E = 1.5 + 4.5 \times (2.67) = 13.5 V$

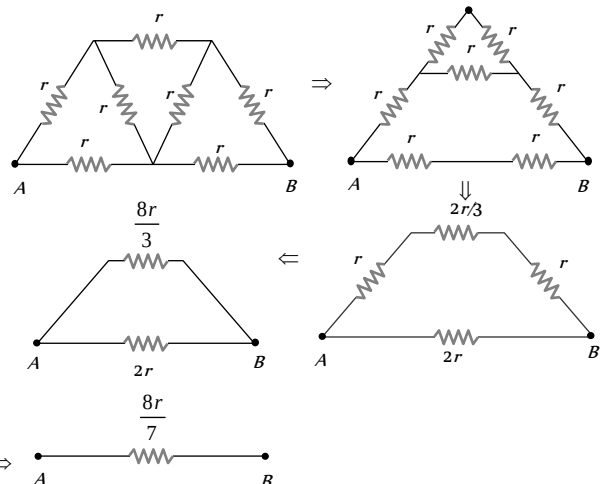
9. (d) Equivalent resistance of the circuit
- $R = 9 \Omega$

$$\therefore \text{Main current } i = \frac{V}{R} = \frac{9}{9} = 1 A$$

After proper distribution, the current through 4Ω resistance is $0.25 A$.

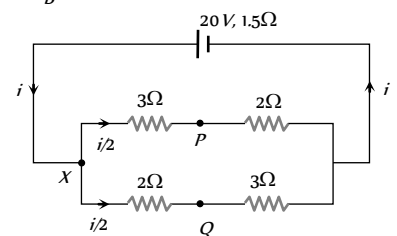
10. (b) Maximum number of resistance =
- $2^{n-1} = 2^{3-1} = 4$

11. (d) The given circuit can be simplified as follows.



12. (d)
- $R_{eq} = \frac{5}{2} \Omega$

$$i = \frac{20}{\frac{5}{2} + 1.5} = 5 A$$

Potential difference between X and P ,

$$V_X - V_P = \left(\frac{5}{2} \right) \times 3 = 7.5 V \quad \dots (i)$$

$$V_X - V_Q = \frac{5}{2} \times 2 = 5V \quad \dots(ii)$$

On solving (i) and (ii) $V_P - V_Q = -2.5 \text{ volt}$; $V_Q > V_P$.

Short Trick : $(V_P - V_Q) = \frac{i}{2}(R_2 - R_1) = \frac{5}{2}(2 - 3) = -2.5$

$$\Rightarrow V_Q > V_P$$

13. (c) $R_{t_1} = R_1(1 + \alpha_1 t)$ and $R_{t_2} = R_2(1 + \alpha_2 t)$

Also $R_{eq.} = R_{t_1} + R_{t_2} \Rightarrow R_{eq} = R_1 + R_2 + (R_1\alpha_1 + R_2\alpha_2)t$

$$\Rightarrow R_{eq} = (R_1 + R_2) \left\{ 1 + \left(\frac{R_1\alpha_1 + R_2\alpha_2}{R_1 + R_2} \right) t \right\}$$

So $\alpha_{eff} = \frac{R_1\alpha_1 + R_2\alpha_2}{R_1 + R_2}$

14. (b) Let the voltage across any one cell is V , then

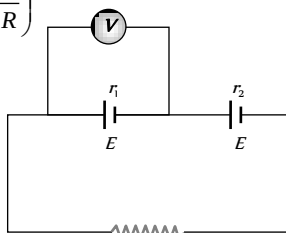
$$V = E - ir = E - r_1 \left(\frac{2E}{r_1 + r_2 + R} \right)$$

But $V = 0$

$$\Rightarrow E - \frac{2Er_1}{r_1 + r_2 + R} = 0$$

$$\Rightarrow r_1 + r_2 + R = 2r_1$$

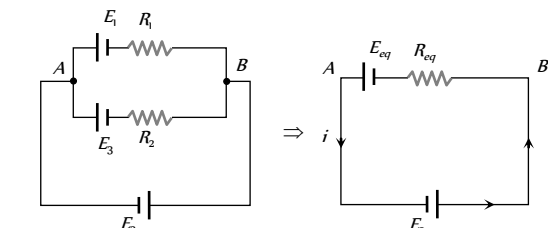
$$\Rightarrow R = r_1 - r_2$$



15. (b) Emf $E = 5V$, Internal resistance $r = \frac{5}{10} = 0.5\Omega$

Current through the resistance $i = \frac{5}{(2 + 0.5)} = 2A$

16. (b) The given circuit can be redrawn

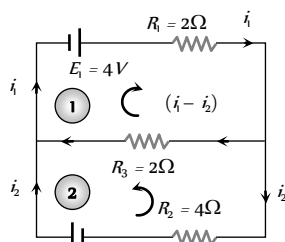


$$E_{eq} = \frac{E_1 R_2 + E_2 R_1}{R_1 + R_2} = \frac{2 \times 4 + 2 \times 4}{4 + 4} = 2V \text{ and}$$

$$R_{eq} = \frac{4}{2} = 2\Omega. \text{ Current } i = \frac{2 + 2}{2} = 2A \text{ from } A \text{ to } B \text{ through } E_2.$$

17. (b) Applying Kirchhoff's law for the loops (1) and (2) as shown in figure

For loop (1)



$$-2i_1 - 2(i_1 - i_2) + 4 = 0 \Rightarrow 2i_1 - i_2 = 2 \quad \dots(i)$$

For loop (2)

$$-2(i_1 - i_2) + 4i_2 - 6 = 0 \Rightarrow -i_1 + 3i_2 = 3 \quad \dots(ii)$$

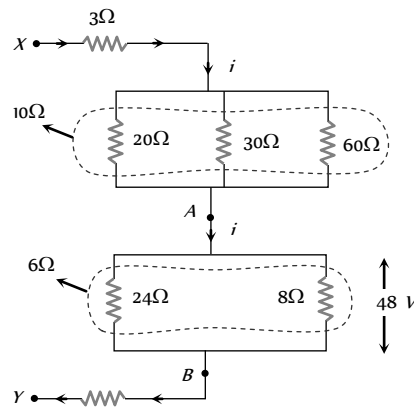
On solving equation (i) and (ii) $i_1 = 1.8A$.

18. (b) To convert a galvanometer into an ammeter, a shunt $S = \frac{I_g}{I - I_g} G$ is connected in parallel with it. To convert a

galvanometer into a voltmeter, a resistance $R = \frac{V}{I_g} - G$ is

connected in series with it.

19. (a) The given circuit can be redrawn as follows



Resistance between A and $B = \frac{24 \times 8}{32} = 6\Omega$

Current between A and $B =$ Current between X and Y
 $= i = \frac{48}{6} = 8A$

Resistance between X and $Y = (3 + 10 + 6 + 1) = 20\Omega$

$\Rightarrow$ Potential difference between X and $Y = 8 \times 20 = 160V$

20. (d) $R_1 + R_2 = R_1(1 + \alpha t) + R_2(1 - \beta t)$

$$\Rightarrow R_1 + R_2 = R_1 + R_2 + R_1\alpha t - R_2\beta t \Rightarrow \frac{R_1}{R_2} = \frac{\beta}{\alpha}$$

21. (d) Current density of drifting electrons $j = nev$

$$n = 5 \times 10^7 \text{ cm}^{-3} = 5 \times 10^7 \times 10^6 \text{ m}^{-3}.$$

$$v = 0.4 \text{ ms}^{-1}, e = 1.6 \times 10^{-19} \text{ C} \Rightarrow j = 3.2 \times 10^{-6} \text{ Am}^{-2}$$

Current density of ions $= (4 - 3.2) \times 10^{-6} = 0.8 \times 10^{-6} \frac{A}{m^2}$

This gives v for ions $= 0.1 \text{ ms}$.

22. (a) In the following figure

Resistance of part PNQ :

$$R_1 = \frac{10}{4} = 2.5\Omega \text{ and}$$

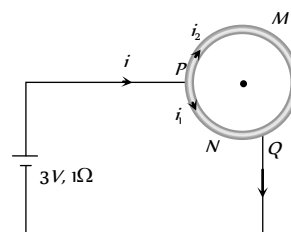
Resistance of part PMQ :

$$R_2 = \frac{3}{4} \times 10 = 7.5\Omega$$

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} = \frac{2.5 \times 7.5}{(2.5 + 7.5)} = \frac{15}{8} \Omega.$$

Main Current $i = \frac{3}{\frac{15}{8} + 1} = \frac{24}{23} A$

So, $i = i \times \left(\frac{R_2}{R_1 + R_2} \right) = \frac{24}{23} \times \left(\frac{7.5}{2.5 + 7.5} \right) = \frac{18}{23} A$



$$\text{and } i_2 = i - i_1 = \frac{24}{23} - \frac{18}{23} = \frac{6}{23} \text{ A.}$$

23. (c) As I is independent of R_6 , no current flows through R_6 this requires that the junction of R_1 and R_2 is at the same potential as the junction of R_3 and R_4 . This must satisfy the condition

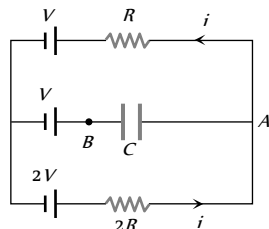
$$\frac{R_1}{R_2} = \frac{R_3}{R_4}, \text{ as in the Wheatstone bridge.}$$

24. (c) Moving anticlockwise from A
 $-iR - V + 2V - 2iR = 0$

$$\text{or } 3iR = V \text{ or } i = \frac{V}{3R}$$

$$V_A - V_B = iR + V - V = iR$$

$$\Rightarrow \text{Potential drop across } C = \frac{V}{3}$$



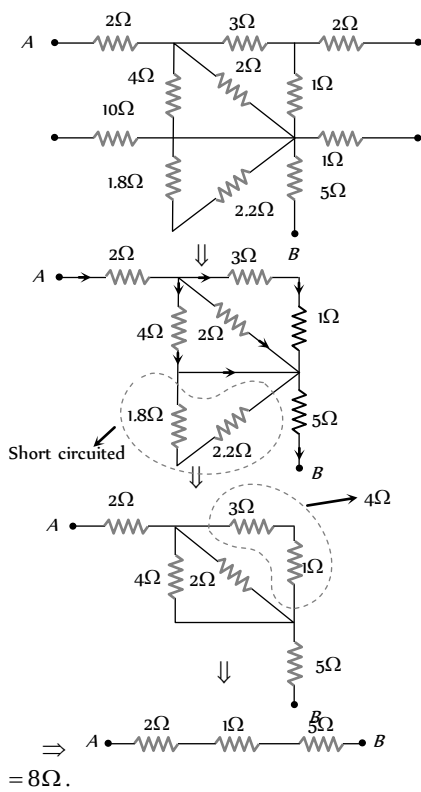
25. (b) Let R and m be the resistance and mass of the first wire, then the second wire has resistance $2R$ and mass $2m$. Let $E = \text{emf of each cell}$, $S = \text{specific heat capacity of the material of the wire}$.

$$\text{For the first wire, current } i_1 = \frac{3E}{R} \text{ and } i_1^2 R t = m S \Delta T$$

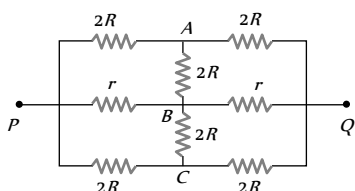
$$\text{For the second wire, } i_2 = \frac{NE}{2R} \text{ and } i_2^2 (2R)t = 2m S \Delta T.$$

$$\text{Thus, } i_1 = i_2 \text{ or } N = 6.$$

26. (b)

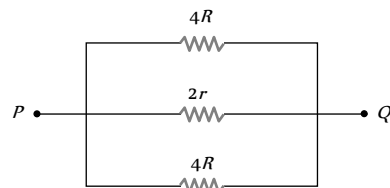


27. (a)



In a circuit, any circuit element placed between points at the same potential can be removed, without affecting the rest of the circuit. Here, by symmetry, points A , B and C are at same potential, for any potential difference between P and Q .

The circuit can therefore be reduced as shown below



$$\text{Effective resistance } R_{eq} = \frac{2Rr}{R+r}.$$

28. (d) Potential difference between A and B

$$V_A - V_B = 1 \times 1.5$$

$$\Rightarrow V_A - 0 = 1.5V \Rightarrow V_A = 1.5V$$

Potential difference between B and C

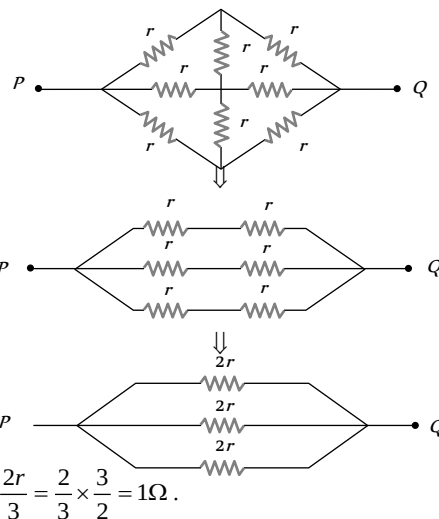
$$V_B - V_C = 1 \times 2.5 = 2.5V$$

$$\Rightarrow 0 - V_C = 2.5V \Rightarrow V_C = -2.5V$$

Potential difference between C and D

$$V_C - V_D = -2V \Rightarrow -2.5 - V_D = -2 \Rightarrow V_D = -0.5V.$$

29. (b) The given circuit can be simplified as follows



$$30. (b) dQ = Idt \Rightarrow Q = \int_{t=2}^{t=3} Idt = \left[2 \int_2^3 t dt + 3 \int_2^3 t^2 dt \right]$$

$$= \left[t^2 \right]_2^3 + \left[t^3 \right]_2^3 = (9 - 4) + (27 - 8) = 5 + 19 = 24C.$$

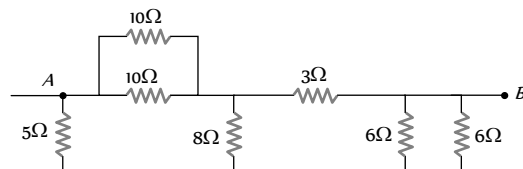
$$31. (d) i = \frac{E_1 + E_2 + E_3 + \dots + E_n}{(r_1 + r_2 + r_3 + \dots + r_n)} = \frac{1.5(r_1 + r_2 + r_3 + \dots + r_n)}{(r_1 + r_2 + r_3 + \dots + r_n)} = 1.5A.$$

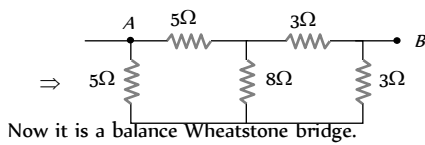
32. (a) Balancing length is independent of the cross sectional area of the wire.

$$33. (a) \frac{R_1}{R_2} = \frac{(1 + \alpha t_1)}{(1 + \alpha t_2)} \Rightarrow \frac{10}{R_2} = \frac{(1 + 5 \times 10^{-3} \times 20)}{(1 + 5 \times 10^{-3} \times 120)} \Rightarrow R_2 \approx 15\Omega$$

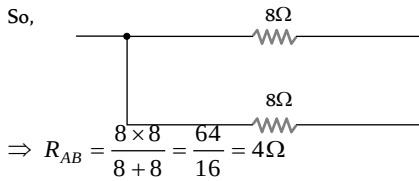
$$\text{Also } \frac{i_1}{i_2} = \frac{R_2}{R_1} \Rightarrow \frac{30}{i_2} = \frac{15}{10} \Rightarrow i_2 = 20 \text{ mA}$$

34. (b) The given circuit can be simplified as follows

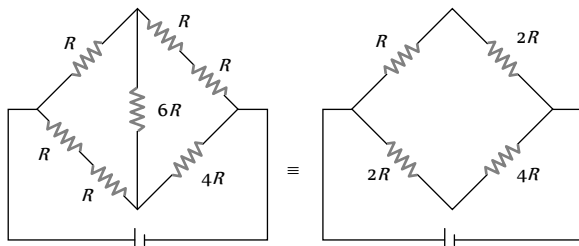




So,



35. (c) The equivalent network is

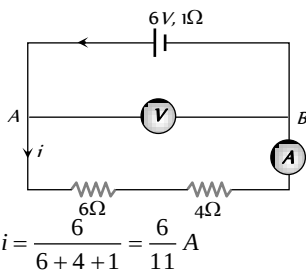


Clearly, the network of resistances is a balanced Wheatstone bridge. So R_{AB} is given by

$$\frac{1}{R_{AB}} = \frac{1}{3R} + \frac{1}{6R} = \frac{2+1}{6R} = \frac{1}{2R} \Rightarrow R_{AB} = 2R$$

For maximum power transfer $2R = 4\Omega \Rightarrow R = \frac{4}{2} = 2\Omega$

36. (c) The given circuit can be redrawn as follows



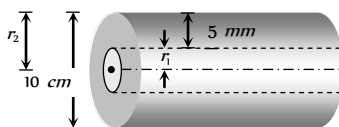
$$\text{Current } i = \frac{6}{6 + 4 + 1} = \frac{6}{11} \text{ A}$$

$$\text{P.D. between A and B, } V = \frac{6}{11} \times 10 = \frac{60}{11} \text{ V.}$$

37. (a) By using $R = \rho \cdot \frac{l}{A}$; here $A = \pi(r_2^2 - r_1^2)$

Outer radius $r_2 = 5 \text{ cm}$

Inner radius $r_1 = 5 - 0.5 = 4.5 \text{ cm}$



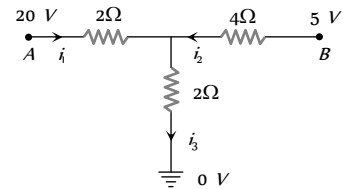
$$\text{So } R = 1.7 \times 10^{-8} \times \frac{5}{\pi \{(5 \times 10^{-2})^2 - (4.5 \times 10^{-2})^2\}} = 5.6 \times 10^{-5} \Omega$$

38. (a) Here $R_{XWY} = \frac{R}{2\pi} \times r(2\pi - \alpha) = \frac{R\alpha}{2\pi} \left(\because \alpha = \frac{l}{r} \right)$

$$\text{and } R_{XZY} = \frac{R}{2\pi} \times r(2\pi - \alpha) = \frac{R}{2\pi} (2\pi - \alpha)$$

$$R_{eq} = \frac{R_{XWY} R_{XZY}}{R_{XWY} + R_{XZY}} = \frac{\frac{R\alpha}{2\pi} \times \frac{R}{2\pi} (2\pi - \alpha)}{\frac{R\alpha}{2\pi} + \frac{R(2\pi - \alpha)}{2\pi}} = \frac{R\alpha}{4\pi^2} (2\pi - \alpha)$$

39. (d) Battery is short circuited so potential difference is zero.
40. (a) Let V be the potential of the junction as shown in figure. Applying junction law, we have

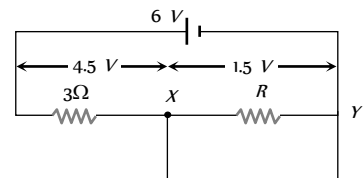


$$\text{or } \frac{20 - V}{2} + \frac{5 - V}{4} = \frac{V - 0}{2}$$

$$\text{or } 40 - 2V + 5 - V = 2V \text{ or } 5V = 45 \Rightarrow V = 9 \text{ V}$$

$$\therefore i_3 = \frac{V}{2} = 4.5 \text{ A}$$

41. (a) $E = x l = i \rho l \Rightarrow i = \frac{E}{\rho l} = \frac{E}{\rho l} = \frac{2.4 \times 10^{-3}}{1.2 \times 5} = 4 \times 10^{-4} \text{ A.}$
42. (b) When bulb glows with full intensity, then voltage across it will be 1.5 V and voltage across 3 Ω resistance will be 4.5 V.



Current through 3 Ω resistance $i = \frac{4.5}{3} = 1.5 \text{ A}$

Same current will flow between X and Y

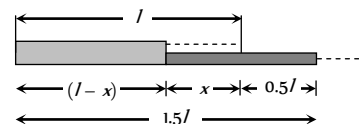
$$\text{So } V_{XY} = i R_{XY} \Rightarrow 1.5 = 1.5 R_{XY} \Rightarrow R_{XY} = 1\Omega$$

43. (a) In figure (b) current through $R_2 = i - \frac{i}{10} = \frac{9i}{10}$
Potential difference across R_2 = Potential difference across R
 $\Rightarrow R_2 \times \frac{9}{10} i = R \times \frac{i}{10}$ i.e. $R_2 = \frac{R}{9} = \frac{11}{9} \Omega$

$$R_{eq} = \frac{R_2 \times R}{(R_2 + R)} = \frac{\frac{11}{9} \times \frac{11}{9}}{\frac{11}{9} + \frac{11}{9}} = \frac{11}{10} \Omega$$

$$\text{Total circuit resistance} = \frac{11}{10} + R_1 = R = 11 \Rightarrow R_1 = 9.9 \Omega$$

44. (a) Let l be the original length of wire and x be its length stretched uniformly such that final length is 1.5 l



$$\text{Then } 4R = \rho \frac{(l-x)}{A} + \rho \frac{(0.5l+x)}{A'} \text{ where } A' = \frac{x}{(0.5l+x)} A$$

$$\therefore 4\rho \frac{l}{A} = \rho \frac{l-x}{A} + \rho \frac{(0.5l+x)^2}{xA}$$

$$\text{or } 4l = l-x + \frac{1}{4} \frac{l^2}{x} + \frac{x^2}{x} + \frac{lx}{x} \text{ or } \frac{x}{l} = \frac{1}{8}$$

45. (b) In series : Potential difference $\propto R$

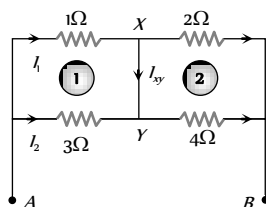
$$\text{When only } S_1 \text{ is closed } V_1 = \frac{3}{4}E = 0.75E$$

$$\text{When only } S_2 \text{ is closed } V_2 = \frac{6}{7}E = 0.86E$$

and when both S_1 and S_2 are closed combined resistance of $6R$ and $3R$ is $2R$

$$\therefore V_3 = \left(\frac{2}{3}\right)E = 0.67E \Rightarrow V_2 > V_1 > V_3$$

46. (c)



$$-i_1 + 0 \times i_{xy} + 3i_2 = 0 \text{ i.e. } i_1 = 3i_2 \text{(i)}$$

$$\text{Also } -2(i_1 - i_{xy}) + 4(i_2 + i_{xy}) = 0$$

$$\text{i.e. } 2i_1 - 4i_2 = 6i_{xy} \text{(ii)}$$

$$\text{Also } V_{AB} - 1 \times i_1 - 2(i_1 - i_{xy}) = 0 \Rightarrow 50 = i_1 + 2(i_1 - i_{xy}) = 3i_1 - 2i_{xy} \text{ (iii)}$$

$$\text{Solving (i), (ii) and (iii), } i_{xy} = 2A$$

47. (b) Let n be the number of wrongly connected cells.

Number of cells helping one another $= (12 - n)$

Total e.m.f. of such cells $= (12 - n)E$

Total e.m.f. of cells opposing $= nE$

Resultant e.m.f. of battery $= (12 - n)E - nE = (12 - 2n)E$

Total resistance of cells $= 12r$

($\because$ resistance remains same irrespective of connections of cells)

With additional cells

(a) Total e.m.f. of cells when additional cells help battery $= (12 - 2n)E + 2E$

Total resistance $= 12r + 2r = 14r$

$$\therefore \frac{(12 - 2n)E + 2E}{14r} = 3 \text{(i)}$$

(b) Similarly when additional cells oppose the battery

$$\frac{(12 - 2n)E - 2E}{14r} = 2 \text{(ii)}$$

Solving (i) and (ii), $n = 1$

48. (a) All the conductors have equal lengths. Area of cross-section of A is $\{(\sqrt{3}a)^2 - (\sqrt{2}a)^2\} = a^2$

Similarly area of cross-section of B = Area of cross-section of C = a^2

Hence according to formula $R = \rho \frac{l}{A}$; resistances of all the conductors are equal i.e. $R_1 = R_2 = R_3$

49. (b) Resistance of CD arm $= 2r \cos 72^\circ = 0.62r$

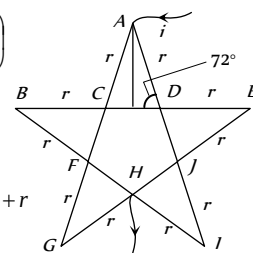
Resistance of CBFC branch

$$\frac{1}{R} = \frac{1}{2r} + \frac{1}{0.62r} = \frac{1}{r} \left(\frac{2.62}{2 \times 0.62} \right)$$

$$\frac{1}{R} = \frac{2.62}{1.24r} \therefore R = \frac{1.24r}{2.62}$$

$$\text{Equivalent } R' = 2R + r = 2 \times \frac{1.24r}{2.62} + r$$

$$= r \left(\frac{2.48}{2.62} + 1 \right) = 1.946r$$

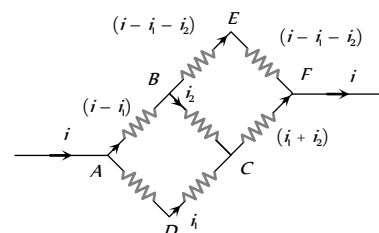


Because the star circuit is symmetrical about the line AH

$\therefore$ Equivalent resistance between A and H

$$\frac{1}{R_{eq}} = \frac{1}{R'} + \frac{1}{R'} \Rightarrow R_{eq} = \frac{R'}{2} = \frac{1.946}{2}r = 0.973r$$

50. (a)



Applying Kirchhoff's law in mesh ABCDA

$$-10(i - i_1) - 10i_2 + 20i_1 = 0 \Rightarrow 3i_1 - i_2 = i \text{(i)}$$

and in mesh BEFCB

$$-20(i - i_1 - i_2) + 10(i_1 + i_2) + 10i_2 = 0$$

$$\Rightarrow 3i_1 + 4i_2 = 2i \text{(ii)}$$

$$\text{From equation (i) and (ii) } i_1 = \frac{2i}{5}, i_2 = \frac{i}{5} \Rightarrow i_{AD} = \frac{2i}{5}$$

51. (d) Let the current in 12Ω resistance is i
Applying loop theorem in closed mesh AEFC A
 $12i = -E + E = 0 \therefore i = 0$

52. (b) Current flowing in the circuit $i = \frac{E}{R} = \frac{10 - 4}{20 + 10} = \frac{1}{5}A$

$$\text{P.D. across AC} = \frac{1}{5} \times 20 = 4V$$

$$\text{P.D. across AN} = 4 + 4 = 8V$$

53. (a) If two resistances are R_1 and R_2 then

$$S = R_1 + R_2 \text{ and } P = \frac{R_1 R_2}{(R_1 + R_2)}$$

$$\text{From given condition } S = nP \text{ i.e. } (R_1 + R_2) = n \left(\frac{R_1 R_2}{R_1 + R_2} \right)$$

$$\Rightarrow (R_1 + R_2)^2 = n R_1 R_2 \Rightarrow (R_1 - R_2)^2 + 4R_1 R_2 = n R_1 R_2$$

$$\text{So } n = 4 + \frac{(R_1 - R_2)^2}{R_1 R_2}. \text{ Hence minimum value of } n \text{ is } 4.$$

54. (b) Voltage sensitivity $= \frac{\text{Current sensitivity}}{\text{Resistance of galvanometer } G}$

$$\Rightarrow G = \frac{10}{2} = 5 \Omega.$$

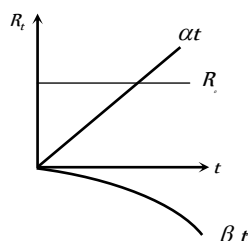
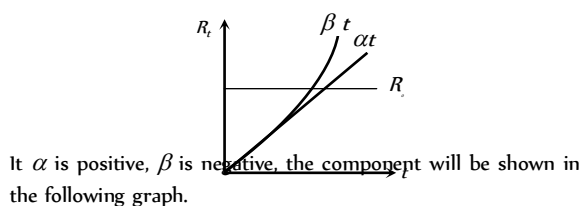
$$\text{Here } i_g = \text{Full scale deflection current} = \frac{150}{10} = 15 \text{ mA}.$$

$$V = \text{Voltage to be measured} = 150 \times 1 = 150 \text{ V}.$$

$$\text{Hence } R = \frac{V}{i_g} - G = \frac{150}{15 \times 10^{-3}} - 5 = 9995 \Omega.$$

Graphical Questions

- (a) For ohmic resistance $V \propto i \Rightarrow V = Ri$ (here R is constant)
- (d) From the curve it is clear that slopes at points A, B, C, D have following order $A > B > C > D$.
and also resistance at any point equals to slope of the $V-i$ curve.
So order of resistance at three points will be $R_A > R_B > R_C > R_D$
- (a) Slope of the $V-i$ curve at any point equal to resistance at that point. From the curve slope for $T_1 >$ slope for T_2 .
 $\Rightarrow R_{T_1} > R_{T_2}$. Also at higher temperature resistance will be higher so $T > T_1$.
- (c) For portion CD slope of the curve is negative i.e. resistance be negative.
- (d) Slope of $V-i$ curve $= R \left(= \frac{\rho l}{A} \right)$. But in given curve axis of i and V are interchanged. So slope of given curve $= \frac{1}{R} \left(= \frac{A}{\rho l} \right)$
i.e. with the increase in length of the wire. Slope of the curve will decrease.
- (c) $E = \frac{iR}{L} = \frac{i\rho}{A} = \frac{neAv_d\rho}{A} \Rightarrow v_d \propto E$ (Straight line)
 $P = i^2 R = \left(\frac{EA}{\rho} \right)^2 R \Rightarrow P \propto E^2$ (Symmetric parabola)
Also $P \propto i^2$ (parabola)
Hence all graphs a, b, d are correct and c is incorrect.
- (b) When we move in the direction of the current in a uniform conductor, the potential difference decreases linearly. When we pass through the cell, from its negative to its positive terminal, the potential increases by an amount equal to its potential difference. This is less than its emf, as there is some potential drop across its internal resistance when the cell is driving current.
- (b) Since the value of R continuously increases, both α and β must be positive.
Actually the components of the given equation are as follows



In this case, the value of R will not increase continuously. Hence the correct option is (c).

- (d) Slope of $V-i$ curve = resistance. Hence $R = \frac{1}{1} = 1 \Omega$
- (a) At point A the slope of the graph will be negative. Hence resistance is negative.
- (b) E.m.f. is the value of voltage, when no current is drawn from the circuit so $E = 2 \text{ V}$. Also $r = \text{slope} = \frac{2}{5} = 0.4 \Omega$
- (d) For conversion of a galvanometer into a voltmeter
 $\frac{V}{R+G} = i_g \Rightarrow \frac{V}{R_V} = i_g$; where $R_V = R + G = \text{Total resistance}$
 $\Rightarrow R_V = \frac{V}{i_g} \Rightarrow R_V \propto V$
- (a) According to ohm's law $V = iR$
 $\Rightarrow \log_e V = \log_e i + \log_e R \Rightarrow \log_e i = \log_e V - \log_e R$
The graph between $\log_e i$ and $\log_e V$ will be a straight line which cut $\log_e V$ axis and its gradient will be positive.
- (c) As we know, for conductors resistance $\propto$ Temperature.
From figure $R \propto T \Rightarrow \tan \theta \propto T \Rightarrow \tan \theta = kT$... (i)
and $R \propto T \Rightarrow \tan (90 - \theta) \propto T \Rightarrow \cot \theta = kT$... (ii)
From equation (i) and (ii) $k(T_2 - T_1) = (\cot \theta - \tan \theta)$
 $(T_2 - T_1) = \left(\frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta} \right) = \frac{(\cos^2 \theta - \sin^2 \theta)}{\sin \theta \cos \theta} = 2 \cot 2\theta$
 $\Rightarrow (T_2 - T_1) \propto \cot 2\theta$
- (b) Let resistivity at a distance ' x ' from left end be $\rho = (\rho_0 + ax)$.
Then electric field intensity at a distance ' x ' from left end will be equal to $E = \frac{i\rho}{A} = \frac{i(\rho_0 + ax)}{A}$ where i is the current flowing through the conductor. It means $E \propto \rho$ or E varies linearly with distance ' x '. But at $x = 0$, E has non-zero value. Hence (b) is correct.
- (d) At an instant approach the student will choose $\tan \theta$ will be the right answer. But it is to be seen here the curve makes the angle θ with the V -axis. So it makes an angle $(90 - \theta)$ with the i -axis.
So resistance = slope = $\tan (90 - \theta) = \cot \theta$.
- (d) Short circuited current $i = \frac{nE}{nr} = \frac{E}{r}$ i.e. i doesn't depend upon n .
- (b) Here internal resistance is given by the slope of graph i.e. $\frac{x}{y}$.
But conductance $= \frac{1}{\text{Resistance}} = \frac{y}{x}$
- (a) $R_{\text{Parallel}} < R_{\text{Series}}$. From graph it is clear that slope of the line A is lower than the slope of the line B . Also slope = resistance, so line A represents the graph for parallel combination.
- (b) To make range n times, the galvanometer resistance should be G/n , where G is initial resistance.

Assertion and Reason

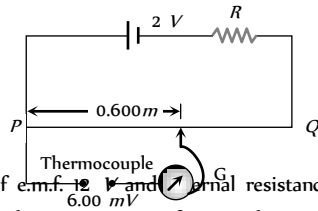
1. (d) Resistivity of a semiconductor decreases with the temperature. The atoms of a semiconductor vibrate with larger amplitudes at higher temperatures thereby increasing its conductivity not resistivity.
2. (d) It is quite clear that in a battery circuit, the point of lowest potential is the negative terminal of the battery and the current flows from higher potential to lower potential.
3. (b) The temperature co-efficient of resistance for metal is positive and that for semiconductor is negative.
In metals free electrons (negative charge) are charge carriers while in P -type semiconductors, holes (positive charge) are majority charge carriers.
4. (a) Here, $E = 2V$, $1 = \frac{2}{2} = 1A$ and $r = 1\Omega$
Therefore, $V = E - ir = 2 - 1 \times 1 = 1V$
5. (a) It is clear that electrons move in all directions haphazardly in metals. When an electric field is applied, each free electron acquires a drift velocity. There is a net flow of charge, which constitutes current. In the absence of electric field this is impossible and hence, there is no current.
6. (c) The metallic body of the electrical appliances is connected to the third pin which is connected to the earth. This is a safety precaution and avoids eventual electric shock. By doing this the extra charge flowing through the metallic body is passed to earth and avoids shocks. There is nothing such as reducing of the heating of connecting wires by three pin connections.
7. (b) On increasing temperature of wire the kinetic energy of free electrons increases and so they collide more rapidly with each other and hence their drift velocity decreases. Also when temperature increases, resistivity increases and resistivity is inversely proportional to conductivity of material.
8. (c) In a conductor there are large number of free electrons. When we close the circuit, the electric field is established instantly with the speed of electromagnetic wave which causes electron drift at every portion of the circuit. Due to which the current is set up in the entire circuit instantly. The current which is set up does not wait for the electrons to flow from one end of the conductor to the other end. It is due to this reason, the electric bulb glows immediately when switch is on.
9. (a) Resistance wire $R = \rho \frac{l}{A}$, where ρ is resistivity of material which does not depend on the geometry of wire. Since when wire is banded, resistivity, length and area of cross-section do not change, therefore resistance of wire also remains same.
10. (c) The resistance of the galvanometer is fixed. In meter bridge experiments, to protect the galvanometer from a high current, high resistance is connected to the galvanometer in order to protect it from damage.
11. (a) Voltmeter measures current indirectly in terms of mass of ions deposited and electrochemical equivalent of the substance $\left(I = \frac{m}{Zt} \right)$. Since value of m and Z are measured to 3rd decimal place and 5th decimal place respectively. The relative error in the measurement of current by voltmeter will be very small as compared to that when measured by ammeter directly.
12. (a) When current flows through a conductor it always remains unchanged, hence no electric field is produced outside it.
13. (b) Here assertion and reason both are correct but the reason is not the correct explanation of assertion.
14. (a) Sensitivity $\propto \frac{1}{\text{Potential gradient}} \propto (\text{Length of wire})$
15. (a) If either the e.m.f. of the driver cell or potential difference across the whole potentiometer wire is lesser than the e.m.f. of the experimental cell, then balance point will not be obtained.
16. (d) Because there is no special attractive force that keeps a person stuck with a high power line. The actual reason is that a current of the order of $0.05 A$ or even less is enough to bring disorder in our nervous system. As a result of it, the affected person may lose temporarily his ability to exercise his nervous control to get himself free from the high power line.
17. (a) Due to high electrical conductivity of copper, it conducts the current without offering much resistance. The copper being a diamagnetic material does not get magnetised due to current through it and hence does not disturb the current in the circuit.

Current Electricity

Self Evaluation Test -19

1. Figure shows a simple potentiometer circuit for measuring a small e.m.f. produced by a thermocouple. The meter wire PQ has a resistance $5\ \Omega$ and the driver cell has an e.m.f. of 2 V . If a balance point is obtained 0.600 m along PQ when measuring an e.m.f. of 6.00 mV , what is the value of resistance R

- (a) $995\ \Omega$
(b) $1995\ \Omega$
(c) $2995\ \Omega$
(d) None of these



2. A car has a fresh battery of e.m.f. 12 V and internal resistance of $0.05\ \Omega$. If the starter motor draws a current of 90 A , the terminal voltage when the starter is on will be

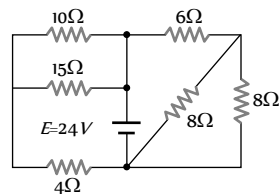
- (a) 12 V
(b) 10.5 V
(c) 8.5 V
(d) 7.5 V

3. If the balance point is obtained at the 35 cm in a metre bridge the resistances in the left and right gaps are in the ratio of

- (a) $7 : 13$
(b) $13 : 7$
(c) $9 : 11$
(d) $11 : 9$

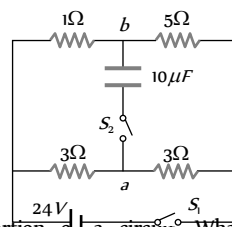
4. Find the equivalent resistance across the terminals of source of e.m.f. 24 V for the circuit shown in figure

- (a) $15\ \Omega$
(b) $10\ \Omega$
(c) $5\ \Omega$
(d) $4\ \Omega$



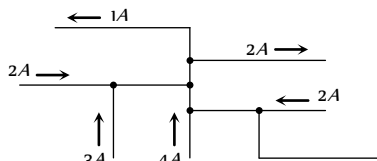
5. In the circuit shown in figure, switch S is initially closed and S_1 is open. Find $V_1 - V_2$

- (a) 4 V
(b) 8 V
(c) 12 V
(d) 16 V



6. The figure here shows a portion of a circuit. What are the magnitude and direction of the current i in the lower right-hand wire

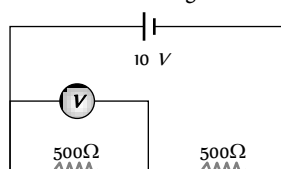
- (a) 7 A
(b) 8 A
(c) 6 A
(d) 2 A



7. A carbon resistor has colour strips as violet, yellow brown and golden. The resistance is

- (a) $641\ \Omega$
(b) $741\ \Omega$
(c) $704\ \Omega$
(d) $407\ \Omega$

8. A voltmeter of resistance $1000\ \Omega$ is connected across a resistance of $500\ \Omega$ in the given circuit. What will be the reading of voltmeter



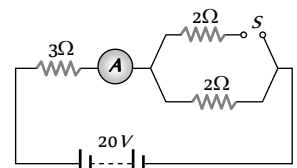
- (a) 1 V
(b) 2 V
(c) 6 V
(d) 4 V

9. A beam contains 2×10^{10} doubly charged positive ions per cubic centimeter, all of which are moving with a speed of 10^6 m/s . The current density is

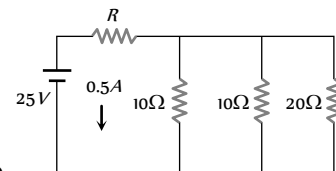
- (a) 6.4 A/m
(b) 3.2 A/m
(c) 1.6 A/m
(d) None of these

10. In the circuit shown, the reading of ammeter when switch S is open and when switch S is closed respectively are

- (a) 3 A and 4 A
(b) 4 A and 5 A
(c) 5 A and 6 A
(d) 6 A and 7 A

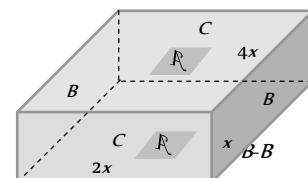


11. In the circuit as shown in figure the



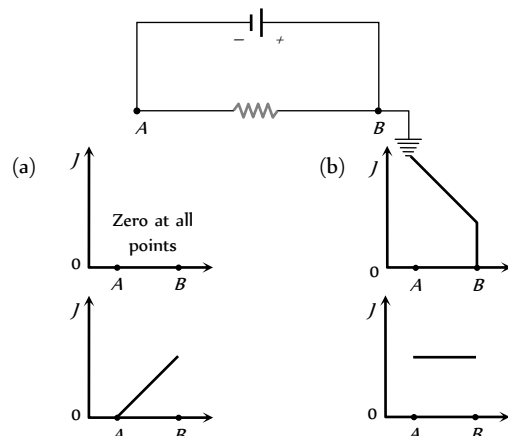
- (a) Resistance $R = 46\ \Omega$
(b) Current through $20\ \Omega$ resistance is 0.1 A
(c) Potential difference across the middle resistance is 2 V
(d) All option are correct

12. In figure shows a rectangular block with dimensions x , $2x$ and $4x$. Electrical contacts can be made to the block between opposite pairs of faces (for example, between the faces labelled $A-A$, $B-B$ and $C-C$). Between which two faces would the maximum electrical resistance be obtained ($A-A$: Top and bottom faces, $B-B$: Left and right faces, $C-C$: Front and rear faces)



- (a) $A-A$
(b) $B-B$
(c) $C-C$
(d) Same for all three pairs

13. A battery is connected to a uniform resistance wire AB and B is earthed. Which one of the graphs below shows how the current density J varies along AB



(c)

(d)

14. A cylindrical metal wire of length l and cross sections area S , has resistance R , conductance G , conductivity σ and resistivity ρ . Which one of the following expressions for σ is valid

(a) $\frac{GR}{\rho}$

(b) $\frac{\rho R}{G}$

(c) $\frac{GS}{l}$

(d) $\frac{Rl}{S}$

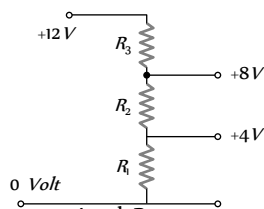
15. A potential divider is used to give outputs of 4 V and 8 V from a 12 V source. Which combination of resistances, (R_1, R_2, R_3) gives the correct voltages ? $R_1 : R_2 : R_3$

(a) 2 : 1 : 2

(b) 1 : 1 : 1

(c) 2 : 2 : 1

(d) 1 : 1 : 2



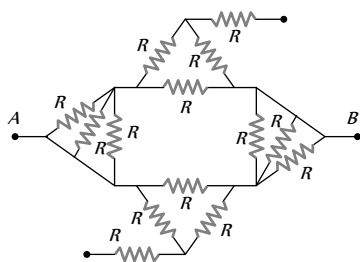
16. Find equivalent resistance between A and B

(a) R

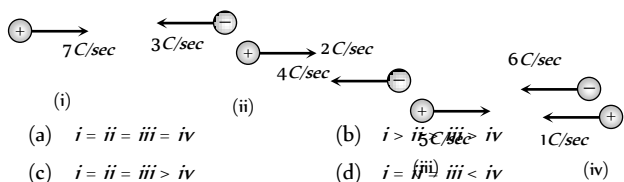
(b) $\frac{3R}{4}$

(c) $\frac{R}{2}$

(d) $2R$



17. Following figure shows four situations in which positive and negative charges moves horizontally through a region and gives the rate at which each charge moves. Rank the situations according to the effective current through the region greatest first



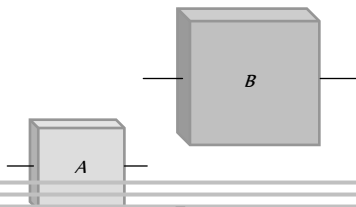
18. A and B are two square plates of same metal and same thickness but length of B is twice that of A. Ratio of resistances of A and B is

(a) 4 : 1

(b) 1 : 4

(c) 1 : 1

(d) 1 : 2



19. A moving coil galvanometer is converted into an ammeter reading upto 0.03 A by connecting a shunt of resistance $4r$ across it and into an ammeter reading upto 0.06 A when a shunt of resistance r is connected across it. What is the maximum current which can be sent through this galvanometer if no shunt is used

(a) 0.01 A

(b) 0.02 A

(c) 0.03 A

(d) 0.04 A

20. Two conductors are made of the same material and have the same length. Conductor A is a solid wire of diameter 1.0 mm. Conductor B is a hollow tube of outside diameter 2.0 mm and inside diameter 1.0 mm. The resistance ratio R_A/R_B will be

(a) 1

(b) 2

(c) 3

(d) 4

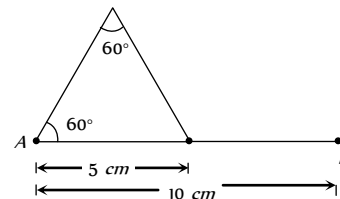
21. A wire has resistance of 24Ω is bent in the following shape. The effective resistance between A and B is

(a) 24Ω

(b) 10Ω

(c) $\frac{16}{3} \Omega$

(d) None of these



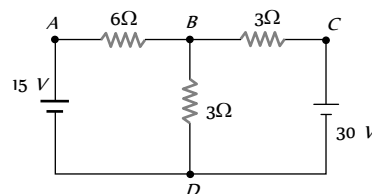
22. In the circuit shown in figure, find the current through the branch BD

(a) 5 A

(b) 0 A

(c) 3 A

(d) 4 A



23. A battery of 24 cells, each of emf 1.5 V and internal resistance 2Ω is to be connected in order to send the maximum current through a 12Ω resistor. The correct arrangement of cells will be

(a) 2 rows of 12 cells connected in parallel

(b) 3 rows of 8 cells connected in parallel

(c) 4 rows of 6 cells connected in parallel

(d) All of these

AS Answers and Solutions

(SET - 19)

1. (a) The voltage per unit length of the metre wire PQ is $\left(\frac{6.00 \text{ mV}}{0.600 \text{ m}}\right)$ i.e. 10 mV/m . Hence potential difference across the metre wire is $10 \text{ mV/m} \times 1 \text{ m} = 10 \text{ mV}$. The

current drawn from the driver cell is $i = \frac{10 \text{ mV}}{5 \Omega} = 2 \text{ mA}$.

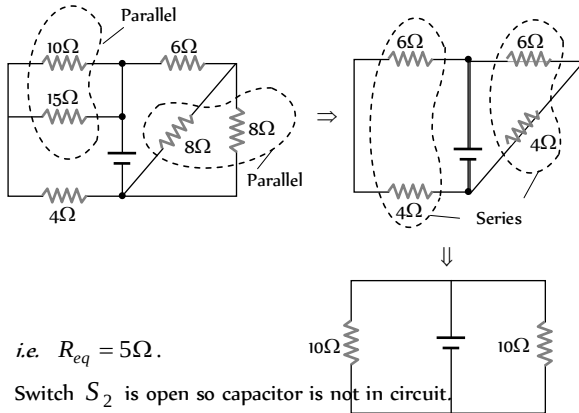
The resistance $R = \frac{(2V - 10 \text{ mV})}{2 \text{ mA}} = \frac{1990 \text{ mV}}{2 \text{ mA}} = 995 \Omega$.

2. (d) $V = E - ir = 12 - 90 \times 0.05 = 12 - 4.5 = 7.5 \text{ V}.$

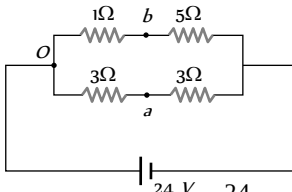
3. (a) Using Wheatstone principle $\frac{P}{Q} = \frac{R}{S} = \frac{R}{100 - l}$

$$= \frac{35}{100 - 35} = \frac{35}{65} = \frac{7}{13}$$

4. (c) Given circuit can be reduced to a simple circuit as shown in figures below



5. (b) Switch S_2 is open so capacitor is not in circuit.



Current through 3Ω resistor $= \frac{24}{3+3} = 4 \text{ A}$

Let potential of point 'O' shown in fig. is V_O

then using ohm's law

$$V_O - V_a = 3 \times 4 = 12 \text{ V} \quad \dots(i)$$

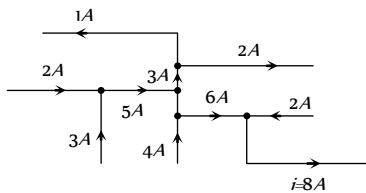
Now current through 5Ω resistor $= \frac{24}{5+1} = 4 \text{ A}$

So $V_O - V_b = 4 \times 1 = 4 \text{ V}$

.....(ii)

From equation (i) and (ii) $V_b - V_a = 12 - 4 = 8 \text{ V}.$

6. (b) By using Kirchoff's junction law as shown below.



7. (b) Using standard colour codes
Violet = 7, yellow = 4, brown = 1 and gold = 5 % (tolerance)

So $R = 74 \times 10^1 \pm 5\% = 740 \pm 5\%$

So its value will be nearest to $741\Omega.$

8. (d) Total current through the circuit

$$i = \frac{10}{\frac{1000}{3} + 500} = \frac{3}{250} \text{ A}$$

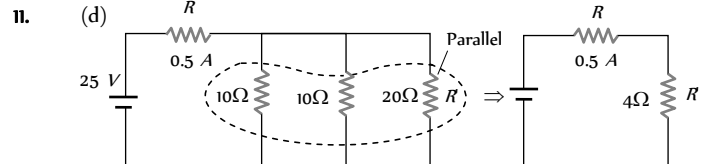
Now voltmeter reading $= i_v \times R_V = \frac{2}{3} \times \frac{3}{250} \times 500 = 4 \text{ V}.$

9. (a) $J = nqv = n(ze)v = \frac{2 \times 10^8 \times 2 \times 1.6 \times 10^{-19} \times 10^5}{(10^{-2})^3} = 6.4 \text{ A/m}$

10. (b) When switch S is open total current through ammeter.

$$i = \frac{20}{(3+2)} = 4 \text{ A}.$$

When switch is closed $i = \frac{20}{3+(2||2)} = 5 \text{ A}.$



$$\frac{1}{R'} = \frac{1}{10} + \frac{1}{10} + \frac{1}{20} \Rightarrow R' = \frac{20}{5} = 4 \Omega$$

Now using ohm's law $i = \frac{25}{R+R'} \Rightarrow 0.5 = \frac{25}{R+4}$

$$\Rightarrow R+4 = \frac{25}{0.5} = 50 \Rightarrow R = 50 - 4 = 46 \Omega$$

Current through 20Ω resistor $= \frac{0.5 \times 5}{20+5} = \frac{2.5}{25} = 0.1 \text{ A}$

Potential difference across middle resistor

$=$ Potential difference across $20\Omega = 20 \times 0.1 = 2 \text{ V}$

12. (c) Let ρ is the resistivity of the material

Resistance for contact A-A

$$R_{AA} = \rho \frac{x}{2x \times 4x} = \frac{\rho}{8x}$$

Similar for contacts B-B and C-C are respectively

$$R_{BB} = \rho \frac{2x}{x \times 4x} = \frac{\rho}{2x} = \frac{4\rho}{8x}$$

$$\text{and } R_{CC} = \rho \frac{4x}{x \times 2x} = \frac{2\rho}{x} = \frac{16\rho}{8x}$$

It is clear maximum resistance will be for contact C-C.

13. (d) Wire AB is uniform so current through wire AB at every across section will be same. Hence current density $J (= i/A)$ at every point of the wire will be same.

14. (a) Conductivity $\sigma = \frac{1}{\rho} \quad \dots(i)$

and conductance $G = \frac{1}{R}$

$$\Rightarrow GR = 1 \quad \dots(ii)$$

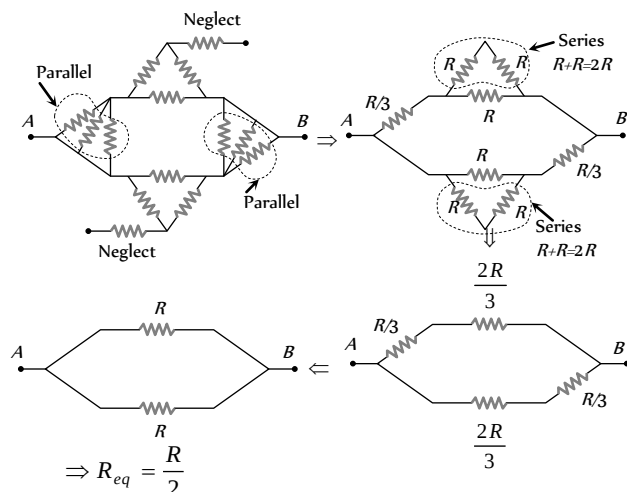
From equation (i) and (ii) $\sigma = \frac{GR}{\rho}$

15. (b) Resistors are connected in series. So current through each resistor will be same

$$\Rightarrow i = \frac{12-8}{R_3} = \frac{8-4}{R_2} = \frac{4-0}{R_1} \Rightarrow \frac{4}{R_3} = \frac{4}{R_2} = \frac{4}{R_1}$$

So, $R_1 : R_2 : R_3 :: 1 : 1 : 1.$

16. (c) Given circuit can be redrawn as follows



17. (c) For figure (i) $i_1 = 7A$
 For figure (ii) $i_2 = 4 + 3 = 7A$
 For figure (iii) $i_3 = 5 + 2 = 7A$
 For figure (iv) $i_4 = 6 - 1 = 5A$

18. (c) $R_A = \frac{\rho l}{l \times t} = \frac{\rho}{t}$ and $R_B = \frac{\rho \times 2l}{2l \times t} = \frac{\rho}{t}$ i.e. $\frac{R_A}{R_B} = 1:1$

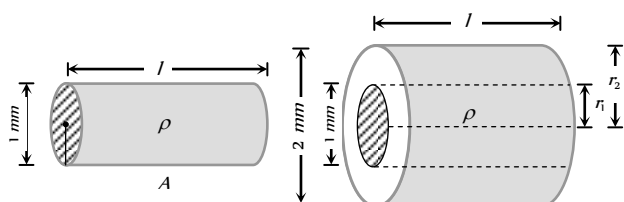
19. (b) $\frac{i_g}{i} = \frac{S}{G+S} \Rightarrow i_g G = (i - i_g)S$
 $\therefore i_g G = (0.03 - i_g)4r$ (i)
 and $i_g G = (0.06 - i_g)r$ (ii)

From (i) and (ii)

$$0.12 - 4i_g = 0.06 - i_g \Rightarrow i_g = 0.02A.$$

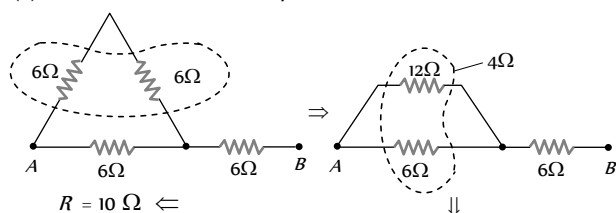
20. (c) For conductor A, $R_A = \frac{\rho l}{\pi r_1^2}$,

$$\text{For conductor B, } R_B = \frac{\rho l}{\pi(r_2^2 - r_1^2)}$$

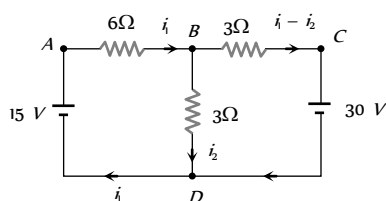


$$\Rightarrow \frac{R_A}{R_B} = \frac{r_2^2 - r_1^2}{r_1^2} = \left(\frac{r_2}{r_1}\right)^2 - 1 = \left(\frac{2}{1}\right)^2 - 1 = 3$$

21. (b) Given resistance of each part will be



22. (a) The current in the circuit are assumed as shown in the fig.



Applying KVL along the loop ABDA, we get

$$-6i - 3i + 15 = 0 \text{ or } 2i + i = 5 \text{(i)}$$

Applying KVL along the loop BCDB, we get

$$-3(i - i) - 30 + 3i = 0 \text{ or } -i + 2i = 10 \text{(ii)}$$

Solving equation (i) and (ii) for i , we get $i = 5A$.

23. (a) Suppose m rows are connected in parallel and each row contains n identical cells (each cell having $E = 15V$ and $r = 2\Omega$) For maximum current in the external resistance R , the necessary condition is $R = \frac{nr}{m}$

$$\Rightarrow 12 = \frac{n \times 2}{m} \Rightarrow n = 6m \text{ (i)}$$

$$\text{Total cells} = 24 = n \times m \text{ (ii)}$$

On solving equations (i) and (ii) $n = 12$ and $m = 2$
 i.e. 2 rows of 12 cells are connected in parallel.



Chapter 20

Heating and Chemical Effect of Current

Joules Heating

When some potential difference V is applied across a resistance R then the work done by the electric field on charge q to flow through the circuit in time t will be $W = qV = Vit = i^2 R t$
 $= \frac{V^2 t}{R}$ Joule. This work appears as thermal energy in the resistor.

Heat produced by the resistance R is
 $H = \frac{W}{J} = \frac{Vit}{4.2} = \frac{i^2 R t}{4.2} = \frac{V^2 t}{4.2 R}$ Cal. This relation is called joules heating.

Electric Power

The rate at which electrical energy is dissipated into other forms of energy is called electrical power *i.e.*

$$P = \frac{W}{t} = Vi = i^2 R = \frac{V^2}{R}$$

(1) **Units** : It's S.I. unit is *Joule/sec* or *Watt*

Bigger S.I. units are *KW*, *MW* and *HP*, remember 1 *HP* = 746 *Watt*

(2) **Rated values** : On electrical appliances (Bulbs, Heater, Geyser *etc.*). Wattage, voltage, *etc.* are printed called rated values *e.g.* If suppose we have a bulb of 40 *W*, 220 *V* then rated power (P_R) = 40 *W* while rated voltage (V_R) = 220 *V*.

(3) **Resistance of electrical appliance** : If variation of resistance with temperature is neglected then resistance of any electrical appliance can be calculated by rated power and rated voltage *i.e.* by using $R = \frac{V_R^2}{P_R}$.

(4) **Power consumed (illumination)** : An electrical appliance (Bulb, heater, *etc.*) consume rated power (P_R) only if applied voltage (V_A) is equal to rated voltage (V_R) *i.e.* If $V_A = V_R$ so $P_{consumed} = P_R$. If $V_A < V_R$ then $P_{consumed} = \frac{V_A^2}{R}$ also we have

$$R = \frac{V_R^2}{P_R} \text{ so } P_{Consumed} \text{ (Brightness)} = \left(\frac{V_A^2}{V_R^2} \right) \cdot P_R$$

(5) **Long distance power transmission** : When power is transmitted through a power line of resistance R , power-loss will be $i^2 R$

Now if the power P is transmitted at voltage V then $P = Vi$

$$\text{i.e. } i = (P/V) \text{ So, Power loss} = \frac{P^2}{V^2} \times R$$

Now as for a given power and line, P and R are constant so
 Power loss $\propto (1/V^2)$

So if power is transmitted at high voltage, power loss will be small and vice-versa. This is why long distance power transmission is carried out at high voltage.

Electricity Consumption

(1) The price of electricity consumed is calculated on the basis of electrical energy and not on the basis of electrical power.

(2) The unit *Joule* for energy is very small hence a big practical unit is considered known as *kilowatt hour (KWH)* or board of trade unit (B.T.U.) or simple unit.

(3) 1 *KWH* or 1 unit is the quantity of electrical energy which dissipates in one hour in an electrical circuit when the electrical power in the circuit is 1 *KW* thus 1 *KWH* = 1000 *W* × 3600 *sec* = 3.6×10^6 *J*.

(4) Important formulae to calculate the no. of consumed units is $n = \frac{\text{Total Watt} \times \text{Total Hours}}{1000}$

Combination of Bulbs

(1) Series combination

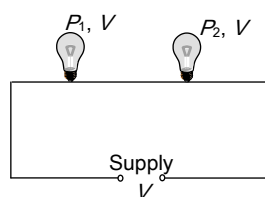


Fig. 20.1

(i) Total power consumed $\frac{1}{P_{total}} = \frac{1}{P_1} + \frac{1}{P_2} + \dots$

(ii) If ' n ' bulbs are identical, $P_{total} = \frac{P}{N}$

(iii) $P_{consumed}$ (Brightness) $\propto V \propto R \propto \frac{1}{P_{rated}}$ i.e. in series

combination bulb of lesser wattage will give more bright light and p.d. appeared across it will be more.

(2) Parallel combination

(i) Total power consumed $P_{total} = P_1 + P_2 + P_3 + P_n$

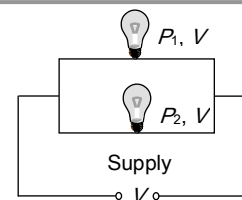


Fig. 20.2

(ii) If ' n ' identical bulbs are in parallel. $P_{total} = nP$

(iii) $P_{consumed}$ (Brightness) $\propto P_R \propto i \propto \frac{1}{R}$ i.e. in parallel

combination, bulb of greater wattage will give more bright light and more current will pass through it.

Chemical Effect of Current

Current can produce or speed up chemical change, this ability of current is called chemical effect (shown by dc not by ac).

(1) **Electrolytes** : The liquids which allows the current to pass through them and also dissociates into ions on passing current through them are called electrolytes *e.g.* solutions of salts, acids and bases in water, *etc.*

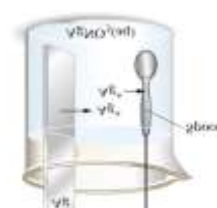
Those liquids which do not allow current to pass through them are called insulators (*e.g.* vegetable oils, distilled water *etc.*)

Solutions of cane sugar, glycerin, alcohol *etc.* are examples of non-electrolytes.

(2) **Electrolysis** : The process of decomposition of electrolyte solution into ions on passing the current through it is called electrolysis.

Practical applications of electrolysis are Electrotyping, extraction of metals from the ores, Purification of metals, Manufacture of chemicals, Production of O_2 and H_2 , Medical applications and electroplating.

(3) **Electroplating** : It is a process of depositing a thin layer of one metal over another metal by the method of electrolysis. The articles of cheap metals are coated with precious metals like silver and gold to make their look more attractive. The



article to be electroplated is made the cathode and the metal to be deposited is made the anode. A soluble salt of the precious metal is taken as the electrolyte. (If gold is to be coated then auric chloride is used as electrolyte).

(4) **Voltameter** : The vessel in which the electrolysis is carried out is called a voltameter. It contains two electrodes and electrolyte. It is also known as electrolytic cell.

Table 20.1 : Types of voltameters

Volatameter	Anode/ cathode	Electrolyte	Deposition
Cu voltameter	Cathode may be of any material but anode must be of Cu	CuSO ₄ or CuCl ₂	At cathode Cu deposited
Ag voltameter	Cathode may be of any material but anode must be of Ag	AgNO ₃	At cathode Ag deposited
Water voltameter	Both electrode	Acidulated water	H ₂ and O ₂ gases are

are made of platinum (Pt)	collects over the cathode and anode respectively in the ratio of 2 : 1
---------------------------	--

Faraday's Law of Electrolysis

(1) **First law** : It states that the mass (m) of substance deposited at the cathode during electrolysis is directly proportional to the quantity of electricity (total charge q) passed through the electrolyte *i.e.* $m \propto q$ or $m = zq = zit$, where the constant of proportionality z is called *electrochemical equivalent (E.C.E.) of the substance*.

Therefore we have $m = zit$. If $q = 1$ coulomb, then we have $m = z \times 1$ or $z = m$

Hence, the electrochemical equivalent of substance may be defined as the mass of its substance deposited at the cathode, when one coulomb of charge passes through the electrolyte.

S.I. unit of electrochemical equivalent of a substance is kilogram coulomb⁻¹ (kg-C⁻¹).

Table 20.2 : E.C.E. for certain substances

Element	Atomic weight	Atomic number	Valency	E.C.E. (Z) in kg / C
Hydrogen	1.0008	1	1	10.4×10^{-9}
Oxygen	15.999	8	2	82.9×10^{-9}
Aluminium	26.982	13	3	93.6×10^{-9}
Chromium	51.996	24	3	179.6×10^{-9}
Nickel	58.710	28	2	304.0×10^{-9}
Copper	63.546	29	2	329.4×10^{-9}
Zinc	65.380	30	2	338.7×10^{-9}
Silver	107.868	47	1	1118×10^{-9}
Gold	196.966	79	3	681.2×10^{-9}

(2) **Second law** : If same quantity of electricity is passed through different electrolytes, masses of the substance

deposited at the respective cathodes are directly proportional to

$$\text{their chemical equivalents i.e. } m \propto E \Rightarrow \frac{m_1}{m_2} = \frac{E_1}{E_2}$$

Let m be the mass of the ions of a substance liberated, whose chemical equivalent is E . Then, according to Faraday's second law of electrolysis, $m \propto E$ or $m = \text{constant} \times E$ or $\frac{m}{E} = \text{constant}$

Chemical equivalent E also known as equivalent weight in gm i.e. $E = \frac{\text{Atomic mass (A)}}{\text{Valancy (V)}}$

(3) **Relation between chemical equivalent and electrochemical equivalent** : Suppose that on passing same amount of electricity q through two different electrolytes, masses of the two substances liberated are m_1 and m_2 . If E_1 and E_2 are their chemical equivalents, then from Faraday's second law, we have $\frac{m_1}{m_2} = \frac{E_1}{E_2}$. Also from Faraday's first law $\frac{m_1}{m_2} = \frac{z_1}{z_2}$

$$\text{So } \frac{z_1}{z_2} = \frac{E_1}{E_2} \Rightarrow z \propto E$$

(4) **Faraday constant** : As we discussed above $E \propto z$
 $\Rightarrow E = Fz \Rightarrow z = \frac{E}{F} = \frac{A}{VF}$. ' F ' is proportionality constant called Faraday's constant.

As $z = \frac{E}{F}$ and $z = \frac{m}{Q}$ so $\frac{E}{F} = \frac{m}{Q}$ hence if $Q = 1 \text{ Faraday}$ then $E = m$ i.e. If electricity supplied to a voltmeter is 1 Faraday then amount of substance liberated or deposited is (in gm) equal to the chemical equivalent.

Electro Chemical Cell

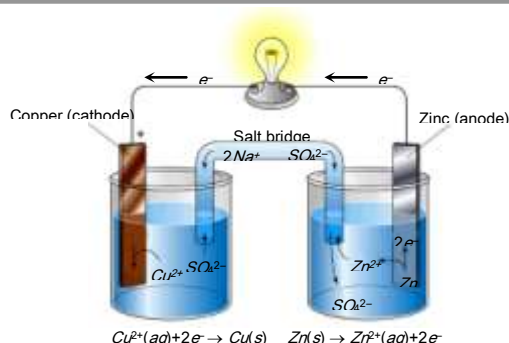


Fig. 20.4

It is an arrangement in which the chemical energy is converted into electrical energy due to chemical action taking place in it.

(1) **Primary cell** : Is that cell in which electrical energy is produced due to chemical energy. In the primary cell, chemical reaction is irreversible. This cell can not be recharged. Examples of primary cells are Voltaic cell, Daniel cell, Leclanche cell and Dry cell etc.

(2) **Secondary cell** : A secondary cell is that cell in which the electrical energy is first stored up as a chemical energy and when the current is taken from the cell, the chemical energy is reconverted into electrical energy. In the secondary cell chemical reactions are reversible. The secondary cells are also called storage cell or accumulator. The commonly used secondary cells is lead accumulator.

(3) **Defects In a primary cell** : In voltaic cell there are two main defects arises.

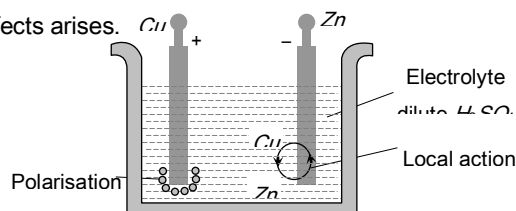


Fig. 20.5

Local action : It arises due to the presence of impurities of iron, carbon etc. on the surface of commercial Zn rod used as an electrode. The particles of these impurities and Zn in contact with sulphuric acid form minute voltaic cell in which small local electric currents are set up resulting in the wastage of Zn even when the cell is not sending the external current.

Removal : By amalgamating Zn rod with mercury (i.e. the surface of Zn is coated with Hg).

Polarisation : It arises, when the positive H^+ ions, which are formed by the action of Zn on sulphuric acid, travel towards the Cu rod and after transferring, the positive charge converted into H_2 gas atoms and get deposited in the form of neutral layer of a gas on the surface of Cu rod. This weakens the action of cell.

Removal : Either by brushing the anode to remove the layer or by using a depolariser (*i.e.* some oxidising agent MnO_2 , $CuSO_4$ etc which may oxidise H_2 into water).

Thermo electric effect of current



If two wires of different metals are joined at their ends so as to form two junctions, then the resulting arrangement is called a "Thermo couple".

Seeback Effect

(1) **Definition** : When the two junctions of a thermo couple are maintained at different temperatures, then a current starts flowing through the loop known as thermo electric current. The potential difference between the junctions is called thermo electric emf which is of the order of a few micro-volts per degree temperature difference ($\mu V/^{\circ}C$).

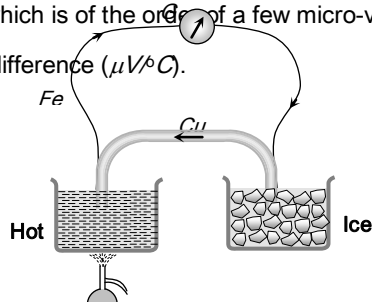


Fig. 20.6

(2) **Seebeck series** : The magnitude and direction of thermo emf in a thermocouple depends not only on the temperature difference between the hot and cold junctions but also on the nature of metals constituting the thermo couple.

(i) Seebeck arranged different metals in the decreasing order of their electron density. Few metals forming the series are as below.

$Sb, Fe, Cd, Zn, Ag, Au, Cr, Sn, Pb, Hg, Mn, Cu, Pt, Co, Ni, Bi$

(ii) Thermo electric emf is directly proportional to the distance between the two metals in series. Farther the metals in the series forming the thermo couple greater is the thermo emf. Thus maximum thermo emf is obtained for **Sb-Bi** thermo couple.

(iii) The current flow at the hot junction of the thermocouple is from the metal occurring later in the series towards that occurring earlier. Thus, in the **copper-iron** thermocouple the current flows from copper (Cu) to iron (Fe) at the hot junction. This may be remembered easily by the **hot coffee**.

(3) **Variation of thermo emf with temperature** : In a thermocouple as the temperature of the hot junction increases keeping the cold junction at constant temperature (say $0^{\circ}C$). The thermo emf increases till it becomes maximum at a certain temperature.

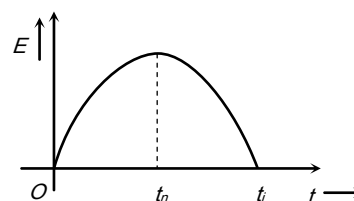


Fig. 20.7

(i) Thermo electric emf is given by the equation $E = \alpha t + \frac{1}{2} \beta t^2$ where α and β are thermo electric constant having units are $\text{volt}/^{\circ}C$ and $\text{volt}/^{\circ}C^2$ respectively (t = temperature of hot junction). For E to be maximum (at $t = t_n$)

$$\frac{dE}{dt} = 0 \quad \text{i.e.} \quad \alpha + \beta t_n = 0 \Rightarrow t_n = -\frac{\alpha}{\beta}$$

(ii) The temperature of hot junction at which thermo emf becomes maximum is called neutral temperature (t_n). Neutral temperature is constant for a thermo couple (*e.g.* for $Cu-Fe$, $t_n = 270^{\circ}C$)

(iii) Neutral temperature is independent of the temperature of cold junction.

(iv) If temperature of hot junction increases beyond neutral temperature, thermo emf start decreasing and at a particular temperature it becomes zero, on heating slightly further, the direction of emf is reversed. This temperature of hot junction is called temperature of inversion (t_i).

(v) Relation between t_n , t_i and t_c is $t_n = \frac{t_i + t_c}{2}$

(4) **Thermo electric power** : The rate of change of thermo emf with the change in the temperature of the hot junction is called thermoelectric power.

It is also given by the slope of parabolic curve representing the variation of thermo emf with temperature of the hot junction, as discussed in previous section.

The thermo electric power $\left(\frac{dE}{dt}\right)$ is also called **Seebeck coefficient**. Differentiating both sides of the equation of thermo emf with respect to t , we have thermoelectric power

$$P = \frac{dE}{dt} = \frac{d}{dt} \left(\alpha t + \frac{1}{2} \beta t^2 \right)$$

$$\Rightarrow P = \alpha + \beta t$$

The equation of the thermo electric power is of the type $y = mx + c$, so the graph of thermo electric power is as shown.

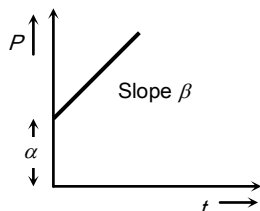


Fig. 20.8

(5) Laws of thermoelectricity

(i) **Law of successive temperature** : If initially temperature limits of the cold and the hot junction are t_1 and t_2 , say the thermo emf is $E_{t_1}^{t_2}$. When the temperature limits are t_2 and t_3 , then say the thermo emf is $E_{t_2}^{t_3}$ then $E_{t_1}^{t_2} + E_{t_2}^{t_3} = E_{t_1}^{t_3}$ where $E_{t_1}^{t_3}$ is the thermo emf when the temperature limits are $E_{t_1}^{t_3}$

(ii) **Law of intermediate metals** : Let A , B and C be the three metals of Seebeck series, where B lies between A and C . According to this law, $E_A^B + E_B^C = E_A^C$

When tin is used as a soldering metal in $Fe-Cu$ thermocouple then at the junction, two different thermo couples are being

formed. One is between iron and tin and the other is between tin and copper, as shown in figure

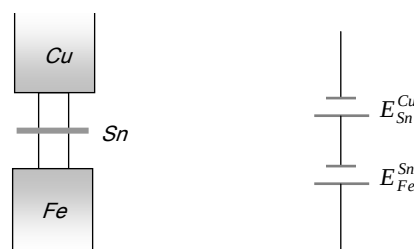


Fig. 20.9

If the soldering metal does not lie between two metals (in Seebeck series) of thermocouple then the resultant emf will be subtractive.

Peltier Effect

When current is passed through a junction of two different metals, the heat is either evolved or absorbed at the junction. This effect is known as Peltier effect. It is the reverse of Seebeck effect. (When a positive charge flows from high potential to low potential, it releases energy and when positive charge flows from low potential to high potential it absorbs energy.)

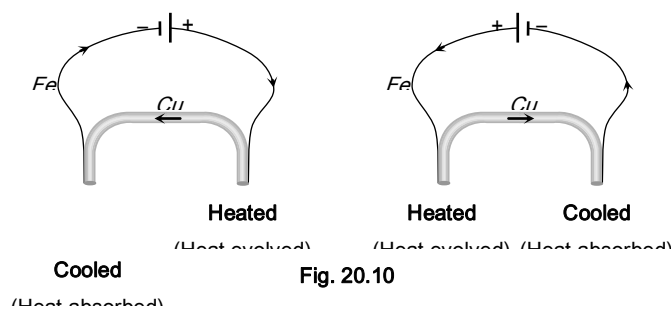


Fig. 20.10

Peltier co-efficient (π) : Heat absorbed or liberated at the junction is directly proportional to the charge passing through the junction i.e. $H \propto Q \Rightarrow H = \pi Q$; where π is called Peltier co-efficient. It's unit is J/C or *volt*.

Peltier co-efficient of a junction is the amount of heat absorbed or liberated per sec. When 1 amp of current is passed to the thermo couple.

It is found that $\pi = T \frac{dE}{dT} = T \times S$; where T is in Kelvin and

$$\frac{dE}{dT} = P = \text{Seebeck coefficient } S$$

Thomson's Effect

In Thomson's effect we deal with only metallic rod and not with thermocouple as in Peltier's effect and Seebeck's effect. (That's why sometimes it is known as homogeneous thermo electric effect. When a current flows through an unequally heated metal, there is an absorption or evolution of heat in the body of the metal. This is Thomson's effect.

(i) **Positive Thomson's effect** : In positive Thomson's effect it is found that hot end is at high potential and cold end is at low potential. Heat is evolved when current is passed from hotter end to the colder end and heat is absorbed when current is passed from colder end to the hotter end. The metals which shows positive Thomson's effect are *Cu, Sn, Ag, Co, Zn... etc.*

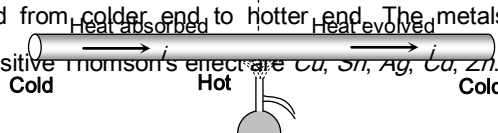


Fig. 28.11

(ii) **Negative Thomson's effect** : In the elements which show negative Thomson's effect, it is found that the hot end is at low potential and the cold end is at higher potential. Heat is evolved when current is passed from colder end to the hotter end and heat is absorbed when current flows from hotter end to colder end. The metals which shows negative. Thomson's effect are *Fe, Co, Bi, Pt, Hg ... etc.*

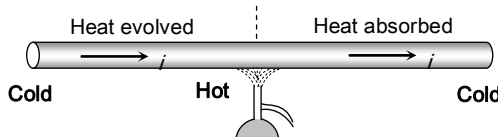


Fig. 20.12

Thomson's co-efficient : In Thomson's effect it is found that heat released or absorbed is proportional to $Q\Delta\theta$ i.e. $H \propto Q\Delta\theta \Rightarrow H = \sigma Q\Delta\theta$ where σ = Thomson's coefficient. Its unit is *Joule/coulomb°C* or *volt/°C* and $\Delta\theta$ = temperature difference.

If $Q = 1$ and $\Delta\theta = 1$ then $\sigma = H$ so the amount of heat energy absorbed or evolved per second between two points of a conductor having a unit temperature difference, when a unit current is passed is known as Thomson's co-efficient for the material of a conductor.

It can be proved that Thomson co-efficient of the material of conductor $\sigma = -T \frac{d^2E}{dT^2} = -T \left(\frac{dS}{dT} \right) = T \times \beta$; where β = Thermo electric constant $= \frac{dS}{dT}$

Application of Thermo Electric Effect

(1) **To measure temperature** : A thermocouple is used to measure very high (2000°C) as well as very low (-200°C) temperature in industries and laboratories. The thermocouple used to measure very high temperature is called pyrometer.

(2) **To detect heat radiation** : A thermopile is a sensitive instrument used for detection of heat radiation and measurement of their intensity. It is based upon Seebeck effect.

A thermopile consists of a number of thermocouples of *Sb-Bi*, all connected in series.

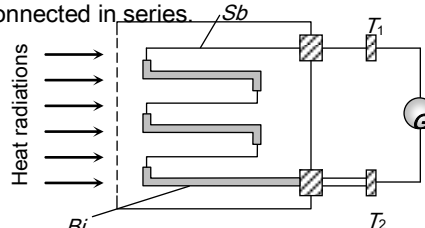


Fig. 20.13

Table 20.3: Heating effect and Thermo-electric effects

S.No.	Joule's effect	Peltier's effect	Seebeck effect	Thomson's effect
1.	Heat produced is directly proportional to the square of the current passing through a	Heat produced or absorbed at a junction is proportional to the current through the junction.	Here temperature difference of junction is used to produce thermo e.m.f. and vice versa.	Thomson's heat is proportional to the current passing through the conductor.

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	conductor.			
2.	This effect is produced due to collision of free electrons with positive ions of the current carrying conductor.	This effect is produced when current is passed through junction of suitable materials.	This effect produced when junctions of a thermocouple are kept at different temperatures.	This effect is produced when parts of same conductor are kept at different temperature.
3.	It is not a reversible effect.	It is a reversible effect	It is a reversible effect	It is a reversible effect
4.	Heat produced depends upon resistance (and thus temperature also) of the conductor.	Heat exchange depends upon nature of conductors and temperature of the junctions.	This effect depends upon nature of materials used to form junctions and temperature of junctions.	This effect depends upon nature of conductor and temperature difference of different parts of the conductor.
5.	It is basically a heating effect	It can be heating as well as cooling effect.	Different junctions are at different temperature.	It is heating as well as cooling effect.

This instrument is so sensitive that it can detect heat radiations from a match stick lighted at a distance of 50 metres from the thermopile.

(3) **Thermoelectric refrigerator** : The working of thermoelectric refrigerator is based on Peltier effect.

(4) **Thermoelectric generator** : Thermocouple can be used to generate electric power using Seebeck effect in remote areas.

(5) **Thermo-couple meter** : The current to be measured passes through a resistance where heat is generated in the amount of $i^2 R$ joule/sec. The hot junction of the thermocouple is in contact with this resistance, and resulting thermoelectric current gives deflection in the galvanometer G .

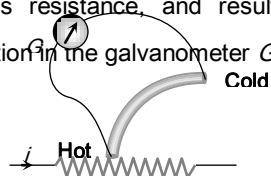
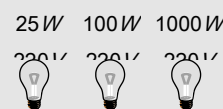


Fig. 20.14

If $V_{Applied} < V_{Rated}$ then % drop in output power of electrical device = $\frac{(P_R - P_{consumed})}{P_R} \times 100$

✍ Different bulbs



- ⇒ Resistance $R_{25} > R_{100} > R_{1000}$
- ⇒ Thickness of filament $t_{1000} > t_{100} > t_{25}$
- ⇒ Brightness $B_{1000} > B_{100} > B_{25}$

✍ Time taken by heater to raise the temperature by $\Delta\theta$ of m kg (or m litre) water is given by $t = \frac{4180(\text{or } 4200)m \Delta\theta}{p}$

✍ Necessary series resistance to glow a bulb, if $V_{Applied} > V_{Rated}$

$$R = \left(\frac{V_{Applied} - V_{Rated}}{P_R} \right) \times V_R \quad (P_R = \text{Rated power of bulb})$$

✍ When some potential difference applied across the conductor then collision of free electrons with ions of the lattice result's in conversion of electrical energy into heat energy

✍ If a heating coil of resistance R , (length l) consumed power P , when voltage V is applied to it then by keeping V constant if it is cut in n equal parts then resistance of each part will be R/n and from $P_{consumed} \propto \frac{1}{R}$, power consumed by each

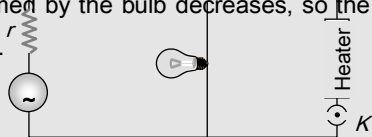
Tips & Tricks

part $P' = nP$.

✍ In series a device of higher power rating consumes less power.

✍ Consider that n bulbs are connected in series across V volt supply. If one bulb gets fused and $(n - 1)$ bulbs are again connected in series across same supply, the illumination will be more with $(n - 1)$ bulbs than n bulbs but risk of fusing of bulbs will increase.

✍ When a heavy current appliance such as motor, heater or geyser is switched on, it will draw a heavy current from the source so that terminal voltage of source decreases. Hence power consumed by the bulb decreases, so the light of bulb becomes less.



✍ If ρ is the density of the material deposited and A is the area of deposition then the thickness (d) of the layer of the material deposited in electroplating process is $d = \frac{m}{\rho A} = \frac{Zit}{\rho A}$; where m = deposited mass, Z = electro chemical equivalent, i = electric current.

✍ Charging current for a secondary cell

$$= \frac{\text{e.m.f. of charger} - \text{e.m.f. of cell}}{\text{Total resistance of the circuit}}$$

✍ Efficiency of a cell is given by $\eta = \frac{R}{r + R}$ where R is external resistance and r is internal resistance.

✍ The efficiency of cell is 50% when the power dissipated in the external circuit is maximum.

✍ Thermo couple can be compared to a heat engine. It absorbs heat at the junction (source) converts heat into electric energy (which appears as the circulating electric current) and rejects the remaining heat to cold junction (Sink).

Heating Effect of Current

- One *kilowatt* hour is equal to [NCERT 1974; MP PMT 2002]
 - 36×10^5 joules
 - 36×10^3 joules
 - 10^3 joules
 - 10^5 joules
- If R_1 and R_2 are respectively the filament resistances of a 200 *watt* bulb and 100 *watt* bulb designed to operate on the same voltage, then [NCERT 1980; CPMT 1991, 97]
 - R_1 is two times R_2
 - R_2 is two times R_1
 - R_2 is four times R_1
 - R_1 is four times R_2
- Two electric bulbs, one of 200 *volt* 40 *watt* and the other 200 *volt* 100 *watt* are connected in a house wiring circuit [NCERT 1971; CBSE PMT 2000]
 - They have equal currents through them
 - The resistance of the filaments in both the bulbs is same
 - The resistance of the filament in 40 *watt* bulb is more than the resistance in 100 *watt* bulb
 - The resistance of the filament in 100 *watt* bulb is more than the resistance in 40 *watt* bulb
- The two bulbs as in the above question are connected in series to a 200 *volt* line. Then
 - The potential drop across the two bulbs is the same
 - The potential drop across the 40 *watt* bulb is greater than the potential drop across the 100 *watt* bulb
 - The potential drop across the 100 *W* bulb is greater than the potential drop across the 40 *W* bulb
 - The potential drop across both the bulb is 200 *volt*
- Forty electric bulbs are connected in series across a 220 *V* supply. After one bulb is fused, the remaining 39 are connected again in series across the same supply. The illumination will be



Ordinary Thinking

Objective Questions

[NCERT 1972; Haryana CEE 1996; DPMT 2001]

- (a) More with 40 bulbs than with 39
- (b) More with 39 bulbs than with 40
- (c) Equal in both the cases
- (d) In the ratio of $49^2 : 39^2$

6. The material of fuse wire should have

[BHU 1999; MH CET 2001; CBSE PMT 2003]

- (a) A high specific resistance and high melting point
- (b) A low specific resistance and low melting point
- (c) A high specific resistance and low melting point
- (d) A low specific resistance and a high melting point

7. Two electric bulbs whose resistances are in the ratio of 1 : 2 are connected in parallel to a constant voltage source. The powers dissipated in them have the ratio

[NCERT 1977; MP PMT 1994, 2000]

- (a) 1 : 2
- (b) 1 : 1
- (c) 2 : 1
- (d) 1 : 4

8. A heater coil is cut into two parts of equal length and one of them is used in the heater. The ratio of the heat produced by this half coil to that by the original coil is

[NCERT 1972; AIEEE 2005; CBSE PMT 2005]

- (a) 2 : 1
- (b) 1 : 2
- (c) 1 : 4
- (d) 4 : 1

9. Resistance of one carbon filament and one tungsten lamp are measured individually when the lamp are lit and compared with their respective resistances when cold. Which one of the following statements will be true [NCERT 972]

- (a) Resistance of the carbon filament lamp will increase but that of the tungsten will diminish when hot
- (b) Resistance of the tungsten filament lamp will increase but that of carbon will diminish when hot
- (c) Resistances of both the lamps will increase when hot
- (d) Resistances of both the lamps will decrease when hot

10. The mechanism of the heat produced in a conductor when an electric current flows through it, can be explained on the basis of

- (a) Viscosity
- (b) Friction
- (c) Free electron theory
- (d) Gauss's theorem

11. Two electric bulbs whose resistances are in the ratio of 1 : 2 are connected in series. The powers dissipated in them have the ratio [NCERT 1977]

- (a) 1 : 2
- (b) 2 : 1
- (c) 1 : 1
- (d) 1 : 4

12. You are given a resistance wire of length 50 cm and a battery of negligible resistance. In which of the following cases is largest amount of heat generated

- (a) When the wire is connected to the battery directly
- (b) When the wire is divided into two parts and both the parts connected to the battery in parallel
- (c) When the wire is divided into four parts and all the four connected to the battery in parallel
- (d) When only half the wire is connected to the battery

13. What is immaterial for an electric fuse wire
[MNR 1984; MP PMT 2002; CPMT 1996, 2003]
- Its specific resistance
 - Its radius
 - Its length
 - Current flowing through it
14. The electric bulbs have tungsten filaments of same length. If one of them gives 60 watt and other 100 watt, then
[NCERT 1979]
- 100 watt bulb has thicker filament
 - 60 watt bulb has thicker filament
 - Both filaments are of same thickness
 - It is possible to get different wattage unless the lengths are different
15. Three equal resistors connected in series across a source of e.m.f. together dissipate 10 watt. If the same resistors are connected in parallel across the same e.m.f., then the power dissipated will be
[CBSE PMT 1998; KCET (Engg.) 1999; MP PMT 2003]
- 10 watt
 - 30 watt
 - 10/3 watt
 - 90 watt
16. How much energy in kilowatt hour is consumed in operating ten 50 watt bulbs for 10 hours per day in a month (30 days).
[NCERT 1978, 80; CPMT 1991]
- 1500
 - 5,000
 - 15
 - 150
17. (1) The product of a volt and a coulomb is a joule.
(2) The product of a volt and an ampere is a joule/second.
(3) The product of volt and watt is horse power.
(4) Watt-hour can be measured in terms of electron volt.
State if
[NCERT 1978; MP PMT 2003]
- All four are correct
 - (1), (2) and (4) are correct
 - (1) and (3) are correct
 - (3) and (4) are correct
18. A 25 W, 220 V bulb and a 100 W, 220 V bulb are connected in parallel across a 440 V line
[CBSE PMT 2001]
- Only 100 watt bulb will fuse
 - Only 25 watt bulb will fuse
 - Both bulbs will fuse
 - None of the bulbs will fuse
19. Two electric lamps of 40 watt each are connected in parallel. The power consumed by the combination will be
[CPMT 1984]
- 20 watt
 - 60 watt
 - 80 watt
 - 100 watt
20. Two heating coils, one of fine wire and the other of thick wire of the same material and of the same length are connected in series and in parallel. Which of the following statement is correct
- In series fine wire liberates more energy while in parallel thick wire will liberate more energy
 - In series fine wire liberates less energy while in parallel thick wire will liberate more energy
 - Both will liberate equally
 - In series the thick wire will liberate more while in parallel it will liberate less energy
21. An electric bulb is rated 220 volt and 100 watt. Power consumed by it when operated on 110 volt is
[CPMT 1986; MP PMT 1986, 94; AFMC 2000]
- 50 watt
 - 75 watt
 - 90 watt
 - 25 watt
22. A 25 watt, 220 volt bulb and a 100 watt, 220 volt bulb are connected in series across a 220 volt lines. Which electric bulb will glow more brightly
[MP PET 1999; MP PMT 1999]
- 25 watt bulb
 - 100 watt bulb
 - First 25 watt and then 100 watt
 - Both with same brightness
23. A resistor R_1 dissipates the power P when connected to a certain generator. If the resistor R_2 is put in series with R_1 , the power dissipated by R_1
[CPMT 1985; MNR 1998]
- Decreases
 - Increases
 - Remains the same
 - Any of the above depending upon the relative values of R_1 and R_2
24. An electric fan and a heater are marked as 100 watt, 220 volt and 1000 watt, 220 volt respectively. The resistance of the heater is
- Zero
 - Greater than that of the fan
 - Less than that of the fan
 - Equal to that of the fan
25. According to Joule's law, if the potential difference across a conductor having a material of specific resistance remains constant, then the heat produced in the conductor is directly proportional to
- ρ
 - ρ^2
 - $\frac{1}{\sqrt{\rho}}$
 - $\frac{1}{\rho}$
26. Two heater wires of equal length are first connected in series and then in parallel. The ratio of heat produced in the two cases is
[MNR 1987; UPSEAT 1999; MP PMT 1996, 2000, 01; AIIMS 2000; MP PET 1999, 2002; BHU 2004; Pb PET 2004]
- 2 : 1
 - 1 : 2
 - 4 : 1
 - 1 : 4
27. Two bulbs of equal wattage, one having carbon filament and the other having a tungsten filament are connected in series to the mains, then
- Both bulbs glow equally
 - Carbon filament bulb glows more
 - Tungsten filament bulbs glows more
 - Carbon filament bulb glows less
28. Two identical heaters rated 220 volt, 1000 watt are placed in series with each other across 220 volt lines. If resistance do not change with temperature, then the combined power is
- 1000 watt
 - 2000 watt
 - 500 watt
 - 4000 watt
29. A 25 watt, 220 volt bulb and a 100 watt, 220 volt bulb are connected in parallel across a 220 volt line. Which bulb will glow more brightly
- 25 watt bulb
 - 100 watt bulb
 - Both will have same brightness

- (d) First 25 watt then 100 watt
30. If two bulbs of wattage 25 and 100 respectively each rated at 220 volt are connected in series with the supply of 440 volt, then which bulbs will fuse [MNR 1988]
 (a) 100 watt bulb (b) 25 watt bulb
 (c) None of them (d) Both of them
31. If current in an electric bulb changes by 1%, then the power will change by [AFMC 1996]
 (a) 1% (b) 2%
 (c) 4% (d) $\frac{1}{2}\%$
32. Two identical batteries, each of e.m.f. 2 volt and internal resistance 1.0 ohm are available to produce heat in an external resistance $R = 0.5 \text{ ohm}$ by passing a current through it. The maximum Joulean power that can be developed across R using these batteries is [CBSE PMT 1990; BHU 1997]
 (a) 1.28 watt (b) 2.0 watt
 (c) $\frac{8}{9}$ watt (d) 3.2 watt
33. A constant voltage is applied between the two ends of a metallic wire. If both the length and the radius of the wire are doubled, the rate of heat developed in the wire [MP PMT 1996]
 (a) Will be doubled (b) Will be halved
 (c) Will remain the same (d) Will be quadrupled
34. The heating coils rating at 220 volt and producing 50 cal/sec heat are available with the resistances 55 Ω , 110 Ω , 220 Ω and 440 Ω . The heater of maximum power will be of [MP PMT 1985]
 (a) 440 Ω (b) 220 Ω
 (c) 110 Ω (d) 55 Ω
35. Which of the following statement is false
 (a) Heat produced in a conductor is proportional to its resistance
 (b) Heat produced in a conductor is proportional to the square of the current
 (c) Heat produced in a conductor is proportional to charge
 (d) Heat produced in a conductor is proportional to the time for which current is passed
36. On an electric heater 220 volt and 1100 watt are marked. On using it for 4 hours, the energy consumed in kWh will be
 (a) 2 (b) 4.4
 (c) 6 (d) 8
37. An electric heater kept in vacuum is heated continuously by passing electric current. Its temperature [MP PET 1993]
 (a) Will go on rising with time
 (b) Will stop after sometime as it will loose heat to the surroundings by conduction
 (c) Will rise for sometime and there after will start falling
 (d) Will become constant after sometime because of loss of heat due to radiation
38. Heat produced in a wire of resistance R due to current flowing at constant potential difference is proportional to [MP PET 1993]
 (a) $\frac{1}{R^2}$ (b) $\frac{1}{R}$
- (c) R (d) R^2
39. The power rating of an electric motor which draws a current of 3.75 amperes when operated at 200 V is about
 (a) 1 H.P. (b) 500 W
 (c) 54 W (d) 750 H.P.
40. An electric bulb of 100 watt is connected to a supply of electricity of 220 V. Resistance of the filament is [EAMCET 1981, 82; MP PMT 1993, 97]
 (a) 484 Ω (b) 100 Ω
 (c) 22000 Ω (d) 242 Ω
41. A cable of resistance 10 Ω carries electric power from a generator producing 250 kW at 10000 volt. The current in the cable is
 (a) 25 A (b) 250 A
 (c) 100 A (d) 1000 A
42. In the above question, the power lost in the cable during transmission is
 (a) 12.5 kW (b) 6.25 kW
 (c) 25 kW (d) 3.15 kW
43. The heat generated through 2 ohm and 8 ohm resistances separately, when a condenser of 200 μF capacity charged to 200 V is discharged one by one, will be [MP PET 1993]
 (a) 4 J and 16 J respectively
 (b) 16 J and 4 J respectively
 (c) 4 J and 8 J respectively
 (d) 4 J and 4 J respectively
44. Two bulbs are in parallel and they together consume 48 W from a battery of 6 V. The resistance of each bulb is
 (a) 0.67 Ω (b) 3.0 Ω
 (c) 4.0 Ω (d) 1.5 Ω
45. The heat developed in an electric wire of resistance R by a current I for a time t is [MP PMT 1993; MP PET 2005]
 (a) $\frac{I^2 R t}{4.2} \text{ cal}$ (b)
 (c) $\frac{I^2 R}{4.2 t} \text{ cal}$ (d) $\frac{R t}{4.2 I^2} \text{ cal}$
46. Two bulbs, one of 50 watt and another of 25 watt are connected in series to the mains. The ratio of the currents through them is
 (a) 2 : 1
 (b) 1 : 2
 (c) 1 : 1
 (d) Without voltage, cannot be calculated
47. The brightness of a bulb will be reduced, if a resistance is connected in
 (a) Series with it
 (b) Parallel with it
 (c) Series or parallel with it
 (d) Brightness of the bulb cannot be reduced
48. A 100 watt bulb working on 200 volt and a 200 watt bulb working on 100 volt have
 (a) Resistances in the ratio of 4 : 1
 (b) Maximum current ratings in the ratio of 1 : 4
 (c) Resistances in the ratio of 2 : 1
 (d) Maximum current ratings in the ratio of 1 : 2

49. There are two electric bulbs of 40 W and 100 W. Which one will be brighter when first connected in series and then in parallel,
(a) 40 W in series and 100 W in parallel
(b) 100 W in series and 40 W in parallel
(c) 40 W both in series and parallel will be uniform
(d) 100 W both in series and parallel will be uniform
50. Two resistances R_1 and R_2 when connected in series and parallel with 120 V line, power consumed will be 25 W and 100 W respectively. Then the ratio of power consumed by R_1 to that consumed by R_2 will be [EAMCET 1983]
(a) 1 : 1 (b) 1 : 2
(c) 2 : 1 (d) 1 : 4
51. A 220 volt and 800 watt electric kettle and three 220 volt and 100 watt bulbs are connected in parallel. On connecting this combination with 220 volt electric supply, the total current will be
(a) 0.15 ampere (b) 5.0 ampere
(c) 5.5 ampere (d) 6.9 ampere
52. You are given three bulbs of 25, 40 and 60 watt. Which of them has lowest resistance [NCERT 1982]
(a) 25 watt bulb (b) 40 watt bulb
(c) 60 watt bulb (d) Information is insufficient
53. The value of internal resistance of an ideal cell is [EAMCET 1989]
(a) Zero (b) 0.5 Ω
(c) 1 Ω (d) Infinity
54. Electric power is transmitted over long distances through conducting wires at high voltage because [MP PET 1994]
(a) High voltage travels faster
(b) Power loss is large
(c) Power loss is less
(d) Generator produced electrical energy at a very high voltage
55. A coil develops heat of 800 cal/sec. When 20 volts is applied across its ends. The resistance of the coil is (1 cal = 4.2 joule)
(a) 1.2 Ω (b) 1.4 Ω
(c) 0.12 Ω (d) 0.14 Ω
56. Resistances R_1 and R_2 are joined in parallel and a current is passed so that the amount of heat liberated is H_1 and H_2 respectively. The ratio $\frac{H_1}{H_2}$ has the value [MP PMT 1994]
(a) $\frac{R_2}{R_1}$ (b) $\frac{R_1}{R_2}$
(c) $\frac{R_1^2}{R_2^2}$ (d) $\frac{R_2^2}{R_1^2}$
57. The internal resistance of a primary cell is 4 ohm. It generates a current of 0.2 amp in an external resistance of 21 ohm. The rate at which chemical energy is consumed in providing the current is
(a) 0.42 J/s (b) 0.84 J/s
(c) 5 J/s (d) 1 J/s
58. A heating coil is labelled 100 W, 220 V. The coil is cut in half and the two pieces are joined in parallel to the same source. The energy now liberated per second is [CBSE PMT 1995]
(a) 200 J (b) 400 J
(c) 25 J [MP PET 1993] (d) 50 J
59. Which of the following is not a correct statement [MP PET 1995]
(a) Resistivity of electrolytes decreases on increasing temperature
(b) Resistance of mercury falls on decreasing its temperature
(c) When joined in series a 40 W bulb glows more than a 60 W bulb
(d) Resistance of 40 W bulb is less than the resistance of 60 W bulb
60. Three light bulbs of 40 W, 60 W and 100 W are connected in series with 220 V source. Which one of the bulbs will glow brightest [MP PMT 1995; UP
(a) 40 W
(b) 60 W
(c) 100 W
(d) All with the same brightness [MP PMT 1975]
61. The energy consumed in 1 kilowatt electric heater in 30 seconds will be
(a) 6×10^2 J (b) 4.99×10^7 J
(c) 9.8×10^6 J (d) 3×10^4 J
62. Two bulbs of 500 watt and 200 watt are manufactured to operate on 220 volt line. The ratio of heat produced in 500 W and 200 W, in two cases, when firstly they are joined in parallel and secondly in series, will be [MP PET 1996; DPMT 1999]
(a) $\frac{5}{2}, \frac{2}{5}$ (b) $\frac{5}{2}, \frac{5}{2}$
(c) $\frac{2}{5}, \frac{5}{2}$ (d) $\frac{2}{5}, \frac{2}{5}$
63. A 60 watt bulb carries a current of 0.5 amp. The total charge passing through it in 1 hour is [MP PMT 1996]
(a) 3600 coulomb (b) 3000 coulomb
(c) 2400 coulomb (d) 1800 coulomb
64. An electric heater of resistance 6 ohm is run for 10 minutes on a 120 volt line. The energy liberated in this period of time is [MP PET 1994]
(a) 7.2×10^3 J (b) 14.4×10^5 J
(c) 43.2×10^4 J (d) 28.8×10^4 J
65. Two bulbs are working in parallel order. Bulb A is brighter than bulb B. If R_A and R_B are their resistance respectively then
(a) $R_A > R_B$ (b) $R_A < R_B$
(c) $R_A = R_B$ (d) None of these
66. Two conductors made of the same material are connected across a common potential difference. Conductor A has twice the diameter and twice the length of conductor B. The power delivered to the two conductors P_A and P_B respectively is such that P_A / P_B equals to
(a) 0.5 (b) 1.0
(c) 1.5 (d) 2.0
67. A heating coil can heat the water of a vessel from 20°C to 60°C in 30 minutes. Two such heating coils are put in series and then used to heat the same amount of water through the same temperature range. The time taken now will be (neglecting thermal capacity of the coils) [MP PMT 1997]
(a) 60 minutes (b) 30 minutes
(c) 15 minutes (d) 7.5 minutes

68. If 2.2 kilowatt power is transmitted through a 10 ohm line at 22000 volt, the power loss in the form of heat will be [MP PMT/PET 1998]
- (a) 0.1 watt (b) 1 watt
(c) 10 watt (d) 100 watt
69. Two resistors having equal resistances are joined in series and a current is passed through the combination. Neglect any variation in resistance as the temperature changes. In a given time interval
- (a) Equal amounts of thermal energy must be produced in the resistors
(b) Unequal amounts of thermal energy may be produced
(c) The temperature must rise equally in the resistors
(d) The temperature must rise unequally in the resistors
70. A 5°C rise in temperature is observed in a conductor by passing a current. When the current is doubled the rise in temperature will be approximately [CBSE PMT 1998]
- (a) 16°C (b) 10°C
(c) 20°C (d) 12°C
71. Watt-hour meter measures [KCET 1994]
- (a) Electric energy (b) Current
(c) Voltage (d) Power
72. An electric lamp is marked 60 W, 230 V. The cost of 1 kilowatt hour of power is Rs. 1.25. The cost of using this lamp for 8 hours is
- (a) Rs. 1.20 (b) Rs. 4.00
(c) Rs. 0.25 (d) Rs. 0.60
73. 4 bulbs marked 40 W, 250 V are connected in series with 250 V mains. The total power is [EAMCET (Engg.) 1995]
- (a) 10 W (b) 40 W
(c) 320 W (d) 160 W
74. Pick out the wrong statement [AMU 1995]
- (a) In a simple battery circuit, the point of lowest potential is the negative terminal of the battery
(b) The resistance of an incandescent lamp is greater when the lamp is switched off
(c) An ordinary 100 W lamp has less resistance than a 60 W lamp
(d) At constant voltage, the heat developed in a uniform wire varies inversely as the length of the wire used
75. Two resistors of 6 Ω and 9 Ω are connected in series to a 120 volt source. The power consumed by the 6 Ω resistor is [SCRA 1994]
- (a) 384 W (b) 576 W
(c) 1500 W (d) 1200 W
76. Electric room radiator which operates at 225 volts has resistance of 50 ohms. Power of the radiator is approximately
- (a) 100 W (b) 450 W
(c) 750 W (d) 1000 W
77. If a power of 100 W is being supplied across a potential difference of 200 V, current flowing is [AFMC 1993]
- (a) 2 A (b) 0.5 A
(c) 1 A (d) 20 A
78. A current of 2 A passing through conductor produces 80 J of heat in 10 seconds. The resistance of the conductor is [CBSE PMT 1993]
- (a) 0.5 Ω (b) 2 Ω
(c) 4 Ω (d) 20 Ω
79. A 4 μF conductor is charged to 400 volts and then its plates are joined through a resistance of 1 kΩ. The heat produced in the resistance is [CBSE PMT 1994]
- (a) 0.16 J (b) 1.28 J
(c) 0.64 J (d) 0.32 J
80. A 10 ohm electric heater operates on a 110 V line. Calculate the rate at which it develops heat in watts [AFMC 1997]
- (a) 1310 W (b) 670 W
(c) 810 W (d) 1210 W
81. A (100 W, 200 V) bulb is connected to a 160 V power supply. The power consumption would be [CBSE PMT 1997; JIPMER 2000]
- (a) 64 W (b) 80 W
(c) 100 W (d) 125 W
82. A battery of e.m.f. 10 V and internal resistance 0.5 ohm is connected across a variable resistance R. The value of R for which the power delivered in it is maximum is given by [BHU 1998; JIPMER 2001, 02; CBSE PMT 2001]
- (a) 2.0 ohm (b) 0.25 ohm
(c) 1.0 ohm (d) 0.5 ohm
83. A piece of fuse wire melts when a current of 15 ampere flows through it. With this current, if it dissipates 22.5 W, the resistance of fuse wire will be [MNR 1998]
- (a) Zero (b) 10 Ω
(c) 1 Ω (d) 0.10 Ω
84. Two wires 'A' and 'B' of the same material have their lengths in the ratio 1 : 2 and radii in the ratio 2 : 1. The two wires are connected in parallel across a battery. The ratio of the heat produced in 'A' to the heat produced in 'B' for the same time is
- (a) 1 : 2 (b) 2 : 1
(c) 1 : 8 (d) 8 : 1
85. A heater draws a current of 2 A when connected to a 250 V source. The rate of energy dissipation is [JIPMER 1999]
- (a) 500 W (b) 1000 W
(c) 250 W (d) 125 W
86. A bulb rated at (100 W – 200 V) is used on a 100 V line. The current in the bulb is [JIPMER 1999]
- (a) $\frac{1}{4}$ amp (b) 4 amp
(c) $\frac{1}{2}$ amp (d) 2 amp
87. A steel wire has a resistance twice that of an aluminium wire. Both of them are connected with a constant voltage supply. More heat will be dissipated in [Roorkee 1999]
- (a) Steel wire when both are connected in series
(b) Steel wire when both are connected in parallel
(c) Aluminium wire when both are connected in series
(d) Aluminium wire when both are connected in parallel
88. A current i passes through a wire of length l , radius of cross-section r and resistivity ρ . The rate of heat generation is [AMU (Med.) 1999]

(a) $\frac{i^2 l \rho}{\pi^2}$ (b) $i^2 \left(\frac{l \rho}{\pi^2} \right)^2$

(c) $i^2 l \rho / r$ (d) $i l \rho / r$

89. Which of the following is not equal to watt [DPMT 1999]

- (a) $(Amp)^2 \times ohm$ (b) $Amp / Volt$
(c) $Amp \times Volt$ (d) $Joule / sec$

90. Two wires with resistances R and $2R$ are connected in parallel, the ratio of heat generated in $2R$ and R is

[DCE 1999, 2000]

- (a) 1 : 2 (b) 2 : 1
(c) 1 : 4 (d) 4 : 1

91. If a high power heater is connected to electric mains, then the bulbs in the house become dim, because there is a

[BHU 1999; Pb. PMT 2000]

- (a) Current drop (b) Potential drop
(c) No current drop (d) No potential drop

92. If three bulbs 60 W, 100 W and 200 W are connected in parallel, then

- (a) 200 W bulb will glow more
(b) 60 W bulb will glow more
(c) 100 W bulb will glow more
(d) All the bulbs will glow equally

93. An expression for rate of heat generated, if a current of I ampere flows through a resistance of $R \Omega$, is [Pb. PMT 2000]

- (a) $I^2 R t$ (b) $I^2 R$
(c) $V^2 R$ (d) $I R$

94. On giving 220 V to a resistor the power dissipated is 40 W then value of resistance is [RPMT 2000]

- (a) 1210 Ω (b) 2000 Ω
(c) 1000 Ω (d) None of these

95. A 60 watt bulb operates on 220 V supply. The current flowing through the bulb is [MP PMT 2000]

- (a) 11/3 amp (b) 3/11 amp
(c) 3 amp (d) 6 amp

96. If two bulbs of wattage 25 and 30, each rated at 220 volts, are connected in series with a 440 volt supply, which bulb will fuse

[MP PET 2000]

- (a) 25 W bulb (b) 30 W bulb
(c) Neither of them (d) Both of them

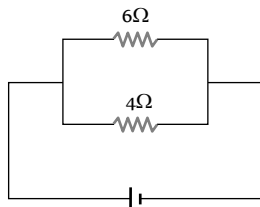
97. Two electric bulbs (60 W and 100 W respectively) are connected in series. The current passing through them is

[AMU (Med.) 2000]

- (a) More in 100 W bulb (b) More in 60 W bulb
(c) Same in both (d) None of these

98. In the circuit shown below, the power developed in the 6 Ω resistor is 6 watt. The power in watts developed in the 4 Ω resistor is

- (a) 16
(b) 9
(c) 6



(d) 4

99. Two wires A and B of same material and mass have their lengths in the ratio 1 : 2. On connecting them to the same source, the rate of heat dissipation in B is found to be 5 W. The rate of heat dissipation in A is [AMU (Engg.) 2000]

- (a) 10 W (b) 5 W
(c) 20 W (d) None of these

100. If two electric bulbs have 40 W and 60 W rating at 220 V, then the ratio of their resistances will be

[BHU 1999; KCET 2001]

- (a) 3 : 2 (b) 2 : 3
(c) 3 : 4 (d) 4 : 3

101. An electric bulb is designed to draw power P_0 at voltage V_0 . If the voltage is V it draws a power P . Then [KCET 2001]

(a) $P = \left(\frac{V_0}{V} \right)^2 P_0$ (b) $P = \left(\frac{V}{V_0} \right)^2 P_0$

(c) $P = \left(\frac{V}{V_0} \right) P_0$ (d) $P = \left(\frac{V_0}{V} \right) P_0$

102. Three bulbs of 40 W, 60 W and 100 W are arranged in series with 220 V. Which bulb has minimum resistance

[AFMC 2001]

- (a) 40 W (b) 60 W
(c) 100 W (d) Equal in all bulbs

103. An electric kettle has two heating coils. When one coil is used, water in the kettle boils in 5 minutes, while when second coil is used, same water boils in 10 minutes. If the two coils, connected in parallel are used simultaneously, the same water will boil in time

- (a) 3 min 20 sec (b) 5 min
(c) 7 min 30 sec (d) 2 min 30 sec

104. An external resistance R is connected to a battery of e.m.f. V and internal resistance r . The joule heat produced in resistor R is maximum when R is equal to [MP PET 2001]

- (a) r (b) $\frac{r}{2}$
(c) $2r$ (d) Infinitely large

105. The amount of heat produced in a resistor when a current is passed through it can be found using [Kerala PET 2001]

- (a) Faraday's Law (b) Kirchhoff's Law
(c) Laplace's Law (d) Joule's Law

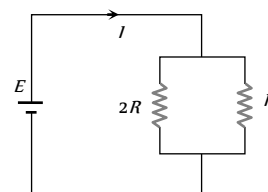
106. Two wires have resistance of 2 Ω and 4 Ω connected to same voltage, ratio of heat dissipated at resistance is

[UPSEAT 2001]

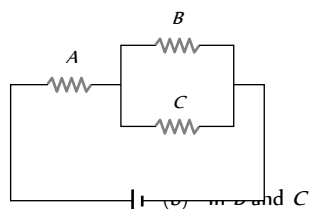
- (a) 1 : 2 (b) 4 : 3
(c) 2 : 1 (d) 5 : 2

107. Two electric bulbs rated P_1 watt V volts and P_2 watt V volts are connected in parallel and V volts are applied to it. The total power will be [MP PMT 2001; MP PET 2002]

- (a) $P_1 + P_2 \text{ watt}$ (b) $\sqrt{P_1 P_2} \text{ watt}$
- (c) $\frac{P_1 P_2}{P_1 + P_2} \text{ watt}$ (d) $\frac{P_1 + P_2}{P_1 P_2} \text{ watt}$
108. n identical bulbs, each designed to draw a power p from a certain voltage supply, are joined in series across that supply. The total power which they will draw is [KCET 2002]
- (a) p/n^2 (b) p/n
- (c) p (d) np
109. A wire when connected to 220 V mains supply has power dissipation P_1 . Now the wire is cut into two equal pieces which are connected in parallel to the same supply. Power dissipation in this case is P_2 . Then $P_2 : P_1$ is [AIEEE 2002]
- (a) 1 (b) 4
- (c) 2 (d) 3
110. An electric bulb marked 40 W and 200 V, is used in a circuit of supply voltage 100 V. Now its power is [AIIMS 2002]
- (a) 100 W (b) 40 W
- (c) 20 W (d) 10 W
111. Electric bulb 50 W-100 V glowing at full power are to be used in parallel with battery 120 V, 10 Ω . Maximum number of bulbs that can be connected so that they glow in full power is
- (a) 2 (b) 8
- (c) 4 (d) 6
112. A bulb has specification of one kilowatt and 250 volts, the resistance of bulb is [MP PMT 2002]
- (a) 125 Ω (b) 62.5 Ω
- (c) 0.25 Ω (d) 625 Ω
113. If a 30 V, 90 W bulb is to be worked on a 120 V line, a resistance of how many ohms should be connected in series with the bulb [MP PMT 2002; KCET 2003]
- (a) 10 ohm (b) 20 ohm
- (c) 30 ohm (d) 40 ohm
114. A fuse wire with radius 1 mm blows at 1.5 amp. The radius of the fuse wire of the same material to blow at 3 A will be [KCET 2003]
- (a) $4^{1/3} \text{ mm}$ (b) $3^{1/4} \text{ mm}$
- (c) $2^{1/2} \text{ mm}$ (d) $3^{1/2} \text{ mm}$
115. Three electric bulbs of rating 60 W each are joined in series and then connected to electric mains. The power consumed by these three bulbs will be [MP PET 2003; CBSE PMT 2004]
- (a) 180 W (b) 60 W
- (c) 20 W (d) $\frac{20}{3} \text{ W}$
116. An electric bulb is rated 60 W, 220 V. The resistance of its filament is [MP PET 2003]
- (a) 708 Ω (b) 870 Ω
- (c) 807 Ω (d) 780 Ω
117. A 220 volt, 1000 W bulb is connected across a 110 volt mains supply. The power consumed will be [AIEEE 2003]
- (a) 1000 W (b) 750 W
- (c) 500 W (d) 250 W
118. Two bulbs of 100 W and 200 W working at 220 volt are joined in series with 220 volt supply. Total power consumed will be approximately. [Pb. PET 2003; BHU 2005]
- (a) 65 watt (b) 33 watt
- (c) 300 watt (d) 100 watt
119. How many calories of heat will be produced approximately in a 210 watt electric bulb in 5 minutes [Pb. PET 2004]
- (a) 80000 cal (b) 63000 cal
- (c) 1050 cal (d) 15000 cal
120. A 5°C rise in the temperature is observed in a conductor by passing some current. When the current is doubled, then rise in temperature will be equal to [BHU 2004]
- (a) 5°C (b) 10°C
- (c) 20°C (d) 40°C
121. If a 2 kW boiler is used everyday for 1 hour, then electrical energy consumed by boiler in thirty days is [BHU 2004]
- (a) 15 unit (b) 60 unit
- (c) 120 unit (d) 240 unit
122. What will happen when a 40 watt, 220 volt lamp and 100 watt, 220 volt lamp are connected in series across 40 volt supply
- (a) 100 watt lamp will fuse (b) 40 watt lamp will fuse
- (c) Both lamps will fuse (d) Neither lamp will fuse [CPMT 2002]
123. What is the ratio of heat generated in R and $2R$ [DCE 2003]
- (a) 2 : 1
- (b) 1 : 2
- (c) 4 : 1
- (d) 1 : 4
124. In an electric heater 4 amp current passes for 1 minute at potential difference of 250 volt, the power of heater and energy consumed will be respectively [DPMT 2003]
- (a) 1 kW, 60 kJ (b) 0.5 kW, 30 kJ
- (c) 10 kW, 600 kJ (d) None of these
125. Some electric bulbs are connected in series across a 220 V supply in a room. If one bulb is fused then remaining bulbs are connected again in series across the same supply. The illumination in the room will [J & K CET 2004]
- (a) Increase (b) Decrease
- (c) Remains the same (d) Not continuous
126. The resistor of resistance ' R ' is connected to 25 V supply and heat produced in it is 25 J/sec. The value of R is [Orissa PMT 2004]
- (a) 225 Ω (b) 1 Ω
- (c) 25 Ω (d) 50 Ω
127. Three bulbs of 40 W, 60 W, 100 W are arranged in series with 220 volt supply which bulb has minimum resistance [Pb. PET 2000]
- (a) 100 W (b) 40 W



- (c) 60 W (d) Equal in all bulbs
128. If two electric bulbs have 40 W and 60 W rating at 220 V, then the ratio of their resistances will be [Pb. PET 2001]
(a) 9 : 4 (b) 4 : 3
(c) 3 : 8 (d) 3 : 2
129. A 10 V storage battery of negligible internal resistance is connected across a $50\ \Omega$ resistor. How much heat energy is produced in the resistor in 1 hour [Pb. PET 2001]
(a) 7200 J (b) 6200 J
(c) 5200 J (d) 4200 J
130. A hot electric iron has a resistance of $80\ \Omega$ and is used on a 200 V source. The electrical energy spent, if it is used for two hours, will be [Pb. PET 2002]
(a) 8000 Wh (b) 2000 Wh
(c) 1000 Wh (d) 800 Wh
131. The heat produced by a 100 watt heater in 2 minute will be equal to [BCECE 2004]
(a) $12 \times 10^3\ J$ (b) $10 \times 10^3\ J$
(c) $6 \times 10^3\ J$ (d) $3 \times 10^3\ J$
132. If two wires having resistance R and $2R$. Both joined in series and in parallel then ratio of heat generated in this situation, applying the same voltage, [BCECE 2004]
(a) 2 : 1 (b) 1 : 2
(c) 2 : 9 (d) 9 : 2
133. Two electric bulbs A and B are rated as 60 W and 100 W. They are connected in parallel to the same source. Then, [KCET 2004]
(a) Both draw the same current
(b) A draws more current than B
(c) B draws more current than A
(d) Current drawn are in the ratio of their resistances
134. Three identical resistances A, B and C are connected as shown in the given figure. The heat produced will be maximum



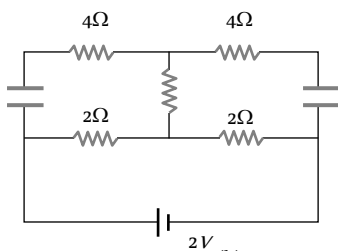
- (a) In B (b) In A and C
(c) In A (d) Same for A, B and C
135. If 2.2 kW power is transmitted through a $100\ \Omega$ line at 22,000 V, the power loss in the form of heat will be [MP PET 2004]
(a) 0.1 W (b) 1 W
(c) 10 W (d) 100 W
136. A heater coil connected to a supply of a 220 V is dissipating some power P_1 . The coil is cut into half and the two halves are connected in parallel. The heater now dissipates a power P_2 . The ratio of power $P_1 : P_2$ is [AFMC 2004]
(a) 2 : 1 (b) 1 : 2
(c) 1 : 4 (d) 4 : 1

137. An electric lamp is marked 60 W, 230 V. The cost of a 1 kWh of energy is Rs. 1.25. The cost of using this lamp 8 hrs a day for 30 day is [Kerala (Med.) 2002]
(a) Rs. 10 (b) Rs. 16
(c) Rs. 18 (d) Rs. 20
138. An electric iron draws 5 amp, a TV set draws 3 amp and refrigerator draws 2 amp from a 220 volt main line. The three appliances are connected in parallel. If all the three are operating at the same time, the fuse used may be of [ISM Dhanbad 1994]
(a) 20 amp (b) 5 amp
(c) 15 amp (d) 10 amp
139. Match the List I with the List II from the combination shown. In the left side (List I) there are four different conditions and in the right side (List II), there are ratios of heat produced in each resistance for each condition : [ISM Dhanbad 1994]
- | List I | List II |
|---|-----------|
| (I) Two wires of same resistance are connected in series and same current is passed through them | (A) 1 : 2 |
| (II) Two wires of resistance R and $2R\ ohm$ are connected in series and same P.D. is applied across them | (B) 4 : 1 |
| (III) Two wires of same resistance are connected in parallel and same current is flowing through them | (C) 1 : 1 |
| (IV) Two wires of resistances in the ratio 1 : 2 are connected in parallel and same P.D. is applied across them | (D) 2 : 1 |
- (a) I – B; II – A; III – C; IV – D
(b) I – C; II – D; III – C; IV – D
(c) I – B; II – D; III – A; IV – C
(d) I – A; II – B; III – D; IV – C [MP PMT 2004]
140. The electric current passing through a metallic wire produces heat because of [BHU 1994]
(a) Collisions of conduction electrons with each other
(b) Collisions of the atoms of the metal with each other
(c) The energy released in the ionization of the atoms of the metal
(d) Collisions of the conduction electrons with the atoms of the metallic wires
141. The maximum current that flows through a fuse wire before it blows out varies with its radius as [SCRA 1998]
(a) $r^{3/2}$ (b) r
(c) $r^{2/3}$ (d) $r^{1/2}$
142. What is immaterial for an electric fuse wire [UPSEAT 1999]
(a) Specific resistance of the wire
(b) Radius of the wire
(c) Length of the wire
(d) Current flowing through the wire
143. The current flowing through a lamp marked as 50 W and 250 V is

- (a) 5 amp (b) 2.5 amp
(c) 2 amp (d) 0.2 amp

144. Find the power of the circuit

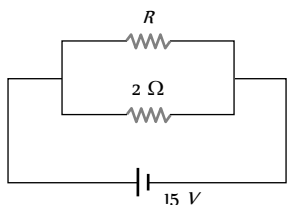
[AIEEE 2002]



- (a) 1.5 W (b) 2 W
(c) 1 W (d) None of these

145. If in the circuit, power dissipation is 150 W, then R is

[AIEEE 2002]



- (a) 2 Ω (b) 6 Ω
(c) 5 Ω (d) 4 Ω

146. Two resistors whose value are in ratio 2 : 1 are connected in parallel with one cell. Then ratio of power dissipated is

[RPMT 2000]

- (a) 2 : 1 (b) 4 : 1
(c) 1 : 2 (d) 1 : 1

147. A heater coil is cut into two equal parts and only one part is now used in the heater. The heat generated will now be

[AIEEE 2005]

- (a) One fourth (b) Halved
(c) Doubled (d) Four times

148. The resistance of hot tungsten filament is about 10 times the cold resistance. What will be the resistance of 100 W and 200 V lamp when not in use

[AIEEE 2005]

- (a) 400 Ω (b) 200 Ω
(c) 40 Ω (d) 20 Ω

149. A 5.0 amp current is setup in an external circuit by a 6.0 volt storage battery for 6.0 minutes. The chemical energy of the battery is reduced by

[KCET 2005]

- (a) $1.08 \times 10^7 J$ (b) $1.08 \times 10^6 volt$
(c) $1.8 \times 10^7 J$ (d) $1.8 \times 10^6 volt$

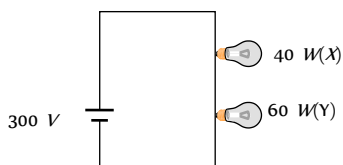
150. A railway compartment is lit up by thirteen lamps each taking 2.1 amp at 15 volts. The heat generated per second in each lamp will be

- (a) 4.35 cal (b) 5.73 cal
(c) 7.5 cal (d) 2.5 cal

151. Two bulbs X and Y having same voltage rating and of power 40 watt and 60 watt respectively are connected in series across a potential difference of 300 volt, then

[Orissa JEE 2005]

- (a) X will glow brighter
(b) Resistance of Y is greater than X



- (c) Heat produced in Y will be greater than X
(d) Voltage drop in X will be greater than Y

152. 3 identical bulbs are connected in series and these together dissipate a power P . If now the bulbs are connected in parallel, then the power dissipated will be

[DPMT 2005]

- (a) $\frac{P}{3}$ (b) $3P$
(c) $9P$ (d) $\frac{P}{9}$

153. A coil takes 15 min to boil a certain amount of water, another coil takes 20 min for the same process. Time taken to boil the same amount of water when both coil are connected in series

- (a) 5 min (b) 8.6 min
(c) 35 min (d) 30 min

Chemical Effect of Current

1. Water can not be made conducting by adding small amount of any of the following except

- (a) Sodium chloride (b) Copper sulphate
(c) Ammonium chloride (d) Sugar

2. The electrochemical equivalent Z of any element can be obtained by multiplying the electrochemical equivalent of hydrogen with

- (a) Atomic weight (b) Molecular weight
(c) Chemical equivalent (d) A constant

3. A silver and zinc voltameter are connected in series and a current i is passed through them for a time t liberating W gm of zinc. The weight of silver deposited is nearly

[NCERT 1973, 76]

- (a) W (b) 1.7 W
(c) 2.4 W (d) 3.5 W

4. To deposit one gm equivalent of an element at an electrode, the quantity of electricity needed is

[IIT 1984; DPMT 1982; MP PET 1998;

MP PMT 1998; 2003]

- (a) One ampere (b) 96000 amperes
(c) 96500 farads (d) 96500 coulombs

5. In an electrolysis experiment, a current i passes through two different cells in series, one containing a solution of $CuSO_4$ and the other a solution of $AgNO_3$. The rate of increase of the weight of the cathodes in the two cells will be

[NCERT 1972]

- (a) In the ratio of the densities of Cu and Ag
(b) In the ratio of the at. weights of Cu and Ag
(c) In the ratio of half the atomic weight of Cu to the atomic weight of Ag
(d) In the ratio of half the atomic weight of Cu to half the atomic weight of Ag

6. To deposit one litre of hydrogen at 22.4 atmosphere from acidulated water, the quantity of electricity that must pass through is

- (a) 1 coulomb (b) 22.4 coulomb
(c) 96500 coulomb (d) 193000 coulomb

7. The amount of substance liberated on electrodes during electrolysis when 1 coulomb of electricity is passed, is
(a) Chemical equivalent
(b) Electrochemical equivalent
(c) Equivalent weight
(d) One mol
8. For goldplating on a copper chain, the substance required in the form of solution is
(a) Copper sulphate
(b) Copper chloride
(c) Potassium cyanide
(d) Potassium aurocyanide
9. On passing the current in water voltameter, the hydrogen
(a) Liberated at anode (b) Liberated at cathode
(c) Does not liberate (d) Remains in the solution
10. In water voltameter, the electrolysis of takes place [DPMT 1999]
(a) H_2O (b) H_2SO_4
(c) H_2O and H_2SO_4 both (d) H_2 and O_2
11. For depositing 1 gm of Cu in copper voltameter on passing 2 amperes of current, the time required will be (For copper $Z = 0.00033 \text{ gm/C}$)
(a) Approx. 20 minutes (b) Approx. 25 minutes
(c) Approx. 30 minutes (d) Approx. 35 minutes
12. A battery of e.m.f. 3 volt and internal resistance 1.0 ohm is connected in series with copper voltameter. The current flowing in the circuit is 1.5 amperes. The resistance of voltameter will be
(a) Zero (b) 1.0 ohm
(c) 1.5 ohm (d) 2.0 ohm
13. According to Faraday's laws of electrolysis, the amount of decomposition is proportional to [MP PMT 1993]
(a) $\frac{1}{\text{Time for which current passes}}$
(b) Electrochemical equivalent of the substance
(c) $\frac{1}{\text{Current}}$
(d) $\frac{1}{\text{Electrochemical equivalent}}$
14. If in a voltaic cell 5 gm of zinc is consumed, then we get how many ampere hours? (Given that E.C.E. of Zn is $3.387 \times 10^{-7} \text{ kg/coulomb}$)
(a) 2.05 (b) 8.2
(c) 4.1 (d) $5 \times 3.387 \times 10^{-7}$
15. The current flowing in a copper voltameter is 1.6 A. The number of Cu^{++} ions deposited at the cathode per minute are
(a) 1.5×10^{20} (b) 3×10^{20}
(c) 6×10^{20} (d) 1×10^{19}
16. In a copper voltameter experiment, current is decreased to one-fourth of the initial value but it is passed for four times the earlier duration. Amount of copper deposited will be [MP PMT 1993]
(a) Same
(b) One-fourth the previous value
(c) Four times the previous value
(d) $\frac{1}{16}$ th of the previous value
17. A certain charge liberates 0.8 gm of O_2 . The same charge will liberate how many gm of silver [MP PET 1999]
(a) 108 gm (b) 10.8 gm
(c) 0.8 gm (d) $\frac{108}{0.8} \text{ gm}$
18. In charging a battery of motor-car, the following effect of electric current is used [MP PET 1993; AFMC 2003]
(a) Magnetic (b) Heating
(c) Chemical (d) Induction
19. The Avogadro's number is 6×10^{23} per gm mole and electronic charge is $1.6 \times 10^{-19} \text{ C}$. The Faraday's number is [DPMT 2001]
(a) $6 \times 10^{23} \times 1.6 \times 10^{-19}$ (b) $\frac{6 \times 10^{23}}{1.6 \times 10^{-19}}$
(c) $\frac{2}{6 \times 10^{23} \times 1.6 \times 10^{-19}}$ (d) $\frac{1.6 \times 10^{-19}}{6 \times 10^{23}}$
20. In $CuSO_4$ solution when electric current equal to 2.5 faraday is passed, the gm equivalent deposited on the cathode is
(a) 1 (b) 1.5
(c) 2 (d) 2.5
21. The atomic weight of silver and copper are 108 and 64. A silver voltameter and a copper voltameter are connected in series and when current is passed 10.8 gm of silver is deposited. The mass of copper deposited will be
(a) 6.4 gm (b) 12.8 gm
(c) 3.2 gm (d) 10.8 gm
22. Faraday's laws of electrolysis are related to [IIT 1983]
(a) The atomic number of positive ion
(b) The equivalent weight of electrolyte
(c) The atomic number of negative ion
(d) The velocity of positive ion
23. In the process of electrolysis, the current is carried out inside the electrolyte by [AMU (Engg.) 1999]
(a) Electrons
(b) Atoms
(c) Positive and negative ions
(d) All the above
24. The mass of ions deposited during a given interval of time in the process of electrolysis depends on [DPMT 2002]
(a) The current (b) The resistance
(c) The temperature (d) The electric power
25. The amount of charge required to liberate 9 gm of aluminium (atomic weight = 27 and valency = 3) in the process of electrolysis is (Faraday's number = 96500 coulombs/gm equivalent) [MP PMT 1994; MP PET 2000]
(a) 321660 coulombs (b) 69500 coulombs
(c) 289500 coulombs (d) 96500 coulombs
26. In an electroplating experiment, m gm of silver is deposited when 4 ampere of current flows for 2 minute. The amount (in gm) of silver deposited by 6 ampere of current for 40 second will be [MNR 1991; UPSEAT 2000]

Pb. PET 2004; Orissa JEE 2005]

- (a) $4m$ (b) $m/2$
(c) $m/4$ (d) $2m$
27. In electrolysis, if the duration of the passage of current is doubled, the mass liberated is [EAMCET 1979]
(a) Doubled (b) Halved
(c) Increased four times (d) Remains the same
28. A current of 16 ampere flows through molten NaCl for 10 minute. The amount of metallic sodium that appears at the negative electrode would be [EAMCET 1984]
(a) 0.23 gm (b) 1.15 gm
(c) 2.3 gm (d) 11.5 gm
29. The mass of a substance liberated when a charge ' q ' flows through an electrolyte is proportional to [EAMCET 1984]
(a) q (b) $1/q$
(c) q^2 (d) $1/q^2$
30. A steady current of 5 amps is maintained for 45 mins. During this time it deposits 4.572 gms of zinc at the cathode of a voltmeter. E.C.E. of zinc is [MP PET 1994]
(a) $3.387 \times 10^{-4} \text{ gm/C}$ (b) $3.387 \times 10^{-4} \text{ C/gm}$
(c) $3.384 \times 10^{-3} \text{ gm/C}$ (d) $3.394 \times 10^{-3} \text{ C/gm}$
31. The relation between faraday constant F , electron charge e and avogadro number N is [MP PET 1995]
(a) $F = N/e$ (b) $F = Ne$
(c) $N = F^2$ (d) $F = N^2e$
32. The electrochemical equivalent of magnesium is 0.126 mg/C . A current of 5 A is passed in a suitable solution for 1 hour. The mass of magnesium deposited will be
(a) 0.0378 gm (b) 0.227 gm
(c) 0.378 gm (d) 2.27 gm
33. Two electrolytic cells containing CuSO_4 and AgNO_3 respectively are connected in series and a current is passed through them until 1 mg of copper is deposited in the first cell. The amount of silver deposited in the second cell during this time is approximately
[Atomic weights of copper and silver are respectively 63.57 and 107.88] [MP PMT 1996]
(a) 1.7 mg (b) 3.4 mg
(c) 5.1 mg (d) 6.8 mg
34. A current I is passed for a time t through a number of voltmeters. If m is the mass of a substance deposited on an electrode and z is its electrochemical equivalent, then [MP PMT 1997]
(a) $\frac{zIt}{m} = \text{constant}$ (b) $\frac{z}{mIt} = \text{constant}$
(c) $\frac{I}{zmt} = \text{constant}$ (d) $\frac{It}{zm} = \text{constant}$
35. For electroplating a spoon, it is placed in the voltmeter at [MP PMT/PET 1998]
(a) The position of anode
(b) The position of cathode
(c) Exactly in the middle of anode and the cathode
(d) Anywhere in the electrolyte
36. If nearly 10^5 coulomb liberate 1 gm equivalent of aluminium, then the amount of aluminium (equivalent weight 9) deposited through electrolysis in 20 minutes by a current of 50 amp will be
(a) 0.6 gm (b) 0.09 gm
(c) 5.4 gm (d) 10.8 gm
37. Electroplating does not help in [AIIMS 1998]
(a) Fine finish to the surface
(b) Shining appearance
(c) Metals to become hard
(d) Protect metal against corrosion
38. When a current is passed through water, acidified with a dilute sulphuric acid, the gases formed at the platinum electrodes are
(a) 1 vol. hydrogen (cathode) and 2 vol. oxygen (anode)
(b) 2 vol. hydrogen (cathode) and 1 vol. oxygen (anode)
(c) 1 vol. hydrogen (cathode) and 1 vol. oxygen (anode)
(d) 1 vol. oxygen (cathode) and 2 vol. hydrogen (anode)
39. The negative Zn pole of a Daniel cell, sending a constant current through a circuit, decreases in mass by 0.13g in 30 minutes. If the electrochemical equivalent of Zn and Cu are 32.5 and 31.5 respectively, the increase in the mass of the positive Cu pole in this time is [AIEEE 2003]
(a) 0.242 g (b) 0.190 g
(c) 0.141 g (d) 0.126 g
40. When a copper voltmeter is connected with a battery of e.m.f. 12 volts. 2 gms of copper is deposited in 30 minutes. If the same voltmeter is connected across a 6 volt battery, then the mass of copper deposited in 45 minutes would be [SCRA 1994]
(a) 1 gm (b) 1.5 gm
(c) 2 gm (d) 2.5 gm
41. [MP PMT 1995] The value of current required to deposit 0.972 gm of chromium in 3 hours if the E.C.E. of chromium is 0.00018 gm per coulomb, is
(a) 1 amp (b) 1.5 amp
(c) 0.5 amp (d) 2 amp
42. The current inside a copper voltmeter [Roorkee 1992]
(a) Is half the outside value
(b) Is the same as the outside value
(c) Is twice the outside value
(d) Depends on the concentration of CuSO_4
43. The resistance of a cell does not depend on [RPET 1996]
(a) Current drawn from the cell
(b) Temperature of electrolyte
(c) Concentration of electrolyte
(d) The e.m.f. of the cell
44. The electrochemical equivalent of a metal is $3.3 \times 10^{-7} \text{ kg/coulomb}$. The mass of the metal liberated at the cathode when a 3 A current is passed for 2 seconds will be
(a) $19.8 \times 10^{-7} \text{ kg}$ (b) $9.39 \times 10^{-7} \text{ kg}$
(c) $6.6 \times 10^{-7} \text{ kg}$ (d) $1.1 \times 10^{-7} \text{ kg}$
45. Faraday's 2nd law states that mass deposited on the electrode is directly proportional to [DCE 1999]
(a) Atomic mass (b) Atomic mass $\times$ Velocity

- (c) Atomic mass/Valency (d) Valency
46. The relation between Faraday constant (F), chemical equivalent (E) and electrochemical equivalent (Z) is [SCRA 1994; AFMC 2000]
- (a) $F = EZ$ (b) $F = \frac{Z}{E}$
- (c) $F = \frac{E}{Z}$ (d) $F = \frac{E}{Z^2}$
47. The electrochemical equivalent of a material in an electrolyte depends on [MP PET 2001]
- (a) The nature of the material
- (b) The current through the electrolyte
- (c) The amount of charge passed through electrolyte
- (d) The amount of material present in electrolyte
48. On passing 96500 coulomb of charge through a solution CuSO_4 the amount of copper liberated is [MP PMT 2001]
- (a) 64 gm (b) 32 gm
- (c) 32 kg (d) 64 kg
49. If 96500 coulombs of electricity liberates one gram equivalent of any substance, the time taken for a current of 0.15 amperes to deposit 20mg of copper from a solution of copper sulphate is (Chemical equivalent of copper = 32) [Kerala (Engg.) 2002]
- (a) 5 min 20 sec (b) 6 min 42 sec
- (c) 4 min 40 sec (d) 5 min 50 sec
50. How much current should be passed through acidified water for 100 s to liberate 0.224 litre of H_2 [DCE 2002]
- (a) 22.4 A (b) 19.3 A
- (c) 9.65 A (d) 1 A
51. Who among the following scientists made the statement –"Chemical change can produce electricity" [DCE 2004]
- (a) Galvani (b) Faraday
- (c) Coulomb (d) Thomson
52. If a steady current of 4 amp maintained for 40 minutes, deposits 4.5 gm of zinc at the cathode and then the electro chemical equivalent will be [MH CET 2003]
- (a) $51 \times 10^{-17} \text{ gm/C}$ (b) $28 \times 10^{-6} \text{ gm/C}$
- (c) $32 \times 10^{-5} \text{ gm/C}$ (d) $47 \times 10^{-5} \text{ gm/C}$
53. The current flowing in a copper voltameter is 3.2 A. The number of copper ions (Cu^{2+}) deposited at the cathode per minute is
- (a) 0.5×10^{20} (b) 1.5×10^{20}
- (c) 3×10^{20} (d) 6×10^{20}
54. A copper voltameter is connected in series with a heater coil of resistance 0.1Ω . A steady current flows in the circuit for twenty minutes and mass of 0.99 g of copper is deposited at the cathode. If electrochemical equivalent of copper is 0.00033 gm/C , then heat generated in the coil is [Pb. PET 2002]
- (a) 750 J (b) 650 J
- (c) 350 J (d) 250 J
55. E.C.E. of Cu and Ag are 7×10^{-6} and 1.2×10^{-6} . A certain current deposits 14 gm of Cu. Amount of Ag deposited is [Orissa PMT 2004]
- (a) 1.2 gm (b) 1.6 gm
- (c) 2.4 gm (d) 1.8 gm
56. The chemical equivalent of silver is 108. If the current in a silver voltameter is 2 Amp., the time required to deposit 27 grams of silver will be [MP PMT 2004]
- (a) 8.57 hrs (b) 6.70 hrs
- (c) 3.35 hrs (d) 12.50 hrs
57. Two voltmeters, one of copper and another of silver, are joined in parallel. When a total charge q flows through the voltmeters, equal amount of metals are deposited. If the electrochemical equivalents of copper and silver are Z_1 and Z_2 respectively the charge which flows through the silver voltameter is
- (a) $q \frac{Z_1}{Z_2}$ (b) $q \frac{Z_2}{Z_1}$
- (c) $\frac{q}{1 + \frac{Z_1}{Z_2}}$ (d) $\frac{q}{1 + \frac{Z_2}{Z_1}}$
58. The chemical equivalent of copper and zinc are 32 and 108 respectively. When copper and silver voltameter are connected in series and electric current is passed through for sometimes, 1.6 g of copper is deposited. Then, the mass of silver deposited will be
- (a) 3.5 g (b) 2.8 g
- (c) 5.4 g (d) None of these
59. Ampere hour is the unit of [Orissa JEE 2005]
- (a) Quantity of charge (b) Potential
- (c) Energy (d) Current

Thermo-Electricity

- The production of e.m.f. by maintaining a difference of temperature between the two junctions of two different metals is known as
 - Joule effect
 - Seebeck effect
 - Peltier effect
 - Thomson effect
- When a current passes through the junction of two different metals, evolution or absorption of heat at the junction is known as
 - Joule effect
 - Seebeck effect
 - Peltier effect
 - Thomson effect
- When a current passes through a wire whose different parts are maintained at different temperatures, evolution or absorption of heat all along the length of wire is known as
 - Joule effect
 - Seebeck effect
 - Peltier effect
 - Thomson effect
- The thermocouple is based on the principle of [MP PET 1984; AFMC 1998; BCECE 2003]
 - Seebeck effect
 - Thomson effect

- (c) Peltier effect (d) Joule effect
5. For a thermocouple, the neutral temperature is 270°C and the temperature of its cold junction is 20°C . If there is no deflection in the galvanometer, the temperature of the hot junction should be
(a) 210°C (b) 540°C
(c) 520°C (d) 209°C
6. Thermocouple is a device for the measurement of
(a) Absolute temperature of a metal
(b) The temperature difference between two substances
(c) The couple acting on a wire
(d) Thermal conductivity of a substance
7. The true statement for thermo e.m.f. of a thermocouple
(a) Depends on the nature of metals
(b) Depends only on temperature of cold junction
(c) Depends only on temperature of hot junction
(d) Depends on the length of the wires used for thermocouple
8. The direction of current in an iron-copper thermocouple is [MP PET 1995]
(a) From copper to iron at the hot junction
(b) From iron to copper at the hot junction
(c) From copper to iron at cold junction
(d) No current will flow
9. Peltier coefficient for the junction of a pair of metals is proportional to [MP PMT 1993; MP PET 1997]
(a) T absolute temperature of the junction
(b) Square of absolute temperature of the junction
(c) $\frac{1}{\text{Absolute temperature of the junction}}$
(d) $\frac{1}{\text{Square of absolute temperature of the junction}}$
10. If for a thermocouple T_n is the neutral temperature, T_c is the temperature of the cold junction and T_i is the temperature of inversion, then [MP PET 2001; AIEEE 2002]
(a) $T_i = 2T_n - T_c$ (b) $T_n = T_i - 2T_c$
(c) $T_i = T_n - T_c$ (d) None of these
11. For a thermocouple, the temperature of inversion is that temperature at which thermo e.m.f. is
(a) Zero (b) Maximum
(c) Minimum (d) None of the above
12. For a given thermocouple, the thermo e.m.f. can be
(a) Zero (b) Positive
(c) Negative (d) All of the above
13. When current is passed in antimony-bismuth couple, then
(a) The junction becomes hot when the current is from bismuth to antimony
(b) The junction becomes hot when current flows from antimony to bismuth
(c) Both junctions become hot
(d) Both junctions become cold
14. A thermocouple is made of Cu and Fe . If a battery is connected in it, then
(a) Both junctions will be at the same temperature
(b) Both junctions will become hot
(c) One junction will be hotter than the other
(d) None of these
15. Thermopile is used for [AMU Engg. 2000]
(a) Collecting the heat energy
(b) The measurement of radiant heat energy
(c) The measurement of current
(d) The change of atomic energy into heat energy
16. When a current of 1 ampere is passed through a conductor whose ends are maintained at temperature difference of 1°C , the amount of heat evolved or absorbed is called
(a) Peltier coefficient (b) Thomson coefficient
(c) Thermoelectric power (d) Thermo e.m.f.
17. In a thermocouple, the temperature that does not depend on the temperature of the cold junction is called
(a) Neutral temperature (b) Temperature of inversion
(c) Both the above (d) None of the above
18. At neutral temperature, the thermoelectric power $\left(\frac{dE}{dT}\right)$ has the value [MP PET 2003; MP PMT 2004]
(a) Zero (b) Maximum but negative
(c) Maximum but positive (d) Minimum but positive
19. In Cu-Fe couple, the flow of current at the temperature of inversion is
(a) From Fe to Cu through the hot junction
(b) From Cu to Fe through the hot junction
(c) Maximum
(d) None of the above
20. In Seebeck series Sb appears before Bi . In a Sb-Bi thermocouple current flows from [MP PET 1994]
(a) Sb to Bi at the hot junction
(b) Sb to Bi at the cold junction
(c) Bi to Sb at the cold junction
(d) None of the above
21. Which of the following statement is correct [MP PET 1994]
(a) Both Peltier and Joule effects are reversible
(b) Both Peltier and Joule effects are irreversible
(c) Joule effect is reversible, whereas Peltier effect is irreversible
(d) Joule effect is irreversible, whereas Peltier effect is reversible
22. For a given temperature difference, which of the following pairs will generate maximum thermo e.m.f. [MP PMT 1994]
(a) Antimony-bismuth (b) Silver-gold
(c) Iron-copper (d) Lead-nickel
23. The cold junction of a thermocouple is maintained at 10°C . No thermo e.m.f. is developed when the hot junction is maintained at 530°C . The neutral temperature is [MP PMT 1994]
(a) 260°C (b) 270°C
(c) 265°C (d) 520°C
24. Which of the following is not reversible [Manipal MEE 1995; DPMT 2001]
(a) Joule effect (b) Peltier effect

- (c) Seebeck effect (d) Thomson effect
25. Neutral temperature of a thermocouple is defined as the temperature at which [MP PMT 1996]
 (a) The thermo e.m.f. changes sign
 (b) The thermo e.m.f. is maximum
 (c) The thermo e.m.f. is minimum
 (d) The thermo e.m.f. is zero
26. As the temperature of hot junction of a thermo-couple is increased (while cold junction is at constant temperature), the thermo e.m.f.
 (a) Increases uniformly at constant rate
 (b) Increases slowly in the beginning and more rapidly at higher temperatures
 (c) Increases more rapidly in the beginning but less rapidly at higher temperatures
 (d) In minimum at neutral temperature
27. As the temperature of hot junction increases, the thermo e.m.f.
 (a) Always increases
 (b) Always decreases
 (c) May increase or decrease
 (d) Always remains constant
28. The e.m.f. in a thermoelectric circuit with one junction at 0°C and the other at $t^{\circ}\text{C}$ is given by $E = At - Bt^2$. The neutral temperature is then [AMU 1995; BCECE 2004]
 (a) $\frac{A}{B}$ (b) $-\frac{A}{2B}$
 (c) $-\frac{B}{2A}$ (d) $\frac{A}{2B}$
29. The temperature of cold junction and neutral temperature of a thermocouple are 15°C and 280°C respectively. The temperature of inversion is [AMU (Engg.) 1999]
 (a) 295°C (b) 265°C
 (c) 545°C (d) 575°C
30. Above neutral temperature, thermo e.m.f. in a thermocouple [AMU (Engg.) 1999]
 (a) Decreases with rise in temperature
 (b) Increases with rise in temperature
 (c) Remains constant
 (d) Changes sign
31. Consider the following two statements *A* and *B*, and identify the correct choice out of given answers
 A. Thermo e.m.f. is minimum at neutral temperature of a thermocouple
 B. When two junctions made of two different metallic wires are maintained at different temperatures, an electric current is generated in the circuit. [EAMCET (Med.) 2000]
 (a) *A* is false and *B* is true (b) *A* is true and *B* is false
 (c) Both *A* and *B* are false (d) Both *A* and *B* are true
32. The temperature at which thermal electric power of a thermo couple becomes zero is called [MP PMT 2001]
 (a) Inversion temperature (b) Neutral temperature
 (c) Junction temperature (d) Null temperature
33. Thomson coefficient of a conductor is $10\mu\text{V/K}$. The two ends of it are kept at 50°C and 60°C respectively. Amount of heat absorbed by the conductor when a charge of 10 C flows through it is
 (a) 1000 J (b) 100 J
 (c) 100 mJ (d) 1 mJ
34. For a thermocouple the neutral temperature is 270°C when its cold junction is at 20°C . What will be the neutral temperature and the temperature of inversion when the temperature of cold junction is increased to 40°C [Kerala PET 2001]
 (a) 290°C , 580°C (b) 270°C , 580°C
 (c) 270°C , 500°C (d) 290°C , 540°C
35. Two ends of a conductor are at different temperatures the electromotive force generated between two ends is [MP PMT 2001; MP PET 2002]
 (a) Seebeck electro motive force (e.m.f.)
 (b) Peltier electro motive force (e.m.f.)
 (c) Thomson electro motive force (e.m.f.)
 (d) None of these
36. The neutral temperature of a thermocouple is 350°C when the cold junction is at 0°C . When the cold junction is immersed in a bath of 30°C , the inversion temperature is [Kerala (Med.) 2002]
 (a) 700°C (b) 600°C
 (c) 350°C (d) 670°C
37. A thermoelectric refrigerator works on [JIPMER 2002]
 (a) Joule effect (b) Seebeck effect
 (c) Peltier effect (d) Thermionic emission
38. If the temperature of cold junction of thermocouple is lowered, then the neutral temperature [JIPMER 2002]
 (a) Increases
 (b) Approaches inversion temperature
 (c) Decreases
 (d) Remains the same
39. Consider the following two statements *A* and *B* and identify the correct choice given in the answers
 (A) Duddells thermo-galvanometer is suitable to measure direct current only
 (B) Thermopile can measure temperature differences of the order of 10^{-3}°C [EAMCET 2003]
 (a) Both *A* and *B* are true (b) Both *A* and *B* are false
 (c) *A* is true but *B* is false (d) *A* is false but *B* is true
40. If $E = at + bt^2$, what is the temperature of inversion [DCE 2003]
 (a) $-\frac{a}{2b}$ (b) $+\frac{a}{2b}$
 (c) $-\frac{a}{b}$ (d) $+\frac{a}{b}$
41. Antimony and bismuth are usually used in a thermocouple, because
 (a) Negative thermal e.m.f. produced
 (b) Constant thermal e.m.f. produced
 (c) Lower thermal e.m.f. produced
 (d) Higher thermal e.m.f. produced [EAMCET 2001]
42. The smallest temperature difference that can be measured with a combination of a thermocouple of thermo e.m.f. $30\mu\text{V}$ per degree and a

galvanometer of $50\ \text{ohm}$ resistance, capable of measuring a minimum current of $3 \times 10^{-6}\ \text{amp}$ is

[MP PET 2000]

- (a) 0.5 degree (b) 1.0 degree
(c) 1.5 degree (d) 2.0 degree

43. $e = \alpha t - \frac{1}{2} \beta t^2$, If temperature of cold junction is 0°C then temperature of inversion is

(if $\alpha = 500.0\ \mu\text{V}/^\circ\text{C}$, $\beta = 5.0\ \mu\text{V}/\text{Square } ^\circ\text{C}$)

[DCE 2001]

- (a) 100 (b) 200
(c) 300 (d) 400

44. If the emf of a thermocouple, one junction of which is kept 0°C is given by $e = at + 1/2 bt^2$ then the neutral temperature will be [J & K CET 2005]

- (a) a/b (b) $-a/b$
(c) $a/2b$ (d) $-1/ab$

Critical Thinking

Objective Questions

1. The resistance of the filament of an electric bulb changes with temperature. If an electric bulb rated 220 volt and 100 watt is connected (220 $\times$ 0.8) volt sources, then the actual power would be

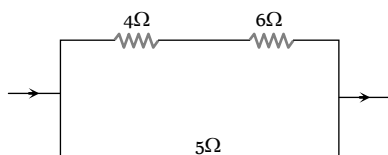
- (a) $100 \times 0.8\ \text{watt}$
(b) $100 \times (0.8)^2\ \text{watt}$
(c) Between $100 \times 0.8\ \text{watt}$ and $100\ \text{watt}$
(d) Between $100 \times (0.8)^2\ \text{watt}$ and $100 \times 0.8\ \text{watt}$

2. An immersion heater is rated 836 watt. It should heat 1 litre of water from 10°C to 40°C in about [AIEEE 2004]

- (a) 200 sec (b) 150 sec
(c) 836 sec (d) 418 sec

3. In the circuit shown in figure, the heat produced in $5\ \text{ohm}$ resistance is 10 calories per second. The heat produced in 4 resistance is [IIT 1981; UPSEAT 2002]

- (a) 1 cal/sec
(b) 2 cal/sec
(c) 3 cal/sec
(d) 4 cal/sec



4. A house is served by 220 V supply line in a unit protected by a 9 ampere fuse. The maximum number of 60 W lamps in parallel that can be turned on, is

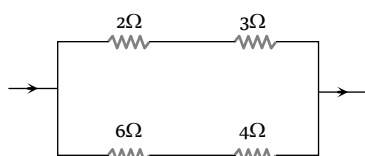
- (a) 44 (b) 20
(c) 22 (d) 33

5. Water boils in an electric kettle in 15 minutes after switching on. If the length of the heating wire is decreased to $2/3$ of its initial value, then the same amount of water will boil with the same supply voltage in [MP PMT 1994]

- (a) 15 minutes (b) 12 minutes
(c) 10 minutes (d) 8 minutes

6. In the circuit as shown in the figure, the heat produced by $6\ \text{ohm}$ resistance due to current flowing in it is 60 calorie per second. The heat generated across $3\ \text{ohm}$ resistance per second will be

- (a) 30 calorie



- (b) 60 calorie
(c) 100 calorie
(d) 120 calorie

7. The resistance of a heater coil is $110\ \text{ohm}$. A resistance R is connected in parallel with it and the combination is joined in series with a resistance of $11\ \text{ohm}$ to a 220 volt main line. The heater operates with a power of 110 watt. The value of R in ohm is

- (a) 12.22
(b) 24.42
(c) Negative
(d) That the given values are not correct

8. A 500 W heating unit is designed to operate from a 115 volt line. If the line voltage drops to 110 volt, the percentage drop in heat output will be [ISM Dhanbad 1994]

- (a) 10.20% (b) 8.1%
(c) 8.6% (d) 7.6%

9. A heater of 220 V heats a volume of water in 5 minute time. A heater of 110 V heats the same volume of water in [AFMC 1993]

- (a) 5 minutes (b) 8 minutes
(c) 10 minutes (d) 20 minutes

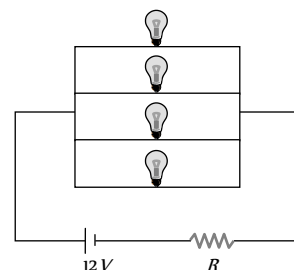
10. An electric kettle takes 4 A current at 220 V. How much time will it take to boil 1 kg of water from room temperature 20°C ? The temperature of boiling water is 100°C [CPMT 1989]

- (a) 6.4 minutes (b) 6.3 minutes
(c) 12.6 minutes (d) 12.8 minutes

11. If a wire of resistance $20\ \Omega$ is covered with ice and a voltage of 210 V is applied across the wire, then the rate of melting of ice is

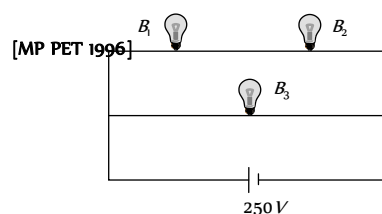
- (a) 0.85 g/s (b) 1.92 g/s
(c) 6.56 g/s (d) All of these

12. Four identical electrical lamps are labelled 1.5 V, 0.5 A which describes the condition necessary for them to operate at normal brightness. A 12 V battery of negligible internal resistance is connected to lamps as shown, then [UPSEAT 2001]



- (a) The value of R for normal brightness of each lamp is $(3/4)\ \Omega$
(b) The value of R for normal brightness of each lamp is $(21/4)\ \Omega$
(c) Total power dissipated in circuit when all lamps are normally bright is 24 W
(d) Power dissipated in R is 21 W when all lamps are normally bright

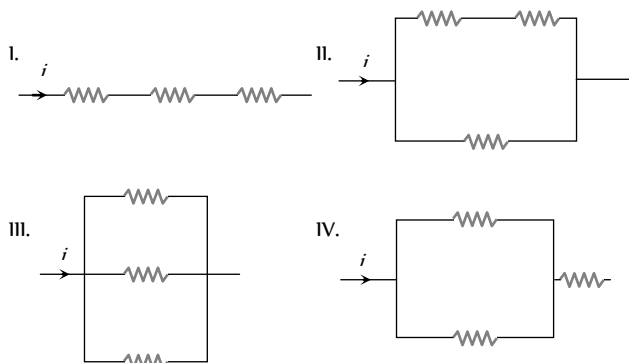
13. A 100 W bulb B_1 and two 60-W bulbs B_2 and B_3 are connected to a 250 V source, as shown in the figure. Now W_1 , W_2 and W_3 are the output powers of the bulbs B_1 , B_2 and B_3 respectively. Then [MP PET 1996]



- (a) $W_1 > W_2 = W_3$ (b) $W_1 > W_2 > W_3$
(c) $W_1 < W_2 = W_3$ (d) $W_1 < W_2 < W_3$

14. The three resistance of equal value are arranged in the different combinations shown below. Arrange them in increasing order of power dissipation

[IIT-JEE (Screening) 2003]



- (a) $III < II < IV < I$ (b) $II < III < IV < I$
(c) $I < IV < III < II$ (d) $I < III < II < IV$

15. Silver and copper voltameter are connected in parallel with a battery of e.m.f. 12 V. In 30 minutes, 1 gm of silver and 1.8 gm of copper are liberated. The power supplied by the battery is

- (a) 24.13 J/sec (b) 2.413 J/sec
(c) 0.2413 J/sec (d) 2413 J/sec

($Z_{Cu} = 6.6 \times 10^{-4} \text{ gm/C}$ and $Z_{Ag} = 11.2 \times 10^{-4} \text{ gm/C}$)

16. A silver voltameter of resistance 2 ohm and a 3 ohm resistor are connected in series across a cell. If a resistance of 2 ohm is connected in parallel with the voltameter, then the rate of deposition of silver

[EAMCET 1983]

- (a) Decreases by 25%
(b) Increases by 25%
(c) Increases by 37.5%
(d) Decreases by 37.5%

17. The expression for thermo e.m.f. in a thermocouple is given by the relation $E = 40\theta - \frac{\theta^2}{20}$, where θ is the temperature difference of two junctions. For this, the neutral temperature will be

- (a) 100°C (b) 200°C
(c) 300°C (d) 400°C

18. For copper-iron (Cu-Fe) couple, the thermo e.m.f. (temperature of cold junction = 0°C) is given by $E = (14\theta - 0.02\theta^2)\mu\text{V}$. The neutral temperature will be

- (a) 350°C (b) 350 K
(c) 560°C (d) 560 K

19. One junction of a certain thermoelectric couple is at a fixed temperature T_r and the other junction is at temperature T . The thermo electromotive force for this is expressed by $E = K(T - T_r) \left[T_0 - \frac{1}{2}(T + T_r) \right]$. At temperature $T = \frac{1}{2}T_0$, the thermoelectric power is [MP PMT 1994]

- (a) $\frac{1}{2}KT_0$ (b) KT_0
(c) $\frac{1}{2}KT_0^2$ (d) $\frac{1}{2}K(T_0 - T_r)^2$

20. The temperature of the cold junction of thermo-couple is 0°C and the temperature of hot junction is $T^\circ\text{C}$. The e.m.f. is $E = 16T - 0.04T^2 \mu\text{ volts}$. The temperature of inversion is

- (a) 200°C (b) 400°C
(c) 100°C (d) 300°C

21. The temperature of the cold junction of a thermocouple is 0°C and temperature of the hot junction is $T^\circ\text{C}$. The thermo e.m.f. is given by the relation $E = AT - \frac{1}{2}BT^2$ (where $A = 16$ and $B = 0.08$). The temperature of inversion is

- (a) 100°C (b) 300°C
(c) 400°C (d) 500°C

22. The thermo e.m.f. of a thermo-couple is $25\mu\text{V}/^\circ\text{C}$ at room temperature. A galvanometer of 40 ohm resistance, capable of detecting current as low as 10^{-5} A , is connected with the thermocouple. The smallest temperature difference that can be detected by this system is [AIEEE 2003]

- (a) 20°C (b) 16°C
(c) 12°C (d) 8°C

23. An electric bulb rated for 500 watts at 100 volts is used in a circuit having a 200-volt supply. The resistance R that must be put in series with the bulb, so that the bulb draws 500 W is

- (a) 10Ω (b) 20Ω
(c) 50Ω (d) 100Ω

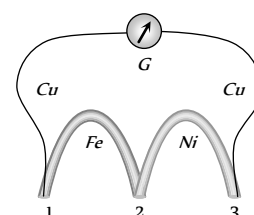
24. A thermo couple develops $200\mu\text{V}$ between 0°C and 100°C . If it develops $64\mu\text{V}$ and $76\mu\text{V}$ respectively between $(0^\circ\text{C} - 32^\circ\text{C})$ and $(32^\circ\text{C} - 70^\circ\text{C})$ then what will be the thermo emf it develops between 70°C and 100°C

- (a) $65\mu\text{V}$ (b) $60\mu\text{V}$
(c) $55\mu\text{V}$ (d) $50\mu\text{V}$

25. A thermo couple is formed by two metals X and Y metal X comes earlier to Y in Seebeck series. If temperature of hot junction increases beyond the temperature of inversion. Then direction of current in thermocouple will so

- (a) X to Y through cold junction
(b) X to Y through hot junction
(c) Y to X through cold junction

- (d) Both (b) and (c)
26. Peltier co-efficient of a thermo couple is 2 nano volts. How much heat is developed at a junction if 2.5 amp current flows for 2 minute
- (a) 6 ergs (b) 6×10^{-7} ergs
(c) 16 ergs (d) 6×10^{-3} erg
27. Resistance of a voltmeter is 2Ω , it is connected in series to a battery of 10 V through a resistance of 3Ω . In a certain time mass deposited on cathode is 1 gm. Now the voltmeter and the 3Ω resistance are connected in parallel with the battery. Increase in the deposited mass on cathode in the same time will be
- (a) 0 (b) 1.5 gm
(c) 2.5 gm (d) 2 gm
28. A current of 1.5 A flows through a copper voltmeter. The thickness of copper deposited on the electrode surface of area 50 cm^2 in 20 minutes will be (Density of copper = 9000 kg/m^3 and E.C.E. of copper = 0.00033 g/C)
- (a) $2.6 \times 10^{-5}\text{ m}$ (b) $2.6 \times 10^{-4}\text{ m}$
(c) $1.3 \times 10^{-5}\text{ m}$ (d) $1.3 \times 10^{-4}\text{ m}$
29. An ammeter, suspected to give inaccurate reading, is connected in series with a silver voltmeter. The ammeter indicates 0.54 A. A steady current passed for one hour deposits 2.0124 gm of silver. If the E.C.E. of silver is $1.118 \times 10^{-3}\text{ gmC}^{-1}$, then the error in ammeter reading is
- (a) + 0.04 A (b) + 0.02 A
(c) - 0.03 A (d) - 0.01 A
30. If 1 A of current is passed through CuSO_4 solution for 10 seconds, then the number of copper ions deposited at the cathode will be about
- (a) 1.6×10^{19} (b) 3.1×10^{19}
(c) 4.8×10^{19} (d) 6.2×10^{19}
31. A silver and a copper voltmeters are connected in parallel across a 6 volt battery of negligible resistance. In half an hour, 1 gm of copper and 2 gm of silver are deposited. The rate at which energy is supplied by the battery will approximately be (Given E.C.E. of copper = $3.294 \times 10^{-4}\text{ g/C}$ and E.C.E. of silver = $1.118 \times 10^{-3}\text{ g/C}$)
- (a) 64 W (b) 32 W
(c) 96 W (d) 16 W
32. A thermocouple of resistance 1.6Ω is connected in series with a galvanometer of 8Ω resistance. The thermocouple develops and e.m.f. of $10\mu\text{V}$ per degree temperature difference between two junctions. When one junction is kept at 0°C and the other in a molten metal, the galvanometer reads 8 millivolt. The temperature of molten metal, when e.m.f. varies linearly with temperature difference, will be
- (a) 960°C (b) 1050°C
(c) 1275°C (d) 1545°C
33. The e.m.f. of a thermocouple, one junction of which is kept at 0°C , is given by $e = at + bt^2$ the Peltier co-efficient will be
- (a) $(t + 273)(a + 2bt)$ (b) $(t + 273)(a - 2bt)$
(c) $(t - 273)(a - 2bt)$ (d) $(t - 273)(a + 2bt)$
34. A coil of wire of resistance 50Ω is embedded in a block of ice. If a potential difference of 210 V is applied across the coil, the amount of ice melted per second will be
- (a) 4.12 gm (b) 4.12 kg
(c) 3.68 kg (d) 2.625 gm
35. The same mass of copper is drawn into two wires 1 mm and 2 mm thick. Two wires are connected in series and current is passed through them. Heat produced in the wire is in the ratio
- (a) 2 : 1 (b) 1 : 16
(c) 4 : 1 (d) 16 : 1
36. The temperature of hot junction of a thermo-couple changes from 80°C to 100°C . The percentage change in thermoelectric power is
- (a) 8% (b) 10%
(c) 20% (d) 25%
37. A thermo couple uses Bismuth and Tellurium as the dissimilar metals. The sensitivity of bismuth is $-72\mu\text{V}/^\circ\text{C}$ and that of the tellurium is $500\mu\text{V}/^\circ\text{C}$. If the difference between hot and cold junction is 100°C , then the maximum output will be
- (a) 50 mV (b) 7.2 mV
(c) 42.8 mV (d) 57.2 mV
38. Three wires of copper, iron and nickel are joined to form three junctions as shown in Fig. When the temperature of junction 1 is kept 50°C with the other two junctions at 0°C , the sensitive galvanometer gives a deflection of 14 divisions. When the temperature of junction 3 is kept 50°C , with the other two junctions at 0°C , the galvanometer gives a deflection of 11 divisions. Then the deflection given by the galvanometer, when temperature of the junction 2 is kept at 50°C , with the other two junctions at 0°C , will be
- (a) 3 div (b) 11 div
(c) 14 div (d) 25 div
39. The wiring of a house has resistance 6Ω . A 100 W bulb is glowing. If a geyser of 1000 W is switched on, the change in potential drop across the bulb is nearly [MNR 1998]
- (a) Nil (b) 23 V



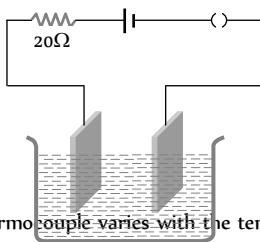
- (c) 32 V (d) 12 V

40. A 12 V lead accumulator is being charged using 24 V supply with an external resistance $2\ \Omega$. The internal resistance of the accumulator is $1\ \Omega$. Find the time in which it will store 360 W-hour energy.

- (a) 1 hr (b) 7.5 hr
(c) 10 hr (d) None of these

41. In a Ag voltameter 2.68 gm of silver is deposited in 10 min. The heat developed in $20\ \Omega$ resistor during the same period will be

- (a) 192 kJ
(b) 192 J
(c) 200 J
(d) 132 kJ

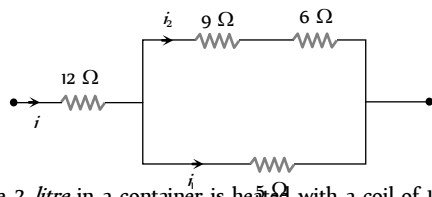


42. The thermo e.m.f. of a thermocouple varies with the temperature θ of the hot junction as $E = a\theta + b\theta^2$ in volts where the ratio a/b is 700°C . If the cold junction is kept at 0°C , then the neutral temperature is [AIIEE 2004]

- (a) 700°C
(b) 350°C
(c) 1400°C
(d) No neutral temperature is possible for this thermocouple

43. In the following circuit, $5\ \Omega$ resistor develops 45 J/s due to current flowing through it. The power developed per second across $12\ \Omega$ resistor is [AMU (Engg.) 1999]

- (a) 16 W
(b) 192 W
(c) 36 W
(d) 64 W

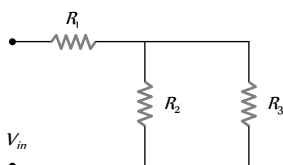


44. Water of volume 2 litre in a container is heated with a coil of 1 kW at 27°C . The lid of the container is open and energy dissipates at rate of 160 J/s. In how much time temperature will rise from 27°C to 77°C [Given specific heat of water is $4.2\ \text{kJ/kg}$]

- (a) 8 min 20 s (b) 6 min 2 s
(c) 7 min (d) 14 min

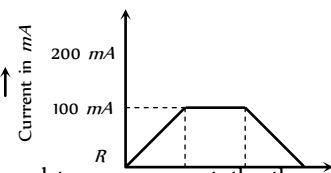
45. For ensuring dissipation of same energy in all three resistors (R_1, R_2, R_3) connected as shown in figure, their values must be related as [AIIMS 2005]

- (a) $R_1 = R_2 = R_3$
(b) $R_2 = R_3$ and $R_1 = 4R_2$
(c) $R_2 = R_3$ and $R_1 = \frac{1}{4}R_2$
(d) $R_1 = R_2 + R_3$



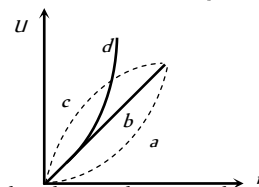
1. In a copper voltameter, mass deposited in 30 second is $m\ \text{gm}$. If the time-current graph is as shown in figure, ECE of copper is

- (a) m
(b) $m/2$
(c) $0.1\ m$
(d) $0.6\ m$

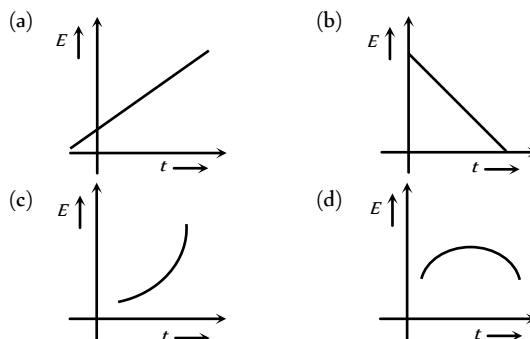


2. Which of the following plots may represent the thermal energy produced in a resistor in a given time as a function of the electric current [MP PMT 1999]

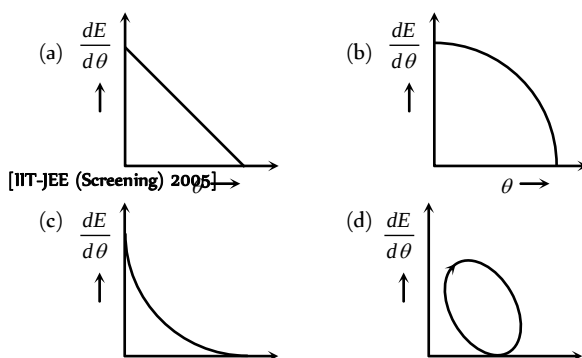
- (a) a
(b) b
(c) c
(d) d



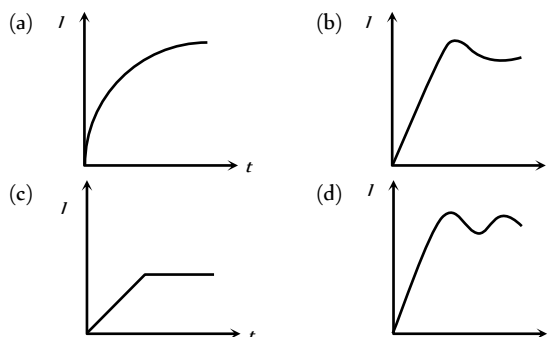
3. Two different metals are joined end to end. One end is kept at constant temperature and the other end is heated to a very high temperature. The graph depicting the thermo e.m.f. is



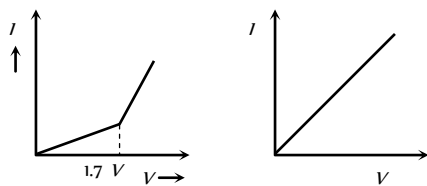
4. Which of the following graphs shows the variation of thermoelectric power with temperature difference between hot and cold junction in thermocouples



5. When an electric heater is switched on, the current flowing through it (i) is plotted against time (t). Taking into account the variation of resistance with temperature, which of the following best represents the resulting curve

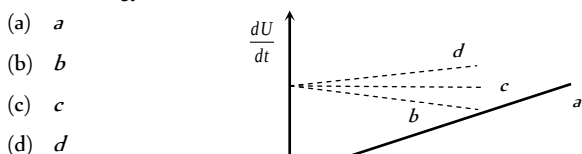


6. The V - i graphs A and B drawn for two voltmeters. Identify each graph



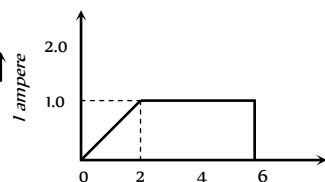
- (a) A for water voltmeter and B for Cu voltmeter
(b) A for Cu voltmeter and B for water voltmeter
(c) Both A and B represents Cu voltmeter
(d) None of these

7. A constant current i is passed through a resistor. Taking the temperature coefficient of resistance into account, indicate which of the plots shown in figure best represents the rate of production of thermal energy in the resistor

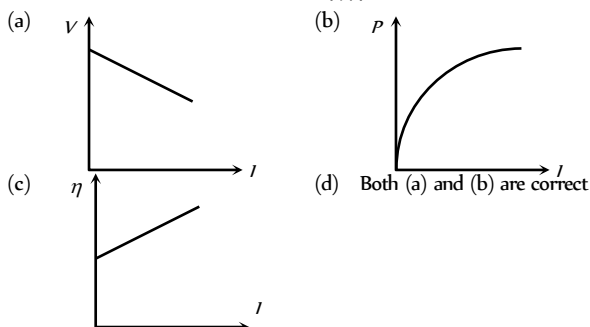
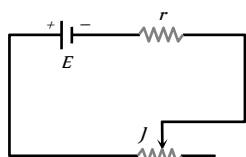


8. In a copper voltmeter, mass deposited in 6 minutes is m gram. If the current-time graph for the voltmeter is as shown here, then the E.C.E of the copper is

- (a) $m/5$
(b) $m/300$
(c) $5m$
(d) $m/18000$



9. Battery shown in figure has e.m.f. E and internal resistance r . Current in the circuit can be varied by sliding the contact J . If at any instant current flowing through the circuit is I , potential difference between terminals of the cell is V , thermal power generated in the cell is equal to η fraction of total electrical power generated in it; then which of the following graph is correct

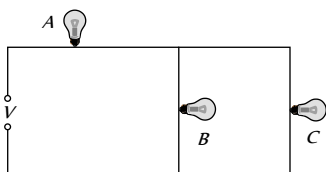


Read the assertion and reason carefully to mark the correct option out of the options given below :

- (a) If both assertion and reason are true and the reason is the correct explanation of the assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of the assertion.
(c) If assertion is true but reason is false.
(d) If the assertion and reason both are false.
(e) If assertion is false but reason is true.

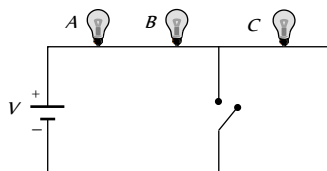
- Assertion : The possibility of an electric bulb fusing is higher at the time of switching ON and OFF
Reason : Inductive effects produce a surge at the time of switch ON and OFF [AIIMS 2003]
- Assertion : The 200 W bulbs glows with more brightness than 100 W bulbs.
Reason : A 100 W bulb has more resistance than a 200 W bulb.
- Assertion : Fuse wire must have high resistance and low melting point.
Reason : Fuse is used for small current flow only.
- Assertion : Two electric bulbs of 50 and 100 W are given. When connected in series 50 W bulb glows more but when connected parallel 100 W bulb glows more.
Reason : In series combination, power is directly proportional to the resistance of circuit. But in parallel combination, power is inversely proportional to the resistance of the circuit.
- Assertion : Two bulbs of same wattage, one having a carbon filament and the other having a metallic filament are connected in series. Metallic bulbs will glow more brightly than carbon filament bulb.
Reason : Carbon is a semiconductor.
- Assertion : An electric bulb is first connected to a dc source and then to a ac source having the same brightness in both the cases.
Reason : The peak value of voltage for an A.C. source is $\sqrt{2}$ times the root mean square voltage.
- Assertion : Current is passed through a metallic wire, heating it red. When cold water is poured on half of its portion, then rest of the half portion become more hot.
Reason : Resistances decreases due to decrease in temperature and so current through wire increases.
- Assertion : Through the same current flows through the line wires and the filament of the bulb but heat produced in the filament is much higher than that in line wires.
Reason : The filament of bulbs is made of a material of high resistance and high melting point.
- Assertion : Neutral temperature of a thermocouple does not depend upon temperature of cold junction.
Reason : Its value is constant for the given metals of the couple.
- Assertion : In practical application, power rating of resistance is not important.
Reason : Property of resistance remain same even at high temperature.
- Assertion : Leclanche cell is used, when constant supply of electric current is not required.
Reason : The e.m.f. of a Leclanche cell falls, if it is used continuously.

12. Assertion : In the given circuit if lamp *B* or *C* fuses then light emitted by lamp *A* decreases.



Reason : Voltage on *A* decreases.

13. Assertion : If three identical bulbs are connected in series as shown in figure then on closing the switches. Bulb *C* short circuited and hence illumination of bulbs *A* and *B* decreases.



Reason : Voltage on *A* and *B* decreases

14. Assertion : Heat is generated continuously in an electric heater but its temperature becomes constant after some time.

Reason : At the stage when heat produced in heater is equal to the heat dissipated to its surrounding the temperature of heater becomes constant.

15. Assertion : Electric appliances with metallic body; e.g. heaters, presses etc, have three pin connections, whereas an electric bulb has a two pin connection. [AIIMS 1996]

Reason : Three pin connections reduce heating of connecting cables.

16. Assertion : A laser beam 0.2 W power can drill holes through a metal sheet, whereas 1000 W torch-light cannot.

Reason : The frequency of laser light is much higher than that of torch light. [AIIMS 1996]

17. Assertion : A domestic electrical appliance, working on a three pin will continue working even if the top pin is removed. [AIIMS 1995]

Reason : The third pin is used only as a safety device.

18. Assertion : In all conductors, for studying the thermoelectric behaviour or metals, lead is taken as a reference metal.

Reason : In lead, the Thomson effect is negative.

19. Assertion : The presence of water molecules makes separation of ions easier in electrolyte.

Reason : The presence of water molecules in electrolyte decreases the resistance of electrolyte.

20. Assertion : Thermocouple acts as a heat engine.

Reason : When two junctions of thermocouple are at different temperature, thermo e.m.f. is produced.

21. Assertion : When temperature of cold junction of a thermocouple is lowered, the value of neutral temperature of this thermocouple is raised.

Reason : When the difference of temperature of two junction is raised, more thermo e.m.f. is produced.

Heating Effect of Current

1	a	2	b	3	c	4	b	5	b
6	c	7	c	8	a	9	b	10	c
11	a	12	c	13	c	14	a	15	d
16	d	17	b	18	c	19	c	20	a
21	d	22	a	23	a	24	c	25	d
26	d	27	c	28	c	29	b	30	b
31	b	32	b	33	a	34	d	35	c
36	b	37	d	38	b	39	a	40	a
41	a	42	b	43	d	44	d	45	a
46	c	47	a	48	b	49	a	50	a
51	b	52	c	53	a	54	c	55	c
56	a	57	d	58	b	59	d	60	a
61	d	62	a	63	d	64	b	65	b
66	d	67	a	68	a	69	a	70	c
71	a	72	d	73	a	74	b	75	a
76	d	77	b	78	b	79	d	80	d
81	a	82	d	83	d	84	d	85	a
86	a	87	a,d	88	a	89	b	90	a
91	b	92	a	93	b	94	a	95	b
96	a	97	c	98	b	99	c	100	a
101	b	102	c	103	a	104	a	105	d
106	c	107	a	108	b	109	b	110	d
111	c	112	b	113	c	114	a	115	c
116	c	117	d	118	a	119	d	120	c
121	b	122	d	123	a	124	a	125	a
126	c	127	a	128	d	129	a	130	c
131	a	132	c	133	c	134	c	135	b
136	c	137	c	138	c	139	b	140	d
141	a	142	c	143	d	144	c	145	b
146	c	147	c	148	c	149	a	150	c
151	a	152	c	153	c				

Chemical Effect of Current

1	d	2	c	3	d	4	d	5	c
6	d	7	b	8	d	9	b	10	a
11	b	12	b	13	b	14	c	15	b
16	a	17	b	18	c	19	a	20	d
21	c	22	b	23	c	24	a	25	d
26	b	27	a	28	c	29	a	30	a
31	b	32	d	33	b	34	a	35	b
36	c	37	c	38	b	39	d	40	b
41	c	42	b	43	d	44	a	45	c
46	c	47	a	48	b	49	b	50	b
51	a	52	d	53	d	54	a	55	c

Answers

56	c	57	d	58	c	59	a
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Thermo-Electricity

1	b	2	c	3	d	4	a	5	c
6	b	7	a	8	a	9	a	10	a
11	a	12	d	13	b	14	c	15	b
16	b	17	a	18	a	19	a	20	b
21	d	22	a	23	b	24	a	25	b
26	c	27	c	28	d	29	c	30	a
31	a	32	b	33	d	34	c	35	c
36	d	37	c	38	d	39	d	40	a
41	d	42	a	43	b	44	b		

Critical Thinking Questions

1	d	2	b	3	b	4	d	5	c
6	d	7	a	8	c	9	d	10	b
11	c	12	b	13	d	14	a	15	a
16	d	17	d	18	a	19	a	20	b
21	c	22	b	23	b	24	b	25	d
26	a	27	b	28	c	29	a	30	b
31	d	32	a	33	a	34	d	35	d
36	d	37	d	38	d	39	b	40	b
41	a	42	d	43	b	44	a	45	c

Graphical Questions

1	b	2	d	3	d	4	a	5	b
6	a	7	d	8	b	9	d		

Assertion and Reason

1	a	2	a	3	c	4	a	5	d
6	e	7	a	8	a	9	b	10	d
11	a	12	a	13	d	14	a	15	c
16	c	17	a	18	c	19	b	20	b
21	d								

Answers and Solutions

Heating Effect of Current

1. (a) $1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ sec} = 36 \times 10^5 \text{ W-sec (or J)}$

$$2. (b) P \propto \frac{1}{R} \Rightarrow \frac{P_1}{P_2} = \frac{R_2}{R_1} \Rightarrow \frac{200}{100} = \frac{R_2}{R_1} \Rightarrow R_2 = 2R_1$$

$$3. (c) P = \frac{V^2}{R} \Rightarrow R_1 = \frac{V_1^2}{P_1} = \frac{(200)^2}{40} = 1000\Omega$$

$$\text{and } R_2 = \frac{V_2^2}{P_2} = \frac{(200)^2}{100} = 400\Omega$$

4. (b) When two bulbs are connected in series, the current will be same in both the bulbs. As a result potential drop will be more in the bulb of higher resistance *i.e.*, bulb of lower wattage.

5. (b) When 1 bulb fuses, the total resistance of the circuit decreases hence the current increases. Since $P = i^2 R$, therefore illumination increases.

6. (c)

$$7. (c) \text{ We know that } \frac{P_1}{P_2} = \frac{R_2}{R_1} = \frac{2}{1}$$

$$8. (a) P = \frac{V^2}{R} \Rightarrow P \propto \frac{1}{R} \text{ and } R \propto l \therefore P \propto \frac{1}{l} \Rightarrow \frac{P_1}{P_2} = \frac{l_1}{l_2} = \frac{2}{1}$$

$$9. (b) R_{\text{wire}} \propto \text{Temperature and } R_{\text{wire}} \propto \frac{1}{\text{Temperature}}$$

10. (c)

11. (a) In series, current is same in both the bulbs, hence

$$P \propto R (P = i^2 R) \therefore \frac{P_1}{P_2} = \frac{R_1}{R_2} = \frac{1}{2}$$

12. (c) In this case, $P = \frac{V^2}{R}$ or $P \propto \frac{1}{R}$ and R will be minimum, when divided four parts are joints in parallel to the battery.

13. (c) Length is immaterial for an electric fuse wire.

$$14. (a) P_{\text{Rated}} \propto \frac{1}{R} \text{ and } R \propto \frac{1}{(\text{Thickness of filament})^2}$$

$$\text{So } P_{\text{Rated}} \propto (\text{Thickness of filament})^2$$

$$15. (d) \text{ In series } P_s = \frac{P}{n} \Rightarrow 10 = \frac{P}{3} \Rightarrow P = 30 \text{ W}$$

$$\text{In parallel } P_p = nP = 3 \times 30 = 90 \text{ W}$$

$$16. (d) \text{ Energy consumed in kWh} = \frac{\text{Watt} \times \text{hour}}{1000}$$

$$\Rightarrow \text{For 30 days, } P = \frac{10 \times 50 \times 10}{1000} \times 30 = 150 \text{ kWh}$$

$$17. (b) W = qV \text{ also } P = i \times V = \frac{W}{t}$$

18. (c) Because given voltage is very high,

$$19. (c) P_p = nP = 2 \times 40 = 80 \text{ W}$$

20. (a) In series, $P \propto R$ (i is same), *i.e.* in series Fine wire (high R) liberates more energy.

$$\text{In parallel, } P \propto \frac{1}{R} (V \text{ is same}) \text{ i.e. thick wire (less } R) \text{ liberates more energy.}$$

$$21. (d) \text{ Resistance of the bulb} = \frac{V^2}{P_{\text{Rated}}} = \frac{220 \times 220}{100} = 484\Omega$$

When connected with 110 V, the power consumed

$$P_{\text{Consumed}} = \frac{V^2}{R} = \frac{110 \times 110}{484} = 25W$$

22. (a) The resistance of 25 W bulb is greater than 100 W bulb. So for the same current, heat produced will be more in 25 W bulb. So it will glow more brightly.

23. (a) Equivalent resistance in the second case $= R_1 + R_2 = R$

Now, we know that $P \propto \frac{1}{R}$

Since in the second case the resistance $(R_1 + R_2)$ is higher than that in the first case (R) .

Therefore power dissipation in the second case will be decreased.

24. (c) For constant voltage, we know that $P \propto \frac{1}{R}$

So higher the power, lower will be the resistance.

25. (d) $P = \frac{V^2}{R}$ but $R = \frac{\rho l}{A} \Rightarrow P = \frac{V^2}{\rho l / A} = \frac{AV^2}{\rho l}$. Since

$\frac{AV^2}{l}$ is constant as per given conditions So $P \propto \frac{1}{\rho}$.

26. (d) Power consumed means heat produced.

For constant potential difference $P_{\text{consumed}} = \text{Heat} \propto \frac{1}{R_{eq}}$

$$\therefore \frac{H_1}{H_2} = \frac{R_2}{R_1} = \frac{R/2}{2R} = \frac{1}{4}$$

(Since $R_2 = \frac{R \cdot R}{R + R} = \frac{R}{2}$ and $R_1 = R + R = 2R$)

27. (c) Resistance of carbon filament decreases with temperature while that of tungsten increases with temperature
In series $P_{\text{Consumed}} \propto R$ i.e. tungsten bulb will glow more brightly
28. (c) Power of the combination $P_s = \frac{P}{n} = \frac{1000}{2} = 500 \text{ W}$
29. (b) For parallel combination
 $P_{\text{Consumed}} \propto \text{Brightness} \propto P_{\text{Rated}}$
30. (b) Resistance of 25 W bulb $= \frac{220 \times 220}{25} = 1936 \Omega$
Its safe current $= \frac{220}{1936} = 0.11 \text{ amp.}$
Resistance of 100 W bulb $= \frac{220 \times 220}{100} = 484 \Omega$
Its safe current $= \frac{220}{484} = 0.48 \text{ amp.}$
When connected in series to 440 V supply, then the current $I = \frac{440}{(1936 + 484)} = 0.18 \text{ amp.}$
Thus current is greater for 25 W bulb, so it will fuse.
31. (b) $P = i^2 R \Rightarrow \frac{\Delta P}{P} = \frac{2\Delta i}{i}$ ($R \rightarrow \text{Constant}$)
 $\Rightarrow \% \text{ change in power} = 2 \times \% \text{ change in current}$
 $= 2 \times 1 = 2\%$
32. (b) $P_{\text{max}} = n \left(\frac{E^2}{4r} \right) = 2 \left(\frac{2 \times 2}{4 \times 1} \right) = 2 \text{ W}$
33. (a) $H \propto \frac{1}{R}$ (If $V = \text{constant}$) $\Rightarrow \frac{H_1}{H_2} = \frac{R_2}{R_1} = \frac{l_2 A_1}{l_1 A_2} = \frac{l_2 r_1^2}{l_1 r_2^2}$
 $\Rightarrow H_2 = 2H_1$
34. (d) $\frac{H}{t} = \frac{V^2}{R} \Rightarrow \frac{H}{t} \propto \frac{1}{R}$
35. (c) $H = i^2 R t$ and $i = \frac{q}{t}$. Hence $H = \frac{q^2 R}{t}$; $\therefore H \propto q^2$
36. (b) $E = \frac{1100 \times 4}{1000} = 4.4 \text{ kWh}$
37. (d) After some time, thermal equilibrium will reach.
38. (b) At constant p.d., heat produced $= \frac{V^2}{R}$ i.e. $H \propto \frac{1}{R}$
39. (a) Power $= 3.75 \times 200 \text{ W} = 750 \text{ W} \approx 1 \text{ H.P.}$
40. (a) $\frac{V^2}{R} = P \Rightarrow R = \frac{V^2}{P} = \frac{220 \times 220}{100} = 484 \Omega$
41. (a) Since $P = VI \Rightarrow I = \frac{P}{V} = \frac{250000}{10000} = 25 \text{ A}$
42. (b) Power lost in cable $= 10 \times (25)^2 = 6250 \text{ W} = 6.25 \text{ kW}$
43. (d) Heat generated in both the cases will be same because the capacitor has the same energy initially
 $= \frac{1}{2} CV^2 = \frac{1}{2} \times 200 \times 10^{-6} \times (200)^2 = 4 \text{ J}$
44. (d) The bulbs are connected in parallel, hence each bulb consumes $\frac{48}{2} = 24 \text{ W}$. Therefore $\frac{V^2}{R} = 24$
 $\Rightarrow R = \frac{6 \times 6}{24} = 1.5 \Omega$
45. (a)
46. (c) The bulbs are in series, hence they will have the same current through them.
47. (a) When resistance is connected in series, brightness of bulb decreases because voltage across the bulb decreases.
48. (b) $R = \frac{V^2}{P} \Rightarrow R_1 = \frac{200 \times 200}{100} = 400 \Omega$ and
 $R_2 = \frac{100 \times 100}{200} = 50 \Omega$. Maximum current rating
 $i = \frac{P}{V}$
So $i_1 = \frac{100}{200}$ and $i_2 = \frac{200}{100} \Rightarrow \frac{i_1}{i_2} = \frac{1}{4}$.
49. (a) $\frac{R_1}{R_2} = \frac{P_2}{P_1} = \frac{100}{40} = \frac{5}{2}$. Resistance of 40 W bulb is $\frac{5}{2}$ times than 100 W. In series, $P = i^2 R$ and in parallel, $P = \frac{V^2}{R}$. So 40 W in series and 100 W in parallel will glow brighter.
50. (a) $P = \frac{V^2}{R} \Rightarrow \frac{P_p}{P_s} = \frac{R_s}{R_p} = \frac{(R_1 + R_2)}{R_1 R_2 / (R_1 + R_2)} = \frac{(R_1 + R_2)^2}{R_1 R_2}$
 $\Rightarrow \frac{100}{25} = \frac{(R_1 + R_2)^2}{R_1 R_2} \Rightarrow \frac{R_1}{R_2} = \frac{1}{1}$
51. (b) Total power $P = (800 + 3 \times 100)$
Also $P = Vi \Rightarrow 1100 = 220 \times i \Rightarrow i = 5 \text{ A}$
52. (c) Because $R \propto \frac{1}{P}$
53. (a) An ideal cell has zero resistance.
54. (c) Power loss in transmission $P_L = \frac{P^2 R}{V^2} \Rightarrow P_L \propto \frac{1}{V^2}$

55. (c) $H = \frac{V^2 t}{4.2 R}$ or $\frac{H}{t} = \frac{V^2}{4.2 R}$
 $\Rightarrow 800 = \frac{20 \times 20}{4.2 \times R} \Rightarrow R = \frac{5}{42} = 0.119 \approx 0.12 \Omega$
56. (a) Heat produced $H = \frac{V^2 t}{4.2 R} = H \propto \frac{1}{R}$ Hence $\frac{H_1}{H_2} = \frac{R_2}{R_1}$
57. (d) $\frac{H}{t} = i^2 R$. Here total $R = (21 + 4) = 25 \Omega$
 $\Rightarrow$ Rate of energy consumed $= 0.2 \times 0.2 \times 25 = 1 \text{ J/s}$
58. (b) When the heating coil is cut into two equal parts and these parts are joined in parallel, the resistance of coil is reduced to one fourth, so power consumed will become 4 times *i.e.* 400 Js^{-1} .
59. (d) The resistance of 40 W bulb will be more and 60 W bulb will be less.
60. (a) In series $P_{\text{Consumed}} \propto \text{Brightness} \propto \frac{1}{P_{\text{Rated}}}$
61. (d) $E = P \times t = 1000 \text{ W} \times 30 \text{ sec} = 3 \times 10^4 \text{ J}$
62. (a) Resistance R_1 of 500 W bulb $= \frac{(220)^2}{500}$
Resistance R_2 of 200 W bulb $= \frac{(220)^2}{200}$
When joined in parallel, the potential difference across both the bulbs will be same.
Ratio of heat produced $= \frac{V^2 / R_1}{V^2 / R_2} = \frac{R_2}{R_1} = \frac{5}{2}$
When joined in series, the same current will flow through both the bulbs.
Ratio of heat produced $= \frac{i^2 R_1}{i^2 R_2} = \frac{R_1}{R_2} = \frac{2}{5}$
63. (d) Charge $q = it = 0.5 \text{ A} \times 3600 \text{ sec} = 1800 \text{ coulomb}$
64. (b) $H = i^2 R t = \frac{V^2 t}{R} = \frac{120 \times 120 \times (10 \times 60)}{6} = 14.4 \times 10^5 \text{ joule}$
65. (b) In parallel $P_{\text{consumed}} \propto \text{Brightness} \propto \frac{1}{R}$
 $P_A > P_B$ (given) $\therefore R_A < R_B$
66. (d) $R = \rho \frac{l}{A}$ and $P \propto \frac{1}{R} \Rightarrow P \propto \frac{A}{l} \Rightarrow P \propto \frac{d^2}{l} \Rightarrow P_A = 2P_B$
67. (a) $t_s = t_1 + t_2 = 30 + 30 = 60 \text{ minutes}$
68. (a) For power transmission power loss in line $P_L = i^2 R$

If power of electricity is P and it is transmitted at voltage V , then $P = Vi \Rightarrow i = \frac{P}{V}$

$$P_L = \left(\frac{P}{V} \right)^2 R = \frac{P^2 R}{V^2} = \frac{2.2 \times 10^3 \times 2.2 \times 10^3 \times 10}{22000 \times 22000} = 0.1 \text{ W}$$

69. (a) $P = i^2 R$ (i and R are same)
So P will be same for given resistors.
70. (c) Since $H \propto i^2$, so on doubling the current, the heat produced and hence the rise in temperature becomes four times.
71. (a) *Watt-hour meter* measures electric energy.
72. (d) Total energy consumed $= \frac{60 \times 8}{1000} = 0.48 \text{ kWh}$
So cost $= 0.48 \times 1.25 = 0.6 \text{ Rs.}$
73. (a) $P_s = \frac{P}{n} = \frac{40}{4} = 10 \text{ W.}$
74. (b) As temperature increases resistance of filament also increases.
75. (a) Current through the combination $i = \frac{120}{(6+9)} = 8 \text{ A}$
So, power consumed by 6Ω resistance
 $P = (8)^2 \times 6 = 384 \text{ W}$
76. (d) $P = \frac{V^2}{R} = \frac{(225)^2}{50} = 1012.5 \approx 1000 \text{ W}$
77. (b) $P = Vi \Rightarrow i = \frac{P}{V} = \frac{100}{200} = 0.5 \text{ A}$
78. (b) $H = i^2 R t \Rightarrow R = \frac{H}{i^2 t} = \frac{80}{4 \times 10} = 2 \Omega$
79. (d) Heat produced = Energy stored in capacitor
 $= \frac{1}{2} C V^2 = \frac{1}{2} \times 4 \times 10^{-6} \times (400)^2 = 0.32 \text{ J}$
80. (d) $P = \frac{V^2}{R} = \frac{(110)^2}{10} = \frac{12100}{10} = 1210 \text{ W}$
81. (a) $P_{\text{consumed}} = \left(\frac{V_A}{V_R} \right)^2 \times P_R = \frac{(160)^2}{(200)^2} \times 100 = 64 \text{ W}$
82. (d) For maximum power $r = R$
83. (d) $P = i^2 R \Rightarrow 22.5 = (15)^2 \times R \Rightarrow R = 0.10 \Omega$
84. (d) $R_1 = \rho \frac{l_1}{A_1}$ and $R_2 = \rho \frac{l_2}{A_2} \Rightarrow \frac{R_1}{R_2} = \frac{l_1}{l_2} \cdot \frac{A_2}{A_1} = \frac{l_1}{l_2} \left(\frac{r_2}{r_1} \right)^2$

$$\text{Given } \frac{l_1}{l_2} = \frac{1}{2} \text{ and } \frac{r_1}{r_2} = \frac{2}{1} \text{ or } \frac{r_2}{r_1} = \frac{1}{2} \Rightarrow \frac{R_1}{R_2} = \frac{1}{8}$$

$$\therefore \text{Ratio of heats } \frac{H_1}{H_2} = \frac{V^2 / R_1}{V^2 / R_2} = \frac{R_2}{R_1} = \frac{8}{1}$$

$$85. (a) P = Vi = 250 \times 2 = 500 \text{ W}$$

$$86. (a) P = \frac{V^2}{R} \Rightarrow 100 = \frac{(200)^2}{R} \Rightarrow R = \frac{4 \times 10^4}{10^2} = 400 \Omega$$

$$\text{Now, } i = \frac{V}{R} = \frac{100}{400} = \frac{1}{4} \text{ amp}$$

$$87. (a, d) R_{\text{steel}} = 2R_{\text{Al}}. \text{ In series } H \propto R \text{ (i is Same)}$$

$$\text{So, } H \text{ will be more in steel wire. In parallel } H \propto \frac{1}{R}$$

(V is Same), so H will be more in aluminium wire.

$$88. (a) H = i^2 R t \Rightarrow \frac{H}{t} = i^2 R = \frac{i^2 \rho l}{\pi r^2}$$

$$89. (b)$$

$$90. (a) H = \frac{V^2}{R} \cdot t \Rightarrow \frac{H_1}{H_2} = \frac{R_2}{R_1} = \frac{R}{2R} = \frac{1}{2}$$

$$91. (b)$$

$$92. (a) \text{ In parallel } P_{\text{Consumed}} \propto P_{\text{Rated}}$$

$$93. (b)$$

$$94. (a) P = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{P} = \frac{(220)^2}{40} = 1210 \Omega$$

$$95. (b) P = Vi \Rightarrow i = \frac{P}{V} = \frac{60}{220} = \frac{3}{11} \text{ amp}$$

$$96. (a) \text{ In series, } P_{\text{Consumed}} \propto \frac{1}{P_{\text{Rated}}} \propto V_{\text{Applied}}$$

i.e. more voltage appears on smaller wattage bulb, so 25 W bulb will fuse

$$97. (c) \text{ Because in series current is same.}$$

$$98. (b) P = \frac{V^2}{R} \Rightarrow \frac{P_1}{P_2} = \frac{R_2}{R_1} \Rightarrow \frac{6}{P_2} = \frac{4}{6} = \frac{2}{3} \Rightarrow P_2 = 9 \text{ W}$$

$$99. (c) \frac{H}{t} = P = \frac{V^2}{R} \Rightarrow P \propto \frac{1}{R} \text{ also } R \propto \frac{l}{A} \propto \frac{l^2 \rho}{A \cdot l \rho}$$

$$\Rightarrow R \propto \frac{l^2}{m} \Rightarrow R \propto l^2 \text{ (for same mass)}$$

$$\text{So } \frac{P_A}{P_B} = \frac{l_B^2}{l_A^2} = \frac{4}{1} \Rightarrow P_A = 20 \text{ W}$$

$$100. (a) P = \frac{V^2}{R} \Rightarrow \frac{R_1}{R_2} = \frac{P_2}{P_1} = \frac{60}{40} = \frac{3}{2}$$

$$101. (b) P \propto V^2 \Rightarrow \frac{P}{P_0} = \left(\frac{V}{V_0} \right)^2 \Rightarrow P = \left(\frac{V}{V_0} \right)^2 P_0$$

$$102. (c) P = \frac{V^2}{R} \Rightarrow R \propto \frac{1}{P}$$

So resistance of the 100 W bulb will be minimum

$$103. (a) \text{ In parallel } \frac{1}{t_p} = \frac{1}{t_1} + \frac{1}{t_2} \Rightarrow t_p = \frac{t_1 t_2}{t_1 + t_2}$$

$$= \frac{5 \times 10}{5 + 10} = \frac{50}{15} = 3.33 \text{ min} = 3 \text{ min. } 20 \text{ sec}$$

$$104. (a) \text{ For maximum joule heat produced in resistor external resistance = Internal resistance.}$$

$$105. (d)$$

$$106. (c) H = \frac{V^2}{R} t \Rightarrow \frac{H_1}{H_2} = \frac{R_2}{R_1} = \frac{4}{2} = \frac{2}{1}$$

$$107. (a) \text{ If resistances of bulbs are } R_1 \text{ and } R_2 \text{ respectively}$$

$$\text{then in parallel } \frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow$$

$$\left(\frac{1}{V^2} \right) = \left(\frac{1}{P_1} \right) + \left(\frac{1}{P_2} \right)$$

$$\Rightarrow P_p = P_1 + P_2$$

$$108. (b)$$

$$109. (b) \text{ When wire is cut into two equal parts then power dissipated by each part is } 2P_1$$

So their parallel combination will dissipate power $P_2 = 2P_1 + 2P_1 = 4P_1$

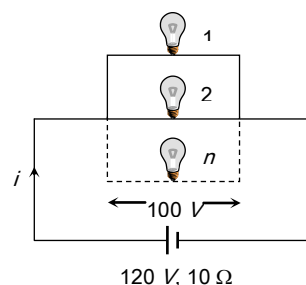
$$\text{Which gives } \frac{P_2}{P_1} = 4$$

$$110. (d) P = \frac{V^2}{R} \Rightarrow \frac{P_2}{P_1} = \frac{V_2^2}{V_1^2} (\because R \text{ is constant})$$

$$\Rightarrow \frac{P_2}{P_1} = \left(\frac{100}{200} \right)^2 = \frac{1}{4} \Rightarrow P_2 = \frac{P_1}{4} = \frac{40}{4} = 10 \text{ W}$$

$$111. (c) \text{ When each bulb is glowing at full power,}$$

$$\text{Current from each bulb } = i = \frac{50}{100} = \frac{1}{2} \text{ A}$$



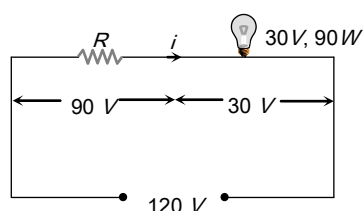
So main current $i = \frac{n}{2} A$

$$\text{Also } E = V + ir \Rightarrow 120 = 100 + \left(\frac{n}{2}\right) \times 10 \Rightarrow n = 4$$

112. (b) $R = \frac{V^2}{P} = \frac{(250)^2}{10^3} = 62.5 \Omega$

113. (c) Suppose resistance R is connected in series with bulb.

Current through the bulb $i = \frac{90}{30} = 3 A$



Hence for resistance $V = iR \Rightarrow 90 = 3 \times R \Rightarrow R = 30 \Omega$

114. (a) $i \propto r^{3/2} \Rightarrow \frac{r_2}{r_1} = \left(\frac{i_2}{i_1}\right)^{2/3} = \left(\frac{3}{1.5}\right)^{2/3} = (4)^{1/3}$

$$\Rightarrow r_2 = (4)^{1/3} \times r_1 = 4^{1/3} (\because r_1 = 1 \text{ mm})$$

115. (c) In series $P' = \frac{P}{n} = \frac{60}{3} = 20 \text{ watts}$

116. (c) $R = \frac{V^2}{P} = \frac{(220)^2}{60} = 807 \Omega$

117. (d) $\frac{P_1}{P_2} = \left(\frac{V_1}{V_2}\right)^2 \Rightarrow \frac{1000}{P_2} = \left(\frac{220}{110}\right)^2 = 4 \Rightarrow P_2 = 250 W$

118. (a) $P_s = \frac{P_1 P_2}{P_1 + P_2} = \frac{100 \times 200}{100 + 200} = \frac{200}{3} \approx 65 \text{ watt}$

119. (d) $H = \frac{V^2 t}{R \times J} \text{ Calories} = \frac{P t}{J} = \frac{210 \times 5 \times 60}{4.2} = 15000 \text{ cal}$

120. (c) Using conservation of energy

Supplied electric energy = absorbed heat energy

$$\Rightarrow i^2 R t = m S T$$

$$\Rightarrow T \propto i^2 \quad (T - \text{change in temperature})$$

i.e. when i is doubled T will be four times i.e.

$$5 \times 4 = 20^\circ C$$

121. (b) Energy = $P \times t = 2 \times 1 \times 30 = 60 \text{ kWh} = 60 \text{ unit}$

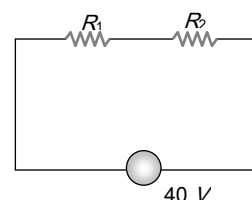
122. (d) Bulb (I) : Rated current $I_1 = \frac{P}{V} = \frac{40}{220} = \frac{2}{11} \text{ amp.}$

$$\text{Resistance } R_1 = \frac{V^2}{P} = \frac{(220)^2}{40} = 1210 \Omega$$

Bulb (II) : Rated current $I_2 = \frac{100}{220} = \frac{5}{11} \text{ amp}$

$$\text{Resistance } R_2 = \frac{(220)^2}{100} = 484 \Omega$$

When both are connected in series across 40 V supply



Total current through supply

$$I = \frac{40}{P_1 + P_2} = \frac{40}{1210 + 484} = \frac{40}{1254} = 0.03 A$$

This current is less than the rated current of each bulb. So neither bulb will fuse.

Short Trick : Since $V_{\text{Applied}} < V_{\text{Rated}}$, neither bulb will fuse.

123. (a) Both R and $2R$ in parallel (V - constant)

$$\text{So using } P = \frac{V^2}{R} \Rightarrow \frac{P_1}{P_2} = \frac{R_2}{R_1} \Rightarrow \frac{H_1}{H_2} = \frac{R_2}{R_1} = \frac{2}{1}$$

124. (a) Power $P = V i t = 250 \times 4 = 1000 W = 1 \text{ kW}$

$$\text{Energy} = P \times t = 1 \text{ kW} \times 60 \text{ sec} = 60 \text{ kJ}$$

125. (a) $P = \frac{V^2}{R} \Rightarrow P \propto \frac{1}{R} \quad (V - \text{constant})$

$\therefore$ When one bulb will fuse out resistance of the series combination will be reduced.

Hence from $P_{\text{Consumed}} \propto \frac{1}{R}$ illumination will increase.

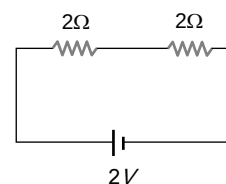
126. (c) $P = \frac{V^2}{R} \Rightarrow R = \frac{V^2}{P} = \frac{25 \times 25}{25} = 25 \Omega$

127. (a) $P_{\text{Rated}} = \frac{V_{\text{Rated}}^2}{R} \Rightarrow R \propto \frac{1}{P_{\text{Rated}}} \quad (V - \text{constant})$

So bulb of high power will have less resistance.

128. (d) $P_{Rated} \propto \frac{1}{R} \Rightarrow \frac{R_1}{R_2} = \frac{P_2}{P_1} = \frac{60}{40} = \frac{3}{2}$
129. (a) Energy $= \frac{V^2}{R} \times t = \frac{10 \times 10}{50} \times 3600 = 7200 J$
130. (c) Energy $\frac{V^2}{R} t = \frac{200 \times 200 \times 2}{80} = 1000 Wh$
131. (a) Energy $= P \times t = 100 \times 2 \times 60 = 12000 J = 12 \times 10^3 J$
132. (c) Heat $H = \frac{V^2 t}{R} \Rightarrow H \propto \frac{1}{R}$ (If V, t constant)
- $$\Rightarrow \frac{H_S}{H_P} = \frac{R_P}{R_S} = \frac{\left(\frac{R \times 2R}{3R}\right)}{(R+2R)} = \frac{2}{9}$$
133. (c) $i \propto \frac{1}{R}$ and $P \propto \frac{1}{R} \Rightarrow i \propto P$ i.e. in parallel bulb of higher power will draw more current.
134. (c) Resistance of A is greater than the resistance of combination of B and C , hence voltage drop across A will be greater than that across B or C . Also $H = \frac{V^2 t}{R}$
- $$\Rightarrow H \propto V^2 \text{ so } H_A > (H_B = H_C) \quad (R = \text{constant})$$
135. (b) $P = Vi \Rightarrow i = \frac{2.2 \times 10^3}{22000} = \frac{1}{10} A$
- Now loss of power $= i^2 R = \left(\frac{1}{10}\right)^2 \times 100 = 1 W$
136. (c) $P = \frac{V^2}{R}$. If resistance of heater coil is R , then resistance of parallel combination of two halves will be $\frac{R}{4}$
- So $\frac{P_1}{P_2} = \frac{R_2}{R_1} = \frac{R/4}{R} = \frac{1}{4}$
137. (c) Total kWh consumed $= \frac{60 \times 8 \times 30}{1000} = 14.4$
- Hence cost $= 14.4 \times 1.25 = 18 Rs$
138. (c) Current capacity of a fuse wire should be slightly greater than the total rated load current.
139. (b)
140. (d) Colliding electrons lose their kinetic energy as heat.
141. (a) It is called safe current and is proportional to $r^{3/2}$.
142. (c)
143. (d) $i = \frac{P}{V} = \frac{50}{250} = 0.2 amp.$

144. (c) In steady state the branch containing capacitors, can be neglected. So reduced circuit is as follows



$$\text{Power } P = \frac{V^2}{R} = \frac{(2)^2}{4} = 1 W.$$

145. (b) $P = \frac{V^2}{R_{eq}} \Rightarrow 150 = \frac{(15)^2}{[2R/(R+2)]} = \frac{225 \times (R+2)}{2R}$
- $$\Rightarrow R = \frac{450}{75} = 6 \Omega.$$
146. (c) $P = \frac{V^2}{R} \Rightarrow \frac{P_1}{P_2} = \frac{R_2}{R_1} = \frac{1}{2}$.
147. (c) $H = \frac{V^2 t}{R} \Rightarrow \frac{H_{Half}}{H_{Full}} = \left(\frac{R_{Full}}{R_{Half}}\right) = \frac{R}{R/2} = 2$
- $$\Rightarrow H_{Half} = 2 \times H_{Full}.$$
148. (c) It is given $R_{Hot} = 10 R_{Cold}$ also resistance at rated temperature $R = \frac{V^2}{P} = \frac{200 \times 200}{100} = 400 \Omega$.
- So resistance when lamp not in use.
- $$R_{Cold} = \frac{R_{Hot}}{10} = \frac{400}{10} = 40 \Omega$$
149. (a) The chemical energy reduced in battery
- $$= VIt = 6 \times 5 \times 6 \times 60 J = 10800 J = 1.08 \times 10^4 J$$
150. (c) The heat generated $= I^2 R t = 2.1 \times 15 \times 1 = 31.5 J$
- $$= 31.5 / 4.2 cal = 7.5 cal. \quad [\because 1 cal = 4.2 J]$$
151. (a) Resistance $\propto \frac{1}{\text{power}}$. Thus, 40 W bulb has a high resistance. Because of which there will be more potential drop across 40 W bulb. Thus 40 W bulb will glow brighter.
152. (c) When bulbs are connected in series, $P = \frac{V^2}{R} = \frac{V^2}{3R}$
- When bulbs are connected in parallel,
- $$P' = \frac{V^2}{R''} = \frac{V^2 \times 3}{R} = 3 \times 3P = 9P.$$
153. (c) Time $t_s = t_1 + t_2 = 35 \text{ min}.$

1. (d) As sugar cannot be decomposed into ions and ions are responsible for conduction.
2. (c) $\because \frac{Z_1}{Z_2} = \frac{E_1}{E_2} \Rightarrow Z_2 = \left(\frac{E_2}{E_1} \right) \cdot Z_1$
3. (d) $\frac{m_{Zn}}{m_{Ag}} = \frac{E_{Zn}}{E_{Ag}} \Rightarrow m_{Ag} = W \left(\frac{E_{Ag}}{E_{Zn}} \right) = 3.3 \text{ W} = 3.5 \text{ W}$
4. (d) 96500 *coulombs* of charge is needed to deposit one gram equivalent of an element at an electrode.
5. (c) As $\frac{m_{Cu}}{m_{Ag}} = \frac{E_{Cu}}{E_{Ag}} = \frac{\frac{1}{2}(\text{Atomic weight})_{Cu}}{(\text{Atomic weight})_{Ag}}$
6. (d) $V_2 = \frac{22.4 \times 1}{1} = 22.4 \text{ litre at NTP}$
 $\therefore 11.2 \text{ litre of } H_2 \text{ is liberated by } 96,500 \text{ C}$
 $\therefore 22.4 \text{ litre of } H_2 \text{ is liberated by } 96500 \times 2 = 1,93,000 \text{ C}$
7. (b) From $m = ZQ$, if $Q = 1 \text{ C} \Rightarrow m = Z$
8. (d)
9. (b) Because H^+ has positive charge.
10. (a) Because H_2O is used as electrolyte.
11. (b) $m = Zit \Rightarrow 1 = 0.00033 \times 2 \times t$
 $\therefore t = \frac{1}{0.00066 \times 60} \text{ min} = \frac{100000}{3960} \approx 25 \text{ min}$
12. (b) $3 = 1.5(1+r) \Rightarrow r = 1\Omega$
13. (b)
14. (c) $m = Zit = Zq$; $q = \frac{5 \times 10^{-3}}{3.387 \times 10^{-7}} \text{ amp-sec}$
or $q = \frac{5 \times 10^{-3}}{3.387 \times 10^{-7} \times 3600} \text{ amp-hr} = 4.1$
15. (b) Charge $Q = It = 1.6 \times 60 = 96 \text{ C}$
Let n be the number of Cu^{+2} ions, then
 $ne = Q \Rightarrow n = \frac{Q}{e} = \frac{96}{2 \times 1.6 \times 10^{-19}} = 3 \times 10^{20}$
16. (a) In the first case, $Zit = m$
In the second case, $Z \times \frac{i}{4} \times 4t = m$
17. (b) $\frac{\text{Mass of } O_2 \text{ ions}}{\text{Mass of Ag ions}} = \frac{\text{Chemical equivalent of } O_2}{\text{Chemical equivalent of Ag}}$
 $\Rightarrow \frac{0.8}{m} = \frac{8}{108} \Rightarrow m = 10.8 \text{ gm}$
18. (c)
19. (a) $F = Ne = 6 \times 10^{23} \times 1.6 \times 10^{-19}$
20. (d) Since 1 *faraday* deposits 1 *gm* equivalent.
21. (c) Equivalent weight of copper $= \frac{64}{2} = 32$
 $\frac{\text{Equivalent weight of Cu}}{\text{Equivalent weight of Ag}} = \frac{\text{Weight of Cu deposited}}{\text{Weight of Ag deposited}}$
Weight of copper deposited $= \frac{10.8 \times 32}{108} = 3.2 \text{ gm}$
22. (b)
23. (c)
24. (a) $m \propto q \Rightarrow m \propto it$
25. (d) Equivalent weight of aluminium $= \frac{27}{3} = 9$
So 1 *faraday* = 96500 *C* are required to liberate 9 *gm* of *Al*.
26. (b) By Faraday's law, $m \propto it$.
 $\therefore \frac{m_1}{m_2} = \frac{i_1 t_1}{i_2 t_2} \Rightarrow \frac{m}{m_2} = \frac{4 \times 120}{6 \times 40} \Rightarrow m_2 = \frac{m}{2}$
27. (a) $m \propto it$
28. (c) Amount of metallic sodium appears
 $m = Zit = \left(\frac{A}{VF} \right) it$
 $= \left(\frac{23}{1 \times 96500} \right) \times 16 \times 10 \times 60 = 2.3 \text{ gm}$
29. (a)
30. (a) $m = Zit \Rightarrow Z = \frac{m}{it} = \frac{4.572}{5 \times 45 \times 60} = 3.387 \times 10^{-4} \text{ gm/C}$
31. (b) Faraday constant = 1 mole electron charge = *Ne*
 $= 6.02 \times 10^{23} \times 1.6 \times 10^{-19} = 96500$
32. (d) $m = Zit = 0.126 \times 10^{-3} \times 5 \times 3600 = 2.27 \text{ gm}$
33. (b) $\frac{m_1}{m_2} = \frac{E_1}{E_2}$ (By faraday law for same current and time)
Where E_1 and E_2 are the chemical equivalents and m_1 and m_2 are the masses of copper and silver respectively.
 $E = \frac{\text{Atomic weight}}{\text{Valency}} \cdot E_1 = \frac{63.57}{2} = 31.79$ and
 $E_2 = \frac{107.88}{1} = 107.88$
 $\therefore \frac{1 \text{ mg}}{m_2} = \frac{31.79}{107.88} \Rightarrow m_2 = \frac{107.88}{31.79} \text{ mg} = 3.4 \text{ mg}$
34. (a) $m = Zit \Rightarrow \frac{m}{Zit} = 1 \text{ (constant)}$

35. (b) Positive ions get deposited on cathode.
36. (c) $m = Zit$ or $m \propto it$
- $$\therefore \frac{m_1}{m_2} = \frac{i_1 t_1}{i_2 t_2} \Rightarrow \frac{9}{m_2} = \frac{10^5}{50 \times 20 \times 60} \Rightarrow m_2 = 5.4 \text{ gm}$$
37. (c) Electroplating only provides a thin deposition of a metal on the surface which in no way can give hardness to the metal.
38. (b)
39. (d) $m = Zit \Rightarrow \frac{m_{\text{Cu}}}{m_{\text{Zn}}} = \frac{Z_{\text{Cu}}}{Z_{\text{Zn}}}$
- $$m_{\text{Cu}} = m_{\text{Zn}} \frac{Z_{\text{Cu}}}{Z_{\text{Zn}}} = 0.13 \times \frac{31.5}{32.5} = 0.126 \text{ g}$$
40. (b) $m = Zit \Rightarrow m = \frac{ZVt}{R} \Rightarrow m \propto Vt \Rightarrow \frac{m_1}{m_2} = \frac{V_1 t_1}{V_2 t_2}$
- $$\Rightarrow \frac{2}{m_2} = \frac{12 \times 30}{6 \times 45} \Rightarrow m_2 = 1.5 \text{ gm}$$
41. (c) $i = \frac{m}{Zt} = \frac{0.972}{0.00018 \times 3 \times 3600} = 0.5 \text{ A}$
42. (b) The current through the voltmeter is same as drawn from the battery outside it.
43. (d) The resistance of the cell is independent of e.m.f.
44. (a) $m = Zit = 3.3 \times 10^{-7} \times 3 \times 2 = 19.8 \times 10^{-7} \text{ kg}$
45. (c) $m = zq$, z = atomic mass / valence
46. (c)
47. (a)
48. (b) 1 faraday (96500 C) is the electricity which liberated that amount of substance which is equal to equivalent wt. So liberated amount of Cu is $\frac{63.5}{2}$
- $$= 31.25 \text{ gm} \approx 32 \text{ gm}$$
49. (b) $m = Zit \Rightarrow 20 \times 10^{-3} = \left(\frac{32}{96500} \right) \times 0.15 \times t$
- $$= 6.7 \text{ min} = 6 \text{ min.} 42 \text{ sec}$$
50. (b) 22.4 litre H_2 = 1 mole H_2 = N molecules of H_2
- $$= 2N \text{ atom of H}$$
- So charge required to liberate 22.4 litre of H_2 = $2Ne$ = $2F$
- Hence charge required to liberate 0.224 litre of H_2
- $$= \frac{2F}{22.4} \times 0.224 = \frac{2F}{100} = 2 \times 965 \text{ C}$$

$$\text{So current } i = \frac{Q}{t} = \frac{2 \times 965}{100} = 19.3 \text{ amp}$$

51. (a)
52. (d) $m = Zit \Rightarrow Z = \frac{m}{it} = \frac{4.5}{4 \times 40 \times 60} = 47 \times 10^{-5} \text{ g/C}$
53. (d) Charge supplied per minute = $3.2 \times 60 = 192 \text{ C}$
- Charge $2e$ liberates one Cu^{+2} ion
- $$\therefore \text{No of } \text{Cu}^{+2} \text{ ion liberate by } 192 \text{ C}$$
- $$= \frac{192}{2e} = \frac{192}{2 \times 1.6 \times 10^{-19}} = 6 \times 10^{20}$$
54. (a) $m = Zit \Rightarrow i = \frac{m}{Zt} = \frac{0.99}{0.00033 \times 1200} = 2.5 \text{ A}$
- Hence heat generated in the coil is
- $$H = i^2 R t = (2.5)^2 \times 0.1 \times 1200 = 750 \text{ J}$$
55. (c) $\frac{m_1}{m_2} = \frac{Z_1}{Z_2} \Rightarrow m_2 = \frac{m_1 Z_2}{Z_1} = \frac{14 \times 1.2 \times 10^{-6}}{7 \times 10^{-6}} = 2.4 \text{ g}$
56. (c) $m = Zit \Rightarrow t = \frac{m}{Zi} = \frac{m \times F}{E \times i} \left(\because Z = \frac{E}{F} \right)$
- $$t = \frac{27 \times 96500}{108 \times 2} = 12062.5 \text{ sec} = \frac{12062.5}{3600} \text{ hr} = 3.35 \text{ hr}$$
57. (d) $m = zq \Rightarrow z \propto \frac{1}{q} \Rightarrow \frac{z_1}{z_2} = \frac{q_2}{q_1} \dots\dots(i)$
- also $q = q_1 + q_2 \Rightarrow \frac{q}{q_2} = \frac{q_1}{q_2} + 1$
- $$\Rightarrow q_2 = \frac{q}{1 + \frac{q_1}{q_2}} \dots\dots(ii)$$
- From equation (i) and (ii) $q_2 = \frac{q}{1 + \frac{z_2}{z_1}}$
58. (c) From Faraday's law, m/E = constant
- where m = mass of substance deposited, E = chemical equivalent.
- $$\therefore \frac{m_2}{m_1} = \frac{E_2}{E_1} \Rightarrow m_2 = \frac{108}{32} \times 1.6 = 5.4 \text{ g}$$
59. (a) $q = it$ = current $\times$ time

Thermo-Electricity

- (b) Production of e.m.f. by temperature difference is known Seeback effect.
- (c) Production of heat at junctions due to current is known as Peltier effect.

3. (d)
4. (a)
5. (c) When there is no deflection, then this temperature is called inversion temperature. It is given by the relation
- $$\theta_n = \frac{\theta_i + \theta_c}{2}$$
- Where θ_c is temperature of cold junction = 20°C and neutral temperature $\theta_n = 270^\circ\text{C}$
- $$\therefore \theta_i = 2\theta_n - \theta_c = 540 - 20 = 520^\circ\text{C}$$
6. (b)
7. (a) Thermo e.m.f. of a thermo couple depends on the nature of metals.
8. (a)
9. (a) According to the definition.
10. (a) $T_n = \frac{T_i + T_c}{2} \Rightarrow T_i = 2T_n - T_c$
11. (a)
12. (d)
13. (b) Based on Peltier effect.
14. (c) Peltier effect
15. (b) Thermopile is used for detection of heat radiation and measurement.
16. (b) $H = \sigma i t \Delta\theta \Rightarrow$ If $i = 1\text{ A}$, $\Delta\theta = 1^\circ\text{C}$, $t = 1\text{ sec}$ then $H = \sigma$.
17. (a) According to Seebeck effect
18. (a) At neutral temperature, $\frac{dE}{dT} = 0$
19. (a) According to Seebeck effect.
20. (b)
21. (d)
22. (a) As a rule, more the metals are separated from each other in the thermoelectric series, the greater will be the thermo emf.
23. (b) $T_n = \frac{T_i + T_c}{2} = \frac{10 + 530}{2} = 270^\circ\text{C}$
24. (a) Joule effect is not reversible.
25. (b)
26. (c)
27. (c) The graph between thermo emf and temperature of hot junction is parabolic in shape.
28. (d) At neutral temperature E is maximum so
- $$\frac{dE}{dt} = 0 \Rightarrow \frac{d}{dt}(At - Bt^2) = 0 \Rightarrow A - 2Bt = 0 \Rightarrow t = \frac{A}{2B}$$
29. (c) $t_n = \frac{t_i + t_c}{2} \Rightarrow 280 = \frac{t_i + 15}{2} \Rightarrow t_i = 545^\circ\text{C}$
30. (a)
31. (a) A is false because at neutral temperature thermo emf is maximum. B is true.
32. (b) Thermo-electric power $P = \frac{dE}{d\theta}$; at t_n , $E \rightarrow$ maximum. So $P \rightarrow$ zero.
33. (d) By using $H = \sigma Q \theta$
- $$\Rightarrow H = (10 \times 10^{-6}) \times 10 \times (60 - 50) = 10^{-3} \text{ J} = 1 \text{ mJ}$$
34. (c) No change in neutral temperature but temperature of inversion is $t_i = 2t_n - t_c \Rightarrow t_i = 2 \times 270 - 40 = 500^\circ\text{C}$
35. (c)
36. (d) $t_i = 2t_n - t_c \Rightarrow t_i = 2 \times 350 - 30 = 670^\circ\text{C}$
37. (c)
38. (d) Neutral temperature is independent of temperature of cold junction.
39. (d)
40. (a) $E = at + bt^2$ at inversion temperature E will be minimum
- $$\text{Thus } \frac{dE}{dt} = 0 \Rightarrow \frac{d}{dt}[at + bt^2] = 0$$
- $$\Rightarrow a + 2bt = 0 \Rightarrow t = -\frac{a}{2b}$$
41. (d)
42. (a) $i = \frac{e}{R} \Rightarrow 3 \times 10^{-7} = \frac{(30 \times 10^{-6}) \times \theta}{50} \Rightarrow \theta = 0.5^\circ$
43. (b) $t_n = \frac{\alpha}{\beta} = \left(\frac{500}{5}\right) = 100^\circ\text{C}$
- $$\text{Also } t_n = \frac{t_i + t_c}{2} \Rightarrow 100 = \frac{t_i + 0}{2} \Rightarrow t_i = 200^\circ\text{C}$$
44. (b) At neutral temperature, thermal emf will be maximum.
- $$\therefore \frac{de}{dt} = a + bt$$
- For maximum or minima, $a + bt_n = 0$
- $$\therefore t_n = -a/b$$

Critical Thinking Questions

1. (d) $P_1 = \frac{(220)^2}{R_1}$ and $P_2 = \frac{(220 \times 0.8)^2}{R_2}$

$$\frac{P_2}{P_1} = \frac{(220 \times 0.8)^2}{(220)^2} \times \frac{R_1}{R_2} \Rightarrow \frac{P_2}{P_1} = (0.8)^2 \times \frac{R_1}{R_2} \quad \text{Here } R_2 <$$

R_1

(because voltage decreases from $220 \text{ V} \rightarrow 220 \times 0.8 \text{ V}$)

It means heat produced $\rightarrow$ decreases)

$$\text{So } \frac{R_1}{R_2} > 1 \Rightarrow P_2 > (0.8)^2 P_1 \Rightarrow P_2 > (0.8)^2 \times 100 \text{ W}$$

$$\text{Also } \frac{P_2}{P_1} = \frac{(220 \times 0.8) i_2}{220 i_1}, \text{ Since } i_2 < i_1 \text{ (we expect)}$$

$$\text{So } \frac{P_2}{P_1} < 0.8 \Rightarrow P_2 < (100 \times 0.8)$$

Hence the actual power would be between $100 \times (0.8)^2 \text{ W}$ and $(100 \times 0.8) \text{ W}$

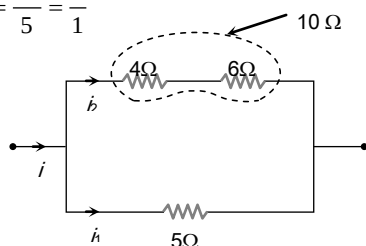
2. (b) $W = JH \Rightarrow P \times t = J \times m \times s \Delta \theta$

$$\Rightarrow t = \frac{J \times m \times s \Delta \theta}{P} \quad (\text{For water } 1 \text{ litre} = 1 \text{ kg})$$

$$\Rightarrow t = \frac{4.2 \times 1 \times 1000 \times (40 - 10)}{836} = 150 \text{ sec}$$

Short Trick : use formula $t = \frac{4200 \times m \times \Delta \theta}{P}$

3. (b) $\frac{i_1}{i_2} = \frac{R_2}{R_1} = \frac{10}{5} = \frac{2}{1}$



Also heat produced per sec i.e. $\frac{H}{t} = P = i^2 R$

$$\Rightarrow \frac{P_5}{P_4} = \left(\frac{i_1}{i_2} \right)^2 \times \frac{5}{4} = \left(\frac{2}{1} \right)^2 \times \frac{5}{4} = \frac{5}{1} \Rightarrow P_4 = \frac{10}{5} = 2 \text{ cal/s}$$

4. (d) $220 \times 9 = n(60) \Rightarrow n = 33$

5. (c) $H = \frac{V^2}{R} t$

Since supply voltage is same and equal amount of heat will produce, therefore

$$\frac{R_1}{t_1} = \frac{R_2}{t_2} \text{ or } \frac{R_1}{R_2} = \frac{t_1}{t_2} \quad \dots(i)$$

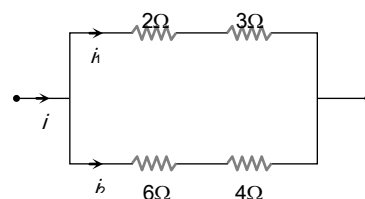
$$\text{But } R \propto l \Rightarrow \frac{R_1}{R_2} = \frac{l_1}{l_2} \quad \dots(ii)$$

$$\text{By (i) and (ii), } \frac{l_1}{l_2} = \frac{t_1}{t_2} \quad \dots(iii)$$

$$\text{Now } l_2 = \frac{2}{3} l_1 \Rightarrow \frac{l_1}{l_2} = \frac{3}{2}$$

$$\therefore \text{By equation (iii), } \frac{3}{2} = \frac{15}{t_2} \Rightarrow t_2 = 10 \text{ minutes}$$

6. (d)



Resistance of upper branch $R_1 = 2 + 3 = 5 \Omega$

Resistance of lower branch $R_2 = 4 + 6 = 10 \Omega$

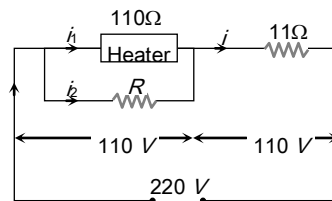
$$\text{Hence } \frac{i_1}{i_2} = \frac{R_2}{R_1} = \frac{10}{5} = 2$$

$$\frac{\text{Heat generated across } 3 \Omega (H_1)}{\text{Heat generated across } 6 \Omega (H_2)} = \frac{i_1^2 \times 3}{i_2^2 \times 6} = \frac{4}{2} = 2$$

$\therefore$ Heat generated across $3 \Omega = 120 \text{ cal/sec}$

7. (a) Power consumed by heater is 110 W so by using

$$P = \frac{V^2}{R}$$



$$110 = \frac{V^2}{110} \Rightarrow V = 110 \text{ V. Also from figure}$$

$$i_1 = \frac{110}{110} = 1 \text{ A and } i = \frac{110}{11} = 10 \text{ A. So } i_2 = 10 - 1 = 9 \text{ A}$$

Applying Ohms law for resistance R , $V = iR$

$$\Rightarrow 110 = 9 \times R \Rightarrow R = 12.22 \Omega$$

8. (c) $P_{\text{consumed}} = \left(\frac{V_A}{V_R} \right)^2 \times P_R = \left(\frac{110}{115} \right)^2 \times 500 = 457.46 \text{ W}$

So, percentage drop in power output

$$= \frac{(500 - 457.46)}{500} \times 100 = 8.6\%$$

9. (d) Heat produced $= \frac{V^2}{R} t$

i.e. when voltage is halved, heat produced becomes one-fourth. Hence time taken to heat the water becomes four times.

10. (b) Electric power consumed by kettle $P = 220 \times 4 \text{ W}$

Heat required

$$H = 1000 \times 1(100 - 20) = 1000 \times 80 \text{ cal} = 4200 \times 80 \text{ J}$$

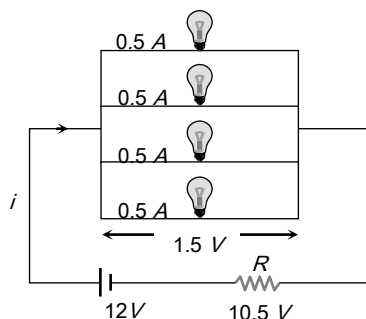
$$P = \frac{H}{t} \Rightarrow H = P \times t$$

$$\therefore 220 \times 4 \times t = 4200 \times 80 \Rightarrow t = 6.3 \text{ minutes}$$

11. (c) $H = \frac{V^2}{R} \times t = \frac{(210)^2}{20} \times 1 = mL$

$$\therefore \frac{(210)^2}{20} = m \times 80 \times 4.2 \Rightarrow m = 6.56 \text{ g/s}$$

12. (b) For normal brightness of each bulb see following circuit. Current through each bulb = 0.5 A



So main current $i = 2 \text{ A}$

Also, voltage across the combination = 1.5 V

So voltage across the resistance = 10.5 V

Hence for resistance $V = iR \Rightarrow 10.5 = 2 \times R$

$$\Rightarrow R = \frac{21}{4} \Omega$$

13. (d) $P = \frac{V^2}{R}$ so $R = \frac{V^2}{P} \Rightarrow R_1 = \frac{V^2}{100}$ and $R_2 = R_3 = \frac{V^2}{60}$

$$\text{Now } W_1 = \frac{(250)^2}{(R_1 + R_2)^2} \cdot R_1, W_2 = \frac{(250)^2}{(R_1 + R_2)^2} \cdot R_2$$

$$\text{and } W_3 = \frac{(250)^2}{R_3}$$

$$W_1 : W_2 : W_3 = 15 : 25 : 64 \text{ or } W_1 < W_2 < W_3$$

14. (a) Power dissipated $\propto R_{\text{equivalent}}$

15. (a) The current taken by the silver voltmeter

$$I_1 = \frac{m}{Zt} = \frac{1}{11.2 \times 10^{-4} \times 30 \times 60} = 0.496 \text{ A}$$

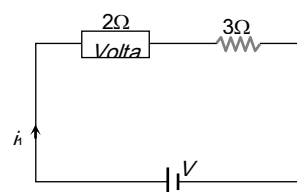
and by copper voltmeter

$$I_2 = \frac{1.8}{6.6 \times 10^{-4} \times 30 \times 60} = 1.515 \text{ A}$$

$$\text{Total current } I = (I_1 + I_2) = 2.011 \text{ A}$$

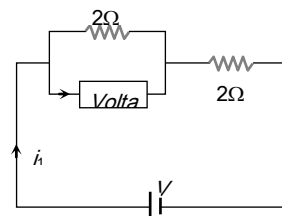
$$\text{Power } P = IV = 2.011 \times 12 = 24.132 \text{ J/sec}$$

16. (d) Initially current through the voltmeter $i_1 = \frac{V}{(3+2)} = \frac{V}{5}$



$$\text{Finally main current } i = \frac{V}{3+1} = \frac{V}{4}$$

$$\text{Hence current through voltmeter } i_2 = \frac{V}{8}$$



$$\therefore \text{Rate of deposition } (R) = \frac{m}{t} = Zi \Rightarrow R \propto i$$

$$\therefore \% \text{ drop in rate} = \frac{R_2 - R_1}{R_1} \times 100 = \frac{i_2 - i_1}{i_1} \times 100$$

$$= \left(\frac{\frac{V}{8} - \frac{V}{5}}{\frac{V}{5}} \right) \times 100 = -37.5\%$$

17. (d) Comparing the given equation with standard equation

$$E = \alpha t + \frac{1}{2} \beta t^2$$

$$\alpha = 40 \text{ and } \frac{1}{2} \beta = -\frac{1}{20} \Rightarrow \beta = -\frac{1}{10}$$

$$\text{Hence neutral temperature } t_n = -\frac{\alpha}{\beta} = \frac{-40}{-1/10}$$

$$\Rightarrow t_n = 400^\circ\text{C}$$

18. (a) Comparing the given equation with standard equation

$$E = \alpha t + \frac{1}{2} \beta t^2, \text{ we get } \alpha = 14 \text{ and } \frac{1}{2} \beta = -0.02$$

$$\Rightarrow \beta = -0.04$$

Hence neutral temperature

$$t_n = -\frac{\alpha}{\beta} = -\frac{14}{-0.04} = 350^\circ\text{C}$$

19. (a) We know that thermoelectric power $S = \frac{dE}{dT}$

$$\text{Given } E = k(T - T_r) \left[T_0 - \frac{1}{2}(T + T_r) \right]$$

By differentiating the above equation w.r.t. T and

$$\text{Putting } T = \frac{1}{2} T_0, \text{ we get } S = \frac{1}{2} k T_0$$

20. (b) Comparing the given equation with $E = \alpha t + \frac{1}{2} \beta t^2$

$$\text{We get } \alpha = 16 \text{ and } \frac{1}{2} \beta = -0.04 \Rightarrow \beta = -0.08$$

$$\Rightarrow t_n = -\frac{\alpha}{\beta} = -\frac{16}{-0.08} = 200^\circ\text{C}$$

$$\text{Also } t_i = 2t_n - t_c \Rightarrow t_i = 2 \times (200) - 0 = 400^\circ\text{C}$$

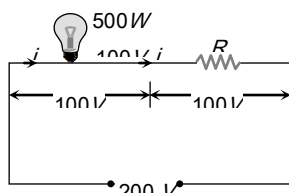
21. (c) $m = Zit \Rightarrow 20 \times 10^{-3} = \left(\frac{32}{96500} \right) \times 0.15 \times t$

$$= 6.7 \text{ min} = 6 \text{ min. } 42 \text{ sec.}$$

22. (b) $e = iR \Rightarrow 25 \times 10^{-6} \times \Delta\theta = 10^{-5} \times 40$

$$\Delta\theta = \frac{40 \times 10^{-5}}{25 \times 10^{-6}} = \frac{400}{25} = 16^\circ\text{C}$$

23. (b)



$$\text{Rated current through the circuit } i = \frac{500}{100} = 5 \text{ A}$$

Potential difference across R ,

$$100 = 5 \times R \Rightarrow R = 20 \Omega$$

24. (b) By using $e_0^{100} = e_0^{32} + e_{32}^{70} + e_{70}^{100}$

$$\Rightarrow 200 = 64 + 76 + e_{70}^{100} \Rightarrow e_{70}^{100} = 60 \mu\text{V}$$

25. (d) In the normal condition current flows from X to Y through cold. While after increasing the temperature of hot junction beyond temperature of inversion. The current is reversed i.e. X to Y through hot junction or Y to X through cold junction.

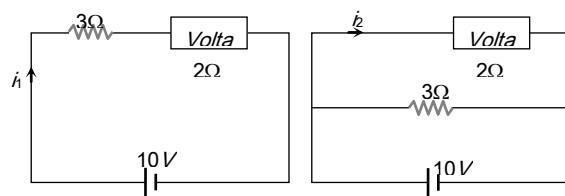
26. (a) $H = \pi it = (2 \times 10^{-9}) \times 2.5 \times (2 \times 60) = 6 \times 10^{-7} \text{ J} = 6 \text{ erg}$

27. (b) Remember mass of the metal deposited on cathode depends on the current through the voltmeter and not on the current supplied by the battery. Hence by

$$\text{using } m = Zit, \text{ we can say } \frac{m_{\text{Parallel}}}{m_{\text{Series}}} = \frac{i_{\text{Parallel}}}{i_{\text{Series}}}$$

$$\Rightarrow m_{\text{Parallel}} = \frac{5}{2} \times 1 = 2.5 \text{ gm.}$$

$$\text{Hence increase in mass} = 2.5 - 1 = 1.5 \text{ gm}$$



28. (c) Mass deposited $m = \text{Density} \times \text{Volume}$ of the metal

$$\Rightarrow m = \rho \times Ax. \text{ Also } m = Zit, \text{ so } Zit = \rho Ax$$

$$\Rightarrow x = \frac{Zit}{A\rho} = \frac{0.00033 \times 10^{-3} \times 1.5 \times 20 \times 60}{(50 \times 10^{-4}) \times 9000} = 1.3 \times 10^{-5} \text{ m}$$

29. (a) $i = \frac{m}{Zt} = \frac{2.0124}{1.118 \times 10^{-3} \times 3600} = 0.5 \text{ A}$

$$\Rightarrow \text{Error} = 0.54 - 0.5 = 0.04 \text{ A}$$

30. (b) Total charge supplied $= 1 \times 10 = 10 \text{ C}$

$\therefore$ 2 electronic charge ($3.2 \times 10^{-19} \text{ C}$) liberates one Cu^{++} ion

$\therefore$ Number of Cu^{++} ions liberated by 10 C charge

$$= \frac{1}{3.2 \times 10^{-19}} \times 10 = 3.1 \times 10^{19}$$

31. (d) $\therefore m = Zit$ or $i = \frac{m}{Zt}$

For silver voltmeter

$$i_1 = \frac{m_1}{Z_1 t} = \frac{2}{1.118 \times 10^{-3} \times 1800} = 0.994 \text{ amp}$$

For copper voltmeter

$$i_2 = \frac{m_2}{Z_2 t} = \frac{1}{3.294 \times 10^{-4} \times 1800} = 1.687 \text{ amp}$$

$$\therefore \text{Power of circuit} = V(i_1 + i_2) = 6 \times (0.994 + 1.687) \\ = 6 \times 2.681 \approx 16 \text{ W}$$

32. (a) Let the temperature of molten metal is $t^\circ \text{C}$.

The thermo-emf $e = 10 \times 10^{-6} t \text{ volt}$

Current in the circuit

$$i = \frac{e}{R + R_G} = \frac{10^{-5} t}{8 + 1.6} = \frac{10^{-5} t}{9.6} \text{ amp.}$$

$$\text{But } i = \frac{V}{R_G} = \frac{8 \times 10^{-3}}{8}$$

$$\therefore \frac{10^{-5} t}{9.6} = \frac{8 \times 10^{-3}}{8} \text{ or } t = \frac{9.6 \times 10^{-3}}{10^{-5}} = 960^\circ \text{C}$$

33. (a) $\therefore$ Peltier coefficient $\pi = T \frac{de}{dT}$ and $t^\circ \text{C} = T - 273$

$$\therefore e = a(T - 273) + b(T - 273)^2$$

$$\text{Differentiating w.r.t. } T \quad \frac{de}{dT} = a + 2b(T - 273)$$

$$\pi = T \frac{de}{dT} = T[a + 2b(T - 273)] \Rightarrow \pi = (t + 273)(a + 2bt)$$

34. (d) $\frac{Q}{t} = \frac{V^2}{4.2 R} = \frac{m}{t} \cdot L$

$$\therefore \frac{m}{t} = \frac{V^2}{4.2 RL} = \frac{(210)^2}{4.2 \times 50 \times 80} \approx 2.625 \text{ gm}$$

35. (d) $H = i^2 RT = i^2 \left(\frac{\rho l}{A} \right) t = \frac{i^2 \rho V t}{A^2}$ ($V = \text{volume} = Al$)

$$\Rightarrow H \propto \frac{1}{r^4} \Rightarrow \frac{H_1}{H_2} = \left(\frac{r_2}{r_1} \right)^4 = \left(\frac{2}{1} \right)^4 = \frac{16}{1}$$

36. (d) Thermoelectric power $P \propto \theta$

$$\Rightarrow \frac{P_{100} - P_{80}}{P_{80}} \times 100 = \frac{100 - 80}{80} \times 100 = 25\%$$

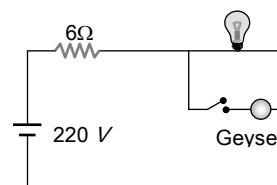
37. (d) The sensitivity of the thermocouple will be $= 500 \mu\text{V}/^\circ \text{C} - (-72 \mu\text{V}/^\circ \text{C}) = 572 \mu\text{V}/^\circ \text{C}$

Therefore for a 100°C temperature difference, the thermo e.m.f. will be

$$E = 572 \times 10^{-6} \times 100 (\text{volt}) = 57.2 \times 10^{-3} = 57.2 \text{ mV.}$$

38. (d) At cold junction, current flows from copper to nickel and from iron to copper, and at hot junction from nickel to iron, thus the contributions add.

39. (b) $R_{\text{Bulb}} = \frac{220^2}{100} = 484 \Omega$, $R_{\text{Geyser}} = \frac{220^2}{1000} = 48.4 \Omega$



- (i) When only bulb is ON,

$$V_{\text{Bulb}} = \frac{220 \times 484}{490} = 217.4 \text{ V}$$

- (ii) When geyser is also switched ON, equivalent resistance of bulb and geyser is

$$R = \frac{484 \times 48.4}{484 + 48.4} = 44 \Omega$$

$$\text{Voltage across the bulb } V_{\text{Bulb}} = \frac{220 \times 44}{50} = 193.6 \text{ V}$$

Hence the potential drop is $217.4 - 193.6 = 23.8 \text{ V}$

40. (b) $i = \frac{24 - 12}{3} = 4 \text{ A}$, Time of charging $t = \frac{360}{V \cdot i}$

$$\Rightarrow t = \frac{360}{12 \times 4} = 7.5 \text{ hours}$$

41. (a) $I = \frac{m}{Zt} = \frac{2.68}{\frac{108}{96500} \times 10 \times 60} = \frac{2.68}{108} \times \frac{965}{6} \approx 4 \text{ A}$

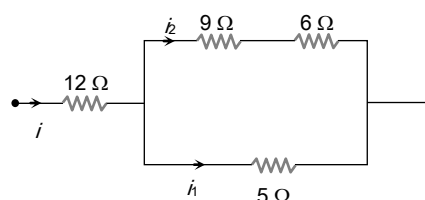
$$\text{Energy} = I^2 R t = 4^2 \times 20 \times 600 = 192 \text{ kJ.}$$

42. (d) Comparing with standard equation $E = \alpha t + \frac{1}{2} \beta t^2$

$$\alpha = a \text{ and } \beta = 2b \Rightarrow t_n = -\frac{a}{2b} = -\frac{1}{2} \times 700 = -350^\circ \text{C}$$

This is not possible.

43. (b) $\frac{i_1}{i_2} = \frac{15}{5} = \frac{3}{1}$... (i)



$$\text{Also } \frac{H}{t} = i^2 R \Rightarrow 45 = (i_1)^2 \times 5$$

$$\Rightarrow i_1 = 3 \text{ A and from equation (i) } i_2 = 1 \text{ A}$$

$$\text{So } i = i_1 + i_2 = 4 \text{ A}$$

Hence power developed in 12Ω resistance

$$P = i^2 R = (4)^2 \times 12 = 192 \text{ W}$$

44. (a) Heat gained by water = Heat supplied by container - heat lost $\Rightarrow mS\Delta\theta = 1000t - 160t$

$$\Rightarrow t = \frac{2 \times 4.2 \times 1000 \times 50}{840} = 8 \text{ min } 20 \text{ sec}$$

45. (c) As the voltage in R_2 and R_3 is same therefore, according to,

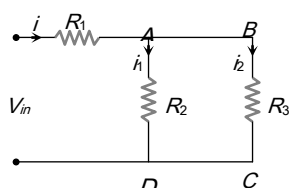
$$H = \frac{V^2}{R} \cdot t, \quad R_2 = R_3$$

Also the energy in all resistance is same.

$$\therefore i^2 R_1 t = i_1^2 R_2 t$$

$$\text{Using } i_1 = \frac{R_3}{R_2 + R_3} i = \frac{R_3}{R_3 + R_3} i = \frac{1}{2} i$$

$$\text{Thus } i^2 R_1 t = \frac{i^2}{4} R_2 t \text{ or, } R_1 = \frac{R_2}{4}$$



Graphical Questions

- (b) Area = $it = 2 \text{ Coulomb}$ and $m = zit \Rightarrow z = \frac{m}{it} = \frac{m}{2}$
- (d) $U \propto i^2$, hence the graph between U and i is parabolic in nature and should be above graph (b).
- (d) $E = \alpha t + \frac{1}{2} \beta t^2$, graph between E and t will be a parabola, such that first emf increases and then decreases.
- (a) Thermo electric power $P = \frac{dE}{d\theta} = \alpha + \beta\theta$
Comparing it with $y = mx + c$, option (a) is correct.
- (b) The filament of the heater reaches its steady resistance when the heater reaches its steady

temperature, which is much higher than the room temperature. The resistance at room temperature is thus much lower than the resistance at its steady state. When the heater is switched on, it draws a larger current than its steady state current. As the filament heats up, its resistance increases and current falls to steady state value.

6. (a) Cu voltameter with soluble electrodes obeys ohms law. In water voltameter, in the beginning when V is small ($< 1.7 \text{ volt}$), very little current flows, the voltameter does not obey ohms law. As soon as V exceeds 1.7 volt (back e.m.f.) the current increases steadily according to ohms law.

7. (d) Thermal energy in resistor is $U = i^2 R t$

$$\text{where } R = R_0(1 + \alpha t) \Rightarrow U = i^2 R_0(1 + \alpha t)t = i^2 R_0 t + i^2 R_0 \alpha t^2$$

$$\text{So } \frac{dU}{dt} = i^2 R_0(1 + \alpha t)$$

With the time temperature increases, hence dU/dt increases. This is best shown by curve (d).

8. (b) $m = Zit$ and it = Area of given curve

$$= \text{Area of triangle} + \text{Area of rectangle}$$

$$\Rightarrow it = \frac{1}{2} \times (2 \times 60) \times 1 + (6 - 2) \times 60 \times 1 = 300$$

$$\therefore Z = \frac{m}{it} = \frac{m}{300}$$

9. (d) Terminal voltage $V = E - Ir$. Hence the graph between V and i will be a straight line having negative slope and positive intercept.

Thermal power generated in the external circuit

$$P = EI - I^2 r. \text{ Hence graph between } P \text{ and } I \text{ will be a parabola passing through origin.}$$

Also at an instant, thermal power generated in the cell = $i^2 r$ and total electrical power generated in the

$$\text{cell} = Ei. \text{ Hence the fraction } \eta = \frac{I^2 r}{EI} = \left(\frac{r}{E} \right) I; \text{ so}$$



$\eta \propto I$. It means graph between η and I will be a straight line passing through origin.

Assertion and Reason

1. (a) The possibility of an electric bulb fusing is higher at the time of switching ON and switching OFF because inductive effect produces a surge at the time of switching ON and OFF.
2. (a) The resistance, $R = \frac{V^2}{P} \Rightarrow R \propto 1/P$
i.e., higher is the wattage of a bulb, lesser is the resistance and so it will glow bright.
3. (c) Assertion is true but reason is false. Fuse wire must have high resistance because in series current remains same, therefore according to Joule's law $H = \frac{i^2 R t}{4.2}$, heat produced is high if R is high. The melting point must be low so that wire may melt with increase in temperature. As the current equal to maximum safe value, flows through the fuse wire, it heats up, melts and break the circuit.
4. (a) Resistance of 50 W bulb is two times the resistance of 100 W bulb. When bulbs are connected in series, 50 W bulb will glow more as $P = i^2 R$ (current remains same in series). In parallel the 100 W bulb will glow more as $P = V^2 / R$ (potential difference remain same in parallel).
5. (d) When two bulbs are connected in series, the resistance of the circuit increases and so the voltage in each decreases, hence the brightness and the temperature also decreases. Due to decrease in temperature, the resistance of the carbon filament will slightly increase while that of metal filament will decrease. Hence, carbon filament bulb will glow more brightly ($P = i^2 R$). Also carbon is not a semiconductor.
6. (e) Voltage of dc source is constant but in ac, peak value of voltage is $\sqrt{2}$ times the *rms.* voltage. Hence bulb will glow with more brightness when connected to an ac source of the same voltage.
7. (a) When cold water is poured on half portion of the wire, its resistance decreases due to decrease in temperature. As a result of this total resistance of circuit decreases *i.e.* current through each portion of wire increases *i.e.* rest of the half portion becomes still more hot.
8. (a) As filament of bulb and line wire are in series, hence current through both is same. Now, because $H = \frac{i^2 R t}{4.2}$ and resistance of the filament of the bulb is much higher than that of line wires, hence heat produced in the filament is much higher than that in line wires.
9. (b) Neutral temperature is the temperature of hot junction, at which the thermo e.m.f. produced in the thermocouple becomes maximum. It is independent of cold junction and depends on the nature of materials of two metals used to form thermocouple.
10. (d) Because of heat production every resistance has a maximum power rating, the maximum power that can be dissipated without overheating the device. When this rating is exceeded, heat is produced, due to which resistance may change unpredictably.
11. (a) The e.m.f. of a Leclanche cell falls, because of the partial polarisation due to accumulation of hydrogen gas. In case, Leclanche cell is used in experiment, where current is drawn after short breaks, then during each break, hydrogen gas escapes and

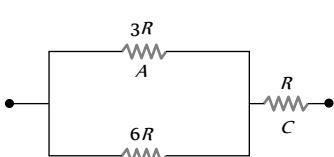
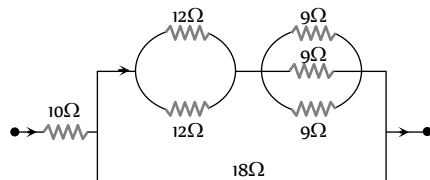


Mn_2O_3 converts into MnO_2 by taking oxygen from the atmosphere. As a result, the cell regains its original e.m.f.

12. (a) When lamp B or C gets fused equivalent resistance of B and C increases. In series voltage distributes in the ratio of resistance, so voltage appears across B increases or in other words voltage across A decreases.
13. (d) When switch S is closed, bulb C is short circuited, so voltage V distributes only in two parts *i.e.* voltage on Bulb A and B increases as compared previously. Hence illumination of Bulb A and B increases.
14. (a)
15. (c) The electrical appliances with metallic body like heater, press *etc.* have three pin connections. Two pins are for supply line and third pin is for earth connection for safety purposes.
16. (c) A laser beam is a beam of light which is light amplification by stimulated emission of radiation.
- The energy per unit area of the laser beam is very high as compared to the torch light.
17. (a) Follow hint of question 15 of this section.
18. (c) Thomson e.m.f. in lead is practically zero.
19. (b) The presence of water molecules reduces force between ions by $1/81$ times because the value of dielectric constant of water is 81. That is why the separation between ions becomes easier.
20. (b) Here reason is not the correct explanation of the assertion, which is correct.
21. (d) Here assertion and reason are not correct.

Heating and Chemical Effect of Current

SET Self Evaluation Test -20

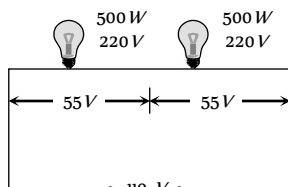
- An electric kettle has two coils. When one of these is switched on, the water in the kettle boils in 6 minutes. When the other coil is switched on, the water boils in 3 minutes. If the two coils are connected in series, the time taken to boil the water in the kettle is
 - 3 minutes
 - 6 minutes
 - 2 minutes
 - 9 minutes
- A 3° rise in temperature is observed in a conductor by passing a certain current. When the current is doubled, the rise in temperature will be
 - 15°C
 - 12°C
 - 9°C
 - 3°C
- Two identical electric lamps marked 500 W, 220 V are connected in series and then joined to a 110 V line. The power consumed by each lamp is
 - $\frac{125}{4}$ W
 - $\frac{25}{4}$ W
 - $\frac{225}{4}$ W
 - 125 W
- When 1 gm hydrogen (e.c.e. = 1.044×10^{-8} kg / C) forms water, 34 kcal heat is liberated. The minimum voltage required to decompose water is
 - 0.75 V
 - 3 V
 - 1.5 V
 - 4.5 V
- In how much time, one litre of H_2 will be collected by 5 A current ? (If $Z = 1 \times 10^{-8}$ kg / C and density of $\text{H}_2 = 0.09$ kg / m^3)
 - 30 minutes
 - 15 minutes
 - 45 minutes
 - 60 minutes
- The three resistances A, B and C have values 3R, 6R and R respectively. When some potential difference is applied across the network, the thermal powers dissipated by A, B and C are in the ratio
 - 2 : 3 : 4
 - 2 : 4 : 3
 - 4 : 2 : 3
 - 3 : 2 : 4
- If the length of the filament of a heater is reduced by 10%, the power of the heater will
 - Increase by about 9%
 - Increase by about 11%
 - Increase by about 19%
 - Decrease by about 10%
- A thermo couple develops 40 μV / kelvin. If hot and cold junctions be at 40°C and 20°C respectively then the emf develops by a thermopile using such 150 thermo couples in series shall be
 - 150mV
 - 80mV
 - 144mV
 - 120mV
- Amount of electricity required to pass through the H_2O voltmeter so as to liberate 11.2 litre of hydrogen will be
 - 1 Faraday
 - $\frac{1}{2}$ Faraday
 - 2 Faraday
 - 3 Faraday
- The resistance of the filament of a lamp increases with the increase in temperature. A lamp rated 100 W, 220 V is connected across 220 V power supply. If the voltage drops by 10% then the power of lamp will be
 - 90 W
 - 81 W
 - Between 90 W and 100 W
 - Between 81 W and 90 W
- In the following circuit, 18Ω resistor develops 2 J/sec due to current flowing through it. The power developed across 10Ω resistance is
 - 125 W
 - 10 W
 - $\frac{4}{5}$ W
 - 25 W
- If resistance of the filament increases with temperature, what will be power dissipated in a 220 V- 100 W lamp when connected to 110 V power supply
 - 25 W
 - < 25 W
 - > 25 W
 - None of these
- Total surface area of a cathode is 0.05m^2 and 1 A current passes through it for 1 hour. Thickness of nickle deposited on the cathode is (Given that density of nickle = 9gm/cc and it's E.C.E. = 3.04×10^{-4} gm / C)
 - 2.4 m
 - $2.4\mu\text{m}$
 - $2.4\mu\text{m}$
 - None of these
- Two bulbs consume same power when operated at 200 V and 300 V respectively. When these bulbs are connected in series across a D.C. source of 500 V, then
 - Ratio of potential difference across them is 3/2
 - Ratio of potential difference across them is 9/4
 - Ratio of power consumed across them is 4/9
 - Ratio of power consumed across them is 2/3

1. (d) In series $\frac{1}{P_s} = \frac{1}{P_1} + \frac{1}{P_2} \Rightarrow \frac{1}{(H_s/t_s)} = \frac{1}{(H_1/t_1)} + \frac{1}{(H_2/t_2)}$
 $\therefore H_s = H_1 = H_2$ So $t_s = t_1 + t_2 = 6 + 3 = 9 \text{ min}$

2. (b) $i^2 R t = C \theta = 3C$; C = Thermal capacity
 when $i_1 = 2i \Rightarrow C \theta_1 = 4i^2 R t = 4 \times 3C \Rightarrow \theta_1 = 12^\circ \text{C}$

3. (a) Voltage across each bulb $V' = \frac{110}{2} = 55 \text{ V}$ so, power consumed by each bulb will be

$$P' = \left(\frac{55}{220} \right)^2 \times 500 = \frac{125}{4} \text{ W}$$



4. (c) $m = Zit \Rightarrow it = \frac{m}{Z} = \frac{1 \times 10^{-3}}{1.044 \times 10^{-8}} \text{ C} = \frac{10^5}{1.044} \text{ C}$

Given $H = 34 \text{ kcal} = 4.2 \times 34 \times 10^3 \text{ J}$

$\Rightarrow$ Heat generated $H = Vit = V \cdot \frac{10^5}{1.044}$

$\Rightarrow V = \frac{4.2 \times 34 \times 1.044}{10^2} = 4.2 \times 0.34 \times 1.044 = 1.5 \text{ V}$

5. (a) $m = zit \Rightarrow 10^{-3} \times 0.09 = 1 \times 10^{-8} \times 5 \times t \Rightarrow t = 30 \text{ min}$

6. (c) Thermal power in $A = P_A = \left(\frac{2i}{3} \right)^2 3R = \frac{4}{3} i^2 R$

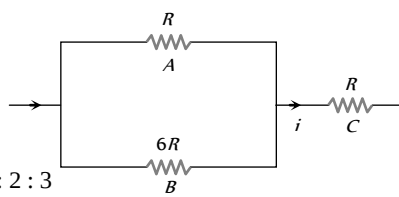
Thermal power in $B = P_B = \left(\frac{i}{3} \right)^2 6R = \frac{2}{3} i^2 R$

Thermal power in

$C = P_C = i^2 R$

$\Rightarrow P_A : P_B : P_C$

$= \frac{4}{3} : \frac{2}{3} : 1 = 4 : 2 : 3$



7. (b) $P \propto \frac{1}{R}$ and $R \propto l \Rightarrow P \propto \frac{1}{l}$

$\Rightarrow \frac{P_1}{P_2} = \frac{l_2}{l_1} \Rightarrow \frac{P_1}{P_2} = \frac{(100-10)}{100} = \frac{90}{100} \Rightarrow P_2 = 1.11 P_1$

% change in power = $\frac{P_2 - P_1}{P_1} \times 100 = 11\%$

8. (d) The temperature difference is $20^\circ \text{C} = 20 \text{ K}$. So that thermo emf developed $E = \alpha \theta = 40 \frac{\mu \text{V}}{\text{K}} \times 20 \text{ K} = 800 \mu \text{V}$

Hence total emf = $150 \times 800 = 12 \times 10^4 \mu \text{V} = 120 \text{ mV}$

9. (a) 22.4 litre H_2 = 1 mole of H_2 = N molecules of H_2
 = $2N$ atoms of H .

So charge required to liberate 22.4 litre of H_2 = $2Ne = 2F$.

Hence charge required to liberate 11.2 litre of H_2 = F .

10. (d) Let the resistance of the lamp filament be R . Then $100 = \frac{(220)^2}{R}$. When the voltage drops, expected power is

$P = \frac{(220 \times 0.9)^2}{R'}$. Here R' will be less than R , because now the rise in temperature will be less. Therefore P is more than $\frac{(220 \times 0.9)^2}{R} = 81 \text{ W}$

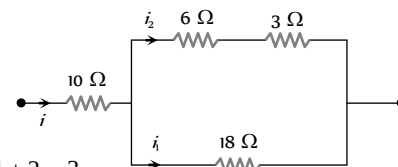
But it will not be 90% of earlier value, because fall in temperature is small. Hence (d) is correct.

11. (b) The given circuit can be redrawn as follows

$\frac{i_1}{i_2} = \frac{9}{18} = \frac{1}{2}$

and $i = i_1 + i_2$

$\Rightarrow \frac{i}{i_1} = 1 + \frac{i_2}{i_1} = 1 + 2 = 3$



From $P = i^2 R \Rightarrow \frac{P_{10\Omega}}{P_{18\Omega}} = \left(\frac{i}{i_1} \right)^2 \times \frac{10}{18} \Rightarrow P_{10\Omega} = 10 \text{ W}$

12. (c) If resistance does not vary with temperature P consumed = $\left(\frac{V_A}{V_R} \right)^2 \times P_R = \left(\frac{110}{220} \right)^2 \times 100 = 25 \text{ W}$. But in second cases resistance decreases so consumed power will be more than 25 W

13. (c) Mass deposited = density $\times$ volume of the metal

$m = \rho \times A \times X$ (i)

Hence from Faraday's first law $m = Zit$ (ii)

So from equation (i) and (ii)

$Zit = \rho \times Ax \Rightarrow x = \frac{Zit}{\rho A}$

$= \frac{3.04 \times 10^{-4} \times 10^{-3} \times 1 \times 3600}{9000 \times 0.05} = 2.4 \times 10^{-6} \text{ m} = 2.4 \mu \text{m}$

14. (c) $P = \frac{V^2}{R} \therefore R = \frac{V^2}{P}$ or $R \propto V^2$ i.e. $\frac{R_1}{R_2} = \left(\frac{200}{300} \right)^2 = \frac{4}{9}$

When connected in series potential drop and power consumed

are in the ratio of their resistances. So, $\frac{P_1}{P_2} = \frac{V_1}{V_2} = \frac{R_1}{R_2} = \frac{4}{9}$



Chapter 21

Magnetic Effect of Current

Oersted found that a magnetic field is established around a current carrying conductor.

Magnetic field exists as long as there is current in the wire.

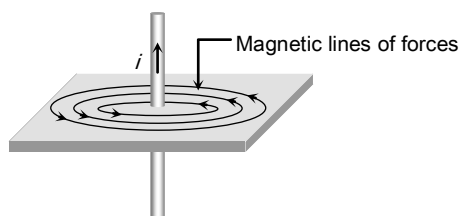


Fig. 21.1

Biot-Savart's Law

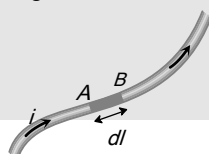
Biot-Savart's law is used to determine the magnetic field at any point due to a current carrying conductor.

This law is although for infinitesimally small conductor yet it can be used for long conductors. In order to understand the Biot-Savart's law, we need to understand the term current-element.

Current element

It is the product of current and length of infinitesimal segment of current carrying wire.

The current element is taken as



a vector quantity. Its direction is same as the direction of current.

Current element $AB = i \vec{dl}$

According to Biot-Savart Law, magnetic field at point 'P' due to the current element $i \vec{dl}$ is given by the expression,

$$d\vec{B} = k \frac{i dl \sin \theta}{r^2} \hat{n} \text{ also } \vec{B} = \int d\vec{B} = \frac{\mu_0 i}{4\pi} \int \frac{dl \sin \theta}{r^2} \hat{n}$$

In C.G.S. $k = 1$ and in S.I. : $k = \frac{\mu_0}{4\pi}$

where μ_0 = Absolute permeability of air or vacuum
 $= 4\pi \times 10^{-7} \frac{\text{Wb}}{\text{Amp} \cdot \text{metre}}$. Its other units are

$$\frac{\text{Henry}}{\text{metre}} \text{ or } \frac{\text{N}}{\text{Amp}^2} \text{ or } \frac{\text{Tesla} \cdot \text{metre}}{\text{Ampere}}$$

$$\text{Vectorially, } d\vec{B} = \frac{\mu_0}{4\pi} \cdot \frac{i(\vec{dl} \times \hat{r})}{r^2} = \frac{\mu_0}{4\pi} \cdot \frac{i(\vec{dl} \times \vec{r})}{r^3}$$

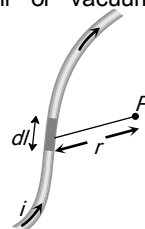


Fig. 21.2

Direction of Magnetic Field

The direction of magnetic field is determined with the help of the following simple laws :

(1) **Maxwell's cork screw rule** : According to this rule, if we imagine a right handed

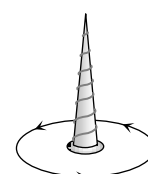


Fig. 21.3

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screw placed along the current carrying linear conductor, be rotated such that the screw moves in the direction of flow of current, then the direction of rotation of the thumb gives the direction of magnetic lines of force.

(2) **Right hand thumb rule** : According to this rule if a straight current carrying conductor is held in the right hand such that the thumb of the hand represents the direction of current flow, then the direction of folding fingers will represent the direction of magnetic lines of force.

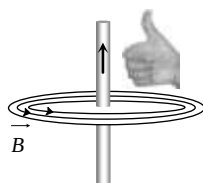


Fig. 21.4

(3) **Right hand thumb rule of circular currents** : According to this rule if the direction of current in circular conducting coil is in the direction of folding fingers of right hand, then the direction of magnetic field will be in the direction of stretched thumb.

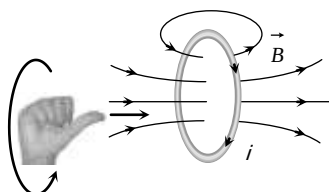


Fig. 21.5

(4) Right hand palm rule

If we stretch our right hand such that fingers point towards the point. At which magnetic field is required while thumb is in the direction of current then normal to the palm will show the direction of magnetic field.

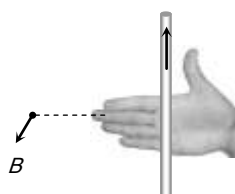


Fig. 21.6

Meaning of Cross $\otimes$ and dot $\odot$

If magnetic field is directed perpendicular and into the plane of the paper it is represented by $\otimes$ (cross) while if magnetic field

is directed perpendicular and out of the plane of the paper it is represented by $\odot$ (dot)

In : Magnetic field is away from the observer or perpendicular inwards.

Out : Magnetic field is towards the observer or perpendicular outwards.

Ampere's Law

Ampere's law gives another method to calculate the magnetic field due to a given current distribution.

Line integral of the magnetic field $\vec{B}$ around any closed curve is equal to μ_0 times the net current i threading through the area enclosed by the curve i.e.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \sum i = \mu_0 (i_1 + i_3 - i_2)$$

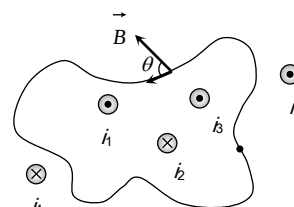


Fig. 21.8

Also using $\vec{B} = \mu_0 \vec{H}$ (where $\vec{H}$ = magnetising field)

$$\oint \mu_0 \vec{H} \cdot d\vec{l} = \mu_0 \sum i \Rightarrow \oint \vec{H} \cdot d\vec{l} = \sum i$$

Total current crossing the above area is $(i_1 + i_3 - i_2)$. Any current outside the area is not included in net current. (Outward $\odot \rightarrow +ve$, Inward $\otimes \rightarrow -ve$)

Table 21.1 : Biot-Savart's law v/s Ampere's law

Biot-Savart's law	Ampere's law
this law is valid for all current distributions	This law is valid for symmetrical current distributions
This law is the differential form of $\vec{B}$ or $\vec{H}$	Basically this law is the integral form of $\vec{B}$ or $\vec{H}$
This law is based only on the principle of magnetism	This law is based on the principle of electromagnetism.

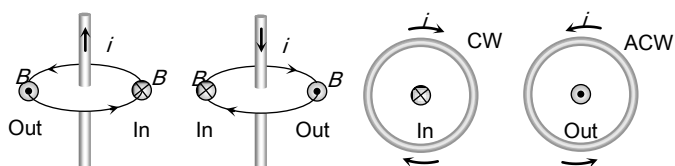


Fig. 21.7

Magnetic Field Due to Circular Current

If a coil of radius r , carrying current i then magnetic field on its axis at a distance x from its centre given by (Application of

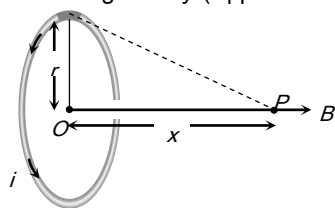


Fig. 21.9

$$(1) B_{axis} = \frac{\mu_0}{4\pi} \cdot \frac{2\pi N i r^2}{(x^2 + r^2)^{3/2}}; \text{ where } N = \text{number of turns in coil.}$$

$$(2) \text{ At centre } x = 0 \Rightarrow B_{centre} = \frac{\mu_0}{4\pi} \cdot \frac{2\pi N i}{r} = \frac{\mu_0 N i}{2r} = B_{max}$$

(3) The ratio of magnetic field at the centre of circular coil and on its axis is given by $\frac{B_{centre}}{B_{axis}} = \left(1 + \frac{x^2}{r^2}\right)^{3/2}$

$$(4) \text{ If } x \gg r \Rightarrow B_{axis} = \frac{\mu_0}{4\pi} \cdot \frac{2\pi N i r^2}{x^3} = \frac{\mu_0}{4\pi} \cdot \frac{2 N i A}{x^3}$$

where $A = \pi r^2 = \text{Area of each turn of the coil.}$

(5) **B-x curve** : The variation of magnetic field due to a circular coil as the distance x varies as shown in the figure.

B varies non-linearly with distance x as shown in figure and is maximum when $x^2 = \min = 0$, i.e., the point is at the centre of the coil and it is zero at $x = \pm \infty$.

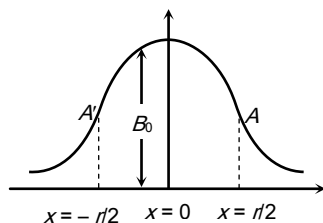


Fig. 21.10

(6) **Point of inflection** (A and A') : Also known as points of curvature change or points of zero curvature.

(i) At these points B varies linearly with $x \Rightarrow \frac{dB}{dx} =$

$$\text{constant} \Rightarrow \frac{d^2B}{dx^2} = 0.$$

(ii) These are located at $x = \pm \frac{r}{2}$ from the centre of the coil

$$\text{and the magnetic field at } x = \frac{r}{2} \text{ is } B = \frac{4\mu_0 N i}{5\sqrt{5} r}$$

(7) Helmholtz coils

(i) This is the set-up of two coaxial coils of same radius such that distance between their centres is equal to their radius.

(ii) At axial mid point O , magnetic field is given by

$$B = \frac{8\mu_0 N i}{5\sqrt{5} R} = 0.716 \frac{\mu_0 N i}{R} = 1.432 B, \text{ where } B = \frac{\mu_0 N i}{2R}$$

(iii) Current direction is same in both coils otherwise this arrangement is not called Helmholtz's coil arrangement.

(iv) Number of points of inflexion $\Rightarrow$ Three (A, A', A'')

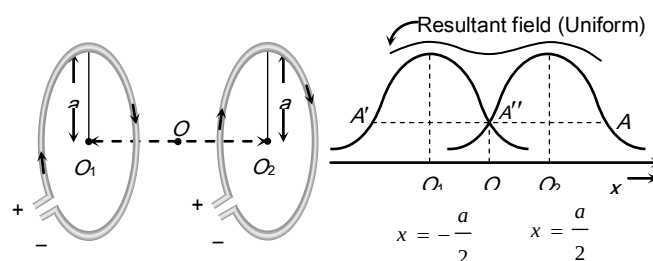
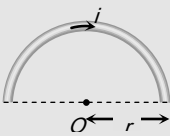
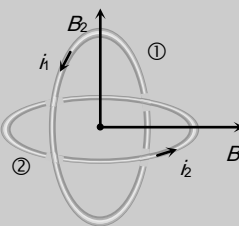
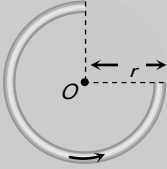
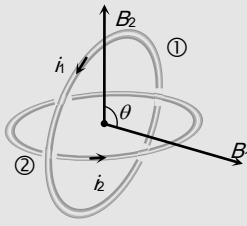
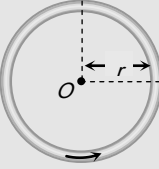
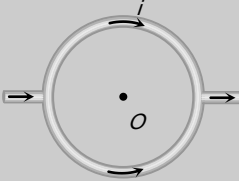
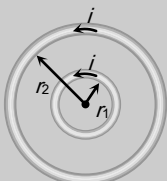
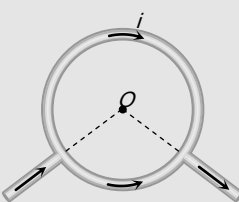
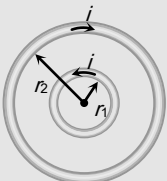


Fig. 21.11

Magnetic Field at Centre O in Different Conditions of Circular Current

Condition	Figure	Magnetic field
Arc subtends angle θ at the centre		$B = \frac{\mu_0}{4\pi} \cdot \frac{\theta i}{r}$
Arc subtends		

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angle $(2\pi - \theta)$ at the centre		$B = \frac{\mu_0}{4\pi} \cdot \frac{(2\pi - \theta)i}{r}$	opposite direction		
Semi-circular arc		$B = \frac{\mu_0}{4\pi} \cdot \frac{\pi i}{r} = \frac{\mu_0 i}{4r}$	Concentric loops but their planes are perpendicular to each other		$B = \sqrt{B_1^2 + B_2^2}$ $= \frac{\mu_0}{2r} \sqrt{i_1^2 + i_2^2}$
Three quarter semi-circular current carrying arc		$B = \frac{\mu_0}{4\pi} \cdot \frac{\left(2\pi - \frac{\pi}{2}\right)i}{r}$ $= \frac{3\mu_0 i}{8r}$	Concentric loops but their planes are at an angle θ with each other		$B = \sqrt{B_1^2 + B_2^2 + 2B_1B_2 \cos \theta}$
Circular current carrying arc		$B = \frac{\mu_0}{4\pi} \cdot \frac{2\pi i}{r}$ $= \frac{\mu_0 i}{2r}$	Distribution of current across the diameter		$B = 0$
Concentric co-planer circular loops carries current in the same direction		$B_1 = \frac{\mu_0}{4\pi} 2\pi i \left(\frac{1}{r_1} + \frac{1}{r_2} \right)$	Distribution of current between any two points on the circumference		$B = 0$
Concentric co-planer circular loops carries current in the		$B_2 = \frac{\mu_0}{4\pi} 2\pi i \left[\frac{1}{r_1} - \frac{1}{r_2} \right]$			

Magnetic Field Due to a Straight Wire

Magnetic field due to a current carrying wire at a point P which lies at a perpendicular distance r from the wire as shown is given as

$$B = \frac{\mu_0}{4\pi} \cdot \frac{i}{r} (\sin\phi_1 + \sin\phi_2)$$

From figure $\alpha = (90^\circ - \phi_1)$

and $\beta = (90^\circ + \phi_2)$

$$\text{Hence } B = \frac{\mu_0}{4\pi} \cdot \frac{i}{r} (\cos\alpha - \cos\beta)$$

(1) **For a wire of finite length** : Magnetic field at a point which lies on perpendicular bisector of finite length wire

$$\phi_1 = \phi_2 = \phi$$

$$\text{So } B = \frac{\mu_0}{4\pi} \cdot \frac{i}{r} (2 \sin\phi)$$

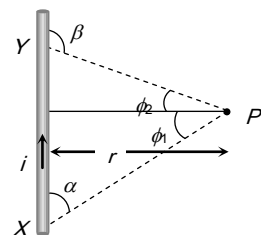


Fig. 21.12

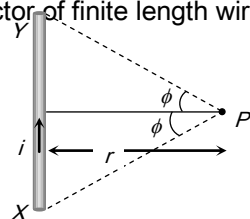


Fig. 21.13

(2) **For a wire of infinite length** : When the linear conductor XY is of infinite length and the point P lies near the centre of the conductor $\phi_1 = \phi_2 = 90^\circ$.

$$\begin{aligned} \text{So, } B &= \frac{\mu_0}{4\pi} \cdot \frac{i}{r} [\sin 90^\circ + \sin 90^\circ] \\ &= \frac{\mu_0}{4\pi} \cdot \frac{2i}{r} \end{aligned}$$

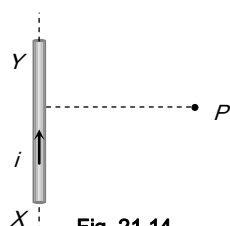


Fig. 21.14

(3) **For a wire of semi-infinite length** : When the linear conductor is of infinite length and the point P lies near the end Y or X . $\phi_1 = 90^\circ$ and $\phi_2 = 0^\circ$

$$\begin{aligned} \text{So, } B &= \frac{\mu_0}{4\pi} \cdot \frac{i}{r} [\sin 90^\circ + \sin 0^\circ] \\ &= \frac{\mu_0 i}{4\pi r} \end{aligned}$$

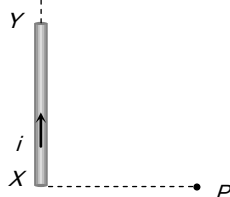


Fig. 21.15

(4) **For axial position of wire** : When point P lies on axial position of current carrying conductor then magnetic field at P



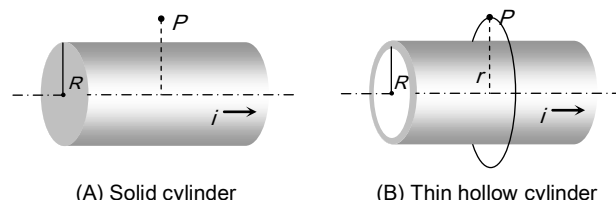
Fig. 21.16

$$B = 0$$

Magnetic Field Due to a Cylindrical Wire

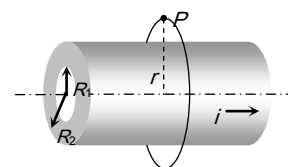
Magnetic field due to a cylindrical wire is obtained by the application of Ampere's law

(1) **Outside the cylinder**



(A) Solid cylinder

(B) Thin hollow cylinder



(C) Thick hollow cylinder

Fig. 21.17

In all above cases magnetic field outside the wire at P

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i \Rightarrow B \int dl = \mu_0 i \Rightarrow B \times 2\pi r = \mu_0 i \Rightarrow B_{out} = \frac{\mu_0 i}{2\pi r}$$

In all the above cases $B_{surface} = \frac{\mu_0 i}{2\pi R}$

(2) **Inside the hollow cylinder** : Magnetic field inside the hollow cylinder is zero.



(A) Thin hollow cylinder

(B) Thick hollow cylinder

Fig. 21.18

(3) **Inside the solid cylinder** : Current enclosed by loop (i) is lesser than the total current (I)

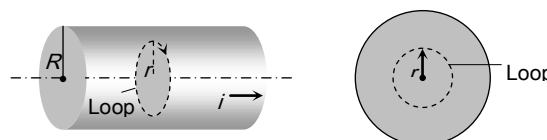


Fig. 21.19

Current density is uniform i.e. $J = J \Rightarrow \vec{i} = i \times \frac{A'}{A} = i \left(\frac{r^2}{R^2} \right)$

Hence at inside point $\oint \vec{B}_{in} \cdot d\vec{l} = \mu_0 \vec{i} \Rightarrow B = \frac{\mu_0}{2\pi} \cdot \frac{ir}{R^2}$

(4) Inside the thick portion of hollow cylinder : Current enclosed by loop is given as $\vec{i} = i \times \frac{A'}{A} = i \times \frac{(r^2 - R_1^2)}{(R_2^2 - R_1^2)}$

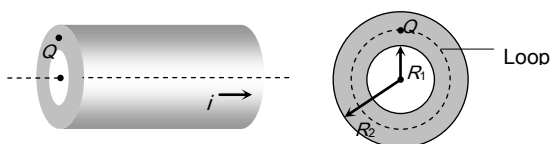


Fig. 21.20

Hence at point Q $\oint \vec{B} \cdot d\vec{l} = \mu_0 \vec{i} \Rightarrow B = \frac{\mu_0 i}{2\pi} \cdot \frac{(r^2 - R_1^2)}{(R_2^2 - R_1^2)}$

Magnetic Field Due to an Infinite Sheet Carrying Current

The figure shows an infinite sheet of current with linear current density j (A/m). Due to symmetry the field line pattern above and below the sheet is uniform. Consider a square loop of side l as shown in the figure.

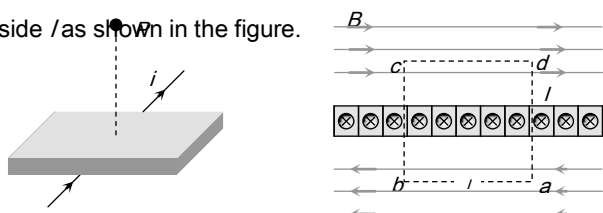


Fig. 21.21

$$\int_a^b \vec{B} \cdot d\vec{l} + \int_b^c \vec{B} \cdot d\vec{l} + \int_c^d \vec{B} \cdot d\vec{l} + \int_d^a \vec{B} \cdot d\vec{l} = \mu_0 i \quad (\text{By Ampere's law})$$

Since $B \perp dl$ along the path $b \rightarrow c$ and $d \rightarrow a$, therefore,

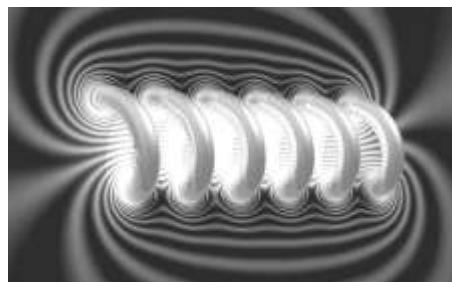
$$\int_b^c \vec{B} \cdot d\vec{l} = 0 ; \int_d^a \vec{B} \cdot d\vec{l} = 0$$

Also, $B \parallel dl$ along the path $a \rightarrow b$ and $c \rightarrow d$, thus

$$\int_a^b \vec{B} \cdot d\vec{l} + \int_c^d \vec{B} \cdot d\vec{l} = 2Bl$$

The current enclosed by the loop is $i = jl$. Therefore, according to Ampere's law $2Bl = \mu_0(jl)$ or $B = \frac{\mu_0 j}{2}$

Solenoid



A cylindrical coil of many tightly wound turns of insulated wire with generally diameter of the coil smaller than its length is called a solenoid.

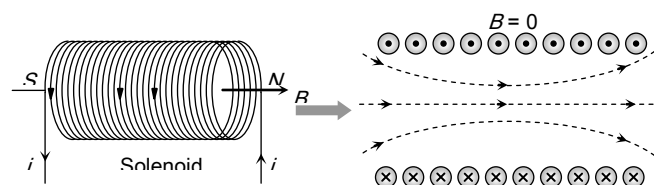


Fig. 21.22

A magnetic field is produced around and within the solenoid. The magnetic field within the solenoid is uniform and parallel to the axis of solenoid.

(1) Finite length solenoid :

If N = total number of turns, l = length of the solenoid, n = number of turns per unit length $= \frac{N}{l}$

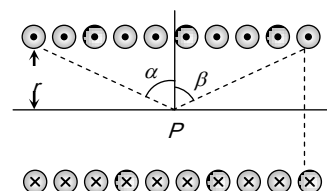


Fig. 21.23

(i) Magnetic field inside the

solenoid at point P is given by $B = \frac{\mu_0}{4\pi} (2\pi ni) [\sin\alpha + \sin\beta]$

(ii) **Infinite length solenoid** : If the solenoid is of infinite length and the point is well inside the solenoid *i.e.* $\alpha = \beta = (\pi / 2)$.

So $B_{in} = \mu_0 ni$

(iii) If the solenoid is of infinite length and the point is near one end *i.e.* $\alpha = 0$ and $\beta = (\pi / 2)$ so $B_{end} = \frac{1}{2}(\mu_0 ni)$
($B_{end} = \frac{1}{2} B_{in}$)

Toroid

A toroid can be considered as a ring shaped closed solenoid. Hence it is like an endless cylindrical solenoid.

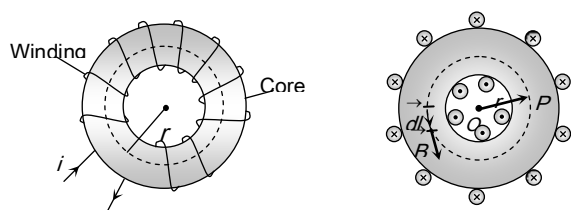


Fig. 21.24

Consider a toroid having n turns per unit length. Magnetic field at a point P in the figure is given as

$$B = \frac{\mu_0 Ni}{2\pi r} = \mu_0 ni \text{ where } n = \frac{N}{2\pi r}$$

Force On a Charged Particle in Magnetic Field

If a particle carrying a positive charge q and moving with velocity v enters a magnetic field B then it experiences a force F which is given by the expression $\vec{F} = q(\vec{v} \times \vec{B}) \Rightarrow F = qvB \sin\theta$

where $\vec{v}$ = velocity of the particle, $\vec{B}$ = magnetic field

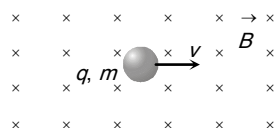


Fig. 21.25

(1) **Zero force** : Force on charged particle will be zero (*i.e.* $F = 0$) if

(i) No field *i.e.* $B = 0 \Rightarrow F = 0$

(ii) Neutral particle *i.e.* $q = 0 \Rightarrow F = 0$

(iii) Rest charge *i.e.* $v = 0 \Rightarrow F = 0$

(iv) Moving charge *i.e.* $\theta = 0^\circ$ or $\theta = 180^\circ \Rightarrow F = 0$

(2) **Direction of force** : The force $\vec{F}$ is always perpendicular to both the velocity $\vec{v}$ and the field $\vec{B}$ in accordance with Right Hand Screw Rule, though $\vec{v}$ and $\vec{B}$ themselves may or may not be perpendicular.

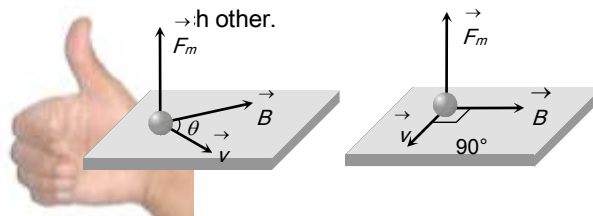


Fig. 21.26

Direction of force on charged particle in magnetic field can also be found by Fleming's Left Hand Rule (FLHR).

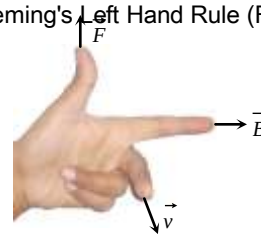


Fig. 21.27

Here, **First finger** (indicates) $\rightarrow$ Direction of magnetic field

Middle finger $\rightarrow$ Direction of motion of positive charge or direction, Opposite to the motion of negative charge.

Thumb $\rightarrow$ Direction of force

Trajectory of a Charged Particle in a Magnetic Field

(1) **Straight line** : If the direction of a $\vec{v}$ is parallel or antiparallel to $\vec{B}$, $\theta = 0$ or $\theta = 180^\circ$ and therefore $F = 0$. Hence the trajectory of the particle is a straight line.

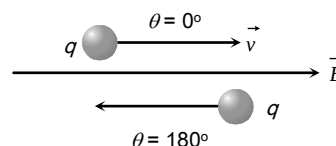


Fig. 21.28

(2) **Circular path** : If $\vec{v}$ is perpendicular to $\vec{B}$ i.e. $\theta = 90^\circ$, hence particle will experience a maximum magnetic force $F_{\max} = qvB$ which acts in a direction perpendicular to the motion of charged particle. Therefore the trajectory of the particle is a circle.

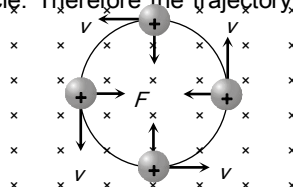


Fig. 21.29

(i) In this case path of charged particle is circular and magnetic force provides the necessary centripetal force i.e.

$$qvB = \frac{mv^2}{r} \Rightarrow \text{radius of path}$$

$$r = \frac{mv}{qB} = \frac{p}{qB} = \frac{\sqrt{2mK}}{qB} = \frac{1}{B} \sqrt{\frac{2mV}{q}}$$

where p = momentum of charged particle and K = kinetic energy of charged particle (gained by charged particle after accelerating through potential difference V) then

$$p = mv = \sqrt{2mK} = \sqrt{2mqV}$$

(ii) If T is the time period of the particle then $T = \frac{2\pi m}{qB}$ (i.e., time period (or frequency) is independent of speed of particle).

(3) **Helical path** : When the charged particle is moving at an angle to the field (other than 0° , 90° , or 180°). Particle describes a path called helix.

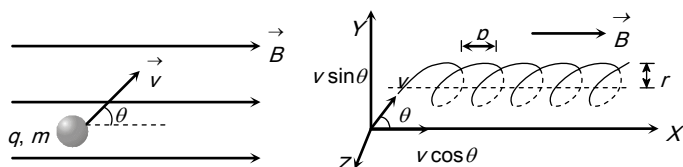


Fig. 21.30

(i) The radius of this helical path is $r = \frac{m(v \sin \theta)}{qB}$

(ii) Time period and frequency do not depend on velocity and so they are given by $T = \frac{2\pi m}{qB}$ and $\nu = \frac{qB}{2\pi m}$

(iii) The *pitch* of the *helix*, (i.e., linear distance travelled in one rotation) will be given by $p = T(v \cos \theta) = 2\pi \frac{m}{qB} (v \cos \theta)$

(iv) If pitch value is p , then number of pitches obtained in length l given as

$$\text{Number of pitches} = \frac{l}{p} \text{ and time required } t = \frac{l}{v \cos \theta}$$

Lorentz Force

When the moving charged particle is subjected simultaneously to both electric field $\vec{E}$ and magnetic field $\vec{B}$, the moving charged particle will experience electric force $\vec{F}_e = q\vec{E}$ and magnetic force $\vec{F}_m = q(\vec{v} \times \vec{B})$; so the net force on it will be $\vec{F} = q[\vec{E} + (\vec{v} \times \vec{B})]$. Which is the famous 'Lorentz-force equation'.

Depending on the directions of $\vec{v}$, $\vec{E}$ and $\vec{B}$ following situations are possible

(i) **When $\vec{v}$, $\vec{E}$ and $\vec{B}$ all the three are collinear** : In this situation the magnetic force on it will be zero and only electric force will act and so $\vec{a} = \frac{\vec{F}}{m} = \frac{q\vec{E}}{m}$

(ii) The particle will pass through the field following a straight-line path (parallel field) with change in its speed. So in this situation speed, velocity, momentum and kinetic energy all will change without change in direction of motion as shown

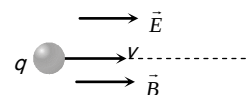


Fig. 21.31

(iii) **$\vec{v}$, $\vec{E}$ and $\vec{B}$ are mutually perpendicular** : In this situation if $\vec{E}$ and $\vec{B}$ are such that $\vec{F} = \vec{F}_e + \vec{F}_m = 0$ i.e., $\vec{a} = (\vec{F}/m) = 0$

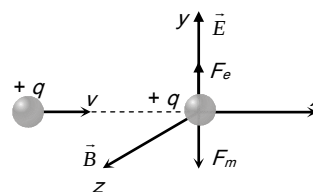


Fig. 21.32

(2) **Maximum energy of particle** : Maximum energy gained

$$\text{by the charged particle } E_{\max} = \left(\frac{q^2 B^2}{2m} \right) r^2$$

where r_0 = maximum radius of the circular path followed by the positive ion.

Hall Effect

The Phenomenon of producing a transverse emf in a current carrying conductor on applying a magnetic field perpendicular to the direction of the current is called Hall effect.

Hall effect helps us to know the nature and number of charge carriers in a conductor.

Consider a conductor having electrons as current carriers. The electrons move with drift velocity $\vec{v}$ opposite to the direction of flow of current

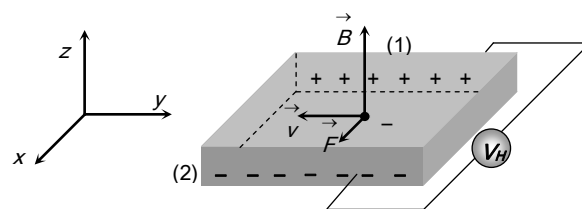


Fig. 21.34

Force acting on electron $F_m = -e(\vec{v} \times \vec{B})$. This force acts along x-axis and hence electrons will move towards face (2) and it becomes negatively charged.

Force On a Current Carrying Conductor In Magnetic Field

In case of current carrying conductor in a magnetic field force experienced by its small length element is $d\vec{F} = i d\vec{l} \times \vec{B}$; $i d\vec{l}$ = current element $d\vec{F} = i(d\vec{l} \times \vec{B})$

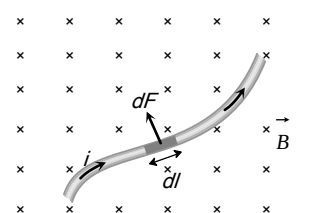


Fig. 21.35

as shown in figure, the particle will pass through the field with same velocity, without any deviation in path.

And in this situation, as $F_e = F_m$ i.e., $qE = qvB$ $v = E/B$

This principle is used in 'velocity-selector' to get a charged beam having a specific velocity.

Cyclotron

Cyclotron is a device used to accelerated positively charged particles (like, α -particles, deuterons etc.) to acquire enough energy to carry out nuclear disintegration etc.

It is based on the fact that the electric field accelerates a charged particle and the magnetic field keeps it revolving in circular orbits of constant frequency.

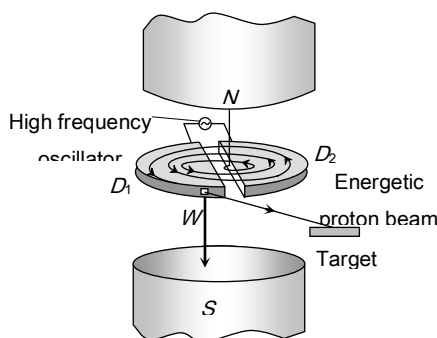


Fig. 21.33

It consists of two hollow D -shaped metallic chambers D_1 and D_2 called dees. The two dees are placed horizontally with a small gap separating them. The dees are connected to the source of high frequency electric field. The dees are enclosed in a metal box containing a gas at a low pressure of the order of 10^{-3} mm mercury. The whole apparatus is placed between the two poles of a strong electromagnet NS as shown in fig. The magnetic field acts perpendicular to the plane of the dees.

(1) **Cyclotron frequency** : Time taken by ion to describe a semicircular path is given by $t = \frac{\pi r}{v} = \frac{\pi m}{qB}$

If T = time period of oscillating electric field then $T = 2t = \frac{2\pi m}{qB}$ the cyclotron frequency $\nu = \frac{1}{T} = \frac{Bq}{2\pi m}$

Total magnetic force $\vec{F} = \int d\vec{F} = \int i(d\vec{l} \times \vec{B})$. If magnetic field is uniform i.e., $\vec{B} = \text{constant}$ $\vec{F} = i[\int d\vec{l}] \times \vec{B} = i(\vec{L} \times \vec{B})$

$\int d\vec{l} = \vec{L}$ = vector sum of all the length elements from initial to final point. Which is in accordance with the law of vector addition is equal to length vector $\vec{L}$ joining initial to final point.

(For a straight conductor $F = Bilsin\theta$)

Direction of force : The direction of force is always perpendicular to the plane containing $i d\vec{l}$ and $\vec{B}$ and is same as that of cross-product of two vectors $(\vec{A} \times \vec{B})$ with $\vec{A} = i d\vec{l}$.

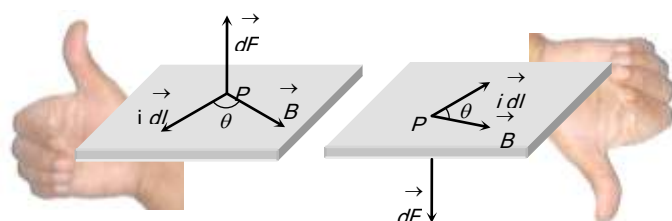


Fig. 21.36

The direction of force when current element $i d\vec{l}$ and $\vec{B}$ are perpendicular to each other can also be determined by applying either of the following rules

Fleming's left-hand rule : Stretch the fore-finger, central finger and thumb of left hand mutually perpendicular. Then if the fore-finger points in the direction of field $\vec{B}$ and the central in the direction of current i , the thumb will point in the direction of force.

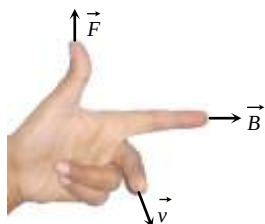


Fig. 21.37

Right-hand palm rule : Stretch the fingers and thumb of right hand at right angles to each other. Then if the fingers point in the direction of field $\vec{B}$ and thumb in the direction of current i , then normal to the palm will point in the direction of force



Fig. 21.38

Force Between Two Parallel Current Carrying Conductors

The force on a length l of each of two long, straight, parallel wires carrying currents i_1 and i_2 and separated by a distance a is

$$F = \frac{\mu_0}{4\pi} \cdot \frac{2i_1 i_2}{a} \times l$$

Hence force per unit length

$$\frac{F}{l} = \frac{\mu_0}{4\pi} \cdot \frac{2i_1 i_2}{a} \left(\frac{N}{m} \right) \text{ or } \frac{F}{l} = \frac{2i_1 i_2}{a} \left(\frac{\text{dyne}}{\text{cm}} \right)$$

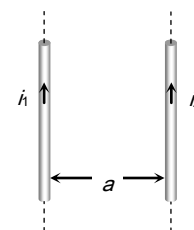


Fig. 21.39

Direction of force : If conductors carries current in same direction, then force between them will be attractive. If conductor carries current in opposite direction, then force between them will be repulsive.

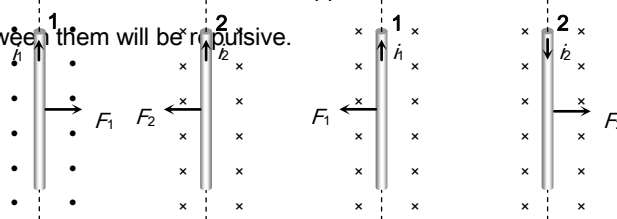


Fig. 21.40

Force Between Two Moving Charges

If two charges q_1 and q_2 are moving with velocities v_1 and v_2 respectively and at any instant the distance between them is r , then

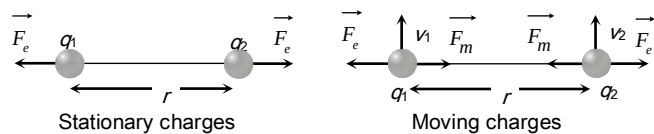


Fig. 21.41

Magnetic force between them is $F_m = \frac{\mu_0}{4\pi} \cdot \frac{q_1 q_2 v_1 v_2}{r^2}$ (i)

and Electric force between them is $F_e = \frac{1}{4\pi\epsilon_0} \cdot \frac{q_1 q_2}{r^2}$ (ii)

From equation (i) and (ii) $\frac{F_m}{F_e} = \mu_0 \epsilon_0 v^2$ but $\mu_0 \epsilon_0 = \frac{1}{c^2}$;

where c is the velocity of light in vacuum. So $\frac{F_m}{F_e} = \left(\frac{v}{c}\right)^2$

As $v < c$ so $F_m < F_e$

Standard Cases For Force on Current Carrying Conductors

Case 1 : When an arbitrary current carrying loop placed in a magnetic field ($\perp$ to the plane of loop), each element of loop experiences a magnetic force due to which loop stretches and open into circular loop and tension developed in it's each part.

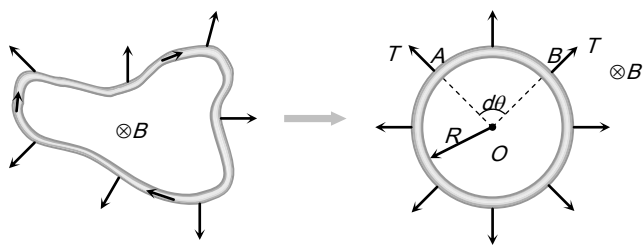


Fig. 21.42

Case 2 : Equilibrium of a current carrying conductor : When a finite length current carrying wire is kept parallel to another infinite length current carrying wire, it can suspend freely in air as shown below

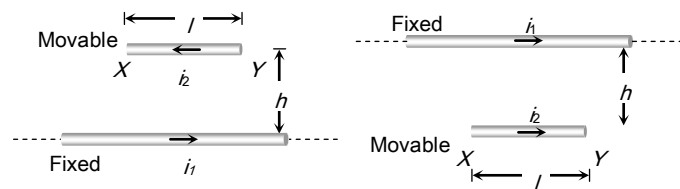
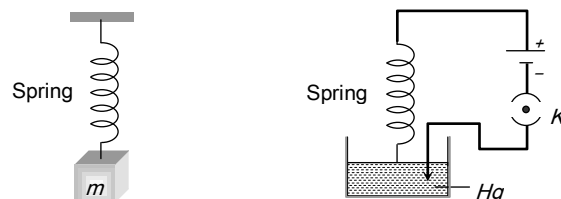


Fig. 21.43

In both the situations for equilibrium of XY it's downward weight = upward magnetic force *i.e.* $mg = \frac{\mu_0}{4\pi} \cdot \frac{2i_1 i_2}{h} \cdot l$

Case 3 : Current carrying spring : If current is passed through a spring, then it will contract because current will flow through all the turns in the same direction.



If current makes to flow through spring, then spring will contract and weight lift up. If switch is closed then current start flowing, spring will execute and weight lift up.

Fig. 21.44

Case 4 : Tension less strings : In the following figure the value and direction of current through the conductor XY so that strings becomes tensionless?

Strings becomes tensionless if weight of conductor XY balanced by magnetic force (F_m).

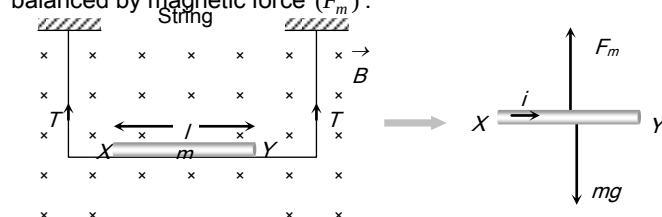


Fig. 21.45

1184 Magnetic Effect of Current

Hence direction of current is from $X \rightarrow Y$ and in balanced

$$\text{condition } F_m = mg \Rightarrow Bil = mg \Rightarrow i = \frac{mg}{Bl}$$

Case 5 : Sliding of conducting rod on inclined rails : When a

conducting rod slides on conducting rails.

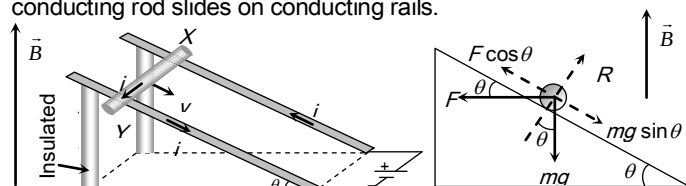


Fig. 21.46

In the following situation conducting rod (X, Y) slides at constant velocity if

$$F \cos \theta = mg \sin \theta \Rightarrow Bil \cos \theta = mg \sin \theta \Rightarrow B = \frac{mg}{il} \tan \theta$$

Current Loop as a Magnetic Dipole

A current carrying circular coil behaves as a bar magnet whose magnetic moment is $M = NiA$; Where N = Number of turns in the coil, i = Current through the coil and A = Area of the coil

Magnetic moment of a current carrying coil is a vector and it's direction is given by right hand thumb rule

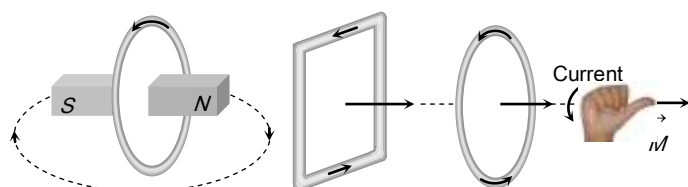


Fig. 21.47

(1) For a given perimeter circular shape have maximum area. Hence maximum magnetic moment.

(2) For a any loop or coil $\vec{B}$ at centre due to current in loop, and $\vec{M}$ are always parallel.



Fig. 21.48

Behaviour of Current Loop in a Magnetic Field

(1) **Torque :** Consider a rectangular current carrying coil $PQRS$ having N turns and area A , placed in a uniform field $\vec{B}$, in such a way that the normal ($\hat{n}$) to the coil makes an angle θ with the direction of $\vec{B}$. the coil experiences a torque given by $\tau = NBiA \sin \theta$. Vectorially $\vec{\tau} = \vec{M} \times \vec{B}$

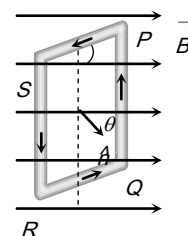


Fig. 21.49

(i) τ is zero when $\theta = 0$, i.e., when the plane of the coil is perpendicular to the field.

(ii) τ is maximum when $\theta = 90^\circ$, i.e., the plane of the coil is parallel to the field $\tau_{\max} = NBiA$

(2) **Workdone :** If coil is rotated through an angle θ from it's equilibrium position then required work. $W = MB(1 - \cos \theta)$. It is maximum when $\theta = 180^\circ \Rightarrow W_{\max} = 2 MB$

(3) **Potential energy :** $U = -MB \cos \theta \Rightarrow U = -\vec{M} \cdot \vec{B}$

Moving Coil Galvanometer

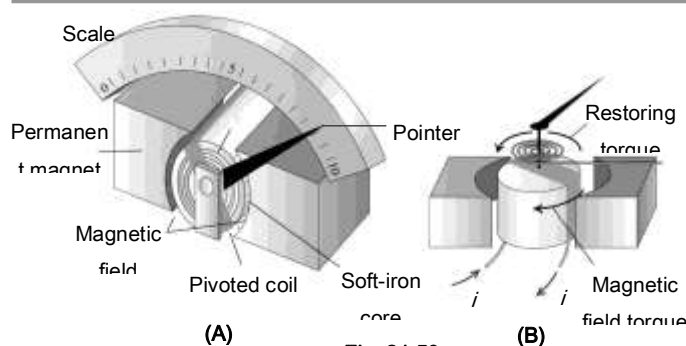


Fig. 21.50

In a moving coil galvanometer the coil is suspended between the pole pieces of a strong horse-shoe magnet. The pole pieces are made cylindrical and a soft iron cylindrical core is placed

within the coil without touching it. This makes the field radial. In such a field the plane of the coil always remains parallel to the field. Therefore $\theta = 90^\circ$ and the deflecting torque always has the maximum value.

$$\tau_{\text{def}} = NBiA \quad \dots\dots(i)$$

Coil deflects, a restoring torque is set up in the suspension fibre. If α is the angle of twist, the restoring torque is

$$\tau_{\text{rest}} = C\alpha \quad \dots\dots(ii)$$

where C is the torsional constant of the fibre.

When the coil is in equilibrium $NBiA = C\alpha \Rightarrow i = K\alpha$,

where $K = \frac{C}{NBA}$ is the galvanometer constant. This linear

relationship between i and α makes the moving coil galvanometer useful for current measurement and detection.

Current sensitivity (S_i) : The current sensitivity of a galvanometer is defined as the deflection produced in the galvanometer per unit current flowing through it.

$$S_i = \frac{\alpha}{i} = \frac{NBA}{C}$$

Voltage sensitivity (S_V) : Voltage sensitivity of a galvanometer is defined as the deflection produced in the galvanometer per unit voltage applied to it.

$$S_V = \frac{\alpha}{V} = \frac{\alpha}{iR} = \frac{S_i}{R} = \frac{NBA}{RC}$$

Tips & Tricks

✍ The device whose working principle based on Halmholtz coils and in which uniform magnetic field is used called as "Halmholtz galvanometer".

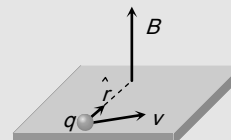
✍ The value of magnetic field induction at a point, on the centre of separation of two linear parallel conductors carrying equal currents in the same direction is zero.

✍ If a current carrying circular loop ($n = 1$) is turned into a coil having n identical turns then magnetic field at the centre

of the coil becomes n^2 times the previous field i.e. $B_{(n \text{ turn})} = n^2 B_{(\text{single turn})}$

✍ When a current carrying coil is suspended freely in earth's magnetic field, its plane stays in **East-West** direction.

✍ Magnetic field ($\vec{B}$) produced by a moving charge q is given by $\vec{B} = \frac{\mu_0}{4\pi} \frac{q(\vec{v} \times \vec{r})}{r^3} = \frac{\mu_0}{4\pi} \frac{q(\vec{v} \times \hat{r})}{r^2}$; where v = velocity of charge and $v \ll c$ (speed of light).



✍ If an electron is revolving in a circular path of radius r with speed v then magnetic field produced at the centre of circular path $B = \frac{\mu_0}{4\pi} \cdot \frac{ev}{r^2} \Rightarrow r \propto \sqrt{\frac{v}{B}}$

✍ The line integral of magnetising field ($\vec{H}$) for any closed path called magnetomotive force (MMF). Its S.I. unit is amp.

✍ Ratio of dimension of e.m.f. to MMF is equal to the dimension of resistance.

✍ The positive ions are produced in the gap between the two dees by the ionisation of the gas. To produce proton, hydrogen gas is used; while for producing alpha-particles, helium gas is used.

✍ Cyclotron frequency is also known as magnetic resonance frequency.

✍ Cyclotron can not accelerate electrons because they have very small mass.

✍ The energy of a charged particle moving in a uniform magnetic field does not change because it experiences a force in a direction, perpendicular to its direction of motion. Due to which the speed of charged particle remains unchanged and hence its K.E. remains same.

✍ Magnetic force does no work when the charged particle

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is displaced while electric force does work in displacing the charged particle.

✍ Magnetic force is velocity dependent, while electric force is independent of the state of rest or motion of the charged particle.

✍ If a particle enters a magnetic field normally to the magnetic field, then it starts moving in a circular orbit. The point at which it enters the magnetic field lies on the circumference. (Most of us confuse it with the centre of the orbit)

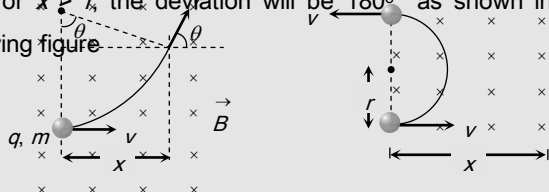
✍ Deviation of charged particle in magnetic field : If a charged particle (q, m) enters a uniform magnetic field $\vec{B}$ (extends upto a length x) at right angles with speed v as shown in figure. The speed of the particle in magnetic field does not change. But it gets deviated in the magnetic field.

$$\text{Deviation in terms of time } t; \theta = \omega t = \left(\frac{Bq}{m}\right)t$$

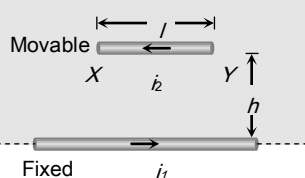
Deviation in terms of length of the magnetic field;

$$\theta = \sin^{-1}\left(\frac{x}{r}\right). \text{ This relation can be used only when } x \leq r.$$

For $x > r$, the deviation will be 180° as shown in the following figure



✍ If no magnetic field is present, the loop will still open into a circle as in its adjacent parts current will be in opposite direction and opposite currents repel each other.



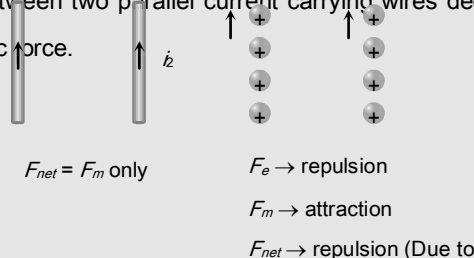
✍ In the following case if wire XY is slightly displaced from its equilibrium position, it executes SHM and its time period

$$\text{is given by } T = 2\pi\sqrt{\frac{h}{g}}.$$

✍ In the previous case if direction of current in movable wire is reversed then its instantaneous acceleration produced is $2g$ ↓.

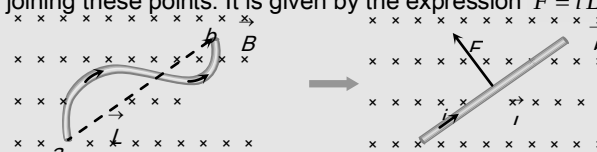
✍ Electric force is an absolute concept while magnetic force is a relative concept for an observer.

✍ The nature of force between two parallel charge beams decided by electric force, as it is dominator. The nature of force between two parallel current carrying wires decided by magnetic force.



✍ If a straight current carrying wire is placed along the axis of a current carrying coil then it will not experience magnetic force because magnetic field produced by the coil is parallel to the wire.

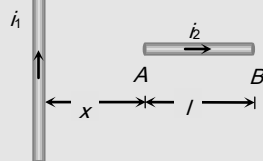
✍ The force acting on a curved wire joining points a and b as shown in the figure is the same as that on a straight wire joining these points. It is given by the expression $\vec{F} = i \vec{L} \times \vec{B}$



✍ If a current carrying conductor AB is placed transverse to a long current carrying conductor as shown then force.

Experienced by wire AB

$$F = \frac{\mu_0 i_1 i_2}{2\pi} \log_e \left(\frac{x+l}{x} \right)$$



Ordinary Thinking

Objective Questions

Biot-savart's Law and Ampere's Law

1. A length L of wire carries a steady current I . It is bent first to form a circular plane coil of one turn. The same length is now bent more sharply to give a double loop of smaller radius. The magnetic field at the centre caused by the same current is [NCERT 1980; AIIMS 1980; MP PMT 1995, 99]
 - (a) A quarter of its first value
 - (b) Unaltered
 - (c) Four times of its first value
 - (d) A half of its first value
2. A vertical straight conductor carries a current vertically upwards. A point P lies to the east of it at a small distance and another point Q lies to the west at the same distance. The magnetic field at P is [MNR 1986; DPMT 2004]
 - (a) Greater than at Q
 - (b) Same as at Q
 - (c) Less than at Q
 - (d) Greater or less than at Q depending upon the strength of the current
3. If a copper rod carries a direct current, the magnetic field associated with the current will be [CPMT 1984]
 - (a) Only inside the rod
 - (b) Only outside the rod
 - (c) Both inside and outside the rod
 - (d) Neither inside nor outside the rod

4. If a long hollow copper pipe carries a direct current, the magnetic field associated with the current will be

[CBSE PMT 1999; AFMC 1999;

CPMT 1984, 2000; Pb. PET 2000; JIPMER 2002]

- (a) Only inside the pipe
(b) Only outside the pipe
(c) Neither inside nor outside the pipe
(d) Both inside and outside the pipe

5. The magnetic field $d\vec{B}$ due to a small current element $d\vec{l}$ at a distance $\vec{r}$ and element carrying current i is,

or

Vector form of Biot-savart's law is

[CBSE PMT 1996; MP PET 2002; MP PMT 2000]

- (a) $d\vec{B} = \frac{\mu_0}{4\pi} i \left(\frac{d\vec{l} \times \vec{r}}{r^3} \right)$ (b) $d\vec{B} = \frac{\mu_0}{4\pi} i^2 \left(\frac{d\vec{l} \times \vec{r}}{r^3} \right)$
(c) $d\vec{B} = \frac{\mu_0}{4\pi} i^2 \left(\frac{d\vec{l} \times \vec{r}}{r^2} \right)$ (d) $d\vec{B} = \frac{\mu_0}{4\pi} i \left(\frac{d\vec{l} \times \vec{r}}{r^3} \right)$

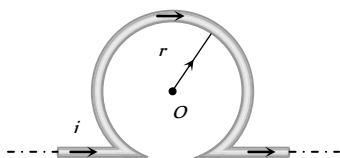
6. A charge q coulomb moves in a circle at n revolutions per second and the radius of the circle is r metre. Then magnetic field at the centre of the circle is

- (a) $\frac{2\pi q}{nr} \times 10^{-7}$ N/amp/metre (b) $\frac{2\pi q}{r} \times 10^{-7}$ N/amp/metre
(c) $\frac{2\pi nq}{r} \times 10^{-7}$ N/amp/metre (d) $\frac{2\pi nq}{r}$ N/amp/metre

7. An infinitely long straight conductor is bent into the shape as shown in the figure. It carries a current of i ampere and the radius of the circular loop is r metre. Then the magnetic induction at its centre will be

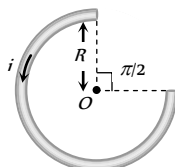
[MP PMT 1999]

- (a) $\frac{\mu_0}{4\pi} \frac{2i}{r} (\pi + 1)$
(b) $\frac{\mu_0}{4\pi} \frac{2i}{r} (\pi - 1)$
(c) Zero
(d) Infinite



8. A current i ampere flows in a circular arc of wire whose radius is R , which subtend an angle $3\pi/2$ radian at its centre. The magnetic induction B at the centre is

- (a) $\frac{\mu_0 i}{R}$
(b) $\frac{\mu_0 i}{2R}$
(c) $\frac{2\mu_0 i}{R}$
(d) $\frac{3\mu_0 i}{8R}$



9. A current i ampere flows along the inner conductor of a coaxial cable and returns along the outer conductor of the cable, then the

magnetic induction at any point outside the conductor at a distance r metre from the axis is

- (a) ∞ (b) Zero
(c) $\frac{\mu_0}{4\pi} \frac{2i}{r}$ (d) $\frac{\mu_0}{4\pi} \frac{2\pi i}{r}$

10. A straight section PQ of a circuit lies along the X -axis from $x = -\frac{a}{2}$ to $x = \frac{a}{2}$ and carries a steady current i . The magnetic field due to the section PQ at a point $X = +a$ will be

- (a) Proportional to a (b) Proportional to a^2
(c) Proportional to $1/a$ (d) Zero

11. A helium nucleus makes a full rotation in a circle of radius 0.8 metre in two seconds. The value of the magnetic field B at the centre of the circle will be

[CPMT 1988; KCET 1998; UPSEAT 2001]

- (a) $\frac{10^{-19}}{\mu_0}$ (b) $10^{-19} \mu_0$
(c) $2 \times 10^{-10} \mu_0$ (d) $\frac{2 \times 10^{-10}}{\mu_0}$

12. A solenoid of 1.5 metre length and 4.0 cm diameter posses 10 turn per cm. A current of 5 ampere is flowing through it. The magnetic induction at axis inside the solenoid is

[CPMT 1990]

- (a) $2\pi \times 10^{-3}$ Tesla (b) $2\pi \times 10^{-5}$ Tesla
(c) $4\pi \times 10^{-2}$ Gauss (d) $2\pi \times 10^{-5}$ Gauss

13. The magnetic induction at a point P which is distant 4 cm from a long current carrying wire is 10^{-8} Tesla. The field of induction at a distance 12 cm from the same current would be

- (a) 3.33×10^{-9} Tesla (b) 1.11×10^{-4} Tesla
(c) 3×10^{-3} Tesla (d) 9×10^{-2} Tesla

14. The strength of the magnetic field at a point r near a long straight current carrying wire is B . The field at a distance $\frac{r}{2}$ will be

- (a) $\frac{B}{2}$ (b) $\frac{B}{4}$
(c) $2B$ (d) $4B$

15. Field at the centre of a circular coil of radius r , through which a current I flows is

[MP PMT 1993]

- (a) Directly proportional to r
(b) Inversely proportional to I
(c) Directly proportional to I
(d) Directly proportional to I^2

16. A current of 0.1 A circulates around a coil of 100 turns and having a radius equal to 5 cm. The magnetic field set up at the centre of the coil is

($\mu_0 = 4\pi \times 10^{-7}$ weber / ampere – metre) [MP PMT 1993]

- (a) $4\pi \times 10^{-5}$ tesla (b) $8\pi \times 10^{-5}$ tesla
(c) 4×10^{-5} tesla (d) 2×10^{-5} tesla

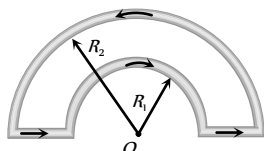
17. The magnetic field B with in the solenoid having n turns per metre length and carrying a current of i ampere is given by

[MP PET 1993]

- (a) $\frac{\mu_0 n i}{e}$ (b) $\mu_0 n i$
(c) $4\pi\mu_0 n i$ (d) $n i$

18. The magnetic induction at the centre O in the figure shown is

- (a) $\frac{\mu_0 i}{4} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$
(b) $\frac{\mu_0 i}{4} \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$
(c) $\frac{\mu_0 i}{4} (R_1 - R_2)$
(d) $\frac{\mu_0 i}{4} (R_1 + R_2)$



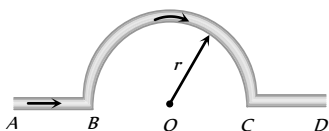
19. Field inside a solenoid is

[MP PMT 1993]

- (a) Directly proportional to its length
(b) Directly proportional to current
(c) Inversely proportional to total number of turns
(d) Inversely proportional to current

20. In the figure, shown the magnetic induction at the centre of there arc due to the current in portion AB will be

- (a) $\frac{\mu_0 i}{r}$
(b) $\frac{\mu_0 i}{2r}$
(c) $\frac{\mu_0 i}{4r}$
(d) Zero



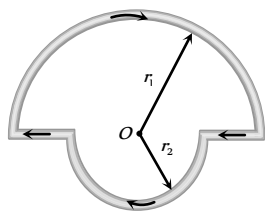
21. In the above question, the magnetic induction at O due to the whole length of the conductor is

[MP PMT/PET 1998; RPET 2002]

- (a) $\frac{\mu_0 i}{r}$ (b) $\frac{\mu_0 i}{2r}$
(c) $\frac{\mu_0 i}{4r}$ (d) Zero

22. In the figure shown there are two semicircles of radii r_1 and r_2 in which a current i is flowing. The magnetic induction at the centre O will be

- (a) $\frac{\mu_0 i}{r} (r_1 + r_2)$
(b) $\frac{\mu_0 i}{4} (r_1 - r_2)$
(c) $\frac{\mu_0 i}{4} \left(\frac{r_1 + r_2}{r_1 r_2} \right)$



(d) $\frac{\mu_0 i}{4} \left(\frac{r_2 - r_1}{r_1 r_2} \right)$

23. The magnetic moment of a current carrying loop is $2.1 \times 10^{-25} \text{ amp} \times \text{m}^2$. The magnetic field at a point on its axis at a distance of $1 \text{ \AA}$ is

- (a) $4.2 \times 10^{-2} \text{ weber / m}^2$ (b) $4.2 \times 10^{-3} \text{ weber / m}^2$
(c) $4.2 \times 10^{-4} \text{ weber / m}^2$ (d) $4.2 \times 10^{-5} \text{ weber / m}^2$

[IIT 1988; KCET 2002]

24. Two straight horizontal parallel wires are carrying the same current in the same direction, d is the distance between the wires. You are provided with a small freely suspended magnetic needle. At which of the following positions will the orientation of the needle be independent of the magnitude of the current in the wires

- (a) At a distance $d/2$ from any of the wires
(b) At a distance $d/2$ from any of the wires in the horizontal plane
(c) Anywhere on the circumference of a vertical circle of radius d and centre halfway between the wires
(d) At points halfway between the wires in the horizontal plane

25. A particle carrying a charge equal to 100 times the charge on an electron is rotating per second in a circular path of radius 0.8 metre . The value of the magnetic field produced at the centre will be ($\mu_0 = \text{permeability for vacuum}$)

[CPMT 1986; KCET 2001; BHU 2001]

- (a) $\frac{10^{-7}}{\mu_0}$ (b) $10^{-17} \mu_0$
(c) $10^{-6} \mu_0$ (d) $10^{-7} \mu_0$

26. A circular coil of radius R carries an electric current. The magnetic field due to the coil at a point on the axis of the coil located at a distance r from the centre of the coil, such that $r \gg R$, varies as [EAMCET 1987]

- (a) $\frac{1}{r}$ (b) $\frac{1}{r^{3/2}}$
(c) $\frac{1}{r^2}$ (d) $\frac{1}{r^3}$

27. In hydrogen atom, an electron is revolving in the orbit of radius $0.53 \text{ \AA}$ with $6.6 \times 10^{15} \text{ rotations/second}$. Magnetic field produced at the centre of the orbit is [MP PET 2003]

- (a) 0.125 wb / m^2 (b) 1.25 wb / m^2
(c) 12.5 wb / m^2 (d) 125 wb / m^2

28. The magnetic induction due to an infinitely long straight wire carrying a current i at a distance r from wire is given by

[MP PET 1994]

- (a) $|B| = \left(\frac{\mu_0}{4\pi} \right) \frac{2i}{r}$ (b) $|B| = \left(\frac{\mu_0}{4\pi} \right) \frac{r}{2i}$
(c) $|B| = \left(\frac{4\pi}{\mu_0} \right) \frac{2i}{r}$ (d) $|B| = \left(\frac{4\pi}{\mu_0} \right) \frac{r}{2i}$

29. Magnetic effect of current was discovered by [MP PET 1994]

- (a) Faraday (b) Oersted
(c) Ampere (d) Bohr

30. Two concentric circular coils of ten turns each are situated in the same plane. Their radii are 20 and 40 cm and they carry respectively

0.2 and 0.3 *ampere* current in opposite direction. The magnetic field in *weber / m²* at the centre is

[MP PMT 1994]

- (a) $\frac{35}{4} \mu_0$ (b) $\frac{\mu_0}{80}$
(c) $\frac{7}{80} \mu_0$ (d) $\frac{5}{4} \mu_0$

31. A long solenoid has a radius a and number of turns per unit length is n . If it carries a current i , then the magnetic field on its axis is directly proportional to [MP PMT 1994]

- (a) ani (b) ni
(c) $\frac{ni}{a}$ (d) n^2i

32. A cell is connected between two points of a uniformly thick circular conductor. The magnetic field at the centre of the loop will be

- (a) Zero (b) $\frac{\mu_0}{2a}(i_1 - i_2)$
(c) $\frac{\mu_0}{2a}(i_1 + i_2)$ (d) $\frac{\mu_0}{a}(i_1 + i_2)$

(Here i_1 and i_2 are the currents flowing in the two parts of the circular conductor of radius ' a ' and μ_0 has the usual meaning)

33. A long solenoid is formed by winding 20 *turns/cm*. The current necessary to produce a magnetic field of 20 *millitesla* inside the solenoid will be approximately

$(\frac{\mu_0}{4\pi} = 10^{-7} \text{ tesla-metre / ampere})$ [MP PMT 1994]

- (a) 8.0 A (b) 4.0 A
(c) 2.0 A (d) 1.0 A

34. A battery is connected between two points A and B on the circumference of a uniform conducting ring of radius r and resistance R . One of the arcs AB of the ring subtends an angle θ at the centre. The value of the magnetic induction at the centre due to the current in the ring is [IIT 1995]

- (a) Proportional to $2(180^\circ - \theta)$
(b) Inversely proportional to r
(c) Zero, only if $\theta = 180^\circ$
(d) Zero for all values of θ

35. A current of 1 *ampere* is passed through a straight wire of length 2.0 *metres*. The magnetic field at a point in air at a distance of 3 *metres* from either end of wire and lying on the axis of wire will be

- (a) $\frac{\mu_0}{2\pi}$ (b) $\frac{\mu_0}{4\pi}$
(c) $\frac{\mu_0}{8\pi}$ (d) Zero

36. A long copper tube of inner radius R carries a current i . The magnetic field B inside the tube is [MP PMT 1995]

- (a) $\frac{\mu_0 i}{2\pi R}$ (b) $\frac{\mu_0 i}{4\pi R}$

- (c) $\frac{\mu_0 i}{2R}$ (d) Zero

37. A straight wire of length (π^2) *metre* is carrying a current of 2A and the magnetic field due to it is measured at a point distant 1 *cm* from it. If the wire is to be bent into a circle and is to carry the same current as before, the ratio of the magnetic field at its centre to that obtained in the first case would be

- (a) 50 : 1 (b) 1 : 50
(c) 100 : 1 (d) 1 : 100

38. The direction of magnetic lines of forces close to a straight conductor carrying current will be

[RPMT 2002; RPET 2003; MP PET 2003]

- (a) Along the length of the conductor
(b) Radially outward
(c) Circular in a plane perpendicular to the conductor
(d) Helical [MP PMT 1994]

39. If the strength of the magnetic field produced 10 *cm* away from a infinitely long straight conductor is 10^{-5} *Weber / m²*, the value of the current flowing in the conductor will be

[MP PET 1996]

- (a) 5 *ampere* (b) 10 *ampere*
(c) 500 *ampere* (d) 1000 *ampere*

40. Due to 10 *ampere* of current flowing in a circular coil of 10 *cm* radius, the magnetic field produced at its centre is 3.14×10^{-3} *Weber / m²*. The number of turns in the coil will be

- (a) 5000 (b) 100
(c) 50 (d) 25

41. There are 50 turns of a wire in every *cm* length of a long solenoid. If 4 *ampere* current is flowing in the solenoid, the approximate value of magnetic field along its axis at an internal point and at one end will be respectively

[MP PET 1996]

- (a) 12.6×10^{-3} *Weber / m²*, 6.3×10^{-3} *Weber / m²*
(b) 12.6×10^{-3} *Weber / m²*, 25.1×10^{-3} *Weber / m²*
(c) 25.1×10^{-3} *Weber / m²*, 12.6×10^{-3} *Weber / m²*
(d) 25.1×10^{-5} *Weber / m²*, 12.6×10^{-5} *Weber / m²*

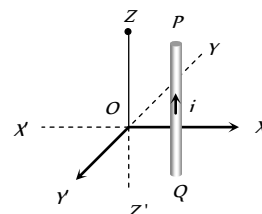
42. A solenoid is 1.0 *metre* long and it has 4250 turns. If a current of 5.0 *ampere* is flowing through it, what is the magnetic field at its centre [$\mu_0 = 4\pi \times 10^{-7}$ *weber / amp-m*] [MP PET 1995]

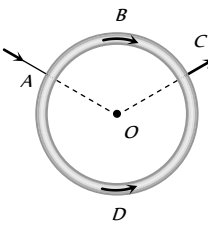
[MP PMT 1996]

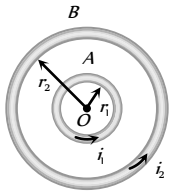
- (a) 5.4×10^{-2} *weber / m²* (b) 2.7×10^{-2} *weber / m²*
(c) 1.35×10^{-2} *weber / m²* (d) 0.675×10^{-2} *weber / m²*

43. A vertical wire kept in $Z-X$ plane carries a current from Q to P (see figure). The magnetic field due to current will have the direction at the origin O along

- (a) OX
(b) OX'



- (c) OY
(d) OY'
44. One metre length of wire carries a constant current. The wire is bent to form a circular loop. The magnetic field at the centre of this loop is B . The same is now bent to form a circular loop of smaller radius to have four turns in the loop. The magnetic field at the centre of this new loop is
(a) $4B$ (b) $16B$
(c) $B/2$ (d) $B/4$
45. In a hydrogen atom, an electron moves in a circular orbit of radius $5.2 \times 10^{-11} \text{ m}$ and produces a magnetic induction of 12.56 T at its nucleus. The current produced by the motion of the electron will be (Given $\mu_0 = 4\pi \times 10^{-7} \text{ Wb/A-m}$)
[MP PET 1997]
(a) $6.53 \times 10^{-3} \text{ ampere}$ (b) $13.25 \times 10^{-10} \text{ ampere}$
(c) $9.6 \times 10^6 \text{ ampere}$ (d) $1.04 \times 10^{-3} \text{ ampere}$
46. An arc of a circle of radius R subtends an angle $\frac{\pi}{2}$ at the centre. It carries a current i . The magnetic field at the centre will be
(a) $\frac{\mu_0 i}{2R}$ (b) $\frac{\mu_0 i}{8R}$
(c) $\frac{\mu_0 i}{4R}$ (d) $\frac{2\mu_0 i}{5R}$
47. At a distance of 10 cm from a long straight wire carrying current, the magnetic field is 0.04 T . At the distance of 40 cm , the magnetic field will be
[MP PMT 1997]
(a) 0.01 T (b) 0.02 T
(c) 0.08 T (d) 0.16 T
48. A uniform wire is bent in the form of a circle of radius R . A current I enters at A and leaves at C as shown in the figure :
If the length ABC is half of the length ADC , the magnetic field at the centre O will be
[MP PMT 1997]
(a) Zero
(b) $\frac{\mu_0 I}{2R}$
(c) $\frac{\mu_0 I}{4R}$
(d) $\frac{\mu_0 I}{6R}$
- 
49. The magnetic induction at any point due to a long straight wire carrying a current is
[MP PMT/PET 1998]
(a) Proportional to the distance from the wire
(b) Inversely proportional to the distance from wire
(c) Inversely proportional to the square of the distance from the wire
(d) Does not depend on distance
50. The expression for magnetic induction inside a solenoid of length L carrying a current I and having N number of turns is
(a) $\frac{\mu_0 N}{4\pi} \frac{N}{LI}$ (b) $\mu_0 NI$
(c) $\frac{\mu_0 ni}{4\pi}$ (d) $\mu_0 ni$
- (c) $\frac{\mu_0}{4\pi} NLI$ (d) $\mu_0 \frac{N}{L} I$
51. In a current carrying long solenoid, the field produced does not depend upon
[MP PET 1999]
(a) Number of turns per unit length
(b) Current flowing
(c) Radius of the solenoid
(d) All of the above three
52. The earth's magnetic induction at a certain point is $7 \times 10^{-5} \text{ Wb/m}^2$. This is to be annulled by the magnetic induction at the centre of a circular conducting loop of radius 5 cm . The required current in the loop is
[MP PET 1999; MP PMT 2002]
(a) 0.56 A (b) 5.6 A
(c) 0.28 A (d) 2.8 A
53. Magnetic field due to 0.1 A current flowing through a circular coil of radius 0.1 m and 1000 turns at the centre of the coil is
(a) $2 \times 10^{-1} \text{ T}$ (b) $4.31 \times 10^{-2} \text{ T}$
(c) $6.28 \times 10^{-4} \text{ T}$ (d) $9.81 \times 10^{-4} \text{ T}$
54. Magnetic field intensity at the centre of coil of 50 turns, radius 0.5 m and carrying a current of 2 A is
[MP PET 2003]
[CBSE PMT 1999; BHU 2002]
(a) $0.5 \times 10^{-5} \text{ T}$ (b) $1.25 \times 10^{-4} \text{ T}$
(c) $3 \times 10^{-5} \text{ T}$ (d) $4 \times 10^{-5} \text{ T}$
55. A circular coil 'A' has a radius R and the current flowing through it is I . Another circular coil 'B' has a radius $2R$ and if $2I$ is the current flowing through it, then the magnetic fields at the centre of the circular coil are in the ratio of (i.e. B_A to B_B)
(a) $4 : 1$ (b) $2 : 1$
(c) $3 : 1$ (d) $1 : 1$
56. The magnetic field at a distance r from a long wire carrying current i is 0.4 Tesla . The magnetic field at a distance $2r$ is
[CBSE PMT 1992; DPMT 2004]
(a) 0.2 Tesla (b) 0.8 Tesla
(c) 0.1 Tesla (d) 1.6 Tesla
57. A current I flows along the length of an infinitely long, straight and thin-walled pipe. Then
[IIT-JEE 1993]
(a) The magnetic field at all points inside the pipe is the same but not zero
(b) The magnetic field at any point inside the pipe is zero
(c) The magnetic field is zero only on the axis of the pipe
(d) The magnetic field is different at different points inside the pipe
58. The magnetic field at the centre of current carrying coil is
[CPMT 1996; RPET 2002, 03]
(a) $\frac{\mu_0 ni}{4\pi}$ (b) $\frac{\mu_0 ni}{2\pi r}$
(c) $\frac{\mu_0 ni}{4r}$ (d) $\mu_0 ni$

59. A straight wire of diameter 0.5 mm carrying a current of 1 A is replaced by another wire of 1 mm diameter carrying the same current. The strength of magnetic field far away is
[CBSE PMT 1997, 99]
- (a) Twice the earlier value
(b) Half of the earlier value
(c) Quarter of its earlier value
(d) Unchanged
60. A neutral point is obtained at the centre of a vertical circular coil carrying current. The angle between the plane of the coil and the magnetic meridian is
[SCRA 1998]
- (a) 0°
(b) 45°
(c) 60°
(d) 90°
61. One *Tesla* is equal to
[AFMC 1998]
- (a) 10^7 gauss
(b) 10^{-4} gauss
(c) 10^4 gauss
(d) 10^{-8} gauss
62. A current carrying wire in the neighborhood produces
[AFMC 1999]
- (a) No field
(b) Electric field only
(c) Magnetic field only
(d) Electric and magnetic fields
63. Tesla is the unit of
[AIIMS 1999]
- (a) Electric flux
(b) Magnetic flux
(c) Electric field
(d) Magnetic field
64. The magnetic induction in air at a point 1 cm away from a long wire that carries a current of 1 A , will be
[BHU 1999]
- (a) $1 \times 10^{-5} \text{ T}$
(b) $2 \times 10^{-5} \text{ T}$
(c) $3 \times 10^{-5} \text{ T}$
(d) $4 \times 10^{-5} \text{ T}$
65. The magnetic field at the centre of coil of n turns, bent in the form of a square of side $2l$, carrying current i , is
[AMU (Engg.) 1999]
- (a) $\frac{\sqrt{2}\mu_0 ni}{\pi l}$
(b) $\frac{\sqrt{2}\mu_0 ni}{2\pi l}$
(c) $\frac{\sqrt{2}\mu_0 ni}{4\pi l}$
(d) $\frac{2\mu_0 ni}{\pi l}$
66. Which of the following gives the value of magnetic field according to, Biot-Savart's law
[BHU 2000]
- (a) $\frac{i\Delta l \sin\theta}{r^2}$
(b) $\frac{\mu_0}{4\pi} \frac{i\Delta l \sin\theta}{r}$
(c) $\frac{\mu_0}{4\pi} \frac{i\Delta l \sin\theta}{r^2}$
(d) $\frac{\mu_0}{4\pi} i\Delta l \sin\theta$
67. A toroid has number of turns per unit length n , current i then the magnetic field is
[RPET 2000]
- (a) $\mu_0 ni$
(b) $\mu_0 n^2 i$
(c) $\mu_0 i / n$
(d) None of these
68. Magnetic field due to a ring having n turns at a distance x on its axis is proportional to (if r = radius of ring)
[RPET 2000]
- (a) $\frac{r}{(x^2 + r^2)}$
(b) $\frac{r^2}{(x^2 + r^2)^{3/2}}$
(c) $\frac{nr^2}{(x^2 + r^2)^{3/2}}$
(d) $\frac{n^2 r^2}{(x^2 + r^2)^{3/2}}$
69. A and B are two concentric circular conductors of centre O and carrying currents i_1 and i_2 as shown in the adjacent figure. If ratio of their radii is $1 : 2$ and ratio of the flux densities at O due to A and B is $1 : 3$, then the value of i_1 / i_2 is
[KCET 2000]
- 
- (a) $\frac{1}{6}$
(b) $\frac{1}{4}$
(c) $\frac{1}{3}$
(d) $\frac{1}{2}$
70. A long straight wire carries an electric current of 2 A . The magnetic induction at a perpendicular distance of 5 m from the wire is
(a) $4 \times 10^{-8} \text{ T}$
(b) $8 \times 10^{-8} \text{ T}$
(c) $12 \times 10^{-8} \text{ T}$
(d) $16 \times 10^{-8} \text{ T}$
71. A straight wire carrying a current 10 A is bent into a semicircular arc of radius 5 cm . The magnitude of magnetic field at the center is
(a) $1.5 \times 10^{-5} \text{ T}$
(b) $3.14 \times 10^{-5} \text{ T}$
(c) $6.28 \times 10^{-5} \text{ T}$
(d) $19.6 \times 10^{-5} \text{ T}$
72. A long solenoid of length L has a mean diameter D . It has n layers of windings of N turns each. If it carries a current ' i ' the magnetic field at its centre will be
[MP PMT 2000]
- (a) Proportional to D
(b) Inversely proportional to D
(c) Independent of D
(d) Proportional to L
73. A circular loop of radius 0.0157 m carries a current of 2.0 amp . The magnetic field at the centre of the loop is
($\mu_0 = 4\pi \times 10^{-7} \text{ weber / amp - m}$)
(a) $1.57 \times 10^{-5} \text{ weber / m}^2$
(b) $8.0 \times 10^{-5} \text{ weber / m}^2$
(c) $2.5 \times 10^{-5} \text{ weber / m}^2$
(d) $3.14 \times 10^{-5} \text{ weber / m}^2$
74. A long solenoid has 200 turns per cm and carries a current of 2.5 amps . The magnetic field at its centre is ($\mu_0 = 4\pi \times 10^{-7} \text{ weber / amp - m}$)
(a) $3.14 \times 10^{-2} \text{ weber / m}^2$
(b) $6.28 \times 10^{-2} \text{ weber / m}^2$

(c) $9.42 \times 10^{-2} \text{ weber / m}^2$

(d) $12.56 \times 10^{-2} \text{ weber / m}^2$

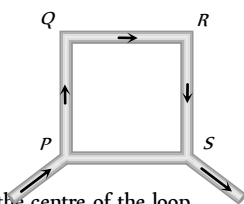
75. Two concentric coplanar circular loops of radii r_1 and r_2 carry currents of respectively i_1 and i_2 in opposite directions (one clockwise and the other anticlockwise.) The magnetic induction at the centre of the loops is half that due to i_1 alone at the centre. If $r_2 = 2r_1$, the value of i_2 / i_1 is

[MP PET 2000]

- (a) 2 (b) $1/2$
(c) $1/4$ (d) 1

76. PQRS is a square loop made of uniform conducting wire the current enters the loop at P and leaves at S. Then the magnetic field will be

[KCET 2000]



- (a) Maximum at the centre of the loop
(b) Zero at the centre of loop
(c) Zero at all points inside the loop
(d) Zero at all points outside of the loop

77. Magnetic fields at two points on the axis of a circular coil at a distance of 0.05 m and 0.2 m from the centre are in the ratio 8 : 1. The radius of the coil is

[KCET 2002]

- (a) 1.0 m (b) 0.1 m
(c) 0.15 m (d) 0.2 m

78. An electric current passes through a long straight wire. At a distance 5 cm from the wire, The magnetic field is B . The field at 20 cm from the wire would be

[CPMT 2001; Pb PET 2002]

- (a) $\frac{B}{6}$ (b) $\frac{B}{4}$
(c) $\frac{B}{3}$ (d) $\frac{B}{2}$

79. A closely wound flat circular coil of 25 turns of wire has diameter of 10 cm and carries a current of 4 ampere . Determine the flux density at the centre of a coil

[AIIMS 2001]

- (a) $1.679 \times 10^{-5} \text{ tesla}$ (b) $2.028 \times 10^{-4} \text{ tesla}$
(c) $1.257 \times 10^{-3} \text{ tesla}$ (d) $1.512 \times 10^{-6} \text{ tesla}$

80. The dimension of the magnetic field intensity B is

[MP PET 2001]

- (a) $MLT^{-2}A^{-1}$ (b) $MT^{-2}A^{-1}$
(c) ML^2TA^{-2} (d) $M^2LT^{-2}A^{-1}$

81. A current of 2 amp. flows in a long, straight wire of radius 2 mm . The intensity of magnetic field on the axis of the wire is

- (a) $\left(\frac{\mu_0}{\pi}\right) \times 10^3 \text{ Tesla}$ (b) $\left(\frac{\mu_0}{2\pi}\right) \times 10^3 \text{ Tesla}$
(c) $\left(\frac{2\mu_0}{\pi}\right) \times 10^3 \text{ Tesla}$ (d) Zero

82. The magnetic field at the centre of a circular coil of radius r carrying current I is B_1 . The field at the centre of another coil of radius $2r$ carrying same current I is B_2 . The ratio $\frac{B_1}{B_2}$ is

- (a) $\frac{1}{2}$ (b) 1
(c) 2 (d) 4

83. A current flows through an infinitely long straight wire. The magnetic field produced at a point 1 metres away from it is

[MP PMT 2001]

- (a) $2 \times 10^{-3} \text{ Tesla}$ (b) $\frac{2}{10} \text{ Tesla}$
(c) $2 \times 10^{-7} \text{ Tesla}$ (d) $2\pi \times 10^{-6} \text{ Tesla}$

84. Two infinitely long parallel wires carry equal current in same direction. The magnetic field at a mid point in between the two wires is

[MP PMT 2001]

- (a) Twice the magnetic field produced due to each of the wires
(b) Half of the magnetic field produced due to each of the wires
(c) Square of the magnetic field produced due to each of the wires
(d) Zero

85. A wire in the form of a square of side ' a ' carries a current i . Then the magnetic induction at the centre of the square wire is (Magnetic permeability of free space $= \mu_0$)

[EAMCET 2001]

- (a) $\frac{\mu_0 i}{2\pi a}$ (b) $\frac{\mu_0 i\sqrt{2}}{\pi a}$
(c) $\frac{2\sqrt{2}\mu_0 i}{\pi a}$ (d) $\frac{\mu_0 i}{\sqrt{2}\pi a}$

86. What should be the current in a circular coil of radius 5 cm to annul $B_H = 5 \times 10^{-5} \text{ T}$

[DCE 2001]

- (a) 0.4 A (b) 4 A
(c) 40 A (d) 1 A

87. A current of 0.1 A circulates around a coil of 100 turns and having a radius equal to 5 cm . The magnetic field set up at the centre of the coil is ($\mu = 4\pi \times 10^{-7} \text{ weber/amp-metre}$)

[DPMT 2002]

- (a) $2 \times 10^{-7} \text{ Tesla}$ (b) $4 \times 10^{-7} \text{ Tesla}$
(c) $8\pi \times 10^{-7} \text{ Tesla}$ (d) $4\pi \times 10^{-7} \text{ Tesla}$

88. An electron moving in a circular orbit of radius r makes n rotation per second. The magnetic field produced at the centre has a magnitude of

[KCET 2001; UPSEAT 2001, 02]

- (a) $\frac{\mu_0 ne}{2r}$ (b) $\frac{\mu_0 n^2 e}{2r}$

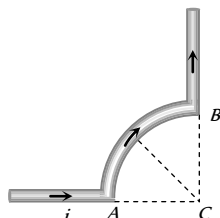
- (c) $\frac{\mu_0 n e}{2\pi}$ (d) Zero

89. A long solenoid has n turns per meter and current I A is flowing through it. The magnetic field at the ends of the solenoid is

- (a) $\frac{\mu_0 n I}{2}$ (b) $\mu_0 n I$
(c) Zero (d) $2\mu_0 n I$

90. A wire carrying current i is shaped as shown. Section AB is a quarter circle of radius r . The magnetic field is directed

[KCET 2002]

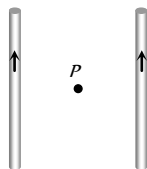


- (a) At an angle $\pi/4$ to the plane of the paper
(b) Perpendicular to the plane of the paper and directed in to the paper
(c) Along the bisector of the angle ACB towards AB
(d) Along the bisector of the angle ACB away from AB

91. Two long straight wires are set parallel to each other. Each carries a current i in the same direction and the separation between them is $2r$. The intensity of the magnetic field midway between them is

[Kerala PET 2002; DCE 2002]

- (a) $\mu_0 i/r$
(b) $4\mu_0 i/r$
(c) Zero
(d) $\mu_0 i/4r$



92. A magnetic field can be produced by [AIEEE 2002]
(a) A moving charge (b) A changing electric field
(c) None of these (d) Both of these

93. Unit of magnetic permeability is [AFMC 2002]
(a) Amp/metre (b) Amp/m^2
(c) Henry (d) $\text{Henry}/\text{metre}$

94. A long straight wire carries a current of π amp. The magnetic field due to it will be $5 \times 10^{-5} \text{ weber}/\text{m}^2$ at what distance from the wire [μ_0 = permeability of air]

[MP PMT 2002]

- (a) $10^4 \mu_0 \text{ metre}$ (b) $\frac{10^4}{\mu_0} \text{ metre}$
(c) $10^6 \mu_0 \text{ metre}$ (d) $\frac{10^6}{\mu_0} \text{ metre}$

95. When a certain length of wire is turned into one circular loop, the magnetic induction at the centre of coil due to some current flowing is B_1 . If the same wire is turned into three loops to make a circular

coil, the magnetic induction at the center of this coil for the same current will be

[MP PMT 2002]

- (a) B_1 (b) $9B_1$
(c) $3B_1$ (d) $27B_1$
96. Gauss is unit of which quantity [MP PET 2002]
(a) H (b) B
(c) ϕ (d) 1

97. On connecting a battery to the two corners of a diagonal of a square conductor frame of side a the magnitude of the magnetic field at the centre will be [MP PET 2002]

- (a) Zero (b) $\frac{\mu_0}{\pi a}$
(c) $\frac{2\mu_0}{\pi a}$ (d) $\frac{4\mu_0 i}{\pi a}$

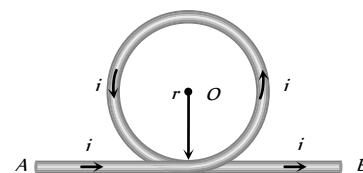
98. The ratio of the magnetic field at the centre of a current carrying coil of the radius a and at a distance ' a ' from centre of the coil and perpendicular to the axis of coil is

[MP PET 2002]

- (a) $\frac{1}{\sqrt{2}}$ (b) $\sqrt{2}$
(c) $\frac{1}{2\sqrt{2}}$ (d) $2\sqrt{2}$

99. A part of a long wire carrying a current i is bent into a circle of radius r as shown in figure. The net magnetic field at the centre O of the circular loop is [UPSEAT 2002]

- (a) $\frac{\mu_0 i}{4r}$
(b) $\frac{\mu_0 i}{2r}$
(c) $\frac{\mu_0 i}{2\pi r}(\pi + 1)$
(d) $\frac{\mu_0 i}{2\pi r}(\pi - 1)$



100. The current in the windings on a toroid is 2.0A. There are 400 turns and the mean circumferential length is 40cm. If the inside magnetic field is 1.0 T, the relative permeability is near to
(a) 100 (b) 200
(c) 300 (d) 400

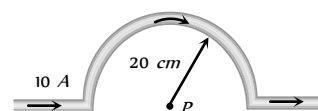
101. "On flowing current in a conducting wire the magnetic field produces around it." It is a law of [RPET 2003]
(a) Lenz (b) Ampere
(c) Ohm (d) Maxwell

102. The magnetic field near a current carrying conductor is given by
(a) Coulomb's law (b) Lenz' law
(c) Biot-savart's law (d) Kirchoff's law

103. A current of 10A is passing through a long wire which has semicircular loop of the radius 20cm as shown in the figure. Magnetic field produced at the centre of the loop is

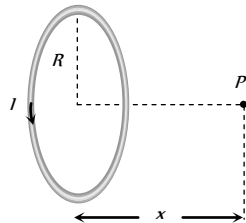
[Orissa JEE 2003]

- (a) $10 \pi \mu T$
(b) $5 \pi \mu T$
(c) $4 \pi \mu T$
(d) $2 \pi \mu T$



104. A wire in the form of a circular loop of one turn carrying a current produces a magnetic field B at the centre. If the same wire is looped into a coil of two turns and carries the same current, the new value of magnetic induction at the centre is [CBSE 2002; KCET 2003]
- (a) $5B$ (b) $3B$
(c) $2B$ (d) $4B$
105. A long solenoid carrying a current produces a magnetic field B along its axis. If the current is doubled and the number of turns per cm is halved, the new value of the magnetic field is [CBSE PMT 2003]
- (a) B (b) $2B$
(c) $4B$ (d) $B/2$
106. A long straight wire carrying current of $30A$ is placed in an external uniform magnetic field of induction $4 \times 10^{-7}T$. The magnetic field is acting parallel to the direction of current. The magnitude of the resultant magnetic induction in *tesla* at a point 2.0 cm away from the wire is [EAMCET 2003]
- (a) 10^{-7} (b) 3×10^{-7}
(c) 5×10^{-7} (d) 6×10^{-7}
107. The earth's magnetic field at a given point is $0.5 \times 10^{-5} \text{ Wb} \cdot \text{m}^{-2}$. This field is to be annulled by magnetic induction at the center of a circular conducting loop of radius 5.0 cm . The current required to be flown in the loop is nearly [AIIMS 2003]
- (a) 0.2 A (b) 0.4 A
(c) 4 A (d) 40 A
108. A coil having N turns carry a current I as shown in the figure. The magnetic field intensity at point P is [BHU 2003; CPMT 2004]

- (a) $\frac{\mu_0 N I R^2}{2(R^2 + x^2)^{3/2}}$
(b) $\frac{\mu_0 N I}{2R}$
(c) $\frac{\mu_0 N I R^2}{(R + x)^2}$
(d) Zero

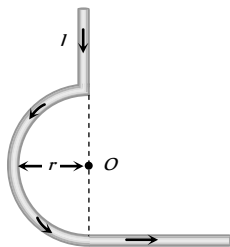


109. Two similar coils are kept mutually perpendicular such that their centres coincide. At the centre, find the ratio of the magnetic field due to one coil and the resultant magnetic field by both coils, if the same current is flown [BHU 2003; CPMT 2004]

- (a) $1 : \sqrt{2}$ (b) $1 : 2$
(c) $2 : 1$ (d) $\sqrt{3} : 1$

110. In the figure, what is the magnetic field at the point O [MP PMT 2004]

- (a) $\frac{\mu_0 I}{4\pi r}$
(b) $\frac{\mu_0 I}{4\pi r} + \frac{\mu_0 I}{2\pi r}$
(c) $\frac{\mu_0 I}{4r} + \frac{\mu_0 I}{4\pi r}$
(d) $\frac{\mu_0 I}{4r} - \frac{\mu_0 I}{4\pi r}$



111. The magnetic moment of a current (i) carrying circular coil of radius (r) and number of turns (n) varies as [AIIMS 2004]
- (a) $1/r$ (b) $1/r$
(c) r (d) r

112. A current flows in a conductor from east to west. The direction of the magnetic field at a points above the conductor is
- (a) Towards north (b) Towards south
(c) Towards east (d) Towards west
113. A long wire carries a steady current. It is bent into a circle of one turn and the magnetic field at the centre of the coil is B . It is then bent into a circular loop of n turns. The magnetic field at the centre of the coil will be [AIEEE 2004]
- (a) nB (b) nB
(c) $2nB$ (d) $2nB$
114. The magnetic field due to a current carrying circular loop of radius 3 cm at a point on the axis at a distance of 4 cm from the centre is $54\text{ }\mu\text{T}$. What will be its value at the centre of the loop
- (a) $250\text{ }\mu\text{T}$ (b) $150\text{ }\mu\text{T}$
(c) $125\text{ }\mu\text{T}$ (d) $75\text{ }\mu\text{T}$
115. The magnetic induction at the centre of a current carrying circular of coil radius r , is [J & K CET 2004]
- (a) Directly proportional to r
(b) Inversely proportional r
(c) Directly proportional to r
(d) Inversely proportional to r
116. The current is flowing in south direction along a power line. The direction of magnetic field above the power line (neglecting earth's field) is [Pb. PMT 2004; Kerala PMT 2004]

- (a) South (b) East
(c) North (d) West

117. Two wires of same length are shaped into a square and a circle. If they carry same current, ratio of the magnetic moment is

- (a) $2 : \pi$ (b) $\pi : 2$
(c) $\pi : 4$ (d) $4 : \pi$

118. When the current flowing in a circular coil is doubled and the number of turns of the coil in it is halved, the magnetic field at its centre will become [DPMT 2003]

- (a) Four times (b) Same
(c) Half (d) Double

119. An electron is revolving round a proton, producing a magnetic field of 16 weber/m in a circular orbit of radius $1\text{ }\text{\AA}$. It's angular velocity will be [RPMT 2002]

- (a) 10^7 rad/sec (b) $1/2\pi \times 10^7\text{ rad/sec}$
(c) $2\pi \times 10^7\text{ rad/sec}$ (d) $4\pi \times 10^7\text{ rad/sec}$

120. 20 ampere current is flowing in a long straight wire. The intensity of magnetic field at a distance 10 cm from the wire will be

- (a) $4 \times 10^{-7}\text{ Wb/m}$ (b) $9 \times 10^{-7}\text{ Wb/m}$
(c) $8 \times 10^{-7}\text{ Wb/m}$ (d) $6 \times 10^{-7}\text{ Wb/m}$

121. The field due to a long straight wire carrying a current I is proportional to [MP PMT 1993]

- (a) I (b) I^3
(c) $\sqrt{I}$ (d) $1/I$

122. Two concentric coils each of radius equal to $2\pi\text{ cm}$ are placed at right angles to each other. 3 ampere and 4 ampere are the currents flowing in each coil respectively. The magnetic induction in

Weber/ m^2 at the centre of the coils will be
($\mu_0 = 4\pi \times 10^{-7} \text{ Wb/A.m}$) [AIEEE 2005]

- (a) 5×10^{-5} (b) 7×10^{-5}
(c) 12×10^{-5} (d) 10^{-5}

123. A wire carrying current I and other carrying $2I$ in the same direction produces a magnetic field B at the mid point. What will be the field when $2I$ wire is switched off [AFMC 2005]

- (a) $B/2$ (b) $2B$
(c) B (d) $4B$

124. Two long parallel wires P and Q are both perpendicular to the plane of the paper with distance 5 m between them. If P and Q carry current of 2.5 amp and 5 amp respectively in the same direction, then the magnetic field at a point half way between the wires is

- (a) $\frac{\sqrt{3}\mu_0}{2\pi}$ (b) $\frac{\mu_0}{\pi}$
(c) $\frac{3\mu_0}{2\pi}$ (d) $\frac{\mu_0}{2\pi}$
(e) $\frac{\sqrt{3}\mu_0}{\pi}$

125. The direction of magnetic lines of force produced by passing a direct current in a conductor is given by [J & K CET 2005]

- (a) Lenz's law (b) Fleming's left hand rule
(c) Right hand palm rule (d) Maxwell's law

126. For the magnetic field to be maximum due to a small element of current carrying conductor at a point, the angle between the element and the line joining the element to the given point must be

- (a) 0° (b) 90°
(c) 180° (d) 45°

Motion of Charged Particle In Magnetic Field

1. A proton moving with a constant velocity passes through a region of space without any change in its velocity. If $\vec{E}$ and $\vec{B}$ represent the electric and magnetic fields respectively, then this region of space may have

[IIT-JEE 1985; AMU 1995; AFMC 2001;
Roorkee 2000; AMU (Med.) 2000]

- (a) $E = 0, B = 0$ (b) $E = 0, B \neq 0$
(c) $E \neq 0, B = 0$ (d) $E \neq 0, B \neq 0$

2. A uniform electric field and a uniform magnetic field are produced, pointed in the same direction. An electron is projected with its velocity pointing in the same direction

[NCERT 1980; CBSE PMT 1993; JIPMER 1997;
AIEEE 2005]

- (a) The electron will turn to its right
(b) The electron will turn to its left
(c) The electron velocity will increase in magnitude
(d) The electron velocity will decrease in magnitude

3. Two particles X and Y having equal charges, after being accelerated through the same potential difference, enter a region of uniform

magnetic field and describes circular path of radius R_1 and R_2 respectively. The ratio of mass of X to that of Y is [IIT-JEE 1988; CBSE PMT 1995; M

- (a) $\left(\frac{R_1}{R_2}\right)^{1/2}$ (b) $\frac{R_2}{R_1}$
(c) $\left(\frac{R_1}{R_2}\right)^2$ (d) $\frac{R_1}{R_2}$

4. A beam of ions with velocity $2 \times 10^5 \text{ m/s}$ enters normally into a uniform magnetic field of $4 \times 10^{-2} \text{ tesla}$. If the specific charge of the ion is $5 \times 10^7 \text{ C/kg}$, then the radius of the circular path described will be [NCERT 1983; BVP 2003]

- (a) 0.16 m (b) 0.16 m
(c) 0.20 m (d) 0.25 m

5. The radius of curvature of the path of the charged particle in a uniform magnetic field is directly proportional to [MNR 1995; UPSEAT 1999, 2000]

- (a) The charge on the particle
(b) The momentum of the particle
(c) The energy of the particle
(d) The intensity of the field

6. An electron has mass $9 \times 10^{-31} \text{ kg}$ and charge $1.6 \times 10^{-19} \text{ C}$ is moving with a velocity of 10^6 m/s , enters a region where magnetic field exists. If it describes a circle of radius 0.10 m , the intensity of magnetic field must be

[NCERT 1982; CPMT 1989; DCE 2005]

- (a) $1.8 \times 10^{-4} \text{ T}$ (b) $5.6 \times 10^{-5} \text{ T}$
(c) $14.4 \times 10^{-5} \text{ T}$ (d) $1.3 \times 10^{-6} \text{ T}$

7. A proton (mass m and charge $+e$) and an α -particle (mass $4m$ and charge $+2e$) are projected with the same kinetic energy at right angles to the uniform magnetic field. Which one of the following statements will be true [NCERT 1983]

- (a) The α -particle will be bent in a circular path with a small radius that for the proton
(b) The radius of the path of the α -particle will be greater than that of the proton
(c) The α -particle and the proton will be bent in a circular path with the same radius
(d) The α -particle and the proton will go through the field in a straight line

8. A charged particle moving in a magnetic field experiences a resultant force [MP PMT 1994]

- (a) In the direction of field
(b) In the direction opposite to that field
(c) In the direction perpendicular to both the field and its velocity
(d) None of the above

9. If the direction of the initial velocity of the charged particle is perpendicular to the magnetic field, then the orbit will be

or

The path executed by a charged particle whose motion is perpendicular to magnetic field is

[MP PMT 1993; CPMT 1996]

- (a) A straight line (b) An ellipse
(c) A circle (d) A helix